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# Crazy Merged 2

**Sections 4–8**

One merged book: the condensed books and the Paper 1–4 answers combined, section by section, with each topic appearing once. Read each chapter top to bottom. Where a topic was answered in the papers, the full paper answer sits in its place (marked "Paper answer", with the papers it was asked in), followed by a dashed box of anything extra the book had. Paper answers not tied to a chapter are at the end of each section, then the section's Rapid Revision Sheet.


| Section | Title |
|---|---|
| 4 | Paediatric Urology 1 |
| 5 | Paediatric Urology 2 |
| 6 | Female Urology |
| 7 | Voiding Dysfunction |
| 8 | Nocturia & Male Incontinence |

SECTION 4

# Paediatric Urology 1

## 41 Management Strategies for Vesicoureteral Reflux

### 1 · Definition and concept

*We treat the infection risk, not the reflux.*

- **Vesicoureteral reflux (VUR)** is retrograde urine flow to the upper tract, linked to **acquired scarring** and **congenital dysplasia**.
- **Infection, not reflux, is the injury**, so aim for sterile urine and good emptying.

### 2 · Embryology and antireflux mechanism

#### LAQ Primary VUR: VUJ embryology/anatomy, pathophysiology, management (incl. bilateral, adult grade 3), reimplantation techniques and STING

Paper answer · P1 · P3 · P4 · merged from 6

*Asked as:* P1 LAQ "Primary (congenital) VUR: VUJ embryology, pathophysiology, natural history and management" (Describe embryological development of VU Junction and outline management of primary VUR?, Summer 2020 Paper I Q2; B/l congenital VUR - Pathophysiology and natural history, Practice Set 1 Paper I Q3f SAQ) · P3 LAQ "Management of bilateral VUR: medical treatment, surgical techniques, selection and long-term outcome" (Describe medical management of VUR. Describe treatment of an 8-yr male child with b/l grade IV VUR, Summer 2015 Paper III Q2; Discuss mgt. of b/l VUR, Practice Set 2 Paper III Q2; Discuss treatment of b/l VUR. Describe surgical techniques and criteria for their selection, Practice Set 2 Paper III Q3d SAQ; Discuss the management of B/l VUR and discuss its long term outcome, Practice Set 6 Paper III Q1) · P4 LAQ "VUR: etiopathogenesis, recent trends in management and STING technique" (Describe etiopathogenesis of VUR and recent trends in management, Practice Set 2 Paper IV Q2; STING technique for VUR, Summer 2020 Paper IV Q3c and Practice Set 5 Paper IV Q3e SAQ) · P3 LAQ "Development and clinical anatomy of Vesico-Ureteric junction and management of grade 3 VUR in adult?" · P3 SAQ "Endoscopic correction of VUR (STING procedure): indications and complications" (Endoscopic correction of VUR : indications and complications, Summer 2019 Paper III Q3e; Sting procedure, Winter 2024 Paper III Q3f) · P3 SAQ "Paquin and Gil-Vernet techniques of ureteric reimplantation"

*"B/l congenital VUR" is read as bilateral primary (congenital) VUR (vesicoureteral reflux), often detected antenatally. STING is taken as subureteric Teflon injection, the endoscopic treatment of VUR.*

**Definition and epidemiology**

- **VUR**: retrograde flow of urine from the bladder into the upper tract.
- **Primary**: congenitally deficient UVJ (ureterovesical junction) flap valve with a short intravesical tunnel and a normal bladder. **Secondary**: high bladder pressure (PUV (posterior urethral valves), neurogenic bladder, BBD (bladder and bowel dysfunction)).

- Found in about **30% of children with UTI (urinary tract infection)**, about 30% of siblings and about 17% of children imaged for hydronephrosis. The younger the child, the likelier VUR.
- Infant boys predominate (higher voiding pressures, more UTI if uncircumcised); overall girls are affected twice as often. Less common in Black children.
- **Bilateral VUR is a high-risk feature:** it resolves less and scars more.
- **Infection, not sterile reflux, injures the kidney.** Infected urine plus intrarenal reflux causes scars, hypertension and CKD (chronic kidney disease); some high-grade "scars" are congenital dysplasia. So management targets infection risk, not the reflux itself.

### Embryology of the VUJ (vesicoureteric junction)

1. **Ureteric bud (weeks 4–6)** arises from the caudal mesonephric (Wolffian) duct: ureter and duct form a "Y", and the common nephric (excretory) duct is the stem.
2. **Reciprocal induction:** the bud enters the metanephric blastema, branches into the collecting system and induces nephrons.
3. **Weeks 5–6:** the urorectal septum divides the cloaca into urogenital sinus (UGS) and anorectal canal.
4. **Absorption:** the common nephric duct is taken into the UGS (current view: removed by **apoptosis**), so ureter and duct open separately.
5. **Rotation (Mackie and Stephens):** the ureteric orifice moves **cranially and laterally** to the bladder base; the duct stays caudal as the ejaculatory duct, ending in the prostatic urethra (Gartner duct in females).
6. **Trigone:** classically ureteral; lineage tracing shows it is mostly bladder muscle. The **Chwalla membrane** over the orifice ruptures; its persistence explains ureterocele.
7. **Maturation:** ureteric muscle joins the detrusor as **Waldeyer's sheath**; the tunnel lengthens with growth, so VUR resolves with age.

| Bud origin | Orifice | Result |
|---|---|---|
| Normal | Trigone corner | Tunnel length to diameter about **5:1** (Paquin); refluxing junctions about 1.4:1 |
| Too low (early budding, early contact with UGS) | High, **lateral** | Short tunnel, so **primary VUR**; abnormal bud–metanephros contact gives **dysplasia** |
| Too high (late budding, late contact) | Caudal (distal), **medial** | **Ectopic ureter** (bladder neck, urethra, Wolffian structures), often obstructed |
| Two buds | Duplex | **Weigert–Meyer rule:** upper pole caudal-medial (ectopia, ureterocele); lower pole lateral (reflux) |

- **Concept:** bud position fixes both orifice and kidney, so this single embryological fault (lateral orifice, short tunnel, dysplasia) explains why high-grade VUR and dysplasia coexist.

### Clinical anatomy of the VUJ

- **Course:** juxtavesical ureter in **Waldeyer's sheath**, then an oblique intramural segment through the detrusor hiatus, then the submucosal tunnel, then the orifice (caudal and medial to the entry).
- **Intramural length 1.2–2.5 cm in adults** (0.5–0.8 cm in neonates). The **narrowest part** of the ureter (about 3–4 mm): the commonest stone impaction site.

- **Trigone:** superficial trigone (ureteric longitudinal muscle, running to the verumontanum); deep trigone (Waldeyer's sheath plus detrusor, to the bladder neck). Its base is the interureteric ridge. Waldeyer's sheath anchors the intramural ureter to the deep trigone.
- **Orifice (Lyon):** position A (medial trigone) to C (lateral); shape cone (normal), stadium, horseshoe, **golf-hole**. The more lateral and open, the more reflux.
- **Relations:** the vas (male) or uterine artery (female) crosses the ureter. **Pelvic plexus nerves lie dorsomedial to the VUJ**; wide extravesical dissection causes voiding dysfunction.
- **Clinical:** Hutch diverticulum at the hiatus; "fish-hook" ureter with BPH (benign prostatic hyperplasia).

### Normal antireflux mechanism

- Balance of three factors: **ureteric peristalsis**, **UVJ anatomy** and **bladder dynamics** (low bladder pressure).
- **Passive flap valve:** the submucosal (intramural) ureter is compressed against firm detrusor backing as the bladder fills. This needs adequate tunnel length and fixation.
- **Paquin:** tunnel length to diameter about **5:1** normally versus 1.4:1 in refluxing units. Orifice shape also matters.
- **Active:** ureteric peristalsis opens the tunnel for each bolus, then it closes.
- **Bladder pressure:** high storage or voiding pressure overwhelms a normal valve.

### Etiopathogenesis

**1. Embryology (Mackie and Stephens):** early budding gives a high, lateral orifice with a short tunnel (VUR) and dysplasia; late budding gives a distal, medial, ectopic and often obstructed ureter (see above).

**2. Primary versus secondary**

| Primary | Secondary |
|---|---|
| Congenitally deficient UVJ: short tunnel, weak detrusor backing, lateral orifice | Normal UVJ overwhelmed by pressure: PUV, neurogenic bladder (VUR in up to 30% of spina bifida), dysfunctional voiding; or UVJ anomaly (ureterocele, duplication, paraureteral diverticulum) |

- They overlap: "primary" VUR often has a treatable secondary component.

**3. Bladder and bowel dysfunction (BBD):** high voiding pressures and incomplete emptying distort the UVJ. Detrusor overactivity is the commonest urodynamic finding. Infant boys have high voiding pressures that settle with maturation.

**4. Genetics:** sibling prevalence about 30%, highest under 3 years; identical twins almost fully concordant; polygenic.

**5. Infection:** cystitis lowers the reflux threshold; *E. coli* endotoxin causes ureteric atony.

### Pathogenesis of renal damage

| | Reflux-associated dysplasia | Acquired scar |
|---|---|---|
| Cause | Abnormal bud (congenital) | Infected urine plus VUR |
| Typical patient | Boy, grade IV–V, antenatal hydronephrosis | Girl, recurrent febrile UTI, BBD |
| Preventable | No | Yes |

- **Intrarenal reflux** through flat compound papillae at the poles needs only 2 mmHg in neonates, so scars are polar.
- **Sterile reflux rarely scars** unless pressures are high (Ransley and Risdon). Most scars follow early infections ("big bang", now debated).
- Sequelae: hypertension in **34–38%** of scarred children; CKD; poor growth; pre-eclampsia.
- **Also know:** massive bilateral VUR gives the megacystis–megaureter association.

### International grading (International Reflux Study, 1981; VCUG (voiding cystourethrogram))

| Grade | Findings |
|---|---|
| I | Non-dilated ureter only |
| II | Ureter, pelvis and calyces reached; no dilatation, sharp fornices |
| III | Mild–moderate dilatation of ureter, pelvis and calyces; minimal forniceal blunting |
| IV | Moderate dilatation and tortuosity; fornices obliterated but **papillary impressions kept** |
| V | Gross dilatation and tortuosity; **papillary impressions lost**; intrarenal reflux |

- **Also know:** grading varies between observers; UDR (ureteral diameter ratio) predicts resolution better.

### Evaluation (children)

- History of febrile UTIs and BBD; urine culture; BP (blood pressure), growth, spine, urine protein, creatinine.
- **RBUS (renal and bladder ultrasound)**: a normal scan does not exclude VUR.
- **VCUG** (gold standard): grade and urethra (excludes PUV). **Nuclear cystogram** (lower dose) for follow-up; **ceVUS (contrast-enhanced voiding urosonography)** avoids radiation.
- **DMSA (dimercaptosuccinic acid) scan**: scars and split function; a defect persisting 6 months is a scar.
- **UDR** predicts resolution and breakthrough UTI better than grade.

### Natural history

- Resolution comes from **elongation of the intravesical ureter** with growth.
- Resolution falls with grade and age: grade II 63–85%, grade III about 50%, **grades IV–V and bilateral grade III rarely** (International Reflux Study). Younger age resolves better; infants with grade IV–V may still resolve fairly well.
- **Bilaterality and older age delay resolution:** persistence is highest in bilateral grade IV (AUA (American Urological Association) 1997: about 10% resolve in 5 years), and in bilateral grade III at 6–10 years.
- **Consequences of bilateral damage:** adult hypertension in 34–38% of scarred children; CKD, with reflux nephropathy a leading paediatric transplant diagnosis; poor growth; pre-eclampsia (worse with bilateral scars). Correction cannot reverse dysplasia.

### Risk stratification (Hidas)

| Low risk | High risk |
|---|---|
| Male, >5 years | Female, <1 year (infant) |
| Unilateral, grade I–II | **Bilateral**, grade III–V |

| Low risk | High risk |
|---|---|
| No BBD, no scars | BBD, scars, multiple febrile UTIs |

- Two-year breakthrough UTI on CAP (continuous antibiotic prophylaxis): 8.6% low, 26% intermediate, 62.5% high risk.

### Recent trends in management

1. **Treat infection risk, not reflux:** risk stratification by sex, age, grade, laterality, BBD, scars and presentation (Hidas model).
2. **Fewer VCUGs:** ultrasound first (AAP (American Academy of Pediatrics) 2011, NICE (National Institute for Health and Care Excellence)); or "top-down" DMSA first. Contrast-enhanced voiding urosonography avoids radiation.
3. **Objective measures:** UDR predicts resolution and breakthrough UTI better than grade; machine-learning models.
4. **Selective CAP:** RIVUR halved recurrent UTI but did not reduce scarring; NNT (number needed to treat) 5 in high-risk and 18 in low-risk children. The Swedish Reflux Trial showed benefit in girls with dilating VUR.
5. **BBD first:** timed voiding, constipation treatment, anticholinergics. Topical steroid for phimosis in boys.
6. **Endoscopic injection:** Dx/HA (dextranomer/hyaluronic acid); HIT and double HIT (hydrodistension implantation technique) give about 90% resolution. Vantris works better but can obstruct late.
7. **RALUR-EV (robot-assisted extravesical reimplantation):** less pain and shorter stay; results approaching the open benchmark of **95.9%**.
8. **Less follow-up imaging:** VCUG may be omitted after surgery for low-grade VUR if ultrasound is normal.

### Medical management

- **Aim:** prevent febrile UTI and scars while VUR resolves.
- **Treat BBD first:** timed voiding, hydration, constipation treatment, anticholinergics if needed. BBD raises breakthrough UTI and lowers resolution and endoscopic success.
- **CAP:** one low dose at night, a quarter of the treatment dose.

| Agent | Age | Dose | Note |
|---|---|---|---|
| Trimethoprim or amoxicillin | <2 months (young infants) | — | Avoid sulfonamide (neonatal jaundice) |
| TMP-SMX (trimethoprim-sulfamethoxazole) | >2 months | 2 mg/kg TMP | Avoid under 2 months (kernicterus); used in RIVUR |
| Nitrofurantoin | >2 months | 1–2 mg/kg | Avoid in G6PD (glucose-6-phosphate dehydrogenase) deficiency; rare lung, liver toxicity |

- **RIVUR (Randomized Intervention for Children with Vesicoureteral Reflux, 2014)**, 607 children: recurrent UTI fell from **23.6% to 12.9%** on TMP-SMX; **scarring unchanged**; most benefit with BBD and in high risk.
- **Swedish Reflux Trial** (grade III–IV, age 1–2 years): febrile UTI in girls 19% on CAP, 23% after endoscopy, 57% on surveillance; **no difference in boys**.

- **IRSC (International Reflux Study in Children) and Birmingham trials:** surgery equalled medical care for (new) scarring. **PRIVENT:** 6% absolute UTI reduction.
- **Circumcision** or topical steroid for phimosis: about 4 boys with VUR circumcised prevent one UTI (NNT about 4).
- After stopping CAP, 8–10% have a febrile UTI.
- Low-risk children: observation without CAP.
- **Guidelines:** AUA 2010/2017 recommends CAP for VUR with BBD and treating BBD first; EAU/ESPU (European Association of Urology/European Society for Paediatric Urology) give high-risk children CAP and correct early after breakthrough infection.

### Indications for correction

- Breakthrough febrile UTI on CAP despite good bladder and bowel habits.
- New scars or falling function.
- Persistent high-grade (IV–V) bilateral VUR (girls traditionally corrected before puberty; benefit unproven).
- Non-adherence or intolerance to CAP; parental choice.
- **Controversy:** asymptomatic bilateral high-grade VUR in an older boy with normal kidneys can be observed.

### Case: 8-year-old boy with bilateral grade IV VUR

- Exclude PUV, neurogenic bladder and BBD; DMSA, creatinine, BP.
- This rarely resolves at 8 years, but older boys have low UTI risk, so treating an **asymptomatic** boy is **controversial**.
- **No UTI, no scars:** observation or CAP with BBD therapy. **Breakthrough UTI, scars or falling function:** correct surgically.
- **Choice here: bilateral Cohen.** `Exclude PUV/neurogenic bladder, treat BBD -> febrile UTI, scars or renal loss? Yes -> bilateral Cohen (or Dx/HA) No -> observe or CAP; correct if UTI/scar`

## Endoscopic correction: STING procedure

### Definition and principle

- **STING (subureteric Teflon injection):** endoscopic injection of a **bulking agent** beneath the ureteric orifice. The mound supports and lengthens the intramural tunnel, restoring the **flap valve**.
- First done with Teflon by Matouschek (1981); popularised by **O'Donnell and Puri, 1986**. **Dx/HA (Deflux)** is now standard.

### Bulking agents

| Agent | Particle / property | Problem |
|---|---|---|
| Teflon (PTFE (polytetrafluoroethylene) paste) | 90% <40 µm, permanent | **Migration**; never FDA (US Food and Drug Administration) approved |
| **Dx/HA (Deflux)** | Microspheres **80–250 µm**; gel resorbed | Volume loss about 23%; late recurrence; FDA approved 2001; standard agent |
| Polyacrylate (Vantris) | 300 µm, permanent | Works better but late ureteric obstruction |

- **Also know:** Macroplastique.

### Indications

- Persistent primary VUR (grades II–IV) with **breakthrough febrile UTI** on prophylaxis, **new scars**, poor adherence or parental preference.
- Older girls; residual VUR after reimplantation; duplex or secondary VUR (lower success).
- **Relative contraindications:** grade V, **UVJ obstruction**, untreated BBD, large diverticulum.

**Technique**

1. Antibiotic at induction; offset-lens cystoscope; rigid 3.7–5 Fr needle; bladder partly filled.
2. **STING:** needle enters **2–3 mm below the orifice at 6 o'clock**, bevel up, advanced **4–5 mm** submucosally.
3. Inject slowly; a mound should appear within **0.1–0.2 mL**, else reposition.
4. Aim for a **"volcano" mound** with a slit-like orifice. Do not cauterise bleeding (the implant extrudes). Day case.

| Technique | Needle entry | Result |
|---|---|---|
| STING | 2–3 mm below the orifice, at 6 o'clock | Baseline |
| **HIT** (hydrodistension implantation technique) | Inside the hydrodistended orifice | Better than STING, all grades |
| **Double HIT** | Two intraureteric sites | About 90% resolution |

**Results**

- Low-grade VUR about 90% after one injection.
- **Elder 2006 meta-analysis** (one injection): grade I–II about 79%, III 72%, **IV 63%**, V 51%; repeat injections raise success.
- **Swedish Reflux Trial:** 20% recurrence at 2 years after initial success.
- Less effective in adults.

**Complications**

- **Failure or recurrence** (commonest): repeat injection once or twice, then reimplant (reimplantation is not harder after Dx/HA).
- **New contralateral VUR** after unilateral injection.
- **Ureteric obstruction:** early or **late-onset**, so long-term ultrasound.
- Febrile UTI despite a negative VCUG.
- **Calcified mound** (over half) mimics a ureteric stone. Teflon migration.

**Follow-up**

- Antibiotics 3 months; ultrasound at 2–6 weeks; ultrasound with or without VCUG at 3 months.

### Ureteric reimplantation: principles and techniques

- **Principles:** exclude secondary VUR; gentle handling; tension-free; straight course; tunnel **5:1** (Paquin's rule).
- Open success **95.9%** overall (Elder 1997; AUA meta-analysis benchmark).

| Technique | Method / tunnel | Strength / weakness |
|---|---|---|
| **Cohen** (intravesical) | Cross-trigonal | Commonest; success over 98%; one cystotomy for both sides; later ureteroscopy or retrograde access hard |
| Politano–Leadbetter (intravesical) | New superomedial hiatus above the old one | Long tunnel; blind hiatus risks bowel injury |
| Glenn–Anderson (intravesical) | Orifice advanced towards the bladder neck | Least kinking; short tunnel |
| **Lich–Gregoir** (extravesical) | Detrusorotomy, ureter laid in | No haematuria or spasm; no cystotomy; **retention up to 10% after bilateral repair** (pelvic plexus injury); stay close to the ureter |
| **RALUR / RALUR-EV** (robot-assisted laparoscopic ureteral reimplantation, extravesical) | Y-shaped detrusorotomy | Less pain, shorter stay; results approaching open |
| **Paquin** | Combined extra- and intravesical, suprahiatal | Long tunnel; for dilated ureters |
| **Gil-Vernet** | Intravesical trigonoplasty, no tunnel | Low-grade bilateral VUR |

**Paquin technique**

- **Combined extravesical–intravesical** reimplant described by Paquin (1959), with the 5:1 rule.
- Steps: 1. Ureter mobilised **extravesically** and divided at the bladder. 2. Bladder opened; a **new hiatus** created higher on the posterior wall (**suprahiatal**). 3. Long **submucosal tunnel** from the new hiatus down to the original orifice site, length about 5 × diameter. 4. Ureter drawn through, spatulated, sutured to mucosa and anchored to detrusor. 5. Old hiatus closed.
- **Advantages:** long tunnel under direct vision; good for **dilated ureters and megaureter** (can combine with tapering).
- **Disadvantages:** kinking or obstruction at the new hiatus.

**Gil-Vernet trigonoplasty**

- **Intravesical, with no tunnelling and no ureteric mobilisation.**
- Steps: 1. Bladder opened. 2. **Transverse mucosal incision** across the trigone between the two orifices. 3. Intramural ureters with their muscle sheath **advanced medially** by one non-absorbable or absorbable stitch across the midline. 4. Mucosa closed.
- **Effect:** lengthens and medialises the intravesical ureters; corrects both sides at once.
- **Indications:** low to moderate grade (I–III), usually bilateral VUR, with non-dilated ureters.
- **Advantages:** quick, simple, low obstruction risk; easy later retrograde access.
- **Disadvantages:** unsuitable for high grade, dilated ureters or paraureteral diverticulum.

| Feature | Paquin | Gil-Vernet |
|---|---|---|
| Approach | Extra- + intravesical | Intravesical |
| Ureter divided | Yes | No |
| Tunnel | New long submucosal | None (advancement) |
| Best for | High grade, dilated | Low grade, bilateral |

### Criteria for selecting the technique

| Situation | Preferred option |
|---|---|
| Grades II–IV, parents want least invasive | Dx/HA injection (lower success at higher grade) |
| **Bilateral VUR, small or trabeculated bladder** | **Cohen** (both sides through one cystotomy) |
| Grade V, dilated ureter | Intravesical reimplant with or without tapering (Politano–Leadbetter or Cohen) |
| Unilateral, older child | Lich–Gregoir or RALUR-EV |
| Bilateral extravesical | Nerve-sparing dissection; warn about retention |
| Duplex system | Common sheath reimplant |
| Failed injection | Reimplant (not harder after Dx/HA) |
| Neurogenic or PUV bladder (secondary VUR) | Treat the bladder first (anticholinergics, CIC (clean intermittent catheterisation)) |

### Complications and follow-up after reimplantation

- Ultrasound at 6–12 weeks.
- Complications: transient obstruction; persistent reflux (usually resolves by 1 year); contralateral VUR (about 1 in 10, usually resolves); leak; retention after bilateral extravesical repair.
- Obstruction early (oedema): stent or nephrostomy; late (ischaemia, kink): redo.
- Lifelong BP and renal checks if scarred; scarred women risk pre-eclampsia.

### Long-term outcome

| Outcome | Facts |
|---|---|
| **Renal scarring** | About 15% get a febrile UTI within 2 years; about 15% of those scar. Risk highest under 1 year. Much "scarring" in boys with high grade is **congenital dysplasia**, not preventable by any treatment |
| **Hypertension** | Reflux nephropathy may be the commonest cause of childhood hypertension after essential hypertension (renin–angiotensin). **34–38%** of scarred children are hypertensive as adults. Correcting VUR does not lower BP; nephrectomy of a small scarred kidney helps selected cases |
| **CKD and ESKD (end-stage kidney disease)** | Now rare: RRT (renal replacement therapy) from reflux nephropathy fell from 1.9 to 0.6 per million (USRDS (United States Renal Data System), 1996–2014). Bilateral scars cause hyperfiltration, proteinuria, concentrating defect, tubular acidosis |
| **Pregnancy** | VUR in late pregnancy: pyelonephritis about one-third versus under 5%. Scars: about **3× hypertension, 8× pre-eclampsia**; bilateral worse. Impaired function: fetal loss. Reimplanted women still get UTIs |

| Outcome | Facts |
|---|---|
| Growth; offspring | Catch-up growth after infection control; about 43% of screened offspring have VUR |

- **Concept:** in adult life **the scar, not the reflux, carries the risk**, so correction does not remove it. Scarred patients need lifelong BP, proteinuria and creatinine checks.

### Grade 3 VUR in an adult

- **Grade 3:** mild–moderate dilatation of ureter, pelvis and calyces, minimal forniceal blunting.
- **Adults rarely lose function from reflux alone**, unless high bladder pressure, stasis and UTI coexist. The **scar**, not the reflux, carries the risk. Spontaneous resolution is unlikely.
- **Causes:** persistent primary VUR; **secondary**: BOO (bladder outlet obstruction), neurogenic bladder, resection near the orifice, ureterocele incision, paraureteral diverticulum, transplant ureter, TB (tuberculosis).
- **Presentation:** recurrent febrile UTI, pyelonephritis (especially in pregnancy), loin pain on voiding, hypertension, proteinuria, CKD.
- **Evaluation:** VCUG (grade, bladder, diverticulum, outlet); ultrasound and **DMSA** for scars and split function; MAG3 (mercaptoacetyltriglycine) renogram if obstruction is suspected; **UDS (urodynamics)**: compliance, voiding pressure, residual; cystoscopy; blood pressure, creatinine, urine protein.
- **Treat secondary causes first** (relieve BOO; CIC and antimuscarinic for neurogenic bladder), then reassess.

| Situation (primary grade 3) | Management |
|---|---|
| Incidental; no UTI, no scars | Observe: timed voiding, bowel care, prompt UTI treatment |
| Recurrent febrile UTI, new scars, falling function | **Correct the reflux** |
| Woman planning pregnancy with scars or prior pyelonephritis | Consider correction before pregnancy |
| Poorly functioning scarred kidney with infection or hypertension | Nephrectomy |

- Low-dose prophylaxis selectively; adult evidence is limited.
- **Correction:** endoscopic Dx/HA injection (day case, repeatable, but less effective in adults); or **ureteroneocystostomy** (open, laparoscopic or robotic). Lich–Gregoir/RALUR-EV suits adults (no cystotomy; stay close to the ureter to protect nerves); Politano–Leadbetter (long tunnel); Cohen (over 98%; later retrograde access hard).
- **Psoas hitch** if length is needed; tapering seldom needed in grade 3.
- Follow-up: ultrasound at 6–12 weeks; lifelong blood pressure and renal checks if scarred.

### Overall management algorithm

```
Treat BBD -> low risk: observe | high risk (bilateral): CAP Breakthrough UTI / new scar / persistent IV-V -> Dx/HA injection or reimplant (Cohen) Scars -> lifelong BP and proteinuria checks
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**Diagrams to draw:**

- "Y" of ureteric bud and mesonephric duct with orifice migration (early-budding Y diagram).

- Long oblique tunnel versus short lateral refluxing tunnel; oblique intramural ureter in Waldeyer's sheath with submucosal tunnel and detrusor backing.
- Grades I–V.
- Cohen cross-trigonal reimplant; Gil-Vernet transverse trigonal incision and medial advancement stitch.
- Needle sites: STING below the orifice, HIT and double HIT inside the tunnel, with volcano mound.

**Key points to remember:**

- Low (early) bud → high lateral orifice → short tunnel → VUR plus dysplasia; high (late) bud → ectopia.
- One embryological fault: lateral orifice, short tunnel, dysplasia.
- Tunnel length to diameter 5:1 normal versus 1.4:1 refluxing (Paquin); adult intramural ureter 1.2–2.5 cm.
- Dysplasia (boys, congenital) versus scar (girls, infection); sterile reflux rarely scars.
- Grade V loses papillary impressions; grade IV–V and bilateral grade III rarely resolve; bilateral grade IV persists most; bilateral VUR is high risk.
- RIVUR: CAP halves recurrent UTI, not scarring.
- Treat BBD and give CAP first; correct for breakthrough febrile UTI, new scars or persistent grade IV–V.
- Trend: risk-stratified care, fewer VCUGs, BBD first, endoscopic or robotic correction.
- STING: O'Donnell and Puri 1986; needle 2–3 mm below orifice at 6 o'clock.
- Dx/HA 80–250 µm; HIT better than STING; double HIT about 90%; Elder 2006: grade IV 63% after one injection.
- Watch for late obstruction; calcified mound mimics a stone.
- Open reimplantation 95.9%; Cohen cross-trigonal (over 98%) suits bilateral VUR through one cystotomy.
- Bilateral extravesical (Lich–Gregoir) repair risks retention up to 10%.
- Paquin: tunnel 5:1; combined, suprahiatal reimplant. Gil-Vernet: trigonoplasty, medial advancement, no tunnel; for low-grade bilateral VUR.
- 34–38% of scarred children become hypertensive; correction does not lower BP; scars bring 8× pre-eclampsia risk.
- In adults the scar, not the reflux, carries the risk; exclude secondary causes first; correct for recurrent pyelonephritis, new scars or pregnancy plans.

**Also, from the condensed book (not in the paper answer):**

- Abnormal bud position also causes hypoplasia, so some "scars" are congenital.

## 3 · Primary vs secondary VUR

*Secondary reflux needs the bladder treated first.*

*See the paper answer **Primary VUR: VUJ embryology/anatomy, pathophysiology, management (incl. bilateral, adult grade 3), reimplantation techniques and STING** (Section 4, Ch 41).*

**Also, from the condensed book (not in the paper answer):**

- In spina bifida, VUR occurs in up to **30%**: start early clean intermittent catheterisation (CIC) and an anticholinergic or β3-agonist.
- **Reflux at low pressure or in early filling means poor resolution.** Treating detrusor overactivity improves resolution.

> **KEY POINT**
>
> Primary and secondary are not exclusive: "primary" VUR often has a treatable secondary component (overactivity, bladder and bowel dysfunction). Remove it and only true primary VUR remains.

## 4 · Epidemiology

- VUR is found in **~30%** of children with UTI and ~17% of those imaged for hydronephrosis. It is present in **38.6%** under 2 years versus 7.6% at 12–21 years.
- Girls are affected ~2× overall, but infant boys predominate.

## 5 · Grading — International Reflux Study (1981)

| Grade | Ureter | Pelvis and calyces |
|---|---|---|
| I | Non-dilated, ureter only | Not reached |
| II | Not dilated | Reached, no dilatation |
| III | Mild–moderate dilatation | Mild–moderate dilatation, minimal forniceal blunting |
| IV | Moderate tortuosity | Moderate dilatation |
| V | Gross dilatation and tortuosity | Gross dilatation; **papillary impressions lost** |

- Grades I–II are non-dilating and III–V dilating. Radionuclide cystography (RNC) uses 3 grades.
- **Ureteral diameter ratio (UDR)**, the largest distal ureteric diameter divided by the L1–L3 vertebral distance, predicts resolution and breakthrough UTI (BUTI) better than grade.

## 6 · Diagnosis

### When to image after UTI

- **AAP 2011**, febrile UTI at 2–24 months: do **ultrasound (US) first**, and a voiding cystourethrogram (VCUG) only if US is abnormal or after a **second febrile UTI**.
- **NICE**: US, with VCUG mainly for atypical or recurrent UTI in younger children.
- **Top-down**: DMSA scan first, VCUG only if abnormal. A normal DMSA leaves a **<1%** chance of high-grade VUR but misses ~1/3 of (mostly low-grade) VUR.

### Cystographic methods

*See the paper answer **VCUG** (Section 1, Ch 5).*

**Also, from the condensed book (not in the paper answer):**

- Direct RNC is more sensitive for grades II–V but **misses grade I**.
- Suspect coexisting **UVJ obstruction** with a distal aperistaltic segment, dilute contrast and delayed drainage. It is a **relative contraindication to endoscopic injection**.

### Upper tract

- **Renal and bladder ultrasound (RBUS)** is the serial mainstay but cannot exclude VUR.
- **DMSA** is the gold standard for cortical defects (sensitivity 98%, specificity 92%); a defect that persists is a scar.

## 7 · Congenital dysplasia vs acquired scar

*Only the acquired scar is preventable.*

*See the paper answer **Primary VUR: VUJ embryology/anatomy, pathophysiology, management (incl. bilateral, adult grade 3), reimplantation techniques and STING** (Section 4, Ch 41).*

**Also, from the condensed book (not in the paper answer):**

- Hypertension develops in **34–38%** of scarred children. **Correcting VUR does not lower blood pressure**; removing a small kidney or segment may.

## 8 · Bladder and bowel dysfunction (BBD)

*BBD is the main modifiable factor.*

- **Treat constipation first**; it causes sphincter co-contraction.
- BBD brings more UTI on prophylaxis and after surgery, less resolution, and **lower endoscopic success, but not open surgical success** (AUA).
- **Phimosis**: in boys with VUR, circumcision cuts the number needed to treat (NNT) to prevent one UTI from ~111 to **~4**. Topical steroid is equivalent.

## 9 · Natural history and resolution

| Grade | Spontaneous resolution |
|---|---|
| I–II | Most; grade II 63–85% |
| III | ~50%; ~90% of resolutions within 4 years |
| IV–V, bilateral III | Few (infants with IV–V resolve well in year 1) |

- Resolution favours **low grade, young age, unilateral reflux, no BBD, voiding-only reflux, higher bladder volume at onset and low UDR**.

## 10 · Medical management — continuous antibiotic prophylaxis (CAP)

*CAP failure is the main trigger for surgery.*

*See the paper answer **Primary VUR: VUJ embryology/anatomy, pathophysiology, management (incl. bilateral, adult grade 3), reimplantation techniques and STING** (Section 4, Ch 41).*

**Also, from the condensed book (not in the paper answer):**

- Give a low dose **at night**; avoid TMP-SMX in neonates (bilirubin displacement).
- **If febrile UTI breaks through CAP despite good bladder and bowel habits, stop medical management and correct the VUR.**

### Risk stratification (Hidas)

| Factor | Low risk | High risk |
|---|---|---|
| Sex / age | Male; >5 y | Female; <1 y |
| Laterality / grade | Unilateral; I–II | Bilateral; III–V |
| BBD / scars | No | Yes |
| Presentation | Prenatal hydronephrosis | Multiple febrile UTIs |

- Two-year BUTI on CAP is **8.6% low, 26% intermediate, 62.5% high** risk. About two-thirds of children are low risk.

## 11 · Practical algorithm (AUA-based, risk-stratified)

**ALGORITHM**

1. Confirm true febrile UTIs; do RBUS. If multiple febrile UTIs or upper-tract findings, then VCUG.
2. In all, treat BBD (constipation first) with timed voiding.
3. If low risk (grade I–II, no BBD, male, older), then observe with or without CAP and treat febrile UTI promptly.
4. If high risk (grade ≥III, girl, BBD, scars, infant), then give CAP.
5. If breakthrough febrile UTI, new scars, or persistent high-grade VUR, then correct (endoscopic or reimplant).

- CAP helps little at grade ≤II without BBD, but helps at ≥III, at least in girls.

## 12 · Surgical correction

*Surgery stops reflux but does not protect kidneys better.*

- Open correction succeeds in **>95%**, but neither open nor robotic surgery beats medical therapy for scarring or growth. The adult UTI risk persists.
- **Endoscopic dextranomer/hyaluronic acid (Dx/HA)**: BBD lowers success; avoid it with UVJ obstruction. Calcified mounds can **mimic ureteric stones**.
- After a cross-trigonal reimplant (Ch 53), **retrograde ureteric access is difficult**.

## 13 · Associated conditions

- **UPJO with VUR**: **correct the obstruction first**; repairing both at once risks the ureteric blood supply.
- In a **duplex** kidney, VUR usually enters the **lower pole** (Weigert–Meyer).
- A **ureter entering a paraureteral diverticulum** will not resolve; a diverticulum beside the orifice resolves like primary VUR.
- Contralateral VUR with a multicystic dysplastic kidney usually resolves.
- **Megacystis–megaureter** (massive VUR recycling urine into a large thin bladder) and **obstructive refluxing megaureter** are treated by reimplant, not injection or outlet surgery.

## 14 · Sibling screening

- Sibling prevalence is ~**30%**, higher when younger.

- The target is **renal damage, not VUR**. AUA: **VCUG for siblings with a cortical abnormality or renal asymmetry on US, or a UTI history**; otherwise observe.

## 15 · Follow-up and adulthood

- If scarred, check BP, proteinuria and renal function for life.
- **Pregnancy**: scars bring ~3× hypertension and **~8× pre-eclampsia**. **Scarring, not VUR, drives morbidity.**

## 16 · Landmark trials

| Trial | Design | Result | Take-home |
|---|---|---|---|
| IRSC | <9 y, high grade; CAP vs open surgery | UTI equal (38%); surgery fewer pyelonephritis; scars, 10-y function equal | Surgery ≈ medical for scarring |
| Birmingham Reflux Study | High grade; medical vs surgical, 5 y | New scars equal, all within 2 y | Equal; supports "big bang" |
| Swedish Reflux Trial | 1–2 y, grade III–IV; CAP vs endoscopic vs surveillance | Girls febrile UTI: 19% CAP, 23% endoscopic, **57%** surveillance; boys no difference | Treat girls with dilating VUR |
| RIVUR | 607 children, grade I–IV after first/second UTI; TMP-SMX vs placebo | Recurrent UTI **12.9% vs 23.6%**; scarring unchanged | CAP prevents UTI, not scars |
| PRIVENT | 576 children with UTI; CAP vs placebo | Absolute UTI reduction 6% (19% to 13%) | Modest benefit |
| French / low-risk trials | Grade I–III; CAP vs none | No significant difference | No benefit at low risk |

> **KEY POINT**
>
> Trials disagree because of population risk: RIVUR was diluted by low-risk children; reanalysis gives NNT 5 high-risk vs 18 low-risk. Neither CAP nor surgery prevents scarring better.

# 49 Anomalies of the Kidney and Collecting System

## 1 · Embryology that explains everything

- The ureteric bud (UB) arises from the caudal wolffian duct (WD) at **5–7 wk**, invades the metanephric mesenchyme (MM) and branches; its tips induce nephrons (**reciprocal induction**).
- In weeks 6–9 the kidney ascends and rotates 90° ventromedially, so the pelvis faces medially.
- Renal anomalies travel with WD (male) and müllerian duct (MD, female) anomalies; the adrenal develops independently.

> **KEY POINT**
>
> Every anomaly of number is a failure of UB–MM induction: timing and extent set the phenotype. An early insult drags the WD/MD with it (genital anomalies). Arrested ascent leaves an unrotated kidney with anterior pelvis and fetal blood supply; fusion freezes kidneys where they meet.

## 2 · Bilateral renal agenesis (BRA)

*BRA is uniformly lethal, so recognise it and counsel.*

- It occurs in **≈1 in 5000** births, **75% male**. First-degree relatives carry ≈1000× risk, so **scan parents and siblings by ultrasound (US)** of any infant with unilateral agenesis (URA), BRA or dysgenesis.
- The ureters are atretic and the bladder absent or hypoplastic. The flat adrenal gives the **"lying down" sign**.
- **Potter facies** and limb deformities are **deformations** from absent amniotic fluid, not malformations. Undescended testes (43%) and gastrointestinal anomalies (≈60%) are common.
- **Pulmonary hypoplasia** is the cause of death. **Peters two-step model**: a kidney-derived "renal factor" acts before 16 wk, then amniotic fluid in the canalicular phase (16–26 wk), so fluid alone only partly restores the lungs.
- **Diagnosis**: prenatally, severe oligohydramnios, no kidneys and a small chest. In a neonate, **anuria >24 h without bladder distension** means BRA until proved otherwise; confirm with US and power Doppler.
- **Prognosis**: ≈40% are stillborn and the rest die in 24–48 h. Amnioinfusion is experimental (RAFT, §9) and needs ethics oversight with full counselling.

> **KEY POINT**
>
> BRA is lethal through the lungs, not the kidneys. Amnioinfusion replaces only the amniotic step, so it converts pulmonary deaths into neonatal end-stage renal disease (ESRD).

## 3 · Unilateral renal agenesis (URA) and the solitary functioning kidney (SFK)

*The issues are genital anomalies and lifelong hyperfiltration.*

- URA occurs in **1 in 1200**, male 1.8:1. Many apparent URAs are an **involuted multicystic dysplastic kidney (MCDK)**.
- Associations include VACTERL, MURCS, Turner, Kallmann, branchio-oto-renal and DiGeorge syndromes. **Maternal diabetes triples the risk**. An isolated preauricular pit or single umbilical artery needs no US; two or more anomalies do.
- **The ipsilateral ureter is absent in ≈60%**. The defect is in the UB, so the gonad is rarely affected.
- **Contralateral kidney**: vesicoureteral reflux (VUR) **≈30%**, pelviureteric junction (PUJ) obstruction ≈11%, vesicoureteric junction (VUJ) obstruction ≈7%.
- **Genital anomalies occur in 20–40%** (females 25–50%, males 10–15%).

| | Male (WD derivatives) | Female (MD derivatives) |
|---|---|---|
| Always present | Testis, head of epididymis | Fimbriated end of tube |
| Commonest | Absent vas/epididymal body-tail/seminal vesicle (≈50%). **79% of men with absent vas lack the ipsilateral kidney** | Unicornuate/bicornuate, didelphys or septate uterus; vaginal atresia/duplication; Gartner duct cyst with blind ectopic ureter |
| Named syndrome | **Zinner**: seminal vesicle cyst and ipsilateral agenesis | **OHVIRA** (obstructed hemivagina and ipsilateral renal anomaly; Herlyn–Werner–Wunderlich): pubertal haematocolpos, pain |
| Trigger to image kidneys | Impalpable vas (infertility, herniotomy/orchidopexy) | Any uterine anomaly; pelvic US after thelarche, before menarche |

- **Magee classification**: type I (before wk 4) gives a unicornuate uterus; type II (early wk 4) gives didelphys with obstructed hemivagina on the agenesis side; type III (after wk 4) gives isolated URA with normal genitalia.
- **MRKH** (46,XX müllerian agenesis): **type II has renal anomalies in 40–60%**; type I rarely does.
- **Work-up**: compensatory hypertrophy is seen from 20 wk in ≈90%. DMSA finds a missed ectopic kidney. A voiding cystourethrogram (VCUG) for contralateral VUR is **not routine**. Sport is not precluded.

## Prognosis and surveillance of the SFK

- **Brenner hyperfiltration**: reduced nephron mass leads to glomerular hypertension, then albuminuria and sclerosis. Risk is highest when the SFK itself has a congenital anomaly (CAKUT), low birth weight or prematurity.
- In KIMONO, ≈⅓ had renal injury by ≈10 y, so follow-up is lifelong.

| Situation | Blood pressure (BP) and first-morning urine albumin | Ultrasound | Creatinine |
|---|---|---|---|
| SFK without CAKUT | Annually | Every 5 y if glomerular filtration rate (GFR) >60 and no medication | Periodic, into adolescence |
| SFK with CAKUT | Twice yearly | As indicated | Periodic |
| GFR <60 or on antihypertensive/antiproteinuric | BP, albumin, creatinine 2–4×/year | | |

Hypertension or microalbuminuria (>30 mg/24 h) calls for an ACE inhibitor, salt limitation and no excess protein.

## 4 · Supernumerary kidney

- It has its **own capsule, collecting system and separate blood supply**, unlike a duplex kidney. About half are hydronephrotic, with obstruction, infection or stones in early adulthood.

## 5 · Ectopia (anomalies of ascent)

*An ectopic kidney is easily mistaken for agenesis or tumour.*

### Simple ectopia

- It occurs in **1 in 900** at autopsy and is bilateral in ≈10%. **Pelvic is commonest**.
- The kidney is small and malrotated (pelvis anterior). **Hydronephrosis is common**, half from obstruction (mostly PUJ). **VUR occurs in 30–50%**, often into the orthotopic kidney.
- The blood supply is anomalous: **map it by CT or MR angiography (MRA) before surgery**.
- Genital anomalies occur in 15–45%. Continuous incontinence in a girl suggests an ectopic ureter from the ectopic kidney.
- **Never remove a "pelvic mass" without imaging**: solitary pelvic kidneys have been excised as tumours.

### Cephalad and thoracic kidney

- Cephalad ectopia is seen with **omphalocele**. A thoracic kidney lies in the **foramen of Bochdalek**, is usually silent, and needs CT or MRU rather than thoracotomy.

## 6 · Fusion anomalies

*Fused kidneys bring obstruction, stones and variable vessels.*

### Crossed renal ectopia

- The kidney lies opposite its ureteric orifice; **90% are fused**, making it the second commonest fusion anomaly. Left-to-right crossing is 3:1, and the ureteric orifice is orthotopic.
- **McDonald–McClellan types**: with fusion (90%), without fusion, solitary crossed and bilateral crossed (rarest).

| Crossed fused form | Key feature |
|---|---|
| **Inferior ectopia** (≈⅔, commonest) | Crossed upper pole fused to normal lower pole; pelves anterior |
| Sigmoid (S-shaped) | Second commonest; pelves face opposite ways |
| Lump (cake) | Lobulated pelvic mass |
| L-shaped (tandem) | Crossed kidney transverse, anterior to L4 |
| Disc (pancake) | Medial fusion, reniform outline kept |
| Superior ectopia | Least common; crossed lower pole to normal upper pole |

- An **anterior pelvis means early fusion**; a medial pelvis means late fusion. VUR occurs in ≈20%.

- **A stone in the crossed ureter hurts on the anephric side**; pyelonephritis or PUJ obstruction hurts where the kidneys lie. Obtain MRU or MRA before major surgery.

### Horseshoe kidney

- It is the **commonest fusion anomaly**: **1 in 400** (0.25%), male 2:1. **Lower-pole fusion occurs in 95%**, before rotation.
- The **isthmus** lies at L3/L4, **just below the IMA**, in front of the aorta and IVC.
- The calyces **point posteriorly**, and the ureter inserts high and bends over the isthmus. **The blood supply is highly variable**, so obtain MRA or CT before surgery.
- **≈60% of Turner patients have a horseshoe kidney**; it is also associated with neural tube, anorectal and VATER anomalies.
- **PUJ obstruction occurs in 13–34%**, but many hydronephroses are not obstructed. VUR occurs in 8–32%.
- **Stones affect ≈⅓ of adults**, so still do a metabolic work-up.
- **Wilms risk is 1.76–7.9×**; preoperative chemotherapy preserved ≈75% of parenchyma (NWTS). Renal cell carcinoma is not increased.
- At least 50% are asymptomatic. Pregnancy is unaffected and the kidney is transplantable.

> **KEY POINT**
>
> † PUJ obstruction in horseshoe kidney: dismembered pyeloplasty with dependent anterior anastomosis; symphysiotomy not routinely needed.

## 7 · Malrotation and vascular anomalies

- Malrotation occurs in ≈1 in 939. **Weyrauch** types: **ventral (no rotation, commonest)**, ventromedial, dorsal (rarest) and lateral (reverse). **The vessel course is the only clue** to direction.
- Hilar fibrosis can cause intermittent PUJ obstruction and stones; suspect malrotation when a stone lies in an unusual position.

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### SAQ Anomalies of renal artery.

Paper answer · P2

**Type:** SAQ (10 marks)
**Source:** Practice Set 1 Paper 2, Q3c

#### Normal pattern

- One renal artery from the aorta at L1–L2; anterior and posterior divisions; **five segmental end arteries** (apical, upper, middle, lower, posterior). The right artery passes behind the IVC (inferior vena cava).

#### Terminology

| Term | Definition |
|---|---|
| **Multiple renal arteries** (preferred) | More than one vessel from the aorta to one kidney |
| **Aberrant** | Arises from a vessel other than aorta or main renal artery (iliac, SMA (superior mesenteric artery), gonadal) |
| **Accessory** | Two or more branches supplying the same segment |

### Incidence

- Single artery in **71–85%**; variants in 25–40%. Accessory arteries 25–28% (aorta T11–L4), usually to the **lower pole**. True aberrant vessels are rare except in **ectopic and horseshoe kidneys**.
- **Also know:** early prehilar branching; right artery anterior to IVC in about 5%.

### Embryology

- The ascending kidney takes successive aortic branches; the lower ones normally degenerate. **Failure of degeneration** gives polar or multiple arteries.

### Clinical significance

- **PUJ (pelviureteric junction) obstruction** from a lower-pole crossing vessel; but crossing vessels occur in a fifth to a quarter of normal PUJs.
- Compression of an infundibulum: hydrocalyx, pain, stones, UTI (urinary tract infection).
- **Surgery:** bleeding at endopyelotomy (incise posterolaterally); polar arteries are end arteries, so ligation infarcts that segment; **donor nephrectomy** needs CT (computed tomography) angiography; aortic surgery.
- Not linked to hypertension.

### Renal artery aneurysm

- About **0.1%**. **Abeshouse types:** saccular (about 93%), fusiform, dissecting, arteriovenous.
- Mostly asymptomatic; hypertension, haematuria, pain; wreath-like calcification.
- **Treat** (endovascular or open) if rupture, **>2 cm**, expanding, **pregnancy or planned pregnancy**, symptoms or emboli.

### Renal arteriovenous fistula

| | **Congenital (cirsoid AVM (arteriovenous malformation))** | **Acquired** |
|---|---|---|
| Share | <25% | Majority (biopsy, PCNL (percutaneous nephrolithotomy), trauma) |
| Features | Bruit, haematuria, high-output failure | Often resolves |
| Treatment | Selective embolisation | Observe; embolise if bleeding |

**Diagram to draw:** kidney with main artery, lower-pole accessory artery crossing the PUJ.

> **Key points to remember:**
> - Single artery in 71–85%; accessory arteries usually lower pole.
> - Crossing vessel can cause PUJ obstruction.
> - Treat aneurysm >2 cm or in pregnancy.

**Also, from the condensed book (not in the paper answer):**

- Treat an aneurysm for rupture, growth, pregnancy, size **>2 cm** or symptoms. An acquired AVF may close within months, so observe it.
- A congenital arteriovenous fistula presents with bruit, haematuria and high-output failure and rarely resolves.

## 8 · Collecting system anomalies

### SAQ Pelvicalyceal anomalies other than PUJ obstruction

Paper answer · P1

**Type:** SAQ (10 marks)
**Source:** Practice Set 14 Paper 1, Q3d

#### Introduction

- Congenital pelvicalyceal variants occur in about 4% of people.
- **Key question:** is there obstruction? Dilatation without obstruction needs no treatment; stasis causes stones and infection.

#### Anomalies

| Anomaly | Features | Management |
|---|---|---|
| **Calyceal diverticulum** | Transitional-lined intrarenal cavity joined to a calyx (Type I, commoner, small) or pelvis (Type II, larger, symptomatic) by a narrow neck; 4.5 per 1000; congenital (persistent 3rd–4th generation ureteric branches) or acquired. **Stones in almost 40%**; milk of calcium | Asymptomatic: observe. Pain, infection, stones: ureteroscopic neck incision, **percutaneous ablation with fulguration**, laparoscopic marsupialisation; SWL (shock wave lithotripsy) ineffective |
| **Hydrocalycosis** | Dilated major calyx connected to the pelvis; infundibular obstruction by vessel, stenosis, scarring or achalasia | Symptomatic only: infundibulotomy or dilatation |
| **Megacalycosis** | **Non-obstructive**; malformed papillae; **12–20 calyces**; pelvis normal; male 6:1; normal diuretic renogram | **None**; may coexist with segmental megaureter |
| **Infundibular / infundibulopelvic stenosis** | Narrow infundibula ± small pelvis; usually bilateral, with VUR (vesicoureteric reflux), dysplasia; over a third reach renal failure (hyperfiltration) | Monitor creatinine and eGFR (estimated glomerular filtration rate); ACE (angiotensin-converting enzyme) inhibitors; surgery only if obstruction harms function: endoscopic, calicocalicostomy, ureterocalicostomy |
| **Extrarenal pelvis** | Large pelvis outside the sinus; mimics dilatation | None |
| **Bifid pelvis** | About 10%; normal variant | None |
| **Unipapillary kidney** | Single papilla and calyx; rare; usually hypoplastic kidney | None |

- **Also know:** acquired infundibular stenosis (tuberculosis, post-PCNL (percutaneous nephrolithotomy)); Fraley syndrome (crossing artery on upper infundibulum).

#### Evaluation

- **Ultrasound:** cyst-like space; milk of calcium layers and shifts with position.
- **CT (computed tomography) urography or MRU (magnetic resonance urography): delayed contrast pooling** in a diverticulum.
- **Diuretic renogram (MAG3 (mercaptoacetyltriglycine)):** separates obstruction (hydrocalycosis) from non-obstruction (megacalycosis, extrarenal pelvis).

- **Retrograde pyelography:** maps the neck.

### Differential diagnosis

- Obstructive hydronephrosis, papillary necrosis, pyelonephritic clubbing, tuberculosis, renal cyst.
- **Stones:** retrograde first (find, dilate or laser-incise the ostium); percutaneous for lower-pole diverticula.

**Diagram to draw:** kidney showing diverticulum with narrow neck, hydrocalyx and megacalyces.

> **Key points to remember:**
> - Diverticulum: Type I to calyx, Type II to pelvis; stones in 40%.
> - Megacalycosis: non-obstructive, 12–20 calyces, no treatment.
> - Infundibulopelvic stenosis: bilateral, renal failure risk.

**Also, from the condensed book (not in the paper answer):**

- Extrarenal pelvis: confirm with diuretic renography before calling it obstruction.
- Infundibulopelvic stenosis: monitor renal function yearly; if it falls, give an ACE inhibitor and restrict protein.

> **KEY POINT**
>
> Collecting-system anomalies split on one question: is there obstruction? Dilatation without obstruction (bifid, extrarenal pelvis, megacalycosis) needs nothing; stasis (diverticulum, hydrocalycosis) is treated for symptoms; infundibulopelvic stenosis threatens function through bilateral dysplasia.

## 9 · Landmark trials

| Trial | Design | Result | Take-home |
|---|---|---|---|
| RAFT (Renal Anhydramnios Fetal Therapy) | Non-randomised multicentre; isolated BRA, serial amnioinfusion before 26 wk | 94% live births; 82% survived ≥14 d with dialysis access; only 35% discharged on peritoneal dialysis; stopped early | Prevents lethal lung hypoplasia but leaves neonatal ESRD — research only |

# 50 Renal Dysgenesis and Cystic Disease of the Kidney

## 1 · Genetics in one table

| Gene | Renal phenotype |
|---|---|
| PAX2 | Renal hypodysplasia or aplasia |
| EYA1 | Branchio-oto-renal (BOR) syndrome |
| HNF1B | Renal cysts and diabetes syndrome |
| PKD1, PKD2 (primary cilium) | Polycystic kidney disease |

## 2 · Dysgenesis: dysplasia vs hypoplasia

*Dysplasia is the leading cause of childhood renal failure.*

- **Dysgenesis** (abnormal size, shape or structure) includes dysplasia and hypoplasia.
- **Dysplasia** is a **histological** diagnosis of arrested metanephric differentiation. Its sine qua non is **primitive ducts** with fibromuscular collars; cartilage may also occur. Not all cysts are dysplastic, and not all dysplasia is cystic.
- Causes are primary UB failure (syndromes such as BOR and renal–coloboma) or fetal outflow obstruction. **It is the leading cause of childhood end-stage renal disease (ESRD).** Ultrasound (US) shows a small echogenic kidney with lost corticomedullary differentiation.
- **Hypoplasia** is reduced mass from too few cells; simple hypoplasia has a normal nephron number.

| Entity | Defining feature | Course / management |
|---|---|---|
| **Oligomeganephronia** | Few, hypertrophied nephrons; usually bilateral | Polyuria, polydipsia, short stature; **proteinuria first sign**; ESRD in second decade. Fluids, correct salt loss/acidosis, protein 1.5 g/kg/d, ACE inhibitor; living related donor acceptable (not familial) |
| **Ask-Upmark** (segmental hypoplasia) | Mostly acquired reflux nephropathy; deep "slit-scars" | Girls ≥10 y, **severe hypertension**; partial/total nephrectomy cures blood pressure (BP); bilateral: medical, dialysis/transplant. Reflux correction does not cure hypertension |
| **Hypodysplasia** | With ectopic orifice, megaureter, ureterocele, posterior urethral valves (PUV), prune-belly | **Dysplasia is proportional to the degree of ureteric ectopia**: lateral orifice gives reflux-associated, medial/ureterocele gives obstruction-associated dysplasia |

> **KEY POINT**
>
> The ureteric orifice predicts the kidney: a bud from the wrong place on the wolffian duct meets the wrong blastema, so ectopia, obstruction and dysplasia travel together.

## 3 · Cystic disease: concepts and classification

*Sort first by inheritance, then by the extrarenal clue.*

- Cysts arise from tubules; the **primary cilium** is central.

- A **multicystic** kidney (MCDK) is dysplastic with no normal nephrons. A **polycystic** kidney starts normal, and its nephrons are progressively replaced by cysts.

| Inheritable | Non-heritable |
|---|---|
| Autosomal recessive polycystic kidney disease (ARPKD) | Multicystic dysplastic kidney (MCDK) |
| Autosomal dominant PKD (ADPKD) | Benign multilocular cyst (cystic nephroma) |
| Juvenile nephronophthisis (recessive) / medullary cystic disease (dominant) | Simple cysts |
| Congenital nephrosis; familial hypoplastic glomerulocystic disease | Medullary sponge kidney; sporadic glomerulocystic disease |
| Syndromes: tuberous sclerosis, von Hippel–Lindau (VHL) | Acquired renal cystic disease; calyceal diverticulum |

**The extrarenal clue decides**: hepatic fibrosis means ARPKD; berry aneurysm, ADPKD; retinitis pigmentosa, nephronophthisis; angiofibroma, tuberous sclerosis complex (TSC); haemangioblastoma, VHL.

## 4 · ARPKD

*Huge echogenic kidneys and an always-fibrotic liver.*

### SAQ Autosomal recessive polycystic kidney disease.

Paper answer · P3

**Type:** SAQ (10 marks)
**Source:** 2018-Summer Paper 3, Q3c

#### Definition and genetics

- **ARPKD (autosomal recessive polycystic kidney disease)**: bilateral, symmetrical renal enlargement from **dilated collecting ducts**, **always with congenital hepatic fibrosis**.
- Gene **PKHD1 on chromosome 6p**; protein **fibrocystin (polyductin)** on the **primary cilium** of tubular and bile-duct cells, so it is a **ciliopathy**.
- Incidence about 1 in 10,000–50,000 live births. Siblings have a 25% risk.
- "Infantile" is a misnomer: a **spectrum**; earlier onset means more severe disease.

#### Pathology

- Radially arranged dilated collecting ducts (**microcysts**).
- **Liver:** periportal fibrosis, biliary ectasia; **Caroli syndrome** if intrahepatic ducts dilate.

#### Clinical features

| Age | Features |
|---|---|
| **Fetus** | Large echogenic kidneys, **oligohydramnios** → Potter facies, limb deformity, **pulmonary hypoplasia** |
| **Neonate** | Huge, smooth, hard flank masses; respiratory distress; rising creatinine; **30–50% die early** |
| **Child** | Hypertension, CKD (chronic kidney disease), growth failure, anaemia, osteodystrophy |
| **Older child** | **Portal hypertension**: varices, splenomegaly, cholangitis; liver function stays normal |

- No association with renal tumours.

### Investigations

- **Ultrasound:** bilateral very large, **homogeneously hyperechoic** kidneys (brighter than liver) from countless microcyst interfaces; discrete cysts appear with age.
- **Parents' ultrasound** normal (excludes ADPKD); **PKHD1 genetic testing**; liver ultrasound.
- **Differential:** neonatal ADPKD (autosomal dominant PKD), bilateral hydronephrosis, MCDK (multicystic dysplastic kidney), glomerulocystic disease, bilateral tumours, renal vein thrombosis.

### Management (no cure)

- **Neonatal:** ventilation, fluid and electrolyte care, antihypertensives (ACE (angiotensin-converting enzyme) inhibitors).
- **CKD care:** nutrition, growth hormone, anaemia and bone treatment; dialysis and **kidney transplantation**.
- **Nephrectomy** (breathing, feeding, blood pressure, room for peritoneal dialysis): cautiously; **very early bilateral nephrectomy** (first 3 months) had the worst neurological outcomes (ARegPKD registry).
- **Liver:** variceal sclerotherapy, portosystemic shunt; **combined liver–kidney transplant** for severe liver disease.
- **Genetic counselling** and prenatal diagnosis.

### ARPKD versus ADPKD

| | **ARPKD** | **ADPKD** |
|---|---|---|
| Gene | PKHD1 (chromosome 6) | PKD1 (16), PKD2 (4) |
| Age | Perinatal | 30–50 years |
| Cysts from | Collecting duct | Whole nephron |
| Ultrasound | Symmetric, homogeneous, echogenic | Large discrete cysts, asymmetric |
| Liver | Fibrosis (always) | Cysts |
| Other | None | Berry aneurysm, mitral prolapse |

**Diagram to draw:** cut kidney with radial dilated collecting ducts.

> **Key points to remember:**
> - PKHD1, chromosome 6, fibrocystin; always congenital hepatic fibrosis.
> - Huge homogeneous echogenic kidneys with oligohydramnios.
> - Neonates die of pulmonary hypoplasia; survivors develop portal hypertension.

**Also, from the condensed book (not in the paper answer):**

- In later-onset disease, portal hypertension dominates.
- Nephrectomy costs nephron mass and brings dialysis earlier, so delay it where possible.

## 5 · ADPKD — paediatric focus

*In children the key finding is hypertension with normal function.*

*See the paper answer **ADPKD (Adult polycystic kidney disease)** (Section 13, Ch 136).*

**Also, from the condensed book (not in the paper answer):**

- **Contiguous gene syndrome**: deletion of PKD1 and the adjacent TSC2 gives TSC with severe early polycystic kidneys.
- A negative US in a young person does **not** exclude PKD2, so use genetic testing after counselling when certainty matters.
- Screen for aneurysm by MR angiography only for family history or high risk. Observe <7 mm; coil or clip if >7–10 mm or symptomatic.

> **KEY POINT**
>
> ADPKD is a volume disease: cyst growth stretches vessels (hypertension), compresses nephrons and hides stones and tumours. Every therapy is judged by whether it slows total kidney volume.

## 6 · Juvenile nephronophthisis (NPH) / medullary cystic kidney disease (MCKD)

*A salt-wasting cause of renal failure that ultrasound often misses.*

- These ciliopathies give tubular atrophy, corticomedullary cysts and interstitial fibrosis in small-to-normal kidneys. **Polyuria and polydipsia occur in >80%**, with **salt wasting**. ESRD follows 5–10 y after presentation.

| | **Juvenile NPH** | **MCKD** |
|---|---|---|
| Inheritance | Autosomal recessive | Autosomal dominant |
| ESRD | Mean 13 y; **10–20% of childhood renal failure** | Third–fourth decade |
| Hypertension | Absent | Present |
| Extrarenal | 20%: **retinitis pigmentosa (Senior–Løken)**, Joubert, Cogan apraxia, hepatic fibrosis, Bardet–Biedl | None; type 2 — hyperuricaemia, gout in teenage males |

- CT or MRI shows the cysts better than US. **Replace sodium, and avoid salt restriction and diuretics**. Transplant works, with no recurrence.

## 7 · Syndromes with renal cysts

### Tuberous sclerosis complex (TSC)

- It is autosomal dominant: **TSC1 (chr 9), TSC2 (chr 16)**. Diagnosis needs 2 major features, or 1 major and 2 minor (angiomyolipoma is major).
- Kidney: cysts and **angiomyolipoma (AML) in 40–80%**, bilateral and multiple. AMLs never metastasise; the risk is haemorrhage. RCC occurs in ≈2%; suspect it in a growing fat-free lesion.
- **MRI is preferred**, because US misses 25–30% of fat-poor AMLs.
- **If an AML is >3 cm and growing, give everolimus (a mammalian target of rapamycin, mTOR, inhibitor) first line**. For an acute bleed, embolise and then give corticosteroids. Kidney-sparing surgery is second line; avoid nephrectomy. Image every 1–3 y, check BP, proteinuria and GFR yearly, and treat hypertension with an ACE inhibitor or ARB.

### Von Hippel–Lindau disease

*See the paper answer **VHL gene and von Hippel–Lindau disease** (Section 13, Ch 137).*

**Also, from the condensed book (not in the paper answer):**

- Partial adrenalectomy for a small phaeochromocytoma avoids lifelong steroids.

## 8 · Multicystic dysplastic kidney (MCDK)

*The MCDK is harmless; watch the opposite kidney.*

- It has non-communicating cysts, **no normal parenchyma, no function** and an atretic ureter ("bunch of grapes").
- It is the **commonest renal cystic disease of the newborn and the second commonest neonatal abdominal mass**: 1 in 1000–4000. Bilateral disease is lethal.
- **The contralateral kidney** hypertrophies and has **vesicoureteral reflux (VUR) in 18–43% and pelviureteric junction (PUJ) obstruction in 3–12%**.
- About 40% involute (later imaged as "agenesis"). Hypertension and Wilms tumour are not increased.

| US feature | MCDK | Severe PUJ obstruction |
|---|---|---|
| Cysts | Haphazard, varied, non-communicating | Calyces around central pelvis, communicating |
| Central large cyst | Absent | Present (pelvis) |
| Isotope uptake | None | Usually some |

**ALGORITHM**

1. For a prenatally detected MCDK, do US on day 2–3 and at 1 month to assess the contralateral kidney and bladder.
2. If the diagnosis is in doubt, do a DMSA or MAG3 scan.
3. If there is a urinary tract infection (UTI) or an abnormal contralateral kidney or bladder, then do a voiding cystourethrogram (VCUG); otherwise do not.
4. Hand over to primary care to check BP, mass and UTI.

- **Nephrectomy** is only for mass effect, hypertension, cyst rupture or **increasing solid tissue**. Routine US surveillance has no proven benefit.

> **KEY POINT**
>
> MCDK removes itself; management looks to the solitary contralateral kidney and its reflux or obstruction.

## 9 · Benign multilocular cyst (cystic nephroma)

*Imaging cannot exclude cystic Wilms tumour, so excise.*

- **It is bimodal: boys <4 y (M 2:1, flank mass) and women >30 y (F 8:1, haematuria from herniation into the pelvis)**.
- **Excision is both diagnosis and treatment** (nephrectomy in children; partial if localised).

## 10 · Simple cysts in children and Bosniak

- **US criteria (all four)**: anechoic, thin smooth wall, posterior enhancement, round or oval. On CT a simple cyst is −10 to +20 HU without enhancement. A hyperdense cyst is benign if non-enhancing, homogeneous and ≤3 cm.
- **Children**: if a simple cyst is <15 mm and asymptomatic, repeat US at 2 y and discharge if stable. Continue follow-up if it is ≥15 mm or complex, or with a family history or genitourinary anomalies.

| Bosniak | Features | Management |
|---|---|---|
| I | Simple | None |
| II | Fine septa, minimal calcification, hyperdense ≤3 cm | None |
| IIF | More septa/calcification, no measurable enhancement | Follow-up (≈11% progress) |
| III | Thick/irregular enhancing walls or septa | Excision (paediatric III–IV: >90% intermediate/malignant) |
| IV | Enhancing solid components | Excision |

Treat a symptomatic benign cyst by aspiration with sclerosis or laparoscopic unroofing.

## 11 · Medullary sponge kidney, glomerulocystic and acquired cystic disease

### SAQ Medullary sponge kidney

Paper answer · P2

**Type:** SAQ (10 marks)
**Source:** Practice Set 9 Paper 2, Q3e

#### Definition

- **MSK (medullary sponge kidney; precalyceal canalicular ectasia):** dilated (ectatic) distal **collecting ducts** with small cysts and diverticula **confined to the medullary pyramids**. Nephrons are initially normal.
- Usually **non-heritable** (sporadic); usually bilateral.

#### Pathogenesis of stones and nephrocalcinosis

- **Stasis and infection** in the ectatic ducts favour crystal deposition.
- **Tubular defects:**
- **Hypercalciuria** in one-third to one-half (mostly absorptive; sometimes renal leak).
- **Incomplete distal RTA (renal tubular acidosis)** in 30–40%, shown on acid loading. This causes **hypocitraturia** and alkaline urine.
- Impaired urinary concentration.
- Mutations in $H^+$-ATPase genes (ATP6V1B1, ATP6V0A4) support an MSK–distal RTA link.
- Result: calcium stones within dilated ducts and medullary **nephrocalcinosis**.

#### Clinical features

- Often silent for life; presents **after 20 years**.
- **Renal colic 50–60%**, UTI (urinary tract infection) 20–33%, gross haematuria 10–18%.

- Found in **2.6–21% of stone formers**, more often women.

### Diagnosis

- **Urographic triad:** 1. Enlarged kidneys, with or without papillary calcification. 2. Elongated papillary tubules or cavities filling with contrast: **"bristles on a brush"** (paintbrush); when calcified, a **"bouquet of flowers"**. 3. Papillary blush and persistent medullary opacification.
- CT urography now replaces IVU (intravenous urography); plain CT shows medullary nephrocalcinosis.
- **Metabolic work-up:** 24-hour urine (calcium, citrate, pH), serum bicarbonate and potassium.
- **Differentials:** papillary necrosis, other medullary nephrocalcinosis (distal RTA, hyperparathyroidism), calyceal diverticulum.

### Management

| Agent | Target | Note |
|---|---|---|
| **Potassium citrate** | Hypocitraturia, hypercalciuria | **First choice**; reduces calciuria and recurrence |
| Thiazide | Hypercalciuria | Useful even without hypercalciuria |
| Inorganic phosphate | Hypercalciuria | If thiazide unsuitable; avoid with urease infection |
| Fluids, low-sodium diet | Volume, calciuria | All patients |
| - Treat infection; stones by SWL (shock-wave lithotripsy) or ureteroscopy; stones inside ducts may be unreachable. | | |
| - **Prognosis:** usually benign; renal failure is rare. | | |

**Diagram to draw:** pyramid with ectatic ducts filled with contrast ("brush") and small calculi.

> **Key points to remember:**
> - Ectatic collecting ducts; sporadic; "bristles on a brush".
> - Hypercalciuria, incomplete distal RTA, hypocitraturia.
> - Potassium citrate is first-line prevention.

**Also, from the condensed book (not in the paper answer):**

- **Sporadic glomerulocystic disease** is a diagnosis of exclusion.

*See the paper answer **Acquired renal cystic disease** (Section 13, Ch 136).*

**Also, from the condensed book (not in the paper answer):**

- Acquired renal cystic disease affects 20–45% of children on dialysis, and a retroperitoneal bleed calls for nephrectomy because an occult RCC is likely.
- **Parapelvic cysts** mimic hydronephrosis on US; delayed contrast CT shows an undilated system, and management is conservative.

## 12 · Landmark trials

| Trial | Design | Result | Take-home |
|---|---|---|---|
| HALT-PKD | Randomised: lisinopril and telmisartan vs lisinopril; rigorous vs standard BP | Dual blockade no benefit; low BP slowed total kidney volume (TKV) growth, lowered left ventricular mass and albuminuria, GFR unchanged | Single renin–angiotensin blocker; tight BP early |
| TEMPO / REPRISE | Randomised, tolvaptan vs placebo | Slower TKV growth and GFR decline | First disease-modifying drug; aquaresis, hepatotoxicity |
| ARegPKD registry | Observational ARPKD cohort | Bilateral nephrectomy <3 months gave the worst neurological outcome | Avoid early bilateral nephrectomy |

# 51 Pathophysiology of Urinary Tract Obstruction

## 1 · Definition and classification

*Function, not dilation, defines obstruction.*

- **Obstruction** is any impediment to flow, calyces to meatus, that injures the kidney or lower tract: **acute or chronic, partial or complete, intrinsic or extrinsic, unilateral or bilateral**.
- It affects ≈ **1 in 500** live births; high-grade obstruction carries a lifetime end-stage renal disease (ESRD) risk of up to **30%**.

| | Lower urinary tract obstruction (LUTO) | Upper tract |
|---|---|---|
| Commonest | Posterior urethral valves (PUV) and urethral atresia ≈ 50% | Ureteropelvic junction obstruction (UPJO), 1 in 1500; M:F 2:1 |
| Others | Anterior urethral valves, megalourethra, prolapsing ureterocele, cloaca, megacystis–microcolon–intestinal hypoperistalsis, trisomy 18/21 | Ureterovesical junction obstruction (UVJO), ectopic ureter, ureterocele, polyp, stones, crossing vessel |
| Outcome driver | ESRD and pulmonary hypoplasia | Unilateral function loss; contralateral kidney buffers |

**Presentation** is prenatal hydronephrosis (mostly at 18–20 wk), or later infection, mass, pain or respiratory failure.

> **KEY POINT**
>
> Trap: **detrusor hypertrophy** causes UVJ obstruction by narrowing the intramural ureter. Neurogenic bladder obstructs without an anatomical lesion.

## 2 · Haemodynamic changes

Rising pressure lowers GFR and renal blood flow (RBF), causing ischaemia. Injury begins within **2 h** of complete obstruction, worse if bilateral or solitary.

| † Phase (complete) | Unilateral | Bilateral / solitary |
|---|---|---|
| I (0–1.5 h) | RBF rises, ureteric pressure rises (afferent dilatation: PGE2, nitric oxide) | Modest RBF rise |
| II (1.5–5 h) | RBF falls, pressure still high (efferent constriction) | RBF falls, pressure rises |
| III (> 5 h) | RBF and pressure fall (afferent constriction: angiotensin II, thromboxane) | Pressure stays high (retained solute, ANP), the basis of post-obstructive diuresis |

- **Hypertension** is renin driven: bilateral ≈ **77%**, unilateral ≈ **20%**.

## 3 · Tubular function

- **Aquaporins and sodium transporters are downregulated**, causing a concentrating defect, salt wasting, polyuria and **distal renal tubular acidosis**.
- Potassium and other solutes are **wasted early and retained late**. FENa is < 1% in acute obstruction and > 1% with tubular injury.

## 4 · Pathology and fibrosis

- Fibrosis is the final common path, driven by stretch and **TGF-β/Smad**, amplified by angiotensin II.
- **Bladder** hypertrophies, compensates, then decompensates with poor compliance. Storage pressure > **40 cm $H_2O$** injures the kidneys.

## 5 · Recovery and compensatory growth

- Relief can reverse damage, but **incomplete recovery is common**.
- Compensatory growth is **glomerulomegaly, not new nephrons**.

## 6 · Post-obstructive diuresis (POD)

### SAQ Post-obstructive diuresis

Paper answer · P1

**Type:** SAQ (10 marks)
**Source:** Practice Set 19 SAQ Sheet, Q3b

#### Definition

- POD (post-obstructive diuresis): marked polyuria after relief of obstruction, as the kidney sheds retained salt, water and urea.
- No agreed rate; a common working definition is **>200 mL/h for 2 consecutive hours or >3 L in 24 hours**. Up to 300 mL/h is common.
- **Physiological** (appropriate, self-limited) versus **pathological** (continues after euvolaemia, causing hypovolaemia and electrolyte loss).

#### Incidence and setting

- Unilateral obstruction: **about 2%** (the other kidney compensates). Bilateral or solitary kidney: **about 50%**.
- Commonest after catheterising **chronic retention with renal impairment**; also bilateral ureteric obstruction and PUV (posterior urethral valves).

#### Mechanisms

1. **Transporter dysregulation:** down-regulation of $Na^+/K^+$-ATPase and aquaporin-2 → concentrating defect (nephrogenic DI (diabetes insipidus)); loss of medullary tonicity.
2. **Volume expansion:** on relief, raised ANP (atrial natriuretic peptide), low aldosterone and sympathetic tone → salt and water loss.
3. **Osmotic diuresis** from retained urea.

#### Risk factors

- Bilateral obstruction, raised creatinine, oedema, heart failure, hypertension, confusion.
- After valve ablation: oligohydramnios, preterm birth, raised creatinine.

#### Clinical features

- Starts within hours of relief; most settle within 24–48 hours.
- Danger signs: postural hypotension, tachycardia, weight loss, hypernatraemia (free-water loss), low potassium, magnesium, phosphate.

**Management**

``` Relieve obstruction (catheter / nephrostomy)

|

Monitor hourly urine output, vitals, weight

|

Alert, can drink -> free oral fluids (most resolve)

|

Output persists / cannot drink / unstable -> admit; electrolytes every 4-6 h -> intravenous 0.45% saline: about 75% of urine output -> replace K, Mg, phosphate; adjust to urine sodium/osmolality ```

- **Do not overcorrect:** replacing every millilitre sustains the diuresis.
- **Gradual bladder decompression has no role**: it does not prevent haematuria or diuresis.
- PCN (percutaneous nephrostomy) for upper-tract obstruction; hydration alone for physiological POD.
- **Also know:** decompression haematuria; neonatal PUV concentrating defect; no approved drug therapy.

**Diagram to draw:** graph of urine output against time after catheterisation.

> **Key points to remember:**
> - Unilateral about 2%, bilateral about 50%.
> - Mechanisms: aquaporin-2 loss, volume expansion, urea osmotic diuresis.
> - Replace about 75% of output with 0.45% saline; do not overcorrect.
> - Gradual decompression has no role.

**Also, from the condensed book (not in the paper answer):**

- Preterm infants have low aquaporin-2, hence their POD risk.

**ALGORITHM**

1. Monitor intake, output, and serum and urine electrolytes.
2. Replace losses and correct electrolytes.
3. **Do not over-replace**: it perpetuates POD.
4. Expect resolution within **24 h**; it seldom lasts > 48 h.

## 7 · Fetus and child

*See the paper answer **Antenatal hydronephrosis, fetal uropathy and prenatal scan anomalies** (Section 5, Ch 34).*
```

**Also, from the condensed book (not in the paper answer):**

- Second-trimester oligo-or anhydramnios carries **60–80%** mortality.
- In the Whitaker test, obstruction is a pelvic pressure > **12 cm $H_2O$** or a continuous rise.
- After birth, use ultrasound first, MAG3 for drainage, DMSA for scars and VCUG for outlet and reflux.

## 8 · Treatment principles

Drain to buy function, then reconstruct.

| Option | Indication / key point |
|---|---|
| Clean intermittent catheterisation (CIC) | Neurogenic bladder; reduces UTI, stones, retention |
| Overnight catheter drainage | Valve bladder with nocturnal polyuria and progressive CKD |
| Vesicostomy | Infant with complex LUTO, unstable or unable to do CIC |
| Percutaneous nephrostomy | Upper tract; needs some hydronephrosis |
| Stent / cutaneous ureterostomy | Temporising; stent excellent for extrinsic obstruction |
| Pyeloplasty / reimplantation | Definitive; anastomose anterior to aberrant vessel; non-refluxing reimplant in children |

- Use NSAIDs only if function is preserved. ESRD needs dialysis or transplant (Ch 61).

# 52 Ectopic Ureter, Ureterocele and Ureteral Anomalies

## 1 · Definitions and classification

- Ectopic ureter and ureterocele are one defect of **ureterotrigonal connection** on a continuum. The appearance changes the technique, not the strategy.
- An **ectopic ureter** is a single or duplex ureter that does not enter the trigone but enters **at or distal to the bladder neck**; merely opening near it is not ectopic.
- A **ureterocele** is **cystic dilation of the distal ureter** within the bladder or spanning bladder neck and urethra. It is never entirely urethral and never attached to wolffian structures.
- In a **duplex system, both lesions always involve the upper pole** ureter.

| | **Ectopic ureter** | **Ureterocele** |
|---|---|---|
| Distal ureter | Dilated or not, opens ectopically | Cystic dilation |
| Position | At/distal to bladder neck; perineum, vagina, rectum (F) or wolffian ducts (M) | Bladder or spanning bladder neck and urethra |
| Wolffian attachment | Yes (male) | Never |
| Incontinence (untreated) | Yes, in females | No |

### Ureterocele classification

#### SAQ Ureterocele

Paper answer · P3

**Type:** SAQ (10 marks)
**Source:** Practice Set 2 Paper 3, Q3c (ectopic ureter also answered in 2018-Summer Paper 3, Q1)

#### Definition

- Cystic dilatation of the distal ureter, in the bladder or spanning bladder neck and urethra; **never wholly urethral**, never joined to Wolffian structures.
- Commoner in girls (about 4:1); about 80% drain the **upper pole of a duplex kidney** (Weigert–Meyer rule: upper pole orifice caudal-medial); about 10% bilateral.
- **Cause:** persistent **Chwalla membrane**; upper pole often dysplastic.

#### Classification

| **System** | **Basis** | **Types** |
|---|---|---|
| **Glassberg (most useful)** | Position | **Intravesical** vs **ectopic (extravesical)**: part at bladder neck or urethra |
| Stephens | Orifice morphology | Stenotic, sphincteric, sphincterostenotic, **cecoureterocele** (cavity extends under urethra), blind, non-obstructed |
| Churchill | Function | Upper pole only / whole kidney / contralateral at risk |

### Clinical features

- Antenatal hydronephrosis (most); febrile UTI (urinary tract infection) or urosepsis; adult single-system type with stone.
- **Prolapse:** smooth interlabial mass in girls; BOO (bladder outlet obstruction).
- **No incontinence when untreated** (unlike female ectopic ureter).

### Evaluation

- **Ultrasound:** thin-walled cyst in bladder, dilated upper pole and ureter.
- **VCUG (voiding cystourethrogram):** lower pole reflux ("drooping lily"), eversion (weak floor), prolapse.
- **DMSA (dimercaptosuccinic acid):** upper pole function; cystoscopy before incision.

### Management

- **Goals:** preserve function; remove infection, obstruction, reflux; keep continence.
- **CAP (continuous antibiotic prophylaxis)** from birth if hydronephrotic.
- **Observation:** small, non-obstructing, low-grade reflux, non-functioning upper pole.

| Option | Indication / result |
|---|---|
| **TUI (transurethral incision)** | Sepsis, neonates, intravesical type. Low transverse full-thickness incision; relieves obstruction in 78–97%; new reflux 0–50%. Intravesical without reflux: 70–80% cured; ectopic: secondary surgery up to 70% |
| Upper pole heminephrectomy | Non-functioning upper pole, no reflux; secondary surgery 40–50% |
| Ureteroureterostomy / pyeloureterostomy | Functioning upper pole |
| Excision + common sheath reimplant ± bladder neck repair | Persistent reflux, ectopic or cecoureterocele; persisting reflux 5–10% |
| Total reconstruction | Older child, non-functioning upper pole, lower pole reflux |

- **Also know:** "watering-can" laser punctures (less reflux); pseudoureterocele; ureterocele disproportion.

**Diagram to draw:** duplex kidney with upper pole ureterocele and lower pole reflux.

> **Key points to remember:**
> - Upper pole of duplex; never wholly urethral; Chwalla membrane.
> - Intravesical vs ectopic predicts TUI success.
> - TUI relieves obstruction in 78–97%; new reflux 0–50%.

**Also, from the condensed book (not in the paper answer):**

- A **cecoureterocele** tunnels submucosally beyond the bladder neck into the urethra; it is often missed before surgery and risks incontinence.
- **Ureterocele disproportion** is a typical ureterocele with a small, dysplastic, non-dilated upper pole, so the kidney looks normal.

## 2 · Embryology essentials

*Bud position explains upper pole obstruction and lower pole reflux.*

- In a duplex, the upper pole bud arises **later and more cephalad** on the wolffian duct, so its orifice is carried **caudally** and becomes ectopic.
- **Weigert–Meyer rule**: **the upper pole orifice lies caudal (and medial) to the lower pole orifice**. So the upper pole gets ectopia or ureterocele (obstruction), and the lower pole gets a short tunnel and vesicoureteral reflux (VUR).
- Classically, a ureterocele results from failure of the **Chwalle membrane** to rupture. Renal dysplasia lies **downstream** of the connection defect.
- An ectopic ureter reaches the vagina, cervix or uterus only through the **Gartner duct**, never directly into müllerian structures.

## 3 · Ectopic ureter — sites and sex asymmetry

### LAQ Embryology and management of ectopic ureter.

Paper answer · P3

**Type:** LAQ (25 marks)
**Source:** 2018-Summer Paper 3, Q1

#### Definition

- An **ectopic ureter** is any ureter, single or duplex, that does **not open on the trigone**. True ectopia means an orifice **at or distal to the bladder neck**.
- Commoner in girls. In girls most ectopic ureters drain the **upper pole of a duplex kidney**; in boys most are **single-system**.

#### Embryology

- **Week 4–6:** the **ureteric bud** arises from the **mesonephric (wolffian) duct** and induces the metanephric blastema to form the kidney.
- The duct segment distal to the bud is absorbed into the **urogenital sinus**; the ureteric orifice then moves cranially and laterally, the wolffian orifice caudally to the verumontanum.
- **Mackie–Stephens bud theory:**

| Bud position on wolffian duct | Absorption | Orifice | Result |
|---|---|---|---|
| Too caudal | Early | High and lateral, short tunnel | VUR (vesicoureteral reflux) |
| Too cranial | Late or never | Low and medial (bladder neck, urethra) or stays on wolffian duct | **Ectopic ureter**, often obstructed |

- A bud from the wrong place meets poor blastema, so **renal dysplasia increases with the degree of ectopia**.
- **Duplex system:** the upper-pole bud is more cranial, absorbed later, and ends **lower and more medial** (**Weigert–Meyer law**): upper pole obstructs; lower pole refluxes.

#### Sites of ectopia (the key sex difference)

| | Female | Male |
|---|---|---|
| Sites (commonest first) | Urethra, vestibule, vagina; rarely cervix, uterus, rectum (via **Gartner duct**) | Posterior urethra, seminal vesicle; also vas, ejaculatory duct |
| Relation to sphincter | Can open **below** the external sphincter | Always **above** the external sphincter |

| | Female | Male |
|---|---|---|
| Continence | **Continuous dribbling with normal voiding** | **Never incontinent** |
| Typical presentation | Constant dampness, perineal pus, UTI (urinary tract infection) | **Epididymo-orchitis**, UTI, flank pain, hydronephrosis |

- **Zinner syndrome:** ectopic ureter into the seminal vesicle with ipsilateral renal agenesis or dysplasia and a seminal vesicle cyst.

### Clinical features

- **Prenatal hydronephrosis**: most are detected this way.
- **Girl:** wet day and night despite normal voids; she is never dry for 30–60 minutes. Often misdiagnosed as voiding dysfunction.
- **Infection:** low-grade fever with spikes; urine culture may be negative because the infected moiety does not drain into the bladder.
- **Boy:** recurrent epididymitis in a young boy demands an upper-tract ultrasound.
- **Also know:** pain then perineal discharge; bilateral single-system ectopia with small bladder and wide bladder neck.

### Evaluation

| Test | What it shows |
|---|---|
| **Ultrasound** (first) | Dilated upper pole and ureter behind the bladder; parenchyma. Bladder view separates it from ureterocele |
| **VCUG** (voiding cystourethrogram) | Lower-pole or contralateral VUR ("drooping lily"), reflux into the ectopic ureter |
| **DMSA** (dimercaptosuccinic acid) | Gold standard for moiety function ("some" versus "none"); MAG3 (mercaptoacetyltriglycine) if drainage needed |
| **MR (magnetic resonance) urography** | Non-dilated upper pole, small dysplastic kidney, complex anatomy |
| **Examination / cystovaginoscopy** | Perineal orifice between urethra and vagina, cannulated for contrast |

- **Also know:** clear urine leaking despite methylene blue in the bladder suggests an ectopic ureter; CT for a small unseen kidney.

### Management

**Goals:** preserve renal function; remove infection, obstruction and reflux; achieve continence; with least morbidity. Start antibiotic prophylaxis in the neonate with significant dilation.

`Neonate with sepsis or massive dilation → end cutaneous ureterostomy (or nephrostomy); DMSA later DUPLEX system Upper pole non-functioning → upper-pole heminephroureterectomy Upper pole worth keeping → ureteroureterostomy (UU) / pyeloureterostomy + lower-pole VUR → common sheath reimplant (or lower-pole reimplant + distal UU) SINGLE system Functioning → ureteral reimplant (± tapering) Non-functioning → nephroureterectomy (laparoscopic) Bilateral → bladder neck reconstruction ± augmentation`

- **Upper-pole heminephroureterectomy** (open, laparoscopic or robotic): upper-pole ureter as a handle, passed behind the hilum; ligate polar vessels; excise the pole; close over fat.

Dissect on the ureter wall to protect the **lower-pole blood supply**. The stump is rarely a problem.

- **Ureteroureterostomy (UU):** end-to-side upper into lower pole ureter, distal (inguinal incision) or proximal (preferred with big size mismatch). **Stent the recipient lower-pole ureter** cystoscopically for identification. Now used even for non-functioning poles; feared hypertension has not been shown.
- **Common sheath reimplantation:** do not separate the two ureters (they share blood supply).

### Complications

- Loss of lower-pole function (most serious); urinoma (up to 20% after minimally invasive partial nephrectomy, mostly harmless); stump infection; new reflux; persistent incontinence.

### Recent advances

- Robotic UU and heminephrectomy; single-site surgery; non-dismembered side-to-side ureterovesicostomy in infants.

**Diagram to draw:** duplex kidney showing Weigert–Meyer orifices, and female versus male ectopic sites relative to the external sphincter.

> **Key points to remember:**
> - Ectopia = orifice at or below the bladder neck; bud too cranial on the wolffian duct.
> - Girls: continuous dribbling with normal voiding; boys: never incontinent, epididymitis.
> - DMSA decides keep (UU/reimplant) versus remove (heminephroureterectomy).
> - Protect the lower-pole blood supply; stent the recipient ureter for UU.

**Also, from the condensed book (not in the paper answer):**

- A **single-system ectopic ureter** may mimic an "absent kidney" on ultrasound; use contrast CT or a renal function study.
- **Bilateral single-system ectopia** (rare) gives a hypoplastic bladder, poor bladder neck and bilateral dysplasia.

> **KEY POINT**
>
> Why does only the girl leak? The female ureter may open distal to the external sphincter and leaks continuously; the male ureter, confined to wolffian derivatives above it, presents as infection or obstruction.

- Male sites are the bladder neck, prostatic urethra, vas, seminal vesicle and ejaculatory duct.

## 4 · Clinical presentation

### Prenatal

- Most are found as prenatal hydronephrosis. A fetal upper pole "cyst" is upper pole hydronephrosis until proven otherwise.
- **The diagnosis is made at the bladder**, because the upper tracts look identical; a large ureterocele can mask the bladder.

- Dilation of the ipsilateral lower pole or contralateral kidney suggests reflux. Dilation of the whole kidney and ureter suggests a single system.

### Incidental and infection

- A "presumed ovarian cyst" may be a dilated ureter, so include it in the differential of a cystic pelvic mass.
- Do ultrasound in every child with urosepsis. If sepsis does not respond, do an **urgent transurethral incision (TUI)** of a ureterocele, or drain an ectopic ureter (usually end ureterostomy).
- An infected ectopic ureter gives low-grade spiking fever, and a **negative culture does not exclude infection**.
- **Any young male with bacterial epididymitis needs upper tract ultrasound**; it is often recurrent before diagnosis.

### Incontinence

- Suspect it in a toilet-trained girl with **continuous low-volume dampness day and night** and normal voiding. Untreated ureteroceles do not cause incontinence.
- The key question is: **can she be dry for 30–60 minutes?** With an ectopic ureter she cannot.
- The leaking moiety may be non-dilated and missed on imaging, so suspicion rests on the history. Exclude voiding postponement and constipation.

### Ureterocele prolapse

- It appears as a **smooth, congested, mucosa-covered interlabial mass**, often with voiding difficulty. A lobulated grape-like mass suggests **sarcoma botryoides** instead.
- Ultrasound confirms it; VCUG shows the prolapse, lower pole reflux and the "drooping lily".

### Late presentation

- Teenagers and adults present with infection or pain; an adult single-system ureterocele often contains a **stone**. Upper pole heminephrectomy is the mainstay.

## 5 · Evaluation

- Examine the perineum of every continuously wet girl. The ectopic orifice lies **between the urethral meatus and the vagina**.

### Imaging

| Test | Answers | Note |
|---|---|---|
| **Ultrasound** | Anatomy; inferred function; bladder | First always; cannot separate ectopic ureter from ureterocele above bladder |
| **Voiding cystourethrogram (VCUG)** | Reflux (lower pole, contralateral, into ureterocele); bladder base, urethra | Optional if obstruction alone dictates incision; repeat after incision |
| **Dimercaptosuccinic acid (DMSA)** | Differential function — gold standard | Only "some vs none" |
| Mercaptoacetyltriglycine (MAG3) renogram | Function and drainage | Observation candidates; after 2 months |

| Test | Answers | Note |
|---|---|---|
| MR urography | Most detailed anatomy and function | Complex anatomy, non-dilated upper pole; needs sedation |
| Intravenous pyelogram (IVP) | Anatomy and qualitative function | Non-dilated ectopic ureter with incontinence |

- A **ureterocele** is a thin-walled cyst within the bladder. A **pseudoureterocele** is a dilated ectopic ureter indenting the bladder; its wall is **bilaminar and thicker, and its lumen lies outside the bladder**.
- A full bladder can efface a ureterocele so it mimics an ectopic ureter, so scan through filling and voiding.
- On VCUG, the **"drooping lily"** is the lower pole pushed down and out by a dilated upper pole.
- † A single-system intravesical ureterocele gives a **"cobra head" (spring-onion)** sign on urography.
- **Preoperative function does not predict function after decompression**.

### VCUG terms

| Term | Meaning | Significance |
|---|---|---|
| Effacement | Flattens with filling | Mimics ectopic ureter |
| Prolapse | Moves into urethra | Outlet obstruction; predicts secondary surgery |
| **Eversion** | Diverticulum-like protrusion of bladder base | **Weak trigonal floor, so open repair likely** |

### Endoscopy

- **Choose the incision site** with little bladder filling, when the lowest part of the ureterocele is seen best.
- Detect any urethral extension (cecoureterocele) so the incision leaves no obstructing segment.
- A patulous bladder neck or cecoureterocele carries a later incontinence risk; consider bladder neck tightening.

## 6 · Management principles

- Goals: **preserve renal function; eliminate infection, obstruction and reflux; maintain continence; minimise total morbidity**.
- Start antibiotic prophylaxis early in significant hydronephrosis, even without reflux.
- Relieving obstruction may **create** reflux, or may let lower pole reflux resolve. This is the central trade-off, and decisions are empirical.
- **Fetal intervention**: none for ectopic ureter; for ureterocele only if it obstructs all units (experimental).

### Neonatal algorithm

**ALGORITHM**

1. After prenatal hydronephrosis, do postnatal ultrasound.
2. If there is significant hydroureteronephrosis, start prophylaxis.
3. Do VCUG (optional if it will not change the decision to decompress), then a renal scan after 2 months.
4. If the ureterocele is small, with minimal hydronephrosis and no obstruction, observe.
5. If obstruction is significant and parenchyma is at risk, do TUI (or cutaneous distal ureterostomy if endoscopy is impossible).
6. If sepsis does not respond, do TUI (ureterocele) or nephrostomy or end ureterostomy (ectopic ureter); elective drainage is still needed later.

- **Observation** suits selected ureteroceles: no obstruction of the lower pole or contralateral kidney and lower pole reflux ≤ grade III, a non-functioning upper pole, or no obstruction on renogram. Counsel about unpredictable acute presentation.

## 7 · Operative options

### Transurethral incision (TUI) of ureterocele

- TUI relieves obstruction in **78–97%**. It is the choice in **sepsis** and in the **infant**.
- Make a **low transverse full-thickness incision as distal and close to the bladder floor as possible** to reduce reflux. For an ectopic ureterocele, extend it into the urethral part so no obstructing flap remains.
- The **"watering-can"** technique (multiple holmium punctures) decompresses equally with less de novo reflux.
- It is a day-case procedure. Do ultrasound and VCUG at **4–6 weeks**.
- New reflux occurs in **0–50%**. **Intravesical** ureteroceles decompress without reflux in 70–80%; **extravesical** ones need secondary surgery in up to 70%. Lower pole reflux (risk ratio 1.74) and an extravesical site (2.78) predict reoperation.

> **KEY POINT**
>
> Why TUI first even if half need more surgery? It is outpatient, may be definitive, defers the second operation to an older child, and decompression often avoids tapering at reimplant. It is less appealing in an older child with a massive non-functioning upper pole, who needs definitive surgery at diagnosis.

### Upper pole partial nephrectomy (heminephrectomy)

- It is for a non-functioning upper pole, by open, laparoscopic or robotic approach. The main concern is **not to damage the viable lower pole**.
- Use the upper pole ureter as a handle, passing it **behind the renal vessels**, ligate the upper pole vessels and resect at the demarcation.
- **Dissect on the wall of the upper pole ureter**, and leave its back wall on the lower pole ureter near the bladder, because they share a blood supply. A small distal stump is left; a large or refluxing one is excised low, sparing the bladder neck and vas.
- Outcomes: for ectopic ureter it is usually curative. For ureterocele, new reflux appears in **15–50%** and secondary surgery is needed in **40–50%**.

### Ureteroureterostomy / pyeloureterostomy

- The upper pole is joined end-to-side into the lower pole system, for functioning and increasingly non-functioning upper poles; feared hypertension has not been substantiated.
- For a **distal** repair, **stent the lower pole ureter cystoscopically first**. A **proximal** repair is preferred for a dilated upper pole or size mismatch. Robotic repair matches open results, and size mismatch is no barrier.

### Lower tract reconstruction

- **Ureterocele excision with common sheath reimplantation** relieves obstruction and reflux at once, and is the fallback for persistent reflux. It risks the bladder neck, sphincter and vagina.
- Mobilise **both ureters as a unit**, excise the ureterocele, **plicate the detrusor defect** and reimplant submucosally.
- A cecoureterocele remnant can act as an obstructing **urethral windsock** flap-valve.
- Reflux persists in **5–10%**, more often after tapering.
- **Total reconstruction** suits an **older child with a massive ureterocele, non-functioning upper pole and significant lower pole reflux**.

### End ureterostomy for ectopic ureter

- It is for a neonate with sepsis or massive dilation. **Do not trim the redundant ureter**, because its length is needed for reimplant. Do DMSA after a few months, then reconstruct or remove.

## 8 · Decision summary

*Upper pole function and lower pole reflux pick the operation.*

### Ectopic ureter (duplex)

**ALGORITHM**

1. If the upper pole is worth keeping and the lower pole refluxes, do a common sheath reimplant, or a lower pole reimplant with distal ureteroureterostomy.
2. If the upper pole is worth keeping and there is no lower pole reflux, do a proximal or distal ureteroureterostomy.
3. If function is unclear, or there is massive dilation or sepsis, do a temporary end ureterostomy, then DMSA, then reconstruct or remove.
4. If removal is chosen, do upper pole nephrectomy with ureterectomy.

### Ureterocele (Table 52.1)

| Procedure | Ideal indication | Advantage | Limitation |
|---|---|---|---|
| TUI | Infant; large ureterocele with VUR | Outpatient; effective; occasionally definitive | Reflux into ureterocele moiety needs bladder surgery |
| Upper pole nephrectomy | Older; large non-functioning upper pole; no VUR | Removes pathology; avoids bladder | Lower pole risk; may still need bladder surgery |

| Procedure | Ideal indication | Advantage | Limitation |
|---|---|---|---|
| Ureteroureterostomy / pyelostomy | Older; functioning upper pole; no VUR | Drains with little risk | Ureterocele left; may reflux |
| Excision and common sheath reimplant | VUR; functioning upper pole without major dilation | Removes obstruction and reflux; no renal risk | Complex; vagina/bladder neck risk; tapering |

**KEY POINT**

VUR is the pivot: without VUR choose an upper tract solution; with VUR, a bladder-level solution, except in the infant, where TUI defers bladder surgery. Upper pole function then picks nephrectomy or ureteroureterostomy.

## 9 · Complications

- **Loss of lower pole function** after heminephrectomy is the most significant; prevent it by preserving the shared blood supply and using gentle traction.
- **Urinoma** occurs in up to 20% after laparoscopic or robotic cases; observe it if asymptomatic.
- **De novo reflux** after TUI or heminephrectomy is observed if there is no infection; otherwise do excision and reimplantation.
- **Voiding dysfunction** comes from poor trigonal support, which causes obstruction, poor emptying and recurrent VUR. Assess with VCUG **including lateral voiding views**, and treat by reconstruction, CIC or bulking.

## 10 · Other ureteral anomalies

### Duplication

- It occurs in ~**1 in 125** (0.8%), F:M 1.6:1, and is mostly unilateral. **Look for contralateral duplication**: an occult second ectopic ureter explains persistent wetting.
- A **bifid ureter** is a partial duplication joining below the ureteropelvic junction (UPJ).
- **The lower pole gets single-system disease** (UPJ obstruction, VUR); **the upper pole gets ureteral-formation disease** (ectopia, ureterocele).
- Lower pole VUR is often high grade and resolves slowly; treat by common sheath reimplant, endoscopic injection or low ureteroureterostomy.

### Triplication

- Smith types: 1, three separate ureters (**35%**, commonest); 2, incomplete with two orifices (21%); 3, trifid with one orifice (31%); 4, two ureters with three orifices (13%).

### Valves, diverticulum, polyp

- † **Ureteral valves** are transverse mucosal folds with smooth muscle and proximal dilation; excise the segment. A ureteral **diverticulum** is usually a **blind-ending bifid ureter**.
- A **fibroepithelial polyp** is commonest at the **UPJ**, **89% male, 78% left**. At the UPJ, do pyeloplasty; elsewhere, **remove it ureteroscopically**.

### Vascular position anomalies

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## LAQ Retrocaval ureter and IVC anomalous development

Paper answer · P1 · P3 · merged from 2

*Asked as:* P1 LAQ "IVC anomalous development and retrocaval ureter" · P3 SAQ "Retrocaval ureter"

## Introduction

- The **IVC (inferior vena cava)** forms between the 6th and 8th weeks from three pairs of fetal veins by anastomosis, regression and persistence. The final IVC is **right-sided**, so long left veins must cross the aorta to reach it.
- Most anomalies are silent CT (computed tomography) findings. They **bleed at surgery, mimic lymph nodes, trap the ureter and alter IVC filter placement**.

## Embryology of the IVC

### Three paired venous systems

- Fetal veins (posterior cardinal, supracardinal, subcardinal) form a venous collar around each ureter and ascending kidney: posterior cardinal and supracardinal veins dorsally, subcardinal veins ventrally.

| Vein | Position | Adult derivatives |
|---|---|---|
| **Posterior cardinal** (first) | Dorsolateral | Iliac confluence; mostly regresses |
| **Subcardinal** | Ventromedial | Right: **suprarenal IVC**; intersubcardinal anastomosis: **left renal vein**; left: adrenal and gonadal veins |
| **Supracardinal** (last) | Dorsomedial | Right: **infrarenal IVC**; thoracic parts: **azygos and hemiazygos** |

### Segments of the definitive IVC (top to bottom)

1. **Hepatic**: right vitelline vein.
2. **Suprarenal**: right subcardinal vein (via the subcardinal–hepatic anastomosis).
3. **Renal**: subcardinal–supracardinal anastomosis.
4. **Infrarenal**: right supracardinal vein.
5. **Iliac confluence**: posterior cardinal veins.

### Renal collar

- A venous ring encircles the aorta at renal level. The **ventral (pre-aortic) limb** persists as the left renal vein; the **dorsal (retro-aortic) limb** regresses.
- Normally the right **supracardinal** vein, which lies **dorsal** to the ureter, forms the infrarenal IVC, so the normal ureter lies in front of the IVC.

`Left-sided channels regress -> right-sided persist -> single right IVC + pre-aortic left renal vein`

**Diagram to draw:** three stages: paired cardinal veins, anastomoses with renal collar, definitive IVC.

## Anomalies of IVC development

| Anomaly | Embryological error | Key features |
|---|---|---|
| **Retrocaval ureter** ("preureteral vena cava") | Infrarenal IVC forms from a vein **ventral** to the ureter | Right only; 1 in 1500 autopsies; male 3:1; presents 3rd–4th decade. **Type I** low loop, fishhook, obstructs; **type II** high loop, sickle, mild |

| Anomaly | Embryological error | Key features |
|---|---|---|
| **Circumaortic left renal vein** | Dorsal limb persists with ventral | Commonest (2–16%); posterior limb runs lower into lumbar IVC |
| **Retroaortic left renal vein** | Ventral limb regresses, dorsal persists | Less common; may be compressed ("posterior nutcracker") |
| **Left IVC** | Left supracardinal persists, right regresses | Crosses to the right at renal level, in front of the aorta |
| **Duplicated IVC** | Both supracardinals persist | Left IVC joins left renal vein; double right cava can trap the ureter |
| **Azygos continuation** | Subcardinal–hepatic anastomosis fails | Large azygos; hepatic veins drain directly to right atrium; polysplenia, heterotaxy |
| **Absent infrarenal IVC** | Supracardinal and posterior cardinal failure | Deep vein thrombosis in the young |

### Retrocaval ureter: definition

- The **right ureter** passes **behind the IVC**, then crosses in front of it from medial to lateral to resume a normal course.
- Better termed "**preureteral vena cava**": the anomaly is venous; the ureter is trapped.
- In P3 the question was also asked as "Retrocecal ureter"; this is taken as a typo for retrocaval (circumcaval) ureter.

### Retrocaval ureter: embryology

- If a vein **ventral** to the ureter persists as the infrarenal IVC, the ureter is trapped behind it.
- Campbell names a persistent **lumbar subcardinal** vein (the lumbar part of the ventral subcardinal vein; Chapter 52) or a **persistent posterior cardinal** vein (Chapter 93, and most texts); both mean that a ventral vein forms the IVC.
- Variant: double right IVC, with the ureter between the limbs.

### Retrocaval ureter: incidence and presentation

- About **1 in 1500** autopsies; **male 3:1**; right side only.
- Presents in the **third–fourth decade** with right flank pain, UTI (urinary tract infection), stones or incidental hydronephrosis.
- Associations: horseshoe kidney, left renal anomalies; bilateral cases occur in situs inversus.

### Retrocaval ureter: classification (Bateson and Atkinson)

| | Type I (low loop), commoner | Type II (high loop) |
|---|---|---|
| Crossing | Lower, at L3–L4 | Higher, at or near renal pelvis level |
| Shape | **J or "fishhook" / reverse-J**, S-curve | Smooth horizontal **sickle** curve |
| Obstruction | Yes, moderate–severe, at the iliopsoas edge | Mild or none (pelvis also crosses) |

### Retrocaval ureter: investigations

- **CT urography (computed tomography urography): investigation of choice**, showing ureter and cava together; MRI (magnetic resonance imaging) is the alternative.
- IVU (intravenous urography): fishhook; ureter not seen beyond the J hook.
- Retrograde pyelography: **S-curve** or fishhook at L3–L4.
- **Diuretic renogram:** grades the degree of obstruction and gives split function.

- Pitfall: mistaken for **PUJ (pelviureteric junction) obstruction**, so pyeloplasty fails.

### Retrocaval ureter: management

- **Non-obstructed (usually type II): observe.**
- **Obstructed: dismembered ureteroureterostomy** (or ureteropyelostomy): divide the ureter, transpose it **anterior to the IVC**, and do a spatulated, tension-free anastomosis over a **double-J stent**. Excise or bypass the **aperistaltic retrocaval segment**; respect and preserve the ureteric blood supply.
- **Laparoscopic or robotic** repair (transperitoneal or retroperitoneal) is now preferred: equal success and faster recovery.
- Nephrectomy if the kidney is non-functioning.
- A pyeloplasty that misses the diagnosis fails.
- **Also know:** retroiliac ureter (L5–S1, with vasal anomalies); mimics of extrinsic ureteric crossing: spermatic / gonadal or lumbar vein, psoas (iliopsoas) tendon.

**Diagram to draw:** right ureter hooking behind the IVC with a fishhook pelvis.

### Clinical implications of IVC anomalies

#### 1. Diagnosis

- Venous-phase CT or MRI. A left or duplicated IVC mimics **para-aortic lymphadenopathy**; a large azygos mimics a mediastinal mass, risking needless biopsy.
- Retrocaval ureter: S-shaped or fishhook ureter at L3–L4 on retrograde pyelography; CT urography confirms; diuretic renography grades obstruction.

#### 2. RPLND (retroperitoneal lymph node dissection)

- Preoperative CT must look for circumaortic or retroaortic veins and retrocaval ureter. A left or duplicated IVC lies in the left template; a retroaortic limb tears when lumbar vessels are divided.

#### 3. Radical nephrectomy

- The hidden posterior limb of a circumaortic vein bleeds when the artery is approached from behind: ligate both limbs. Caval thrombus in a duplicated IVC needs rerouting of the left IVC.

#### 4. Living donor nephrectomy

- The left kidney is preferred for its long vein. A circumaortic or retroaortic vein shortens it and adds bleeding risk; CT angiography is mandatory, and the right kidney may be chosen.

#### 5. Retrocaval ureter repair

- As above: observe non-obstructed type II; for obstruction, divide, transpose in front of the IVC and re-anastomose over a stent, excising the aperistaltic segment.

#### 6. Interventional and vascular

- **IVC filter:** with duplication, emboli bypass a single filter, so filter both cavae or place the filter suprarenally.
- Aortic surgery risks retroaortic vein injury. A retroaortic vein can cause haematuria or left varicocele.
- **Also know:** variable left adrenal vein entry (relevant to adrenal vein sampling); right gonadal vein drains into the right renal vein in up to 10%.

**Key points to remember:**

- Infrarenal IVC = right supracardinal; suprarenal = right subcardinal; hepatic = vitelline.
- Circumaortic left renal vein is commonest (2–16%): ligate both limbs.
- Retrocaval ureter: venous anomaly (persistent subcardinal vein); right side only; 1 in 1500; male 3:1; transpose anterior to IVC.
- Type I fishhook obstructs; type II sickle often does not.
- CT urography; anterior transposition and ureteroureterostomy.
- Duplicated IVC: filter both cavae; do not mistake it for nodes.

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## SAQ Retroileal Ureter

Paper answer · P2

**Type:** SAQ (10 marks)
**Source:** 2025-Winter Paper 2, Q3b

*"Retroileal" is taken as **retroiliac ureter**, the nearest real entity; retrocaval ureter is compared below (full answer: 2017-Winter Paper 3, Q3d).*

### Definition

- The ureter passes **behind the common iliac artery** and is compressed there.
- Better called "**preureteral iliac artery**": the anomaly is presumed vascular; the ureter is trapped.

### Embryology

- Normally the **ventral root of the umbilical artery** is replaced by a more **dorsal** branch, which lies behind the ureter.
- If the **ventral root persists** and the dorsal root fails, the ureter is trapped behind the artery.

### Features

- **Very rare**; either side; **bilateral** cases reported.
- Obstruction at **L5–S1**.
- **Associated anomalies are common**, especially **vasal and mesonephric duct ectopia**: ureter and vas draining together into the posterior urethra; vasa ending in the ureters; ectopic vaginal ureter.

### Presentation and diagnosis

- Flank pain, UTI (urinary tract infection) or hydroureteronephrosis down to the iliac crossing.
- **CT urography (computed tomography urography) with arterial phase** or MRI (magnetic resonance imaging) shows ureter and artery together; **diuretic renogram** for function and drainage.
- Caution: a vessel indenting an already dilated ureter may not be the cause.

### Management

- Non-obstructed: observe.
- Obstructed: **divide the ureter, re-route it in front of the artery** and do a spatulated ureteroureterostomy (or reimplantation) over a double-J stent; open, laparoscopic or robotic.

### Comparison with retrocaval ureter

| | Retroiliac | Retrocaval (circumcaval) |
|---|---|---|
| Vessel | Common iliac artery | IVC (inferior vena cava) |
| Embryology | Persistent ventral umbilical artery root | Persistent subcardinal vein |
| Side | Either, may be bilateral | Right |
| Level | L5–S1 | L3–L4, "fishhook" |
| Associations | Vasal/mesonephric ectopia | Horseshoe kidney, left renal anomalies |
| - **Also know:** distal ureter obstruction by uterine, umbilical, obturator or hypogastric vessels. | | |

**Diagram to draw:** ureter behind the common iliac artery at L5–S1, dilated above.

**Key points to remember:**

- Retroiliac = preureteral iliac artery; persistent ventral umbilical artery root.
- Obstruction at L5–S1; either side or bilateral.
- Look for vasal and ureteric ectopia.
- Retrocaval: right, L3–L4, fishhook, 1 in 1500.

**Also, from the condensed book (not in the paper answer):**

- A retrocaval ureter is always right-sided and mimics UPJ obstruction, so a simple pyeloplasty fails.

# 53 Surgery of the Ureter in Children

## 1 · Pelviureteric junction (PUJ) obstruction — definition and aetiology

*See the paper answer **Congenital PUJ obstruction, failed pyeloplasty and salvage procedures** (Section 11, Ch 93).*

**Also, from the condensed book (not in the paper answer):**

- Paediatric hydronephrosis is a pelvic anteroposterior diameter (APD) of **12 mm**, but hydronephrosis is not obstruction.
- With a crossing vessel, **dismembering and transposing the ureter anterior to the vessel prevents recurrence**. For secondary PUJ obstruction from high-grade reflux, avoid reimplant and pyeloplasty at one sitting; stage them.
- For **lower-moiety PUJ obstruction in a duplex**, do ureteropyelostomy or pyeloplasty; if the moiety has ≤ **5–10%** function, do partial nephrectomy.

## 2 · Diagnosis and indications

*Decide on function trend and dilatation, not on washout alone.*

- Most are found on antenatal ultrasound (US) at ≈ 20 weeks.
- Neonatal renogram washout has false positives up to **44%**, so decide on the trend in differential renal function (DRF) and on dilatation, not on half-time (T½) alone.
- **Pyeloplasty indications**: increasing APD; low or falling DRF; breakthrough infection on prophylaxis; symptoms (pain; failure to thrive in infants; haematuria; rarely hypertension).

## 3 · Anderson–Hynes dismembered pyeloplasty

It is the **gold standard** for any access, with success ≈ **95%**. It preserves a crossing vessel, excises the adynamic segment and allows pelvic reduction.

**ALGORITHM**

1. Expose the PUJ and free the lower pole, looking for an accessory vessel.
2. Use bipolar diathermy sparingly and place stay sutures.
3. Decompress the pelvis with a 21-G needle, then dismember, with or without pelvic reduction.
4. **Spatulate laterally** into healthy ureter, keeping the stenotic segment as a "no-touch" handle.
5. Bring the apex of the spatulation to the inferior pelvis and make a tension-free, watertight 6-0/5-0 absorbable anastomosis.
6. Excise the stenotic segment last, irrigate out clot and cover with perinephric fat.

- For a long gap, mobilise the whole kidney. A drain is usually unnecessary.

| Alternative | Best indication | Point |
|---|---|---|
| Foley Y-V-plasty | High insertion, **no vessel** | V base on dependent medial pelvis, apex at PUJ |
| Spiral flap | Long proximal stricture, dependent PUJ | Length:width ≤ 3:1 |
| Vertical flap | Dependent PUJ, large box-shaped extrarenal pelvis | Flap length must match gap |

| Alternative | Best indication | Point |
|---|---|---|
| Vascular hitch (Hellström) | Anterior lower-pole vessel, **no intrinsic stenosis**: moderate hydronephrosis, no calyceal dilatation, good cortex and split function; peristalsing PUJ at surgery | Vessel fixed cranially in pelvic wall; no hypertension seen |
| Ureterocalicostomy | Failed repair, short scarred ureter, dilated lower calyx with thin cortex | Lower-pole nephrectomy exposes calyx; spatulated ureter joined over stent |

Use flaps only when a tension-free primary anastomosis is impossible. Non-dismembered repair is inferior.

> **KEY POINT**
>
> Every variant answers two questions: is there a crossing vessel (dismember), and is the gap too long for tension-free anastomosis (flap, mobilisation)?

> **KEY POINT**
>
> One fixed principle, many routes: dismember, excise, spatulate, anastomose without tension. Failure is managed by repeating the principle.

## 4 · Access

- In **infants**, open surgery is preferred; **posterior lumbotomy** suits only infants and toddlers, while the flank approach suits any age.
- Laparoscopic results equal open, and robotic surgery has a shorter learning curve.

| | Transperitoneal | Retroperitoneal |
|---|---|---|
| For | Easier landmarks; left transmesocolic | No bowel injury; leak extraperitoneal; shorter stay |
| Against | Bowel risk; urinary ascites causing ileus | Learning curve; small space |
| Note | Preferred for horseshoe kidney, retrocaval ureter | One randomised trial favoured it; table **not flexed** |

- **Endopyelotomy** does poorly long-term under 18 years; its niche is **failed pyeloplasty** without a vessel, with a double-J stent for 6 weeks.

## 5 · Stents, outcomes, failure

| Stent | Removal | Point |
|---|---|---|
| Internal double-J | Second anaesthetic unless string/magnet | Catheter 24–48 h; string gives more febrile UTI (girls) |
| Externalised nephroureteric | 7–14 days, no anaesthesia | Fewer complications, shorter stay in infants |
| Stentless | — | Needs flawless technique |

- Stent duration of 1–4 weeks does not change outcome.
- For emergency diversion before repair, use a **nephrostomy, not a retrograde stent**, because a stent inflames the PUJ.
- Success is 90–100%; the commonest reason for conversion is bleeding.

*See the paper answer **Congenital PUJ obstruction, failed pyeloplasty and salvage procedures** (Section 11, Ch 93).*

## 6 · Primary megaureter

*Most resolve; the challenge is spotting the truly obstructed.*

### SAQ Primary obst. megaureter.

Paper answer · P3

**Type:** SAQ (10 marks)
**Source:** Practice Set 2 Paper 3, Q3f

*Assumed meaning: primary obstructive megaureter (POM).*

#### Definition

- **Megaureter:** ureter diameter **≥7 mm** (normal up to 16 years is 5.0–6.5 mm). It is a descriptive term.
- **Classification:** primary (intrinsic) or secondary (neurogenic bladder, outlet obstruction); and **obstructed, refluxing, non-refluxing non-obstructed, or refluxing and obstructed**.
- **POM:** an **adynamic juxtavesical segment** at the UVJ (ureterovesical junction) blocks peristalsis and causes functional obstruction above.

#### Pathogenesis

- Ureteric muscle matures cranially to caudally; the juxtavesical segment matures last, even after birth. This explains spontaneous resolution.
- Excess collagen, thick muscle sleeve, abnormal interstitial cells of Cajal.

#### Clinical features

- 10–23% of antenatal upper-tract dilatation. Boys, **left side**; **bilateral 25%**; contralateral dysplasia or obstruction 10–15%.
- Mostly asymptomatic; or UTI (urinary tract infection), pain, haematuria, stones.

#### Evaluation

- **Ultrasound:** hydroureteronephrosis.
- **VCUG (voiding cystourethrogram):** excludes reflux and PUV (posterior urethral valves).
- **MAG3 (mercaptoacetyltriglycine) renogram:** split function; washout T½ unreliable (false positive up to 44%).
- MRU (magnetic resonance urography) for anatomy.

#### Management

##### Conservative (most)

- CAP (continuous antibiotic prophylaxis) in infants; serial ultrasound and renograms; most resolve.
- Diameter <10 mm: virtually never needs surgery; ≥10 mm: a substantial minority do.
- Follow to early adulthood.

#### Indications for surgery

- Recurrent febrile UTI; progressive dilatation; **DRF (differential renal function) <40%**; **fall in DRF ≥5%**.

**Infant (small bladder, huge ureter): temporise**

- Cutaneous ureterostomy; refluxing reimplant; nephrostomy; **endoscopic balloon dilatation with double-J** for 2–6 months (70–80% success; complications up to a third).

**Definitive repair (child)**

- Excise stenotic segment → straighten → **taper** → **non-refluxing reimplant at 5:1** (Cohen or Politano–Leadbetter; extravesical for huge ureters). Stent afterwards.

| Tailoring | Method | Use |
|---|---|---|
| Starr plication | Imbricate lateral wall | Moderate (<1.75 cm) |
| Kalicinski folding | Exclude and fold lumen | Moderate |
| **Hendren excision** | Excise lateral wall | Massive; risk of ischaemia |

- Success over 90%.
- **Complications:** obstruction (stent first), VUR (vesicoureteral reflux; observe or inject), persistent dilatation.

**Diagram to draw:** dilated ureter above narrow adynamic segment; tapered reimplant.

**Key points to remember:**

- Megaureter ≥7 mm; POM = adynamic juxtavesical segment.
- Surgery: UTIs, progression, DRF <40% or fall ≥5%.
- Excise, taper, reimplant 5:1; success >90%.

**Also, from the condensed book (not in the paper answer):**

- In infancy the small bladder cannot take a safe tunnel for a massive ureter, so temporise with a **distal loop cutaneous ureterostomy** or a **refluxing reimplant**; balloon dilatation has weak evidence.
- Follow into adulthood, since some worsen at puberty.

**KEY POINT**

Why not rely on the renogram? Neonatal washout depends on immature function, hydration and a huge ureter's capacity, so serial DRF, diameter and infection history drive surgery. Megaureter is the PUJ problem moved to the bottom of the ureter: an immature adynamic segment that usually matures.

**ALGORITHM**

1. Excise the stenotic distal segment and straighten the ureter.
2. Taper if needed.
3. Reimplant non-refluxing at a **5:1** tunnel length:diameter.
4. Stent with a feeding tube for 7 days or a double-J for 4 weeks.

- Plication and folding preserve the blood supply; Hendren excisional tapering risks ischaemic stricture.
- Treat postoperative obstruction with a stent first; an ischaemic stricture needs redo.

## 7 · Surgery for vesicoureteral reflux

**KEY POINT**

**Principles**: exclude or treat bladder causes first; handle ureter and bladder gently; mobilise without tension; keep a tunnel length:diameter of **5:1** (Paquin); ensure a straight course with the bladder empty and full.

- Techniques are intravesical or extravesical, with the tunnel suprahiatal or infrahiatal.

| Technique | Approach / tunnel | Strength | Weakness |
|---|---|---|---|
| Politano–Leadbetter | Intravesical; suprahiatal neo-hiatus, vertical tunnel | Long tunnel for high grade | Blind neo-hiatus (bowel, vas, vagina); kink if too lateral/anterior |
| Glenn–Anderson | Intravesical; infrahiatal toward bladder neck | Least kinking | Short tunnel |
| **Cohen** | Intravesical; cross-trigonal | Commonest; success **98%**; thick bladder; with bladder-neck reconstruction | Difficult later retrograde catheterisation |

| Technique | Approach / tunnel | Strength | Weakness |
|---|---|---|---|
| **Lich–Gregoir** | Extravesical detrusorotomy ≈ 5 cm | No haematuria/spasm; short catheter | Transient voiding dysfunction/retention up to **10%**, esp. bilateral |
| Robotic extravesical (RALUR-EV) | Y-shaped detrusorotomy 4–5 cm | Less pain, spasm, stay | Early series: leak, obstruction, ureteric injury |

### Cohen — key steps

**ALGORITHM**

1. Open the bladder vertically and put a feeding tube in the orifice as a handle.
2. Circumscribe the orifice and free the ureter, preserving its distal blood supply, until it reaches the contralateral wall without tension.
3. Make a submucosal tunnel across the trigone, with the new orifice above the contralateral one.
4. Suture mucosa to mucosa with 6-0, with a detrusor hitch at 6 o'clock.

### Lich–Gregoir — key steps

**ALGORITHM**

1. With the bladder one-third full, divide the obliterated umbilical artery.
2. Free the ureter **hugging it**, only as far as the UVJ.
3. Distend the bladder and mark a 5-cm tunnel; incise the detrusor down to mucosa and raise wide flaps.
4. Lay the ureter in and close the detrusor caudocranially, catching adventitia at the apex.
5. Leave a Foley catheter for 12–48 h.

**KEY POINT**

Why voiding dysfunction? The pelvic plexus lies ≈ 1.5–2 cm dorsomedial to the UVJ, so stay on the ureter and stop at the UVJ.

- In the American Urological Association (AUA) meta-analysis, open success is **95.9%**; minimally invasive results are converging.
- **Follow-up** is US at 6–12 weeks. A VCUG is needed for high grade, new hydronephrosis, UTI or a dysfunctional bladder.

### Complications of reimplantation

| | Early | Long-term |
|---|---|---|
| Reflux | Ipsilateral low grade usually resolves by 1 year. Contralateral ≈ 1 in 10 after unilateral repair, mostly resolves, so observe; intervene for pyelonephritis. Routine bilateral repair not justified | Short tunnel/untapered ureter, or **unrecognised secondary reflux** (neurogenic/valve bladder): treat the bladder first (anticholinergic and clean intermittent catheterisation) |
| Obstruction | Oedema, clot, haematoma, kink within 2 weeks: double-J or nephrostomy; most resolve | Ischaemia, angulation (hiatus too lateral/anterior), tight tunnel, stenosis: balloon and stent, then re-reimplant |

- Contralateral reflux risk rises with age < 6 years, capacity < 50% of expected and a refluxing duplex.
- At **reoperation**, excise ischaemic ureter back to healthy tissue; for a short ureter use a **psoas hitch**, Boari flap or transureteroureterostomy.

**KEY POINT**

Early complications are tissue reactions, so drain and wait. Late ones are design errors (blood supply, hiatus position, tunnel ratio) or an untreated bladder.

### Endoscopic injection

*See the paper answer **Primary VUR: VUJ embryology/anatomy, pathophysiology, management (incl. bilateral, adult grade 3), reimplantation techniques and STING** (Section 4, Ch 41).*

**Also, from the condensed book (not in the paper answer):**

- After failure, repeat the injection once or twice, then reimplant; the encapsulated implant does not make reimplantation harder.

## 8 · Other ureteric lesions

- † **Retrocaval ureter** is right-sided with a fish-hook shape; re-anastomose it in front of the cava.
- A **congenital mid-ureteric stricture** is often misread as megaureter; excise it and do a ureteroureterostomy.
- A **fibroepithelial polyp** (upper third, male, left) causes pain and haematuria; remove it endoscopically.
- **Retroperitoneal fibrosis** is treated with steroids, then azathioprine, then a stent, and finally ureterolysis.

# 56 Hypospadias

## 1 · Definition, diagnosis and embryology

- The **triad** is an ectopic ventral meatus, ventral curvature and a dorsal hooded foreskin (the last two not always present). Incidence is ≈ **1 in 150–300** male births.
- **Megameatus with intact prepuce (MIP)**, ≈ 5%, hides under a normal foreskin, so retract before newborn circumcision. Avoid circumcision when curvature or ventral skin deficiency is present.

### LAQ Proximal (perineoscrotal) hypospadias: penile embryology and management including Koyanagi repair

Paper answer · P2 · P3 · merged from 2

*Asked as:* P3 LAQ "Embryology of the penis and management of proximal hypospadias (including Koyanagi repair)" · P2 SAQ "Perineo-scrotal hypospadias"

#### Embryology of the penis

##### Indifferent stage (up to 8–9 weeks)

- Cloacal folds form the **genital tubercle**; after the cloaca divides (7 weeks), **urethral folds** flank the urogenital membrane, with **labioscrotal swellings** laterally. Both sexes look identical until about 9 weeks.

##### Hormonal control

- SRY (sex-determining region Y) drives the testis, whose **Leydig cells (about 8 weeks)** secrete testosterone.
- **Type 2 5α-reductase** in the tubercle converts it to **DHT (dihydrotestosterone)**, which acts on androgen receptors. **DHT at 8–12 weeks** is the key mediator; the critical window is weeks 8–16. Without androgen action the default is female.
- First sign: longer anogenital distance, then tubercle elongation (sonic hedgehog-driven).

| Primordium (germ layer) | Male | Female |
|---|---|---|
| Genital tubercle (mesoderm: corpora; ectoderm: skin, prepuce) | Glans, corpora | Clitoris |
| Urethral folds | Fuse to form penile urethra, median raphe | Labia minora |
| Labioscrotal swellings | Scrotum | Labia majora |
| Urethral plate (endoderm) | Whole urethra, including glanular | Vestibule |

#### Urethral formation: the "double zipper"

1. **Opening zipper** (both sexes): the urethral plate canalises proximal to distal into a urethral groove.
2. **Closing zipper** (androgen-dependent, male only): the folds fuse in the ventral midline, forming the tubular urethra and raphe. - The **glanular urethra** forms by direct canalisation of the endodermal plate; Glenister's ectodermal ingrowth theory (1954) is outdated. - Timetable: coronal sulcus 11–12 weeks; shaft fused by 16 weeks; complete by about 18 weeks; fetal ventral curvature gone by 20 weeks. - **Prepuce** fuses dorsally first; ventral

fusion awaits urethral closure, hence the **dorsal hood** in hypospadias. - Dorsal nerves run at 11 and 1 o'clock, leaving a **12 o'clock nerve-free zone** (safe plication site). - **Hypospadias = arrest of the closing zipper**: the meatus stops where closure stopped, and unfused dysplastic ventral tissue tethers the shaft into curvature.

**Diagram to draw:** double-zipper urethral formation.

### Definition of proximal (perineoscrotal) hypospadias

- Meatus on the proximal shaft, at the penoscrotal junction, scrotum or perineum (10–25% of hypospadias).
- Perineoscrotal hypospadias is a severe form, with severe ventral curvature, a short penis, bifid scrotum and often **penoscrotal transposition**.
- Cause: failed androgen-dependent urethral fold fusion (8–16 weeks).

### Evaluation

- Severity is judged **after degloving**: meatus, plate and curvature (GMS (glans–meatus–shaft) score).
- Examine stretched penile length, glans width, penoscrotal transposition, bifid scrotum and **both testes**.
- **DSD (disorder of sex development) work-up** if hypospadias plus an undescended testis (especially non-palpable), perineal meatus or ambiguity: karyotype, ultrasound for Müllerian remnants, testosterone, LH (luteinising hormone), FSH (follicle-stimulating hormone), AMH (anti-Müllerian hormone).
- DSD risk is 64% in proximal versus 7% in distal cases. A 46,XY karyotype can still hide 5α-reductase deficiency or androgen insensitivity.
- Prostatic utricle is commoner; cystoscopy in severe cases. No routine VCUG (voiding cystourethrogram) or renal ultrasound.

### Preparation

- **Timing:** 6–18 months (AAP (American Academy of Pediatrics): 6–12 months).
- **Preoperative androgen** if glans width <14–15 mm: e.g. testosterone 2 mg/kg IM (intramuscular) at 5 and 2 weeks before surgery (about 4 mm glans gain). Side effects are transient; evidence is mixed, but two RCTs (randomised controlled trials) showed fewer complications.

### Curvature correction (decides one- versus two-stage)

```
Deglove + excise dysplastic dartos + artificial erection ├─ straight → urethroplasty ├─ residual <30° → dorsal midline plication (12 o'clock) └─ >30° → divide urethral plate └─ still >30° → ventral corporotomy ± graft → staged repair
```

| | **Dorsal plication (Nesbit, Baskin)** | **Ventral lengthening** |
|---|---|---|
| Method | Shortens the convex dorsal side | 1–3 transverse corporotomies, or one full-thickness 9-to-3 o'clock corporotomy with dermis, tunica vaginalis or SIS (small intestinal submucosa) graft |
| Use | Mild (≤15–30°) | Moderate–severe |
| Recurrence | 28% | 9% |

**Diagram to draw:** curvature algorithm.

### Urethroplasty options

| Approach | Technique | Comment |
|---|---|---|
| One-stage **proximal TIP (tubularised incised plate)** | Plate kept, incised and tubularised | Recurrent curvature, stricture: largely abandoned for severe cases |
| One-stage **onlay island flap** | Inner preputial flap on preserved plate | Diverticulum, fistula; higher complications |
| One-stage **TPIF (transverse preputial island flap, Duckett tube)** | Tube after plate division | High complications |
| One-stage **Koyanagi** | Meatus-based parameatal foreskin flap (see below) | No proximal circular anastomosis |
| **Two-stage Bracka / STAG (staged tubularised autograft)** | Stage 1: correct curvature, divide plate, quilt inner-prepuce or **buccal graft**. Stage 2 (≥6 months): tubularise, glansplasty, barrier layer | **Preferred for most proximal cases**; STAG lowest complications (about 23%) |
| **Two-stage Byars** | Dorsal preputial flaps moved ventrally, tubularised later | Supple flap bed; complications 43–81% |

- **Why stage?** A urethroplasty laid over a fresh corporal graft fails more often; staging gives a supple, vascular bed.
- Principles: magnification, tension-free closure, offset suture lines, **vascularised barrier layer** (dartos, tunica vaginalis), fine absorbable sutures, no hair-bearing skin. Stent, compressive dressing, oxybutynin; **scrotoplasty** for transposition.

**Diagram to draw:** two-stage graft repair: stage 1 graft bed, stage 2 tubularisation.

### Koyanagi repair: definition and principle

- **One-stage repair of proximal hypospadias** with a **meatus-based parameatal foreskin flap** (Koyanagi). Skin strips round the meatus continue into the inner prepuce on both sides; they are swung ventrally and tubularised.
- **Principle:** the neourethra stays attached to the native meatus, so there is **no circular proximal anastomosis**, and the urethral plate can be divided to correct severe chordee.

### Koyanagi repair: indications and contraindications

- **Indications:** proximal hypospadias (penoscrotal, scrotal, perineal) with **severe curvature** needing plate transection, and an **ample, unoperated prepuce**.
- **Contraindications:** prior circumcision or failed repair; small glans (give preoperative testosterone); curvature needing a ventral corporal graft (stage instead).

### Koyanagi repair: procedure

`Stay suture; artificial erection → incise round the meatus, along both sides of the plate to the corona and round the inner prepuce (meatus-based "wings") → deglove; divide the plate; excise dysplastic dartos → residual curvature <30° → dorsal midline plication → split the dorsal prepuce; bring both wings ventrally on dartos pedicles → join inner edges, then outer edges, over a catheter → tube`

→ glanuloplasty, meatus at the tip → dartos barrier layer; skin cover ± scrotoplasty → stent 7–10 days; compressive dressing

- Modified and "neo-modified" versions add flap fixation and barrier layers to reduce fistula and diverticulum.

**Diagram to draw (Koyanagi):** meatus-based foreskin wings split dorsally and joined ventrally into a tube.

### Koyanagi repair: complications

| Complication | Cause |
|---|---|
| **Fistula** (commonest) | Long suture line; repair ≥6 months later |
| Meatal stenosis, stricture | Distal flap ischaemia |
| **Diverticulum** | No spongiosum; flap not fixed to the shaft |
| Glans dehiscence | Tension |
| Residual or recurrent curvature | Plication alone in severe cases |
| Flap necrosis, infection, poor cosmesis | Pedicle injury |

- Because proximal repairs overall have **23–68%** complications, many now prefer **two-stage repair** (STAG, Bracka).

### Complications and outcomes of proximal repair

- Complications **23–68%**; 25–75% need further surgery after planned staging.
- **Fistula** is the commonest (3–45%; repair ≥6 months later; exclude distal obstruction).
- Others: glans dehiscence, meatal stenosis (<8 Fr before puberty), stricture, **diverticulum** (no spongiosum), **recurrent curvature >30°**, BXO (balanitis xerotica obliterans), hair-bearing urethra.
- A long, rigid neourethra raises resistance (Poiseuille's law), causing fistula and diverticulum.

### Follow-up

- Follow through puberty; patient-reported outcomes (PPS (Penile Perception Score), HOPE (Hypospadias Objective Penile Evaluation) score).

> **Key points to remember:**
> - DHT at 8–12 weeks; opening then closing zipper; 12 o'clock nerve-free zone.
> - Hypospadias (including perineoscrotal) plus undescended testis: DSD work-up.
> - <30° plication; >30° divide plate ± ventral graft and stage.
> - STAG/Bracka staged repair preferred; fistula commonest complication.
> - Koyanagi: one stage; meatus-based parameatal foreskin flap; plate divided; no proximal circular anastomosis.
> - Koyanagi: fistula commonest; diverticulum, meatal stenosis.

**Also, from the condensed book (not in the paper answer):**

- The critical window is **weeks 8–16**. Ventral prepuce fusion awaits urethral fusion, so the foreskin is incomplete in **95%**. MCQ

> **KEY POINT**
>
> Hypospadias is an arrest of the androgen-dependent closing zipper: the meatus stops where closure stopped, the prepuce cannot close over an open urethra, and dysplastic ventral tissue tethers the shaft into curvature.

## 2 · Aetiology

It is **multifactorial**, but all causes mean too little androgen action in weeks 8–16.

- Placental insufficiency and growth restriction double the risk; assisted reproduction is an independent risk.
- Twin concordance is 9–27%, and chromosomal abnormality is found in up to 30%; about 200 syndromes include it. Endocrine disruptors are unproven in humans.
- It belongs to **testicular dysgenesis syndrome**, and a shorter anogenital distance (AGD) means more severe disease.

## 3 · Classification and severity

- Distal forms are **70–85%**. The meatus is the least reliable marker, so **classify after degloving** by meatus, curvature and plate.

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## LAQ Hypospadias: present concepts, single-stage vs staged repairs, and hypospadias fistula

Paper answer · P1 · P4 · merged from 2

*Asked as:* P4 LAQ "Hypospadias: present concepts in management, and hypospadias fistula" · P1 SAQ "Hypospadias repairs – single-stage versus staged techniques"

### Introduction

- Read "present concepts" as the present (current) concepts in the management of hypospadias.
- Hypospadias is an arrest of the androgen-dependent "closing zipper" of urethral fold fusion (weeks 8–16). Triad: ventral ectopic meatus, ventral curvature, dorsal hooded prepuce.
- 1 in 150–300 male births; distal 70–85%, proximal 10–25%.
- **Goals:** a straight erection, a urethra to the tip (standing voiding, semen delivery) and acceptable appearance, with the fewest operations and a result lasting into adulthood.
- The choice of repair depends on **curvature after degloving**, plate quality and glans size, not on meatal position alone.

### Aetiology and associations

- **Aetiology:** causes converge on deficient androgen action (placental insufficiency, assisted reproduction, genes, endocrine disruptors).
- Associations: hernia or hydrocele 9–16%, cryptorchidism about 7% (10% in proximal), enlarged utricle.

### Evaluation

- **DSD (disorder of sex development) work-up** if hypospadias plus undescended testis (especially non-palpable), perineal meatus or ambiguity: karyotype, ultrasound for Müllerian remnants, testosterone, LH (luteinising hormone), FSH (follicle-stimulating hormone), AMH (anti-Müllerian hormone).

- No routine VCUG (voiding cystourethrogram) or renal ultrasound.

### Classification: severity is judged in theatre, not by the meatus

- A "distal" meatus may hide absent spongiosum and severe curvature, so classify **after degloving**.

| GMS (glans–meatus–shaft) score | Glans | Meatus | Shaft curvature |
|---|---|---|---|
| 1 | Good size, deep groove | Glanular | None |
| 2 | Adequate, grooved | Coronal | <30° |
| 3 | Small; narrow or flat plate | Mid or distal shaft | 30–60° |
| 4 | Very small; flat plate | Proximal shaft, penoscrotal | >60° |

- Total 3–12; a higher score predicts more complications.
- **POST (plate objective scoring tool):** a low ratio of neomeatus length to glans fusion length means a poor plate, which favours an inlay graft over plain TIP (tubularised incised plate).

### Timing and preparation

- Repair at **6–18 months** (AAP (American Academy of Pediatrics): 6–12 months), before genital awareness at about 18 months. One short anaesthetic appears low-risk.
- A straight penis that voids standing may gain little from surgery: counsel honestly.
- **Preoperative androgen** if glans width <14–15 mm: testosterone 2 mg/kg IM (intramuscular) at 5 and 2 weeks before surgery (about 4 mm glans gain). Evidence mixed; two RCTs (randomised controlled trials) showed fewer complications.

### Curvature decides the operation

```
Deglove + excise dysplastic dartos + artificial erection ├─ straight → urethroplasty (single-stage) ├─ residual <30° → dorsal midline plication (12 o'clock nerve-free zone), single-stage └─ >30° → divide plate → still >30° → ventral corporotomy ± graft → staged repair
```

- Recurrence 28% after plication versus 9% after grafting. Under-correction causes repeated surgery, so correct fully first time.

### Choice of technique

| Situation | Current choice |
|---|---|
| Glanular meatus, no curvature | Advancement: MAGPI (meatal advancement and glanuloplasty), MIV (M inverted V) glansplasty; selectively |
| Distal or midshaft, good plate | **TIP (Snodgrass)**: >90% of distal, about 80% of midshaft repairs; Thiersch–Duplay; GAP (glans approximation procedure) for a deep groove |
| Narrow or shallow plate | TIP with **dorsal inlay graft**; onlay island flap; Mathieu (SLAM modification) |
| Proximal, curvature >30° | **Two-stage**: STAG (staged tubularised autograft) or Bracka graft; Byars flaps; distal-first (Ulaanbaatar) |
| Proximal, curvature corrected, good tissue | One-stage onlay or TPIF (transverse preputial island flap); proximal TIP abandoned |

### Single-stage techniques

| Group | Examples | Indication |
|---|---|---|
| Advancement | **MAGPI** (Duckett 1981); MIV | Glanular meatus, no curvature |
| Tubularisation | **TIP** (Snodgrass 1994); Thiersch–Duplay; GAP | Distal–midshaft; TIP in >90% of distal |
| Flap | Mathieu (flip-flap); **OIF** (onlay island flap) | Thin plate |
| Proximal one-stage | TPIF (Duckett tube); proximal OIF | Curvature corrected; good tissue |
| Graft | Dorsal buccal inlay | Narrow plate |

- Pros: one anaesthetic.
- Proximal TIP is abandoned: devascularised plate, stricture, recurrent curvature.

### Staged techniques

- Now the approach of choice for most proximal hypospadias.
- **Rationale (why stage):** a urethroplasty laid over a fresh corporal graft fails more often; stage 1 builds a supple, vascular bed. Stage 2 after **≥6 months**.

| Type | Stage 1 | Stage 2 (≥6 months) |
|---|---|---|
| **Bracka / STAG** (staged tubularised autograft) | Correct curvature, divide plate, quilt inner preputial or buccal graft | Tubularise, glansplasty, tunica vaginalis cover |
| **Byars flaps** | Split dorsal prepuce, transpose ventrally | Tubularise |
| Distal-first (Ulaanbaatar) | Curvature and distal urethra | Join proximal urethra |

- Repeat the artificial erection before stage 2.

### Single-stage versus staged: comparison of outcomes

| Measure | Single-stage | Staged |
|---|---|---|
| Distal success | 83–95% | – |
| Proximal complications | About 60% (long TIP, mixed) | STAG about 23% (lowest); Byars 43–81% |
| Recurrent curvature | Higher after plication | 0–10% |
| Penile length, perception | Lower | Better |
| Extra procedures | Fewer planned | 25–75% need more |

### Operative principles: protect blood supply

- Magnification, tension-free closure, offset suture lines, **vascularised barrier layer** (dartos, tunica vaginalis), rapidly absorbed fine sutures, no hair-bearing skin, haemostasis.
- Postoperative: stent 3–21 days by complexity, compressive dressing, oxybutynin; antibiotic prophylaxis debated.

### Complications and outcomes (standardised)

- Definitions: glans dehiscence (fusion <2 mm); meatal stenosis (symptoms plus <8 Fr before puberty, <12 Fr after); recurrent curvature >30°.
- TIP meta-analysis (Pfistermuller 2015): complications 10.6%, fistula 5.7%. Proximal complications 23–68%; STAG about 23% (lowest); Byars 43–81%.
- Fistula is the commonest complication. A long, rigid, spongiosum-free neourethra raises resistance (Poiseuille's law), causing fistula and diverticulum.

- **Also know:** BXO (balanitis xerotica obliterans): excise and stage with buccal graft; redo repairs use buccal (lip, inside glans) mucosa, Cecil scrotal parasitisation, tissue expansion.

### Hypospadias fistula: definition and incidence

- A UCF (urethrocutaneous fistula) is a urethral leak anywhere below the meatus: the **commonest complication** of hypospadias repair, just under 10% at short-term follow-up.
- After TIP: 5.7% (**Pfistermuller 2015**); after proximal repair 3–45%. Usually at the original meatus or coronal margin.

### Hypospadias fistula: aetiology

- **Distal obstruction** (meatal stenosis, stricture): raises voiding pressure and turbulence; the **key correctable cause**.
- **Ischaemia**, oedema, infection, haematoma, tension.
- **Technical:** overlapping suture lines, poorly inverted epithelium, slowly absorbed sutures, **no barrier layer**.
- **Patient:** proximal meatus, glans <14 mm, redo surgery. A long, rigid neourethra raises resistance (Poiseuille's law).

### Hypospadias fistula: prevention

- **Vascularised barrier** (dartos; **tunica vaginalis** in proximal or redo), offset suture lines, stent in complex repairs.

### Hypospadias fistula: evaluation

- Split stream / multiple streams, usually within 2 years.
- Under anaesthesia: **bougie calibration** for distal narrowing, **urethroscopy** if irregular, and a **stress test** (compress proximally and distally, inject saline or methylene blue) for hidden tracts.

### Hypospadias fistula: management

- Rarely closes spontaneously. **Wait 6–12 months**; **correct distal obstruction at the same sitting**.

| Fistula | Repair |
|---|---|
| Small, shaft | Excise the epithelialised tract, inverting subepithelial closure with fine absorbable suture, multiple offset layers, **dartos barrier** |
| Large or thin tissue | Trap-door or island skin flap plus second layer |
| Coronal margin (glans dehiscence) | Implies distal narrowing or poor tissue: **redo urethroplasty and glansplasty** |
| With diverticulum | Excise, or reuse the redundant wall as a flap |

- Recurrence about **20%**; reduce with interposed vascularised layers (dartos, tunica vaginalis, Cecil modification when tissue is scarce).
- Catheter: none for simple repairs, 7–14 days for complex.
- **Also know:** preputial fistula does not involve the urethra; congenital urethral fistula is closed the same way; follow through puberty.

```
UCF → wait 6–12 months → calibrate, urethroscopy, stress test ├─ distal obstruction → correct at same sitting ├─ small → excise + layered closure + barrier flap └─ large/coronal → flap repair or redo urethroplasty
```

### Follow-up and patient-reported outcomes

- About three-quarters of complications appear by 3 years; stricture and curvature appear late. Follow proximal cases **through puberty**.
- Tools: HOSE (Hypospadias Objective Scoring Evaluation, surgeon), PPS (Penile Perception Score), HOPE (Hypospadias Objective Penile Evaluation, photo panels). Decision aids reduce parental regret.

### Recent advances

- Registries, AI (artificial intelligence) video review, genomic risk profiling, a quality-of-life measure, engineered tissue.

**Diagrams to draw:** curvature algorithm; STAG stage 1 graft and stage 2 tubularisation; fistula repair showing neourethra, dartos barrier and skin with offset suture lines.

> **Key points to remember:**
>
> - Classify after degloving (GMS); curvature, not meatus, decides one versus two stages: curvature >30° after degloving means stage.
> - TIP for distal; staged graft repair (STAG/Bracka) for severe proximal; STAG has the lowest proximal complications.
> - Stage 2 after ≥6 months.
> - Hypospadias plus undescended testis: DSD work-up.
> - Follow through puberty; use patient-reported outcomes.
> - Fistula: commonest complication, just under 10%; TIP 5.7%.
> - Exclude distal obstruction; wait 6–12 months.
> - Layered closure with barrier flap; recurrence about 20%.

**Also, from the condensed book (not in the paper answer):**

- The glans–meatus–shaft (GMS) score (Merriman 2013) is scored in theatre after degloving.

## 4 · Associated anomalies and DSD work-up

- Hernia or hydrocele occurs in 9–16%, and undescended testis (UDT) in ≈ 7%.
- A **DSD work-up** is needed for UDT with proximal hypospadias, a perineal meatus or atypical genitalia. DSD risk is ≈ **50%** with a non-palpable vs 15% with a palpable testis, and 64% in proximal vs 7% in distal cases. Since 2006 severe hypospadias is classed as 46,XY DSD.
- Do ultrasound for Müllerian remnants, karyotype, testosterone, FSH, LH and AMH; a 46,XY karyotype does not exclude DSD (Ch 59).
- A large **prostatic utricle** causes stasis, UTI and difficult catheterisation. Routine VCUG and renal ultrasound are not needed.

> **KEY POINT**
>
> Hypospadias with cryptorchidism signals a systemic androgen problem, hence karyotype before surgery.

## 5 · Pre-operative assessment

- The goals are a straight erection, a urethra to the tip, good cosmesis and the fewest procedures. Exclude micropenis, and note that a deflected stream may justify surgery even when distal.
- **Timing**: the AAP advises 6–12 months, and the authors **6–18 months**, before genital awareness; younger repair has fewer complications.
- A glans width **≤14 mm** means more complications. Give testosterone cypionate **2 mg/kg intramuscularly at 5 and 2 weeks** before surgery if the glans is <15 mm, which gains about 4 mm. Two randomised trials showed fewer complications, but meta-analyses are inconclusive; side effects resolve within 6 months.

## 6 · Intra-operative principles

- Use a regional block, ketorolac 0.5 mg/kg, antibiotics, and a tourniquet or epinephrine. Cystoscopy is only for severe proximal or redo cases.
- Rapidly absorbed fine sutures beat polydioxanone, whose slow absorption favours fistulae. Close tension-free with non-overlapping lines, and **never use hair-bearing skin**.

## 7 · Curvature

*Curvature decides one-stage or staged repair.*

It comes from ventral skin tethering, dysplastic dartos, a short urethra and corporal disproportion.

**ALGORITHM**

1. Deglove to the penoscrotal junction and excise dysplastic dartos.
2. Make an **artificial erection**.
3. If straight, then proceed to urethroplasty.
4. If residual curvature is <30°, then do dorsal plication.
5. If >30°, then divide the urethral plate.
6. If still >30°, then do ventral corporal lengthening (usually staged).

- Mild is <30°, moderate 30–45° and severe ≥45°. Eyeballing underestimates curvature.
- Plication risks nerve injury, recurrence and shortening. Lengthening uses corporotomies, with or without a dermis or tunica vaginalis graft.
- Recurrence is 28% after plication vs 9% after grafting, and not dividing the plate is an independent risk, so correct fully the first time.

## 8 · Urethroplasty

Tubularised incised plate (TIP) repair is used for >90% of distal repairs.

| Technique | Indication | Key point / pitfall |
|---|---|---|
| **MAGPI** | Glanular, mobile meatus, no curvature | Used more proximally gives a retrusive meatus |
| Urethromeatoplasty | Stenotic distal meatus | Heineke–Mikulicz |
| GAP (glans approximation) | Wide deep groove (MIP) | Two-layer neourethra |

| Technique | Indication | Key point / pitfall |
|---|---|---|
| Thiersch–Duplay | Wide plate | Tubularise over 12 Fr |
| **TIP / Snodgrass** | Distal, midshaft; narrow plate; mild curvature | Midline relaxing incision; stop 2–3 mm short; meatal stenosis; inlay graft if poor plate |
| **Mathieu** (flip-flap) | Distal, shallow plate | Narrow base causes ischaemia, fistula; fewer stenoses than TIP |
| **Onlay island flap** | Plate kept but narrow | Diverticulum if too wide; less stenosis |
| **Duckett tube** | Proximal; plate divided | Higher complications; diverticulum |
| **Two-stage Bracka** | Proximal severe curvature, redo, poor plate | Stage 1 quilted preputial/buccal graft; stage 2 (≥6 months) tubularise |
| **Two-stage Byars** | Proximal severe curvature | Dorsal prepuce flaps; highest staged complications |

- Staged repair (STAG) has the lowest proximal complication rate (≈ 23%), because urethroplasty over a fresh corporotomy or graft fails. Repeat the artificial erection at stage 2. Proximal TIP has been abandoned because of devascularisation and recurrent curvature.

> **KEY POINT**
>
> Why proximal repairs fail (Poiseuille): resistance rises with length / radius⁴. A long, rigid tube ending in a non-distensible glans raises proximal pressure, causing fistula, diverticulum and stricture. 25–75% need extra procedures after planned two-stage repair.

- Glansplasty needs a tension-free **two-layer closure**, and a **barrier layer** (dartos or tunica vaginalis) reduces fistula.

## 9 · Post-operative care

- Leave a compressive dressing for ≈ **72 h**, and divert urine for **3–21 days** according to complexity. Short distal repairs need no stent. Give oxybutynin until the stent is out, with paracetamol and ibuprofen.

## 10 · Complications

| Complication | Standard definition |
|---|---|
| Fistula | Urethral leak below meatus |
| Glans dehiscence | Wing separation; glans fusion <2 mm |
| Meatal stenosis | Obstructive symptoms and **<8 Fr** prepubertal / <12 Fr postpubertal |
| Diverticulum | Segmental sacculation on voiding |
| Recurrent curvature | **>30°** on erection |

*See the paper answer **Hypospadias: present concepts, single-stage vs staged repairs, and hypospadias fistula** (Section 4, Ch 56).*

**Also, from the condensed book (not in the paper answer):**

- **Balanitis xerotica obliterans** needs complete excision and a staged graft. Recurrence is up to **40%**.
- **Meatal stenosis** is commoner after TIP than Mathieu and is treated by dilatation, then meatotomy or redo. A late **stricture** needs buccal urethroplasty, often staged.

> **KEY POINT**
>
> Nearly every complication traces to distal obstruction, ischaemia or tension, so repair begins by calibrating the distal urethra.

### Reoperation

- Excise scar widely and transfer tissue rather than doing "small operations".
- **Buccal mucosa** is the graft of choice. Make it 10% larger, then tubularise ≥6 months later; lip mucosa suits the glans. Always do cystoscopy and an artificial erection.
- The **Cecil–Culp** procedure buries the penis in the scrotum when skin is short.

## 11 · Outcomes and follow-up

- A TIP meta-analysis (4,675 boys) found complications in 10.6% and reoperation in 4.5%. Distal success is 83–95%, while proximal complications are 23–68%.
- About 75% of complications occur by 3 years, so follow proximal cases **through puberty**. Recurrent UTI needs imaging for diverticulum, stricture or utricle.
- Proximal cases have worse adult sexual outcomes, and penile length is the chief complaint.

# 57 Etiology, Diagnosis and Management of the Undescended Testis

## 1 · Definitions

*A retractile testis is watched; an undescended one is operated on.*

### LAQ Testicular descent and undescended testis: evaluation, treatment algorithm and orchidopexy methods

Paper answer · P1 · P2 · P3 · merged from 4

*Asked as:* P1 LAQ "Testicular development, gonadal descent, abnormalities of descent and bilateral undescended testis" (Write a note on the embryological descent of the gonads, explaining the various phases of descent. Outline the evaluation and management of bilateral undescended testis in a 3 year old child, Winter 2024 Paper I Q2; Describe developmental embryology of testes and its physiological descent. Mention various abnormalities associated with testicular descent, Practice Set 14 Paper I Q1) · P3 SAQ "Maldescended (undescended) testes and treatment algorithm" (Treatment algorithm of undescended testis, Summer 2018 Paper III Q3a; Maldescended testes, Practice Set 5 Paper III Q3b) · P2 SAQ "Undescended testis" · P1 SAQ "Methods of orchidopexy"

*Maldescent is taken as any testis not in a stable scrotal position: undescended, ectopic, retractile and acquired.*

#### Development of the testis

- **Indifferent gonad:** the gonadal (germinal) ridge forms at **4 weeks** on the medial urogenital ridge. It has four components: coelomic epithelium, underlying mesenchyme, adjacent mesonephros and **PGCs (primordial germ cells)**.
- PGCs arise in the **yolk sac wall in week 3** and migrate along the dorsal mesentery of the hindgut to the ridge (about 6 weeks).
- **SRY (sex-determining region Y)** on the short arm of Y, partly activated by SF1 (steroidogenic factor 1), switches on **SOX9** in pre-Sertoli cells. Sertoli cells then appear (about 40 days). Without SRY no Sertoli cells form and the gonad becomes an ovary.
- Sertoli cells enclose germ cells as **testicular cords (about 6 weeks)**, which canalise into seminiferous tubules.
- **Rete testis** forms with the cords and joins **5–12 mesonephric tubules**, which become efferent ductules. Full continuity (tubules, rete, efferent ductules, epididymis) is established by the **third month**.
- The testis turns round and a thick **tunica albuginea** forms. Testes become spherical after 7–8 weeks; ovaries stay oblong.

| Hormone | Cell | Action | Male remnants |
|---|---|---|---|
| **AMH (anti-Müllerian hormone)** | Sertoli | Müllerian regression, weeks 8–10 | Appendix testis, prostatic utricle |
| **Testosterone** | Leydig (from 8 weeks; driven by placental hCG (human chorionic gonadotrophin), later fetal LH (luteinising hormone)) | Wolffian duct to epididymis, vas, seminal vesicle (weeks 8–12) | Appendix epididymis, paradidymis |

| Hormone | Cell | Action | Male remnants |
|---|---|---|---|
| **DHT (dihydrotestosterone)** | Formed locally by 5α-reductase type 2 | External genitalia, prostate, scrotum (fused by 10–12 weeks) | – |
| **INSL3 (insulin-like peptide 3)** | Leydig | Gubernacular swelling (via RXFP2 receptor) | – |

### Gonadal descent: starting point

- At **7–8 weeks** the testis lies near the kidney (upper lumbar), held by the **cranial suspensory ligament (CSL)** above and the **gubernaculum** below. The gubernaculum appears at 7 weeks, attaches to the **epididymis** and runs to the future inguinal canal.
- **Female:** the CSL persists; the gubernaculum becomes the ovarian and round ligaments; the ovary descends only into the pelvis.

### Phases of testicular descent (classic two-phase model)

| | Transabdominal phase | Inguinoscrotal phase |
|---|---|---|
| Timing | Weeks 10–15 | From the 7th month; inguinal passage rarely before 22 weeks; most testes scrotal after 27 weeks |
| Events | **CSL regresses**; **gubernaculum swells** (outgrowth); its cord shortens, holding the testis at the internal ring while the kidney and abdomen grow away (the kidney moves cranially) | Gubernaculum migrates **3–5 cm** to the scrotum; **processus vaginalis** forms within it; the bulb then regresses; the processus closes |
| Control | CSL regression by **androgen**; swelling by **INSL3** from Leydig cells, via its receptor RXFP2 (occurs even in androgen resistance) | **Androgen**, mainly indirectly via the **genitofemoral nerve (GFN)** releasing **CGRP (calcitonin gene-related peptide)**; intra-abdominal pressure |
| Failure gives | Intra-abdominal testis | Canalicular or ectopic testis |

- **Modern view:** the "transabdominal descent" is largely relative, not true migration. Barteczko and Jacob describe **five phases** (5 weeks to 28 weeks). The gubernaculum **swells to make room** (outgrowth), then **shrinks to let the testis pass** (regression). Testosterone (peak 14–16 weeks) and INSL3 (15–17 weeks) act together on gubernacular and cremaster growth (myogenesis).
- Spontaneous postnatal descent occurs mainly in the first 3–6 months.
- **Also know:** AMH defects (persistent Müllerian duct syndrome) cause cryptorchidism; Insl3-knockout mice are cryptorchid; the scrotum forms by 10–12 weeks under DHT.

| Failed step | Result |
|---|---|
| CSL retained | Intra-abdominal testis |
| Abnormal or ectopic gubernacular attachment | Canalicular testis; ectopic (superficial inguinal pouch, perineal, femoral, prepenile) |
| Processus not closed | Hernia, hydrocele (patent in 87% of unilateral cases) |

| Failed step | Result |
|---|---|
| Fibrous remnant of processus / processus not fully involuted | Acquired (ascending) testis; acquired cryptorchidism |

### Abnormalities of descent: definitions and classification

| Type | Description |
|---|---|
| **Undescended (cryptorchid)** | Arrested on the normal path: abdominal, "peeping" at the ring, canalicular, high scrotal/prescrotal |
| **Ectopic** | Off the normal path: **superficial inguinal pouch** (commonest), perineal, femoral, prepubic, penile, contralateral scrotum (transverse, crossed) |
| **Retractile** | Scrotal testis pulled up by the cremaster; stays down after sustained traction; review yearly (risk of ascent); some become acquired |
| **Acquired (ascending)** | Documented scrotal earlier, ascends later; diagnosed at 8–11 years; fibrous processus remnant |
| Recurrent, secondary, iatrogenic | Re-ascent after spontaneous descent; after hernia repair or orchidopexy |
| **Vanishing** vs agenesis | Lost after formation, in utero or perinatally (torsion; monorchia, anorchia) vs never formed |

- Clinically: **palpable** (canalicular, high scrotal, ectopic) or **non-palpable** (intra-abdominal, "peeping", vanished/absent).

### Epidemiology and aetiology

- 1–4% of term boys (P2: 2–4% at birth, 1–2% at 3 months); up to 45% of preterm boys. Most descend by 6 months.
- 75–80% palpable; 60–70% unilateral.
- Descent needs Leydig hormones: **INSL3** (transabdominal phase) and **androgen** (inguinoscrotal phase), with the genitofemoral nerve/CGRP guiding the gubernaculum.
- Risk factors: prematurity, low birth weight, syndromes (prune belly, cerebral palsy).

### Associated anomalies and sequelae

- **Testicular: TTE (transverse testicular ectopia)**, polyorchidism, splenogonadal fusion; epididymal non-fusion.
- **PMDS (persistent Müllerian duct syndrome):** mutation of AMH or its receptor (AMHR2); bilateral cryptorchidism or TTE.
- **Genital and DSD (differences of sex development):** hypospadias (12–24%); bilateral non-palpable testes need karyotype and 17-OHP (17-hydroxyprogesterone) to exclude **CAH (congenital adrenal hyperplasia)**.
- **Syndromes:** prune belly (all), Klinefelter, androgen insensitivity, hypogonadotrophic hypogonadism, cerebral palsy, gastroschisis, omphalocele, myelomeningocele, PUV (posterior urethral valves).
- **Sequelae:** subfertility (worse if bilateral), **TGCT (testicular germ cell tumour) 2–5-fold**, torsion, hernia, trauma.

### Evaluation

- **History:** prematurity, testes once scrotal (acquired), hernia, family history, salt-wasting illness.

- **Examination:** warm room; supine and frog-leg, then cross-legged or squatting. Sweep ("milk") from the anterior superior iliac spine down the canal to the scrotum to find the lowest position; check the superficial inguinal pouch.
- Sustained traction fatigues the cremaster: a **retractile** testis stays in the scrotum after gentle traction.
- Note penis size, meatus (hypospadias), scrotal development, syndromic features. Contralateral hypertrophy suggests a vanished testis.
- **No ultrasound or MRI (magnetic resonance imaging)**: inaccurate (ultrasound sensitivity about 45% for non-palpable testes); they never change the operation or remove the need for surgery, and delay surgery.

### Bilateral undescended testis at 3 years

- This is **late presentation**; surgery is advised from 6 months, preferably by 12 months and by 18 months at the latest.
- **Bilateral palpable:** no investigations needed; plan orchidopexy, both sides usually in one sitting.
- **Bilateral non-palpable (or with hypospadias, small phallus or micropenis): treat as DSD until proven otherwise**, with urgent work-up:
- **Karyotype**; electrolytes and **17-OHP** to exclude **CAH** in a virilised 46,XX child.
- **Hormones:** LH, FSH (follicle-stimulating hormone), testosterone, **AMH and inhibin B**. Measurable AMH or inhibin B confirms testicular tissue; high gonadotrophins with absent AMH suggest **anorchia**. The **hCG stimulation test** (testosterone rise) is now less used.
- Pelvic ultrasound for Müllerian structures.
- **Both testes very high or of doubtful viability:** stage the sides, the second at 6 months, to avoid bilateral atrophy.
- **Biopsy** only if DSD or dysgenesis is suspected. **Bilateral anorchia:** testosterone at puberty, prostheses later.
- Adjuvant LHRH (buserelin) in bilateral cases is experimental.

### Management principles (AUA (American Urological Association) 2014; EAU (European Association of Urology) similar)

- **Observe to 6 months** (corrected age) for spontaneous descent; refer if still undescended.
- **Orchidopexy from 6 months, preferably by 12 months, and by 18 months at the latest.** Reason: germ-cell and Leydig-cell loss starts in infancy; early surgery restores testicular growth and lowers cancer risk. **Kollin RCT (randomised controlled trial):** surgery at 9 months gave better testicular growth than at 3 years.
- **No hormonal therapy** (hCG or LHRH (luteinising hormone-releasing hormone)) to induce descent: descent in about 20% only; hCG may increase germ-cell apoptosis.
- **Goals of surgery:** tension-free **dependent scrotal** testis with intact blood supply; ligate the patent processus.

### Treatment algorithm

```
Boy with empty scrotum ├─ Bilateral non-palpable ± abnormal phallus → DSD work-up (karyotype, hormones, electrolytes) └─ Otherwise → re-examine at 6 months corrected age Still undescended → refer → examination under anaesthesia (EUA) ├─ Retractile → yearly review (risk of ascent) ├─ Palpable (undescended or
```

ectopic) → inguinal or scrotal (Bianchi) orchidopexy └─ Non-palpable → diagnostic laparoscopy ├─ Testis seen → laparoscopic orchidopexy │ vessels short (testis not within 2–4 cm of ring) → Fowler–Stephens │ (clip testicular vessels; testis lives on vasal collaterals; │ one-stage, or two-stage with second stage at 6 months) or Shehata traction ├─ Vessels + vas enter ring → inguinal or scrotal exploration (excise nubbin) └─ Blind-ending vessels → stop (vanished testis)

- **Scrotal nubbin with contralateral hypertrophy** may start with scrotal exploration.
- **Postpubertal intra-abdominal testis with a normal contralateral testis: orchidectomy** (poor function, cancer risk); offer a prosthesis.

### Methods of orchidopexy: palpable testis

- **Inguinal orchidopexy** (steps): 1. Groin incision. 2. Open external oblique (spare the ilioinguinal nerve). 3. Free the cord; divide the gubernaculum. 4. Separate and **ligate the sac (patent processus) at the internal ring**. 5. Retroperitoneal mobilisation; divide lateral bands. 6. **Prentiss manoeuvre** if needed (medial transposition under the inferior epigastric vessels) for extra length. 7. **Subdartos pouch.** Transalbugineal sutures are unnecessary.
- **Trans-scrotal (Bianchi) orchidopexy** for low testes: high scrotal incision; sac ligated through it; inguinal conversion 3.5–4.4%; atrophy 0.3–0.6%; less pain.

### Methods of orchidopexy: non-palpable testis

Examination under anaesthesia, then laparoscopy.

| Method | Principle | Success |
|---|---|---|
| **Primary laparoscopic** | Mobilise vessels, preserve vasal collaterals, new hiatus medial to epigastric vessels | >90% (97%); primary orchidopexy, when possible, is most successful |
| **Open transabdominal (Jones and Bagley)** | Preperitoneal high mobilisation | 95% |
| **Fowler–Stephens** (one- or two-stage) | Divide testicular vessels high; testis survives on **vasal and gubernacular collaterals**; second stage after 6 months | 60–97% (one-stage 63%, two-stage 68%) |
| **Shehata traction** | Fix testis near the opposite anterior superior iliac spine; vessels lengthen; orchidopexy at 12 weeks | 84–85%; atrophy 10% |
| **Microvascular autotransplantation** | Testicular vessels to inferior epigastric vessels | 96% |
| **Orchidectomy** | Postpubertal, dysgenetic or atrophic testis | — |

- Fowler–Stephens is for testes **not within 2–4 cm of the internal ring**.
- **Also know:** staged (two-stage) inguinal orchidopexy; Dessanti PTFE (polytetrafluoroethylene) sleeve traction; gubernaculum-sparing technique; redo approaches (Redman lateral, Cartwright fascial patch).

### Complications of orchidopexy

- **Atrophy** (highest after Fowler–Stephens), retraction or re-ascent, vas or vessel injury, haematoma, infection; torsion very rare after subdartos fixation.

**Follow-up and prognosis**

- Check size and position at 6 months. Offer a prosthesis for an atrophic testis.
- **Fertility:** paternity about 80% (four-fifths) after unilateral and one-third after bilateral; worse when bilateral (oligospermia 42%, azoospermia 31% in one large review).
- **Cancer:** 2–5-fold TGCT risk (P1: about 3-fold), higher with late surgery; orchidopexy after 10–11 years carries relative risk 5.8 (Walsh).
- Counsel on fertility and cancer risk; teach **self-examination** after puberty.

**Diagrams to draw:**

- Sagittal fetus showing CSL, gubernaculum, processus vaginalis and both phases of descent.
- Line of descent with ectopic sites.
- Laparoscopic findings at the internal ring: testis, vessels entering the ring, blind-ending vessels.
- Fowler–Stephens collateral supply: testicular artery clipped, deferential and cremasteric arteries intact.

> **Key points to remember:**
>
> - SRY → SOX9 → Sertoli cells (AMH); Leydig cells → testosterone and INSL3.
> - Transabdominal: CSL regression plus INSL3-driven gubernacular swelling.
> - Inguinoscrotal: androgen via genitofemoral nerve (GFN) and CGRP; most testes scrotal after 27 weeks.
> - Ectopic = off the path; commonest site superficial inguinal pouch.
> - Operate from 6 months, preferably by 12 months, and by 18 months at the latest.
> - No imaging, no hormones.
> - Non-palpable: EUA then diagnostic laparoscopy; laparoscopy decides.
> - Bilateral non-palpable: DSD work-up first (karyotype, 17-OHP, AMH) to exclude CAH/DSD.
> - Subdartos pouch; Prentiss manoeuvre gives extra length.
> - Fowler–Stephens relies on vasal collaterals.

| Term | Key point (beyond the papers) |
|---|---|
| **Undescended testis (UDT)** | Not in the normal scrotal position (midpoint at or below mid-scrotum). It is palpable or non-palpable; non-palpable means undescended or absent. |
| High scrotal testis | Stable, above the scrotal midpoint; a mixed group, not routinely called undescended. |
| **Retractile** | One that rarely stays down or rests high may progress to acquired UDT. |
| Recurrent | Undescended at birth, descended spontaneously, then re-ascended (10–22%, at 1–5 years). |
| Secondary / testicular retraction | Suprascrotal after hernia repair / after orchidopexy (iatrogenic). |
| Agenesis | Never formed; comes with ipsilateral Müllerian persistence (extremely rare). |

## 2 · Embryology of descent

*See the paper answer **Testicular descent and undescended testis: evaluation, treatment algorithm and orchidopexy methods** (Section 4, Ch 57).*

**Also, from the condensed book (not in the paper answer):**

- The gubernaculum does not pull the testis; it swells, then regresses.

| Phase | Timing | Event | Control |
|---|---|---|---|
| Transabdominal (relative) | 10–15 weeks | Gubernacular swelling (outgrowth) from 12 weeks; cranial mesonephric ligament regresses by 13 weeks | INSL3 and androgen act together |
| Inguinoscrotal | **20–28 weeks** | The canal widens and the processus forms; rapid passage through the canal; the gubernaculum then regresses. Most are scrotal after 27 weeks. | Androgen via the genitofemoral nerve (**GFN**), which releases calcitonin gene-related peptide (**CGRP**) |

> **KEY POINT**
>
> Outgrowth creates the canal; regression lets the testis follow, and a defect in either gives the same phenotype. INSL3 and androgen act on one target (gubernacular/cremaster myogenesis), hence one clinical picture whichever fails.

**Minipuberty** is a hormonal surge peaking at 1–3 months, when gonocytes become **adult dark (Ad) spermatogonia**, the stem-cell pool. This is where a positional defect becomes a germ cell deficit.

## 3 · Epidemiology, aetiology and associations

*Spontaneous descent ends by about 6 months.*

- Prevalence is **2–4% at birth** (term), falling to **1–2% at 3 months**; up to 45% in preterm boys.
- About 88% of high scrotal testes descend spontaneously, but only about half of extrascrotal ones.
- 80–85% are isolated. Risk factors are **prematurity and low birth weight**, family history, maternal diabetes and smoking. Diethylstilbestrol doubles risk.
- It is a polygenic threshold trait, and part of **testicular dysgenesis syndrome** with hypospadias, infertility and testicular germ cell tumour (TGCT).
- No hormone predicts germ cell number, so hormones are not tested routinely outside DSD and anorchia.

| Associated condition | Cryptorchidism | Implication |
|---|---|---|
| Prune-belly syndrome | All (bilateral abdominal) | Abdominal orchidopexy |
| Spigelian hernia | 53–80% | Absent gubernaculum/canal |
| Cerebral palsy | 25–54% | Often bilateral |
| Gastroschisis | 5–25% | **50–55% descend spontaneously — observe** |
| Omphalocele, imperforate anus, myelomeningocele, posterior urethral valves | 12–33% | Caudal-field anomalies are ipsilateral in 90% |
| Congenital diaphragmatic hernia | 7–30% | Always ipsilateral |

| Associated condition | Cryptorchidism | Implication |
|---|---|---|
| Klinefelter (47,XXY) | > 50% of prepubertal Klinefelter boys | < 2% of non-syndromic UDT; more with hypospadias |
| Persistent Müllerian duct syndrome (PMDS) | Bilateral 59%; hernia uteri inguinalis; transverse ectopia 22% | AMH or AMHR2 mutation; adult infertility ≥ 80%, malignancy risk |

### Congenital vs acquired

| Feature | Congenital | Acquired |
|---|---|---|
| Diagnosis | Birth/neonatal | Mean 8–11 years |
| Position | Often higher | Lower; often superficial inguinal pouch (SIP) |
| Processus vaginalis | Patent (87% unilateral) | Usually closed (patent 45–50%) |
| Epididymal anomalies | 37–60% | 11–31% |

An acquired UDT is a never-fully-descended testis unmasked by growth, with semen results like congenital cases. Retractile testes share this pathology and ascend in about 2–3%, so they need **annual examination through childhood**.

## 4 · Histology, associated anomalies and fertility

*Germ cells are lost from late infancy, so operate early.*

- Germ cell loss begins **after infancy** and progresses; the contralateral testis is affected less.
- The UDT is smaller at birth, so the defect is intrinsic and worsened by position.
- Spermatogonia per tubule (S/T) is 2.5 at birth, with a nadir of 1.2 at 2–3 years.
- A **patent processus** is present in 87% of unilateral and 71% of bilateral cases. A UDT is at higher torsion risk.

| Anomaly | Key features | Management |
|---|---|---|
| Polyorchidism | Usually left, scrotal in 75%; Kumar classification by drainage (A with vas, B without) | Observe a normal scrotal testis; orchidopexy if undescended with intact drainage |
| Splenogonadal fusion | Left side; the continuous form comes with limb defects and micrognathia | Recognise it to avoid a needless orchidectomy |
| Transverse testicular ectopia | Hernia with a contralateral non-palpable testis; PMDS in 20–50% | Laparoscopy; trans-septal placement if vasa fused |

| Trial | Design | Result | Take-home |
|---|---|---|---|
| **Kollin et al.** | Randomised trial: orchidopexy at 9 months vs 3 years | Early: more Sertoli and germ cells, better testicular growth (to 4 years) | Operate in first year; age at surgery, not hormones, predicts germ cells |

### Fertility outcome

| Outcome | Former unilateral | Former bilateral |
|---|---|---|
| Oligospermia / azoospermia | 31% / 14% | 42% / 31% |
| Paternity | **≈ 90% of those attempting — as controls** | ≈ 65% of those attempting (≈ 1/3 overall) |

- In unilateral cases the normal testis carries fertility, so semen analysis overstates infertility; in bilateral cases it reflects true infertility.
- Ad spermatogonia at biopsy predict adult sperm counts; if they are absent, all counts are subnormal.
- Even after first-year surgery, 20–25% remain at infertility risk. Surgery **before 18 months** reduced assisted reproduction, and each 6-month delay added 5%.

## 5 · Malignancy

*Orchidopexy lowers cancer risk only partly.*

- TGCT risk is **2–5×** (pooled odds ratio (OR) ≈ 4). About 1% of cryptorchid men get TGCT, and 5–10% of TGCT patients had a UDT.
- In the **Swedish population study** the OR was about 2.8 if treated before puberty vs about 5.0 after. † Pettersson 2007: relative risk (RR) 2.23 if operated before 13 years vs 5.40 at 13 or older. A meta-analysis gave RR **5.8** for orchidopexy after 10–11 years.
- Untreated adults have OR ≈ 14.
- A persistently cryptorchid testis develops **seminoma** (74%), whereas tumours after orchidopexy are mostly non-seminoma (63%).
- Germ cell neoplasia in situ occurs in 2–3% of formerly cryptorchid men, so biopsy is for DSD and chromosomal cases. Biopsy at orchidopexy does not raise TGCT risk.
- Postpubertal UDT with a normal contralateral testis needs **orchidectomy** (up to age 50), with a prosthesis at the same sitting.
- **Microlithiasis** is not an independent TGCT risk, so routine ultrasound surveillance is not needed.

## 6 · Diagnosis

*Examination decides; imaging adds nothing.*

### Examination — the gold standard

**ALGORITHM**

1. Warm room and hands. Examine supine, then cross-legged or squatting.
2. With lubricated fingers, sweep one hand from the anterior superior iliac spine (ASIS) along the canal and palpate the lowest position with the other.
3. Compress the canal and invaginate the scrotum to feel the SIP.
4. Apply sustained traction to fatigue the cremaster. If it **stays, it is retractile; if it springs back, it is undescended**.

- About **80% palpable**, and 60–70% are unilateral. At surgery a non-palpable testis is abdominal or peeping in 25–50%, vanished in 15–40% and a missed extra-abdominal testis in 10–30%.
- Examination under anaesthesia (EUA) is the final arbiter.

*See the paper answer **Testicular descent and undescended testis: evaluation, treatment algorithm and orchidopexy methods** (Section 4, Ch 57).*

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**SAQ** **Vanishing testes**

Paper answer · P1

**Type:** SAQ (10 marks)
**Source:** Practice Set 13 Paper 1, Q3b

### Definition

- A testis that **developed normally and was then lost**, usually from **torsion or a vascular accident in fetal life**, after genital masculinisation but before scrotal fixation.
- Unilateral loss gives **monorchia**; bilateral loss gives **anorchia** (testicular regression syndrome), which is rare.
- Excised "nubbins" contain haemosiderin.

### Incidence

- **15–40% of nonpalpable testes** at surgery.

### Clinical features

- **Contralateral compensatory hypertrophy:** volume >2 mL (sensitivity 72%, specificity 100%) or length about 19–20 mm suggests a solitary testis.
- No palpable intrascrotal "appendage" tissue; sometimes a scrotal nubbin.
- These aid counselling but **do not replace exploration**.
- **Anorchia:** micropenis in nearly half.

### Evaluation

- **Imaging** (ultrasound, MRI [magnetic resonance imaging]) is not recommended: it does not change the operation.
- **Bilateral nonpalpable:** exclude DSD (difference of sex development) and CAH (congenital adrenal hyperplasia) first.
- Anorchia: high FSH (follicle-stimulating hormone) and LH (luteinising hormone); low AMH (anti-Müllerian hormone) and inhibin B; no rise after hCG (human chorionic gonadotrophin). Most reliable during mini-puberty.

### Management: diagnostic laparoscopy (after examination under anaesthesia)

| Laparoscopic finding | Action |
|---|---|
| **Blind-ending vessels and vas** above a closed ring | Diagnosis confirmed; stop |
| Vessels and vas entering the internal ring | Inguinal or scrotal exploration; excise the nubbin |
| Testis seen | Orchidopexy (single stage or Fowler–Stephens) |
| Neither vas nor vessels | Search up to the kidney (agenesis very rare) |

- **Excise the remnant:** viable germ cells or tubules in 5–15%.
- **Scrotal-first exploration** is reasonable with a palpable nubbin and contralateral hypertrophy (length ≥1.8 cm); laparoscopy is then needed in about a quarter.
- **Contralateral fixation:** debated; not required by the AUA (American Urological Association) algorithm.
- **Prosthesis:** offer at least 6 months after scrotal surgery or after puberty, inguinal approach.
- **Anorchia:** testosterone from puberty. **Also know:** agenesis shows ipsilateral Müllerian persistence.

**Diagram to draw:** laparoscopic view of blind-ending vessels at a closed ring.

**Key points to remember:**

- Prenatal torsion after masculinisation; 15–40% of nonpalpable testes.
- Contralateral hypertrophy (>2 mL) suggests it, but explore anyway.
- Blind-ending vessels confirm; excise remnant (germ cells 5–15%).
- Anorchia: high FSH/LH, low AMH and inhibin B.

### Imaging — not recommended

**The AUA (2014) advises no ultrasound or MRI.** Ultrasound sensitivity is only about 45%, and imaging merely delays surgery.

### Bilateral non-palpable or UDT with hypospadias: DSD work-up

- A newborn with bilateral non-palpable testes needs an **urgent karyotype and hormones** to exclude **congenital adrenal hyperplasia (CAH)**, because of salt-wasting.
- Proximal hypospadias with UDT carries a chromosomal abnormality in 32–47% (see Ch 56).
- Micropenis with bilateral UDT suggests hypogonadotrophic hypogonadism or anorchia.

| Marker | Anorchia | Bilateral abdominal UDT |
|---|---|---|
| FSH, LH | Raised | Normal |
| **AMH, INHB** (best prepubertal Sertoli markers) | Undetectable/low | Present |
| Basal testosterone | Low in first year only | Normal |
| Testosterone after hCG | No rise | Rises |

hCG stimulation is no longer the test of choice, and **laparoscopy remains the confirmatory test**.

## 7 · Management principles and timing

- **Timing**: observe to **6 months** (corrected age), then refer; operate at **6–18 months**, preferably by 12.
- Retractile testes need no surgery but yearly examination. Acquired UDT should be operated on, especially if bilateral.
- A pubertal or postpubertal boy with a normal contralateral testis needs orchidectomy, with or without a prosthesis.

### Hormonal therapy — not recommended (AUA)

- Intranasal LHRH gives descent in only about 20%, and hCG was never placebo-tested. hCG may increase germ cell apoptosis and reduce adult volume.
- Adjuvant LHRH for bilateral UDT with absent Ad spermatogonia (Hadziselimovic) has pilot data only.

**KEY POINT**

Hormones fail on both claims for one reason: position, not endocrinology, is the modifiable variable. Descent is not androgen-limited after fetal life, and germ cells track age at scrotal placement.

## 8 · Surgery for the palpable testis

*The goal is a tension-free scrotal testis with intact vessels.*

### Inguinal orchidopexy — key steps

**ALGORITHM**

1. EUA, skin-crease incision, check the SIP.
2. Open the external oblique, sparing the ilioinguinal nerve.
3. Free the testis; divide the gubernaculum distal to the sac (protect a long-looping vas).
4. **Separate the patent processus from vas and vessels without skeletonising; high-ligate it at the internal ring** (main length gain).
5. If short, divide lateral spermatic bands and dissect retroperitoneally; then the **Prentiss manoeuvre** (medial to the inferior epigastric vessels).
6. Transverse scrotal incision and **subdartos pouch**; pass the testis untwisted.
7. Close dartos around the cord; transalbugineal sutures are unnecessary.

- If the testis cannot reach the scrotum, stage the procedure. Orchidectomy is only for an atrophic or abnormal testis, a postpubertal boy or an insufficient vas.
- Biopsy is not routine; it is indicated in DSD and considered in bilateral UDT or after 12–13 years.

### Scrotal (Bianchi) orchidopexy

The testis is delivered through a high scrotal incision, the processus ligated and the testis fixed in a subdartos pouch. It is increasingly first choice for palpable, especially low, testes.

## 9 · The non-palpable testis

*Laparoscopic findings at the ring decide the next step.*

**ALGORITHM**

1. EUA: if now palpable, do an inguinal or scrotal orchidopexy.
2. If **still non-palpable, do diagnostic laparoscopy** (ring patency, vessels, vas, testis).
3. If **blind-ending vessels** (with or without vas) lie above a closed ring, the testis has vanished: stop (the only unequivocal endpoint).
4. If **vessels and vas enter the ring**, explore the groin/scrotum and excise the nubbin (viable tubules 5–15%).
5. If the testis is **intra-abdominal**, do a primary laparoscopic or Fowler–Stephens (one-or two-stage) orchidopexy; orchidectomy if postpubertal or dysgenetic.
6. If it is peeping at the ring, do a laparoscopic or inguinal orchidopexy.
7. If neither vas nor vessels are seen, search the perivesical area and retroperitoneum to the kidney.

- An open ring usually, but not always, means a viable distal testis, and apparently blind-ending vessels may coexist with an ectopic testis, so keep exploring if equivocal.
- The AUA does not require fixing the contralateral solitary testis.

### Laparoscopic orchidopexy

**ALGORITHM**

1. Empty bladder and stomach. Open (Hasson) umbilical port; CO2 at 8–12 mmHg. Inspect both rings and add working ports.
2. Divide the gubernaculum distal to the ring and use it as a handle.
3. Incise peritoneum lateral to the vessels and distal to the vas; mobilise the vessels, **preserving the peritoneal triangle between vas and vessels** (keeps Fowler–Stephens possible).
4. Make a new hiatus medial to the inferior epigastric vessels.
5. Bring the testis through a trans-scrotal port to a subdartos pouch without tension.

- Ring closure is unnecessary, and the robot offers no clear advantage.

### Fowler–Stephens (FS) and alternatives

*See the paper answer **Testicular descent and undescended testis: evaluation, treatment algorithm and orchidopexy methods** (Section 4, Ch 57).*

**Also, from the condensed book (not in the paper answer):**

- Two-stage FS: clip the vessels high, wait about **6 months**, then mobilise on a wide peritoneal pedicle. Two-stage is slightly favoured.
- Keeping peritoneum over the vasal vessels and **sparing the gubernaculum** lowers atrophy. Avoid dividing vessels when possible, because it reduces germ cells.
- Operate on bilateral high testes about 6 months apart.

**KEY POINT**

Every abdominal technique answers one question: how to gain length without losing blood supply. FS sacrifices the testicular artery; traction gains length over time while keeping it, so vessel division is the last resort.

## 10 · Complications and follow-up

| Complication | Cause / prevention | Management |
|---|---|---|
| **Atrophy** (≈ 2% modern; historically 24% abdominal/peeping vs 10% distal) | Vessel injury, skeletonisation, FS, tension | Offer a **prosthesis** ≥ 6 months after scrotal surgery or after puberty, inguinal route, fixed to dartos |
| **Retraction / non-scrotal** (≈ 2–10%) | Inadequate retroperitoneal mobilisation, unligated processus | Redo: a prescrotal testis by the scrotal approach; otherwise a lateral approach through unscarred tissue or a fascial patch on the cord. Results equal primary surgery. |
| **Vas injury** | Long-looping vas cut with the gubernaculum | Divide the gubernaculum distal to the sac; avoid skeletonising |
| Torsion after orchidopexy | Rare | Prevented by subdartos fixation |
| Laparoscopic | Bladder/vessel injury, hypercapnia, bowel obstruction | Hasson entry, bladder drainage |

**Follow-up**: examine at ≥ **6 months** for position and size. Counsel on fertility and cancer, and teach self-examination from puberty.

# 61 Urologic Considerations in Pediatric Renal Transplantation

## LAQ Urological evaluation and management of pediatric kidney transplant patient.

Paper answer · P1

**Type:** LAQ (25 marks)
**Source:** 2022-Summer Paper 1, Q2

### Introduction

- Children needing KRT (kidney replacement therapy) often have CAKUT (congenital anomalies of the kidney and urinary tract). The urologist must ensure that the bladder that destroyed the native kidneys does not destroy the graft.
- **Five goals:** unobstructed drainage; a capacious, low-pressure reservoir with social continence; volitional emptying; no infection; the fewest operations.

### Causes of ESKD (end-stage kidney disease) in children

- Commonest age at transplant 13–17 years.
- **Younger:** dysplasia, hypoplasia, aplasia and obstructive uropathy, mainly PUV (posterior urethral valves). **Older:** FSGS (focal segmental glomerulosclerosis) and neuropathic bladder.

### Pretransplant evaluation

- **Every child:** urological history, renal ultrasound, uroflow with PVR (post-void residual) if toilet-trained.
- **VCUG (voiding cystourethrogram)** only with neuropathic bladder, PUV, VUR (vesicoureteral reflux), previous lower-tract surgery, recurrent febrile UTI (urinary tract infection) or unexplained hydronephrosis.
- **Urodynamics** for neuropathic bladder, PUV or large ureterocele, refractory voiding dysfunction, high-grade hydronephrosis or recurrent UTI. They measure capacity, compliance, detrusor activity, emptying and sphincter function, and are repeated.

```
History positive → ultrasound + VCUG → urodynamics Poor capacity/compliance → anticholinergic, CIC, cycling → still poor → augment Poor emptying → CIC ± catheterisable stoma Adequate → clear for transplant History negative → ultrasound + PVR → normal → clear
```

- An anuric child with no urological history needs no further tests.

### Pretransplant management

**1. Bladder optimisation (drugs → CIC → cycling → augmentation)**

- Commonest problem: low capacity, detrusor overactivity, poor compliance. High pressure harms the kidney even without reflux.
- **Target:** age-expected capacity at detrusor pressure **<40 cm $H_2O$**.
- Antimuscarinics (oxybutynin titrated; solifenacin), mirabegron, intradetrusor botulinum toxin; α-blockers aid emptying. Better storage may worsen emptying, so add **CIC (clean intermittent catheterisation)**. Prove volitional emptying by repeated low PVRs.
- A family's adherence to CIC predicts adherence to immunosuppression.

**2. Valve bladder versus neuropathic bladder**

- PUV bladder moves from overactivity to **myogenic failure**: postpone augmentation. Neuropathic bladder may deteriorate under high post-transplant urine output: reconstruct earlier.

**3. Defunctionalised bladder** (anuria or diversion)

- **Bladder cycling:** catheter instillation of increasing saline volumes, with drugs, until age-expected capacity; confirm by urodynamics. **Never use the graft to test the bladder.**

**4. Augmentation**

- Indication: cannot reach **>75% expected capacity at <40 cm $H_2O$** with CIC every 3–4 hours despite maximal therapy. It does not clearly raise transplant complications.

| Segment | Point |
|---|---|
| Dilated ureter (ureterocystoplasty) | Explore first: urothelium, no mucus or acidosis |
| Ileum, sigmoid | Most used; mucus, stones, infection, malignancy; lifelong follow-up |
| Stomach | Haematuria–dysuria syndrome in anuric patients; rarely used |

- Continent stoma (Mitrofanoff) if the urethra is sensate or difficult; site it medial to the planned incision. Allow **6–12 weeks** healing before transplant.

**5. Infection, nephrectomy and dialysis**

- Treat stasis (double voiding, CIC); distinguish colonisation from UTI; consider circumcision in boys with valves.
- **Native nephrectomy** (default is to keep): uncontrolled hypertension, nephrotic syndrome, massive polyuria, recurrent pyelonephritis, large stones, malignancy risk, very large kidneys, massive VUR. Retroperitoneal route keeps PD (peritoneal dialysis).
- Keep a normal native ureter for later ureteroureterostomy. Intraperitoneal surgery on PD needs temporary haemodialysis.

### At transplantation

- Retroperitoneal Gibson incision; children **<15 kg** may need aorta–caval anastomosis, sometimes transperitoneal.
- **Ureter:** extravesical non-refluxing ureteroneocystostomy (3–4 cm detrusorotomy) preferred; Barry tunnel; ureteroureterostomy to a normal native ureter.
- **Stent:** selective; double-J removed at 4–6 weeks, or early removal at 5–7 days.

### Post-transplant complications

| Complication | Rate | Management |
|---|---|---|
| Ureteric obstruction | 6% | Usually distal ischaemic stenosis; >half within 100 days. Stent or nephrostomy; balloon dilatation; open repair (native ureter, bladder flap) |
| Urine leak | 3% | Early; drain creatinine; catheter, nephrostomy, stent, then reimplantation |
| Graft VUR | 2% (significant) | Febrile UTI → reimplant; asymptomatic → observe; injection works in one-third |
| Stones | 2% | Treat if >1 cm and growing, obstructing or infected |

- **Rising creatinine with hydronephrosis:** catheter drainage 1–2 weeks (bladder cause?) → biopsy if rejection is suspected → stent or nephrostomy. Obstruction and rejection can coexist.
- Also: lymphocele, **PTLD (post-transplant lymphoproliferative disorder)**, graft thrombosis (a leading cause of early graft loss), late infection reflecting bladder dysfunction.

**Follow-up**

- Lifelong ultrasound, PVR and bladder surveillance; recurrent UTIs lower graft GFR (glomerular filtration rate).

**Diagram to draw:** extravesical ureteroneocystostomy with detrusor flaps closed over the ureter.

> **Key points to remember:**
> - Bladder target: age-expected capacity at <40 cm $H_2O$; augment if <75% despite CIC.
> - PUV bladder heads to myogenic failure; neuropathic bladder needs earlier reconstruction.
> - Never test a bladder with a graft; cycle the defunctionalised bladder.
> - Stenosis 6%, leak 3%, stones 2%, significant reflux 2%.

**Also, from the condensed book (not in the paper answer):**

- Graft reflux matters more than native reflux, because the kidney is solitary and immunosuppressed.

## 1 · Causes of ESKD in children

- The commonest cause of end-stage kidney disease (ESKD) is congenital anomalies of the kidney and urinary tract (**CAKUT**), then obstructive uropathy, mainly posterior urethral valves (**PUV**).
- Older children have FSGS or neuropathic bladder. The commonest age at transplant is **13–17 years**.

> **KEY POINT**
>
> **Five goals**: unimpeded drainage · capacious low-pressure reservoir with social continence · volitional emptying · no infection · fewest operations. Chronic rejection is unavoidable, but every added insult accelerates it.

## 2 · Pre-transplant evaluation

- Every child needs a history and renal ultrasound, plus uroflow with post-void residual if toilet-trained.
- VCUG and urodynamics are selective: neuropathic bladder, PUV, reflux, prior surgery, febrile UTI or worsening hydronephrosis.

**ALGORITHM**

1. Positive history: US and VCUG. If dynamics are abnormal, urodynamics.
2. Treat with drugs and clean intermittent catheterisation (CIC), with or without bladder cycling, then reassess.
3. If still unsafe, augment; then clear for transplant.

## 3 · Bladder management

*The bladder must be safe before the graft goes in.*

- The commonest pattern is **low capacity, overactivity and poor compliance**, and high pressure harms the kidney even without reflux.
- The **target** is expected capacity at detrusor pressure **< 40 cm $H_2O$**, with social continence.
- Use antimuscarinics, mirabegron or botulinum toxin, with CIC as needed.
- Cycle a **defunctionalised bladder** with rising saline volumes, then repeat urodynamics. **Never use the allograft to test a bladder.**
- **Augment** if the bladder cannot reach > **75%** of expected capacity at < 40 cm $H_2O$ with CIC no more often than every **3–4 h** despite maximal therapy. Postpone this in PUV, but augment earlier in neuropathic bladder.
- A dilated ureter is the best segment. Bowel augments need lifelong malignancy surveillance.
- Site a **Mitrofanoff medial to the planned transplant incision**.
- Reconstruct **before** transplant, allowing **6–12 weeks** to heal, except when transplanting before 2 years.

## 4 · Native nephrectomy

*See the paper answer **Pre transplant nephrectomy** (Section 15, Ch 91).*

**Also, from the condensed book (not in the paper answer):**

- Combine it with the transplant only to make room, as in a small child with huge kidneys.

## 5 · Transplant into the abnormal bladder

- A Gibson retroperitoneal incision is standard; a child < **15 kg** may need the aorta and vena cava through a transperitoneal approach.
- The preferred join is an **extravesical non-refluxing ureteroneocystostomy**. **Ureteroureterostomy** to the native ureter is the alternative and the salvage.
- Stent if difficult, removing the double-J at **4–6 weeks**.

## 6 · Urological complications

| Complication | Rate | Timing / management |
|---|---|---|
| Ureteric stenosis | **6%** | Most within 100 days; distal, ischaemic. Focal: balloon and stent for 4–6 weeks; otherwise open repair (native ureter, bladder flap); do not delay. |
| Urine leak | **3%** | Early. Drain-fluid creatinine; check catheter, then US, then MAG3 scan or cystogram. Small: observe; large: drain or explore. Short necrosis: reimplant; long: native ureter, psoas hitch or flap. |

| Complication | Rate | Timing / management |
|---|---|---|
| Significant VUR | 2% | See below |
| Stones | 2% | Treat if > 1 cm and growing, obstructing or infected; one-stage modality |

### Rising creatinine with hydronephrosis

**ALGORITHM**

1. Indwelling catheter for 1–2 weeks: if creatinine falls, the bladder is the cause.
2. If not, stent and keep the catheter; biopsy if rejection is suspected.
3. If retrograde stenting fails, nephrostomy and antegrade study.

- The graft is unusually sensitive to obstruction, and obstruction and rejection coexist, so stent at a low threshold.
- An **allograft mass is post-transplant lymphoproliferative disorder (PTLD) until proven otherwise**.
- Recurrent febrile UTI from graft reflux with an optimised bladder needs an **open or robotic reimplant**; asymptomatic reflux is observed.

## 7 · Outcomes

- Urological complications rarely lose the graft, but **≥ 2 UTIs** lower graft GFR at 1 and 4 years.

# 63 Paediatric Urologic Oncology: Renal and Adrenal

## 1 · Wilms tumour (WT): epidemiology and genetics

*Genetics decide who is screened and who needs kidney-sparing surgery.*

### LAQ Wilms tumour, including bilateral Wilms tumour

Paper answer · P2 · P3 · merged from 2

*Asked as:* P2 LAQ "Wilms tumour: classification, staging and management" · P3 SAQ "Bilateral Wilms tumour: treatment strategies"

#### Introduction

- Read as "Wilms tumour" (nephroblastoma); "Wilm's bladder" in one P2 wording is taken as a typo.
- WT (Wilms tumour) is an embryonal tumour of metanephric blastema. It forms 95% of kidney cancers under 15 years; median age 3.5 years; over 80% present before 5 years.

#### Genetics, syndromes and precursors

- **Genetics:** mostly sporadic; familial 1–2%. Genes: **WT1 (11p13)**, IGF2/11p15 (commonest change, about 70%), WTX, CTNNB1; **TP53 in anaplasia (in most anaplastic tumours)**.

| Syndrome | WT risk |
|---|---|
| **WAGR** (Wilms, aniridia, genitourinary anomalies, intellectual disability / retardation; WT1 deletion) | 50% |
| **Denys–Drash** (46,XY DSD [disorder of sex development], mesangial sclerosis; WT1 point mutation) | 50% |
| **BWS (Beckwith–Wiedemann syndrome)**, hemihypertrophy (11p15) | 5–10% |

- **Screening:** ultrasound every 3–4 months until 7 years when risk is >1%.
- **Nephrogenic rests** (precursors) occur in over one-third of WT kidneys and over 90% of bilateral cases.

#### Classification: histology

| Group | Features | Significance |
|---|---|---|
| **Favourable histology** | **Triphasic:** blastema, epithelium (tubules), stroma; pseudocapsule | Survival >90% |
| **Anaplasia** (unfavourable) | Nuclei ≥3× neighbours (larger), hyperchromasia, abnormal mitoses; **focal** or **diffuse** (diffuse worse than focal) | Chemoresistant; poor even at stage I |

- **SIOP (International Society of Paediatric Oncology) post-chemotherapy risk groups:**

| Risk | Histology |
|---|---|
| Low | Completely necrotic |
| Intermediate | All others (regressive, mixed, epithelial, stromal, focal anaplasia) |
| High | **Blastemal-predominant**; diffuse anaplasia |

- Residual blastema predicts relapse; stromal and epithelial tumours shrink little but are cured by excision.

### Staging

#### COG (Children's Oncology Group): surgical-pathological, before chemotherapy

| Stage | Description |
|---|---|
| I | Confined to kidney, completely resected, capsule intact, no renal sinus vessel invasion |
| II | Beyond kidney (renal sinus vessels, perirenal fat, vein thrombus) but completely resected |
| III | Residual abdominal disease: **positive nodes, any spill, prior biopsy**, positive margin, peritoneal implants |
| IV | Haematogenous metastases (lung commonest, liver, bone, brain) |
| V | Bilateral at diagnosis (each side staged separately) |

- **SIOP staging** uses the same five stages but is applied **after preoperative chemotherapy**. Necrotic, chemotherapy-induced change in the sinus or perirenal fat does not upstage.
- **Biology:** combined **1p and 16q LOH (loss of heterozygosity)** predicts relapse; 1q gain is the next stratifier.

### Clinical features

- Abdominal mass; haematuria about 20%; hypertension about 25% (renin).
- Look for aniridia, hemihypertrophy and varicocele. Varicocele or hepatomegaly suggests IVC (inferior vena cava) thrombus.
- Acquired von Willebrand disease in up to 8%.

### Investigations

- Ultrasound with Doppler (IVC), then contrast CT (computed tomography) or MRI (magnetic resonance imaging) abdomen, then **chest CT** (lung is the commonest metastatic site; do it before anaesthesia).
- Blood count, renal and liver function, coagulation; syndrome review.

### Management: two philosophies

| | COG (North America) | SIOP (Europe) |
|---|---|---|
| Sequence | **Upfront transperitoneal radical nephrectomy**, then chemotherapy | **Preoperative VCR + AMD** (4 weeks; add DOX for 6 weeks if metastatic), no biopsy, then nephrectomy |
| Advantage | Untreated histology and biology for risk assignment | Tumour shrinks; spill almost zero |
| Risk basis | Stage + histology + LOH | Post-chemotherapy histology + stage |

(VCR = vincristine; AMD = actinomycin D (dactinomycin); DOX = doxorubicin; RT = radiotherapy; NSS = nephron-sparing surgery.)

### Surgical principles (radical nephrectomy)

- Transperitoneal approach; palpate renal vein and IVC before ligation; spare a tumour-free adrenal.
- **Avoid spill** (it makes the tumour stage III).

- **Sample at least 5 lymph nodes** (hilar, paracaval, para-aortic, interaortocaval); this is essential, formal dissection is not. Omitted sampling is the commonest protocol violation.
- IVC thrombus at or above the hepatic veins: preoperative chemotherapy to avoid bypass.

### Postoperative therapy (COG, favourable histology)

| Group | Treatment |
|---|---|
| Very low risk (<2 years, <550 g, stage I) | Nephrectomy alone (AREN0532) |
| Stage I–II | VCR + AMD |
| Stage III | VCR + AMD + DOX + flank RT |
| Stage IV | Three drugs + whole-lung RT; lung RT omitted if lung nodules clear by week 6 (AREN0533) |
| Anaplasia or 1p/16q LOH | Intensified regimens (regimen M adds cyclophosphamide + etoposide) |

### Bilateral WT (stage V): principles

- **Synchronous bilateral WT** is **5–7%**: **COG stage V**. Each kidney is staged separately.
- Linked to WAGR, Denys–Drash (about 50% risk) and BWS; **nephrogenic rests in over 90%**.
- Aim: **cure while preserving renal tissue (nephrons)**. Renal failure is about 15% at 15 years; ESRD (end-stage renal disease) mainly follows bilateral nephrectomy.
- **No upfront biopsy and no upfront nephrectomy.**

### Bilateral WT: COG approach (AREN0534)

```
VCR + AMD + DOX × 6 weeks (no biopsy) → imaging at week 6 ├─ resectable / NSS feasible → bilateral NSS └─ not → 6 more weeks chemotherapy → surgery by week 12 poor response → bilateral OPEN biopsy → further therapy by each kidney's stage and histology ± flank RT
```

- **Why stop at 12 weeks:** in NWTS-4 (National Wilms Tumor Study 4), prolonged chemotherapy in poor responders delayed surgery and some tumours became anaplastic. A non-shrinking tumour is often differentiated and is cured by excision.
- **Why an open, bilateral biopsy:** anaplasia is discordant between kidneys in about 4 of 5.

### Bilateral WT: surgery

- **Bilateral NSS** (partial nephrectomy or enucleation); frozen section; lymph node sampling; avoid spill.
- Compressed normal parenchyma is often underestimated, so try NSS in every case. Expert centres achieve bilateral NSS in over 90%.
- Enucleation: positive margins 23–36%, good outcome with flank radiotherapy.
- Unresectable both sides: bilateral nephrectomy and dialysis are rare and only for refractory tumour; transplant after **1–2 years** tumour-free.

### Bilateral WT: adjuvant therapy and SIOP approach

- Adjuvant therapy is guided by the worst histology and margins: anaplasia or positive margin means intensified chemotherapy and flank radiotherapy.
- **SIOP:** same principle (no biopsy, NSS by week 12) but **two drugs, VCR + AMD**, if non-metastatic.

### Unilateral nephron-sparing surgery

- For syndromic children and solitary kidney; SIOP UMBRELLA criteria are volume <300 mL, a substantial remnant and negative nodes.

### Prognosis and outcomes

- Favourable histology: survival >90%.
- **AREN0534 (bilateral):** surgery by week 12 in 84%; bilateral NSS 39%; **4-year EFS (event-free survival) 82.1%, OS (overall survival) 94.9%**, versus 56% and 80.8% on NWTS-5.
- Salvage survival after relapse: 70–80% after VCR/AMD only; about 10% for anaplastic or blastemal relapse.

### Late effects and follow-up

- **Late effects:** cardiotoxicity (DOX), second cancers (RT), infertility, renal failure (bilateral).
- **Follow-up (bilateral):** lifelong blood pressure, proteinuria, creatinine (renal function) and imaging, as late relapse occurs.

**Diagrams to draw:** COG staging diagram showing capsule, renal sinus, nodes and spill; bilateral tumours with planned partial nephrectomies; AREN0534 timeline (weeks 0, 6, 12).

> **Key points to remember:**
>
> - WT1 at 11p13 (WAGR, Denys–Drash); BWS at 11p15.
> - Anaplasia (≥3× nuclei, hyperchromasia, abnormal mitoses) is unfavourable; TP53.
> - Any spill, biopsy or positive node = stage III; sample ≥5 nodes.
> - COG operates first; SIOP gives chemotherapy (VCR + AMD) first.
> - Bilateral: stage V, 5–7%; no upfront nephrectomy or biopsy; three drugs for 6 weeks (reassess at 6 weeks), surgery by week 12, NSS; SIOP uses two drugs.
> - AREN0534: 4-year OS about 95%; bilateral NSS to avoid ESRD.

**Also, from the condensed book (not in the paper answer):**

- WT is **6–7%** of childhood cancers and **95%** of kidney cancers under 15 y; median age **3.5 y**, younger if bilateral or syndromic. Refer for genetics if bilateral, under 2 y, with rests, syndromic features or family history.
- WAGR (Wilms, aniridia, genitourinary anomalies, intellectual disability) is a WT1 deletion; **Denys–Drash** syndrome (DDS) is a **dominant-negative** point mutation with early end-stage renal disease. Type I tumours (WT1) are stromal with intralobar rests; Type II (11p15, Beckwith–Wiedemann syndrome, BWS) are blastemal/epithelial with perilobar rests.
- Combined 1p and 16q loss of heterozygosity (~5%) earns the intensified **regimen M**; 1q gain (25–30%) is the next stratifier.

| Syndrome risk | Examples |
|---|---|
| High (> 20%) | WAGR, DDS (**50%**); Fanconi D1 (BRCA2); Perlman (DIS3L2); mosaic variegated aneuploidy |
| Moderate (5–10%) | BWS, isolated hemihypertrophy, Frasier (WT1), Simpson–Golabi–Behmel (GPC3) |

| Syndrome risk | Examples |
|---|---|
| Low | Li–Fraumeni, neurofibromatosis 1, Sotos, trisomy 18, Bloom |

**KEY POINT**

Trap: BWS is the **commonest** predisposing syndrome but only moderate risk (21% bilateral); WAGR/DDS are rarer and highest.

**ALGORITHM**

1. If WT risk is > 1%, do a renal ultrasound (US) every **3–4 months** until **7 y**.
2. If a lesion looks suspicious, do computed tomography (CT) or magnetic resonance imaging (MRI).

- In BWS, hemihypertrophy and similar syndromes, add α-fetoprotein (AFP) 3-monthly to 4 y for hepatoblastoma.
- Screening lowers stage and allows nephron-sparing surgery (NSS), but survival benefit is unproven.

## 2 · Pathology

- Favourable histology is **triphasic**; the tumour is friable and spills easily.
- **Anaplasia** (chemoresistant): nuclei ≥ **3×** neighbours, hyperchromasia and abnormal mitoses; it is commoner in older children (~13% over 5 y).
- Nephrogenic rests are found in > 1/3 of WT kidneys and **> 90%** of bilateral cases; most regress. Perilobar rests are peripheral (BWS); intralobar rests lie anywhere (WAGR/DDS).

## 3 · Clinical features and work-up

- A varicocele persisting supine, or hepatomegaly, suggests inferior vena cava (IVC) thrombus. Gross haematuria needs cystoscopy at surgery.
- **Acquired von Willebrand disease** occurs in up to 8%; give desmopressin.
- Ascites beyond the cul-de-sac best predicts rupture; operate urgently only for bleeding. Imaging cannot judge nodes or separate WT from renal cell carcinoma (RCC) or clear cell sarcoma (CCSK).

## 4 · Staging and COG vs SIOP

*COG operates first; SIOP gives chemotherapy first.*

- Children's Oncology Group (COG) stages I–V are surgical–pathological. Stage III means **positive nodes, any spill**, implants, a positive margin or piecemeal removal.
- The International Society of Paediatric Oncology (SIOP) gives pre-operative vincristine (VCR) and dactinomycin (AMD) without biopsy.

**KEY POINT**

COG stage is surgical-pathological, so the surgeon, not the scan, decides chemotherapy intensity and irradiation: one spill converts stage I/II to III.

## 5 · Surgical principles

- Do a transperitoneal radical nephrectomy and inspect the liver, peritoneum and nodes.

- Sample nodes (COG **≥5 nodes**); omission is the commonest protocol violation and understages the tumour.
- Spill (~10%) causes local relapse; complications (~11%) are mainly haemorrhage and small-bowel obstruction.
- COG favourable histology: stage I–II get VCR and AMD; stage III–IV add doxorubicin (DOX) with flank/abdominal (and lung) radiotherapy (RT). Cyclophosphamide and etoposide are added in regimen M and for CCSK.

### Bilateral WT (stage V) and NSS

*See the paper answer **Wilms tumour, including bilateral Wilms tumour** (Section 4, Ch 63).*

**Also, from the condensed book (not in the paper answer):**

- Use frozen section and a normal-kidney margin in NSS.

**ALGORITHM**

1. Synchronous bilateral WT: **no upfront nephrectomy, no biopsy**.
2. Give VCR, AMD and DOX for 6 weeks, then re-image.
3. If NSS is feasible, operate.
4. If not, give more chemotherapy, but do **surgery by week 12**.
5. If the tumour does not respond, do a **bilateral open biopsy**.

- Unilateral NSS suits syndromic children or a solitary kidney; UMBRELLA criteria are **<300 mL**, an adequate remnant and negative nodes.

### IVC thrombus

- It occurs in ~6–10%. Below the hepatic veins, do upfront nephrectomy with **en bloc** thrombectomy (piecemeal means stage III).
- **At or above the hepatic veins, give pre-operative chemotherapy** to avoid bypass.

## 6 · Relapse and late effects

| Relapse group | Definition | Salvage |
|---|---|---|
| Standard | Non-anaplastic, previously VCR/AMD only | 70–80% |
| High | Non-anaplastic, after ≥ 3 drugs | 40–50% |
| Very high | Anaplastic/blastemal | ~10% |

- Late effects track **RT and anthracycline**: heart failure follows DOX (rare below **150 mg/m²**). Renal failure after unilateral nephrectomy is only **0.6%** at 20 y.

## 7 · Other renal tumours

| Tumour | Key facts | Management |
|---|---|---|
| Congenital mesoblastic nephroma (CMN) | Commonest **neonatal** renal tumour; ETV6–NTRK3 fusion in the cellular type | Radical nephrectomy alone; adjuvant only if incompletely resected |
| Clear cell sarcoma (CCSK) | 3%; **bone** and brain metastases; late relapse; never bilateral | DOX-containing chemotherapy; RT even at stage I |

| Tumour | Key facts | Management |
|---|---|---|
| Rhabdoid tumour (RTK) | 2%; **most lethal**; SMARCB1/INI1 loss; associated brain tumours | No standard; tazemetostat on trial |
| Cystic nephroma / cystic partially differentiated nephroblastoma (CPDN) | Identical on imaging; CPDN has blastema in the septa; DICER1 | Excision cures |
| RCC | Outnumbers WT above 13–14 y. **Translocation RCC (TFE3/TFEB, Xp11.2)** is commonest. | Complete resection; resistant to chemotherapy and RT |
| Angiomyolipoma | Tuberous sclerosis; bleeding risk if > 3 cm and growing | Mammalian target of rapamycin (mTOR) inhibitor first; embolise for bleeding |

**KEY POINT**

Age points to diagnosis: a neonate has CMN, a toddler WT or rhabdoid, an adolescent RCC. Metastatic site sets staging tests: lung (WT), bone (CCSK), brain (rhabdoid/CCSK).

## 8 · Neuroblastoma

*Biology, not radical surgery, decides outcome.*

- It is the commonest **extracranial solid tumour** and commonest infant malignancy; median age is 19 months, 50% are adrenal and ~70% are metastatic at diagnosis.
- Paraneoplastic syndromes include watery diarrhoea from vasoactive intestinal peptide and **opsoclonus–myoclonus–ataxia**.
- Urinary vanillylmandelic acid and homovanillic acid (**VMA/HVA**) are raised in **90–95%**. **Calcification and vessel encasement** favour neuroblastoma over WT.
- 123I-meta-iodobenzylguanidine (MIBG) scanning is required for staging; MRI is best for intraspinal extension. Screening caused **overdiagnosis** without fewer deaths.

| International Neuroblastoma Risk Group (INRG) stage | Definition |
|---|---|
| L1 | Localised, **no** image-defined risk factor (IDRF) |
| L2 | Localised, ≥ 1 IDRF (vessel/nerve encasement) |
| M | Distant metastases |
| MS | Age **<18 months**; skin, liver, marrow (< 10%) only; no bone (INSS 4S: < 12 months) |

- INRG is imaging-based, before treatment; the International Neuroblastoma Staging System (INSS) depends on resection and is surgeon-dependent.
- **MYCN amplification** (> 10 copies; ~20%) means high risk at any stage. M at ≥ 18 months is always high risk.
- Adverse features are 1p36 deletion, 11q LOH, 17q gain and **diploidy**; favourable features are **hyperdiploidy**, TrkA expression and age < 18 months.

**KEY POINT**

Trap: diploid is bad, hyperdiploid good. Stage 4S/MS has OS 80–88% with frequent regression; infants < 2 months may die of massive hepatomegaly or coagulopathy.

- **Low risk**: resection alone (stage 1 > 90% disease-free); chemotherapy only for symptoms, progression or adverse biology. Never remove the kidney or vital structures.
- **4S**: resecting the primary is not mandatory. **Intermediate**: chemotherapy, then resection.
- **High risk**: induction chemotherapy, delayed surgery after 3–4 courses, myeloablative consolidation with autologous stem-cell rescue, RT 15–30 Gy, and **13-cis-retinoic acid with anti-GD2 antibody**.
- In stage 4, radical resection improves local control but not survival.
- **Cord compression**: give chemotherapy first; laminectomy only for progressive deficit, as it causes scoliosis.

> **KEY POINT**
>
> Surgical value runs inversely with biology: favourable tumours are cured by little or no resection; unfavourable ones are not rescued by radical resection, so vital structures are never traded for margin.

## 9 · Adrenocortical tumours

*See the paper answer **Adrenal mass evaluation, incidentaloma and adrenocortical carcinoma** (Section 14, Ch 146).*

**Also, from the condensed book (not in the paper answer):**

- Offer **TP53** testing to every child. Androgen excess causing **virilisation** is the commonest presentation.
- Work-up includes DHEA-S, testosterone, 17-hydroxyprogesterone, cortisol tests, a **urinary steroid profile**, metanephrines, CT/MRI and chest CT.
- Weight > 200 g and spill are adverse. Give **perioperative steroids** for cortisol-secreting tumours.

> **KEY POINT**
>
> Why open? Cure needs complete removal of an intact tumour; capsular rupture seeds the retroperitoneum, with poor salvage.

## 10 · Landmark trials

| Trial | Design | Result | Take-home |
|---|---|---|---|
| AREN0532/0533 | 1p/16q LOH treated with regimen M | Stage III–IV event-free survival (EFS) ~90% vs 61% | Biology-driven intensification |
| AREN0533 (lung) | Lung nodule response at week 6 | RT omitted, EFS and overall survival (OS) preserved | Response-adapted RT omission |
| AREN0534 | Bilateral WT: 3-drug, surgery by week 12 | EFS 82%, OS 95% (vs 56%/81% in NWTS-5) | Fixed clock, maximise NSS |
| COG P9641 | Low-risk neuroblastoma, surgery alone | EFS ~89%, OS ~97% | Chemo only if symptomatic |
| COG anti-GD2 | Anti-GD2 + cytokines + retinoid vs retinoid | Antibody superior | Immunotherapy standard in high risk |

| Trial | Design | Result | Take-home |
|---|---|---|---|
| German NB97 | Extent of resection, stage 4 > 18 months | No EFS/OS benefit | Radical resection not essential |

# 64 Paediatric Urological Oncology: Pelvis, Bladder, Testis and Genitalia

## 1 · Rhabdomyosarcoma (RMS): biology

*Fusion status, not the microscope, now sets risk.*

- RMS is the **commonest childhood sarcoma**; 20–30% are genitourinary (GU), at bladder/prostate (BP), paratesticular (PT) or vulvovaginal (VV) sites. GU-RMS 5-y survival is **>80%**.
- Syndromic cases (Li–Fraumeni, DICER1, neurofibromatosis 1, Costello) are usually embryonal; diagnosis under 3 y should prompt a germline search.

| | **Embryonal (ERMS)** | **Alveolar (ARMS)** |
|---|---|---|
| Site | GU (commonest); the **botryoid** variant protrudes from bladder or vagina | Trunk, extremities, perineum |
| Age | Younger (~70% < 10 y) | Older, adolescents |
| Fusion | Usually negative | ~80% **PAX3/7–FOXO1** [t(2;13), t(1;13)] |
| Prognosis | Better | Fusion-positive worse; fusion-negative ARMS behaves like ERMS |

**KEY POINT**

Fusion status predicts outcome better than histology, so trials assign risk by **fusion status**.

## 2 · Presentation and work-up

- BP tumours cause haematuria or retention; PT tumours are a painless mass separate from the testis; VV tumours cause bloody discharge or a prolapsing mass.

**ALGORITHM**

1. Start with US.
2. CT or MRI of the primary and its nodal basin: BP, pelvic and retroperitoneal nodes up to the renal vessels; PT, retroperitoneal; VV, inguinal, pelvic and retroperitoneal.
3. **Chest CT** (lung is the commonest metastatic site).
4. Bone scan and bilateral marrow sampling; omit these in fusion-negative ERMS without nodal or lung disease.

## 3 · Stage, group and risk

The **stage** is pretreatment tumour–node–metastasis plus site. The **clinical group** records what the surgeon left. The **risk group** adds fusion status and age.

| **Children's Oncology Group (COG) stage** | **Site** | **Size / nodes** |
|---|---|---|
| 1 | Favourable: **paratesticular, female GU**, orbit, head–neck | Any, M0 |
| 2 | Unfavourable: **bladder/prostate**, extremity, parameningeal, trunk | ≤ 5 cm, N0/NX |

| Children's Oncology Group (COG) stage | Site | Size / nodes |
|---|---|---|
| 3 | Unfavourable | ≤ 5 cm N1, or > 5 cm any N |
| 4 | Any | M1 |

N1 means a node ≥ 1 cm in short axis. BP-RMS is never below stage 2.

| Clinical group | Definition |
|---|---|
| I | Localised, completely resected, nodes negative |
| IIa / IIb / IIc | IIa: microscopic residual. IIb: involved nodes resected, most distal node negative. IIc: microscopic residual or most distal node positive. |
| III | Gross residual or **biopsy only** |
| IV | Distant metastases |
| Risk | Criteria |
| Low | Fusion-negative; stage 1 group I–II (III orbit only), or stage 2 group I–II |
| Intermediate | Fusion-negative stage 1–2 group III (non-orbit), stage 3 group I–II; any fusion-positive non-metastatic; stage 4 fusion-negative < 10 y |
| High | Stage 4 fusion-positive, or ≥ 10 y |

**KEY POINT**

An organ-preserving **biopsy** places BP-RMS in group III, which means **intermediate risk**; PT-RMS (resected by orchidectomy) is almost always low risk. Age < 1 y or > 10 y is independently adverse. Trap: most distal node positive = IIc, not IIb.

## 4 · Treatment principles

*Chemotherapy does most of the work; surgery aims to save the organ.*

- **Chemotherapy is universal**; local control is by radiotherapy (RT) and/or surgery. Complete resection (except PT) is done only if the organ is preserved.
- Primary exenteration fell from almost all cases in IRS-I to **13%** in IRS-IV, with overall survival unchanged.
- **Pretreatment re-excision** (before chemotherapy, without organ loss) improves GU survival and can change the group. **Delayed primary excision** cannot change the group and is done only if a negative-margin resection is likely.

### Bladder/prostate RMS

- Biopsy is preferably **cystoscopic**; relieve obstruction with a **urethral catheter** first and avoid a suprapubic tube (seeding).
- An **end-of-therapy residual mass** is usually stroma or mature rhabdomyoblasts, so observe it if stable. Viable tumour after chemoradiation may need cystectomy.
- Only ~40% of preserved bladders are "normal", so do **urodynamics at 6 months** and renal US yearly for ≥ 10 y.

### Paratesticular RMS

- Do a **radical inguinal orchidectomy**; after a scrotal approach, excise the remaining cord inguinally.
- Do an **ipsilateral nerve-sparing retroperitoneal lymph node dissection (RPLND)** before chemotherapy if **age ≥10 y** or nodes are radiologically N1.

> **KEY POINT**
>
> Why age, not CT? Older boys have far more retroperitoneal disease and CT has a high false-negative rate for nodes.

### Female GU, perineal, infants

- Vaginoscopy and biopsy; chemotherapy with or without brachytherapy controls most, and exenteration is reserved for unresponsive disease.
- Perineal tumours are mostly fusion-positive alveolar, with 5-y OS ~50%. RT causes most late effects, including bladder and sexual dysfunction, infertility and second cancers.

## 5 · Other bladder tumours

- **Urothelial** tumours in children are mostly low grade; treat by transurethral resection and follow with US. A recurrence should prompt a search for a syndrome (Lynch).
- **Spina bifida/exstrophy**: lifetime bladder cancer risk is **2–4%**, aggressive, and **not increased by augmentation**; evaluate haematuria promptly.

| Benign lesion | Key point | Treatment |
|---|---|---|
| Inflammatory myofibroblastic tumour | ALK rearrangement; intermediate malignancy | Complete resection |
| Nephrogenic adenoma | Follows trauma, infection, surgery or transplant | TURBT and treat the cause; recurs |
| Eosinophilic cystitis | Biopsy essential; blood eosinophilia often absent | Steroids, antihistamines |
| Haemangioma | Severe haematuria; Klippel–Trenaunay, Sturge–Weber | Propranolol, laser; partial cystectomy |

## 6 · Testicular tumours

*Prepubertal tumours are mostly benign, so the testis can often be spared.*

- Before puberty, **teratoma** is the commonest tumour (benign) and **yolk sac tumour (YST)** the commonest malignant one.
- Work-up is Doppler US with LDH, **α-fetoprotein (AFP)** and β-hCG; an **"onion-skin" appearance suggests epidermoid cyst**.

| | AFP | β-hCG |
|---|---|---|
| Source | **YST invariably** | Embryonal carcinoma, mixed tumours; rare before puberty |
| Half-life | **5 days** | **24 h** |
| Caveat | Normally high in infancy; reaches adult level (< 10 ng/mL) by **6–9 months** | — |

**ALGORITHM**

1. Use an inguinal approach.
2. If markers are normal and US looks benign, do a **testis-sparing partial orchidectomy** with frozen section.
3. If benign, close.
4. If frozen section shows germ cell malignancy, or AFP is raised, do a radical inguinal orchidectomy.

- Prepubertal teratoma needs partial orchidectomy and **no follow-up**; postpubertal teratoma is malignant and needs radical orchidectomy.
- **YST**: radical inguinal orchidectomy; **90%** are stage I, managed by **surveillance without RPLND or chemotherapy**. Stage III–IV disease gets cisplatin, etoposide and bleomycin.
- **Stromal** tumours (Leydig, Sertoli, juvenile granulosa) are treated by partial orchidectomy if markers are normal. Testicular adrenal rests in congenital adrenal hyperplasia respond to more steroid.

### Disorders of sex development (DSD)

Tumour risk rises with Y material in dysgenetic gonads: high in gonadal dysgenesis, Frasier, Denys–Drash and partial androgen insensitivity (PAIS); intermediate in complete androgen insensitivity (CAIS); low without Y. Gonadoblastoma is premalignant.

> **KEY POINT**
>
> Trap: **complete** androgen insensitivity is intermediate (not high) risk; partial is high.

## 7 · Other pelvic and genital tumours

- **Sacrococcygeal teratoma** causes neurogenic bladder in 36–51%, so follow with renal–bladder US and urodynamics for higher types. The **Currarino triad** also causes neurogenic bladder.
- A Sertoli–Leydig ovarian tumour should prompt **DICER1 screening**.
- **Urethral prolapse** mimics RMS in prepubertal girls; treat with topical oestrogen.

# ★ Putting it together: common paediatric decisions

## A · VUR after febrile UTI

**ALGORITHM**

1. Do renal–bladder US; add a voiding cystourethrogram (VCUG) if UTIs recur or US is abnormal.
2. Treat bladder and bowel dysfunction (BBD) and constipation first.
3. Low risk (grade I–II, older boy) is observed; high risk (grade ≥ III, girl, BBD, scars, infant) gets continuous antibiotic prophylaxis.
4. Breakthrough UTI or new scars: inject or reimplant (see Ch 41, 53).

## B · Dilated upper tract

| Condition | Key test | Operate when | Operation |
|---|---|---|---|
| Pelviureteric junction (PUJ) obstruction | MAG3 diuretic renogram | Function falling or < 40%, symptoms | Anderson–Hynes pyeloplasty |
| Primary megaureter | US, MAG3, VCUG | Falling function, infection; most resolve | Tapered reimplant |
| Ureterocele / ectopic ureter | US, VCUG, functional scan | Obstruction, infection, incontinence | Incision, heminephrectomy or ureteroureterostomy (see Ch 52) |
| Multicystic dysplastic kidney | US and DMSA scan | Rarely; most involute | Observe |

## C · Undescended testis

**ALGORITHM**

1. **Palpable**: orchidopexy at **6–18 months**; no imaging.
2. **Non-palpable**: examination under anaesthesia, then laparoscopy to find and fix the testis or confirm it is absent.
3. **Bilateral non-palpable, or with hypospadias**: DSD work-up first (see Ch 59).

## D · Acute scrotum

*Never let a scan delay exploring torsion.*

| | Testicular torsion | Appendage torsion | Epididymitis |
|---|---|---|---|
| Typical | Sudden pain, high-riding testis | Upper-pole tenderness, "blue dot" | Gradual onset |
| Action | **Explore at once**; fix both sides, remove if non-viable | Analgesia, rest | Culture; antibiotics if infected |

## E · Newborn with atypical genitalia

> **KEY POINT**
>
> Bilateral impalpable gonads with atypical genitalia = salt-wasting congenital adrenal hyperplasia until proven otherwise. Check electrolytes, 17-hydroxyprogesterone and karyotype; crisis comes at days 5–7.

## F · Wilms tumour vs neuroblastoma

| | **Wilms tumour** | **Neuroblastoma** |
|---|---|---|
| Age | Median 3.5 years | Median ~18 months |
| Imaging | Intrarenal claw sign; IVC extension | Calcification, vessel encasement; crosses the midline |
| Markers | WT1, 11p15; WAGR, Denys–Drash | VMA/HVA raised in 90–95%; MYCN amplification = high risk |
| Strategy | COG: nephrectomy first; SIOP: chemotherapy first | By INRG risk group |

# Other paper questions in this section

*Asked in the papers but not tied to a specific place in the book above.*

## SAQ TIP (Snodgrass) repair for hypospadias

Paper answer · P1 · P3 · merged from 2

*Asked as:* P3 SAQ "Snodgrass (TIP) repair for hypospadias, including extended indications" · P1 SAQ "TIP repair in hypospadias"

### Introduction

- "Shodgrass" in one P3 wording is taken as the Snodgrass TIP repair, and "distal hypos." as distal hypospadias.
- **TIP (tubularised incised plate) urethroplasty** was described by Snodgrass in 1994, built on Rich's hinging incision (1989).
- Designed for distal hypospadias, it is now used for **over 90% of distal and about 80% of midshaft repairs**.
- Distal hypospadias forms 70–85% of cases.

### Principle

- A **deep midline relaxing incision** of the urethral plate widens a narrow plate so that it can be tubularised **without tension**.
- The incised raw area **heals by re-epithelialisation** without a graft, giving a **vertical slit-like meatus**.

### Indications, prerequisites and assessment

- Distal and midshaft hypospadias, and redo cases with a healthy plate, when curvature is mild (**<30°** after degloving) and corrected by degloving or dorsal plication **without dividing the plate**.
- Supple, well-vascularised plate of adequate width and depth, and an adequate glans.
- **Glans width <14 mm** raises complications (glans width ≥14 mm is needed): give preoperative testosterone to enlarge it.
- **POST (plate objective scoring tool):** a low ratio (AB/BC ratio) of neomeatus length to glans fusion length means a poor plate, which favours a dorsal inlay graft.
- Timing: 6–12 months of age.
- Not for severe curvature needing plate division; proximal TIP is largely abandoned.

### Extended indications

| Indication | Rationale | Caveat |
|---|---|---|
| **Midshaft hypospadias** | Same principle | Good results |
| **Proximal hypospadias with mild curvature**, good plate | Single stage; no flaps | More fistula, stricture, recurrent curvature; Campbell-Walsh authors now prefer staging |
| **Redo or failed repairs** with preserved plate | Plate often survives | Worse than primary; inlay graft if scarred |
| **MIP (megameatus intact prepuce)** | Wide, deep plate | GAP (glans approximation procedure) alternative |

| Indication | Rationale | Caveat |
|---|---|---|
| **Narrow or shallow plate**: TIP + **dorsal inlay graft** ("Snodgraft") | Preputial or buccal graft fills the incision, so it cannot contract | Less meatal stenosis |
| **Adults, circumcised** boys | No preputial tissue needed | Success similar |

- **Also know:** in redo cases with poor blood supply a dorsal buccal inlay beats onlay or tube; incise-and-graft is also used in adult urethral stricture (Asopa dorsal inlay) and female urethroplasty.

### Contraindications

- Curvature >30° needing plate division; absent or scarred plate; penoscrotal transposition. Use staged repair instead (Bracka, STAG [staged tubularised autograft], Byars).

### Operative steps

1. Glans stay (traction) suture; tourniquet.
2. U-incision around the plate and meatus; circumscribe and deglove; artificial erection.
3. Separate (mobilise) the glans wings deeply from the plate.
4. **Midline incision of the plate** from inside the meatus to the end of the plate, down to the corpora.
5. Tubularise over a **6–8 Fr stent** with subepithelial 6-0/7-0 absorbable sutures, from distal to proximal, placing the first stitch about 3 mm (2–3 mm) short of the plate end to leave an oval meatus.
6. **Dartos flap** (dorsal preputial) over the neourethra as a barrier layer.
7. Two-layer glansplasty; mucosal collar; skin closure with Byars flaps, or preputioplasty.
8. Stent 5–7 days; compressive dressing; oxybutynin for bladder spasm.

### Complications and outcomes

- Pfistermuller meta-analysis (2015, 4,675 boys): complications 10.6%, reoperation 4.5%; **fistula 5.7%**, **meatal stenosis 3.6%**, stricture 1.3%.
- **Meatal stenosis is commoner than after Mathieu**; glans dehiscence also occurs.
- Distal repair success 83–95%.
- **Proximal and redo cases do significantly worse**; a dartos barrier layer improves results and reduces fistula.

### Modifications

- **Dorsal inlay graft (Snodgraft)**: inner preputial skin graft in the incised bed for a shallow plate.
- Lateral plate incisions to 12 mm width.
- **Also know:** MAGPI (meatal advancement and glanuloplasty incorporated), Mathieu, Thiersch–Duplay, onlay flap.

**Diagram to draw:** cross-sections showing plate incision, inlay graft, tubularisation and dartos cover.

**Key points to remember:**

- Snodgrass 1994; incise and tubularise; heals by epithelialisation.
- Slit-like meatus; dartos barrier reduces fistula.
- Glans <14 mm and poor plate raise complications.
- Fistula 5.7%, meatal stenosis 3.6% (Pfistermuller 2015).
- Extended to midshaft, selected proximal, redo, MIP and adults.
- Key condition: curvature <30° without plate division.
- Poor plate: dorsal inlay graft ("Snodgraft").
- Proximal and redo TIP fail more often.

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## SAQ Glandular hypospadias

Paper answer · P1

**Type:** SAQ (10 marks)
**Source:** Practice Set 11 Paper 1, Q3b

*"Glandular" is read as glanular hypospadias: meatus on the glans.*

### Definition

- The mildest **distal (anterior)** hypospadias: the meatus opens on the ventral glans, proximal to the normal tip.
- Hypospadias occurs in 1 in 150–300 male births; distal forms are 70–85%.
- Usual features: **dorsal hooded foreskin**, ventral glans tilt, little or no curvature. **MIP (megameatus intact prepuce)** variant (up to 5%): wide meatus with normal foreskin.

### Embryology

- Urethral folds close from proximal to distal during weeks 8–16 under **DHT (dihydrotestosterone)**.
- Glanular urethra: ectodermal ingrowth (classic) or endodermal plate ("two-zipper", current view). Arrest at the last step gives a glanular meatus.

### Evaluation

- Examine meatus, plate, glans size, curvature, torsion; retract the foreskin before any **newborn circumcision**.
- **GMS (glans–meatus–shaft) score** (each 1–4, total 3–12) predicts complications.
- No workup needed for isolated distal hypospadias; DSD (difference of sexual development) workup if a testis is undescended.

### Indications and timing

- Deflected or spraying stream, meatal stenosis, curvature, cosmesis. A straight penis that voids standing may need no surgery; counsel parents.
- **Age 6–18 months** (AAP, American Academy of Pediatrics, 6–12 months).

### Operations

| Procedure | Suitable for | Key point |
|---|---|---|
| Urethromeatoplasty | Stenotic meatus, blind pit | Heineke–Mikulicz: vertical incision, horizontal closure |

| Procedure | Suitable for | Key point |
|---|---|---|
| **MAGPI** (meatal advancement and glanuloplasty, Duckett 1981) | Glanular meatus, mobile skin | Advance meatus, two-layer glanuloplasty; retracts if used more proximally |
| MIV (M inverted V) glansplasty | Glanular or coronal | Avoids MAGPI retraction |
| **GAP** (glans approximation) | Wide deep groove, fish-mouth meatus (MIP) | Tubularise and close glans wings |
| **TIP** (tubularised incised plate, Snodgrass 1994) | Narrow plate | Most used; slit-like meatus |
| Mathieu flip-flap; Thiersch–Duplay | Selected | Stenosis, fistula |

- Preputioplasty or circumcision; dartos barrier layer.

### Complications

- Meatal stenosis or retraction, fistula, **glans dehiscence**, residual curvature, poor cosmesis. Distal repairs succeed in about 80%.

**Diagram to draw:** MAGPI steps (meatal advancement, glans wings closed beneath).

> **Key points to remember:**
> - Glanular = most distal hypospadias; dorsal hood, minimal curvature.
> - MAGPI only for a truly glanular meatus; GAP for MIP; TIP is the workhorse.
> - Repair at 6–18 months; may be left alone if function is normal.

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## SAQ Retractile testis?

Paper answer · P2

**Type:** SAQ (10 marks)
**Source:** 2020-Summer Paper 2, Q3b

### Definition

- A **descended testis** that moves up out of the scrotum because of an active **cremasteric reflex**. It can be brought to the bottom of the scrotum by hand and **stays there**, at least for a while, after release.
- It is not undescended, but may be a mild form of the same disease and can later become an acquired undescended testis.

### Epidemiology

- Common: suprascrotal in up to one-third of 4-year-olds; 11–15% of boys up to 11 years. Nearly all are scrotal after 12 years.

### Examination (diagnosis is clinical)

- Warm room and hands, relaxed child, soap or gel; supine frog-leg (thigh abduction inhibits the cremasteric reflex), cross-legged or squatting.
- One hand sweeps from the internal ring down the canal; the other pulls the testis to the scrotal bottom. **Sustained gentle traction tires the cremaster.**
- Note size, mobility, contralateral testis and hernia.

| Testis | Finding | Action |
|---|---|---|
| **Retractile** | Reaches scrotal bottom and stays | Observe yearly |
| Gliding (unstable high scrotal) | Pulled down but springs back | Treat as undescended |
| Undescended (palpable) | Cannot reach mid-scrotum | Orchidopexy by 18 months |
| Ascending (acquired) | Previously scrotal, now not | Orchidopexy |

### Natural history

- Ascent: 2.6% in a prospective birth cohort; 7–32% in specialist series.
- Operated retractile testes show small volume, patent processus vaginalis and epididymal anomalies, suggesting a continuum with cryptorchidism.

### Management

- No ultrasound (adds nothing); no hormones (hCG [human chorionic gonadotrophin] neither separates nor treats).
- Reassure parents; **examine yearly** (AUA [American Urological Association] 2014) until puberty, because some testes ascend.
- **Orchidopexy** if it ascends, cannot be brought down, rests mostly high or becomes smaller. If unsure, examine under anaesthesia.

**Diagram to draw:** inguinoscrotal sketch: testis drawn by traction to the scrotal bottom.

> **Key points to remember:**
> - Retractile = comes down and stays down.
> - Gliding testis springs back: treat as undescended.
> - Yearly examination until puberty; operate if it ascends.

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## SAQ Megameatus intact prepuce

Paper answer · P2

**Type:** SAQ (10 marks)
**Source:** Practice Set 15 Paper 2, Q3f

### Definition

- **MIP (megameatus intact prepuce)** is a hypospadias variant (up to 5%) with a **wide, "fish-mouth" glanular or coronal meatus**, a **completely formed prepuce** and **no significant curvature**. Described by Duckett and Keating (1989).

### Embryology

- Normally the prepuce closes ventrally only after the urethral folds fuse (about 20 weeks), so 95% of hypospadias have a hooded, incomplete foreskin.
- In MIP the foreskin is complete, showing that **preputial and urethral development can dissociate**.

### Clinical features

- Hidden under a normal foreskin; often found on **retraction before circumcision** or during circumcision.
- Wide, deep glans groove with a large, patulous meatus; well-developed spongiosum.
- Symptoms: spraying or deflected stream; cosmetic concern.

- **Advice:** retract the foreskin before every newborn circumcision. If found, the circumcision may be completed; prior circumcision does not harm later repair.

### Evaluation

- Examine meatal site and size, glans and plate, curvature (artificial erection at surgery). GMS (glans–meatus–shaft) score.
- No routine upper-tract imaging for isolated distal hypospadias.

### Management

- **Indication:** cosmesis and a straight stream; repair is optional. Usual timing 6–18 months.
- **Pyramid procedure (Duckett and Keating):** 1. Glans traction suture; retract prepuce. 2. Incise around the megameatus in a **pyramid shape**, apex on the ventral shaft. 3. Dissect the wide distal urethra free of the glans wings. 4. **Excise a ventral wedge** of redundant urethra to taper it. 5. Tubularise over a catheter; cover with dartos. 6. Approximate glans wings in two layers (glansplasty); preserve or reconstruct the prepuce.
- **Alternatives:** GAP (glans approximation procedure; pyramid = GAP with intact prepuce), TIP (tubularised incised plate) urethroplasty, Thiersch–Duplay tubularisation, Mathieu.

### Complications

- Fistula, meatal stenosis or regression, glans dehiscence; generally low after distal repair.

**Diagram to draw:** fish-mouth glanular meatus with intact prepuce, and the pyramid incision.

> **Key points to remember:**
> - Wide glanular meatus, intact prepuce, no curvature; up to 5% of hypospadias.
> - Often found at circumcision; retract foreskin first.
> - Pyramid repair: dissect, taper and tubularise; glansplasty.

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## SAQ Techniques of chordee correction

Paper answer · P2

**Type:** SAQ (10 marks)
**Source:** Practice Set 17 Paper 2, Q3b (related: Practice Set 10 Paper 1, Q3d)

### Definition and causes

- Chordee: **ventral penile curvature**, with or without hypospadias. It, not meatal position, decides one-stage versus staged repair.
- Causes: tethered ventral skin, fibrous dysplastic dartos and Buck's fascia, short urethral plate, **corporal disproportion**.
- Devine–Horton (1973) graded the dysplastic layers: I spongiosum, II Buck's and dartos, III dartos only, IV corporal disproportion.

### Assessment

- Parents' photographs; **artificial erection** in theatre: Gittes–McLaughlin (tourniquet at the base plus saline), corporal compression, or prostaglandin. A tourniquet can hide basal curvature.
- Grade: mild <30°, moderate 30–45°, severe ≥45° (goniometer; eye underestimates).

### Stepwise correction

```
Deglove to penoscrotal junction + excise dysplastic dartos → erection test ├─ Straight → urethroplasty ├─ Residual <30° → dorsal plication └─ >30° → divide urethral plate → still >30° → ventral lengthening (stage)
```

| | Dorsal shortening (plication) | Ventral lengthening (corporotomy ± graft) |
|---|---|---|
| Methods | **Nesbit:** excise dorsal tunical ellipses, close transversely. **Baskin:** single midline plication in the **12 o'clock nerve-free zone**. Heineke–Mikulicz dorsal incision closed transversely; 16-dot plication | 1–3 transverse **full-thickness corporotomies** (STAG, staged tubularised autograft); single **9-to-3 o'clock corporotomy** with graft: **dermis, tunica vaginalis**, SIS (small intestinal submucosa); dermal graft 20% larger than defect |
| Use | ≤15–30° | Moderate–severe after plate division |
| Recurrence | 28% (one series) | 9% |
| Risk | Shortening; dorsal nerve injury off midline | Longer, staged; theoretical erectile dysfunction |

- **Plate division** (transection) releases a short plate; not dividing it is an independent risk factor for recurrence. Repair is then staged.
- **Also know:** "fairy cuts" (superficial tunical incisions); lateral or dorsal curves need multipoint plication; correct **torsion** by dartos flap rotation.
- **Rule:** correct fully the first time and repeat the erection test before urethroplasty. Curvature may recur at puberty.

**Diagram to draw:** penile cross-section with dorsal nerves at 11 and 1 o'clock and midline plication at 12.

> **Key points to remember:**
> - Artificial erection after degloving decides the technique.
> - <30°: dorsal plication; >30°: divide plate ± ventral graft and stage.
> - Plication recurrence 28% vs grafting 9%.

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## SAQ Congenital ureteral anomalies

Paper answer · P3

**Type:** SAQ (10 marks)
**Source:** 2017-Summer Paper 3, Q3f

### Embryology basis

- The **ureteric bud** arises from the **mesonephric (wolffian) duct** at about week 5 and induces the metanephric blastema.
- Anomalies arise from **abnormal budding**, **failed distal canalisation**, **abnormal muscle maturation** or **persistent embryonic vessels**.

### Classification

| Category | Anomaly | Key points |
|---|---|---|
| **Number** | **Duplication**: incomplete (bifid, Y-ureter) or complete | About 1 in 125; unilateral 6× bilateral. Lower pole: VUR (vesicoureteric reflux), PUJ (pelviureteric junction) obstruction. Upper pole: ectopia, ureterocele |

| Category | Anomaly | Key points |
|---|---|---|
| **Termination (orifice)** | **Ectopic ureter** | Opens at or below the bladder neck. **Females**: bladder neck to vagina, uterus; **continuous dribbling with normal voiding**. **Males**: always above external sphincter (posterior urethra, seminal vesicle, vas); **never incontinent**; epididymitis. Zinner syndrome |
| **Structure** | **Ureterocele** | Cystic dilation of distal submucosal ureter (Chwalle membrane). Intravesical vs ectopic (Glassberg); cecoureterocele (Stephens) |
| | **PUJ obstruction** | Intrinsic aperistaltic segment (infants) or crossing lower-pole vessel (older children) |
| | **Megaureter** (≥7 mm) | Obstructed, refluxing, non-refluxing non-obstructed, refluxing-obstructed; primary obstructive = adynamic distal segment |
| | **Primary VUR** | Short submucosal tunnel, lateral orifice |
| **Position** | **Preureteral (retrocaval) vena cava** | Persistent subcardinal vein traps the right ureter; fishhook (S) shape at L3–L4; type I obstructs, type II does not |
| | Retroiliac ureter; herniation | Rare |

### Weigert–Meyer law

- In complete duplication the **upper-pole ureter opens lower and more medial** (ectopic, obstructs, ureterocele); the **lower-pole ureter opens higher and lateral** (refluxes).
- Reason: the upper-pole bud arises more cephalad on the wolffian duct, is incorporated later and ends more caudally.
- **Also know:** blind-ending ureter, inverted-Y ureter, triplication, congenital stricture, ureteral valves, fibroepithelial polyp.

### Evaluation

- Ultrasound, VCUG (voiding cystourethrogram), **DMSA (dimercaptosuccinic acid)** scan, MR (magnetic resonance) urography.

### Management principles

- **Ureterocele:** endoscopic incision, upper-pole heminephrectomy or ureteroureterostomy.
- **Ectopic ureter:** heminephroureterectomy if non-functioning; otherwise ureteroureterostomy or reimplantation.
- **PUJ obstruction:** pyeloplasty. **Megaureter:** observe most; taper and reimplant (5:1) if function falls.
- **Retrocaval ureter:** divide, bring anterior to the cava, excise the aperistaltic segment, reanastomose.

**Diagram to draw:** complete duplication showing Weigert–Meyer orifice positions.

> **Key points to remember:**
> - Upper pole obstructs (ectopia, ureterocele); lower pole refluxes.
> - Female ectopic ureter: continuous dribbling with normal voiding; male never incontinent.
> - Retrocaval ureter is a venous anomaly; always right-sided.

# Rapid Revision Sheet · Section 4

### Primary VUR: VUJ embryology/anatomy, pathophysiology, management (incl. bilateral, adult grade 3), reimplantation techniques and STING (LAQ)

- Low (early) bud → high lateral orifice → short tunnel → VUR plus dysplasia; high (late) bud → ectopia.
- One embryological fault: lateral orifice, short tunnel, dysplasia.
- Tunnel length to diameter 5:1 normal versus 1.4:1 refluxing (Paquin); adult intramural ureter 1.2–2.5 cm.
- Dysplasia (boys, congenital) versus scar (girls, infection); sterile reflux rarely scars.
- Grade V loses papillary impressions; grade IV–V and bilateral grade III rarely resolve; bilateral grade IV persists most; bilateral VUR is high risk.
- RIVUR: CAP halves recurrent UTI, not scarring.
- Treat BBD and give CAP first; correct for breakthrough febrile UTI, new scars or persistent grade IV–V.
- Trend: risk-stratified care, fewer VCUGs, BBD first, endoscopic or robotic correction.
- STING: O'Donnell and Puri 1986; needle 2–3 mm below orifice at 6 o'clock.
- Dx/HA 80–250 µm; HIT better than STING; double HIT about 90%; Elder 2006: grade IV 63% after one injection.
- Watch for late obstruction; calcified mound mimics a stone.
- Open reimplantation 95.9%; Cohen cross-trigonal (over 98%) suits bilateral VUR through one cystotomy.
- Bilateral extravesical (Lich–Gregoir) repair risks retention up to 10%.
- Paquin: tunnel 5:1; combined, suprahiatal reimplant. Gil-Vernet: trigonoplasty, medial advancement, no tunnel; for low-grade bilateral VUR.
- 34–38% of scarred children become hypertensive; correction does not lower BP; scars bring 8× pre-eclampsia risk.
- In adults the scar, not the reflux, carries the risk; exclude secondary causes first; correct for recurrent pyelonephritis, new scars or pregnancy plans.

### Anomalies of renal artery. (SAQ)

- Single artery in 71–85%; accessory arteries usually lower pole.
- Crossing vessel can cause PUJ obstruction.
- Treat aneurysm >2 cm or in pregnancy.

### Pelvicalyceal anomalies other than PUJ obstruction (SAQ)

- Diverticulum: Type I to calyx, Type II to pelvis; stones in 40%.
- Megacalycosis: non-obstructive, 12–20 calyces, no treatment.
- Infundibulopelvic stenosis: bilateral, renal failure risk.

### Autosomal recessive polycystic kidney disease. (SAQ)

- PKHD1, chromosome 6, fibrocystin; always congenital hepatic fibrosis.
- Huge homogeneous echogenic kidneys with oligohydramnios.
- Neonates die of pulmonary hypoplasia; survivors develop portal hypertension.

**Medullary sponge kidney (SAQ)**

- Ectatic collecting ducts; sporadic; "bristles on a brush".
- Hypercalciuria, incomplete distal RTA, hypocitraturia.
- Potassium citrate is first-line prevention.

**Post-obstructive diuresis (SAQ)**

- Unilateral about 2%, bilateral about 50%.
- Mechanisms: aquaporin-2 loss, volume expansion, urea osmotic diuresis.
- Replace about 75% of output with 0.45% saline; do not overcorrect.
- Gradual decompression has no role.

**Ureterocele (SAQ)**

- Upper pole of duplex; never wholly urethral; Chwalla membrane.
- Intravesical vs ectopic predicts TUI success.
- TUI relieves obstruction in 78–97%; new reflux 0–50%.

**Embryology and management of ectopic ureter. (LAQ)**

- Ectopia = orifice at or below the bladder neck; bud too cranial on the wolffian duct.
- Girls: continuous dribbling with normal voiding; boys: never incontinent, epididymitis.
- DMSA decides keep (UU/reimplant) versus remove (heminephroureterectomy).
- Protect the lower-pole blood supply; stent the recipient ureter for UU.

**Retrocaval ureter and IVC anomalous development (LAQ)**

- Infrarenal IVC = right supracardinal; suprarenal = right subcardinal; hepatic = vitelline.
- Circumaortic left renal vein is commonest (2–16%): ligate both limbs.
- Retrocaval ureter: venous anomaly (persistent subcardinal vein); right side only; 1 in 1500; male 3:1; transpose anterior to IVC.
- Type I fishhook obstructs; type II sickle often does not.
- CT urography; repair by anterior transposition and ureteroureterostomy.
- Duplicated IVC: filter both cavae; do not mistake it for nodes.

**Retroileal Ureter (SAQ)**

- Retroiliac = preureteral iliac artery; persistent ventral umbilical artery root.
- Obstruction at L5–S1; either side or bilateral.
- Look for vasal and ureteric ectopia.
- Retrocaval: right, L3–L4, fishhook, 1 in 1500.

**Primary obst. megaureter. (SAQ)**

- Megaureter ≥7 mm; POM = adynamic juxtavesical segment.
- Surgery: UTIs, progression, DRF <40% or fall ≥5%.
- Excise, taper, reimplant 5:1; success >90%.

**Proximal (perineoscrotal) hypospadias: penile embryology and management including Koyanagi repair (LAQ)**

- DHT at 8–12 weeks; opening then closing zipper; 12 o'clock nerve-free zone.
- Hypospadias (including perineoscrotal) plus undescended testis: DSD work-up.
- <30° plication; >30° divide plate ± ventral graft and stage.
- STAG/Bracka staged repair preferred; fistula commonest complication.

- Koyanagi: one stage; meatus-based parameatal foreskin flap.
- Koyanagi: plate divided; no proximal circular anastomosis.
- Koyanagi: fistula commonest; diverticulum, meatal stenosis.

**Hypospadias: present concepts, single-stage vs staged repairs, and hypospadias fistula (LAQ)**

- Classify after degloving (GMS); curvature, not meatus, decides one versus two stages.
- Curvature >30° after degloving: stage.
- TIP for distal; staged graft repair (STAG/Bracka) for severe proximal; STAG has the lowest proximal complications.
- Stage 2 after ≥6 months.
- Hypospadias plus undescended testis: DSD work-up.
- Follow through puberty; use patient-reported outcomes.
- Fistula: commonest complication, just under 10%; TIP 5.7%.
- Exclude distal obstruction; wait 6–12 months.
- Layered closure with barrier flap; recurrence about 20%.

**Testicular descent and undescended testis: evaluation, treatment algorithm and orchidopexy methods (LAQ)**

- SRY → SOX9 → Sertoli cells (AMH); Leydig cells → testosterone and INSL3.
- Transabdominal: CSL regression plus INSL3-driven gubernacular swelling.
- Inguinoscrotal: androgen via genitofemoral nerve (GFN) and CGRP; most testes scrotal after 27 weeks.
- Ectopic = off the path; commonest site superficial inguinal pouch.
- Operate from 6 months, preferably by 12 months, and by 18 months at the latest.
- No imaging, no hormones.
- Non-palpable: EUA then diagnostic laparoscopy; laparoscopy decides.
- Bilateral non-palpable: DSD work-up first (karyotype, 17-OHP, AMH) to exclude CAH/DSD.
- Subdartos pouch; Prentiss manoeuvre gives extra length.
- Fowler–Stephens relies on vasal collaterals.

**Vanishing testes (SAQ)**

- Prenatal torsion after masculinisation; 15–40% of nonpalpable testes.
- Contralateral hypertrophy (>2 mL) suggests it, but explore anyway.
- Blind-ending vessels confirm; excise remnant (germ cells 5–15%).
- Anorchia: high FSH/LH, low AMH and inhibin B.

**Urological evaluation and management of pediatric kidney transplant patient. (LAQ)**

- Bladder target: age-expected capacity at <40 cm $H_2O$; augment if <75% despite CIC.
- PUV bladder heads to myogenic failure; neuropathic bladder needs earlier reconstruction.
- Never test a bladder with a graft; cycle the defunctionalised bladder.
- Stenosis 6%, leak 3%, stones 2%, significant reflux 2%.

**Wilms tumour, including bilateral Wilms tumour (LAQ)**

- WT1 at 11p13 (WAGR, Denys–Drash); BWS at 11p15.
- Anaplasia (≥3× nuclei, hyperchromasia, abnormal mitoses) is unfavourable; TP53.
- Any spill, biopsy or positive node = stage III; sample ≥5 nodes.

- COG operates first; SIOP gives chemotherapy (VCR + AMD) first.
- Bilateral: stage V, 5–7%; no upfront nephrectomy or biopsy.
- Bilateral: COG three drugs for 6 weeks (reassess at 6 weeks), surgery by week 12, NSS; SIOP two drugs.
- AREN0534: 4-year OS about 95%.
- Bilateral NSS to avoid ESRD.

**TIP (Snodgrass) repair for hypospadias (SAQ)**

- Snodgrass 1994; incise and tubularise; heals by epithelialisation.
- Slit-like meatus; dartos barrier reduces fistula.
- Glans <14 mm and poor plate raise complications.
- Fistula 5.7%, meatal stenosis 3.6% (Pfistermuller 2015).
- Extended to midshaft, selected proximal, redo, MIP and adults.
- Key condition: curvature <30° without plate division.
- Poor plate: dorsal inlay graft ("Snodgraft").
- Proximal and redo TIP fail more often.

**Glandular hypospadias (SAQ)**

- Glanular = most distal hypospadias; dorsal hood, minimal curvature.
- MAGPI only for a truly glanular meatus; GAP for MIP; TIP is the workhorse.
- Repair at 6–18 months; may be left alone if function is normal.

**Retractile testis? (SAQ)**

- Retractile = comes down and stays down.
- Gliding testis springs back: treat as undescended.
- Yearly examination until puberty; operate if it ascends.

**Megameatus intact prepuce (SAQ)**

- Wide glanular meatus, intact prepuce, no curvature; up to 5% of hypospadias.
- Often found at circumcision; retract foreskin first.
- Pyramid repair: dissect, taper and tubularise; glansplasty.

**Techniques of chordee correction (SAQ)**

- Artificial erection after degloving decides the technique.
- <30°: dorsal plication; >30°: divide plate ± ventral graft and stage.
- Plication recurrence 28% vs grafting 9%.

**Congenital ureteral anomalies (SAQ)**

- Upper pole obstructs (ectopia, ureterocele); lower pole refluxes.
- Female ectopic ureter: continuous dribbling with normal voiding; male never incontinent.
- Retrocaval ureter is a venous anomaly; always right-sided.

SECTION 5

# Paediatric Urology 2

## 32 Embryology of the Human Genitourinary Tract

### 1 · Overview

- Development is mostly complete by the second trimester. The **indifferent stage** lasts to about **8–9 wk**; then androgens drive the sexes apart.

### 2 · Kidney and ureter

*Bud position decides kidney quality and orifice site.*

#### Three successive kidneys

| Kidney | Features |
|---|---|
| †Pronephros | Cervical, week 4; non-functional and regresses |
| Mesonephros | †Transient, but its **Wolffian duct** persists; tubules become efferent ductules (male) or epoöphoron (female) |
| Metanephros (permanent) | †Week 5; ureteric bud and metanephric blastema form the nephrons |

#### Reciprocal induction †

- The **ureteric bud** grows from the Wolffian duct into the metanephric mesenchyme and forms the ureter and collecting system.
- Mesenchymal **GDNF** binds **RET** on the bud to drive branching. **WT1** makes the mesenchyme competent, and loss of **PAX2** causes agenesis or hypoplasia (renal-coloboma syndrome).
- Nephrogenesis ends at about 34–36 wk; no nephrons form after birth.

> **KEY POINT**
>
> One defect gives both dysplasia and an abnormal orifice: bud position on the Wolffian duct sets where it meets the blastema (kidney quality) and where its orifice lands in the bladder.

#### Ascent and rotation †

- The kidney ascends in weeks 6–9 and its hilum rotates medially; failure gives a pelvic or malrotated kidney. Fused lower poles make a **horseshoe kidney**, trapped under the inferior mesenteric artery.

### 3 · Cloaca, bladder, trigone and urethra

#### LAQ Bladder embryology and exstrophy-epispadias complex

Paper answer · P1 · P2 · P3 · merged from 3

*Asked as:* P3 LAQ "Bladder and ureter development, exstrophy–epispadias complex and its management" · P1 LAQ "Describe embryology of bladder. What are the proposed theories for

extrophy epispadias complex? Describe skeletal abnormalities." · P2 SAQ "Embryogenesis of urinary bladder"

### Introduction

- Assumed meanings: "ectopia vesicae" means classic bladder exstrophy (CBE); "single stage repair of exstrophy of UB" means single-stage (complete primary) repair of exstrophy of the urinary bladder, mainly CPRE (complete primary repair of exstrophy, Mitchell).
- **EEC (exstrophy–epispadias complex)** is one spectrum: glanular epispadias, then **CBE (classic bladder exstrophy)** (over 50%), then cloacal exstrophy.
- CBE incidence about 1 in 46,000; male:female 2.3:1. Recurrence in offspring of an affected parent: 1 in 70.

### Development of the ureter

- **Weeks 4–6:** the **ureteric bud** arises from the distal **Wolffian (mesonephric) duct**. GDNF (glial cell line-derived neurotrophic factor) from the metanephric mesenchyme induces a single bud (extra buds cause duplication).
- **Reciprocal induction:** the bud invades the metanephric blastema. The bud stalk forms the ureter; its branches form the pelvis, calyces and collecting ducts; the blastema forms the nephrons.
- **Wall:** urothelium, lamina propria, smooth muscle and serosa, the same plan as the bladder.
- **Chwalla membrane** at the future orifice normally ruptures; failure is the classic explanation of ureterocele.

### Development of the bladder: division of the cloaca

- The **cloacal membrane** (bilaminar ectoderm–endoderm) occupies the infraumbilical wall. **Mesenchyme grows in** between its layers and forms the lower abdominal muscles and pelvic bones.
- **Cloacal division (weeks 5–6; P2: weeks 4–6):** the **urorectal septum** (Tourneux and Rathke folds) grows caudally and divides the endodermal cloaca into a dorsal **anorectal canal** (rectum, posterior) and a ventral **UGS (urogenital sinus)** (bladder, anterior). It meets the cloacal membrane, which perforates to form the urogenital and anal openings.
- Paired genital tubercles fuse **cephalad** to the membrane.
- **Wolffian (mesonephric) duct entry** is the landmark that divides the UGS:
- **Cranial vesicourethral canal** (continuous with the allantois) forms the **bladder and pelvic urethra**.
- **Caudal (definitive) UGS** forms the phallic urethra in males, and the vestibule and distal vagina in females.

### Development of the bladder: wall layers

- **Urothelium:** UGS endoderm.
- **Lamina propria, detrusor, adventitia / serosa:** surrounding splanchnic mesenchyme, by epithelial–mesenchymal signalling (epithelium and mesenchyme induce each other).

### Trigone and ureterovesical junction (weeks 5–6)

- The **common excretory duct** (Wolffian duct distal to the bud) dilates and is absorbed into the UGS; **apoptosis** separates the ureter from the Wolffian duct.

- The arms rotate (**Mackie–Stephens**): the ureteric orifice moves **cranially and laterally**; the Wolffian duct ends caudally as the ejaculatory duct at the verumontanum.
- **Trigone muscle:** classic view, ureteric extension; current view (mouse lineage tracing), mostly bladder smooth muscle.

| Bud origin | Orifice | Result |
|---|---|---|
| Too low (early contact) | High, lateral, short tunnel | VUR (vesicoureteral reflux) |
| Too high (late contact) | Caudal, medial | Ectopic ureter, often obstructed |
| Either | Abnormal bud–blastema contact | Dysplasia (VUR and dysplasia coexist) |

- **Weigert–Meyer:** in duplex systems the upper-pole orifice is caudal and medial.

### Allantois, urachus and fetal bladder function

- **Months 4–5:** the bladder descends into the pelvis; the allantoic duct elongates and obliterates as the **urachus** (median umbilical ligament).
- Urine production starts at about **week 10**. The fetal bladder is seen in about half at 10 weeks and almost all by 13 weeks; it empties every 15–20 minutes. After **16 weeks** amniotic fluid is mainly fetal urine.

### Bladder anomalies explained by embryology

| Anomaly | Basis |
|---|---|
| **Exstrophy** | Cloacal membrane not reinforced by infraumbilical mesoderm |
| Urachal (patent urachus, sinus, cyst, diverticulum) | Failed obliteration at different levels |
| Agenesis, duplication | UGS or trigone failure; septation |
| Congenital (Hutch) diverticulum | Detrusor deficiency at the ureteric hiatus |
| VUR, ectopic ureter | Abnormal bud position |

### Developmental basis of EEC: core defect

- The **cloacal membrane is not reinforced by mesodermal ingrowth**. This mesoderm normally forms the lower abdominal muscles and pelvic bones, so the thin membrane ruptures (**Marshall–Muecke** overdevelopment theory).
- **Timing and extent of rupture decide the variant:** before septum descent gives cloacal exstrophy; after gives CBE; limited rupture gives epispadias.

### Developmental basis of EEC: theories

| Theory | Proposed defect | Result |
|---|---|---|
| **Cloacal membrane overdevelopment (Marshall–Muecke)** | Abnormally large membrane in week 4 blocks medial mesenchymal migration | Rupture before septum descent: cloacal exstrophy; after: bladder exstrophy; limited: epispadias |
| Lateral body-wall fold failure | Folds fail to meet in midline | Abdomen and pelvis: cloacal; pelvis alone: classic |
| Caudal genital hillocks | Genital tubercles fuse **below** rather than above the membrane | Dorsal urethral defect (epispadias) |

| Theory | Proposed defect | Result |
|---|---|---|
| Abnormal body-stalk insertion | Caudal insertion prevents mesenchyme interposition | Superficial unstable membrane disintegrates |
| **Bony pelvis first** | Pelvic ring primordium fails to rotate | Pelvic structures cannot meet, so the bladder herniates |

- All theories converge on **absent infraumbilical midline mesoderm**.

### Aetiology of EEC

- Multifactorial (inconsistent twin concordance). **22q11.2 duplication** is the commonest variant; **ISL1** susceptibility gene; p63 under-expressed in exstrophic urothelium.
- **IVF (in vitro fertilisation)** raises risk about 7.5-fold and early high-dose progesterone about tenfold; maternal smoking is linked to cloacal exstrophy; folate to the mildest form.

### Resulting anatomy in classic exstrophy

- Exposed (open) bladder plate; epispadias; low umbilicus; anterior anus; wide levator hiatus; inguinal hernias.
- **Male genitalia:** anterior corpora about 50% shorter; short, upturned penis with dorsal chordee.
- **Female:** bifid clitoris, short vagina, anterior stenotic introitus; later prolapse.
- **Bladder:** immature, not neuropathic (raised type III collagen falls after closure). Ureters enter with no obliquity, so **VUR (vesicoureteric reflux) occurs in 100%** after closure.

### Skeletal abnormalities: classic bladder exstrophy

#### 3D (three-dimensional) CT (computed tomography) studies

- **Rotational:** posterior pelvis externally rotated (mean 12° each side); anterior segment externally rotated (mean 18° each side); sacroiliac joint rotated; acetabular and femoral retroversion.
- **Dimensional: pubic diastasis** (mean 4.8 cm); **anterior pubic segment shortened by 30%**; intertriradiate distance increased by about 31%; posterior segment length normal; larger sacrum.
- Bone histology is normal, so growth potential is normal.
- **Clinical effects:** short pendular penis (anterior segment short and splayed); wide hips, waddling gait and out-toeing in early childhood. These cause little disability and improve with growth.
- **Pelvic floor:** levator ani displaced posteriorly and flattened; anterior anus; later pelvic organ prolapse.
- **Spine:** minor variants (spina bifida occulta, lumbarisation, sacralisation) fairly common; scoliosis and dysraphism uncommon.

### Skeletal abnormalities: cloacal exstrophy and isolated epispadias

- **Cloacal exstrophy:** widest, **asymmetric** diastasis (about twice CBE), sacroiliac malformation, hip dislocation.
- **Spinal anomalies in 85–100%** (lumbar myelodysplasia commonest; lipomeningocele); MRI (magnetic resonance imaging) for all.
- Limb defects: **clubfoot** (commonest), limb shortening, tibial and fibular deformity.

- **Isolated epispadias:** male symphysis widened (less than CBE); female symphysis usually closed or narrow.

### Management: prenatal and at birth

- **Prenatal:** prenatal diagnosis in only about 25%. Ultrasound shows no bladder filling, low umbilicus, wide pubic rami, small genitalia and an enlarging lower abdominal mass. Counsel; deliver at an exstrophy centre.
- **At birth:** tie the cord with silk (no clamp); cover the plate with plastic wrap; saline irrigation at nappy change. Renal ultrasound; multidisciplinary team.
- **Assess:** template size and polyps (**≥3 × 3 cm** suits newborn closure), diastasis and pelvic malleability under anaesthesia, urethral plate and penile length, associated anomalies.
- Small template, penoscrotal duplication, hypoplastic bladder: **delayed closure at 6–9 months** (delay closure rather than do a bad early closure).

### Pelvic osteotomy and immobilisation

- **Purpose:** closing the pelvic ring places the bladder neck deep in the pelvis, reduces tension and helps continence.
- **Indications:** closure **after 72 hours**, diastasis **>4 cm**, stiff (poorly malleable) pelvis, failed closure, or cloacal exstrophy (always).
- **Combined bilateral anterior innominate + vertical iliac osteotomy** (most used) with external fixator and Buck traction for 4–6 weeks.
- Under 72 hours with malleable pubes: closure without osteotomy, with modified Bryant traction for 4 weeks.
- Spica casting fails more often (failure with external fixation plus traction 7.5% versus spica 47%).

### Modern staged repair of exstrophy (MSRE)

- Aim of closure: convert exstrophy into incontinent epispadias, with enough outlet resistance for bladder growth but not enough to harm the kidneys.

| Stage | Age | Procedure | Purpose |
|---|---|---|---|
| 1 | Newborn | Bladder, posterior urethra and abdominal wall closure; ± osteotomy | Converts to complete epispadias; outlet resistance stimulates bladder growth |
| 2 | 6–12 months | Epispadias repair (**modified Cantwell-Ransley**), after testosterone | Raises capacity; straightens penis |
| 3 | 5–7 years (another source: 4–7 years) | BNR (bladder neck reconstruction): modified **Young-Dees-Leadbetter (YDL)** + bilateral **Cohen** reimplantation | Continence; needs capacity **≥100 mL** and a motivated child |

- **Closure keys: radical division of the urogenital diaphragm fibres** so the vesicourethral unit sits deep in the pelvis (missing this is the commonest error); ureteric stents; suprapubic tube 4 weeks; pubic nylon mattress suture.
- **Alternatives:** CPRE (Mitchell) with penile disassembly; Kelly radical soft-tissue mobilisation; Erlangen total repair (single stage); Mainz ureterosigmoidostomy. Most still need an outlet procedure.

### Single-stage repair (CPRE and others)

- Single-stage repairs aim for fewer operations and continence without bladder neck repair.

| Repair | Features |
|---|---|
| **CPRE (Mitchell)** | Newborn closure + penile disassembly + bladder neck tailoring |
| Erlangen ("total") | All in one at about 8 weeks, no osteotomy |
| Combined closure + epispadias repair | Delayed or failed closures, boys >6 months |

- **Selection:** template ≥3 × 3 cm, elastic, few polyps; experienced team. Not for penoscrotal duplication, hypoplastic bladder or bilateral hydronephrosis.
- **CPRE technique:** 1. Combined general and epidural anaesthesia; **osteotomy if >72 hours** (3 days) old, diastasis >4 cm or stiff pelvis. 2. Incise around the plate; free the bladder; **divide the urogenital diaphragm fibres** to the levator hiatus so the unit sits deep. 3. **Penile disassembly:** right and left hemicorporoglanular bodies plus a urethral wedge on spongiosum (now often left attached to the glans). 4. Urethra placed ventrally between the corpora; corpora joined; meatus usually **hypospadiac**. 5. Bladder and neck closed and tapered; ± bilateral ureteric reimplantation (BUR-CPRE). 6. Pubic No. 2 nylon mattress suture; suprapubic catheter 4 weeks, ureteric stents. 7. Immobilise: external fixation plus traction (failure 7.5% versus spica 47%).
- **Complications:** same frequency as MSRE but **more serious**: dehiscence, prolapse, vesicocutaneous fistula, **glans or corporal loss** (ischaemia), early reimplantation for VUR or hydronephrosis; retention in girls.

### Follow-up, failures and salvage

- **Follow-up:** prophylaxis; ultrasound; yearly cystogram and cystoscopy under anaesthesia.
- Dehiscence or prolapse: wait 4–6 months, then reclose with osteotomy.
- Failed BNR (no 3-hour dry interval by 2 years): bulking, redo YDL, or **augmentation + bladder neck closure + Mitrofanoff** (continent catheterisable stoma).

### Cloacal exstrophy (OEIS) management

- OEIS: omphalocele, exstrophy, imperforate anus, spinal defects.
- Stabilise; neurosurgery (spinal defects 85–100%); separate bowel from hemibladders; colostomy; osteotomy always; gender congruent with karyotype; continence mostly by augmentation with CIC (clean intermittent catheterisation).

### Outcomes and long-term issues

- Daytime dryness after MSRE about **70–74%** (another source: 70–75%), versus about 56% after CPRE with YDL; most still need bladder neck surgery after CPRE.
- **Bladder capacity at neck repair is the main predictor**; each failed closure lowers it. **Best continence predictors: template size at birth and a successful first closure.**
- **Also know:** variants (superior vesical fissure, duplicate exstrophy, pseudoexstrophy); adenocarcinoma risk (cytology, cystoscopy; colonoscopy after ureterosigmoidostomy); reduced male fertility and uterine prolapse in adults; female epispadias; psychosocial support.

**Diagrams to draw:** cloaca divided by the urorectal septum (vesicourethral canal, UGS, allantois) with failed mesodermal ingrowth of the cloacal membrane; Wolffian duct with

ureteric bud and its absorption into the UGS; exstrophy pelvis with diastasis; MSRE timeline; penile disassembly into two hemicorpora and urethral wedge.

**Key points to remember:**

- Bud from Wolffian duct; position decides VUR or ectopia.
- Urorectal septum weeks 5–6 (P2: 4–6); mesonephric duct entry is the landmark; vesicourethral canal forms the bladder; urachus from allantois.
- Trigone: common excretory duct absorption; mostly bladder muscle (current view).
- Urachus forms at months 4–5; urine from week 10; amniotic fluid mainly urine after 16 weeks.
- Marshall–Muecke: overdeveloped cloacal membrane, failed mesodermal reinforcement; rupture timing decides variant.
- CBE: external rotation 12°/18°, anterior pubis 30% short, diastasis about 4.8 cm; VUR in 100%.
- Cloacal exstrophy: spinal anomalies 85–100%; osteotomy always.
- MSRE: closure at birth, epispadias repair 6–12 months, YDL / BNR at 5–7 years at ≥100 mL.
- Osteotomy if closure after 72 h or diastasis >4 cm.
- CPRE: closure + penile disassembly in newborn. Risks: glans loss, hypospadias, VUR.
- Capacity ≥100 mL, template size and a successful first closure predict continence.

**Also, from the condensed book (not in the paper answer):**

- Anorectal malformations arise from the **cloacal membrane**, not the urorectal septum (weeks 5–6).

### Weigert–Meyer rule (duplex kidneys)

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## SAQ Posterior urethral valves: embryology, management and valve bladder

Paper answer · P1 · P2 · P3 · merged from 3

*Asked as:* P3 SAQ "Embryology and Management of P.U. valve patient." · P1 SAQ "Embryology of Posterior Urethral Valves." · P2 SAQ "Valve bladder: management"

### Introduction

- PUV (posterior urethral valves): the commonest cause of bladder outlet obstruction in boys (1.4–2.2 per 10,000 births).
- Morgagni described them (1769); Hugh Hampton Young named and classified them (1919).

### Normal development

- **Weeks 5–6:** the **urorectal septum** divides the cloaca into the anorectal canal (rectum) and the **urogenital sinus**.
- The **Wolffian (mesonephric) ducts** join the cloaca just before division. Their orifices normally migrate to the **verumontanum**, leaving small folds (plicae colliculi).
- The **urogenital membrane** normally disappears completely.
- Fetal urine starts at 10–14 weeks; after 16 weeks amniotic fluid is mainly fetal urine. So obstruction in this period damages bladder, kidneys and lungs.

### Embryology of the valves: Young types and their origin

| Type | Anatomy | Embryology |
|---|---|---|
| **I (about 95%)** | Leaflets from the verumontanum fuse anteriorly just proximal to the external sphincter | Abnormal anterior / anterolateral insertion of the Wolffian duct ends into the cloacal wall: hypertrophied inferior urethral crest |
| **II** | Folds from verumontanum to bladder neck | Not real (Stephens 1955): secondary bladder-neck hypertrophy |
| **III (5–10%)** | Annular (ring) diaphragm with a central hole | **Persistent urogenital membrane** |

### Other theories and genetics

- Hypertrophied mucosal folds (oldest); cloacal remnants; Wolffian–Müllerian maldevelopment.
- **COPUM (congenital obstructing posterior urethral membrane, Dewan):** one oblique membrane, torn by the first catheter into apparent type I leaflets. Plausible but unproven.
- **Cobb's collar (Moormann's ring):** a congenital ring-like narrowing of the proximal bulbar urethra, just distal to the external sphincter; some authors group it with type III valves.
- **Genetics:** mostly sporadic; chromosome 11 duplications; copy number variants (worse renal outcome); TBX5, PTK7.

**Diagram to draw:** posterior urethra with type I leaflets versus a type III membrane, with keyhole VCUG.

### Consequences of obstruction

- Thick bladder, dilated posterior urethra (**keyhole sign**), hydroureteronephrosis, **renal dysplasia**, oligohydramnios, pulmonary hypoplasia.
- Pop-offs (VUR (vesicoureteral reflux), urinoma, ascites) relieve pressure.
- Dysplasia is fixed in utero; ablation cannot reverse it.

### Antenatal management

- Bilateral hydroureteronephrosis, thick bladder, **keyhole sign**, oligohydramnios.
- **Vesicoamniotic shunt** helps survival (lungs), not kidneys (PLUTO trial, 2013).

### Neonatal management

1. **Drain the bladder:** 5–7 Fr feeding tube (coudé if it curls); minimal balloon inflation.
2. **Stabilise:** ventilation; correct **post-obstructive diuresis**, hyponatraemia, acidosis, hyperkalaemia; antibiotics.
3. **Confirm:** VCUG (voiding cystourethrogram), renal ultrasound, MAG3 (mercaptoacetyltriglycine) at 2–3 months.
4. **Primary endoscopic valve ablation** (preferred): laser, Bugbee or hook at 5 and 7 o'clock ± 12; VCUG within 1 month.
5. **Vesicostomy** if the urethra is too small or drainage fails. **Upper-tract diversion** rarely: no proven renal benefit.
6. **Circumcision:** cuts febrile UTI (urinary tract infection) sharply. - **VUR** (50–80%) is secondary; treat the bladder, not the ureter.

### Prognosis

| Nadir creatinine (first year) | ESRD (end-stage renal disease) risk |
|---|---|
| < 0.4 mg/dL | Nil |
| 0.4–0.69 | 2% |
| 0.7–0.99 | 27% |
| > 1.0 | Up to 100% |

- Lifetime ESRD risk 20–50%.

### Valve bladder: definition

- **Valve bladder syndrome** (Mitchell, 1982): upper tracts and renal function worsen after successful PUV ablation, without residual obstruction, because of bladder dysfunction.

### Valve bladder: pathophysiology (vicious cycle)

``` Detrusor hypertrophy, collagen, low compliance → poor sensation, rising residual

- polyuria (nephrogenic diabetes insipidus) → bladder never empty → pressure transmitted to dilated upper tracts → they refill the bladder → renal damage ```
- Three processes: **polyuria, poor sensation, residual urine**; together they keep the upper tracts dilated.
- Age pattern: hyperreflexia in infancy, better compliance in childhood, then a large atonic bladder (myogenic failure) after puberty.

**Diagram to draw:** vicious-cycle diagram of the valve bladder.

### Valve bladder: clinical features and evaluation

- Late continence (about 5.5 years), incontinence, UTI, worsening hydronephrosis, rising creatinine. **New incontinence after being dry is a red flag.**
- Uroflowmetry and PVR (post-void residual) at each visit; ultrasound; creatinine; voiding diary.
- **Videourodynamics** when the picture changes and before transplantation.

### Valve bladder: management (escalating ladder)

1. **Behavioural:** adequate fluids, timed and double voiding, biofeedback.
2. **Drugs:** anticholinergic (oxybutynin 0.1 mg/kg twice daily) for a small high-pressure bladder; α-blocker for high residual. Stop oxybutynin if residual rises.
3. **CIC (clean intermittent catheterisation)** if residual rises despite drugs.
4. **Overnight bladder drainage** (catheter for 7–10 hours) for worsening hydronephrosis, UTI or renal function; it gives bladder and kidneys a daily rest.
5. **Mitrofanoff appendicovesicostomy** if urethral CIC is difficult (sensate urethra, high bladder neck).
6. **Augmentation** (rare) for a small, high-pressure bladder despite the above: **ureterocystoplasty** with the dilated ureter is preferred, avoiding bowel (mucus, acidosis, stones). - Reflux is treated through the bladder; reimplantation is rarely needed.

### Long-term care and transplantation

- **CKD (chronic kidney disease)** care.
```

- ESRD in 20–50%. Before **transplantation**, optimise the bladder (videourodynamics, CIC or overnight drainage); graft survival is then similar.

**Key points to remember:**

- Type I (95%): abnormal Wolffian duct insertion; type III: persistent urogenital membrane; type II not real.
- Urorectal septum divides the cloaca at weeks 5–6.
- COPUM: one membrane torn into leaflets by catheterisation.
- Dysplasia is fixed in utero; ablation cannot reverse it.
- Catheter, stabilise, then primary valve ablation; vesicostomy if not possible.
- Nadir creatinine below 0.7 mg/dL predicts good renal outcome.
- Valve bladder: polyuria, poor sensation, residual urine; overnight drainage and CIC break the cycle.
- Ureteric augmentation rather than bowel; optimise before transplant.

**Also, from the condensed book (not in the paper answer):**

- Male ectopia stays above the external sphincter, so there is no continuous wetting; †female ectopia below it causes continuous dribbling with normal voids.
- With an **ectopic vas**, the vas joins the ureter and both drain into the prostatic urethra.

## 4 · Gonads

*Two testicular hormones explain every disorder of sex development (DSD).*

*See the paper answer **Testicular descent and undescended testis: evaluation, treatment algorithm and orchidopexy methods** (Section 4, Ch 57).*

**Also, from the condensed book (not in the paper answer):**

- **SRY** makes Sertoli cells enclose primordial germ cells in cords at about **6 wk**; Leydig cells appear from 8 wk.
- **Müllerian-inhibiting substance (MIS, AMH)** from Sertoli cells makes the Müllerian ducts regress at **8–10 wk**; without it, the uterus and tubes persist.
- **Testosterone** from Leydig cells turns the Wolffian duct into epididymis, vas and seminal vesicle.

### Ovary — the default

- Without SRY there is no MIS or androgen, so development is female.

**KEY POINT**

Two independent testicular signals act on two independent duct systems: MIS decides Müllerian, testosterone decides Wolffian and external genitalia. Every DSD is a combination of the two.

## 5 · Gonadal descent

*See the paper answer **Testicular descent and undescended testis: evaluation, treatment algorithm and orchidopexy methods** (Section 4, Ch 57).*

## 6 · Genital ducts

### Male accessory glands

- The **prostate** buds at **10 wk** from urogenital sinus (UGS) epithelium, mostly lateral to the **verumontanum**.
- Androgen acts via the **androgen receptor in mesenchyme**; stromal type 2 5α-reductase makes **dihydrotestosterone (DHT), which has about 10× the AR affinity** of testosterone. Its deficiency therefore gives a rudimentary prostate and severe hypospadias.

### Female tract

- The Müllerian ducts use the **Wolffian duct as a guidewire**, which explains combined Müllerian and renal anomalies.
- The **entire vaginal epithelium and hymen are UGS-derived** (Bulmer view). Female Wolffian remnants include the **Gartner duct**.

## 7 · External genitalia

*See the paper answer **Proximal (perineoscrotal) hypospadias: penile embryology and management including Koyanagi repair** (Section 4, Ch 56).*

**Also, from the condensed book (not in the paper answer):**

- The tunica is thickest at the 12 o'clock nerve-free zone, the dorsal plication site for mild to moderate curvature.
- The shaft is fused by **16 wk**. In females the folds do not fuse, leaving an open vestibule.

## 8 · Developmental step and resulting anomaly

| Step that fails | Anomaly |
|---|---|
| †No bud or no induction (GDNF–RET, PAX2) | Renal agenesis (absent hemitrigone) |
| †Abnormal bud–blastema contact | Dysplasia, hypoplasia, MCDK |
| †Early bifurcation or two buds | Bifid ureter or complete duplication |
| Bud too low / too high | VUR / ectopic ureter |
| †Ascent or rotation fails | Pelvic, malrotated or horseshoe kidney |
| Cloacal membrane or septation | Anorectal malformation; †cloacal anomalies, exstrophy |
| Allantois persists | Urachal remnants |
| Fold fusion arrests | Hypospadias |
| Urethral plate splits | Urethral duplication |
| Androgen excess in 46,XX | Virilisation (CAH) |
| Cranial suspensory ligament retained | Intra-abdominal testis |
| Gubernacular attachment abnormal | Canalicular or ectopic testis |
| Gubernacular migration fails | Congenital cryptorchidism |
| Processus fails to close / to involute | Hydrocele, hernia / ascending testis |
| One Müllerian duct absent | Unicornuate uterus |
| Ducts fail to fuse | Uterus didelphys |

| Step that fails | Anomaly |
|---|---|
| Septum persists | Uterine obstruction or duplication |
| Vaginal plate fails to canalise | Transverse vaginal septum |
| Inferior end fails to open | Imperforate hymen, the commonest vaginal obstruction |
| Müllerian and Wolffian both abnormal | OHVIRA |

## 9 · Clinical correlations

- **Hypospadias** (about 1 in 250) is an **arrest, not a deformation**.
- **CAH** may make a 46,XX infant look fully male. Suspect it with **impalpable gonads and hyperpigmentation**, since it mimics hypospadias with bilateral impalpable testes.
- A **streak gonad** makes no MIS or testosterone, so the child is female; give hormones and **remove the streak if 46,XY**, because of gonadoblastoma risk.
- **Persistent Müllerian duct syndrome** is 46,XY with uterus and tubes, usually from an MIS-receptor defect, found at exploration for bilateral abdominal testes.
- **OHVIRA** presents after menarche with cyclic pain or pyocolpos. The kidney is often **dysplastic with an ectopic ureter into the obstructed hemivagina**; if missed, leakage persists after septum resection.

> **KEY POINT**
>
> Homologues: Littre and Skene; Cowper and Bartholin; spongiosum and vestibular bulb; gubernaculum and the ovarian plus round ligaments; labioscrotal swellings and labia majora. Traps: the vagina is UGS-derived; the appendix testis is Müllerian and the appendix epididymis Wolffian.

# 33 Urologic Aspects of Pediatric Nephrology

## 1 · Renal function in the child

- CAKUT is the **commonest cause of paediatric end-stage kidney disease (ESKD)**.
- GFR reaches the adult level by **2 years**. Creatinine in the first 48 h is **maternal**; by day 7 it is the neonate's own (**<0.5 mg/dL** at term).
- †Schwartz: estimated GFR = 0.413 × height (cm) / creatinine (mg/dL).

> **KEY POINT**
>
> The neonatal kidney is "hormonally deaf": maternal creatinine, salt wasting, hyperkalaemia, low bicarbonate and poor concentration reflect immaturity, not disease.

## 2 · Haematuria

- Microscopic haematuria is **>5 red blood cells (RBCs)/high-power field**; it affects about 1% of school children and mostly resolves.
- **Fresh microscopy is critical**: red urine is non-glomerular, while brown urine with RBC casts is glomerular. A positive dipstick without RBCs means haemoglobin or myoglobin.
- **UTI is the commonest cause of gross haematuria**. Trauma, pain or clots need urological work-up, often CT; suspected GN needs complement, ASO and nephrology referral.
- In isolated microscopic haematuria, **VCUG and cystoscopy are not indicated**.

## 3 · Proteinuria

- Normal urine protein-to-creatinine ratio (UPCR) is **<0.2 mg/mg** (<0.7 in newborns).
- **Transient** (fever, exercise) and **orthostatic** proteinuria are benign; **fixed** proteinuria needs full evaluation with or without biopsy.

## 4 · Glomerular disease

Under 1 year causes are genetic; at 1–10 years think of minimal change, post-streptococcal GN, HSP and HUS; from 7 years, lupus, IgA, MPGN and Alport.

### Nephrotic syndrome

- Proteinuria **>40 mg/m²/h or UPCR >2.0** with hypoalbuminaemia and hyperlipidaemia.
- In primary disease give **steroids first, without biopsy**, since minimal change is 90% under 10 years and mostly steroid-sensitive.

| Category | Definition | Action |
|---|---|---|
| Steroid-sensitive | Remission within 8 weeks | ~50% relapse |
| Frequently relapsing | ≥2 relapses in 6 months or ≥4 in 12 | Steroid-sparing: cyclophosphamide, mycophenolate, rituximab, calcineurin inhibitors |
| Steroid-dependent | 2 consecutive relapses on steroids | |
| Steroid-resistant | No remission after 6–8 weeks | Biopsy; usually FSGS, which recurs after transplant |

### Nephritic syndrome

Complement sorts the causes:

| C3 | C4 | Diagnoses |
|---|---|---|
| Low | Normal | Post-infectious GN, C3 glomerulopathy |
| Low | Low | Lupus nephritis, MPGN |
| Normal | Normal | IgA, HSP, Alport, ANCA GN, anti-GBM |

- In **post-streptococcal GN**, C3 normalises by **8–12 weeks**; if it stays low, biopsy.
- **IgA nephropathy** gives painless gross haematuria **during** a respiratory infection. **Alport** adds sensorineural deafness.
- **HUS** is the commonest cause of severe AKI in a young child; atypical (complement) HUS gets eculizumab.

## 5 · Tubulopathies

*See the paper answer **Renal tubular acidosis (distal and type 2)** (Section 11, Ch 95).*

**Also, from the condensed book (not in the paper answer):**

- Type 4 RTA (**hyperkalaemia**) from obstruction improves after urological correction.

## 6 · Cystic disease

- **MCDK** has no function, and bilateral disease is lethal. In MCDK, VUR or PUJ obstruction occurs in up to 25%, so do a VCUG, especially with hydroureteronephrosis. Follow with US and BP until involution, then yearly.
- **ARPKD** causes enlarged kidneys and hepatic fibrosis, with >75% ESKD by 5 years. Nephronophthisis causes polyuria and ESKD by adolescence.

## 7 · Hypertension

- Hypertension is repeated BP **>95th percentile** for age, sex and height.
- Infants have coarctation or renovascular disease, young children Wilms or neuroblastoma, school-age children **renal parenchymal disease including reflux nephropathy**, and obese adolescents essential hypertension.
- Drugs are for stage 2, symptoms, end-organ damage or a secondary cause. Target <90th percentile, or **<50th** in CKD.

## 8 · Acute kidney injury

| KDIGO stage | Creatinine | Urine output |
|---|---|---|
| I | Rise ≥0.3 mg/dL in 48 h or 1.5–1.9× baseline | <0.5 mL/kg/h for 6–12 h |
| II | 2.0–2.9× | <0.5 for ≥12 h |
| III | ≥3× or ≥4.0 mg/dL or clearance <35 (<18 years) | <0.5 for ≥24 h or <0.3 for 12 h |

- Creatinine lags GFR by 24–72 h. FeNa **<1%** means prerenal and >3% intrinsic.
- Dialysis is for diuretic-resistant overload, refractory hyperkalaemia or acidosis, or uraemia.

## 9 · Chronic kidney disease

- CKD stages by GFR are I ≥90, II 60–89, III 30–59, IV 15–29 and V **<15** or dialysis.
- Complications include growth failure from growth hormone resistance, bone disease, anaemia and acidosis.
- **Do not restrict protein**; strict BP control is the only proven way to slow progression.

## 10 · Dialysis and transplantation

*See the paper answer **Urological evaluation and management of pediatric kidney transplant patient.** (Section 4, Ch 61).*

**Also, from the condensed book (not in the paper answer):**

- Peritoneal dialysis is first choice if **<9 years / <20 kg**; omphalocele, gastroschisis, diaphragmatic hernia and exstrophy contraindicate it.
- Living donors give less delayed graft function (5% vs 15%) and a longer graft half-life (20 vs 12.5 years).
- Many centres await **≥10 kg and 65 cm** before transplant because of graft thrombosis.

# 34 Perinatal Urology

## 1 · Fetal renal development and prenatal imaging

*Amniotic fluid is fetal urine, so obstruction threatens the lungs.*

### LAQ Antenatal hydronephrosis, fetal uropathy and prenatal scan anomalies

Paper answer · P1 · P3 · merged from 2

*Asked as:* P3 LAQ "Antenatal hydronephrosis (UTD) and fetal uropathy (LUTO)" · P1 SAQ "Anomalies detected on Prenatal scan between 16th and 20th weeks of gestation"

#### Introduction and terminology

- Assumed meaning: HN (hydronephrosis) and ANH (antenatal hydronephrosis) are now called **UTD (urinary tract dilation)**.
- Major congenital anomalies occur in about **2–3% of births**. Urinary tract anomalies are among the commonest detected prenatally (commonly quoted as 20–30% of all prenatal anomalies).
- **UTD is the second commonest prenatal anomaly after cardiac defects, in 1–3% of pregnancies.**
- Mostly UTD on the 20-week scan, usually mild and transient: mild 57–88%, moderate 10–30%, severe 1.5–13% of UTD. The serious group is **LUTO (lower urinary tract obstruction)**: PUV (posterior urethral valves), urethral atresia, PBS (prune-belly syndrome).

#### The prenatal anomaly scan (16–20 weeks)

- The **anomaly scan is done at 18–20 weeks** (ACOG (American College of Obstetricians and Gynecologists)); by 20 weeks most urological anomalies are visible.
- Screening US (ultrasound) at 18–20 weeks detects dilation well (sensitivity about 90%) but predicts the diagnosis poorly.
- **Why this window:** fetal urine production begins at about 14 weeks, and after about 16 weeks amniotic fluid is mainly fetal urine, so **oligohydramnios reflects renal failure or outlet obstruction** and causes **pulmonary hypoplasia**. The bladder should be seen filling and emptying.
- Survey kidneys, collecting system, ureters, bladder, genitalia, amniotic fluid, abdominal wall and spine. AFI (amniotic fluid index): normal 8–18 cm; oligohydramnios <5–6 cm.

#### Anomalies detected and their signs

| Anomaly | Signs |
|---|---|
| **PUV (posterior urethral valves)** | Male; thick bladder; **keyhole sign**; bilateral hydroureteronephrosis; oligohydramnios |
| **Bilateral renal agenesis** | **Anhydramnios after 16 weeks**, no bladder, no renal arteries: lethal Potter sequence |
| **MCDK (multicystic dysplastic kidney)** | Non-communicating cysts, no central pelvis |
| **ARPKD (autosomal recessive polycystic kidney disease)** | Large, homogeneously **bright** kidneys; oligohydramnios |

| Anomaly | Signs |
|---|---|
| **Bladder exstrophy** | **Bladder never seen**, low cord insertion, lower abdominal mass, wide pubis |
| **Prune belly** | Thin-walled large bladder, empty scrotum |
| **Myelomeningocele** | Spinal defect, lemon and banana signs |

- **Also know:** duplex kidney with ureterocele (intravesical cyst); cloacal exstrophy (adds omphalocele, spinal defects); hydrocolpos; ambiguous genitalia; renal ectopia.

**Diagram to draw:** fetal keyhole sign.

### Causes of antenatal UTD (SFU [Society for Fetal Urology] consensus)

| Cause | Incidence |
|---|---|
| Transient/physiological dilation | 50–70% |
| UPJ (ureteropelvic junction; PUJ, pelviureteric junction) obstruction pattern | 10–30% |
| VUR (vesicoureteral reflux) | 10–40% |
| UVJ (ureterovesical junction) obstruction/megaureter | 5–15% |
| MCDK | 2–5% |
| Duplex system, ureterocele, ectopic ureter | 5–7% |
| LUTO (PUV, prune-belly) | 0.1–0.5% |

- The chance of postnatal pathology rises with grade: about 12% (mild) to 88% (severe). UPJ obstruction and PUV track grade; **VUR is not predicted by grade**.

### Grading systems

**APD (anteroposterior renal pelvic diameter, APRPD):** transverse image at the hilum. APD normal <4 mm before 28 weeks, <7 mm after, <10 mm postnatally.

| Grade | Second trimester | Third trimester |
|---|---|---|
| Mild | 4 to <7 mm | 4 to <9 mm |
| Moderate | 7–10 mm | 9–15 mm |
| Severe | >10 mm | >15 mm |

**SFU (postnatal only):** 1 pelvis barely split; 2 pelvis and major calyces; 3 minor calyces uniformly dilated; 4 plus parenchymal thinning.

**UTD consensus 2014 (risk stratification):** six parameters: APD, central/peripheral calyces, parenchymal thickness, parenchymal appearance, ureter, bladder (plus amniotic fluid prenatally).

| Prenatal parameter | A1 low risk | A2–3 increased risk (any one) |
|---|---|---|
| APD 16–27 weeks | 4 to <7 mm | ≥7 mm |
| APD ≥28 weeks | 7 to <10 mm | ≥10 mm |
| Calyces | Central or none | Peripheral |
| Parenchyma, ureter, bladder | All normal | Abnormal |
| Amniotic fluid | Normal | Unexplained oligohydramnios |

- **A1:** one repeat US after 32 weeks. **A2–3:** US every 4 weeks to delivery. Both: postnatal US after 48 hours.

### Prenatal algorithm (unilateral UTD, male fetus)

```
Anomaly scan: unilateral UTD → Detailed scan: contralateral kidney, ureter, bladder wall and emptying, posterior urethra (keyhole), amniotic fluid, other anomalies ├─ Bilateral / thick bladder / keyhole / oligohydramnios ─► LUTO pathway │ (fetal medicine; karyotype; Ruano staging; possible shunt) ├─ A1 ─► repeat US after 32 weeks └─ A2–3 ─► US every 4 weeks to delivery → Normal delivery at local centre; no fetal intervention for unilateral UTD → Postnatal US after 48 hours (earlier if solitary kidney or suspected LUTO)
```

- Repeat scans because the lesion may progress and a second anomaly is more likely. **Fetal MRI** (magnetic resonance imaging) only for complex anatomy.
- Counsel the parents: most cases resolve; prognosis with a normal opposite kidney is excellent.

### Antenatal management of fetal uropathy (suspected LUTO): evaluation

- Male fetus, thick-walled "keyhole" bladder, bilateral hydroureteronephrosis, oligohydramnios: LUTO until proved otherwise.
- Anomaly scan, **fetal echocardiogram, karyotype** (one-fifth of megacystis is aneuploid); fetal MRI if unclear.
- **Serial fetal urine** (3 samples, 48–72 hours apart). Favourable: Na <100 mEq/L, Cl <90 mEq/L, osmolality <200 mOsm/L, β2-microglobulin <6 mg/L; not validated.
- **Ruano staging:** I mild, normal fluid: observe. **II severe, function preserved** (oligohydramnios, favourable urine): intervene. III cysts, dysplasia, unfavourable urine: intervention for the lungs only.

### Antenatal management of fetal uropathy: options

| Option | Notes |
|---|---|
| Observation, serial scans | Most cases |
| **VAS (vesicoamniotic shunt)** | Dislodgement, blockage, preterm labour |
| **Fetal cystoscopy + laser valve ablation** | Diagnoses and treats; fistula up to 10% |
| Serial amnioinfusion | Lungs in anhydramnios; under trial |
| Early delivery / termination | Mature lungs, falling fluid / poor prognosis |

- Shunt or fetal cystoscopy is done at 18–24 weeks in severe LUTO.
- **PLUTO (2013):** shunt versus conservative (31 fetuses): better perinatal survival, **no renal benefit**.

### Antenatal management of fetal uropathy: counselling and delivery

- Multidisciplinary team: fetal medicine, paediatric urology, nephrology, neonatology.
- Intervention improves survival, not renal function; complications **over 40%**.
- Deliver where the bladder can be decompressed at once; prophylaxis, then VCUG (voiding cystourethrogram).

### Postnatal algorithm

- **First US after 48 hours**, because neonatal dehydration under-reads dilation. Postnatal normal: APD <10 mm, no calyceal or ureteral dilation.

| | P1 (SFU 1–2) | P2 (SFU 3) | P3 (SFU 4) |
|---|---|---|---|
| Findings | APD 10–15 mm, central calyces | APD ≥15 mm, peripheral calyces or dilated ureter | Abnormal parenchyma or bladder |
| Repeat US | 3–6 months | Individualised | 1 month |
| Antibiotic prophylaxis | No | Discretion; **yes if ureter >7 mm** | Yes |
| VCUG | No | Discretion | Yes |
| MAG3 (mercaptoacetyltriglycine) renogram | No | Discretion | If persistent |

- **Male-specific points:** VCUG confirms a normal posterior urethra. **Circumcision** protects (NNT [number needed to treat] 10). UTI (urinary tract infection) risk: female sex, uncircumcised boys, high grade, dilated ureter.
- Prophylaxis: amoxicillin or a first-generation cephalosporin from birth; nitrofurantoin and trimethoprim only after 8 weeks.
- MAG3: DRF (differential renal function) <40%, T½ >20 minutes, or delayed retention suggest obstruction. MRU (magnetic resonance urography) for anatomy; DMSA (dimercaptosuccinic acid) for MCDK or duplex function.
- **Surgery indications (UPJ):** rising APD, DRF <40% with poor drainage, falling DRF, symptoms (UTI, stone, pain). Megaureter surgery: DRF <40%, febrile UTI; defer reimplant beyond 1 year.
- Outcome: SFU 1 resolves in nearly all by 2.5 years; SFU 2 about 95% by 4 years.

### Bilateral dilation and PUV: how they differ

- Bilateral hydroureteronephrosis, thick bladder, keyhole sign and oligohydramnios mean LUTO until proved otherwise.
- At birth: 5 Fr feeding tube, prophylaxis, creatinine, early VCUG; prenatal intervention improves survival but not renal function (PLUTO).

**Diagram to draw:** UTD A1/A2–3 prenatal and P1–P3 postnatal flowchart; keyhole bladder with bilateral hydroureteronephrosis and shunt.

**Key points to remember:**

- UTD: 1–3% of pregnancies; second commonest anomaly after cardiac.
- Anomaly scan 18–20 weeks; after 16 weeks amniotic fluid = fetal urine.
- ANH = UTD; APD normal <4 mm before 28 weeks, <7 mm after, <10 mm postnatally.
- UTD A2–3 = APRPD ≥7 mm (16–27 weeks) or any abnormal feature.
- Transient dilation 50–70% (commonest cause); UPJ obstruction commonest surgical cause; VUR is not predicted by grade.
- First postnatal US after 48 hours; A1: one scan after 32 weeks; A2–3 scan every 4 weeks.
- Risk-based care: P1 no prophylaxis or VCUG; prophylaxis if ureter >7 mm or P3; P3 prophylaxis, VCUG, MAG3.
- Keyhole + oligohydramnios = PUV; absent bladder = exstrophy or agenesis.
- ARPKD large bright kidneys; MCDK non-communicating cysts.
- Favourable urine: Na <100, Cl <90, osmolality <200, β2-microglobulin <6.
- PLUTO: survival up, renal function unchanged.

**Also, from the condensed book (not in the paper answer):**

- The metanephric kidney appears at **day 28** and urine production starts at **14 weeks**. After **~16 weeks** amniotic fluid is mainly urine, so obstruction causes oligohydramnios and pulmonary hypoplasia.
- The amniotic fluid index (AFI) is normally 8–18 cm; oligohydramnios is **<5–6 cm**.
- Ultrasound severity tracks PUJ obstruction and posterior urethral valves (PUV), but not VUR or duplex. Fetal MRI is used only for anatomy, such as cloaca or exstrophy.

**KEY POINT**

Prenatal detection is for triage and counselling; there is no proof it improves outcome.

## 2 · Grading systems

- **Anteroposterior pelvic diameter (APD)** is measured transversely at the hilum and is **meaningless without its gestational age**. Severe is >10 mm in the 2nd trimester and >15 mm in the 3rd.
- Postnatal pathology rises from about 12% (mild) to 88% (severe), but **VUR shows no trend**.
- **SFU** grading (postnatal): 1, sinus barely split; 2, major calyces; 3, minor calyces with parenchyma spared; 4, parenchymal thinning. All SFU 3 and about 70% of SFU 4 kidneys keep DRF >40%.
- Pyramidal thickness **≤3 mm** predicts pyeloplasty and reduced function.
- The **urinary tract dilation (UTD)** classification uses six parameters plus oligohydramnios and covers the whole tract, pre- and postnatally.

## 3 · Prenatal follow-up

**ALGORITHM**

1. Normal APD ceiling, all else normal: **<4 mm (16–27 weeks) → <7 mm (≥28 weeks) → <10 mm (postnatal >48 h)**.
2. If A1 (APD 4 to <7 mm at 16–27 wk, or 7 to <10 mm at ≥28 wk), repeat US once after 32 weeks.
3. If A2–3 (any abnormal parameter), scan every 4 weeks until delivery.
4. In both, do a postnatal US after 48 h.

- Exstrophy, cloaca, ARPKD, outlet obstruction and myelomeningocele need a multidisciplinary prenatal consult.

## 4 · Fetal LUTO and oligohydramnios

*Fetal intervention saves lungs, not kidneys.*

- The picture is a male fetus with bilateral hydroureteronephrosis, a non-cycling bladder, the **"keyhole" sign** and oligohydramnios. Only fetal cystoscopy can separate PUV, urethral atresia and prune-belly syndrome (PBS).
- Prognosis is poor with oligohydramnios **before 27 weeks**, cortical cysts or echogenicity.
- Fetal urine biomarkers are **not validated** (European consensus 2022). †Classic favourable values were Na <100 mEq/L, Cl <90 mEq/L, osmolality <200 mOsm/L and β2-microglobulin <4 mg/L.

| Ruano stage | Fluid / kidneys | Urine biochemistry | Intervention |
|---|---|---|---|
| I — mild | Normal fluid and echogenicity | Favourable | Not indicated |
| II — severe, function preserved | Oligo/anhydramnios; hyperechoic; no cysts | Favourable within 3 samples | **Indicated** — for lungs and kidneys |
| III — severe, function abnormal | Anhydramnios; hyperechoic, with or without cysts and dysplasia | Unfavourable after 3 samples | Maybe — for lungs only |

### Options

- Options are early delivery, vesicoamniotic shunt (VAS), fetal cystoscopy and serial amnioinfusion; the window is **18–24 weeks**.
- Complications occur in about **44%**, most often shunt failure, followed by preterm labour and membrane rupture.

| Feature | VAS | Fetal cystoscopy |
|---|---|---|
| Diagnoses cause | No | Yes |
| Mechanism | Bladder-to-amniotic bypass | Laser valve ablation |
| Specific complications | Blockage, migration, bowel herniation | Urethral fistula (≤10%) |
| Survival / renal function (PUV) | Improved / not improved | Improved / short-term only |

- After VAS, over half have renal insufficiency and a third need renal replacement therapy.

**KEY POINT**

Counsel: intervention improves perinatal survival (lungs); complications exceed 40%; it does not reliably prevent renal replacement, because dysplasia is often established first.

## 5 · Postnatal imaging

- Do the **first US after 48 h**, because neonatal dehydration under-reads dilation.
- Postnatal P1 (APD 10 to <15 mm, central calyces), P2 (≥15 mm, peripheral calyces or abnormal ureter) and P3 (abnormal parenchyma or bladder) roughly equal SFU 1–2, 3 and 4. **Classify by the highest-risk feature**.
- **VCUG** finds VUR in up to 31%; order it by **UTI risk** (females, uncircumcised males, high-grade dilation, dilated ureter, abnormal bladder).
- Contrast-enhanced voiding urosonography avoids radiation but shows the urethra poorly.
- **MAG3** is preferred over DTPA. Consider surgery if **DRF <40%**, T½ >20 min or retention persists upright. Pitfalls are dehydration and poorly timed furosemide.
- **DMSA** shows scars, dysplasia and duplex upper-pole function; **MRU** shows ectopic ureters and crossing vessels but needs sedation.

## 6 · Postnatal management, UTI and prophylaxis

- **P1** needs only a US at 3–6 months; **P2** needs CAP **if the ureter is >7 mm**; **P3** needs CAP, VCUG and MAG3 if dilation persists.
- In P3 without VUR, give CAP until pyeloplasty or **1 year**, as about 90% of UTIs occur in year 1.
- UTI occurs in 8–22%; agreed risks are **female sex, high-grade dilation and intact foreskin**. Circumcision has a number needed to treat of **10**.
- Evidence for CAP is weak. Use amoxicillin or a cephalosporin from birth; nitrofurantoin or trimethoprim only from **≥8 weeks**.

**ALGORITHM**

1. US after 48 h: if normal, the dilation was transient.
2. If P1, repeat US at 3–6 months.
3. If P2, give CAP if the ureter is >7 mm; VCUG and MAG3 at discretion.
4. If P3, give CAP, do a VCUG and repeat US at about 1 month; if dilation persists, do a MAG3.
5. If bilateral severe dilation, treat as outlet obstruction: decompress, CAP, urgent evaluation.

## 7 · Specific antenatal diagnoses

- **Transient** dilation (50–70%) is commonest, and mild dilation resolves in 90–98%, mostly by 2–4 years.

### PUJ obstruction pattern

- It is the commonest **surgical** cause. Third-trimester APD **15 mm** has about 80% sensitivity and specificity for pyeloplasty.

- Serial US suffices if P2, APD <20 mm and pyramidal thickness >3 mm, since DRF >40% is likely.
- Operate for rising dilation, DRF <40% with poor drainage, falling DRF, upright retention or symptoms.

### Vesicoureteral reflux

- Prenatally detected VUR is typically male, high-grade and bilateral, with congenital dysplasia; VUR after febrile UTI is typically female, lower-grade and acquired. Screen girls with high-grade UTD and uncircumcised boys.

### Megaureter / ureterovesical junction obstruction

- Most resolve; surgery is predicted by a ureter **>13 mm** and DRF <30%.
- BAPU: a prenatal ureter **>7 mm after 30 weeks** needs postnatal US, CAP and VCUG.
- Operate for DRF <40%, febrile UTI or pain. **Under 1 year, temporise** with ureterostomy or a refluxing reimplant, and reimplant after 1 year.

> **KEY POINT**
>
> In PUJ, VUR and megaureter, dilation alone is not an indication: operate for function loss, infection or progression; grade sets surveillance.

### LUTO conditions

- **PUV** is the commonest LUTO. **PBS** combines megacystis, a deficient abdominal wall and impalpable testes.
- **Urethral atresia** means a catheter will not pass; it is managed by vesicostomy, then progressive dilation of the anterior urethra (PADUA). Massive reflux and MMIHS also cause megacystis.

> **KEY POINT**
>
> Bilateral dilation with normal fluid means urine still leaves the fetus (reflux, megacystis); with low fluid it means true outlet obstruction.

### Neonatal management of LUTO

**ALGORITHM**

1. Have the team at delivery and resuscitate the lungs first.
2. Pass a 5 Fr feeding tube, with a finger in the rectum to guide it.
3. Confirm position on a plain film (KUB).
4. If it fails, place a US-guided suprapubic catheter.

- **Do not use a balloon catheter**, which obstructs the posterior urethra and ureteric orifices. Judge decompression by the **creatinine nadir**, and warn of post-obstructive diuresis.
- Oliguria after decompression suggests a thick bladder obstructing the ureters; treat with antimuscarinics.

### Multicystic dysplastic kidney

- VUR occurs in about 20%. There is **no Wilms association**, so no routine nephrectomy; follow with US and blood pressure.

### Duplex, ureterocele, ectopic ureter

- By **Meyer–Weigert**, the upper-pole orifice is inferomedial and obstructs, while the lower-pole orifice is superolateral and refluxes.
- Suspect a duplex when the ureter is more dilated than the pelvis.
- Endoscopic puncture of a ureterocele is increasingly first-line, and mandatory for outlet obstruction or sepsis. High-grade VUR usually needs ureterocele excision and common-sheath reimplant.

### Exstrophy

- Both forms show a non-visualised bladder, low umbilicus and pubic diastasis. **Cloacal exstrophy (OEIS)** adds omphalocele, imperforate anus and spinal anomalies.

### Cloaca, urogenital sinus, CAH

- A **cloaca** shows a single channel, a pelvic cyst and calcified meconium; a **urogenital sinus** has no bowel obstruction and normal external genitalia.
- **CAH** (21-hydroxylase, autosomal recessive) is 75% salt-wasting. Only **1 in 8** at-risk pregnancies is an affected girl, so prenatal treatment is controversial.

### Myelomeningocele

Fetal closure before 26 weeks reduces shunting and improves motor outcome, but gives **no bladder benefit**.

## 8 · Neonatal urological emergencies

### Urinary ascites and retention

- Urine causes up to 30% of neonatal ascites, and **PUV is the commonest cause**.
- Spontaneous bladder rupture heals with a catheter, but injury from an umbilical catheter needs repair.
- Retention >24 h suggests PUV or atresia in boys, and urogenital sinus or ureterocele in girls.

### Abdominal masses

- **Congenital mesoblastic nephroma** is the commonest solid renal mass under 6 months.
- **Adrenal haemorrhage** has no Doppler flow, shrinks on serial US and calcifies like an eggshell, whereas **neuroblastoma** is vascular and persists. Haemorrhage may track down the processus and mimic an acute scrotum.

### Anorectal malformation and ambiguous genitalia

- Anorectal malformation has genitourinary or spinal anomalies in up to **75%**, so get renal and spinal US.
- In ambiguous genitalia, defer gender assignment and **exclude salt-wasting first**; the commonest cause is CAH.

### Interlabial masses

| Mass | Position | Management |
|---|---|---|
| Imperforate hymen | Midline bulging membrane | Hymenal incision |
| Urethral prolapse | Midline collar at meatus | Conservative |
| Prolapsed ureterocele | Off-midline from meatus; hymen intact | US and VCUG |

| Mass | Position | Management |
|---|---|---|
| Paraurethral cyst | Off-midline, displaces meatus | Most resolve |

### Gross haematuria

First exclude an oestrogen-withdrawal bleed and urate crystals, then get a Doppler US.

| Feature | Renal vein thrombosis | Renal artery thrombosis | Cortical necrosis |
|---|---|---|---|
| Risk | Maternal diabetes, prematurity, umbilical venous catheter, polycythaemia | Umbilical artery catheter | Hypoperfusion |
| Presentation | **Triad**: haematuria, flank mass, thrombocytopenia | Hypertension, heart failure | Renal failure |
| Treatment | Hydration, with or without anticoagulation | Remove catheter, with or without thrombolysis | Supportive |

### Scrotal masses and torsion

- **Prenatal torsion** gives a hard, non-tender testis at birth with almost no salvage, so do urgent contralateral orchidopexy.
- **Postnatal torsion** (first 30 days) has salvage up to 44%, so explore immediately and fix the other side.

## 9 · Landmark trials

| Trial | Design | Result | Take-home |
|---|---|---|---|
| PLUTO | RCT, VAS vs conservative in fetal LUTO; 31 randomised, stopped early | Fewer perinatal deaths; no renal benefit; 44% complications | Shunting saves lungs, not kidneys |
| Great Ormond Street pyeloplasty RCT | APD ≥15 mm, DRF >40%: pyeloplasty vs observation | Most observed kidneys stable or improved; a minority (APD 20–40 mm) lost function | Observation safe with monitoring |
| MOMS† | RCT, prenatal vs postnatal myelomeningocele closure | Less shunting, better motor outcome; no urodynamic benefit | Fetal closure is not for the bladder |

# 35 Urologic Evaluation of the Child

## 1 · Triage by acuity

Triage asks: **what is lost by waiting** — an organ, a life, or nothing?

| Tier | Conditions |
|---|---|
| Emergent | Acute scrotum; acute abdomen; AKI; obstruction with UTI and/or AKI; retention; incarcerated hernia; paraphimosis; priapism; trauma; newborn ambiguous genitalia, exstrophy, renal vein thrombosis |
| Urgent (24 h) | Gross haematuria **without infection**; newborn bilateral or solitary-kidney hydronephrosis; renal or testicular tumour; spina bifida |
| Semiurgent (1–3 wk) | After treated febrile UTI; postnatal dilation (no outlet obstruction); reducible hernia; interlabial mass; amenorrhoea |
| Routine | Hydrocele; circumcision; incontinence or enuresis; hypospadias; labial adhesions; meatal stenosis; microscopic haematuria; recurrent UTI; undescended testis; varicocele |

**KEY POINT**

Context shifts the tier: a hernia is emergent if incarcerated but semiurgent if reducible; hydronephrosis is urgent if bilateral or in a solitary kidney.

## 2 · History

- For hydronephrosis, ask about prenatal history, progression and **amniotic fluid**, a proxy for fetal renal function.
- Obtain **consent from the guardian and assent from the child**; adolescents get a private history.

## 3 · Examination

*Children hide shock until late.*

- Minimum systolic BP is **≥60** in neonates, ≥70 at 1 month–1 year, **70 + 2×age** at 1–10 years and ≥90 above 10 years.
- **Shock shows first as tachycardia and tachypnoea; hypotension means decompensation.**

*See the paper answer **Testicular descent and undescended testis: evaluation, treatment algorithm and orchidopexy methods** (Section 4, Ch 57).*

**Also, from the condensed book (not in the paper answer):**

- **Micropenis** is a stretched length below mean − 2.5 SD, which is **1.9 cm** at term.
- In **Tanner staging**, the first male sign is testicular enlargement (stage II), not hair.
- A **simple sacral dimple** (<5 mm, within the cleft, <25 mm above the anus) needs no imaging; image for a lipoma, tag, deep dimple or deviated cleft.

## 4 · Urine and blood tests

- In febrile children under 24 months, use a **catheter or suprapubic aspirate**; **bag specimens have high false-positive rates** and help only if negative.

- The AAP defines UTI as pyuria and/or bacteriuria with **≥50,000 colony-forming units (CFU)/mL** (catheter or aspirate) in a symptomatic child.

| Test | Threshold |
|---|---|
| Microscopic haematuria | **>5 red cells (RBC)/high-power field** (adult ≥3), on >2 occasions |
| Calcium/creatinine ratio (hypercalciuria) | >0.4 at age 2–5 years; >0.2 above 5 years |
| Normal protein/creatinine ratio | <0.7 in the first year; <0.4 up to 4 years; <0.2 after 5 years; >3.0 is nephrotic |
| Specific gravity | <1.010 dilute (water excess, diabetes insipidus); >1.020 dehydration |

- Always **confirm dipstick blood by microscopy**, and test proteinuria on a **first-morning** sample.

## 5 · Imaging

Only US and MRI avoid radiation; keep doses as low as reasonably achievable.

| Modality | Answers |
|---|---|
| US | Kidney, bladder, scrotum, torsion, spine |
| VCUG | Reflux, urethra, genitogram |
| CT | Severe trauma, stones |
| MRI | Tumours, ectopia, duplication, Müllerian or sinus anatomy |
| MAG3 renogram | **Obstruction** and differential function |
| DMSA scan | **Scarring** or pyelonephritis, and differential function |

## 6 · Emergent presentations

### Acute scrotum

- The commonest cause is **torsion of the appendix testis**, with point tenderness in a prepubertal boy; treat it supportively.
- **Cord torsion** (mostly over 10 years) gives severe pain, vomiting, a high-riding testis and an absent cremasteric reflex.
- It is a **clinical diagnosis**: explore without imaging and fix the other side.

| TWIST item | Points | Score and action |
|---|---|---|
| Testicular swelling | 2 | 0: no US |
| Hard testis | 2 | 1–5: scrotal US |
| Absent cremasteric reflex; nausea or vomiting; high-riding testis | 1 each | **≥6**: explore without US |

### Other emergencies

- In **ambiguous genitalia**, exclude **salt-wasting CAH** (21-hydroxylase in about 95%). Crisis comes on day **4–15**, so do not discharge early; check **17-hydroxyprogesterone and electrolytes**.
- In **exstrophy**, tie the cord with silk (no clamp), cover the plate with cling film and transfer.

- **Infected obstruction** needs prompt drainage by stent or nephrostomy.
- In **paraphimosis**, exclude a **hair tourniquet**, then compress the glans under analgesia; a dorsal slit is the fallback.
- Ischaemic **priapism** is usually from **sickle cell disease**; give oxygen, fluids and analgesia.
- In **trauma**, the **mechanism** decides CT. After straddle injury consider **sexual abuse**; meatal blood needs a retrograde urethrogram, not a blind catheter.
- In **retention**, expected capacity (mL) is **(age + 2) × 30**. Try analgesia and disimpaction before a catheter.

## 7 · Common referral presentations

### Antenatal hydronephrosis

- Scan **before discharge** if bilateral, in a solitary kidney or with suspected outlet obstruction; otherwise after **48 h**.
- A boy with bilateral dilation and a thick bladder needs a **VCUG before discharge**, catheterised meanwhile.

### Haematuria

- Glomerular haematuria is tea-coloured with casts and protein; urological haematuria is bright red with clots.
- Investigate all gross haematuria with urinalysis, culture and US. Isolated microhaematuria is usually benign, so repeat it over 2–3 weeks.

### Bladder bowel dysfunction and enuresis

- **Always examine the back**, and treat stepwise, starting with behavioural therapy.

### Scrotal, inguinal and genital

- A communicating **hydrocele** is smaller in the morning and is treated by high ligation at the internal ring.
- An **undescended testis** with hypospadias or micropenis suggests a DSD.
- For **varicocele**, the Lambert volume is L×W×H×**0.71**; a non-decompressing or right-sided one needs a retroperitoneal US.
- Prepubertal **testicular tumours** are mostly benign, and tumour markers guide testis-sparing surgery.
- **Meatal stenosis** is treated by meatotomy under EMLA, and symptomatic **labial adhesions** by oestrogen or steroid cream.

> **KEY POINT**
>
> Red flags: absent cremasteric reflex with acute pain; ambiguous genitalia with impalpable gonads; bilateral hydronephrosis in a boy; fever with obstruction; hypotension; abdominal mass; meatal blood; lumbosacral lipoma or deviated cleft.

## 8 · Child comfort

- Distraction, midazolam, intranasal ketamine and EMLA cream can avoid general anaesthesia.

# 36 Pediatric Urogenital Imaging

## 1 · Principles — ALARA and the three harms

- Under **ALARA**, **every child is treated as radiation sensitive**. **Sedation risk usually exceeds contrast risk**, and ultrasonography (US) is **safe at all ages and gestations**.

> **KEY POINT**
>
> Every choice trades **radiation, sedation and contrast** against yield; the long lifespan tips it towards US, and the question is which test fills US's functional gap (obstruction, reflux, pyelonephritis) most cheaply.

## 2 · Ultrasonography

*US comes first, but shows anatomy, not function.*

### SAQ Ultrasonography in neonates

Paper answer · P1 · P3 · merged from 2

*Asked as:* P1 SAQ "USG in neonates" · P3 SAQ "HSG in neonates"

#### Introduction

- Assumed meaning: ultrasonography (USG) of the neonatal urinary tract, including antenatal hydronephrosis follow-up. "HSG" (hysterosalpingography) in the P3 wording is not done in neonates, so it is taken as USG.
- First-line imaging, used for antenatal **UTD (urinary tract dilation)** (1–3% of pregnancies), febrile UTI (urinary tract infection), mass and poor stream.

#### Why ultrasound first

- No radiation or sedation; repeatable. Shows **anatomy, not function**.

#### Timing and technique

- **First scan after 48 hours**: earlier, physiological (relative) dehydration under-reads dilation.
- **Scan early if outlet obstruction (PUV, posterior urethral valves) is suspected or the kidney is solitary**; catheterise in suspected obstruction.
- Higher-frequency probes than in adults; kidneys supine and prone; length against age nomograms; bladder full and after voiding; Doppler.
- Measure **APD (anteroposterior pelvic diameter)** on a transverse mid-kidney view; assess calyces, parenchyma, ureters and bladder wall (thick if over 3 mm when full).

#### Normal neonatal kidney (pitfalls)

- Cortex **iso- or hyperechoic** to liver.
- **Prominent hypoechoic pyramids** are normal and mimic dilated calyces or cysts (not hydronephrosis).
- Normal postnatal **APD <10 mm**, with no calyceal or ureteric dilation.

#### Grading

- **SFU (Society for Fetal Urology) grades:** 0 none; 1 pelvis only; 2 major calyces; 3 major and minor calyces; 4 grade 3 plus parenchymal thinning.

- **UTD 2014, prenatal:** A1 (APD 4 to <7 mm before 28 weeks, 7 to <10 mm after) versus A2–3 (larger, or any other abnormality).

### Antenatal hydronephrosis follow-up (UTD 2014, postnatal)

| UTD risk | Findings | Action |
|---|---|---|
| **P1** (SFU 1–2) | APD 10 to <15 mm, central calyces | Repeat US at 3–6 months; no VCUG (voiding cystourethrogram) or prophylaxis |
| **P2** (SFU 3) | APD ≥15 mm, peripheral calyces or dilated ureter | Repeat US; VCUG and prophylaxis at discretion / optional (give prophylaxis if ureter >7 mm) |
| **P3** (SFU 4) | Abnormal parenchyma or bladder | Repeat US at 1 month; VCUG, prophylaxis, MAG3 (mercaptoacetyltriglycine) scan |

**Diagram to draw:** SFU grades 0–4 as kidney outlines.

### Specific neonatal findings

| Condition | Ultrasound finding |
|---|---|
| PUJO (pelviureteric junction obstruction) | Dilated pelvis, normal ureter; pyramids under 3 mm predict pyeloplasty |
| VUR (vesicoureteral reflux) | Normal first scan in up to 25% |
| **PUV** | Thick bladder, dilated posterior urethra (**keyhole sign**), bilateral hydroureteronephrosis; VCUG before discharge |
| Duplex anomalies | Ureterocele (thin-walled intravesical cyst), ectopic ureter, megaureter |
| MCDK (multicystic dysplastic kidney) | Non-communicating cysts of varying size, no central pelvis, scant parenchyma |
| ARPKD (autosomal recessive polycystic kidney disease) | Bilateral large, bright (echogenic) kidneys |
| Renal vein thrombosis | Large echogenic kidney, no venous flow; haematuria, mass, thrombocytopenia, hypertension |
| Adrenal haemorrhage | Avascular, evolving suprarenal mass (neuroblastoma is vascular) |
| Other masses | Mesoblastic nephroma |
| Spine | Occult dysraphism before arch ossification (up to 6 months) |
| DSD (differences of sex development) | Müllerian structures, gonads |

- **Also know:** hydrometrocolpos, scrotal masses, twinkling artefact (stones).

### Limitations

- Normal US does not exclude VUR. Function needs MAG3 renography (after renal maturation) or DMSA (dimercaptosuccinic acid).

### Recent advances

- ceVUS (contrast-enhanced voiding urosonography).

**Key points to remember:**

- First postnatal scan after 48 h, except suspected PUV; normal APD <10 mm.
- Echogenic cortex and prominent pyramids are normal in neonates.
- UTD: P1 APD 10 to <15 mm; P2 15 mm or more; P3 abnormal parenchyma or bladder.
- Keyhole sign + thick bladder = PUV; spinal US only up to 6 months.

**Also, from the condensed book (not in the paper answer):**

- Fetal urine is made from **10–12 weeks**; by **20 weeks** US detects most anomalies.
- SFU grade 4 is defined by parenchymal thinning, not more dilatation.
- **Bilateral hydroureteronephrosis with a thick, poorly emptying bladder needs a VCUG** for posterior urethral valves (PUV); megaureter or ureterocele needs diuretic renography or MRU.

## Cysts, masses, stones

| Lesion | Side | US hallmark |
|---|---|---|
| MCDK (multicystic dysplastic) | Unilateral | Non-communicating random cysts, scant echogenic parenchyma |
| ARPKD (recessive polycystic) | Bilateral | Large homogeneously **bright** kidneys, not discrete cysts |
| ADPKD (dominant) | Bilateral | Parenchyma almost replaced by cysts |
| Multilocular cystic nephroma | Unilateral | Like MCDK, more parenchyma; infancy |
| Cystic Wilms | Unilateral | Overlaps; 2–4 years |

- A neonatal solid mass is usually **mesoblastic nephroma**; a toddler's is Wilms. Neonatal enlargement with acute kidney injury and hypertension suggests **renal vein thrombosis**.
- **Stones** are echogenic **with** shadowing (fat is without), and Doppler **twinkling artefact** confirms them.

## Infection and trauma

- In UTI, US is not routine at the start; **do US if there is failure to improve on antibiotics**. It misses low-grade VUR and scars, where DMSA is better. **Normal US neither risk-stratifies APN nor predicts VUR.**

## Contrast-enhanced voiding urosonography (CeVUS)

- Microbubbles grade VUR on five grades with **no radiation**. Area under curve is **0.97**, so it is recommended for VUR.

## Bladder/urethra

- The **keyhole sign** (thick bladder and dilated posterior urethra) means PUV.
- To tell clot from tumour, scan supine then prone, because a **clot moves**.

## Scrotum and genitalia

- **Never delay exploration for suspected torsion**.
- The **whirlpool sign** gives torsion sensitivity ~96% vs ~76% for colour Doppler, with specificity ~99%.

- Appendage torsion is an avascular nodule with surrounding hyperaemia. **Teratoma** is the commonest prepubertal tumour; epidermoid shows **onion rings**.
- **Do not image a routine undescended testis** (AUA Choosing Wisely) except in an obese child.

## 3 · Voiding cystourethrography

*VCUG is the reference test for reflux and urethral anatomy.*

- **After UTI**, do it once the child is improving on antibiotics; a 4–6-week wait is unnecessary, though very early VCUG may overgrade VUR.
- Indications are hydronephrosis with **dilated ureter(s)**, febrile or recurrent UTI, suspected PUV, epididymitis in boys, and UTI in a circumcised boy.
- **Occult VUR**: VCUG finds VUR in only ~40% of DMSA-proven APN. **PIC cystography** (positional instillation of contrast at the orifice) is used when VCUG is negative but febrile UTI recurs.
- **Trauma**: with meatal blood, do a **retrograde urethrogram (RUG) before any catheter**. Capacity (mL) = **(age + 2) × 30**, max 400. A "pie-in-the-sky" bladder means pelvic fracture with extraperitoneal injury.

## 4 · Nuclear medicine

### DMSA

- DMSA is taken up by proximal tubules, so APN shows photon-deficient areas.

| Feature | Acute pyelonephritis | Scar |
|---|---|---|
| Contour | Preserved | Volume loss, concavity |
| Timing | Within days; resolves over ~5 months | Lesion at **≥6 months** = scar |

- A third to a half of acute lesions scar.

### MAG3 diuretic renography

*See the paper answer **Radionuclide studies and diuretic renography in PUJ obstruction** (Section 11, Ch 93).*

**Also, from the condensed book (not in the paper answer):**

- Give furosemide **1 mg/kg IV** at maximal filling; if given too early, drainage is measured before the system is full.
- There is **no validated paediatric T½ cut-off**, so decide on age, function, curve, upright drainage and serial studies. A **biphasic curve** and poor upright drainage predict progression.

### Radionuclide cystography and testicular scan

- **Direct** cystography is more sensitive and lower-dose than VCUG but grades coarsely, so it suits follow-up. **Indirect** cystography is insensitive and not routine.

## 5 · MR urography

- MRU gives anatomy and function without radiation and suits complex anomalies.

- Protocol: hydration and catheter, then **furosemide 1 mg/kg (max 20 mg)**, T2, gadolinium and dynamic T1; function is measured by **Patlak** analysis.
- It needs sedation under 6 years and is best for **bladder rhabdomyosarcoma** and **local staging of renal tumours**.

## 6 · CT and radiography

- CT is chosen for speed in trauma and staging, not for hydronephrosis.
- **In trauma, arterial-phase images miss collecting-system injury, so delayed excretory images are mandatory.** CT cystography separates intra- from extraperitoneal rupture.
- CT rarely changes stone care. IVP is obsolete.

## 7 · Choice of test by clinical question

| Question | First line | Next | Avoid |
|---|---|---|---|
| Hydronephrosis — obstruction? | US | MAG3; MRU | CT, IVP |
| Hydroureteronephrosis, with or without thick bladder | US | VCUG | — |
| VUR | VCUG | CeVUS; direct cystogram; PIC if occult | Indirect cystogram |
| APN / scar | DMSA | — | Grey-scale US; MRI |
| Stones | US | Low-dose CT; KUB | Reflex CT |
| Renal mass | US | Contrast CT; MRI | — |
| Bladder mass | US (full bladder) | MRI | — |
| Trauma | CT with delayed phase; US | RUG; cystography | Catheter before RUG if meatal blood |
| Acute scrotum | Clinical, with or without Doppler/whirlpool | Scintigraphy if equivocal | Imaging delaying exploration |
| Undescended testis | Examination | US only if obese | Routine US, MRI |
| DSD | US and genitography | MRI | CT |

# 37 Infection and Inflammation of the Paediatric Genitourinary Tract

## 1 · Overview and the paradigm shift

- Urinary tract infection (UTI) affects ~**1% of boys, 3–5% of girls**, and 30–50% recur.
- Congenital dysplasia now explains more scarring than reflux does, so the approach is **selective imaging, prompt treatment and prophylaxis only for the high-risk child**.

## 2 · Pathogenesis and virulence

- Infection is **ascending**: gut bacteria colonise the perineum, enter the bladder, adhere to umbrella cells and may ascend. Uropathogenic E. coli (UPEC) causes most UTIs.

| Factor | Mechanism | Association |
|---|---|---|
| Type 1 fimbriae (mannose-sensitive) | FimH binds uroplakins, allowing entry | Cystitis, asymptomatic bacteriuria (ASB) |
| P fimbriae (mannose-resistant) | Bind Galα1–4Galβ (P blood group) | **Pyelonephritis** |
| Aerobactin | Siderophore | Growth in iron-poor urine |
| α-Haemolysin, CNF1 | Cell lysis, cytoskeletal damage | Inflammation, invasion |
| Capsule (K antigen) | Antiphagocytic, anticomplementary | Pyelonephritis |
| Biofilm, flagella | Shield; motility | Recurrence; ascent |

- **Intracellular bacterial communities** (IBCs) form a **quiescent reservoir** that reseeds the lumen for months, causing same-strain recurrence despite adequate antibiotics.

> **KEY POINT**
> IBCs and biofilms shield bacteria from neutrophils and antibiotics; a biofilm survives 1000–1500× normal killing concentrations.

## 3 · Host risk factors

- In the first year UTI affects boys **3.7%** vs girls 2%; by 6 years, boys 2% and girls 7%.
- **Circumcision** lowers UTI in the first 6 months **~10-fold**. The number needed to circumcise is **111** normal boys, 11 with recurrent UTI and **4** with high-grade VUR.
- The **faecal reservoir** recycles the same strain, and antibiotics such as TMP-SMX destroy protective flora.
- **VUR** is found in 1–2% of newborns and **25–40%** after a first UTI. Most pyelonephritis occurs without VUR, but grade III–IV doubles pyelonephritic change.
- Treating **bladder and bowel dysfunction (BBD)** cuts recurrence and aids VUR resolution.
- In **neurogenic bladder**, 40–80% on CIC are bacteriuric; **ASB without VUR needs no antibiotics**, and frequent catheterisation prevents UTI.
- **Catheter-associated UTI** is the commonest nosocomial infection, so remove catheters early. Early infancy (IgG nadir) is the highest-risk period.

**KEY POINT**

Host factors either raise inoculum/dwell time (prepuce, faecal flora, stasis, constipation, catheter) or lower defence (immature immunity, receptor-bearing blood groups); prevention reverses whichever applies.

## 4 · Classification

- UTI is **complicated** (anomaly, dysfunction, foreign body; all neonatal UTI) or uncomplicated; upper or lower; first or recurrent.
- Febrile UTI is presumed pyelonephritis, but only **50–66%** of febrile children show acute changes on DMSA scan, the gold standard.

| Recurrence | Between episodes | Cause | Remedy |
|---|---|---|---|
| Unresolved bacteriuria | Never sterile | Non-compliance, malabsorption, resistance | Culture-directed retreatment |
| Bacterial persistence | Sterile; same organism | Nidus: stone, anomaly, papilla, catheter, stent | **Remove nidus** |
| Reinfection | Sterile; new organism | New ascent | Prevention |

### Asymptomatic bacteriuria

- Kass definition: two cultures of **≥$10^5$ colony-forming units (CFU)/mL** of the same organism, without symptoms.
- Infants with ASB are treated and scanned. In school-age girls without VUR or anomaly, **do not screen or treat**, because this selects resistance.

## 5 · Diagnosis

### Clinical

- Fever is ≥38 °C. A toxic look, age <30 days and ≥39.4 °C predict bacteraemia.
- Under 2 years, risk factors are fever >40 °C, previous UTI, suprapubic tenderness and an **uncircumcised penis**. **Another fever source, such as otitis, only halves UTI risk.**

### Urine collection

| Method | Contamination | Role |
|---|---|---|
| Bag | 50–60% (false-positive culture up to 75%) | **Reliable only when negative**; never culture |
| Clean catch | ~26% | Toilet-trained; preferred by NICE |
| Catheter | 10–15% | Sensitivity 95%, specificity 99%; **standard <2 years non-toilet-trained** |
| Suprapubic aspiration (SPA) | <1% | Reference; US guidance gives 80–90% success |

- Collect **before antibiotics** unless delay is dangerous. For 2–24 months the AAP accepts bag urinalysis; if positive, culture a catheter or SPA sample.

| Test | Sensitivity | Specificity | Use |
|---|---|---|---|
| Leukocyte esterase (LE) | ~79% | ~87% | Negative helps **rule out** |
| Nitrite | ~50% | **~98%** | Positive **rules in** |
| LE or nitrite or microscopy | 99.8% | 70% | Best rule-out |
| Pyuria ≥5 WBC/HPF | ~74% | ~86% | No advantage over LE |
| Gram stain, fresh urine | ~91% | 92–96% | Most reliable rapid test |

- **Nitrite is falsely negative** with short bladder dwell (infants), dilute urine and gram-positives (Enterococcus).
- UTI without pyuria is rare, which separates it from ASB. A threshold of **≥10 WBC/HPF** works better under 2 years.
- **Culture is essential** unless urinalysis is entirely normal.

| Guideline | Specimen | Significant count (CFU/mL) |
|---|---|---|
| Kass | Voided | $\geq 10^5$ |
| AAP 2011 (2–24 months) | Catheter/SPA | **≥50,000 + pyuria** |
| EAU | Midstream/catheter | $10^3$–$10^4$ monoculture |
| RCH Melbourne | SPA / catheter | Any gram-negative / $>10^3$ |
| ISPN (Italy) | Catheter / midstream | $>10^4$ / $>10^5$ |

- **The cleaner the specimen, the lower the accepted count.** Mixed growth and skin flora suggest contamination.
- CRP is sensitive, not specific. **Procalcitonin ≥0.5 ng/mL** best predicts acute pyelonephritis (APN) and later scarring.

## 6 · Imaging after UTI

- Routine imaging has not been shown to prevent scarring or renal failure, so **order a test only if it changes management**.
- US is abnormal in ~15% after a first febrile UTI, but only 1–2% need action. It is **urgent if the child is unwell or not responding at 48 h**.
- VCUG is the gold standard for VUR, done with sterile urine in a well child; CeVUS is an alternative.
- DMSA shows APN best within **1 week**; a defect at **≥6 months** is a scar, which cannot be told from dysplasia. CT is only for occult stones or mass vs inflammation.

| Guideline | Ultrasound | VCUG | Late DMSA |
|---|---|---|---|
| AAP (2–24 months) | Yes, after first febrile UTI | If US abnormal (or recurrence) | No |
| NICE <6 months | Yes | Atypical/recurrent | Atypical/recurrent |
| NICE >6 months | Atypical/recurrent | No | Atypical/recurrent |
| Top-down | No | If acute DMSA abnormal | If acute DMSA abnormal |
| RCH Melbourne | If <3 months or seriously ill | No | No |

- **NICE "atypical" UTI** means seriously ill, poor urine flow, abdominal or bladder mass, raised creatinine, septicaemia, no response within **48 h**, or a non-E. coli organism. Late DMSA is at 4–6 months.
- NICE and AAP miss about half of grade III–V VUR but spare most children a VCUG, because high-grade VUR declares itself at recurrence.

## 7 · Treatment

- **Delay matters**: new-scar odds double if antibiotics start after 72 h rather than within 48 h, so start empirically after urine collection.
- **Oral therapy equals intravenous** for non-toxic children over 2 months who tolerate oral drugs.
- **Admit** if under 1 month, toxic, dehydrated, vomiting or of doubtful compliance. Neonates need a septic work-up and IV therapy, switching to oral after ≥24 h afebrile.
- 90% are afebrile by 48 h. **If not improving at 48 h, do RBUS and broaden cover**.

| Infection | Duration |
|---|---|
| Febrile UTI | **7–14 days** (shorter is inferior) |
| Afebrile cystitis | **2–4 days** (beats single dose; equals 7–14 days) |
| Lobar nephronia | ≥3 weeks |
| Renal abscess | Antibiotics; drain if no response |
| Pyonephrosis | Antibiotics and drainage |

- Choose by the **local antibiogram**. TMP-SMX and amoxicillin are poor empirical agents; prefer **cephalexin**, and nitrofurantoin for cystitis. **Proteus is a boys' organism**.
- **Neonates** need Enterococcus cover, so give **ampicillin with a third-generation cephalosporin or an aminoglycoside**.
- Use broad cover after previous UTI, recent antibiotics or admission, or with an anomaly. Fluoroquinolones are reserved for complicated UTI.

| Agent | Dose (mg/kg/day) | Restriction |
|---|---|---|
| Cephalexin | 50–100 in 4 | — |
| Cefixime | 8 once daily | — |
| Amoxicillin–clavulanate | 20–40 in 3 | — |
| TMP-SMX | 6–12 (TMP) in 2 | **<6 weeks/premature**: kernicterus |
| Nitrofurantoin | 3–5 in 2 | **<3 months**, GFR <50%, G6PD deficiency; **never for febrile UTI** |
| Ceftriaxone (IV) | 75 once daily | — |
| Gentamicin (IV) | 7.5 (once daily acceptable) | Nephro-/ototoxicity |

No repeat culture is needed if the organism was sensitive.

## 8 · Renal scarring and sequelae

- Scars favour the **poles**, where compound papillae allow intrarenal reflux, and risk rises with VUR grade.

| Pyelonephritic episode | 1st | 2nd | 3rd | 4th | 5th |
|---|---|---|---|---|---|
| Scar risk | 5% | 10% | 20% | 40% | 60% |

- Childhood UTI causes **<1% of CKD**. Scars lead to hypertension and pre-eclampsia, so bilateral scars need lifelong checks of BP, function and proteinuria.

## 9 · Prevention and prophylaxis

- Recurrence is **10–30%**, highest in the first 3–6 months.
- **Continuous antibiotic prophylaxis (CAP)** is not given after a first UTI (AAP, NICE). It is for the **high-risk** child: girls with dilating VUR (≥III), BBD or recurrent febrile UTI.
- Give about **¼ of the therapeutic** dose at bedtime, using a different agent from treatment. Options are TMP-SMX (not under 6 weeks), nitrofurantoin and a cephalosporin in young infants.
- With VUR and BBD, recurrence is ~45% vs 15%; treat BBD with timed voiding and laxatives, then anticholinergics, biofeedback or CIC.
- **Circumcision** helps high-risk boys (recurrent febrile UTI, high-grade VUR, hydroureteronephrosis). Cranberry and probiotics are at most adjuncts.

## 10 · Special infections

*See the paper answer **Xanthogranulomatous pyelonephritis.** (Section 3, Ch 26).*

**Also, from the condensed book (not in the paper answer):**

- In children, XGP has calculi in only 38–70% and presents with failure to thrive and anaemia.
- **Lobar nephronia** mimics a mass on CT and needs antibiotics for ≥3 weeks. **Pyonephrosis** must be **drained**.
- **Viral haemorrhagic cystitis** is due to adenovirus 11; **BK virus** after marrow transplant is treated with **cidofovir**.
- **Prepubertal epididymitis** is usually sterile; if culture-positive or recurrent, look for ectopic ureter or utricle. **Balanoposthitis** needs circumcision only if recurrent.

## 11 · Landmark trials

| Trial | Design | Result | Take-home |
|---|---|---|---|
| RIVUR | 607 children, VUR I–IV, TMP-SMX vs placebo, 2 years | Recurrent UTI ~halved (number needed to treat 8); **no difference in scarring** | Prevents recurrence, not scars; most benefit with BBD, febrile index UTI |
| Swedish Reflux Trial | Randomised, dilating VUR, 1–2 years | Prophylaxis reduced recurrence in girls | CAP for girls with dilating VUR |

# 38 Core Principles of Perioperative Management in Children

## 1 · Perioperative surgical home and enhanced recovery

- **Enhanced recovery after surgery (ERAS)** combines counselling, shorter fasting, judicious IV fluid, no routine tubes or drains, opioid-sparing analgesia and early feeding and mobilisation. It shortens stay with no more complications.

## 2 · Preoperative assessment

- Risk is raised by upper respiratory tract infection (URTI), asthma exacerbation, dehydration, uncontrolled diabetes, prematurity and fever.
- **Preoperative anxiety** causes more pain and emergence delirium; explanation and play help.
- **Investigations are indication-driven**, with no routine bloods, chest X-ray or ECG. Check urine hCG in females of menstrual age; it can be falsely positive after augmentation cystoplasty.
- Send a **urine culture** before any procedure opening the urinary tract and reschedule until sterile. Treat asymptomatic bacteriuria before endourology.
- **URTI** leaves the airway hyperreactive for **6–8 weeks**, risking laryngospasm and bronchospasm. Postpone elective surgery at least 4, ideally 6 weeks.
- **Sepsis**: give antibiotics within **1 h** of septic shock and avoid neuraxial blocks. In **trauma**, assume a full stomach.

## 3 · Fasting (NPO, nil by mouth)

| Minimum fast | Material |
|---|---|
| **2 h** | Clear liquids, carbohydrate drinks (European and Canadian 2022: 1 h) |
| **4 h** | Breast milk |
| **6 h** | Infant formula, non-human milk, light meal, tube feeds |
| **8 h** | Regular meal |

- Non-human milk counts as a light meal. List the youngest first, because they dehydrate faster.
- **Diabetes**: list the child first, stop metformin 24 h before and halve the long-acting insulin the night before; a long case needs IV insulin.
- Omit ACE inhibitors that morning, stop warfarin and aspirin 7–10 days before, and never stop clonidine abruptly.

## 4 · Anaesthesia considerations

### Neonate and preterm infant

- **Postoperative apnoea** peaks at 12–23 h, so give 23-h monitoring if postconceptional age is **< 50–60 weeks** (ex-preterm) or **< 44 weeks** (term). Defer elective surgery beyond these ages.

### Anaesthetic neurotoxicity

- The FDA warning (2016) says repeated or lengthy (**> 3 h**) anaesthesia at **< 3 years** may affect brain development. Human trials of a single short exposure are negative (§11), so do not delay necessary surgery.

### Comorbidities

- **Asthma**: give a β2-agonist before induction. **Chronic kidney disease**: avoid surgery within 4–6 h of haemodialysis and give no NSAIDs in CKD 3–4.
- **Sickle cell disease**: above all avoid dehydration, and also hypoxia, cold and tourniquets; acute chest syndrome is the main cause of death.
- **Phaeochromocytoma**: give α-blockade before β-blockade, with volume loading, and expect hypoglycaemia after resection.
- **Congenital adrenal hyperplasia**: give IV stress-dose hydrocortisone with glucose-containing fluid.

### Spina bifida

- Confirm VP shunt function. **Latex allergy** can cause anaphylaxis, so care should be latex-free from birth.
- **Autonomic dysreflexia** (lesion at or above **T6**): bladder distension causes severe hypertension and bradycardia. Stop the stimulus, drain the bladder, sit the child up and give a rapid antihypertensive.

## 5 · Fluid and electrolyte management

*Hypotonic fluids can cause fatal hyponatraemia.*

- **Holliday–Segar 4-2-1 rule**: **4 mL/kg/h** for the first 10 kg, plus **2 mL/kg/h** for the next 10 kg, plus **1 mL/kg/h** per kg above 20. For 22 kg this is 62 mL/h.
- Add dextrose under 2–4 months; give no potassium if dialysed.

| Maintenance fluid | Historical | Current |
|---|---|---|
| Tonicity | Hypotonic (0.45% saline) | Isotonic (0.9% saline or lactated Ringer) with dextrose |
| Potassium | 10 mEq/L if < 10 kg; 20 mEq/L if > 10 kg | |

> **KEY POINT**
>
> **Why isotonic?** Surgical stress, pain, nausea and hypoxia trigger non-osmotic ADH release, so hypotonic fluid adds free water and causes hyponatraemia (< 135 mmol/L). A child's large brain-to-skull ratio makes the resulting encephalopathy potentially fatal.

## 6 · Blood loss and transfusion

- Estimated blood volume (EBV) is 80–90 mL/kg in a term neonate and 70–75 mL/kg in a child. Maximum allowable blood loss = EBV × (starting − target haematocrit) / starting haematocrit.
- Transfuse at haemoglobin **< 7 g/dL** in stable children; 10–15 mL/kg raises haemoglobin by about 2–3 g/dL.

## 7 · Temperature

- Infants cool quickly, causing acidosis, coagulopathy and apnoea, so use a warm theatre, forced-air warming and warmed fluids.

## 8 · Antibiotic prophylaxis and intraoperative crises

- **Timing**: give within **60 min** of incision; vancomycin and fluoroquinolones within **120 min**.
- Doses: cefazolin **30 mg/kg** (redose at 4 h); ampicillin 50 mg/kg (redose at 2 h); gentamicin 2.5 mg/kg, vancomycin 15 mg/kg, metronidazole 15 mg/kg and ciprofloxacin 10 mg/kg (not redosed).

| | **Anaphylaxis** | **Malignant hyperthermia (MH)** |
|---|---|---|
| Triggers | Neuromuscular blockers, β-lactams, latex | Succinylcholine, volatile agents (not nitrous oxide) |
| First sign | Collapse, bronchospasm, hives minutes after induction | Unexplained hypercarbia, tachycardia, masseter spasm; late — acidosis, hyperkalaemia, rigidity |
| Treatment | **Adrenaline**, 100% oxygen, fluids; send tryptase | Stop triggers, **dantrolene**, clean circuit; admit 24–48 h (recrudescence) |

## 9 · Analgesia

- Under 3 years, use the FLACC pain score.
- **Regional blocks under general anaesthesia** are safe and opioid-sparing (penile, **caudal**, epidural). Caudal 0.25% bupivacaine is 0.5–1 mL/kg; maximum bupivacaine is 2.5 mg/kg.
- Give **multimodal, scheduled** analgesia: acetaminophen 15 mg/kg 6-hourly and ibuprofen 10 mg/kg 8-hourly. Ketorolac is limited to 72–96 h, and no NSAIDs are given under 6 months.
- **Codeine is contraindicated**, because ultrarapid metabolisers make lethal amounts of morphine. Limit discharge opioids to 3–5 days.

## 10 · Postoperative care, VTE and day-case discharge

- **Emergence delirium** occurs at 2–8 years, within 30 min of waking; the child does not recognise parents, unlike a child in pain.
- **Venous thromboembolism** is rare before puberty; at-risk adolescents need mobilisation and mechanical prophylaxis, with LMWH if several factors are present.
- **Day-case** surgery suits ASA I–II children outside the apnoea-risk ages having a short, low-blood-loss procedure.

## 11 · Landmark trials

| **Trial** | **Design** | **Result** | **Take-home** |
|---|---|---|---|
| GAS | Randomised; infant hernia, awake regional vs sevoflurane < 1 h | No cognitive difference at 2 and 5 years | Single short anaesthetic safe |
| PANDA | Sibling-matched; single exposure < 3 years | No IQ difference at 8–15 years | Repeated exposure untested |
| MASK | Cohort; exposure < 3 years | Negligible deficit after single exposure | Reassuring |

# 39 Principles of Laparoscopic and Robotic Surgery in Children

## 1 · Evolution and indications

### SAQ Discuss diagnostics and operative pediatric laparoscopy in urology

Paper answer · P3

**Type:** SAQ (10 marks)
**Source:** Practice Set 3 Paper 3, Q3a

#### Evolution

- Diagnostic (non-palpable testis, 1976) → nephrectomy (1994) → pyeloplasty (1995) → robotic (2002).

#### Diagnostic laparoscopy

- **Non-palpable testis** (procedure of choice, after examination under anaesthesia): note vessels, vas, testis and internal ring.

| Finding | Meaning |
|---|---|
| Blind-ending vessels and vas, closed ring | Vanishing testis: stop |
| Vessels and vas entering ring | Explore groin; excise remnant |
| Intra-abdominal testis | Orchidopexy (primary or Fowler-Stephens) |
| - **DSD (differences of sex development):** gonadal inspection and biopsy, Müllerian structures. | |

#### Operative laparoscopy

- **Testis:** primary orchidopexy (>90% success); Fowler-Stephens one- or two-stage; gonadectomy in DSD.
- **Kidney:** nephrectomy, heminephrectomy, **pyeloplasty** (commonest MIS (minimally invasive surgery) procedure).
- **Ureter and bladder:** extravesical or vesicoscopic (cross-trigonal) reimplantation, Mitrofanoff, augmentation, bladder neck reconstruction, utricle and urogenital sinus.
- **Also know:** RPLND (retroperitoneal lymph node dissection) for paratesticular rhabdomyosarcoma.

#### Technical principles

- **Access:** open Hasson or Veress at umbilicus (Palmer's point if scarred); equal safety.
- 3–5 mm ports; robots use 8 mm. Transperitoneal (most space), retroperitoneal (spares peritoneum, useful in peritoneal dialysis), vesicoscopic.
- **Pressures:** $CO_2$ absorption, reduced venous return, raised airway pressure; reversible oliguria (no fluid bolus).

| Age | 0–1 month | 2–12 months | 1–2 years | 2–8 years |
|---|---|---|---|---|
| Max IAP (intra-abdominal pressure) | ≤6 mmHg | ≤8 | ≤10 | ≤12 |

| Age | 0–1 month | 2–12 months | 1–2 years | 2–8 years |
|---|---|---|---|---|
| - **Positioning:** modified flank for kidney; supine steep Trendelenburg for pelvis. Infants: minimise inhalational anaesthetic, rectal tube to reduce bowel distension. | | | | |
| - **Only absolute contraindication:** inability to tolerate pneumoperitoneum. | | | | |

### Complications

- Right common iliac vessel injury (behind umbilicus; commonest major), inferior epigastric (commonest minor), bowel injury, subcutaneous emphysema, **port-site hernia 0.5–3%** (close fascia).

### Pros and cons

- Shorter stay, better cosmesis, deep pelvic access; pain benefit unproven. Robot: cost, learning curve; infant working space only about 1 L.

**Diagram to draw:** laparoscopic view of internal ring with blind-ending vessels.

> **Key points to remember:**
> - Diagnostic laparoscopy is the procedure of choice for non-palpable testis.
> - Pressure ceiling 6–12 mmHg by age.
> - Right common iliac at Veress entry; close port fascia.

**Also, from the condensed book (not in the paper answer):**

- The first paediatric urological laparoscopy was for non-palpable testis in **1976**. The wristed robot removed the suturing bottleneck, so **reconstruction** grew, with pyeloplasty the commonest.
- The only strict contraindication is **inability to tolerate pneumoperitoneum**; age and size are not contraindications.

## 2 · Physiology of pneumoperitoneum

P1 *Pneumoperitoneum physiology; laparoscopic access and gases — pressure vs CO2 effects on heart, lungs, kidney, ICP; vagal bradycardia; CO2 choice; gas embolism signs and treatment (left lateral head-down, oxygen, aspirate)*

- Effects depend on pressure and reverse on desufflation, so use the **lowest age-appropriate pressure**: intra-abdominal pressure (IAP) < 12 mm Hg, and about 7 mm Hg is feasible.
- Infant oliguria is transient and is not a guide to resuscitation, so give **no fluid boluses**.

| Age | 0–1 month | 2–12 months | 1–2 years | 2–8 years |
|---|---|---|---|---|
| Maximum IAP | **≤ 6** | **≤ 8** | **≤ 10** | **≤ 12 mm Hg** |

- **Cardiovascular**: IAP > 10 mm Hg lowers venous return and cardiac index. Insufflation can trigger a **vagal reflex**, so desufflate and reinsufflate slowly.
- **Respiratory**: infants have a compliant chest wall and low FRC, so tidal volume falls, worse with Trendelenburg, and CO2 absorption affects them most.

- Laparoscopy is generally safe with **VP shunts**. Infants cool fast, so use warmed humidified gas.

## 3 · Access

- Open (Hasson) and closed (Veress) entry do not differ in injury rates (Cochrane); confirming intraperitoneal position is what protects.
- After scarring or failed umbilical entry, use **Palmer's point** (left midclavicular line, about 3 fingerbreadths below the costal margin).

## 4 · Ports, instruments and placement

- **3- and 5-mm** laparoscopic ports suit infants, whereas robots need **8 mm**. Robotic wristed instruments have 7 degrees of freedom against 4 for laparoscopic ones.
- Triangulation is essential in laparoscopy, while the robot tolerates linear placement. **HIdES** puts ports below the Pfannenstiel line.

## 5 · Approaches

| | **Transperitoneal** | **Retroperitoneal** | **Vesicoscopic** |
|---|---|---|---|
| Space | Largest; familiar | Limited | Limited; small bladder predicts complications |
| Strength | Complex reconstruction | Avoids peritoneum; preserves peritoneal dialysis | Intravesical access |
| Limitation | Colon mobilised or mesenteric window | Unfamiliar view; peritoneal tear collapses space | Learning curve; port-site closure |
| Uses | Testis, kidney, ureter, bladder | Pyeloplasty, (hemi)nephrectomy, lymph node dissection | Cross-trigonal reimplantation |

- In **left pyeloplasty**, a transmesenteric window avoids colonic mobilisation.

## 6 · Robotics in children

- **Working space** is about **1 L at 1 year vs 6 L in adults**, so arms collide; wide port spacing and bowel preparation help.
- **Limitations**: there are no 5-mm robotic ports, and single-port needs a 25-mm cannula and about **10 cm** to the target, which is prohibitive in small children.
- **Benefits** are shorter stay and less narcotic use; incisions under 15 mm contract with growth.

## 7 · Complications

- The major vascular injury is to the **right common iliac** artery at Veress entry; the minor one is the **inferior epigastric**, so insert ports under vision.
- Port-site hernia peaks around 1 year, so **close fascia** at every port.
- Experience is the strongest predictor, with fewer complications above 12 cases a year.

**KEY POINT**

**Ergonomics**: set the table at 70–80% of elbow height and the monitor at eye level. The robot offloads the arm but fixes the neck, so use the clutch to keep the elbows on the armrest.

# 40 Clinical and Urodynamic Evaluation of Lower Urinary Tract Dysfunction in Children

## 1 · Principles and terminology

- Non-invasive assessment excludes anatomical or neurological causes, then defines **storage vs emptying** dysfunction (ICCS 2014).
- **Expected bladder capacity (EBC)**: infant ≤1 y = **weight (kg) × 7 mL**; older child = **(age in years + 2) × 30 mL**. Small is <65% EBC; large is >150% (some use >130%).

## 2 · History

- **Secondary enuresis** is wetting after **≥6 months** dry, so look for stressors.
- In girls, poor **toilet posture** activates the pelvic floor and causes incomplete emptying; a footstool helps.
- **Holding manoeuvres** (potty dance, squatting) suppress detrusor overactivity (DO).
- Bowel dysfunction may be the sole cause of urinary symptoms. Target a painless **Bristol 3–4 stool daily**.
- **Rome IV functional constipation** (developmental age ≥4 y, ≥1 month) needs **≥2 of**: ≤2 stools/week; ≥1 soiling/week; retentive posturing; painful or hard stools; large rectal mass; stools blocking the toilet.
- **Neurological red flags** are perineal numbness, or gait or limb change with new LUTS.

**ALGORITHM**

1. **Bowel and bladder dysfunction (BBD)**
2. Apply Rome IV; obtain a 7-day diary, urinalysis and pelvic US.
3. **Treat the bowel first.**
4. If LUTS persist, do uroflow with EMG and start urotherapy.
5. If that fails, add an anticholinergic, then a β3-agonist; then botulinum toxin, biofeedback or neuromodulation.

- The **48-h frequency–volume chart** shows capacity and polyuria and is **mandatory** when first-line therapy fails.
- Scores: **DVSS** best tracks treatment, and **DVISS** is best for diagnosis.

## 3 · Physical examination

- Measure height, weight and BP at **every visit**.
- **Neurological**: include perineal sensation, anal tone and sacral reflexes (anal wink, bulbocavernosus); their absence means the S2–S4 arc is interrupted.
- **Spine**: a simple sacral dimple (<2.5 cm from anus) is low risk and gets spinal US up to 6 months. High-risk signs (hair tuft, lipoma, vascular malformation, multiple stigmata, VACTERL) need **MRI**.
- **Genitalia**: **continuous incontinence in a girl suggests an ectopic ureteric orifice.**

## 4 · Laboratory tests

- Do a **urinalysis for every child with LUTS**; **culture is not routine.** On CIC, culture only with fever, pain, new incontinence or spasm.

## 5 · Non-invasive urodynamics

*See the paper answer **Uroflowmetry** (Section 7, Ch 107).*

**Also, from the condensed book (not in the paper answer):**

- Uroflow is valid only if toilet-trained and voiding **≥50% EBC** (replacing the adult ≥150 mL rule), with more than one curve.
- A staccato curve never reaches zero, whereas an interrupted one does.

> **KEY POINT**
>
> The curve reflects detrusor against outlet. Too much detrusor gives a tower; too much outlet gives a plateau; intermittent outlet activity gives staccato; intermittent detrusor effort gives an interrupted curve. Patch EMG tells which side is at fault.

- **Pelvic US**: the bladder wall is abnormal if **>3 mm full / >5 mm empty**. A rectal diameter **>30 mm** without urge means faecal loading.
- **PVR** is abnormal at 4–6 y if **>30 mL or >21%** of capacity and at 7–12 y if **>20 mL or >15%**. It is valid only **<5 min** after voiding with the bladder at 50–115% EBC, so repeat it.

## 6 · Invasive urodynamics

*Filling pressure, not wetting, decides renal risk.*

### Indications

- **Neurogenic**: spinal dysraphism, tethered cord, sacral agenesis, transverse myelitis or spinal cord injury (SCI).
- **Severe obstruction**: posterior urethral valves (PUV) or urethral atresia.
- Non-neurogenic LUTS: study only after failed 2nd/3rd-line therapy, and only if it will change management.

### Technique

- **Avoid sedation or general anaesthesia**; if unavoidable, use ketamine rather than midazolam.
- Routine prophylaxis is not needed, because post-study UTI is about **1.4%**.
- Use a **6 Fr dual-lumen** urethral and a rectal catheter, and **fill at 5–10% of EBC per minute** with warm fluid.
- **Stop filling** at strong urge, discomfort, voiding, leak greater than fill rate, Pdet persistently **>40 cm$H_2O$**, or volume **>150% EBC**.
- Report the first fill, which gives the highest Pdet. **Videourodynamics** adds trabeculation, VUR with its onset volume and pressure, and bladder-neck shape.

### Storage phase

- **Pdet = Pves − Pabd**, and normal filling is flat. **DO** means phasic involuntary rises.

- **Compliance = ΔV/ΔPdet**: **>20 normal; 10–20 moderately; <10 severely impaired**. In small contracted bladders absolute Pdet (<30–40) is the better guide.

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## SAQ Leak point pressures and ALPP

Paper answer · P1 · P2 · merged from 2

*Asked as:* P1 SAQ "Leak point pressures (VLPP, ALPP, DLPP) in SUI: clinical implications" · P2 SAQ "ALPP"

### Definition

- **LPP (leak point pressure):** the pressure at which leakage is first seen, spontaneous or provoked. There are two independent types:

| Feature | ALPP (abdominal LPP), incl. VLPP (Valsalva LPP) and CLPP (cough LPP) | DLPP (detrusor LPP) |
|---|---|---|
| Definition | Total Pves (vesical pressure) at leak during raised abdominal pressure, **no detrusor contraction** | Lowest Pdet (detrusor pressure) at leak, **without contraction or straining** |
| Measures | Sphincter strength against abdominal pressure | Bladder pressure against outlet resistance |
| Bad result | **Low = bad** (weak sphincter) | **High = bad**: >40 $cmH_2O$ means upper-tract risk; hydronephrosis or reflux in 85% (McGuire 1981) |
| Used in | SUI (stress urinary incontinence) | Neurogenic or low-compliance bladder |

- **ALPP (abdominal leak point pressure):** the **intravesical pressure** at which urine leaks because of raised abdominal pressure, **without a detrusor contraction**. It measures the urethra's ability to resist abdominal pressure: **lower ALPP = weaker sphincter**.
- Provoked by Valsalva (**VLPP**, the intravesical pressure at which urine leaks during Valsalva) or cough (**CLPP**).
- Seen only in SUI; there is **no normal value** (continent women never leak).
- Newer term: "urodynamic stress test" (ICS, International Continence Society).

### Technique (VLPP/ALPP)

1. Small dual-lumen urethral catheter plus rectal catheter; zero to atmosphere.
2. Patient **standing or sitting**; fill to **150 mL** (original method; some use 200 mL).
3. Ask for a **graded Valsalva** or cough; record the **lowest total Pves at visible leak** (include baseline, not the change).
4. No leak: repeat every 50 mL to capacity; then, at capacity, **remove the urethral catheter** and read pressure from the rectal line.
5. Report volume, catheter size and provocation (Valsalva or cough).

### Interpretation (McGuire 1993)

| VLPP / ALPP ($cmH_2O$) | Inference |
|---|---|
| **<60** | **ISD (intrinsic sphincter deficiency)** |
| **60–90** | Equivocal; mixed |
| **≥90** (>90) | **Urethral hypermobility**, little ISD |

### Clinical implications and uses of ALPP

- Separates ISD from hypermobility; a low VLPP/ALPP (ISD) traditionally favours a sling or bulking agent over colposuspension.
- Post-prostatectomy incontinence: many men leak only once the urethral catheter is removed.

### Limitations

- Not standardised: falls with **larger catheters** and **higher volumes**.
- **VLPP is lower than CLPP** in the same woman (VLPP < CLPP), so state the method.
- Invalid during DO (detrusor overactivity) or poor compliance; prolapse raises it or masks the leak (reduce it).
- ISD and hypermobility coexist; VLPP/ALPP alone **does not predict surgical outcome** well.
- **VALUE trial (Nager 2012):** for uncomplicated SUI, office evaluation was non-inferior to urodynamics before surgery. AUA (American Urological Association)/SUFU (Society of Urodynamics, Female Pelvic Medicine and Urogenital Reconstruction) 2023 SUI guideline: urodynamics optional in the index patient.
- **EAU (European Association of Urology):** do not use leak point pressure to grade severity.

### DLPP: clinical implications

- **McGuire (1981):** DLPP **>40 $cmH_2O$** in myelodysplastic children leads to hydronephrosis or VUR (vesicoureteral reflux) in **85%**.
- A strong outlet keeps the patient dry but transmits pressure to the kidneys; a weak outlet "pops off" safely.
- **Hostile bladder** (≥40 $cmH_2O$): CIC (clean intermittent catheterisation) plus antimuscarinic; botulinum toxin, augmentation or vesicostomy if needed.
- **Caveat:** 40 is not a validated "safe" value; 20 $cmH_2O$ is more sensitive in children.

### Also know

- DO leak point pressure (ICS 2017); MUCP (maximum urethral closure pressure) ≤20 $cmH_2O$ suggests ISD.

### Diagram to draw

- Pves, Pabd and Pdet traces marking ALPP (leak on Valsalva with flat Pdet) and DLPP.

> **Key points to remember:**
> - VLPP/ALPP: Pves at leak with Valsalva or strain, no detrusor contraction; start at 150 mL.
> - ALPP low = weak sphincter; DLPP high = upper-tract danger.
> - VLPP <60 ISD; 60–90 equivocal; ≥90 (>90) hypermobility.
> - VLPP < CLPP; catheter size and volume alter it.
> - DLPP >40: upper-tract danger; 85% upper-tract change (McGuire 1981).
> - Lower DLPP is safer; no validated safe cut-off.

**Also, from the condensed book (not in the paper answer):**

- A DLPP **>40 $cmH_2O$**, or no leak at high pressure, signals danger; the adult ALPP cut-off (<60 $cmH_2O$) has no paediatric norms.

**KEY POINT**

**McGuire (1981)**, myelodysplasia: with leak pressure >40 $cmH_2O$, hydronephrosis or secondary VUR developed in **85%**. Aim for the lowest achievable pressure.

### Voiding phase

- **High Pdet with low Qmax means obstruction.**
- Anatomical obstruction has a **quiet EMG**; functional obstruction (DSD) has an **active EMG**.
- Normal Pdet at Qmax is about 66 $cmH_2O$ in boys and 57 in girls; infant boys reach 118 physiologically MCQ.

**KEY POINT**

**Upper-tract threats**: compliance <20 (especially <10); DLPP >40 $cmH_2O$; sustained high filling Pdet; DSD with poor emptying; high voiding pressure with poor flow.

- **UMPIRE** (spina bifida) studies at about **3 months**, then at 1, 2 and 3 years, then yearly to 5 y if VUR or a hostile bladder. Hydronephrosis is a late sign, so re-study on any change and before reconstruction.

## 7 · Evaluation in specific conditions

| Condition | Typical findings | When to study |
|---|---|---|
| Non-neurogenic LUT dysfunction / enuresis | DO, dysfunctional voiding | Only if refractory |
| Spinal dysraphism | UMPIRE risk by DLPP/end-fill Pdet: **<25 low; 25–39 or DO intermediate; ≥40 high** | Baseline, serial, pre-surgery |
| Sacral agenesis | Areflexia with weak sphincter, or DO with DSD; level does not predict | If voiding dysfunction |
| Anorectal malformation | Tethered cord common | Baseline about 1 y after repair |
| PUV | DO, then poor compliance, then myogenic failure ("valve bladder" about 15%) | Serial |
| Cerebral palsy | Upper motor neuron; DO, DSD rare | Only if conservative therapy fails |
| SCI | Spinal shock causes retention; higher lesions have more DSD | Video-urodynamics 3–6 months after injury |
| Transverse myelitis | DO most often; motor recovery does not predict bladder | Early, with CIC |

**KEY POINT**

UMPIRE dropped DSD from high risk because reviewers disagreed on who had it, not because DSD is benign. Neurogenic bladder goals are normal capacity, low storage pressure, complete emptying and upper-tract protection.

## 8 · Landmark trials

| Trial | Design | Result | Take-home |
|---|---|---|---|
| McGuire 1981 | Myelodysplasia urodynamic cohort | DLPP >40 $cmH_2O$ led to upper-tract damage in 85% | 40 $cmH_2O$ = safety threshold |

# 42 Bladder Anomalies in Children

## 1 · Classification and embryology

### LAQ Embryological development of urinary bladder and management of Hutch Diverticulum?

Paper answer · P3

**Type:** LAQ (25 marks)
**Source:** 2021-Summer Paper 3, Q1 (bladder diverticulum also answered in 2016-Summer Paper 2, Q3c)

#### A. Embryology of the bladder

**1. Division of the cloaca (weeks 4–6)**

- The **urorectal septum** (Tourneux and Rathke folds) divides the endodermal cloaca into a dorsal **anorectal canal** and a ventral **UGS (urogenital sinus)**. The cloacal membrane perforates to form the urogenital and anal openings.
- The **Wolffian (mesonephric) duct entry** is the landmark:
- **Cranial vesicourethral canal** (continuous with the allantois) → bladder and pelvic urethra.
- **Caudal (definitive) UGS** → phallic urethra in males; distal vagina and vestibule in females. The whole female urethra comes from the pelvic part.

**2. Wall layers**

- **Urothelium:** endoderm of the UGS.
- **Lamina propria, detrusor and adventitia:** surrounding splanchnic mesenchyme. Epithelium and mesenchyme induce each other (epithelial–mesenchymal signalling).

**3. Trigone (weeks 5–6)**

- The **common excretory ducts** (mesonephric duct distal to the ureteric bud) dilate and are absorbed into the UGS. **Apoptosis** separates the ureters from the mesonephric ducts.
- Ureteric orifices migrate **cranially and laterally**; the Wolffian ducts end caudally as ejaculatory ducts at the verumontanum.
- Classic view: trigone muscle is ureteric (Wolffian) tissue. Current view (mouse lineage tracing): **mostly bladder smooth muscle**, minor ureteric contribution.
- **Bud position (Weigert–Meyer):** bud too low → lateral orifice, short tunnel → VUR (vesicoureteral reflux); too high → caudal, medial, ectopic orifice.

**4. Allantois and urachus**

- **Months 4–5:** the bladder descends into the pelvis. The allantoic duct elongates and obliterates into the **urachus** (median umbilical ligament).

**5. Function**

- Urine production starts at **week 10**. Fetal bladder is seen from 10 weeks, almost always by 13 weeks, and empties every 15–20 minutes. After **16 weeks** amniotic fluid is mainly fetal urine.

### B. Anomalies explained by embryology

| Anomaly | Embryological basis |
|---|---|
| Agenesis | Cranial UGS atrophy, or trigone not formed; survival only with ectopic ureters |
| Exstrophy | Cloacal membrane not reinforced by mesoderm |
| Duplication | Complete or incomplete; sagittal or coronal |
| Urachal (patent urachus, sinus, cyst, vesicourachal diverticulum) | Urachus not obliterated at different levels |
| Congenital diverticulum | Focal detrusor deficiency, often at the ureteric hiatus |

### C. Hutch diverticulum: definition

- **Paraureteral diverticulum**: mucosa herniates through the **ureteric hiatus**, between the intravesical ureter and the roof of the hiatus, superolateral to the orifice, sparing the trigone. **Hutch** divided them into primary and secondary.

| | Primary (congenital) | Secondary (acquired) |
|---|---|---|
| Cause | Congenitally deficient detrusor at the hiatus | High pressure: PUV (posterior urethral valves), BOO (bladder outlet obstruction), neurogenic bladder (described in paraplegics), dysfunctional voiding, prior surgery |
| Bladder | Smooth; single, often intermittent | Trabeculated; one of many |

### D. Pathophysiology

- No muscle backing → **VUR**; no muscle in the sac → cannot empty → stasis, UTI (urinary tract infection), stones; tumour in adults.
- An enlarging sac damages **Waldeyer's sheath** and pulls the ureter out → UVJ (ureterovesical junction) dysfunction, reflux, later **obstruction** and renal damage.
- Large sacs compress the bladder neck → retention (vicious circle).

### E. Evaluation

- **US (ultrasound):** sac, residual urine, hydronephrosis.
- **VCUG (voiding cystourethrogram), gold standard:** oblique and post-void films show size, VUR and whether the ureter enters the sac. Repeat if suspicion persists (sac is dynamic).
- **DMSA (dimercaptosuccinic acid):** scars; **MAG3 (mercaptoacetyltriglycine) renogram** if obstruction.
- **Urodynamics** if secondary; cystoscopy (orifice, neck); in adults enter the sac, cytology.

### F. Management

**1. Treat the cause first (secondary):** valve ablation, BOO surgery, CIC (clean intermittent catheterisation) and antimuscarinics. Remodelling may make excision unnecessary.

**2. Primary diverticulum:**

| Situation | Management |
|---|---|
| Small, asymptomatic, no VUR | Observe |

| Situation | Management |
|---|---|
| VUR, ureter beside the sac | As primary VUR (resolves like VUR of the same grade); many excise if it persists. Dx/HA (dextranomer/hyaluronic acid) injection possible |
| Ureter enters the sac | VUR will not resolve → surgery |
| Breakthrough febrile UTI, new scars, large sac with residual, stones, obstruction, retention | **Diverticulectomy + ipsilateral reimplantation** |

**3. Operative steps (open, laparoscopic or robotic):** `Cystoscopy ± ureteric catheter → cystotomy → evert sac or dissect it from its pseudocapsule (0.5 cm mucosal cuff round the neck) → close detrusor defect in two layers (firm backing) → reimplant ureter: Cohen cross-trigonal or Politano–Leadbetter (tunnel 5:1) → catheter ± suprapubic tube, drain`

- Extravesical or combined approaches suit selected cases. In adults, endoscopic outlet surgery is done first.

### G. Complications and follow-up

- Ureteric injury, leak, persistent VUR, obstruction, recurrence (syndromic patients).
- US at 6–12 weeks; VCUG for febrile UTI; urodynamics if voiding is abnormal.

**Also know:** posterolateral multiple diverticula (Ehlers–Danlos, Williams, Menkes); hypoplasia; megacystis.

**Diagram to draw:** cloacal division by the urorectal septum; Hutch sac beside the ureteric orifice.

> **Key points to remember:**
> - Urorectal septum weeks 4–6; vesicourethral canal → bladder; urachus at months 4–5.
> - Hutch: herniation through the ureteric hiatus → VUR.
> - Ureter entering the sac: VUR will not resolve.
> - Secondary: treat the outlet first; else diverticulectomy + reimplantation.

**Also, from the condensed book (not in the paper answer):**

- Anomalies are **structural** (exstrophy, duplication, agenesis, urachal, diverticula) or **functional**; prune-belly syndrome (PBS) is both.

## 2 · Fetal bladder on ultrasound

- The fetal bladder empties every **15–20 min**, so if unseen, **repeat in the same sitting**. Amniotic fluid is mainly fetal urine after **16 weeks**.
- In exstrophy there is no filling with **normal amniotic fluid**.

| | Agenesis | Hypoplasia | Exstrophy |
|---|---|---|---|
| US | Absent after repeat scan | Small, never reaches capacity | No filling; bladder template on abdominal wall |
| Core defect | Cranial UGS atrophy or no trigone | Inadequate filling or storage | Open bladder plate |

| | Agenesis | Hypoplasia | Exstrophy |
|---|---|---|---|
| Outcome | Survives only with ectopic ureteric drainage | May grow; augmentation often | Reconstruction (see Ch 43) |

- **Agenesis** is very rare (about 1:600,000, mostly female); ureters drain into müllerian structures or the rectum, and it is treated by ureterosigmoidostomy or stoma.

## 3 · Megacystis

- In classic congenital megacystis, **massive VUR causes the dilation**; today the term means any prenatally dilated bladder.

| Gestation | Normal | Abnormal |
|---|---|---|
| 10–14 weeks | Length ≤6 mm; <10% of CRL | **≥8 mm**; ≥13% of CRL |
| 15–40 weeks | Length about GA − 5 mm | > GA + 12 mm |
| Later (functional) | Empties within 45 min | No emptying in 45 min |

- **Causes**: LUTO, with PUV in about 50%, urethral atresia or anterior valves; non-obstructive causes are PBS, MMIHS and cloacal anomalies. Male:female is 8:1.
- **Aneuploidy is about 1/5** (trisomy 13/18), so karyotype.
- In euploid fetuses, 7–15 mm mostly resolves, while **>15 mm** never does. **Persistent megacystis with oligohydramnios means LUTO.**

### MMIHS

- MMIHS is a visceral myopathy (**ACTG2**), mostly female, with megacystis and normal or increased fluid. The bladder is acontractile and most need **lifelong CIC**; TPN complications are the leading cause of death.

> **KEY POINT**
>
> A female fetus with megacystis cannot have PUV, so think of MMIHS, cloacal anomaly or urethral atresia.

## 4 · Bladder duplication

- **Complete sagittal duplication is commonest**, with external genital duplication in up to 90% and lower GI duplication in up to 42%.

**ALGORITHM**

1. Early: relieve obstruction and prevent infection.
2. Later, if both sphincters are competent, then join the distal urethras.
3. If one is incompetent, then close that bladder neck and excise its urethra.
4. Join duplicated vaginas, with vulvoplasty.

- **Sphincter competence, not anatomy, decides reconstruction.**

## 5 · Bladder diverticula

- A primary (Hutch) diverticulum is observed if small and excised if there is VUR. For a secondary one, **treat the outlet first**.

- **Syndromic** diverticula (**Ehlers–Danlos, Williams, Menkes**) are multiple and giant; resect only for UTI or stones, as recurrence is common.

## 6 · Urachal anomalies

| | Patent urachus | Sinus | Cyst | Vesicourachal diverticulum |
|---|---|---|---|---|
| Open | Whole canal | Umbilical end | Isolated segment | Bladder apex |
| Presents | Urine from the umbilicus in a neonate | Draining umbilicus | Infection (S. aureus), painful mass | Incidental |
| Imaging | US; fistulogram or VCUG | Sinugram | US; CT/MRI if extensive | Incidental |
| Treat | Excise with cuff | Complete excision | Drain, then excise | Only if a narrow neck causes stones or UTI |

- The differential is a patent **omphalomesenteric duct**. Incidental remnants are increasingly excised only if symptomatic.

**ALGORITHM**

1. If infected, drain and give antibiotics first.
2. Infraumbilical incision; circumscribe the tract, preserving the umbilicus.
3. Dissect extraperitoneally and excise with a **bladder cuff**.
4. Two-layer closure; catheter overnight.

## 7 · Acquired lesions

| Lesion | Key features | Treatment |
|---|---|---|
| Haemangioma | Klippel–Trénaunay, Sturge–Weber; haematuria | Fulguration or Nd:YAG laser |
| Fibroepithelial polyp | 2nd commonest paediatric bladder tumour | Endoscopic resection |
| Nephrogenic adenoma | Female 5:1; after infection, surgery, augmentation | TUR; **80% recur**, so surveillance cystoscopy |
| Cystitis cystica | Almost all female; mimics tumour | Long-term antibiotic prophylaxis |
| Eosinophilic cystitis | Male | **Biopsy mandatory**; steroids and antihistamines |
| Haemorrhagic cystitis | Cyclophosphamide or ifosfamide; BK virus, adenovirus | Mesna prevents; pentosan polysulfate |

# 43 Exstrophy–Epispadias Complex

## 1 · Spectrum, epidemiology and embryology

*Rupture of the cloacal membrane explains the whole spectrum.*

*See the paper answer **Bladder embryology and exstrophy-epispadias complex** (Section 5, Ch 32).*

**Also, from the condensed book (not in the paper answer):**

- Spica cast fails in 66% without osteotomy, and diastasis >6 cm needs staged closure.
- Rupture timing explains the phenotypes: early, extensive rupture opens bladder **and** hindgut (cloacal exstrophy); later rupture opens the bladder only; rupture at the genital tubercle gives epispadias.

## 2 · Anatomy of classic bladder exstrophy

- **Pelvis**: the acetabula are retroverted and the posterior segment is normal. The resulting waddling gait and out-toeing improve with growth.
- **Pelvic floor**: the levator ani lies posterior and flattened. Continent patients have the shortest intersymphyseal distance and deepest bladder neck, hence **deep placement, osteotomy and pelvic fixation**.
- The **intrasymphyseal band** tethers the posterior vesicourethral unit to the pubic rami. It contains **no striated sphincter**, so radical incision down to the levator hiatus is essential.
- **Indirect inguinal hernia** is common, often bilateral in males, so repair it at closure.
- **Anorectum**: the anus is anterior. **Rectal prolapse before closure** resolves with closure, but prolapse **after closure** suggests outlet obstruction and needs urgent cystoscopy.
- **Male**: the **anterior corpora are about 50% shorter** and 30% wider. The prostate is never circumferential. Cavernous nerves lie laterally, so potency is retained, though retrograde ejaculation may follow BNR.
- **Female**: the vagina is short with a stenotic anterior orifice, the **cervix sits in the anterior vaginal wall** and the clitoris is bifid. A wide levator hiatus causes **uterine prolapse**, even in nulliparous women, and **osteotomy does not prevent it**.
- **Bladder**: it is **immature, not neuropathic**. The exposed plate develops polyps, squamous metaplasia and cystitis glandularis, which carry an **adenocarcinoma risk**. Compliance falls after BNR, so follow-up is lifelong.
- The ureters enter laterally without obliquity, so **vesicoureteral reflux (VUR) occurs in 100%** of closed bladders and is corrected by reimplantation at BNR.

## 3 · Variants

*Except superior vesical fissure, variants are managed like CBE.*

| Variant | Feature | Management |
|---|---|---|
| Skin-covered (split symphysis), the commonest | Musculoskeletal defect; urinary tract intact | Formal closure at birth |
| Duplicate exstrophy | Everted mucosal patch plus a closed pelvic bladder | Excise patch, close wall |

| Variant | Feature | Management |
|---|---|---|
| Superior vesical fissure | Rupture only at the top; sphincter intact | Toilet-trains normally; no BNR |
| Pseudoexstrophy | Musculoskeletal defect only; low umbilicus | — |
| Epispadias with bladder prolapse | Prolapse may show only on Valsalva | Repair, with or without osteotomy |

## 4 · Prenatal diagnosis

- Only about 25% are diagnosed prenatally. Signs are **absent bladder filling**, a low umbilicus, wide pubic rami, small genitalia and an **enlarging lower abdominal mass**. Plan **delivery at an exstrophy centre**.

## 5 · Newborn care and selection

*A good first closure best predicts continence.*

**ALGORITHM**

1. Tie the cord with 2-0 silk (no clamp, which traumatises mucosa).
2. Cover the plate with non-adherent plastic wrap; irrigate with saline at each nappy change.
3. Assess heart and lungs; obtain a renal US.
4. Transfer to an exstrophy centre for MDT counselling.

- **Assess** the template (≥**3 × 3 cm** is suitable), diastasis, pelvic malleability under anaesthesia and penile length. A plate that **bulges on crying** is adequate.
- A plate <3 cm, polyp-covered or fibrotic is **inadequate**. Penoscrotal duplication, hypoplastic bladder and bilateral hydronephrosis preclude primary closure.
- A small plate is closed **later, at 6–9 months, with osteotomy**, because a bad closure ruins continence. If it has not grown by 6–12 months, excise it and divert. After delayed closure a **catheterisable channel is needed far more often**.

## 6 · Surgical philosophies

| Repair | Stages / feature | Weakness |
|---|---|---|
| **Modern staged repair (MSRE)** | Closure at birth; epispadias repair at 6–12 months; BNR with reimplant at 4–7 years | Multiple operations |
| **Complete primary repair (CPRE, Mitchell)** | Newborn closure with penile disassembly and neck tailoring | Hypospadias; glans loss; early reimplant |
| **Kelly** | Closure, then radical soft-tissue mobilisation at 3–6 months, then hypospadias repair; avoids osteotomy | Fistula, stenosis |
| Erlangen | Total repair at about 8 weeks without osteotomy | — |
| Mainz | Closure, then sigma pouch at about 2 years | Colonic malignancy |

**KEY POINT**

Erlangen, Kelly and CPRE claim to avoid BNR, yet almost all still need an outlet procedure. Compressing the reconstruction adds ischaemic and outlet risk without removing that need.

## 7 · Primary closure (MSRE stage 1)

*Closure creates an incontinent epispadias with enough outlet for growth.*

**ALGORITHM**

1. General anaesthesia with **epidural** (catheter to T10–T12).
2. Outline a 2-cm mucosal strip from the trigone to mid-shaft (boy) or vaginal orifice (girl); incise the bladder–skin junction.
3. Free the bladder from the rectus and strip peritoneum off the dome.
4. **Divide the urogenital diaphragm fibres sharply down to the levator hiatus**; free the corpora from the rami.
5. Close bladder, neck and posterior urethra; **taper over a 12-Fr sound**.
6. Place a suprapubic Malecot (4 weeks) and ureteric stents; leave the **urethra unstented**.
7. Close the pubis with a **No. 2 nylon horizontal mattress** suture, knot away from the neourethra; fashion a neoumbilicus.

- **Tapering** gives enough resistance for bladder growth and against prolapse, but not enough to harm the kidneys.
- **Ureteric stents** stay 10–14 days because oedema can obstruct the ureters; a urethral stent causes necrosis.
- The **commonest error** is inadequate urogenital diaphragm dissection, after which the unit is pulled forward and prolapses.

## 8 · Osteotomy and immobilisation

*Osteotomy and rigid immobilisation cut closure failure several-fold.*

- In a **newborn <72 h** whose malleable pubes meet easily, osteotomy is not needed. **If in doubt, osteotomise.**
- Osteotomise if closure is **>72 h** after birth, diastasis is **>4 cm**, the pelvis is stiff, the template is large, after **failed closure**, and in **cloacal exstrophy**. Diastasis >6 cm needs staged closure with a fixator.

| Type | Position | Features |
|---|---|---|
| **Combined bilateral anterior innominate and vertical iliac** (preferred) | Supine | Best malleability; least recurrence |
| Bilateral posterior iliac (original) | Prone | Worse approximation, more recurrence |
| Anterior innominate alone | Supine | Less mobility |

- Aftercare is **modified Buck traction for about 4 weeks with an external fixator for 4–6 weeks**. Without osteotomy, use **modified Bryant traction** (hips 90°) for 4 weeks.

| Immobilisation (CBE) | Failure overall | With osteotomy | Without osteotomy |
|---|---|---|---|
| Spica cast | 47% | 30% | **66%** |

| Immobilisation (CBE) | Failure overall | With osteotomy | Without osteotomy |
|---|---|---|---|
| External fixation with traction | **7.5%** | 8% | — (few) |

- Failure falls with traction plus fixator for **>4 weeks**; osteotomy type and epidural make no difference. Complications are pin-site inflammation, transient femoral nerve palsy and delayed union.

## 9 · After primary closure

- At 4 weeks calibrate the outlet, scan and check the residual. Give **continuous antibiotic prophylaxis**, because all have VUR.
- Perform **yearly cystoscopy and cystogram under anaesthesia** to measure capacity. If capacity is **<30 mL at 1–2 years**, a continence procedure may not be feasible.
- If the **outlet is too tight**, escalate from dilation to incision and revision. Recurrent UTI with a distended bladder suggests **intrapubic stitch erosion**.

## 10 · Epispadias repair (stage 2)

*Urethroplasty raises outlet resistance, so the bladder grows.*

- It is done at 6–12 months, before BNR, because urethroplasty **increases capacity**. Give **testosterone enanthate 2 mg/kg intramuscularly (IM) at 5 and 2 weeks** before every urethroplasty.
- **Modified Cantwell–Ransley** (preferred): the plate is freed but its **distal 1 cm stays on the glans**; it is tubularised over 8 Fr and the **corpora are rotated over the neourethra**, which then lies ventral.
- **Chordee** is corrected by dorsomedial corporotomy with cavernocavernostomy; skin cover uses ventral preputial flaps.
- Early fistula occurs in about 1 in 6, usually at the base; **coronal fistula is rare**.
- **Mitchell–Bägli disassembly** risks plate ischaemia, and most end **hypospadiac**. Glans or corporal loss is linked to closure **without osteotomy**, so if ischaemia is seen, osteotomise immediately.
- **CPRE**: about half need reimplant in year 1, 65–80% need an outlet procedure, and most end on CIC.
- **Kelly**: unaided voided continence is under 1 in 5, with glans loss reported.
- **Combined closure and epispadias repair** suits failed or delayed closures and boys >6 months, but risks ischaemic penile loss.
- **Female closure**: bladder, urethra and vagina move deep as one unit, with **monsplasty and joining of the clitoral halves**.

## 11 · Bladder neck reconstruction (stage 3)

*Capacity at BNR best predicts dryness.*

- **Prerequisites**: capacity **≥100 mL** under anaesthesia, stable urodynamics, a motivated child of 5–7 years and a voiding programme. Reimplant earlier for pyelonephritis or worsening hydronephrosis.

**ALGORITHM**

1. **Modified YDL**
2. Open the bladder neck transversely.
3. Perform **Cohen cross-trigonal reimplantation**.
4. Outline a posterior mucosal strip **15 × 30 mm**; denude the lateral muscle triangles.
5. Tubularise the strip and **double-breast the muscle flaps** over it.
6. Suspend to the rectus fascia; suprapubic tube for 3 weeks, then clamp trials.

- After MSRE, about **70–75%** are dry by day (intervals >3 h); CPRE with YDL is lowest (about 56%).
- **Predictors** are template size, **successful primary closure** and, most importantly, **capacity at BNR**. Closure within 72 h or with osteotomy makes continence likely, whereas delayed closure without osteotomy gives about 10%. After two closures, fewer than 1 in 5 void continently.
- Day dryness comes within about 12 months and night dryness takes 2–3 years; **not dry by 2 years means failure**.

| Category (after BNR) | Definition |
|---|---|
| Continent | Dry day and night, interval ≥3 h |
| Socially continent | Dry 3 h by day, occasional wet nights |
| Wet | Dry <3 h by day, wet at night |

## 12 · Failures and complications

*Each failed closure shrinks the bladder.*

- **Failed closure** means dehiscence, **bladder prolapse**, outlet stricture or vesicocutaneous fistula. Causes are incomplete pelvic-diaphragm mobilisation, inadequate immobilisation, infection and tube malfunction.
- In expert series, major dehiscence or prolapse occurs in about 3% and femoral palsy in about 2%.
- **Reclosure** waits at least 4–6 months and uses **osteotomy**, rigid immobilisation and radical urogenital diaphragm division. After secondary closure, voided continence is only about 1 in 6.
- Outlet stricture presents with UTI, an enlarging bladder, stones or rectal prolapse; obstruction for more than 6 months usually means augmentation.
- **Failed BNR** means no 3-h dry interval by 2 years. **Redo YDL** only if the neck is patulous, capacity >100 mL and urodynamics stable; otherwise do **bladder neck transection, augmentation and a Mitrofanoff channel**.
- **Genital**: urethrocutaneous fistula is commonest; glans or corporal loss may need **radial forearm phalloplasty**.

## 13 · Diversion and augmentation

- **Ureterosigmoidostomy** needs anal continence. It risks pyelonephritis, hyperchloraemic acidosis and **colonic adenocarcinoma about three decades later**, so **yearly colonoscopy** is required.

- If the bladder is too small, use a **non-refluxing colon conduit**. After failed BNR, do **augmentation with an appendiceal channel, with or without neck closure**; stones are the commonest complication.
- **Continent diversion** is indicated for closures that do not grow, inadequate capacity or wetness after BNR, and upper-tract decompensation.

## 14 · Long-term issues

- **Malignancy**: augmented exstrophy bladders develop adenocarcinoma, so survey endoscopically from 5 years after reconstruction.
- **Male**: libido, erection and orgasm are preserved. Fertility is impaired by retrograde ejaculation and oligospermia; about 1/4 achieve paternity, half via ICSI.
- **Female**: a stenotic introitus needs **perineal flap vaginoplasty**, because a cutback worsens prolapse. **Uterine prolapse** is treated by sacral suspension. **Pregnancy** ends in a **planned caesarean with a urologist present**.
- Psychosocial problems need lifelong support and transition care.

## 15 · Epispadias

### LAQ Development of the urethra, congenital urethral anomalies and management of epispadias (including isolated penopubic epispadias)

Paper answer · P3

**Asked as:**

- Discuss development of urethra. Classify congenital anomalies and discuss management of epispadias — *Practice Set 6 Paper III Q2 (LAQ)*
- Isolated penopubic Epispadias. — *Summer 2022 Paper III Q3f (SAQ)*

**Type:** LAQ (25 marks); also asked as SAQ (10 marks)
(isolated penopubic epispadias also answered in 2022-Summer Paper 3, Q3f)

#### Development of the urethra

**1. Division of the cloaca (weeks 5–6)**

- The urorectal septum divides the cloaca into the dorsal anorectal canal and the ventral **UGS (urogenital sinus)**.
- Wolffian duct entry is the landmark. Above it, the **vesicourethral canal** forms the bladder and pelvic urethra. Below it, the **definitive UGS** forms the phallic urethra in males, and the urethra and vestibule in females.
- UGS also forms the prostate (buds at 10 weeks) and Cowper and Littre glands; the **utricle** is Müllerian.
- The epithelium of the **entire male urethra is endodermal** (UGS origin).

**2. Penile urethra: the "double zipper" (androgen-dependent)**

- Indifferent genital tubercle until 8–9 weeks. Testosterone (Leydig cells, 8 weeks) is converted to DHT (dihydrotestosterone) by **type 2 5α-reductase**.
- The **urethral plate** (endoderm) forms in the tubercle.
- **Opening zipper:** the plate canalises proximal to distal into a urethral groove bounded by the urethral folds.

- **Closing zipper:** the folds fuse in the ventral midline, forming the tubular urethra and median raphe.
- **Glanular urethra:** formed by direct canalisation of the plate inside the glans (current endodermal view; Glenister's ectodermal ingrowth theory is outdated).
- **Timetable:** coronal sulcus 11–12 weeks; fusion complete 16 weeks; urethra complete 18 weeks; fetal curvature gone 20 weeks.
- The prepuce forms with the glanular urethra, so hypospadias has a hooded foreskin.
- **Female:** the vestibular plate canalises but the folds do not fuse, giving an open vestibule and labia minora.

### Classification of congenital urethral anomalies

| Mechanism | Anomaly |
|---|---|
| Failed ventral fold fusion | **Hypospadias** (1 in 250 males; commonest at shaft–glans junction); congenital urethral fistula |
| Dorsal defect | **Epispadias** (exstrophy–epispadias complex) |
| Obstruction | **PUV (posterior urethral valves)**, Young type 1 (95%) and type 3; **anterior urethral valves** with diverticulum; urethral atresia or hypoplasia; meatal stenosis |
| Duplication | **Urethral duplication**: sagittal; ventral channel is functional; Effmann type I blind, II complete, III with caudal duplication |
| Dilatation | **Megalourethra**: scaphoid (spongiosum deficient) or fusiform (all erectile tissue deficient, more severe); prune-belly association |
| Cystic or remnant | Cowper's syringocele; enlarged utricle (proximal hypospadias); parameatal cyst |

### Epispadias (including isolated penopubic epispadias): definition, embryology, classification and clinical features

- Part of the **exstrophy–epispadias complex**: a **dorsal urethral defect** (defect of the dorsal urethral wall): the roof is replaced by a broad (dorsal) mucosal strip (urethral plate). Cause (embryology): absent mesodermal reinforcement of the cloacal membrane; the genital tubercles fuse **caudal** to it, giving a dorsal urethral defect.
- **Male (1 in 117,000; incidence of isolated male epispadias about 1 in 117,000), classification by meatus:** glanular, penile, **penopubic** (subsymphyseal) (about 70% of male cases are complete (penopubic), with incontinence). **Female (1 in 484,000), Davis:** vesicular (patulous), subsymphyseal, retrosymphyseal (incontinent).
- **Features (clinical):** penopubic meatus; outlet may admit a finger → **total incontinence**. **Short, broad, upturned penis with dorsal chordee**, like exstrophy; pubic diastasis (less than exstrophy); small bladder from no outlet resistance; **VUR (vesicoureteral/vesicoureteric reflux) 30–40% (male), 30–75% (female)**. Female (Davis retrosymphyseal type): bifid clitoris, depressed mons, incontinence.

### Evaluation

- (Renal) ultrasound; VCUG (voiding cystourethrogram); cystoscopy and capacity under anaesthesia.
- **Key decision:** epispadias or exstrophy pathway. Treat as exstrophy (osteotomy and closure) if pubic diastasis is over 3 cm, the intrasymphyseal bar is poor, sphincter is absent or capacity is tiny (tiny bladder).

### Management of epispadias

- **Aims:** continence, safe upper tracts, a straight functional penis of good length.
- **Preoperative testosterone** 2 mg/kg intramuscularly (intramuscular), 5 and 2 weeks before urethroplasty.

### Stage 1: penile reconstruction and urethroplasty (6–12 months)

- Release chordee: divide suspensory ligaments and scar; free corpora from the inferior pubic rami.
- **Modified Cantwell–Ransley:** outline an 18-mm plate to the tip; reverse MAGPI (meatal advancement and glanuloplasty) brings the meatus to the tip; free the plate from the corpora (distal 1 cm left on the glans); tubularise over 8 Fr; transpose it (the tubularised plate is moved) **ventral to the corpora**, which are rotated medially ± cavernocavernostomy; close glans wings; Z-plasty at the base; stent 10–12 days.
- **Mitchell–Bägli complete penile disassembly:** reliable chordee correction, but often leaves hypospadias.
- Outlet resistance rises, so capacity grows.

### Stage 2: bladder neck reconstruction (4–5 years, capacity about 100 mL)

- **YDL (Young–Dees–Leadbetter)** bladder neck repair + Marshall–Marchetti–Krantz suspension + **ureteric reimplantation**. Osteotomy usually unnecessary.
- **Good capacity:** single-stage repair possible.

**Female epispadias:** at 6–12 months excise the urethral roof, close over 10 Fr, unite clitoral halves and labia minora, monsplasty; YDL at 4–5 years.

### Outcomes and complications

- Continence after bladder neck reconstruction: **males 47–82%, females 70–87%**; capacity is the main predictor.
- Early fistula up to (about) one-fifth (at the penile base); long-term fistula and stricture about 5% (stricture rare); erections and paternity (usually) preserved.
- Better than exstrophy (bladder never exposed or scarred). Failures (failure): bulking injection (injectables), augmentation, continent stoma.

**Diagram to draw:** double zipper (canalisation then fusion); penopubic epispadias (dorsal penopubic meatus and plate).

> **Key points to remember:**
> - Whole male urethra is endodermal UGS; fold fusion complete at 16, urethra at 18 weeks.
> - About 70% of male epispadias are penopubic and incontinent; VUR 30–40%.
> - Diastasis over 3 cm (>3 cm) → exstrophy pathway.
> - Cantwell–Ransley at 6–12 months; YDL (Young–Dees–Leadbetter) at about 100 mL capacity.
> - Continence after YDL: males 47–82%, females 70–87%.

**Also, from the condensed book (not in the paper answer):**

- Glanular epispadias is never incontinent.

| | Male | Female |
|---|---|---|
| Incidence | 1 in 117,000 | 1 in 484,000 |
| Classification | Glanular, penile, **penopubic** (about 70%, incontinent) | Davis: vesicular, subsymphyseal, retrosymphyseal (incontinent), in rising severity |
| VUR | 30–40% | 30–75% |
| Presentation | At birth | May present only with incontinence |

**ALGORITHM**

1. **Female repair (6–12 months)**
2. Incise vertically through the urethral wall at 9 and 3 o'clock.
3. Excise the urethral roof (dorsal wedge taper) and close over a 10-Fr Foley.
4. Unite the clitoral halves and labia minora; bring mons fat over the suture line.
5. Catheter 5–7 days; BNR at 4–5 years.

- Treat as exstrophy (osteotomy and closure) if diastasis is >3 cm, the sphincter is scant or capacity is small.
- In **males** with a small bladder, do urethroplasty first, then BNR at about 100 mL (4–5 years).
- Female continence after BNR is about **87%**, as good as in CBE girls, because the bladder was never exposed.

## 16 · Cloacal exstrophy

*A multisystem defect; neurological level sets disability.*

- Incidence is 1 in 200,000–400,000. **OEIS** stands for Omphalocele, Exstrophy, Imperforate anus and Spinal defects. **Two hemibladders flank exstrophied ileocaecal bowel**, with a rudimentary hindgut and separated hemiphalluses.

| Feature | CBE | Cloacal |
|---|---|---|
| Spinal defects | Uncommon | **85–100%** (lipomeningocele commonest) |
| Omphalocele | Rare | Near-universal |
| Upper-tract anomalies | Uncommon | 41–60%; solitary kidney commonest |
| Osteotomy | Selective | Always |
| Continence | Mostly voiding | Mostly augmentation with CIC |

- Only about 1 in 5 has a sacral skin sign, so all need spinal MRI. A 46,XY child may have an **intravesical phallus**, so biopsy before calling it aphallia.
- Prenatally, the **"elephant trunk" sign** is prolapsed ileum. **Gender** follows **karyotype**.
- **Bowel**: never discard hindgut; tubularise the caecal plate with an end colostomy.

**ALGORITHM**

1. Stage 1 (newborn): close the omphalocele, separate the caecal plate, join the hemibladders and form a colostomy.
2. Stage 2: bladder closure with **combined osteotomy** and fixation for 6–8 weeks.
3. At 5–7 years: YDL only if capacity >100 mL (rare); usually augmentation with a continent stoma.

- Combined osteotomy gives about 90% closure success but no continence gain; staged closure beats single-stage.
- Primary closure fails in about 31–43%. Continence relies on augmentation with CIC, and the **neurological level** determines disability.

# 44 Prune-Belly Syndrome

## LAQ Prune belly syndrome

Paper answer · P1 · P3 · merged from 2

*Asked as:* P1 LAQ "Prune Belly Syndrome. Antenatal diagnosis, pathophysiology and postnatal management." · P3 SAQ "Prune belly syndrome"

### Definition

- **PBS (prune belly syndrome)**, also called Eagle–Barrett or triad syndrome: **deficient abdominal wall muscles**, **bilateral intra-abdominal testes** and a **dilated, poorly contracting urinary tract**.
- 1 in 29,000–44,000 live births; **95% male**.
- **Survival depends on renal dysplasia and lung hypoplasia, not on the abdominal wall.**

### Antenatal diagnosis

- **Ultrasound** may suggest PBS from 11–14 weeks, but the classic signs are not consistent even at 30 weeks.
- **Findings:** huge bladder filling the abdomen; bilateral hydroureteronephrosis; urachal diverticulum; dilated posterior urethra; echogenic kidneys with cortical cysts (dysplasia); irregular abdominal circumference; fetal ascites; oligohydramnios when severe.
- **Differential diagnosis:** PUV (posterior urethral valves), urethral atresia, megacystis–megaureter.
- Fetal dilatation does **not** predict postnatal renal function.
- **Prenatal management:** intervention has no proven renal benefit. Vesicoamniotic shunting is justified only for **urethral atresia with progressive oligohydramnios**, or to decompress the abdomen and avoid dystocia. Termination is hard to justify except with early, severe oligohydramnios. Plan delivery in a centre with neonatal intensive care.

**Diagram to draw:** fetal ultrasound sketch: huge bladder, dilated posterior urethra, bilateral hydroureteronephrosis.

### Pathogenesis (theories, none proven)

| Theory | Mechanism |
|---|---|
| Early (fetal) urethral obstruction | Back-pressure dilates the tract and stretches the wall; but most have no obstruction after birth |
| Lateral plate mesoderm defect | This mesoderm forms ureters, bladder, prostate, urethra and gubernaculum, so one defect explains the triad |
| Intrinsic urinary tract (ureteric) defect | Primary ureteric dilatation and fetal ascites |
| Yolk sac defect | Primary defect of the yolk sac |

- **Genetics:** mostly sporadic, normal karyotype; smooth-muscle genes **MYOCD (myocardin)** and **FLNA (filamin A)** in multiplex families.

### Pathophysiology and features by organ

- **Kidneys:** dysplasia in about 50% (Potter type II from mesenchymal defect; type IV cystic from obstruction).
- **Ureters:** long and tortuous, worse distally. Collagen replaces smooth muscle, so walls do not coapt and peristalsis fails: **a failed pump, not a blocked pipe**. The result is stasis and infection. **VUR (vesicoureteric reflux) in about 75%.** The proximal ureter has better muscle.
- **Bladder:** large, smooth, collagen-rich; urachal pseudodiverticulum (urachus patent in 25–30%); splayed trigone. Normal compliance but poor contractility, so voiding is "unbalanced", with residual urine.
- **Prostate and urethra:** prostatic hypoplasia dilates the posterior urethra; distal obstruction in about 20%; urethral atresia; **megalourethra** (scaphoid: spongiosum deficient; fusiform: all erectile bodies, more severe).
- **Testes:** intra-abdominal; azoospermia usual; paternity only by sperm retrieval and ICSI (intracytoplasmic sperm injection).
- **Extra-genitourinary (about 75%):** pulmonary hypoplasia; cardiac about 10%; gastrointestinal at least 30% (malrotation); orthopaedic up to 65% (talipes, hip dysplasia).

### Classification (Woodard)

| Category | Features | Outlook | Management |
|---|---|---|---|
| I | Oligohydramnios, **pulmonary hypoplasia**, dysplasia, urethral atresia | Perinatal death | Supportive; bladder drainage only |
| II | Full triad, moderate or unilateral dysplasia / moderate uropathy, **no pulmonary hypoplasia** | Risk of CKD (chronic kidney disease) | Individualised (controversial, debated group) |
| III | Mild or incomplete; **stable function** | Good; majority | Surveillance |

- **Also know:** RUBACE score (Renal, Ureter, Bladder, Abdominal wall, Cryptorchidism, Extra-genitourinary) grades morbidity / severity in survivors; "pseudo-prune" presents late with renal failure.

### Postnatal management: neonatal

Team: neonatology, nephrology, urology, cardiology.

1. Heart and lungs first: **chest radiograph** for pneumothorax and hypoplasia.
2. Creatinine, BUN (blood urea nitrogen), electrolytes; follow the **trend**, because the first value reflects the mother.
3. Renal and bladder ultrasound once stable.
4. Antibiotic prophylaxis; **circumcision**; strict asepsis.
5. **VCUG (voiding cystourethrogram)** only with renal insufficiency or suspected outlet obstruction.
6. **DMSA (dimercaptosuccinic acid) at 4–6 weeks** for split function; MAG3 (mercaptoacetyltriglycine) for drainage.
7. Outlet obstruction: catheter or suprapubic tube.

**Diagram to draw:** VCUG showing large bladder and dilated posterior urethra.

### Postnatal management: urinary tract

- **Conservative** for category III: surveillance, treat infection.
- **Surgery for:** recurrent upper tract infection, sepsis, progressive dilatation or rising creatinine / renal decline, high-grade VUR, true obstruction.
- **Temporary diversion:** wide **cutaneous vesicostomy (Blocksom)**. Avoid high ureterostomy, which wastes the good proximal ureter.
- **Reconstruction after 3–6 months** (lung maturity): excise the distal ureter, taper and reimplant the healthier proximal ureter; reduction cystoplasty gives little lasting benefit (capacity recurs). **CIC (clean intermittent catheterisation)** or Mitrofanoff for poor emptying.

### Orchidopexy

- **Bilateral transabdominal at about 6 months**, often with other surgery; later, **staged Fowler–Stephens** (better than single-stage) or laparoscopic.

### Abdominoplasty

- Aims: cosmesis; better cough, voiding / emptying and defecation.

| Technique | Key feature |
|---|---|
| Randolph | Transverse; excises the worst lower wall |
| Ehrlich | Vertical; umbilicus kept; vest-over-pants overlap |
| Monfort | Vertical; umbilicus kept; central fascial plate with lateral flaps overlapped; transperitoneal access, good exposure for combined surgery |

- **Also know:** Furness (extraperitoneal plication); Dénes; corset.

### Prognosis and follow-up

- About 20% die perinatally; **up to 30% develop chronic renal failure**.
- **Nadir creatinine <0.7 mg/dL at 1 year** predicts stable function.
- Lifelong timed voiding, cultures and imaging; transplant when needed.

> **Key points to remember:**
> - Triad; 95% male; dysplasia and lungs decide survival.
> - Ureter is a failed pump (collagen); VUR about 75%; keep the proximal ureter.
> - Woodard I–III; vesicostomy for drainage; orchidopexy at about 6 months.
> - Nadir creatinine <0.7 mg/dL is favourable; up to 30% reach renal failure.

**Also, from the condensed book (not in the paper answer):**

- Renal failure is driven by pyelonephritis, dysplasia and polyuria; transplant outcomes are equal.
- Female PBS is ~5%, with normal gonads.

## 1 · Definition and aetiology

- The **triad** (Eagle–Barrett) is deficient abdominal wall muscle, bilateral intra-abdominal testes and a dilated urinary tract. Incidence is ~**1 in 29,000–44,000** and **95% male**; it is commoner in twins and African ancestry.

- **Survival follows the kidneys and lungs, not the belly**; the wall is least prognostic.
- There is no inheritance pattern; multiplex families show the smooth-muscle genes **MYOCD** and **FLNA** MCQ .
- Theories are early urethral obstruction, a lateral plate mesoderm defect, an intrinsic urinary tract defect and a yolk sac defect.

> **KEY POINT**
>
> Weakness of the obstruction theory: most have **no obstruction after birth**. The tract is **dilated, collagen-rich and largely non-obstructed**, and this drives management.

## 2 · Genitourinary pathology

- **Kidneys**: dysplasia in ~**50%**, worst with outlet obstruction without urachal decompression. Dilation does not correlate with dysplasia; **infection, not obstruction, is the main threat**.
- **Ureters**: the distal ureter is worse, so the **proximal ureter is preserved** in reconstruction. Vesicoureteric reflux (VUR) occurs in ~**75%**. The ureter is a **failed pump, not a blocked pipe**.
- **Bladder**: massive, thick but smooth, with a urachal pseudodiverticulum; the urachus is patent at birth in **25–30%**. It has normal compliance, large capacity and poor sensation; about half void normally but may deteriorate.
- **Prostatic hypoplasia** dilates the posterior urethra and causes ejaculatory failure. **Urethral atresia** is lethal unless the urachus is patent. **Megalourethra** is scaphoid (spongiosum only, milder) or fusiform (cavernosa too, more severe).
- **Testes** lie over the iliac vessels. Infant germ cells are normal, so the abdominal environment destroys later potential, hence early orchidopexy.

## 3 · Extragenitourinary anomalies

About **75%** have them, and nearly **half premature**. Pulmonary problems (>50% of survivors) include hypoplasia, pneumothorax and pneumonia; cardiac (~10%), gastrointestinal (≥30%, malrotation) and orthopaedic (up to 65%, talipes, hip dysplasia, knee dimples) anomalies are common.

> **KEY POINT**
>
> Two mechanisms: **oligohydramnios** (lung hypoplasia, limb deformity, knee dimples) and **absent abdominal wall pressure** (weak cough, atelectasis, constipation).

## 4 · Presentation and spectrum

- **Prenatal** appearance mimics posterior urethral valves: massive bladder and hydroureteronephrosis. Dilation does **not** predict function.

| Woodard category | Kidneys / lungs | Course | Urological management |
|---|---|---|---|
| **I** | Dysplasia; **pulmonary hypoplasia**; oligohydramnios, urethral atresia | Stillbirth or early death (~20% perinatal mortality overall) | Supportive; simple bladder drainage |

| Woodard category | Kidneys / lungs | Course | Urological management |
|---|---|---|---|
| **II** | Full triad; minimal or unilateral dysplasia; no lung hypoplasia | Risk of progressive renal insufficiency | Individualised — **controversial group** |
| **III** | Incomplete or mild; no dysplasia; stable | Majority; good | Surveillance; surgery for UTI or deterioration; orchidopexy in year 1 |

- The abdominal wall is not a sorting criterion. **Pseudo-prune** (partial flaccidity, unilateral undescended testis) is diagnosed late and has higher late renal failure.

## 5 · Evaluation

**ALGORITHM**

1. Team; **heart and lungs first** (chest X-ray).
2. Creatinine trend; the day-1 value is maternal.
3. Renal–bladder ultrasound.
4. Prophylaxis and circumcision; then selective imaging.

- A voiding cystourethrogram is done only for renal insufficiency or suspected outlet obstruction, under antibiotic cover. A **DMSA** scan is done at **4–6 weeks**; MAG3 renograms are unreliable with massive dilation. Outlet obstruction is drained by a percutaneous suprapubic tube.

## 6 · Management philosophy

| | Conservative | Early reconstruction |
|---|---|---|
| Premise | Dilation is non-obstructive and may improve | Stasis, reflux and infection drive renal loss |
| Operate when | Proven obstruction, intractable infection | Progressive dilation, high-grade VUR, recurrent upper tract infection, declining function |
| Best for | Category III | Selected category II |

- Delay major reconstruction to **≥3–6 months** for the lungs. Single-stage comprehensive surgery adds no morbidity.

## 7 · Urinary tract surgery

- **Blocksom vesicostomy** (sepsis, renal failure, atresia): make the stoma larger than usual, as it tends to stenose. Avoid proximal ureterostomy.
- **Ureteroplasty**: excise the distal ureter, taper the upper part and reimplant. Reduction cystoplasty is reserved for a large urachal diverticulum, as capacity recurs; clean intermittent catheterisation empties better. Circumcision reduces UTI.

## 8 · Orchidopexy

- A broad mesorchium defeats the inguinal approach. Do a **bilateral transabdominal orchidopexy at ~6 months**, often with other surgery. Later, a **staged Fowler–Stephens** procedure is preferred.

## 9 · Abdominoplasty

- Cosmetic gain is certain; functional gains (cough, defecation, emptying) are plausible. Operate from ~6 months with other surgery.

| Technique | Incision / umbilicus | Key step / note |
|---|---|---|
| **Randolph** | Transverse, 12th rib to 12th rib | Excision of the lower wall; lateral bulge persists |
| **Ehrlich** | Vertical; umbilicus on a pedicle | Vest-over-pants advancement |
| **Monfort** | Vertical ellipse; umbilicus in situ | Central fascial plate with lateral flaps; **transperitoneal**, so ideal with reconstruction |
| Dénes | Fusiform | Modified Ehrlich |
| Furness | Midline, limited | Plication; peritoneum not opened |

**ALGORITHM**

1. **Monfort**
2. Skin ellipse, umbilicus spared; incise the central plate at the lateral rectus borders.
3. Do the genitourinary surgery.
4. Suture plate edges to scored lateral peritoneum.
5. Overlap lateral flaps over drains.

## 10 · Long-term outcome

- Creatinine **<0.7 mg/dL at 1 year** predicts stable function. Up to **30%** develop chronic renal failure, driven by pyelonephritis, dysplasia and polyuria. Transplant outcomes are equal.
- **Bladder**: timed and double voiding; if these fail, use CIC.
- **Fertility**: puberty and erections are normal, but **azoospermia means no spontaneous paternity**; sperm retrieval with ICSI is used.

> **KEY POINT**
>
> Long-term care is a contest against **stasis**: lowering residual urine and infection protects the kidneys, while dysplasia and polyuria set a ceiling surgery cannot raise.

# 45 Posterior Urethral Valves

## 1 · Definition and classification

- **Posterior urethral valves (PUV)** are the commonest cause of bladder outlet obstruction in infant boys. Ablation relieves obstruction but **not the embryological injury** to bladder and kidney, so care is lifelong.
- Lower urinary tract obstruction (LUTO) affects ~2.2/10,000 births; PUV is commonest (~1.4/10,000), then urethral atresia and prune-belly.

*See the paper answer **Posterior urethral valves: embryology, management and valve bladder** (Section 5, Ch 32).*

**Also, from the condensed book (not in the paper answer):**

- Type III resembles a congenital stricture; it is contentious but accepted.

| Young type | Anatomy | Status |
|---|---|---|
| **I (95%)** | Leaflets from verumontanum fuse anteriorly just proximal to external sphincter | Obstructing; the lesion treated |
| II | Folds from verumontanum to bladder neck | Non-obstructing; probably not real |
| III (5–10%) | Annular membrane, central aperture (persistent urogenital membrane) | Obstructing |

- Copy number variant burden predicts higher nadir creatinine and CKD stage 5.

## 2 · Pathophysiology

- Obstruction causes detrusor hypertrophy, then **fibrosis replaces smooth muscle**. The bladder is the **driver**: renal dysfunction, reflux (VUR) and voiding dysfunction act through it.
- The voiding cystourethrogram (VCUG) triad is a **dilated posterior urethra, hypertrophied bladder neck and flattened verumontanum**.
- Ureters dilate from **pressure transmission, VUR (up to 70%) and polyuria**, then lose peristalsis, causing stasis and infection.
- Renal injury is **obstructive uropathy** (partly reversible) plus **dysplasia** (fixed at birth, which sets the ceiling). Tubular injury causes **nephrogenic polyuria** and post-obstructive diuresis.
- Angiotensin II induces TGF-β1 and fibrosis; ACE inhibitors may help selected children.

| Organ / lesion | Effect | Natural history |
|---|---|---|
| Lung — hypoplasia | Respiratory failure | Commonest cause of neonatal death |
| Kidney — obstructive uropathy | Reversible insufficiency | Improves; may recur with bladder dysfunction |
| Kidney — dysplasia | Irreversible insufficiency | Progressive failure, hypertension |
| Kidney — tubular injury | Salt and water wasting | Nephrogenic diabetes insipidus; worsens with age |

| Organ / lesion | Effect | Natural history |
|---|---|---|
| Bladder | Poor sensation, hypercontractility, low compliance; later myogenic failure | Lifelong, changes with age |
| Ureters | Poor coaptation, stasis | Most retain hydronephrosis — **not** a sign of residual obstruction |

### "Pop-off" mechanisms

- **VURD** (unilateral high-grade VUR into a non-functioning dysplastic kidney) was thought protective. This is **refuted**: most with normal creatinine at 2 years lose it by 8–10 years, and urodynamics are worse. Urinoma, ascites and diverticula are debated but managed the same way.

> **KEY POINT**
>
> Every pop-off is false reassurance: normal creatinine in a growing infant hides an abnormal GFR. Follow VURD as closely as any PUV child.

## 3 · Diagnosis

### Antenatal

- Ultrasound shows a **thick distended bladder, bilateral hydroureteronephrosis, "keyhole sign" and oligohydramnios**; an echogenic or cystic cortex means dysplasia. It is only ~50% accurate for the cause.
- About half are diagnosed antenatally, but this has **not** improved renal outcome. A male fetus with these findings needs postnatal US and a **VCUG before discharge**.

### Postnatal

- **US** comes first: the wall is thick if **>3 mm distended / >5 mm empty**.
- **VCUG** is definitive, showing a dilated elongated posterior urethra with **abrupt funnelling** at the valve; about half have high-grade VUR. Use a 5–8 Fr feeding tube and **withdraw the catheter during voiding**.
- A **MAG3** renogram is done at **2–3 months**. Creatinine in the first 48 h is maternal; follow the trend to the **nadir** in year 1.

## 4 · Fetal intervention

- Decisions hinge on **oligohydramnios, hydronephrosis degree and laterality, and gestational age**. Favourable features are oligo- (not an-)hydramnios, normal karyotype, no cortical cysts and favourable urine.

| Fetal urine (after 20 weeks) | Favourable |
|---|---|
| Sodium / chloride | **<100 / <90 mEq/L** |
| Osmolality | **<200 mOsm/L** |
| β2-microglobulin | **<6 mg/L** (most sensitive) |

- **Triple serial sampling** 48–72 h apart beats one sample, though prediction was poor on systematic review.
- A **vesicoamniotic shunt** may prevent **pulmonary hypoplasia** but gives **no renal benefit**. Fetal cystoscopic laser ablation is no better. Both carry **40–50%** pregnancy loss.

**KEY POINT**

Fetal intervention can rescue the lung, not the kidney: fluid deficiency is still in progress, but dysplasia is already established.

## 5 · Neonatal presentation and initial management

- **Pulmonary hypoplasia** is the commonest early death and the usual cause of delayed surgery.

**ALGORITHM**

1. Stabilise the airway.
2. Pass a **5–7 Fr feeding tube**. If it curls: coudé, curved stylet, or rectal finger.
3. Confirm on US; minimal balloon fill.
4. Creatinine, electrolytes, fluids.
5. US, then VCUG; antibiotic prophylaxis.
6. Ablate when stable.

- **Post-obstructive diuresis** needs monitoring of output, weight and electrolytes with IV replacement. **Hyponatraemia** is transient pseudohypoaldosteronism, treated with sodium.
- **Urinoma or ascites** (3–10%) resolves with bladder drainage.
- **Delayed presentation** (UTI, incontinence, renal failure) is common: over 40% present after 6 months, and late does not mean milder.

## 6 · Surgery

### Primary valve ablation (preferred)

- Ablation restores flow and **normal bladder cycling**, superior to passive diversion. Use a **7.5–9 Fr cystoscope** with holmium laser or Bugbee.
- Incise at **5 and 7 o'clock, with or without 12 o'clock**. Valves are thin and avascular, so **incise, don't resect**.
- For a small urethra, use vesicostomy or **PADUA** (progressive urethral dilation with up-sized tubes). Leave a catheter ≥24 h and **repeat VCUG within 1 month**.

| Complication | Cause / prevention | Management |
|---|---|---|
| Urethral stricture | Over-resection, hot loop, oversized instrument | Dilation / urethrotomy; urethroplasty if dense |
| Meatal injury | Resectoscope through small meatus | Smaller scope / PADUA |
| Residual valve | Incomplete incision | Persistent funnel on VCUG: re-ablate |
| Sphincter injury | Ablating distal to valves | Incontinence — precise incision |

- **Bladder neck incision** gives no benefit and risks retrograde ejaculation; it is not recommended.

### Vesicostomy

- It is used for a very small or premature infant, failed decompression, or persistent deterioration. It **does not defunctionalise** the bladder and matches ablation in final outcome.

## SAQ Vesicostomy.

Paper answer · P4

**Type:** SAQ (10 marks)
**Source:** 2016-Summer Paper 4, Q3f

### Definition

- **Cutaneous vesicostomy**: the bladder dome is brought to the lower abdominal skin as an incontinent stoma draining into the nappy. It "vents" the bladder, protecting the upper tracts; usually a **temporary** bridge to reconstruction.

### Indications

- **PUV (posterior urethral valves)**: infant too small for a resectoscope; upper tract deterioration despite drainage or ablation.
- **Prune-belly syndrome**: acute renal failure, urosepsis, urethral atresia. A larger stoma is needed because stenosis is common.
- **Neurogenic bladder**: high pressure despite CIC (clean intermittent catheterisation) and drugs, or CIC impossible; worsening hydronephrosis; febrile UTI (urinary tract infection) with high-grade reflux.
- **Also know:** urethral atresia; MMIHS (megacystis-microcolon-intestinal hypoperistalsis syndrome); urogenital sinus anomalies.

### Blocksom technique (Duckett modification)

1. Transverse 2 cm incision midway between umbilicus and pubis. (In prune-belly, excise any urachal diverticulum.)
2. Incise the rectus sheath; separate the recti.
3. Fill the bladder; mobilise the **dome** up to the urachus, sweeping peritoneum off.
4. Bring the **dome** (not the anterior wall) to the wound. This reduces prolapse of the posterior wall.
5. Anchor detrusor to rectus fascia; stoma should take about 24 Fr.
6. Open the dome; suture mucosa to skin. No catheter needed.

### Advantages

- Simple; no catheter or appliance.
- The bladder still fills and empties through the stoma at low leak pressure, so compliance is **not** lost; it remains usable for reconstruction or transplantation.

### Complications

- **Stomal stenosis**, mucosal **prolapse**, peristomal dermatitis, stones, UTI; leakage harder to manage as the child grows.

### Closure

- When the cause is treated (valve ablation, urethral growth) or at definitive reconstruction; check bladder function first (cystogram or urodynamics).
- Excise the stoma, free the bladder from fascia, close in two layers, catheter drainage for a few days.

### Alternatives

| Option | Use |
|---|---|
| Cutaneous ureterostomy or pyelostomy | Rare; persistent upper tract obstruction |
| Ileovesicostomy | Unable to catheterise; 10–15 cm ileum |
| Appendicovesicostomy (Mitrofanoff, 1980) | Continent catheterisable channel |

**Diagram to draw:** sagittal view of bladder dome sutured to fascia and skin.

> **Key points to remember:**
> - Blocksom: dome-based stoma midway between umbilicus and pubis.
> - Bladder keeps cycling; compliance preserved.
> - Stenosis and prolapse are the main complications.

**Also, from the condensed book (not in the paper answer):**
- The urachus is excised with a small dome cuff; using the dome keeps the posterior wall taut.

### Upper tract diversion, circumcision and nephrectomy

- Upper tract diversion gives **no renal protection**; use it only for deterioration **despite full lower tract drainage**.
- UTI risk is 50–60% in PUV; circumcision cuts it by over 80%, so offer it to all.
- VURD nephroureterectomy is rare; **keep the ureter** for augmentation.

## 7 · VUR in PUV

- VUR occurs in 50–80% and is **secondary**. It resolves after ablation in 25–40% and does not predict renal outcome.
- Treat the **bladder**. Reimplant only for recurrent febrile UTI despite circumcision and optimal bladder care.

> **KEY POINT**
> VUR in valves is a pressure gauge, not a lesion: fix the pressure, not the reading.

## 8 · Bladder dysfunction and valve bladder syndrome

*See the paper answer **Posterior urethral valves: embryology, management and valve bladder** (Section 5, Ch 32).*

**Also, from the condensed book (not in the paper answer):**
- Stop oxybutynin if residual and capacity rise, as myogenic failure can follow.

**ALGORITHM**
1. High-pressure voiding causes fibrosis and weaker contraction.
2. Post-void residual (PVR) rises; nephrogenic polyuria adds volume.
3. Upper tracts store urine and refill the bladder, so it is never empty.
4. Pressure is back-transmitted; renal function declines, causing more polyuria.

- Continence comes at ~5.5 vs 2.9 years. Do **uroflow and PVR** in every toilet-trained boy; new incontinence after dryness is a red flag.
- An α-blocker suits early bladder-neck obstruction and an antimuscarinic the overactive wall; both harm once the bladder is hypocontractile. Do **videourodynamics** serially.

## 9 · Prognosis for renal function

- Lifetime end-stage renal disease (ESRD) is **20–50%**. Risks are dysplasia, **year-1 nadir creatinine**, recurrent UTI and bladder dysfunction.

| Nadir creatinine (1 year) | ESRD risk |
|---|---|
| <0.8 mg/dL | Minimal (best single predictor) |
| >1.2 mg/dL | High |
| <0.4 / 0.4–0.69 / 0.7–0.99 / >1.0 mg/dL | **0% / 2% / 27% / 100%** |

- **0.7 mg/dL is the watershed**, where risk leaps. On US, echogenicity, cysts and a smaller renal parenchymal area predict a worse outcome.

## 10 · CKD, ESRD and transplantation see Ch 61

- Do **videourodynamics in every candidate** and optimise with CIC or overnight drainage.
- Graft and patient survival are equal if the bladder is managed; late graft loss follows bladder dysfunction. Augmentation can be done before or after transplant.

## 11 · Long-term follow-up, sexual function and fertility

- Transition to adult urologists who know valves. Adult LUTS are 2–3 times commoner; erectile function and fertility are similar overall but worse with incontinence or renal insufficiency.
- Lifelong checks are blood pressure, creatinine, urine protein, US, and uroflow with PVR.

## 12 · Related urethral lesions

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### SAQ Anterior urethral valve

Paper answer · P2

**Type:** SAQ (10 marks)
**Source:** Practice Set 17 Paper 2, Q3c

#### Definition and epidemiology

- AUV (anterior urethral valve): a congenital obstructing fold or cusp in the anterior urethra (bulbar, penoscrotal or penile). It is the **commonest congenital obstructive lesion of the anterior urethra** but **25–30 times less common than PUV (posterior urethral valves)**.
- Often linked to an **anterior urethral diverticulum**: the distal lip of the diverticulum fills and acts as a valve, or a semilunar fold hangs from the roof.

#### Embryology (unclear)

- Hypoplastic corpus spongiosum over the segment suggests a place on the **hypospadias spectrum**, or faulty union of urethral mucosa and fossa navicularis epithelium; a ruptured dilated Cowper (bulbourethral) gland duct is also proposed.

### Clinical features (vary with severity and age)

- Antenatal hydronephrosis, LUTO (lower urinary tract obstruction) in severe cases.
- **Ventral penile bulge** during voiding, with urine expressed by compressing the shaft; post-void dribbling, weak stream, incontinence.
- Palpable bladder, UTI (urinary tract infection), renal insufficiency.

### Evaluation

- **VCUG (voiding cystourethrogram):** dilated anterior urethra proximal to the valve, diverticulum, trabeculated bladder, bladder diverticula, often VUR (vesicoureteric reflux).
- Ultrasound, creatinine; urethroscopy confirms.
- **Differential:** Cowper syringocele, megalourethra, acquired diverticulum, stricture.

### Management

- **Premature or small infant:** vesicostomy until the urethra takes a cystoscope.
- **Most cases: endoscopic valve ablation** (Bugbee electrode or laser), usually definitive.
- **Large diverticulum:** open excision and urethroplasty over a catheter.
- Then manage VUR, infection and bladder dysfunction.

### AUV versus PUV

| Feature | AUV | PUV |
|---|---|---|
| Frequency | 25–30 times rarer | Commonest infant outlet obstruction |
| Site | Bulbar, penoscrotal, penile | From the verumontanum |
| Sign | Penile bulge | Weak stream, palpable bladder |
| Bladder dysfunction | Up to 80% | Near universal |
| Renal outcome | 78% normal function | ESRD (end-stage renal disease) 20–50% |

### Prognosis

- Depends on pre-treatment creatinine and GFR (glomerular filtration rate). Azotaemia, VUR and UTI together raise the risk of poor renal outcome 25-fold.

**Diagram to draw:** VCUG sketch: dilated bulbar urethra with diverticulum ending at a valve lip.

> **Key points to remember:**
> - 25–30 times rarer than PUV; diverticulum common.
> - Penile bulge on voiding; diagnosis by VCUG.
> - Endoscopic ablation usually definitive; 78% keep normal renal function.

**Also, from the condensed book (not in the paper answer):**

- **Urethral atresia** needs decompression to survive; treat with vesicostomy, then PADUA.
- **Urethral duplication** (Effmann I–III): the **ventral** channel has the sphincter, so it is the one moved to the glans.
- **Urethrorrhagia** in peripubertal boys resolves in up to 92%; treat voiding dysfunction and reserve cystoscopy for atypical cases.
- Anorectal malformation often has a rectourinary fistula; catheterise before PSARP.

## 13 · Landmark trials

| Trial | Design | Result | Take-home |
|---|---|---|---|
| PLUTO | RCT: vesicoamniotic shunt vs conservative in LUTO | Trend to better 28-day survival; poor long-term survival and renal function in both | Shunt may help lungs, not kidneys |
| Circumcision RCT | Circumcision and prophylaxis vs prophylaxis | Febrile UTI 3% vs 49% over 2 years | Circumcise boys with PUV |
| Oxybutynin RCT | Oxybutynin to toilet training vs observation post-ablation | Better hydronephrosis/VUR; no change in creatinine, UTI, continence | Selective, monitored use |

# 46 Neuromuscular Dysfunction of the Lower Urinary Tract in Children

## 1 · Definition and causes

- **Neurogenic lower urinary tract (LUT) dysfunction** is bladder and/or sphincter dysfunction from aberrant innervation.
- The commonest cause is **neural tube defect (NTD)**, mainly spina bifida (SB). Survival is about 90% to 10 years, so **renal failure** is a leading late morbidity; **10%** of SB deaths are urological. Clean intermittent catheterisation (CIC, Lapides 1971) was the pivotal advance.

| Congenital | Acquired |
|---|---|
| Myelomeningocele; occult NTD (lipomeningocele, tethered cord); sacral agenesis; anorectal malformation (ARM) | Brain: cerebral palsy (CP), tumours, infarcts. Cord: trauma, transverse myelitis. Pelvis: extensive surgery |

## 2 · Myelodysplasia (NTD)

### SAQ Paediatric neurogenic bladder (myelodysplasia): assessment, management and bladder myoplasty

Paper answer · P1 · P3 · merged from 2

*Asked as:* P3 SAQ "Neurogenic bladder: prenatal/neonatal assessment and bladder myoplasty" · P1 SAQ "Neurogenic bladder in myelodysplasia"

#### Introduction

- Myelodysplasia (NTD [neural tube defect], mainly **myelomeningocele**, an open NTD) is the commonest cause of neurogenic LUT (lower urinary tract) dysfunction in children; other causes are occult dysraphism, sacral agenesis and anorectal malformation.
- Lesions: lumbosacral 47%, lumbar 26%, sacral 20%.
- **Bony or motor level does not predict bladder function**, and the lesion can progress (re-tethering), so every newborn needs urodynamic testing.
- About 10% of deaths in SB (spina bifida) are urological. Folic acid 400 µg daily before conception prevents 50–70% of NTDs.

#### Patterns of dysfunction

- Detrusor: overactive (NDO, neurogenic detrusor overactivity) or underactive. Outlet: competent, incompetent or dyssynergic.
- Newborns: most have detrusor contractions; over a third are acontractile. Needle EMG (electromyography): intact sacral arc about 40%, partial denervation 25%, complete 35%.

#### Prenatal assessment

- **Maternal serum AFP (alpha-fetoprotein)** is raised in open defects; amniotic AFP and acetylcholinesterase confirm.
- **Ultrasound:** spinal defect; "lemon" and "banana" signs of Chiari II; ventriculomegaly; hydronephrosis, thick bladder. Fetal MRI (magnetic resonance imaging) for level.
- Multidisciplinary counselling at a fetal centre.

- **MOMS (Management of Myelomeningocele Study) 2011:** closure **before 26 weeks** reduced shunt need (about 40% versus 82%) and improved motor and cognitive scores, but caused prematurity (34 versus 37 weeks) and uterine thinning. **No proven bladder benefit**: CIC (clean intermittent catheterisation) and reconstruction rates unchanged.

### Neonatal assessment (baseline; EAU [European Association of Urology] 2022)

| Test | Timing / purpose |
|---|---|
| Examination | Spine, legs, anal tone, perineal sensation, bladder |
| **Renal and bladder ultrasound** | Before discharge, then 6-monthly for 2 years (serially); hydronephrosis, wall thickness |
| Residual urine | Start CIC if not emptying |
| Creatinine | Within 3 months (cystatin C later; better with low muscle mass) |
| **VCUG (voiding cystourethrogram) or video-UDS** | Reflux; dilating VUR (vesicoureteric reflux) in about 7% |
| **UDS (urodynamics)** with EMG | Within 3 months, once stable after closure |
| DMSA (dimercaptosuccinic acid) | Baseline scarring |

### Urodynamic risk (McGuire 1981; UMPIRE [Urologic and Renal Protocol for the Newborn and Young Child with Spina Bifida])

| Risk | DLPP (detrusor leak point pressure) or end-filling pressure |
|---|---|
| High (hostile) | 40 cm $H_2O$ or more |
| Intermediate | 25–39 cm $H_2O$, NDO or poor compliance |
| Low (safe) | Under 25 cm $H_2O$ |

- Poor compliance and DSD (detrusor–sphincter dyssynergia) also predict upper-tract damage.
- Repeat studies: the lesion can change (re-tethering).

**Diagram to draw:** neonatal UDS trace with DLPP over 40 cm $H_2O$.

### Management: kidneys first, then continence

- **Early CIC (Lapides 1971) plus oxybutynin** for high-risk (hostile) bladders: hydronephrosis 0–15% versus 18–80% expectant; ESRD (end-stage renal disease) falls from about 30% to under 2% (P1: 0–1.6%).
- Other drugs: solifenacin, mirabegron; **botulinum toxin A** about 10 U/kg (maximum 360 U), continence 65–87%.
- Surgery: vesicostomy (bridge); **augmentation cystoplasty**; Mitrofanoff channel; bladder-neck sling, reconstruction or artificial sphincter.
- VUR: CIC and antimuscarinic first; avoid Credé; reimplant during augmentation if grade ≥III.
- **Also know:** latex allergy; bowel programme; sexual dysfunction (about 75% of men have ED [erectile dysfunction]; sildenafil helps); precocious puberty.

### Bladder myoplasty: definition

- The P3 wording "bladder myeloplsty" is assumed to mean "bladder myoplasty" (a muscle procedure on the bladder wall).

- **Detrusor myoplasty:** wrapping a **vascularised, innervated skeletal muscle flap** round the bladder so that voluntary contraction of that muscle empties an **acontractile bladder**.

### Bladder myoplasty: indications

- **Acontractile or underactive detrusor** with catheter dependence: lower motor neuron lesions (spina bifida, cauda equina injury), myogenic failure.
- Needs good bladder capacity and compliance, and a competent but not obstructing outlet.

### Latissimus dorsi detrusor myoplasty (LDDM; Stenzl)

- **Free latissimus dorsi flap:** artery and vein joined to the **inferior epigastric vessels**; motor nerve joined to an **intercostal nerve**.
- Muscle wrapped **spirally over more than three-quarters of the bladder** and anchored to pelvic floor fascia and ligaments.
- The patient voids by contracting the transferred muscle (abdominal effort).
- **Results:** most voided with low residual; bladder contractility index rose to the strong range.
- **Complications** in about one-third: thromboembolism, abscess, wound infection; none long term.
- **Rectus abdominis detrusor myoplasty:** simpler, less morbid; small spinal cord injury series with spontaneous voiding.
- **Status: investigational.** Standard care stays **clean intermittent catheterisation**.

**Diagram to draw:** latissimus flap spiralled round the bladder with vascular and nerve anastomoses.

### Other "myoplasty": detrusor myotomy/myectomy (autoaugmentation)

- Purpose is the opposite: **increase capacity and compliance** in a small, poorly compliant or overactive bladder.
- **Technique (Cartwright and Snow, 1989):** strip and excise the detrusor over the dome, leave the mucosa intact, and anchor the detrusor edges (psoas hitch) so the mucosa bulges as a **wide-mouthed diverticulum**.
- **Pros:** urothelial lining, no mucus, no metabolic change, no bowel.
- **Cons:** seldom adds significant capacity; compliance gain may not last; **not recommended for adult detrusor overactivity**.
- **Also know:** seromuscular enterocystoplasty covers the bulge with demucosalised sigmoid.

### Alternatives

- **Also know:** sacral neuromodulation, sacral anterior root stimulation, reduction cystoplasty (abandoned), muscle-derived stem cells (research).

**Key points to remember:**

- Level does not predict bladder: UDS within 3 months.
- DLPP 40 cm $H_2O$ or more = hostile (McGuire 1981); early CIC + antimuscarinic protects kidneys.
- MOMS: fewer shunts, no bladder benefit.
- LDDM: free latissimus to inferior epigastric vessels and intercostal nerve; for acontractile bladder.
- Investigational; clean intermittent catheterisation remains standard.
- Autoaugmentation: detrusor excised, mucosa bulges; little lasting gain.

**Also, from the condensed book (not in the paper answer):**

- Trospium (a quaternary amine) has few central effects.

## Classification and prevention

- **Myelomeningocele** is open, commonest and most severe; **SB occulta** is closed and usually benign.
- The neural tube closes by **day 35**. **Folic acid** cuts NTDs by 50–70%: all women of child-bearing age take **400 µg/day** from 4 weeks before conception, and women with SB take **4–5 mg/day**.
- Recurrence is **5%** after one affected child. Almost all open NTDs have **Chiari II**, and most have hydrocephalus.
- **LUT function is not predictable from bony level or the legs**, so urodynamics defines every bladder.

**KEY POINT**

Management of Myelomeningocele Study (**MOMS**): prenatal repair (<26 weeks) reduced death or shunt need (RR 0.70) and improved motor and cognitive scores at 30 months, at the cost of prematurity, uterine-scar thinning or dehiscence, and more re-tethering.

## Pathophysiology

- In **detrusor–sphincter dyssynergia (DSD)**, the sphincter fails to relax during detrusor contraction; it is diagnosed by electromyography.

**KEY POINT**

Every pattern is detrusor (over/under) × outlet (competent/incompetent/dyssynergic). Dangerous: **overactive or poorly compliant detrusor against a closed or dyssynergic outlet**, where storage pressure rises and causes reflux and hydroureteronephrosis. An incompetent outlet "pops off": the child is wet but the kidneys are safe.

### Newborn evaluation

**ALGORITHM**

1. Before discharge: renal–bladder US and post-void residual (need for CIC?).
2. After week 1: serum creatinine.
3. Once stable: urodynamics, **safe from ≈3 months**.
4. Cystography: ideally videourodynamics, otherwise VCUG.
5. Optional baseline DMSA scan (EAU).
6. US at 6, 12, 18 and 24 months, then yearly.

- Guidelines come from ICCS 2012, BAPU 2016 (most selective), SBA 2018 and EAU 2022. In the UMPIRE newborn cohort, dilating VUR was **7%**.
- **Cystatin C** beats creatinine, because low muscle mass hides a falling GFR. High-grade VUR acts as a pop-off, so urodynamic pressures underestimate risk.

### Urodynamic risk factors

| Category | Definition |
|---|---|
| **Hostile / high** | Detrusor leak point pressure (DLPP) or end-filling pressure **≥40 $cmH_2O$**; NDO with DSD (DSD removed in UMPIRE revision) |
| Intermediate | NDO, reduced compliance, 25–39 $cmH_2O$ |
| Abnormal but safe | <25 $cmH_2O$ |
| Normal | Normal capacity, compliance >15 $mL/cmH_2O$, no NDO/DSD, minimal residual |

- Poor compliance independently predicts renal injury.

### Tethered cord

- Growth stretches a low-lying cord, so gait, sensation and bladder deteriorate. Suspect it with new wetting, rising pressure, new UTIs, worsening scoliosis or back pain. Do urodynamics **before untethering and 3–4 months after**.

## 3 · Proactive (early) management

Priorities in order are **kidneys, then continence, then UTI avoidance, then sexual function and fertility**.

- **Early CIC and an antimuscarinic** (Bauer) treat hostile dynamics. All paediatric societies endorse this; there is **no benefit if the child empties spontaneously**. If hydronephrosis persists, add overnight drainage.
- Give **no** antibiotic prophylaxis on CIC; treat bacteriuria only if symptomatic.

| Outcome | Expectant | Early CIC + antimuscarinic |
|---|---|---|
| Hydroureteronephrosis | 18–80% | 0–15% |
| Renal scarring | 27–50% | 4–12% |
| End-stage renal disease (ESRD) | ≈30% | 0–1.6% |
| Augmentation | — | 18–24% fewer |

**KEY POINT**

Proactive won because it treats the cause (pressure, failure to empty) before the consequence (reflux, dilatation, scarring, renal failure).

## 4 · Management ladder

**ALGORITHM**

1. CIC and antimuscarinic, with or without a β3-agonist.
2. If that fails: intradetrusor botulinum toxin.
3. If CIC is impossible: vesicostomy as a bridge.
4. Then augmentation, with or without a catheterisable channel and an outlet procedure see Ch 48.

### Pharmacotherapy

- **Antimuscarinics** are first-line. FDA-licensed drugs are **oxybutynin** (≥5 y; 0.1–0.6 mg/kg/day, max 5 mg/dose; oral, patch or intravesical), **solifenacin** (≥2 y), **fesoterodine** (≥6 y, >25 kg) and **mirabegron** (≥3 y; may cause hypertension).
- α-Sympathomimetics for low-outlet incontinence are unlicensed and help a minority.

### Botulinum toxin A

- Under general anaesthesia, inject **≈10 U/kg (max 360 U)** into about 30 sites, **sparing the trigone**. Continence is 65–87%, lasting 6–10 months, and repeats stay effective. Poor pre-existing compliance predicts poor response.

### Surgery

- **Vesicostomy** is a temporary bridge when CIC is unmanageable. **Augmentation** (ileum preferred) treats the refractory bladder, with a channel or outlet procedure as needed see Ch 48. After ileum, check B12 yearly.

### VUR in the neurogenic bladder

- VUR is secondary, and treating the bladder resolves 30–55%. **Avoid Credé** with a reactive sphincter, because it worsens VUR.
- Antireflux surgery is for breakthrough febrile UTI despite low pressure. At augmentation, **reimplant grade ≥III**.

## 5 · Bowel management and latex allergy

- Neurogenic bowel is managed stepwise with diet, laxatives and transanal irrigation, then a **Malone antegrade continence enema (ACE)**, often at augmentation.
- Latex sensitisation risks intraoperative anaphylaxis, so care is latex-free **from birth**.

## 6 · Sexual function and transition

- Function is **inversely proportional to neurological level**. Central precocious puberty affects about half of girls.
- **Males** have erectile dysfunction in about 75%; **sildenafil** works in about 80%. **Females** conceive in 70–80%; avoid the combined pill and IUD. Offer a structured transition.

## 7 · Other causes

- **Sacral agenesis** is linked to maternal diabetes; look for flat buttocks and a short gluteal cleft. Level does not predict the bladder, and about a third present late.
- **Spinal cord injury**: CIC during spinal shock; a **suprasacral lesion later gives NDO with DSD**. **Autonomic dysreflexia** (lesion **≥T6**) causes severe hypertension with bladder filling, so monitor blood pressure during urodynamics.
- **Transverse myelitis** causes acute retention in over 95%; **late CIC leads to poorer compliance**, so do early urodynamics and CIC.
- **Cerebral palsy** gives coordinated overactivity; upper-tract risk is **<5%**, so urodynamics is only for failed conservative care or recurrent UTI.
- **ARM** has a cord anomaly in 18–50%, so do spinal US before 3 months; neuropathy comes from the cord, not the PSARP. Pelvic surgery can injure pelvic nerves.

> **KEY POINT**
>
> Lesion level predicts pattern. **Suprapontine** (CP) gives coordinated overactivity and low risk. **Suprasacral cord** (SCI, myelitis, tethering) gives NDO with DSD and high risk. **Sacral/peripheral** gives underactivity. Myelodysplasia is mixed and unpredictable.

## 8 · Landmark trials

| Trial | Design | Result | Take-home |
|---|---|---|---|
| MOMS | RCT, prenatal vs postnatal repair | Less shunting, better motor score; more prematurity, uterine dehiscence | Fetal repair in selected centres |
| McGuire 1981 | Myelodysplasia cohort | DLPP >40 $cmH_2O$ led to upper-tract damage | 40 $cmH_2O$ = safety line |
| Sildenafil in SB | Double-blind placebo RCT | ≈80% improved erections | First-line for ED |

# 47 Functional Disorders of the Bladder and Bowel in Children

## 1 · Normal bladder and bowel function

### LAQ Voiding dysfunction in children: neurophysiology and differential diagnosis of frequency/urgency

Paper answer · P1 · P3 · merged from 2

*Asked as:* P1 LAQ "Describe neurophysiology of lower urinary tract and various types of voiding dysfunction in children" · P3 SAQ "DD of frequency and urgency in 3 yr male child and treatment"

#### Introduction

- The LUT (lower urinary tract) has two opposite functions: **storage** at low pressure with a closed outlet, and **emptying** by a coordinated detrusor contraction with a relaxed outlet. Bladder and sphincter act reciprocally.

#### Neurophysiology: peripheral innervation

| | Parasympathetic | Sympathetic | Somatic |
|---|---|---|---|
| Origin | **S2–S4** (sacral parasympathetic nucleus) | Thoracolumbar (T10–L2) | **Onuf nucleus** (S2–S4 ventral horn) |
| Nerve | Pelvic | Hypogastric | Pudendal |
| Bladder | Contracts (ACh [acetylcholine] on **M3**; ATP) | Relaxes (noradrenaline on **β3**) | – |
| Outlet | Relaxes urethra (**NO**, nitric oxide) | Contracts bladder neck (**α1**) | Contracts EUS (external urethral sphincter; nicotinic) |
| Role | Voiding | Storage | Voluntary closure |

- All three nerves carry afferents. **Aδ fibres** (tension receptors in muscle) signal normal fullness. **Unmyelinated C fibres** are "silent" normally; after inflammation, SCI (spinal cord injury) or obstruction they are recruited and drive overactivity (the basis of capsaicin and resiniferatoxin).
- Detrusor has more M2 receptors, but **M3 mediates contraction**.

#### Neurophysiology: reflex and central control

- **Storage (spinal reflexes):** low afferent activity triggers sympathetic outflow (detrusor inhibition, bladder-neck closure) and the **guarding reflex** (pudendal; sphincter activity rises with filling).
- **Voiding (spinobulbospinal reflex):** high afferent activity reaches the **PAG (periaqueductal grey)** and then the **PMC (pontine micturition centre, Barrington nucleus)**; the sphincter relaxes first, then the detrusor contracts. The PMC is an on–off switch.
- **Forebrain decides:** the insula senses fullness; the **medial prefrontal cortex** tonically holds the switch "off" until voiding is socially appropriate.
- Lesion patterns: above the pons, overactivity with a coordinated sphincter; between pons and sacral cord, overactivity with **DSD (detrusor–sphincter dyssynergia)**; sacral or

peripheral, underactive bladder.

- **Central transmitters:** glutamate excitatory; GABA (γ-aminobutyric acid) and glycine inhibitory; serotonin inhibits voiding; dopamine tonically inhibits the PMC (lost in Parkinson disease).

`Filling → Aδ afferents → sacral cord → PAG low traffic: spinal storage reflexes (sympathetic + guarding) high traffic + cortex "permits": PMC → EUS relaxes → detrusor contracts`

**Diagram to draw:** bladder with pelvic, hypogastric and pudendal nerves, plus brainstem loop (PAG–PMC).

### Maturation in children

- Bladder visible on fetal ultrasound by 14 weeks. Infants void reflexly but **wake before voiding** (cortical arousal gates voiding).
- Infant voiding is often interrupted (transient dyscoordination), with detrusor pressure often **>100 cmH2O**; frequent small voids largely disappear by **2 years**.
- Capacity rises: **EBC (expected bladder capacity) = (age + 2) × 30 mL** (infants: weight in kg × 7 mL).
- Toilet training: bowel control, then daytime bladder, then night-time bladder. Persisting dyscoordination after toddlerhood is dysfunction, not immaturity.

### Definitions: frequency and urgency

- **Frequency:** ≥8 voids a day (ICCS, International Children's Continence Society). **Urgency:** sudden, compelling need to void.
- At 3 years toilet training may still be ongoing, so decide first whether the pattern is abnormal.

### Types of voiding dysfunction: non-neurogenic (functional) LUT dysfunction (ICCS)

| Type | Key feature |
|---|---|
| **OAB (overactive bladder)** | Urgency ± frequency (≥8 voids a day), urgency incontinence; tower flow |
| **Voiding postponement** | Habitual holding; holding manoeuvres; infrequent voids (≤3 a day) |
| **Dysfunctional voiding** | Habitual sphincter/pelvic-floor contraction during voiding in a neurologically normal child; **staccato** flow |
| **Underactive bladder** | Needs abdominal straining to void; interrupted flow |
| Giggle incontinence | Complete emptying with laughter |
| Vaginal reflux | Small leak just after voiding in girls |
| **Enuresis** | Wetting in sleep, ≥5 years; monosymptomatic or non-monosymptomatic |

- **BBD (bladder and bowel dysfunction):** LUT dysfunction plus constipation; rectal loading worsens symptoms and UTI (urinary tract infection).
- **Hinman (non-neurogenic neurogenic bladder):** the severe end of dysfunctional voiding, with trabeculated bladder, VUR (vesicoureteric reflux) and renal damage.

### Types of voiding dysfunction: neurogenic LUT dysfunction

- Causes: **myelomeningocele** (commonest), occult dysraphism / **tethered cord**, sacral agenesis, anorectal malformation, cerebral palsy, SCI, transverse myelitis, pelvic tumour surgery (sacrococcygeal teratoma).
- Pattern = detrusor (overactive or underactive) × outlet (competent, incompetent or dyssynergic). Bony level does not predict bladder function.
- **Hostile bladder: DLPP (detrusor leak point pressure) ≥40 cmH2O** (McGuire 1981), poor compliance, or NDO (neurogenic detrusor overactivity) with DSD.

### Types of voiding dysfunction: anatomical

- **PUV (posterior urethral valves)** "valve bladder": overactivity, then poor compliance, then myogenic failure; ectopic ureter.

### Differential diagnosis of frequency and urgency (3-year-old boy)

| Group | Causes and clues |
|---|---|
| **Infection** | **UTI**: fever, dysuria; balanoposthitis; viral haemorrhagic cystitis |
| **Functional** | **BBD**/OAB; voiding postponement; **extraordinary daytime urinary frequency** (sudden very frequent small voids, no nocturia or wetting, often after stress; self-limiting) |
| **Bowel** | **Constipation**: rectal mass presses on bladder; commonest associated cause |
| **Obstruction** | **PUV**, mild late form; **meatal stenosis** (circumcised boys; thin deflected stream); severe phimosis |
| **Neurogenic** | Occult spinal dysraphism, tethered cord (sacral dimple, hair tuft, gait or foot abnormality) |
| **Bladder irritants** | Stone, foreign body, hypercalciuria, pinworms, rarely **rhabdomyosarcoma** |
| **Polyuria** | Diabetes mellitus, diabetes insipidus, chronic kidney disease |
| **Behavioural** | Anxiety, family stress, developmental disorders |

### Evaluation

- **History:** onset, nocturia, wetting, stream, fever, **bowel habit (Bristol stool chart)**, thirst, stressors.
- **Examination:** abdomen, meatus and foreskin, **lumbosacral spine**, lower-limb neurology.
- **Urinalysis and culture**; glucose; specific gravity >1.005 makes diabetes insipidus unlikely.
- **Bladder and bowel diary.**
- **Ultrasound** of kidneys and bladder: wall thickness, residual, dilation, stones.
- **Uroflowmetry with PVR (post-void residual) and patch EMG (electromyography)** if the child cooperates: bell (normal), tower (OAB), staccato (dysfunctional voiding), interrupted (underactive/DSD), plateau (obstruction).
- **Selective:** VCUG (voiding cystourethrogram) for suspected PUV or febrile UTI with abnormal ultrasound; spinal MRI (magnetic resonance imaging) for neurological signs; urine calcium/creatinine ratio.
- Invasive (video)urodynamics only for neurogenic, anatomical or refractory cases.

**Diagram to draw:** flowchart: urinalysis, then ultrasound with residual, then targeted tests.

### Management

- **Treat constipation first:** disimpaction, then maintenance **polyethylene glycol**, diet and toilet routine.
- **Standard urotherapy** for 4 weeks: timed voiding every 2–3 hours, relaxed posture, adequate fluids, avoid caffeine.
- **UTI:** antibiotics per culture.
- **OAB persisting after urotherapy: oxybutynin 0.3–0.6 mg/kg/day** in 2–3 doses (maximum 15 mg/day; watch for constipation); mirabegron; botulinum toxin; parasacral TENS (transcutaneous electrical nerve stimulation).
- **Dysfunctional voiding: biofeedback**; α-blocker; sphincter botulinum toxin if refractory.
- **Extraordinary daytime frequency:** reassurance; resolves in weeks to months.
- **Enuresis:** alarm or desmopressin.
- **Meatal stenosis:** meatotomy. **PUV:** endoscopic valve ablation. **Tethered cord:** neurosurgical release.
- **Neurogenic: CIC (clean intermittent catheterisation; Lapides 1971) + antimuscarinic**, then intradetrusor botulinum toxin, then augmentation.

**Key points to remember:**

- S2–S4 pelvic (voiding), T10–L2 hypogastric (storage), Onuf/pudendal (EUS).
- PMC = switch; medial prefrontal cortex holds it off.
- EBC = (age + 2) × 30 mL; staccato = dysfunctional voiding.
- DLPP ≥40 cmH2O = hostile.
- Always ask about bowels; treat constipation first.
- Frequency/urgency: rule out UTI, PUV, meatal stenosis and occult dysraphism.
- Extraordinary daytime frequency: no nocturia, normal tests, self-limiting.

**Also, from the condensed book (not in the paper answer):**

- The International Children's Continence Society (ICCS) now uses expected bladder capacity = (age + 1) × 30 mL; the paper's (age + 2) is the older formula†.
- In **infancy**, voiding pressures are often **>100 cm H2O** and interrupted voiding is normal; both are gone by **~2 yr**. Dyscoordination later is **dysfunction, not immaturity**.
- Late toilet training is linked to incontinence, enuresis and urinary tract infection (UTI).

**KEY POINT**

Bladder and bowel are one unit, sharing pelvic space and afferents. A loaded rectum causes overactivity and poor emptying, so treating constipation improves lower urinary tract symptoms (LUTS).

## 2 · Definitions — ICCS

**Bladder and bowel dysfunction (BBD)** is lower urinary tract (LUT) dysfunction plus constipation and/or faecal incontinence in a neurologically normal child.

| Group | Symptom | ICCS definition |
|---|---|---|
| Storage | Increased frequency | **≥8 voids/day** |

| Group | Symptom | ICCS definition |
|---|---|---|
| | Decreased frequency | **≤3 voids/day** |
| | Urgency | Sudden immediate need to void |
| | Nocturia | Waking at night to void |
| Voiding | Hesitancy | Difficulty initiating when ready |
| | Straining | Valsalva to initiate and maintain |
| | Intermittency | Non-continuous stream |
| | Retention | Cannot void despite effort; distended bladder |
| Other | Holding manoeuvres; incomplete emptying; recurrent UTI | Incomplete-emptying sensation counts even with minimal post-void residual (PVR); recurrent UTI is itself a BBD symptom |

- **Daytime urinary incontinence (DUI)** is leakage while awake at age **≥5 yr**. **Enuresis** is intermittent leakage while asleep at ≥5 yr: **monosymptomatic (MNE)** has no other LUTS, **non-monosymptomatic (NMNE)** has some.
- **Faecal incontinence** is mostly **retentive** (overflow); ~1 in 5 is non-retentive.
- † **Expected bladder capacity (EBC)** = (age in years + 1) × 30 mL (ICCS, to ~12 yr).

## Daytime LUT conditions

| Condition | Definition / mechanism | Key features |
|---|---|---|
| **Overactive bladder (OAB)** | Urgency, with or without frequency, nocturia, incontinence; no UTI or other pathology | Urgency incontinence, holding manoeuvres |
| **Dysfunctional voiding** | Habitual sphincter/pelvic-floor contraction **during voiding**, neurologically intact; staccato flow | Learned; high PVR, UTI, secondary vesicoureteral reflux (VUR) |
| **Underactive bladder** | Raises abdominal pressure to initiate, maintain or complete voiding | Infrequent, interrupted flow, PVR |
| **Voiding postponement** | † Habitual deferral with holding until urgency incontinence | † Low frequency; oppositional defiant disorder |
| **Giggle incontinence** | Extensive emptying during or after laughing; LUT otherwise normal | Distinguish from laugh-provoked OAB leak and vaginal reflux |
| **Vaginal reflux** | Vagina fills retrograde during voiding | Prepubertal girl; small leak just after voiding |
| **Extraordinary daytime frequency** | † ≥ hourly small voids, no incontinence or nocturia | † Benign, self-limiting; exclude UTI, diabetes |
| **Hinman syndrome** (non-neurogenic neurogenic bladder) | End stage of dysfunctional voiding | Trabeculation, VUR, hydronephrosis, renal damage, hypertension |

**KEY POINT**

**Trap:** dysfunctional voiding (neurologically intact) is not detrusor–sphincter dyssynergia (DSD), which needs a neurological lesion. Urgency incontinence is not always OAB: a voiding postponer also leaks with urge, and anticholinergics help briefly, then worsen emptying.

## 3 · Clinical presentations and associations

- **Acute retention** (peak 2–4 yr) follows genital inflammation, instrumentation or **faecal impaction**; try **analgesia, sitz bath and/or enema before a catheter**.
- **Functional constipation** affects ~14% and is driven by a painful-stool **withholding cycle**. Exclude organic causes such as anorectal malformation (ARM), Hirschsprung disease and spinal causes.
- **ADHD, autism and intellectual disability** have the highest BBD rates and respond worse.

## 4 · Evaluation

*Evaluation mainly **excludes organic disease**; avoid over-investigation.*

### Warning signs

Red flags are rectal bleeding, weight loss or vomiting (organic disease); poor growth or hypertension (chronic kidney disease, CKD); a lumbosacral skin lesion or abnormal leg neurology (spinal MRI); an abnormal anus (ARM); low specific gravity (SG) or glycosuria (diabetes insipidus, DI, or diabetes mellitus); and constant dribbling in a girl (ectopic ureter).

- Ask about bowels in **every** child with LUTS, and use a **3-day bladder diary** and a **bowel diary**. Bristol types 1–2 indicate constipation.
- A **gush on separating the labia** means vaginal reflux. Routine rectal examination is **not** needed.

### Investigations (stepwise, non-invasive)

- **ICCS response**: complete 100%, partial 50–99%, none <50%.
- **Urinalysis**: SG >1.005 makes DI unlikely. Persistent proteinuria needs a first-morning protein:creatinine ratio (>**0.2 mg/mg** is abnormal).
- Measure **PVR** within 5 min of voiding at 50–115% EBC.
- **Uroflow** (3 flows, >50% EBC): bell is normal, tower suggests OAB, staccato dysfunctional voiding, interrupted underactive bladder, and plateau obstruction.
- Pelvic floor **EMG** active during voiding means dysfunctional voiding; an **EMG lag time <2 s** means detrusor overactivity. **Video urodynamics** is only for treatment failure or suspected obstruction or upper-tract risk.

### Bowel — Rome IV functional constipation (child ≥4 yr)

**ALGORITHM**

1. ≥2 of, ≥1/week for ≥1 month, irritable bowel syndrome (IBS) criteria not met:
2. ≤2 defecations in toilet/week; ≥1 faecal incontinence/week; retentive posturing/withholding; painful or hard stools; large rectal faecal mass; large-diameter stools that may block the toilet.

- Pain is **not** a Rome IV criterion.
- No routine labs or **plain radiograph**. A rectal diameter **≥30 mm** on US suggests constipation (European Association of Urology/European Society for Paediatric Urology, EAU/ESPU).
- If refractory, use a transit study, contrast enema and **anorectal manometry** (absent RAIR needs rectal biopsy for Hirschsprung).

## 5 · Management of daytime incontinence

**ALGORITHM**

1. History, exam, urinalysis, PVR, with or without uroflow.
2. Exclude neurogenic causes, diabetes, CKD and anatomical causes.
3. Optimise neurodevelopmental disorders; **treat constipation first**.
4. If wetting is only at night, treat as MNE.
5. Otherwise, **standard urotherapy** (≥4 wk); re-evaluate.
6. Then **specific urotherapy** by diagnosis.
7. If refractory: botulinum toxin or neuromodulation.

### Standard urotherapy

- Give information, fluid and diet advice and diaries. Sit with hips and knees at ~90° and feet supported, and use **timed voiding** 6–7 times a day. **Reward adherence, not dryness**. Urotherapy cures DUI in more than half.

### Dysfunctional voiding

- Diagnose by **staccato flow with active pelvic floor EMG**.

| Renal risk | Management |
|---|---|
| High (hydronephrosis, VUR, rising creatinine) | Video urodynamics; lower storage pressure; clean intermittent catheterisation (CIC) past the sphincter |
| Low | Urotherapy, double voiding, treat constipation, short-course prophylaxis for recurrent UTI |

- **Biofeedback is the first-line specific therapy**. **α-blockers** (doxazosin) lower PVR. If refractory, give **intrasphincteric onabotulinumtoxinA** 50–300 units.

### Overactive bladder

Use drugs only after urotherapy and bowel management.

| Agent | Dose / point |
|---|---|
| **Oxybutynin** (non-selective) | Immediate release **0.3–0.6 mg/kg/day** in 2–3 doses (max 15 mg); extended release (ER) 5–20 mg/day; patch 3.9 mg/day. Only US-approved drug for paediatric OAB; causes dry mouth and constipation. |
| Tolterodine | 1–2 mg twice daily or ER 2–4 mg; ER randomised controlled trial (RCT) negative |
| **Solifenacin** (M3-selective) | 2.5–10 mg daily; positive phase III RCTs (LION, LEOPARD) |
| Trospium (quaternary amine) | Does not cross the blood–brain barrier |

| Agent | Dose / point |
|---|---|
| **Mirabegron** (β3-agonist) | 25–50 mg if >35 kg; approved ≥3 yr for **neurogenic** detrusor overactivity only; safe add-on |

- Check PVR and stools before and during antimuscarinics.
- **Intradetrusor onabotulinumtoxinA** (refractory): **5 units/kg, max 100 units**; it lasts ~12 months.

Neuromodulation options are parasacral TENS over S2–S3 (beat oxybutynin, but a sham RCT was negative), the weaker tibial nerve stimulation, and off-label sacral nerve stimulation (SNS), which helps bladder and bowel.

### Other conditions

- **Underactive bladder**: timed and double voiding and biofeedback; CIC if PVR is high.
- **Postponement**: timed voiding and treat comorbidity; † avoid anticholinergics.
- **Giggle**: **methylphenidate** or **biofeedback**.
- **Vaginal reflux**: void with legs abducted (sit backwards on the toilet). **Extraordinary frequency**: † reassure.

## 6 · Nocturnal enuresis

### SAQ Nocturnal enuresis

Paper answer · P1 · P3 · merged from 2

*Asked as:* P1 SAQ "Nocturnal enuresis" · P3 SAQ "Enuresis"

#### Definition (ICCS [International Children's Continence Society])

- Intermittent urinary incontinence **during sleep** in a child **aged 5 years or more**.
- **Primary:** never dry. **Secondary:** wetting after at least **6 months** dry (look for stressors, diabetes, UTI [urinary tract infection], neurological disease).
- **MNE (monosymptomatic):** no daytime LUTS (lower urinary tract symptoms). **NMNE (non-monosymptomatic):** with daytime LUTS.

#### Epidemiology

- About **5–10% at 7 years**; falls with age, with spontaneous resolution about 15% a year; about 2% of adults; familial.

#### Pathophysiology: three-way mismatch

1. **Nocturnal polyuria** (poor or reduced night-time ADH [antidiuretic hormone] rise).
2. **Small (reduced) nocturnal bladder capacity** or DO (detrusor overactivity).
3. **Failure to arouse** from sleep. - **Why sleep apnoea matters:** obstructed breathing makes arousal difficult and releases atrial natriuretic peptide, which raises urine output. - Comorbidities: constipation, sleep-disordered breathing, ADHD (attention deficit hyperactivity disorder).

**Diagram to draw:** triad: nocturnal polyuria, small bladder capacity, arousal failure.

#### Evaluation

- **History:** day symptoms, bowels, snoring, fluids, family history.
- **Examination:** spine, genitalia, abdomen for faecal loading; urinalysis.

- **Bladder diary:** daytime voided volumes against EBC (expected bladder capacity) = (age + 2) × 30 mL in the chapter (ICCS uses [age + 1] × 30 mL); night urine output by weighing nappies.
- Uroflowmetry, PVR (post-void residual) and ultrasound only for NMNE or before adding anticholinergics. No invasive tests in MNE.

### Management: general

- **Treat daytime LUTS and constipation first** (in NMNE).
- Education removes blame; standard urotherapy. Waking, fluid restriction alone and punishment do not work.

### Management: first line

| | **Alarm** | **Desmopressin** |
|---|---|---|
| Target | Arousal (conditioning) | Night urine output |
| Use | Nightly for months; parent wakes child; continue to 14 dry nights | 0.2–0.4 mg tablet or 120–240 µg melt, 1 hour before bed |
| Response | 50–70%; durable | Rule of thirds; immediate but relapses off the drug |
| Caution | Needs motivated family | Restrict / limit evening fluids (hyponatraemia) |

- Overlearning (extra evening fluid once dry) reduces alarm relapse.

### Management: failure and resistant cases

- **Failure:** switch or combine.
- **Second line:** add an anticholinergic (oxybutynin) after checking PVR and bowels.
- **Resistant:** imipramine (cardiotoxic; ECG [electrocardiogram]), off-label β3-agonists; treat sleep apnoea (adenotonsillectomy).

### Adult enuresis

- New enuresis in an adult may signal high-pressure chronic retention: catheterise and treat the obstruction.

> **Key points to remember:**
> - ≥5 years; secondary = after 6 months dry.
> - Alarm (durable) and desmopressin (rule of thirds) are first line.
> - Treat daytime symptoms and constipation first.

**Also, from the condensed book (not in the paper answer):**

- Prevalence is **5–10% at 7 yr**; ~15% resolve each year, and ~2% persist into adulthood.
- The **triad** is nocturnal polyuria (relative antidiuretic hormone deficiency), reduced nocturnal capacity or overactivity, and failure to arouse.
- In **NMNE**, treat bothersome daytime LUTS and constipation first. In **MNE**, treat from **~6 yr**.

| | Enuresis alarm | Desmopressin |
|---|---|---|
| Use | Nightly; the parent wakes the child; continue until **14 consecutive dry nights** | **0.2–0.4 mg** tablet or 120–240 µg melt ~1 h before bed; stop if no effect after ~1 week |
| Response | 50–70%; slow | Immediate; rule of thirds; best with nocturnal polyuria |
| Durability | **Durable**; overlearning lowers relapse | **Not durable**; drug holidays or special nights |
| Safety | Nil | Hyponatraemia: † no fluids from 1 h before dose |

**ALGORITHM**

1. MNE at ≥6 yr: start the alarm or desmopressin.
2. If it fails, check use, then switch or **combine**.
3. Re-evaluate: hidden daytime LUTS, constipation, sleep-disordered breathing, neurodevelopmental disorder; uroflow and PVR.
4. Then **add an anticholinergic** to desmopressin.
5. If refractory: imipramine, a β3-agonist or specialist drugs.

- **Imipramine** 25–50 mg is **cardiotoxic** (overdose, long QT), so do an electrocardiogram if there is syncope, palpitations or a family history of sudden death.

## 7 · Functional constipation — treatment

### SAQ Management of Defecation Disorders resulting into LUTS

Paper answer · P4

**Type:** SAQ (10 marks)
**Source:** 2025-Winter Paper 4, Q3f

*Assumed meaning: bladder and bowel dysfunction, where constipation or defecation disorders cause LUTS (lower urinary tract symptoms).*

#### Why the bowel causes LUTS

- **Shared space:** a loaded rectum compresses the bladder, reducing capacity and obstructing outflow.
- **Shared nerves:** bladder and bowel afferents converge on the same spinal segments; cross-organ sensitisation causes urgency and detrusor overactivity.
- **Shared muscle:** a pelvic floor that contracts during defecation (**dyssynergic defecation**) also contracts during voiding (**dysfunctional voiding**) → high PVR (post-void residual), UTI (urinary tract infection), reflux.
- Result: urgency, daytime incontinence, enuresis, retention, recurrent UTI.

#### Evaluation

- Stool frequency, pain, **Bristol chart**; 3-day bladder and bowel diary.
- **Rome IV (child):** at least 2, weekly for 1 month: ≤2 defecations/week; faecal incontinence weekly; retentive posturing; painful or hard stools; large rectal mass; large-diameter stools.
- **Rome IV (adult), for 3 months:** ≥2 of straining, hard stools, incomplete evacuation, blockage, manual manoeuvres (each in >25% of defecations), <3 stools/week.
- Examine abdomen, spine and perineum; exclude organic causes (Hirschsprung disease, hypothyroidism, neurogenic).

- **Also know:** rectal diameter ≥30 mm on ultrasound; no routine X-ray; manometry if refractory.

### Management

**Rule: treat the bowel first (EAU, European Association of Urology/ESPU, European Society for Paediatric Urology).**

1. **Disimpaction:** oral PEG (polyethylene glycol) 3350, 1–1.5 g/kg/day for 3–6 days; enemas if needed.
2. **Maintenance:** PEG 0.4 g/kg/day (range 0.2–0.8), at least 2 months; lactulose alternative; stimulants (senna) for expulsion.
3. **Urotherapy:** toileting after meals (gastrocolic reflex); footstool posture; normal fluid and fibre.
4. **Biofeedback, pelvic floor physiotherapy** for dyssynergia.
5. **Refractory: transanal irrigation** (water via rectal cone; a quarter to a third stop); **ACE (antegrade continence enema, Malone)**; botulinum toxin to the sphincter; sacral or tibial neuromodulation. - **Adults:** laxatives, prucalopride or linaclotide, biofeedback; rectocele repair; irrigation in neurogenic bowel. - **Caution:** anticholinergics worsen constipation; LUTS return.

**Diagram to draw:** pelvic cross-section showing loaded rectum compressing bladder.

> **Key points to remember:**
> - Treat the bowel first.
> - Rome IV: ≥2 criteria for 1 month (child).
> - PEG disimpaction 1–1.5 g/kg/day for 3–6 days.

**Also, from the condensed book (not in the paper answer):**

- Daily enemas for 3–6 days are an alternative to PEG.

**ALGORITHM**

1. If impacted, **disimpact** with oral polyethylene glycol (PEG) or enemas.
2. Give education, a diary, toilet sitting and **maintenance PEG**.
3. Review at 2 wk; if failing, check adherence, dose, impaction, mental health.
4. Continue for **≥2 months**. Wean when ≥1 month symptom-free, with ≥3 stools/week and Rome IV not met.
5. Relapse: reinstate, or refer to gastroenterology.

| Laxative | Dose |
|---|---|
| PEG 3350 — disimpaction | **1–1.5 g/kg/day × 3–6 days** (>3 yr) |
| PEG 3350 — maintenance | **0.2–0.8 g/kg/day**, start 0.4 (>6 months) |
| Lactulose — maintenance | 1–2 g/kg once or twice daily (safe in infants) |

- **Titrate** PEG: increase it if there is no stool in 24 h, and add a **stimulant** (senna, bisacodyl) if watery accidents occur. There is **no dependence**.
- Refractory cases may need biofeedback, transanal irrigation or SNS. Before an antegrade continence enema (**ACE**), do **colonic manometry**, because diffuse dysmotility may need

resection (see Ch 48).

## 8 · Recurrent UTI and BBD

- In RIVUR/CUTIE, recurrence was **BBD + VUR 51% > BBD alone 35% > VUR alone 20%**.
- Secondary VUR often resolves once BBD is treated.
- **EAU/ESPU**: exclude BBD in every toilet-trained child with febrile or recurrent UTI, and give prophylaxis during BBD treatment. The **AUA** advises prophylaxis for BBD with VUR.

## 9 · Landmark trials

| Trial | Design | Result | Take-home |
|---|---|---|---|
| RIVUR + CUTIE (BBD analysis) | Prophylaxis vs placebo, <6 yr after febrile UTI, with or without VUR | BBD + VUR highest recurrence; BBD alone > VUR alone | Treat BBD; prophylaxis for BBD + VUR |
| LION / LEOPARD | Phase III placebo RCTs, solifenacin | Improved all efficacy measures; mild–moderate side effects | Best paediatric antimuscarinic evidence |

# 48 Lower Urinary Tract Reconstruction in Children

## 1 · Goals and principles

- The priorities, in order, are: **protect the kidneys, then prevent urinary tract infection (UTI), then achieve continence**.
- The indication is a hostile bladder (poor compliance, high pressure, poor emptying) despite medical care, mostly in spina bifida; also exstrophy, posterior urethral valves and bilateral ectopic ureters.
- **Clean intermittent catheterisation (CIC)** is the key enabler, and the main determinant of outcome is **patient and family commitment**.

> **KEY POINT**
>
> Every reconstructive step trades storage for emptying. A continent reservoir that is not catheterised causes hydronephrosis, sepsis and perforation, so incontinence can be "protective". CIC and motivation are prerequisites.

## 2 · Patient selection and preoperative evaluation

- **Storage**: a chronic filling pressure **>40 cm H2O** causes hydroureteronephrosis, a falling glomerular filtration rate (GFR) and acquired vesicoureteral reflux (VUR). Start anticholinergics and CIC; persistent poor compliance needs **augmentation**.
- **Outlet**: resistance must exceed bladder pressure during all activity; a deficient outlet needs a procedure.
- **Emptying**: nothing predicts voiding after augmentation, so every patient practises CIC beforehand.

| Urodynamic finding | What it decides |
|---|---|
| Capacity and fluoroscopic bladder shape | Length of bowel segment |
| Detrusor leak point pressure (DLPP), end-filling pressure, open bladder neck | Need for a bladder neck procedure |
| VUR with a history of UTI | Reimplantation |
| Sphincter electromyography (EMG) | Dyssynergia |

- Fill at **≤10% of predicted capacity per minute**, because fast filling falsely lowers compliance.
- Every child has ultrasound (US). Hydronephrosis needs a renogram **with a bladder catheter** and voiding cystography or videourodynamics.
- Check electrolytes, blood urea nitrogen (BUN) and creatinine. If raised, involve nephrology and measure 24-h urine **volume**: the reservoir must hold **3–4 h** of output.
- In **renal insufficiency**, acidosis **always worsens** with ileum or colon, so consider stomach. In **liver failure**, ammonia reabsorption is dangerous.
- Low-grade secondary **reflux** usually resolves after augmentation; correct high-grade reflux at reconstruction.
- In short gut, diarrhoea or faecal incontinence, spare the ileocaecal valve; add an antegrade continence enema (ACE) if needed.

- A sensate urethra, wheelchair use or prior bladder neck repair favour an abdominal stoma. Do all steps in **one operation** where possible.
- **Sterile urine** is mandatory; bowel preparation is optional. With a ventriculoperitoneal shunt, infection is rare but shunt failure increases.

## 3 · Managing the ureters

- Reimplant into the native bladder, with or without a **psoas hitch** (avoid the genitofemoral and femoral nerves). A short ureter needs transureteroureterostomy.
- Into bowel: **colon** under a taenia; **stomach** submucosally; **ileum** needs a **nipple valve** (hardest).

## 4 · Bladder augmentation (enterocystoplasty)

*See the paper answer **Intestinal segments in urology and orthotopic neobladder** (Section 12, Ch 133).*

**Also, from the condensed book (not in the paper answer):**

- Base deficit is the most sensitive marker; height can fall even without frank acidosis.
- Mucus: colon more than ileum more than stomach; stomach forms uric acid stones.

### Principles

- **Bivalve the native bladder widely**; a narrow anastomosis makes a diverticulum.
- **Always detubularise** and **reconfigure into a sphere**. Intact bowel contracts at **60–100 cm $H_2O$**, and a sphere gives the most volume per surface area.
- Length: ileum **20–40 cm**, colon **~20 cm**; err on the large side.
- Use **no permanent sutures or staples**, because they seed stones.

### Segments

- **Ileocystoplasty**: take ileum at least **15–20 cm proximal to ileocaecal valve** (to protect B12 absorption) and open it antimesenterically into a U or W.
- **Gastrocystoplasty** suits **renal insufficiency with acidosis, short gut and irradiated bowel**. **Jejunum** is avoided (salt-losing acidosis).

| Segment | Advantages | Disadvantages |
|---|---|---|
| **Ileum** (default) | Least pressure activity; plentiful, mobile mesentery | Hyperchloraemic acidosis; B12 loss; mucus; hard to make antireflux |
| Ileocaecal | Taenial reimplant; appendix; imbricated valve | Diarrhoea in neurogenic bowel; more mucus |
| Sigmoid | Taenial reimplant; close to bladder | Strongest contractions; most mucus; more reaugmentation |
| Stomach | No acidosis; least mucus; fewer struvite stones; usable in renal failure | Hypokalaemic hypochloraemic alkalosis; haematuria–dysuria syndrome; tumours |
| Ureter | No mucus, no metabolic effect | Needs a massively dilated ureter of a non-functioning kidney; limited volume |
| Autoaugmentation (detrusorectomy) | Extraperitoneal; no bowel | Seldom adds capacity; compliance may not last |

- Seromuscular and tissue-engineered augments are experimental. Technique matters more than segment, so use ileum unless there is a reason not to.

> **KEY POINT**
>
> Augmentation is needed less now, thanks to early anticholinergics, CIC, botulinum toxin and neuromodulation (immediate vs expectant treatment: 17% vs 41% augmented). This is not trial-proven, but it is the rationale for treating at diagnosis.

### Incontinent alternatives

- **Vesicostomy** is an infant bridge for worsening hydronephrosis, febrile UTI with high-grade VUR, or high pressure. **Ileovesicostomy** suits those who cannot catheterise.

## 5 · Continent catheterisable channels

*See the paper answer **Continent urinary diversion: continence principles, Mitrofanoff, orthotopic vs non-orthotopic** (Section 12, Ch 134).*

**Also, from the condensed book (not in the paper answer):**

- In a **flap valve**, a narrow tube tunnelled into a **low-pressure** reservoir is **compressed** as pressure rises; the appendix needs only **~2 cm** of tunnel. Nipple valves (Kock, Mainz) are **effaced** by pressure (about 85% vs more than 90–95% continent). A flap valve has **no pop-off**.

**ALGORITHM**

1. Detach the appendix with a caecal cuff and preserve the mesoappendix.
2. Tunnel the **distal end** submucosally.
3. Bring the caecal end to skin without tension or twist, with a **skin flap** in the stoma.
4. Keep the channel **short and straight**; fix the bladder to the abdominal wall.
5. Test catheterisation after each step.
6. Stent with 6–12 French (Fr) for 10–14 days; the surgeon does the first catheterisation.
7. Then CIC at least 4-hourly by day.

- Site the stoma at the umbilicus or right lower quadrant. The appendix can be **split** for both an ACE and a channel.
- **Yang–Monti**: a 1–2 cm ileal ring is retubularised **transversely** over 12 Fr; it needs about twice the revisions of the appendix. The **Casale** bladder flap spares bowel but has high stenosis.
- **Stomal stenosis** is the commonest complication and occurs early. Stomal incontinence (rare) means a short tunnel or high reservoir pressure.
- **Ureterosigmoidostomy and Mainz II** are largely abandoned because of acidosis, pyelonephritis and cancer.

## 6 · Bladder outlet procedures

Outlet procedures sacrifice voiding and pop-off, so neurogenic repairs need **concurrent augmentation** and surveillance for **≥5 yr**.

| Procedure | Mechanism | Voiding | Main problem / place |
|---|---|---|---|
| **Fascial sling** | Coaptation and elevation | Rarely adequate, so CIC | Better in girls; only predictor of success is augmentation; AUS still possible later |
| **Artificial urinary sphincter (AUS)** | Inflatable bladder-neck cuff | **Preserved** | Infection, erosion, malfunction; best if voiding |
| Bulking injection | Submucosal | Preserved | Poor durability |
| Young–Dees–Leadbetter | Tubularised trigone | Possible | Exstrophy (see Ch 43); poor in neurogenic |
| **Kropp** | Anterior bladder tube under trigone (flap valve) | Abolished | Highest continence; hard urethral CIC (boys); new VUR |
| **Pippi Salle** | Anterior flap onlaid on trigonal strip | Abolished | Easier CIC; lower night continence; fistula |
| **Bladder neck closure** | Complete separation and continent stoma | Abolished | After failed repairs; no pop-off |

- The **sling** suits a marginal bladder neck in an augmented child on CIC.
- The **AUS** cuff goes at the bladder neck in girls and prepubertal boys and is left **deactivated ~6 weeks**; most neurogenic children still need CIC. New hydronephrosis means unmasked bladder hostility.
- **Kropp** and **Salle** repairs often need later augmentation. Bladder neck **closure** with **interposed omentum** keeps fistula low, and suits reliable catheterisers only.

> **KEY POINT**
>
> Choosing the outlet: a child who voids gets an AUS (the only option preserving voiding). A child on CIC who is being augmented gets a sling (girls) or flap-valve lengthening with a channel. After failed repairs, do bladder neck closure with a Mitrofanoff. Always re-study the bladder afterwards.

## 7 · Malone antegrade continence enema (MACE)

- † An appendicocaecostomy gives scheduled antegrade washouts and faecal continence in spina bifida, anorectal malformation and refractory constipation. Plan it with the urinary reconstruction.

## 8 · Postoperative care

- Irrigate the suprapubic tube **≥3 times daily**. Do a **cystogram at ~3 weeks**, then start CIC 2–3-hourly by day, **never beyond 4–5 h**.

## 9 · Complications of augmentation

*Up to ⅓ need further surgery: **bowel is not bladder**.*

| Segment | Derangement | Management |
|---|---|---|
| Ileum, colon | **Hyperchloraemic metabolic acidosis** (overt mainly with renal impairment) | Bicarbonate; **base deficit** is most sensitive |

| Segment | Derangement | Management |
|---|---|---|
| Jejunum | Hyponatraemic, hypochloraemic, hyperkalaemic acidosis | Avoid |
| Stomach | **Hypokalaemic hypochloraemic alkalosis** with vomiting; **haematuria–dysuria** | Fluids, H2 blockers or proton pump inhibitor |
| Stomach and bowel | Near neutral | Complex patients |

- Acid load demineralises bone and lowers height, so treat base deficit early. Check B12 long term.
- **Bacteriuria** is expected with CIC; treat only symptomatic infection and **urea-splitters**.
- **Stones** are mostly **struvite**; prevent them with **daily irrigation ≥240 mL** saline, which also clears mucus.

### Delayed spontaneous perforation

- Counsel a **~9%** risk, mostly in myelodysplasia. **Outlet procedures double the risk**, as do missed CIC and substance abuse.
- Signs are pain, distension, fever, sepsis and shoulder-tip pain; poor sensation means **late presentation** and deaths. Do a **cystogram** with a post-drain film, then a CT cystogram if negative. Treat by **immediate laparotomy and repair**; manage conservatively only if stable, sterile and contained.

### Malignancy

*See the paper answer **Malignancy in augmented bladder and after urinary diversion** (Section 12, Ch 133).*

**Also, from the condensed book (not in the paper answer):**

- After ureterosigmoidostomy, adenocarcinoma arises at the anastomosis after **~26 yr**, from N-nitroso compounds.
- After augmentation it is reported from **4 yr**, is highest after **gastrocystoplasty**, and is aggressive. Bladder cancer in spina bifida is ~2–4%. There is no consensus on annual cystoscopy; consider it after gastrocystoplasty, in exstrophy and after transplant.

### Pregnancy

- Vaginal delivery is possible; caesarean after major bladder neck repair or AUS, with a urologist present.

## 10 · Follow-up and transition

| When | Tests |
|---|---|
| 3, 6, 12 months | US; electrolytes, BUN, creatinine, urine culture (with or without urodynamics by 6 months) |
| Yearly, lifelong | US, chemistry (with or without blood gas, B12), CIC and irrigation compliance; urodynamics if new hydronephrosis, incontinence or VUR |
| Symptomatic | † Haematuria needs cystoscopy; pain or sepsis needs perforation excluded |

- Late chronic kidney disease after augmentation is driven by **CIC non-compliance**, so adult urology must police CIC and the upper tracts.

## 11 · Landmark trials

| Trial | Design | Result | Take-home |
|---|---|---|---|
| Early vs expectant neurogenic bladder care | Cohort, treated at diagnosis vs expectantly | Augmentation 17% vs 41% | Treat hostile bladder early |
| Daily bladder irrigation | Augmented cohort, ≥240 mL saline daily | Marked fall in stones | Routine daily irrigation |
| Tissue-engineered bladder | Industry multicentre trial | No gain in compliance or capacity | Bowel remains standard |

# ★ Putting it together: common paediatric pathways

## A · Antenatal urinary tract dilation (UTD)

**ALGORITHM**

1. **Prenatal:** A1 before 28 weeks: ultrasound (US) after 32 weeks. A2–3: US 4-weekly. Male, thick bladder, bilateral: suspect posterior urethral valves (PUV).
2. **Postnatal:** first US **after 48 hours**, except suspected PUV or bilateral severe dilation.
3. Normal (<10 mm): discharge; mild: repeat US; moderate–severe or dilated ureter: voiding cystourethrogram (VCUG), MAG3, consider prophylaxis (→ Ch 34).

## B · First febrile urinary tract infection (UTI)

**ALGORITHM**

1. **Diagnose:** catheter or suprapubic sample if not toilet-trained; bag only rules out.
2. **Treat** within 48 h (>72 h doubles new scarring); oral if well.
3. **Image:** US all; VCUG if US abnormal, atypical or recurrent; DMSA at 4–6 months if needed.
4. **Prevent:** treat bladder and bowel dysfunction and constipation; prophylaxis if high risk (→ Ch 37, 41).

## C · Newborn with PUV

**ALGORITHM**

1. Stabilise breathing → 5–7 Fr feeding tube → creatinine, electrolytes, fluids.
2. US, then VCUG; prophylaxis.
3. When stable, **endoscopic ablation**; if the urethra is small, vesicostomy.
4. Prognosis: first-year nadir creatinine (<0.8 mg/dL good); lifelong renal follow-up (→ Ch 45).

## D · Neurogenic bladder in spina bifida

| Step | What | Why |
|---|---|---|
| Birth | US, post-void residual, creatinine; latex-free care | Baseline |
| ≈3 months | Urodynamics | Hostile: leak point or end-filling ≥40 $cmH_2O$, poor compliance, dyssynergia |
| Hostile bladder | Early clean intermittent catheterisation (CIC) and antimuscarinic | Kidney protection |
| Failing | Botulinum toxin, vesicostomy (infant), then augmentation | Safe storage (see Ch 46, 48) |
| Always | Bowel programme; watch for re-tethering | Safety |

## E · Daytime wetting and bedwetting

**ALGORITHM**

1. Urinalysis, residual with or without uroflow; exclude organic causes.
2. **Constipation first**, then urotherapy ≥4 weeks.
3. Antimuscarinic for overactive bladder, biofeedback for dysfunctional voiding; toxin or neuromodulation if refractory.
4. **Monosymptomatic enuresis:** alarm or desmopressin; combine if partial (→ Ch 47).

## F · Augmentation segment

| Segment | Advantages | Disadvantages |
|---|---|---|
| Ileum | Default; compliant | Mucus, hyperchloraemic acidosis, B12 loss |
| Ileocaecum / colon | Appendix for channel | Diarrhoea in neurogenic bowel; acidosis |
| Stomach | Suits renal failure; little mucus | Haematuria–dysuria, alkalosis |
| Ureter | No mucus or metabolic effect | Needs a massive ureter |

**KEY POINT**

Reconstruction trades storage for emptying: an unreliably catheterised reservoir causes sepsis and perforation, so CIC commitment comes first.

# Other paper questions in this section

*Asked in the papers but not tied to a specific place in the book above.*

## SAQ CSIC.

Paper answer · P3

**Type:** SAQ (10 marks)
**Source:** 2018-Summer Paper 3, Q3e

*Taken to mean **CISC – clean intermittent self-catheterisation**.*

### Definition and principle

- Regular, **clean (not sterile)** passage of a catheter by the patient (or carer) to empty the bladder, then removal.
- **Lapides (1971–72)**: infection comes from **stasis and overdistension** (poor bladder-wall blood flow), not from bacteria alone. Regular complete emptying protects the bladder wall and upper tracts, even though bacteriuria develops.
- Gold standard for chronic emptying failure.

### Indications

- **Neurogenic bladder:** spina bifida, spinal cord injury, multiple sclerosis, cauda equina, diabetic cystopathy.
- **Underactive or acontractile detrusor**; chronic retention; post-operative retention.
- **Detrusor–sphincter dyssynergia**: CISC bypasses the obstructed sphincter.
- After **augmentation or continent diversion**; **valve bladder**; high residual after botulinum toxin.
- **Contraindications:** poor hand function or cognition without a carer, urethral false passage or stricture, unwillingness.

### Technique

- Wash hands → clean meatus with water → lubricated catheter (or pre-lubricated **hydrophilic** catheter) → pass until urine flows → drain fully → withdraw slowly.
- Sizes: adults about 12–14 Fr; children by age (8–10 Fr).
- **Frequency:** 4–6 times a day, keeping each volume **below about 400–500 mL** (child: below expected bladder capacity).
- Combine with **antimuscarinics** for overactivity; bladder diary.

### Catheters

| Type | Note |
|---|---|
| Plain PVC (polyvinyl chloride), reusable after washing | Cheap; common in India |
| **Hydrophilic-coated** single-use | Less urethral trauma; longer time to first UTI (urinary tract infection) |
| Compact or closed-system | Convenience, outdoor use |

### Children

- Start **early (infancy)** in spina bifida with hostile bladder; families adapt best; child involved from 3–5 years.

- **Mitrofanoff appendicovesicostomy** (1980) if urethral catheterisation is difficult (wheelchair, sensate urethra, girls).

### Complications

- **Asymptomatic bacteriuria** (common): **do not treat or give prophylaxis**; treat only symptomatic UTI.
- Urethral trauma, bleeding, **false passage**, stricture, epididymitis, bladder stones (hair, mucus), rarely lost catheter.

### Outcome

- Less pyelonephritis, epididymo-orchitis and urosepsis than indwelling catheters; preserves the upper tracts; with drugs, continence in many.

**Diagram to draw:** neurogenic bladder management ladder: CISC + antimuscarinic → botulinum toxin → augmentation + channel.

> **Key points to remember:**
> - Lapides: overdistension, not bacteria, causes infection; clean is enough.
> - 4–6 times daily, under about 500 mL each.
> - Do not treat asymptomatic bacteriuria.
> - Mitrofanoff when urethral access is hard.

# Rapid Revision Sheet · Section 5

### Bladder embryology and exstrophy-epispadias complex (LAQ)

- Bud from Wolffian duct; position decides VUR or ectopia.
- Urorectal septum weeks 5–6 (P2: weeks 4–6); mesonephric duct entry is the landmark.
- Vesicourethral canal forms the bladder; urachus from allantois.
- Trigone: common excretory duct absorption; mostly bladder muscle (current view).
- Urachus forms at months 4–5; urine from week 10; amniotic fluid mainly urine after 16 weeks.
- Marshall–Muecke: overdeveloped cloacal membrane, failed mesodermal reinforcement; rupture timing decides variant.
- CBE: external rotation 12°/18°, anterior pubis 30% short, diastasis about 4.8 cm; VUR in 100%.
- Cloacal exstrophy: spinal anomalies 85–100%; osteotomy always.
- MSRE: closure at birth, epispadias repair 6–12 months, YDL / BNR at 5–7 years at ≥100 mL.
- Osteotomy if closure after 72 h or diastasis >4 cm.
- CPRE: closure + penile disassembly in newborn. Risks: glans loss, hypospadias, VUR.
- Capacity ≥100 mL predicts continence; template size and first closure success predict continence.

### Posterior urethral valves: embryology, management and valve bladder (SAQ)

- Type I (95%): abnormal Wolffian duct insertion; type III: persistent urogenital membrane; type II not real.
- Urorectal septum divides the cloaca at weeks 5–6.
- COPUM: one membrane torn into leaflets by catheterisation.
- Dysplasia is fixed in utero; ablation cannot reverse it.
- Catheter, stabilise, then primary valve ablation; vesicostomy if not possible.
- Nadir creatinine below 0.7 mg/dL predicts good renal outcome.
- Valve bladder: polyuria, poor sensation, residual urine.
- Valve bladder: overnight drainage and CIC; overnight drainage breaks the cycle.
- Ureteric augmentation rather than bowel; optimise before transplant.

### Antenatal hydronephrosis, fetal uropathy and prenatal scan anomalies (LAQ)

- UTD: 1–3% of pregnancies; second commonest anomaly after cardiac.
- Anomaly scan 18–20 weeks; after 16 weeks amniotic fluid = fetal urine.
- ANH (antenatal hydronephrosis) = UTD; APD normal <4 mm before 28 weeks, <7 mm after, <10 mm postnatally.
- UTD A2–3 = APRPD ≥7 mm or any abnormal feature.
- Transient dilation 50–70% (commonest cause); UPJ obstruction commonest surgical cause; VUR is not predicted by grade.
- First postnatal US after 48 hours; A1: one scan after 32 weeks; A2–3 scan every 4 weeks.
- Risk-based care: P1 no prophylaxis or VCUG; prophylaxis if ureter >7 mm or P3; P3 prophylaxis, VCUG, MAG3.
- Keyhole + oligohydramnios = PUV; absent bladder = exstrophy or agenesis.

- ARPKD large bright kidneys; MCDK non-communicating cysts.
- Favourable urine: Na <100, Cl <90, osmolality <200, β2-microglobulin <6.
- PLUTO: survival up, renal function unchanged.

**Ultrasonography in neonates (SAQ)**

- First postnatal scan after 48 h, except suspected PUV; normal APD <10 mm.
- Echogenic cortex and prominent pyramids are normal in neonates.
- UTD: P1 APD 10 to <15 mm; P2 15 mm or more; P3 abnormal parenchyma or bladder.
- Keyhole sign + thick bladder = PUV; spinal US only up to 6 months.

**Discuss diagnostics and operative pediatric laparoscopy in urology (SAQ)**

- Diagnostic laparoscopy is the procedure of choice for non-palpable testis.
- Pressure ceiling 6–12 mmHg by age.
- Right common iliac at Veress entry; close port fascia. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

**Leak point pressures and ALPP (SAQ)**

- VLPP/ALPP: Pves at leak with Valsalva or strain, no detrusor contraction; start at 150 mL.
- ALPP low = weak sphincter; DLPP high = upper-tract danger.
- VLPP <60 ISD; 60–90 equivocal; ≥90 (>90) hypermobility.
- VLPP < CLPP; catheter size and volume alter it.
- DLPP >40: upper-tract danger; 85% upper-tract change (McGuire 1981).
- Lower DLPP is safer; no validated safe cut-off.

**Embryological development of urinary bladder and management of Hutch Diverticulum? (LAQ)**

- Urorectal septum weeks 4–6; vesicourethral canal → bladder; urachus at months 4–5.
- Hutch: herniation through the ureteric hiatus → VUR.
- Ureter entering the sac: VUR will not resolve.
- Secondary: treat the outlet first; else diverticulectomy + reimplantation.

**Development of the urethra, congenital urethral anomalies and management of epispadias (including isolated penopubic epispadias) (LAQ)**

- Whole male urethra is endodermal UGS; fold fusion complete at 16, urethra at 18 weeks.
- About 70% of male epispadias are penopubic and incontinent; VUR 30–40%.
- Diastasis over 3 cm (>3 cm) → exstrophy pathway.
- Cantwell–Ransley at 6–12 months; YDL (Young–Dees–Leadbetter) at about 100 mL capacity.
- Continence after YDL: males 47–82%, females 70–87%.

**Prune belly syndrome (LAQ)**

- Triad; 95% male; dysplasia and lungs decide survival.
- Ureter is a failed pump (collagen); VUR about 75%; keep the proximal ureter.
- Woodard I–III; vesicostomy for drainage; orchidopexy at about 6 months.
- Nadir creatinine <0.7 mg/dL is favourable; up to 30% reach renal failure.

**Vesicostomy. (SAQ)**

- Blocksom: dome-based stoma midway between umbilicus and pubis.
- Bladder keeps cycling; compliance preserved.

- Stenosis and prolapse are the main complications.

**Anterior urethral valve (SAQ)**

- 25–30 times rarer than PUV; diverticulum common.
- Penile bulge on voiding; diagnosis by VCUG.
- Endoscopic ablation usually definitive; 78% keep normal renal function. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

**Paediatric neurogenic bladder (myelodysplasia): assessment, management and bladder myoplasty (SAQ)**

- Level does not predict bladder: UDS within 3 months.
- DLPP 40 cm $H_2O$ or more = hostile (McGuire 1981); early CIC + antimuscarinic.
- Early CIC plus antimuscarinic protects kidneys.
- MOMS: fewer shunts, no bladder benefit.
- LDDM: free latissimus to inferior epigastric vessels and intercostal nerve; for acontractile bladder.
- Investigational; clean intermittent catheterisation remains standard.
- Autoaugmentation: detrusor excised, mucosa bulges; little lasting gain.

**Voiding dysfunction in children: neurophysiology and differential diagnosis of frequency/urgency (LAQ)**

- S2–S4 pelvic (voiding), T10–L2 hypogastric (storage), Onuf/pudendal (EUS).
- PMC = switch; medial prefrontal cortex holds it off.
- EBC = (age + 2) × 30 mL; staccato = dysfunctional voiding.
- DLPP ≥40 cmH2O = hostile; treat bowel first.
- Always ask about bowels; treat constipation first.
- Rule out UTI, PUV, meatal stenosis and occult dysraphism.
- Extraordinary daytime frequency: no nocturia, normal tests, self-limiting.

**Nocturnal enuresis (SAQ)**

- ≥5 years; secondary = after 6 months dry.
- Alarm (durable) and desmopressin (rule of thirds) are first line.
- Treat daytime symptoms and constipation first.

**Management of Defecation Disorders resulting into LUTS (SAQ)**

- Treat the bowel first.
- Rome IV: ≥2 criteria for 1 month (child).
- PEG disimpaction 1–1.5 g/kg/day for 3–6 days. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

**CSIC. (SAQ)**

- Lapides: overdistension, not bacteria, causes infection; clean is enough.
- 4–6 times daily, under about 500 mL each.
- Do not treat asymptomatic bacteriuria.
- Mitrofanoff when urethral access is hard.

SECTION 6

# Female Urology

## 101 Surgical, Radiographic and Endoscopic Anatomy of the Female Pelvis

### 1 · Perineum and external genitalia

- The meatus lies **1 cm** anterior to the vaginal opening.
- **Skene's (paraurethral) glands** sit beside the meatus and are the prostate homologue; obstructed periurethral glands form a diverticulum or cyst.
- **Bartholin's gland** lies under the posterior end of each vestibular bulb and may form a cyst or abscess.
- The **deep perineal space** holds the urethral sphincter, urethrovaginal sphincter, compressor urethrae and deep transverse perineal; the **superficial space** (under Colles fascia) holds the superficial perineal muscles, clitoris, vestibular bulbs and Bartholin's glands.
- The perineal muscles and external anal sphincter converge at the **perineal body**, so episiotomy is angled laterally to spare the sphincter.
- The **labia majora have a dual supply** (superficial external pudendal from above, posterior labial from below), so the **Martius flap** is reliable.

### 2 · Bony pelvis and ligaments

- The **ischial spine** marks apical sutures.

| Fixation point | Used for | Adjacent danger |
|---|---|---|
| **Sacrospinous ligament (SSL)**: ischial spine to lower sacrum/coccyx | SSL fixation (apical prolapse) | Pudendal nerve/vessels near the spine; sacral plexus just posterior |
| **Cooper's ligament** (medial superior pubic ramus) | Burch colposuspension for stress urinary incontinence (SUI) | Obturator vessels; aberrant obturator |
| **Obturator foramen**; canal superolateral | Transobturator tape (TOT) trocar | Obturator nerve/vessels, causing groin pain or bleeding |
| **Anterior longitudinal ligament (ALL)** over S1 | Sacrocolpopexy mesh | Presacral veins, middle sacral vessels, left common iliac vein, right ureter |
| **Uterosacral ligament** | Uterosacral (apical) suspension | Ureter (lateral, closest at the cervix); sacral nerve entrapment |

- The pudendal nerve leaves through the **greater** sciatic foramen and re-enters through the **lesser** to reach Alcock's canal.

## 3 · Pelvic floor muscles

- The pelvic diaphragm is the **levator ani** and coccygeus; the levator is supplied directly by **S3–S5 (mainly S4)**.

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### LAQ Female pelvic floor anatomy and pelvic organ prolapse: evaluation and surgical repair

Paper answer · P1 · P2 · merged from 3

*Asked as:* P1 LAQ "Female pelvic floor anatomy: prolapse, SUI surgery and fertility-sparing treatment of procidentia" · P1 SAQ "Describe clinical presentation of pelic floor prolapse. diagnostic evaluation and brief discussion of various surgical options." · P2 SAQ "Vaginal prolapse repair."

#### Introduction and definitions

- Assumed meanings: "procedentia" is read as procidentia (complete uterine prolapse); "fertility sparing" / "fertility preserving" means uterus-preserving treatment for a woman who wants future pregnancy; "pelic floor prolapse" and "vaginal prolapse repair" mean pelvic organ prolapse (POP) in women and the surgical repair of vaginal wall and vault prolapse.
- **SUI (stress urinary incontinence):** involuntary leakage on effort, coughing or sneezing.
- **POP (pelvic organ prolapse):** descent of the anterior wall, posterior wall, uterus/cervix or vault. Anterior is commoner than posterior, which is commoner than apical. It is a disease only if symptomatic: a quality-of-life condition.
- Pelvic organs are supported by the **levator ani**, the **endopelvic fascia and its ligaments** and the **perineal membrane and perineal body**, anchored to the **bony pelvis**. Failure causes POP and SUI. Every SUI operation uses one of these anchors.

#### Bony pelvis and surgical anchors

- **Bony landmarks** and dangers: **sacrospinous ligament** (pudendal bundle), **Cooper's ligament** (bleeding), **sacral promontory** (left common iliac vein, right ureter).
- **Sacrospinous ligament (SSL)**, 7–8 cm, ischial spine to sacrum: the pudendal nerve and vessels pass behind the spine, the sacral plexus just posterior.
- **Sacral promontory:** left common iliac vein and right ureter each about 2.7 cm away.

#### Pelvic floor muscles: pelvic diaphragm (levator ani + coccygeus)

| Muscle | Attachment | Function |
|---|---|---|
| Puborectalis | Back of pubis, loops (as a sling) behind the anorectal junction | Anorectal angle; anorectal sling |
| Pubococcygeus (pubovaginal part) | Posterior pubis and arcus tendineus to viscera (vagina, perineal body, anus) and anococcygeal raphe | Slings urethra and vagina; closes hiatus, lifts urethra |
| Iliococcygeus | ATLA (arcus tendineus levator ani), ischial spine to raphe, coccyx | Horizontal **levator plate** |
| Coccygeus (not levator; separate) | Ischial spine to sacrum, coccyx | Lies on sacrospinous ligament |

- Levator tone (tonic contraction) closes the **urogenital hiatus**, through which all POP passes; the uterus and upper vagina rest on the levator plate.

- Levator is palpable through the lateral vaginal walls for PFMT (pelvic floor muscle training) assessment.
- **Muscle fails first, fascia second:** once injury (birth injury, avulsion) opens the hiatus, ligaments carry the load and fail; the fascia then carries the whole load and stretches.
- Levator defects in up to **20% of primiparous women**; avulsion predicts advanced prolapse.

### Endopelvic fascia and ligaments

- **Cardinal ligament** (side wall to cervix; carries the uterine artery) and **uterosacral ligament** (sacrum to cervix): the **cardinal–uterosacral complex** suspends the apex.
- The ureter is 4 cm from the uterosacral ligament near the sacrum but 0.9 cm near the cervix.
- **Pubocervical (pubovesical) fascia** attaches laterally to the **ATFP (arcus tendineus fasciae pelvis)**, the "white line" from the lower pubis to just above the ischial spine.
- **Rectovaginal fascia** supports the posterior wall.
- **Pubourethral ligaments** suspend the urethra from the pubis and anchor the mid-urethra; **urethropelvic ligaments** anchor it to the ATFP.

### Perineal membrane, perineal body and urethra

- The perineal membrane spans the pubic arch and is pierced by the urethra and vagina; above it lies the striated urogenital sphincter. The perineal body anchors the distal vagina.
- **Striated urogenital sphincter:** sphincter urethrae (mid-urethra), **compressor urethrae** and **urethrovaginal sphincter**.
- Urethra 4–5 cm; the continence zone is the **proximal and mid-urethra**. The **oestrogen-sensitive vascular submucosa** gives up to about 30% of closure force.

### DeLancey's levels of support (1992)

| Level | Region | Structures | Loss produces |
|---|---|---|---|
| **I – suspension** | Cervix/upper third | Cardinal–uterosacral complex | **Uterine or vault prolapse**, enterocele |
| **II – attachment** | Middle third | Paravaginal attachment to ATFP, levator fascia | Cystocele (central or paravaginal), rectocele; anterior or posterior wall prolapse |
| **III – fusion** | Distal third | Urethra, perineal membrane, perineal body | **Urethral hypermobility**, perineal descent, distal rectocele |

- **Central cystocele:** midline weakness, rugae lost. **Paravaginal cystocele:** ATFP detachment (Richardson 1981), rugae preserved.
- Level I is the keystone: always correct the apex.

**Diagram to draw:** sagittal female pelvis with levator plate and DeLancey levels I–III, with sacrospinous and sacrocolpopexy fixation points; hammock cross-section.

### Innervation

- **Pudendal nerve (S2–S4):** sphincters and perineal muscles; runs through Alcock's canal near the ischial spine; stretched in delivery.
- **Nerve to levator ani (S3–S5):** runs on the muscle's pelvic surface, separate from the pudendal; neuropathy in about a quarter of primiparous women.

- Autonomic pelvic splanchnic and hypogastric nerves: injured in radical pelvic surgery.

### Theories of continence and the operations they produced

| Theory | Mechanism | Surgical answer |
|---|---|---|
| **Pressure transmission (Enhorning 1961)** | Hypermobile urethra drops out of the abdominal pressure zone | Restore position: Burch, MMK (Marshall–Marchetti–Krantz) |
| **Hammock (DeLancey 1994)** | Urethra lies on anterior vaginal wall and endopelvic fascia attached to ATFP and levator; abdominal pressure compresses it against this firm backing; repositioning not essential | Restore backing |
| **Integral theory (Petros & Ulmsten 1990)** | Pubourethral ligament laxity at the **mid-urethra** | Tension-free mid-urethral sling |
| **Intrinsic sphincter deficiency (ISD) (McGuire 1978)** | Urethra fails to close | Sling, bulking, AUS (artificial urinary sphincter) |

- ISD: VLPP (Valsalva leak point pressure) <60 $cmH_2O$ or maximal urethral closure pressure <20 $cmH_2O$. Most SUI includes some ISD. So slings work in hypermobility, but Burch does poorly in severe ISD with a fixed urethra.

### Why prolapse occurs: risk factors (Bump & Norton)

- Muscle fails first (birth trauma, levator avulsion, denervation). The hiatus widens and the ligaments carry the whole load and stretch.
- **Predisposing:** genetics, collagen disorders (Ehlers–Danlos, Marfan).
- **Inciting: vaginal delivery** (strongest; forceps worse), parity, hysterectomy.
- **Promoting:** obesity, constipation, cough.
- **Decompensating:** ageing.

### Clinical presentation of prolapse

- **Vaginal bulge** (most specific), pressure, heaviness, worse by evening.
- **Splinting/digitation** to void or defecate (high-grade POP).
- Urinary: voiding difficulty (urethral kinking), OAB (overactive bladder); SUI in about 40%, which may be **occult**.
- Bowel (rectocele): obstructed defecation, stool trapping.
- Sexual dysfunction, discharge, ulceration; procidentia may cause hydronephrosis.

### Examination

- Lithotomy (supine) and **standing** with Valsalva / strain; Sims or split speculum for each compartment; lateral sulci (paravaginal defect); rectovaginal examination.
- **Cough stress test**, repeated with prolapse reduced (**occult SUI**); **Q-tip** angle change >30° = hypermobility; levator strength and avulsion.

### POP-Q (Pelvic Organ Prolapse Quantification, Bump 1996)

- Nine points in cm relative to the **hymen (0)**: inside (above) negative, outside positive; measured on maximal strain, recorded on a 3 × 3 grid.

| Point | Definition |
|---|---|
| Aa / Ap | Anterior / posterior wall 3 cm inside hymen (−3 to +3) |
| Ba / Bp | Most distal point of anterior / posterior wall (−3 if no prolapse) |
| C | Cervix or vault cuff |
| D | Posterior fornix (omitted after hysterectomy) |
| gh, pb, tvl | Genital hiatus, perineal body, total vaginal length (at rest) |

| Stage | Most distal point (leading edge) |
|---|---|
| 0 | No prolapse |
| I | >1 cm above hymen (< −1 cm) |
| II | Within ±1 cm of hymen (−1 to +1 cm) |
| III | >1 cm below hymen (> +1 cm) but < (tvl − 2) cm |
| IV | Complete eversion (≥ tvl − 2) |

- **Also know:** Baden–Walker grades 0–4; stage I–II often normal (POSST study).

### Investigations

- Urinalysis, PVR (post-void residual), bladder diary.
- **Cough test or urodynamics with prolapse reduced** for occult SUI; urodynamics also if ISD or mixed symptoms.
- Renal ultrasound in procidentia (hydronephrosis).
- Dynamic MRI or translabial ultrasound if complex or recurrent.
- A pessary trial confirms that the POP causes the symptoms.

### Principles of prolapse repair

- Operate for bother, not anatomy. The first operation gives the best chance.
- Identify each compartment by POP-Q. **The apex is the cornerstone:** failing to support it greatly raises recurrence; uncorrected level I loss causes recurrent prolapse after SUI repair.
- **Three categories: restorative** (native tissue), **compensatory** (graft or mesh, for weak tissue or failed repair), **obliterative**.
- Vaginal route: fewer complications, quicker recovery. Abdominal route: more durable.
- Screen for **occult SUI** with prolapse reduced and consider a concomitant sling (OPUS, CARE trials). Finish with **cystoscopy** to confirm ureteric efflux.

### Surgical options by compartment

| Compartment | Operations | Key points |
|---|---|---|
| **Anterior** (cystocele) | **Anterior colporrhaphy** (midline plication of pubocervical fascia); **paravaginal repair** (fascia reattached to the ATFP) | Central versus lateral defect; does not support the apex |
| **Apical, vaginal** (uterus, vault) | **Uterosacral suspension** (sutures; ureter injury 1–11%; cystoscopy); **sacrospinous fixation** (suture 1.5–3 cm medial to the ischial spine, avoiding the pudendal bundle; gluteal pain 15%); iliococcygeus suspension; McCall culdoplasty; vaginal hysterectomy; uterus-sparing hysteropexy; Manchester | OPTIMAL trial: uterosacral = sacrospinous at 2 years |

| Compartment | Operations | Key points |
|---|---|---|
| **Apical, abdominal** (compensatory) | **Sacrocolpopexy** (open, laparoscopic, robotic): macroporous polypropylene mesh from vaginal apex to the anterior longitudinal ligament over S1 / sacral promontory | Most durable (>90%); mesh exposure about 3–10% (P2: about 10% long term, CARE) |
| **Posterior** (rectocele) | **Midline fascial plication** without levatorplasty; site-specific repair; perineorrhaphy (levels II–III) | Levator plication causes dyspareunia; constipation may persist |
| **Obliterative** | **Le Fort** (partial, uterus in situ, lateral channels) or total colpocleisis | Elderly, not sexually active; success >90%; coital regret ≤13% |

- Cochrane 2016: vaginal apical repair has more recurrence and dyspareunia than sacrocolpopexy, but suits many older women.
- **April 2019:** the US FDA (Food and Drug Administration) ordered transvaginal synthetic mesh for (anterior) POP withdrawn; slings unaffected.
- Anterior colporrhaphy is not an SUI operation.

### Importance of anatomy in SUI surgery

| Operation | Anatomy used | Anatomical hazards |
|---|---|---|
| **Burch colposuspension** | Space of Retzius (avascular: symphysis in front, bladder behind, endopelvic fascia below). Paravaginal fascia at mid-urethra and bladder neck sutured to **Cooper's (iliopectineal) ligament** | Bleeding from Cooper's ligament and perivesical veins; ureteric kinking (cystoscopy); needs a mobile vagina |
| Paravaginal repair / VOS (vagino-obturator shelf) | Lateral vaginal sulcus to **ATFP** or obturator fascia (level II) | Obturator vessels and nerve |
| **Retropubic MUS (mid-urethral sling), TVT (tension-free vaginal tape; Ulmsten 1995)** | Tape under mid-urethra (integral theory), through **endopelvic fascia** and Retzius; suprapubic exits about 2 cm from midline | Bladder perforation (4.5%), vessels, bowel |
| **TOT (transobturator tape; Delorme 2001)** | Through obturator foramen / membrane, **obturator internus and externus**; exit at adductor longus tendon notch | Groin pain (6.4%), obturator neurovascular bundle; bladder injury rare (0.6%) |
| **SIMS (single-incision mini-sling)** | Anchors in obturator internus or endopelvic fascia | Works by restricting mobility |
| **Autologous PVS (pubovaginal sling)** | Rectus fascia at the **bladder neck** (proximal urethra), arms to rectus | Voiding dysfunction; best for ISD (compression) |
| Bulking agents | **Submucosa** coaptation | Repeated injections |

- **Why mid-urethra:** it is the zone of maximal closure pressure and pubourethral support. A loose tape lets the posterior urethral wall be compressed and kinked during stress.
- **TOMUS (2010):** retropubic 80.8% versus transobturator 77.7% objective success, not different.
- **SISTEr:** autologous sling more successful than Burch at 2 years, with more voiding dysfunction.

- **Also know:** needle suspensions abandoned (sutures pulled through).

**Diagram to draw:** cross-section of urethra on the vaginal hammock (DeLancey) with Burch, bladder-neck sling and mid-urethral sling sites.

### Procidentia: definition and assessment

- **Procidentia:** complete uterine (or vault) prolapse with the uterus wholly outside the hymen, **POP-Q stage IV**.
- Assess all compartments, occult SUI with prolapse reduced, PVR, hydronephrosis (ureteric kinking at the hiatus), cervical length and ulceration.
- Abnormal bleeding or cervical pathology needs gynaecological evaluation first, because it may argue against keeping the uterus.

### Fertility-sparing treatment of procidentia

**Principle:** restore **level I** support while keeping uterus and cervix. Defer definitive surgery until the family is complete where possible, because pregnancy and delivery cause recurrence.

**1. Conservative (first line)**

- PFMT, weight loss, treat constipation.
- **Pessary:** space-filling type (Gellhorn, cube) for procidentia; removed for conception; review about 6-monthly for discharge and erosion.

**2. Uterus-preserving surgery**

| Operation | Technique | Fertility points |
|---|---|---|
| **Sacrospinous hysteropexy** (vaginal) | Cervix sutured into the SSL **1.5–3 cm medial to the ischial spine** | Non-inferior to vaginal hysterectomy with uterosacral suspension (SAVE U trial); later data suggest superiority. Buttock pain, pudendal injury |
| **Uterosacral hysteropexy** (vaginal or laparoscopic) | Shorten and reattach uterosacral ligaments to cervix, sutures lateral to medial | No mesh; cystoscopy for ureteric kinking |
| **Sacrohysteropexy** (open, laparoscopic, robotic) | Mesh from posterior cervix to anterior longitudinal ligament over the promontory | Most durable apical support; later delivery usually by caesarean |
| **Manchester (Fothergill)** | Cervical amputation, cardinal ligaments sutured in front of the stump, anterior colporrhaphy | Suits elongated cervix, but risks **cervical stenosis, incompetence, miscarriage, preterm labour**: avoid if pregnancy wanted |

- **Also know:** Indian abdominal (cervicopexy) slings for young nulliparous prolapse (Shirodkar, Purandare, Khanna, Virkud). **Le Fort colpocleisis** keeps the uterus but not fertility (elderly only).

**3. Choosing** `Procidentia, wants pregnancy -> PFMT + pessary -> fails -> sacrospinous hysteropexy or laparoscopic sacrohysteropexy (avoid Manchester) -> recurrence after childbearing -> hysterectomy + vault suspension or sacrocolpopexy`

### Complications and follow-up

- Bladder, ureter or rectal injury; retention; de novo SUI; dyspareunia.

- Recurrence (mostly posterior overall; anterior after sacrospinous fixation).
- Mesh exposure: 3.4–5.4% after sacrocolpopexy in meta-analyses (P1 SAQ: about 3–10%; P2: about 10% long term, CARE); ureteric kinking.
- The retained uterus needs cervical screening.

**Key points to remember:**

- Levator ani = puborectalis, pubococcygeus, iliococcygeus; S3–S5.
- DeLancey: I suspension (uterosacral–cardinal; failure causes uterine prolapse), II attachment (ATFP), III fusion (urethral hypermobility); level I is the keystone: always restore apical support.
- Enhorning 1961, DeLancey 1994, Petros & Ulmsten 1990, McGuire ISD 1978.
- Integral theory: mid-urethral sling; hammock: backing, not repositioning.
- Burch to Cooper's ligament; TOT through obturator foramen; PVS at bladder neck.
- Bulge is most specific; splinting means high grade.
- Central cystocele rugae lost, paravaginal preserved; POP-Q stage II = −1 to +1 cm (±1 cm of hymen).
- Sacrocolpopexy most durable; colpocleisis for frail elderly.
- Cystoscopy after repair; screen for occult SUI.
- Pessary first; sacrospinous hysteropexy 1.5–3 cm medial to the ischial spine.
- Manchester amputates the cervix: poor choice for future pregnancy.

**Also, from the condensed book (not in the paper answer):**

- The **levator hiatus** transmits the urethra, vagina and rectum, and the **levator plate** is a horizontal visceral shelf; injury leaves the hiatus lax, the plate tilts and **prolapse** follows.

## 4 · Endopelvic fascia and visceral ligaments

- The **intermediate** fascial stratum (parametrium, paracolpium, pubocervical fascia) **gives most organ support**.
- The **pubourethral ligaments** (puboprostatic homologue) and anterior vaginal wall form a mid-urethral hammock; weakness causes SUI.
- A vesicopelvic fascia defect at the arcus tendineus gives a paravaginal cystocele.
- The **cardinal (Mackenrodt's) ligament** carries the pelvic plexus, so dividing it at hysterectomy can cause bladder dysfunction.
- Distinguish the **infundibulopelvic ligament** (ovarian vessels, ureter beneath) from the **ovarian ligament** (ovary to uterus) MCQ.

### DeLancey levels of vaginal support †

*See the paper answer **Female pelvic floor anatomy and pelvic organ prolapse: evaluation and surgical repair** (Section 6, Ch 101).*

**KEY POINT**

Support ligaments are connective tissue condensed around vessels and nerves, so the best anchors (uterosacral, cardinal) are exactly where sutures injure the ureter, pelvic plexus and sacral nerves. See Ch 108

## 5 · Urethra and continence anatomy

- The urethra is **4.7–5.1 cm** long and hangs from the posterior pubourethral ligaments on the anterior vaginal wall.
- **The mucosa and spongy vascular submucosa generate most closure pressure and are oestrogen-dependent**, which is the rationale for vaginal oestrogen.
- The **internal sphincter** is smooth and autonomic at the bladder neck; the **external sphincter** (sphincter urethrae, **compressor urethrae**, **urethrovaginal sphincter**) is striated, voluntary (pudendal) and covers the distal two-thirds.
- **Hypermobility** is one cause of SUI. An autologous pubovaginal sling sits at the **bladder neck**; a synthetic mid-urethral sling (MUS) sits at the **mid-urethra**.

## 6 · Bladder and ureter

*The ureter is the structure most often injured.*

- The **dome** is peritonealised, so intraperitoneal injury needs formal repair. Injury to the **base** causes vesicovaginal or vesicouterine fistula.
- The **pelvic ureter** crosses the iliac bifurcation, passes behind the ovary, runs **under the uterine artery** ("water under the bridge"), then crosses the anterior fornix to the bladder.

**ALGORITHM**

1. **Ureteric danger points**:
2. Pelvic brim, at the infundibulopelvic ligament.
3. Under the uterine artery (hysterectomy).
4. Beside the cardinal/uterosacral ligaments near the cervix.
5. Anterior vaginal fornix (vaginal surgery).
6. Sacral promontory: the right ureter (sacrocolpopexy).

## 7 · Nerves at surgical risk

- The **pudendal nerve (S2–S4)** crosses the SSL **close to the ischial spine** into Alcock's canal; pudendal block is given transvaginally just behind the spine.
- In SSL fixation, place sutures about 2 cm medial to the spine, through (not around) the ligament †.

| Nerve | Procedure / mechanism | Deficit |
|---|---|---|
| Common peroneal | Lithotomy: compression at the fibular head | Foot drop |
| Sacral plexus (L4–S4) | SSL fixation (plexus exits just posterior to the SSL) | Posterior leg motor/sensory loss |
| Pudendal | SSL fixation near the spine | Perineal pain, sensory loss, sphincter dysfunction |
| Sacral trunks S1–S4 | Uterosacral suspension (deep sutures) | Buttock/posterior thigh pain |
| Ilioinguinal (L1), genitofemoral (L1–L2) | Hernia repair, caesarean, hysterectomy | Mons/labial numbness or pain |

| Nerve | Procedure / mechanism | Deficit |
|---|---|---|
| Obturator (L2–L4) † | TOT trocar, obturator lymphadenectomy | Groin pain, adductor weakness |
| Posterior femoral cutaneous (S1–S3) | Gluteal pressure, pelvic tumour | Perineal sensory loss |

**KEY POINT**

Post-lithotomy foot drop is peripheral (fibular head), not pelvic. Leg pain and swelling after long lithotomy suggest **compartment syndrome**, an emergency († fasciotomy).

## 8 · Vessels and lymphatics

| Internal iliac | Anterior division | Posterior division |
|---|---|---|
| Visceral | Obliterated umbilical, superior vesical, uterine, vaginal, middle rectal | None |
| Parietal | Obturator, internal pudendal, inferior gluteal | Iliolumbar, lateral sacral, superior gluteal |

- The **internal iliac** has rich collaterals, so ligation or embolisation is tolerated; ligating the **external iliac** threatens the limb.
- The **presacral plexus** is valveless and under high pressure, and a 1–2 mm vein can bleed more than **1 L/min**.

**ALGORITHM**

1. **Sacrocolpopexy**
2. Expose the ALL at the promontory over the L5/S1 disc.
3. Open the peritoneum distally.
4. Suture into the ALL over **S1** (strongest; less discitis).
5. Avoid the left common iliac vein, middle sacral vessels, presacral plexus and right ureter (each about 2.7 cm from the mid-promontory).

| Organ | First nodes |
|---|---|
| Distal ⅓ urethra, lower ⅓ vagina, vulva | Inguinal, then pelvic |
| Proximal ⅔ urethra, upper ⅔ vagina | Iliac, obturator (pelvic) |
| Bladder, uterus | Perivisceral, then pelvic, presacral and great-vessel nodes |
| Ovary MCQ | Along the gonadal vessels to para-aortic nodes (bypasses the pelvis) |

## 9 · Endoscopic anatomy

**ALGORITHM**

1. Distal urethra: glands, diverticular ostia, mesh, stones.
2. Mid-urethral coaptation (external sphincter).
3. Bladder neck coaptation.
4. Trigone and orifices (30° lens).
5. Survey the whole mucosa.

- Squamous metaplasia and cystitis cystica are benign, but biopsy if in doubt.

## 10 · Imaging

- **Translabial ultrasound** shows bladder-neck mobility, prolapse, and **mesh, MUS and bulking agents**.
- On **endoanal ultrasound**, an internal sphincter defect is a bright break in dark muscle; an external sphincter defect is a dark lesion in bright muscle.
- **T2-weighted MRI is best for urethral diverticulum** and separates other cysts such as Gartner, Skene and Bartholin cysts.
- **MR defecography** grades descent below the **pubococcygeal line** and is used for defaecatory dysfunction or symptoms out of proportion to examination.

# 108 Urinary Incontinence and Pelvic Prolapse: Epidemiology and Pathophysiology

## 1 · Definitions and terminology

*The type of UI decides treatment.*

- **Urinary incontinence (UI)** is complaint of involuntary urine loss (ICS/IUGA, 2010).
- Symptoms are non-specific, so combine them with signs and observations (diary, urodynamics [UDS]). **Bother drives treatment**: without quality-of-life impact, none may be needed.

| Type | Complaint of loss… | Urodynamic correlate / trap |
|---|---|---|
| **Stress UI (SUI)** | On effort, cough, sneeze | Urodynamic SUI: leak with abdominal pressure rise **and no detrusor contraction** |
| **Urgency UI (UUI)** | With urgency | **Detrusor overactivity (DO) not required**; DO incontinence (DOI) is a separate UDS diagnosis |
| **Mixed UI (MUI)** | With urgency and effort | Urge- or stress-predominant, or equal |
| Overflow UI | Frequent small-volume loss with retention | Underactivity or obstruction; high post-void residual (PVR) |
| Functional UI † | Impaired mobility/cognition, normal lower urinary tract | Frail elderly, long-term care |
| Continuous UI | Day and night, few voids | Suspect fistula |
| Nocturnal enuresis | During sleep | Differs from nocturnal UUI (wakes, cannot reach toilet) |

- **Urgency** is a sudden compelling desire that is hard to defer (pathological), whereas **urge** is normal desire.
- **Overactive bladder (OAB)** is urgency with or without UUI, usually with frequency and nocturia; it is **defined by urgency**. Frequency ≥ **8 voids/24 h** is abnormal once polyuria is excluded.

## 2 · Epidemiology of female UI

- Remission reaches about 12%/year and is **lowest for MUI**.
- **Pregnancy** is the commonest cause of new UI in young women (about **40%**, mostly SUI). UI at 3 months postpartum usually persists, and pelvic floor muscle training (PFMT) during and after pregnancy reduces it.
- **Hysterectomy** can cause or cure UI; vaginal and laparoscopic routes carry more risk than open.

## 3 · Risk factors for female UI

| Factor | Key point |
|---|---|
| **Ageing** | Strongest independent non-modifiable factor. SUI peaks at 40–59 years; UUI/MUI dominate after 60. Over 40% in long-term care at admission |

| Factor | Key point |
|---|---|
| **Vaginal delivery** | Causes later **SUI, not UUI**. Caesarean section (CS) protects only if **every** delivery is by CS; number needed to treat (NNT) is **13** to prevent one moderate–severe UI. Elective CS protects more than CS in labour. Baby ≥4 kg: OR about 1.5 |
| **Parity** | First delivery OR about 1.5, then modest rise; strongest for SUI |
| **Obesity** | Body mass index (BMI) ≥30 more than doubles risk (SUI, MUI); **reversible** with weight loss. Little effect on SUI surgery success |
| **Menopause / oestrogen** | **Systemic oestrogen increases UI** (new UI nearly doubled vs placebo). Vaginal oestrogen is not linked to SUI and treats atrophy and recurrent UTI |
| Race | SUI commoner in White than Black/Asian women (higher closure pressure in Black women) |
| Smoking | Urgency, frequency, UUI (dose-dependent) |
| Caffeine | UUI, MUI, OAB, but not SUI |
| Diabetes | Up to 2×; neuropathy and macroalbuminuria are independent factors |

> **KEY POINT**
>
> Obstetric injury and loss of support cause **stress** UI; ageing, neurological, metabolic and lifestyle factors cause **urgency**. So the dominant type moves from stress to urge with age.

### Male UI (contrast)

- In men **UUI is commoner than MUI, which is commoner than SUI**. Male SUI is nearly always iatrogenic (radical prostatectomy [RP]) or neurological, and remission reaches **40%**.
- Post-RP UI is 8–60% depending on definition; after TURP it is about 1%.

## 4 · Physiology of continence

- During filling, **pelvic** parasympathetic outflow is suppressed, **hypogastric** sympathetic β-action relaxes the detrusor and α-action closes the bladder neck, and the **pudendal** nerve (Onuf's nucleus) keeps the sphincter closed.
- In **women**, the continence zone is the **proximal and mid-urethra**; the bladder neck is often incompetent in continent women. The oestrogen-sensitive spongy submucosa gives up to **30%** of closure.
- **Support**: the anterior vaginal wall is the firm backing. Lax vaginal support with intact pubourethral ligaments causes **shear** in the continence zone.
- **Male vs female**: men tolerate losing one outlet unit (TURP) because the bladder neck is powerful; in women the bladder neck is weak, so external sphincter deficiency causes UI.

> **KEY POINT**
>
> The female outlet relies on a **single** mid-urethral sphincter against a mobile backing, so continence needs support, intrinsic urethral properties and muscle together.

## 5 · General pathophysiology: bladder vs outlet

| Mechanism | Causes | Result |
|---|---|---|
| Loss of suprapontine inhibition | Stroke, multiple sclerosis (MS), Parkinson; early diabetes mellitus (DM) | Neurogenic DO, causing UUI |
| Outlet obstruction | Women: after anti-incontinence surgery; men: bladder outlet obstruction (BOO) | DO with new (de novo) UUI |
| Detrusor underactivity | Conus/cauda lesion, neuropathy (DM, alcohol), radical hysterectomy, abdominoperineal resection (APR), opioids including tramadol | Retention, then **overflow UI** |
| Poor compliance | Pelvic radiotherapy | High detrusor leak point pressure (LPP), causing UI |
| Sphincter deficiency | RP, urethral/anti-incontinence surgery, cord injury, α-blockers, muscle relaxants | Intrinsic sphincter deficiency (ISD), causing SUI |

- In **DM**, early polyuria and UUI give way to lost sensation and contractility, with poor emptying, UTI and overflow.
- In multiple system atrophy ≥20% leak after TURP.
- In post-RP UI, **ISD predominates (>70%)**, and DO or poor compliance contributes in >30%.
- Each failed operation adds fibrosis and ISD; **labour** causes fascial injury and **pudendal denervation**.

## 6 · Pathophysiology of female SUI

### LAQ Female SUI: continence mechanism, classification, evaluation (UDS) and management, including male SUI surgery

Paper answer · P3 · merged from 2

*Asked as:* P3 LAQ "Female pelvic floor, continence mechanism, classification and management of SUI" · P3 LAQ "SUI: evaluation in females, role of UDS, and surgical therapy in males"

#### Definition and mechanism

- **SUI (stress urinary incontinence)**: involuntary leakage on effort, exertion, coughing or sneezing (ICS, International Continence Society).
- Continence needs **urethral pressure greater than bladder pressure** at rest and during stress.
- Support comes from the **levator ani, endopelvic fascia and ligaments, perineal membrane and perineal body**.
- Two mechanisms: **urethral hypermobility** (loss of the suburethral support, the DeLancey "hammock") and **ISD (intrinsic sphincter deficiency)**. Most women have some ISD.

#### Anatomy of the female pelvic floor

**1. Pelvic diaphragm**

| Muscle | Attachment | Function |
|---|---|---|
| Puborectalis | Back of pubis, loops behind anorectal junction | Anorectal angle |
| Pubococcygeus (pubovaginal) | Pubis, ATFP (arcus tendineus fasciae pelvis) to viscera, raphe | Slings urethra and vagina |
| Iliococcygeus | Arcus tendineus levator ani, ischial spine to coccyx | Horizontal **levator plate** |
| Coccygeus (not levator) | Ischial spine to sacrum, coccyx | Lies on sacrospinous ligament |
| - Nerve to levator ani **(S3–5)** on its pelvic surface. Resting tone closes the **urogenital hiatus**; widening of the hiatus or tilting of the plate causes prolapse. Muscle fails first, then fascia carries the load. | | |

**2. Perineal membrane and perineal body**

- The perineal membrane spans the pubic arch; urethra and vagina pierce it. Above it lies the **striated urogenital sphincter**: sphincter urethrae (mid-urethra), **compressor urethrae** and **urethrovaginal sphincter**. The perineal body anchors the distal vagina.

**3. Endopelvic fascia and ligaments**

- **Pubocervical fascia** attaches laterally to the **ATFP** (pubis to ischial spine); **rectovaginal fascia** posteriorly; **cardinal–uterosacral complex** at the apex; **pubourethral ligaments** anchor the mid-urethra.

**4. DeLancey levels (1992)**

| Level | Structures | Defect |
|---|---|---|
| I suspension | Cardinal–uterosacral | Apical prolapse, enterocele |
| II attachment | ATFP, levator fascia | Cystocele (central or paravaginal), rectocele |
| III fusion | Perineal membrane, perineal body | **Urethral hypermobility**, perineal descent |

### Continence mechanism in the female

**1. Intrinsic urethral factors** (smooth and striated sphincters, vascular submucosa)

- **Mucosal seal**: folded urothelium coapts. The **oestrogen-dependent submucosal vascular plexus** provides up to **30%** of closure force.
- **Smooth muscle** (internal sphincter, bladder neck and proximal urethra): involuntary tone. The female bladder neck is weak and often open in continent women.
- **Striated urogenital sphincter**: the rhabdosphincter, mostly **slow-twitch** fibres, gives tonic mid-urethral closure. Distal parts are the **compressor urethrae** and **urethrovaginal sphincter**.
- The **continence zone is the proximal and mid-urethra.**

**2. Support (extrinsic)**

- The urethra lies on the **anterior vaginal wall and endopelvic fascia**, attached laterally to the **ATFP** and levator ani.

- **Pubourethral ligaments** anchor the mid-urethra; **levator ani** (pubococcygeus) tone keeps the backing firm.

### 3. Neural

- **Pudendal nerve (S2–4)** drives the sphincter (**guarding reflex**). Urethral pressure rises **just before** a cough reaches it, so closure is active, not just transmitted.

### 4. Theories of continence and SUI

| Theory | Mechanism | Surgical implication |
|---|---|---|
| **Pressure transmission (Enhorning 1961)** | Hypermobile urethra drops out of the abdominal pressure zone | Restore position (Burch) |
| **Hammock (DeLancey 1994)** | Abdominal pressure compresses the urethra against a firm suburethral layer; a lax layer fails | Restore backing |
| **Integral theory (Petros & Ulmsten 1990)** | Pubourethral ligament laxity at the mid-urethra | Tension-free MUS (mid-urethral sling) |
| **ISD (McGuire & Lytton 1978)** | Urethra itself fails to close (VLPP [Valsalva leak point pressure] <60, MUCP [maximum urethral closure pressure] <20 $cmH_2O$) | Sling, bulking, AUS (artificial urinary sphincter) |
| - Most SUI has some ISD. **Hypermobility alone never causes SUI.** | | |

## Classification of SUI

### 1. Blaivas & Olsson (1988), on video-urodynamics

| Type | Bladder neck at rest | On stress |
|---|---|---|
| **0** | Closed, above lower border of symphysis | Descends and opens, but no leak seen (history of SUI) |
| **I** | Closed, above symphysis | Descends **<2 cm**; leaks; little or no cystocele |
| **IIA** | Closed, above symphysis | Descends **>2 cm** with cystourethrocele; leaks |
| **IIB** | Closed but **at or below** lower border of symphysis | Proximal urethra opens; leaks with or without further descent |
| **III** | **Open at rest** without detrusor contraction | Leaks with minimal stress = **ISD** |

### 2. Functional: hypermobility versus ISD

| | Urethral hypermobility | ISD |
|---|---|---|
| Q-tip test | Angle change **>30°** | Often fixed urethra |
| VLPP / ALPP | **≥90 $cmH_2O$** | **<60 $cmH_2O$** (60–90 equivocal) |
| MUCP | Normal | **≤20 $cmH_2O$** |
| Cause | Childbirth, ageing, obesity | Previous surgery, radiation, neuropathy, ageing |

### 3. Other

- **Severity (Stamey):** grade 1 leak on cough or sneeze; grade 2 on minimal effort (walking, rising); grade 3 at rest.
- **Index (uncomplicated)** versus **complicated** SUI (recurrent, prolapse, neurogenic, high PVR [post-void residual], radiation); pure versus mixed incontinence.

## Evaluation of SUI in women

### History

- Leakage with cough or activity means SUI; after urgency, UUI (urgency urinary incontinence); both, MUI (mixed). A positive SUI history predicts urodynamic SUI in about three-quarters.
- Severity (pads/day), bother; voiding symptoms, prolapse, obstetric history, prior surgery or radiation, neurological disease, drugs.
- **Questionnaires:** ICIQ-UI SF (International Consultation on Incontinence Questionnaire–UI Short Form); UDI-6 (Urogenital Distress Inventory: bother); IIQ-7 (Incontinence Impact Questionnaire: impact). **Bladder diary** (3 days).

### Examination (comfortably full bladder)

- Neurological (S2–S4 sensation, anal tone, bulbocavernosus reflex); atrophy; prolapse by **POP-Q (Pelvic Organ Prolapse Quantification)**.
- **Cough stress test:** cough supine, then standing if negative. Leakage synchronous with cough confirms SUI. Reduce prolapse to unmask **occult SUI**.
- **Q-tip test:** swab to bladder neck, Valsalva; rotation **>30°** = hypermobility. Visual mobility examination is now routine.

### Tests

| Test | Method and use |
|---|---|
| Urinalysis | Required; exclude infection, haematuria |
| PVR | Required before surgery; ultrasound; repeat if high (no agreed cut-off) |
| 1-hour pad test (ICS) | 500 mL fluid, standard exercises; >1.4 g positive; 1–10 g mild, 11–50 g moderate, >50 g severe |
| 24-hour pad test | Normal activities; more reproducible |
| Cystoscopy | Haematuria, irritative symptoms, previous mesh |

## Role of urodynamics (UDS) in SUI

- UDS (urodynamic studies) assess storage and voiding. The key question in SUI is: **will the result change the operation?** Pure, demonstrable SUI gains little. **UDS is not routine.**

| Parameter | Method and meaning |
|---|---|
| USI (urodynamic stress incontinence) | Leak with raised abdominal pressure, no detrusor contraction |
| ALPP/VLPP (abdominal/Valsalva leak point pressure) | Standing, from 150 mL, Valsalva or cough; **<60 cm$H_2O$ = ISD**, 60–90 equivocal, ≥90 = little ISD (McGuire 1993) |
| MUCP | Urethral pressure profile; a low MUCP (≤20 cm$H_2O$) also suggests ISD; does not diagnose or grade SUI |
| Filling cystometry | DO (detrusor overactivity), compliance, capacity; cough-induced DO mimics SUI |
| Pressure–flow study | Voiding dysfunction, poor contractility (retention risk after sling) |
| Prolapse reduction | Occult SUI |
| - If no leak with the catheter in place, repeat with it removed. | |

### Evidence

- **VALUE (ValUE) trial (2012), 630 women** with uncomplicated SUI: office evaluation alone was **non-inferior** to office evaluation plus UDS before surgery (success 77.2% vs 76.9%).
- **Dutch VUE study:** immediate midurethral sling was non-inferior to UDS-tailored treatment.
- **SISTEr:** UDS did not predict outcome.

### Guidelines

- **AUA (American Urological Association)/SUFU (Society of Urodynamics, Female Pelvic Medicine and Urogenital Reconstruction) 2023:** UDS may be omitted in the index patient with demonstrable SUI.
- **UDS advised in non-index patients:** unclear diagnosis or SUI not demonstrated; mismatch between symptoms and findings; urgency-predominant MUI; neurogenic bladder; raised PVR or voiding dysfunction; prolapse stage ≥3 without demonstrable SUI; failed previous anti-incontinence or prolapse surgery.
- **EAU (European Association of Urology):** no routine UDS for uncomplicated SUI; do not use leak point or urethral pressure to grade severity.
- **Men:** before AUS or sling for post-prostatectomy incontinence, UDS confirms USI and finds DO, poor compliance or weak contractility. The MASTER trial recruited men with USI.

### Conservative management (first line)

- Weight loss, fluid and caffeine adjustment.
- **Supervised PFMT (pelvic floor muscle training)** for at least 3 months (Cochrane: more cure than no treatment).
- **Pessary**; vaginal oestrogen for atrophy.
- **Duloxetine** (serotonin–noradrenaline reuptake inhibitor): increases Onuf's nucleus drive to the sphincter; approved in Europe, not the USA.

### Surgical options for female SUI: principles

| Operation | Principle (how it works) | Best for | Key facts |
|---|---|---|---|
| **MUS**: retropubic (TVT, tension-free vaginal tape, Ulmsten 1995), transobturator (TOT; Delorme 2001; inside-out de Leval), single-incision | Tension-free type I polypropylene tape under the **mid-urethra** acts as a new pubourethral ligament; the urethra kinks against it during stress | First-line index SUI | **First line** (AUA/SUFU strong recommendation). **TOMUS**: retropubic 80.8% vs transobturator 77.7% cure (equivalent); retropubic often preferred in ISD. Retropubic: more bladder perforation; transobturator: more groin pain |
| **SIMS (single-incision mini-sling)** | Short tape, vaginal only | Option | Non-inferior to TOT for subjective cure in most studies; conditional recommendation (2023) |

| Operation | Principle (how it works) | Best for | Key facts |
|---|---|---|---|
| **Autologous pubovaginal sling (PVS)** (rectus fascia or fascia lata) | At the **bladder neck**, minimal tension; restores the hammock, giving bladder-neck support and compression | **All types, including ISD**, recurrent SUI, neurogenic SUI, failed MUS, after mesh removal, mesh-averse | Gold standard. **SISTEr**: sling beat Burch (47% vs 38%) but more voiding dysfunction and UTI (urinary tract infection) |
| **Burch colposuspension** (open, laparoscopic or robotic) | Paravaginal fascia beside the mid-urethra and bladder neck sutured to **Cooper's ligament**; lifts the bladder neck back into the pressure zone and gives a stable backing | Hypermobility; women declining mesh; at abdominal surgery | 85–90% at 1 year, about 70% at 5 years. Needs a mobile urethra; poor in ISD |
| Paravaginal repair, VOS (vagino-obturator shelf) | Vagina to ATFP or obturator fascia | | Flatter, less obstructive |
| **MMK (Marshall–Marchetti–Krantz)** | Bladder neck to symphysis periosteum | Historical | Osteitis pubis 0.7–2.5%; abandoned |
| **Urethral bulking agents** | Submucosal injection at bladder neck or proximal urethra improves coaptation | ISD with fixed urethra; frail patients; minimal surgery | Less durable; repeat injections |
| **AUS** | Bladder-neck cuff; constant circumferential closure | Severe refractory ISD, fixed urethra | Second-line; erosion, infection, revision |
| - **Also know:** needle suspensions abandoned (sutures pull through); anterior colporrhaphy is not an SUI operation; CARE (prophylactic Burch at sacrocolpopexy); Ward–Hilton (TVT = colposuspension). | | | |

**Burch steps (algorithm):** Pfannenstiel → space of Retzius → stay lateral to urethra → 2–4 sutures each side (mid-urethra, then bladder neck) → Cooper's ligament → tie loosely (two fingers behind pubis) → **cystoscopy** for ureteric efflux.

### Retropubic vs transobturator MUS

| | Retropubic | Transobturator |
|---|---|---|
| Bladder perforation | 4.5% | 0.6% |
| Voiding dysfunction (TOMUS surgery) | More (2.7% vs 0%) | Less |
| Groin pain | 1.3% | 6.4% |
| ISD, long-term durability | Often preferred | Possibly more repeat surgery |

### Complications of female SUI surgery

- **MUS complications:** perforation, retention, de novo urgency, mesh exposure (about 2%), pain.
- Bladder trocar injury: re-pass and continue (cystoscopy after every pass); **urethral injury: abandon mesh**.
- Voiding dysfunction (sling too tight or proximal): urethrolysis or sling incision; de novo urgency; UTI; haematoma.
- **Mesh complications** (IUGA/ICS [International Urogynecological Association/International Continence Society] terms): exposure, extrusion, perforation into bladder or urethra, pain; manage by excision. FDA (Food and Drug Administration) 2019 stopped transvaginal mesh for prolapse, not slings.

### Male SUI: surgical therapy

- **Aetiology:** commonest after **radical prostatectomy (RP)**, which removes the internal sphincter, leaving continence to the rhabdosphincter. Also TURP (transurethral resection of the prostate), radiation, pelvic fracture.
- **Work-up:** 24-h pad weight (mild <150 g, moderate <400 g, severe above); standing cough test; **cystoscopy** (anastomotic stricture, sphincter coaptation); UDS if mixed or poor contractility.
- **Timing:** PFMT first; no surgery for 6–12 months, as continence keeps improving (other P3 answer: continence improves for up to 2 years, but do not delay surgery beyond 6 months for severe leak).

| Device | Mechanism | Use and results |
|---|---|---|
| **Transobturator sling (AdVance XP)** | Repositions the bulb 3–4 cm proximally; needs residual sphincter | Mild–moderate leakage, no radiotherapy; cure about 60% |
| Quadratic sling (Virtue) | Compression plus elevation | Mild; limited data |
| Adjustable sling (ATOMS, adjustable transobturator male system) | Cushion adjusted via scrotal port | Mild–moderate; about two-thirds dry |
| ProACT balloons | Bladder-neck compression | More erosion |
| Bulking agents | Submucosal bulk | Small leaks, unfit men; rarely durable; not advised |
| **AUS** | Circumferential compression | **Gold standard**; all severities, moderate–severe, after radiation; about 80% continent |

### AMS 800 AUS

- **Components:** urethral **cuff** (bulbar, usually 4.0–4.5 cm); **PRB (pressure-regulating balloon) 61–70 cm$H_2O$** (51–60 high risk; 71–80 bladder neck) in a preperitoneal pocket; scrotal **pump**. Filled with 23 mL.
- **Technique:** perineal incision; cuff on the proximal bulbar urethra; **deactivate for 4–6 weeks**.
- **Needs** dexterity and cognition. Social continence (0–1 pad) 55–77%; satisfaction about 90%; lifespan 7–10 years.

- **Complications:** infection 1–3% (explant; reimplant after 3–6 months); **erosion** 2–15% (explant, catheter about 3 weeks); urethral **atrophy** (late recurrent leak); mechanical failure; retention.
- **Erosion risks:** radiation, prior erosion, urethroplasty; use a **transcorporal cuff** (tunica adds bulk) or lower PRB. Atrophy: smaller or tandem cuff.

**Algorithm (AUA/SUFU 2019)** `Severe SUI or any radiotherapy -> AUS Moderate, no radiotherapy -> AUS or male sling Mild, no radiotherapy -> sling (or PFMT, AUS) Sling failure -> AUS`

- **MASTER RCT (randomised controlled trial):** sling was **non-inferior** to AUS for continence at 12 months; AUS remains preferred for heavy leak (>250 g/day) or radiation.

### Diagrams to draw

- DeLancey levels and hammock cross-section.
- Sites of Burch sutures, bladder-neck sling and mid-urethral tape; sagittal pelvis showing retropubic and transobturator tape paths.
- Cystometrogram trace with abdominal pressure spike and leak (ALPP).
- AMS 800 (cuff, pump, PRB) and male transobturator sling path.

**Key points to remember:**

- Levator ani = puborectalis, pubococcygeus, iliococcygeus; S3–5.
- DeLancey: I suspension, II attachment (ATFP), III fusion.
- Continence zone is the mid-urethra; submucosal plexus gives up to 30% of closure.
- Blaivas types 0, I, IIA, IIB, III; type III = ISD.
- Q-tip >30° = hypermobility; VLPP / ALPP <60 $cmH_2O$ = ISD; MUCP ≤20 $cmH_2O$.
- Burch corrects position; slings and AUS correct function.
- VALUE (ValUE) trial: UDS not needed in uncomplicated (index) SUI.
- Use UDS when it changes treatment: mixed, failed surgery, voiding dysfunction.
- PFMT first; MUS first-line surgery.
- SISTEr (fascial sling > Burch); TOMUS (retropubic = transobturator).
- PVS for ISD; urethral injury → no mesh.
- Men: AUS gold standard; sling for mild, non-irradiated.
- AUS PRB 61–70 $cmH_2O$; deactivate 4–6 weeks.
- Radiation or severe male SUI: AUS first; MASTER: sling non-inferior.

**Also, from the condensed book (not in the paper answer):**

- Support does not explain everything: some women with marked hypermobility are dry, and some with a fixed urethra leak severely.

### Hypermobility vs ISD

- ISD equals **type III SUI** (Blaivas) MCQ and is usually seen after previous surgery.
- A **"pipestem" urethra** is fixed and non-coapting; subtler ISD coexists with hypermobility in most SUI.
- **Valsalva leak point pressure (VLPP)** is the abdominal pressure that causes leakage. **VLPP < 60 cm $H_2O$ means ISD**, historically treated by bulking or pubovaginal sling.

**KEY POINT**

**Slings work in hypermobility but Burch fails in ISD**. Almost all SUI includes some ISD, so sphincter-directed procedures (pubovaginal sling, mid-urethral sling [MUS], bulking) work in hypermobile women. Burch corrects only hypermobility, so it is weak with severe ISD and a fixed urethra.

## 7 · OAB pathophysiology (brief)

- Neurogenic causes (§5) and outlet obstruction (BOO, post-sling) both lead to DO and de novo urgency.
- With pelvic organ prolapse (POP), urgency is mainly due to obstruction from kinking. **OAB improves after POP repair in 60–80%**, more after anterior/apical repair, with no reliable predictor.

## 8 · Insensible incontinence

- It is usually ISD or DOI; if treatment fails, seek rarer causes. Post-void dribble suggests diverticulum (pelvic MRI), continuous loss with normal voids suggests ectopic ureter (upper-tract imaging), and frequent small losses suggest overflow (UDS and PVR).

## 9 · Pelvic organ prolapse: definition and staging

- **POP** is descent of the uterus/vault and/or vaginal walls from loss of support. **Name the vaginal segment** (anterior, apical, posterior), not the presumed organ.
- The most specific symptom is a **visible or palpable bulge**, which correlates best once past the hymen.
- Never attribute bleeding or pain to POP; relief with a **pessary trial** predicts benefit from surgery.

| POP-Q stage | Most distal point on maximal strain |
|---|---|
| 0 | Aa, Ba, Ap, Bp = −3; C/D ≤ −(tvl − 2) |
| 1 | > 1 cm above hymen (< −1) |
| 2 | Within 1 cm of hymen (−1 to +1) |
| 3 | > +1 but < +(tvl − 2) |
| 4 | Complete eversion: ≥ +(tvl − 2) |

*See the paper answer **Female pelvic floor anatomy and pelvic organ prolapse: evaluation and surgical repair** (Section 6, Ch 101).*

**Also, from the condensed book (not in the paper answer):**

- The stage is set by the single most distal point, and measuring right after pessary removal underestimates it.
- POP-Q is the IUGA/ICS-recommended system; Baden–Walker grades have poor interobserver reliability.

## 10 · POP epidemiology and risk factors

- Loss of support is seen in 43–90%, and **asymptomatic stage 1–2 is normal**; symptomatic POP affects 3–12%.

- **Anterior POP is commoner than posterior, which is commoner than apical**. Apical descent accompanies most significant anterior POP and is universal at stage 4. **So when planning cystocele repair, assess the apex and consider vault suspension.**
- POP progresses slowly; progression predictors are age >60, Ba ≥4 cm and genital hiatus (gh) ≥5 cm.

| Risk factor | POP | UI (contrast) |
|---|---|---|
| **Vaginal childbirth** | Strongest; rises steeply with parity; forceps worse. Elective CS risk is about nulliparous; CS in labour about vaginal | SUI, not UUI |
| Age | About 40% rise per decade; surgery peaks in the 7th decade | Shift to UUI/MUI |
| Obesity | About 1.4–2.5-fold | >2-fold, reversible |
| Hysterectomy | Raises risk, most if done for POP (no vault suspension) | Cause or cure |
| Race | Hispanic highest, Black lowest | White highest |
| Oestrogen / menopause | Not consistently associated | Oral oestrogen increases UI |
| Other | Family history (about 5× in siblings), connective-tissue disease, chronic constipation (2.5×) | — |

- Recurrence after surgery is predicted by preoperative stage 3–4.

## 11 · Pathophysiology of POP

- The **levator ani** is the primary support; the uterosacral–cardinal ligaments hold the organs in position over it.

*See the paper answer **Female pelvic floor anatomy and pelvic organ prolapse: evaluation and surgical repair** (Section 6, Ch 101).*

**Also, from the condensed book (not in the paper answer):**

- The commonest cystocele defect at surgery is detachment of the arcus tendineus fasciae pelvis (ATFP) from the ischial spine.

**ALGORITHM**

1. **POP sequence**
2. Labour causes levator trauma or denervation.
3. The muscle loses tone and the urogenital hiatus widens.
4. Fascia and ligaments now carry the whole load.
5. They stretch and fail (ageing, straining, hysterectomy): POP.
6. POP widens the hiatus further.

- MRI shows levator defects in up to **20%** of primiparas and none in nulliparas; **avulsion** predicts advanced, multicompartment POP.
- Levator neuropathy follows vaginal delivery or CS in labour but almost never elective CS, so **labour itself is the injury**.
- An altered collagen type I:III ratio and Ehlers–Danlos or Marfan syndrome increase risk.

## 12 · POP and UI interaction

*Prolapse can hide SUI until repair.*

- 40–55% of women with SUI have significant anterior POP, as do about 20% with OAB.
- **Occult SUI** is SUI seen only with the prolapse reduced (up to a quarter of advanced POP).

**ALGORITHM**

1. A cystocele kinks the urethra, and the obstruction masks SUI.
2. POP repair relieves the kink, so SUI is unmasked.
3. Detect it by UDS or stress test **with the prolapse reduced** (pessary or speculum).

- De novo SUI after POP surgery is predicted by age >66, DM and low urethral closure pressure.
- Examine for POP in every UI evaluation, because POP missed at sling surgery means later prolapse surgery.

## 13 · Landmark trials

| Trial | Design | Result | Take-home |
|---|---|---|---|
| CARE | Stress-continent women, abdominal sacrocolpopexy with or without Burch | Burch reduced postoperative SUI; SUI on UDS with prolapse reduced predicted it | Prophylactic Burch at sacrocolpopexy prevents de novo SUI |
| OPUS | Stress-continent women, vaginal POP repair with prophylactic MUS vs sham | UI at 1 year 27% vs 43% (NNT about 6); more bladder perforation, UTI, bleeding, retention | Fewer de novo SUI, more adverse events, so counsel |

# 105 Evaluation and Management of Women With Urinary Incontinence and Pelvic Organ Prolapse

## 1 · Definitions and risk factors

*Anatomical prolapse is common; disease means symptoms.*

- Risk factors for **urinary incontinence (UI)** are age, obesity, smoking, comorbidity and postmenopausal hormones; vaginal delivery causes stress UI (SUI).
- **Pelvic organ prolapse (POP)** is commonest anteriorly, then posteriorly, then apically. Half of parous women have anatomical POP, but disease means **symptoms attributable to it**. A **vaginal bulge** is the most specific symptom.

| Category | POP risk factors |
|---|---|
| Primary only | Older age, high body mass index (BMI), high birth weight, levator defects, low education, poverty |
| Recurrence only | **Younger age**, preoperative Pelvic Organ Prolapse Quantification (POP-Q) stage 3–4 |
| Both | Vaginal delivery, parity, levator hiatal ballooning |
| Protective (primary) | Caesarean delivery; **smoking** MCQ |

- **ICS UI terms** follow the trigger: SUI, urgency UI (UUI) and mixed UI (MUI). **Insensible** UI occurs without awareness, and **occult SUI** appears only after POP reduction.

## 2 · History

- Ask directly and exclude reversible causes with **DIAPPERS**: Delirium, Infection, Atrophic vaginitis, Psychological, Pharmacological, Excess urine output, Restricted mobility, Stool impaction.
- **SUI is urethral failure, UUI is bladder failure, and MUI is both**. In MUI, treat the more **bothersome** component first.
- **POP** gives a bulge and pressure that worsen later in the day. **Splinting** suggests high-grade POP. Anterior POP kinks the urethra and causes voiding difficulty; posterior POP obstructs defecation.
- SUI coexists in about **40%** of women with POP, and many expect repair to cure leakage, so counsel them.
- **Occult SUI** is found in about ⅓ before vaginal POP repair, and **de novo SUI** can appear after POP surgery; both need a shared decision on concurrent versus staged surgery.
- Ask about neurological disease, diabetes, radiation, pelvic or mesh surgery and obstetric trauma. **Local** oestrogen may improve UI and **systemic** oestrogen may worsen it.
- **Drugs**: **α-blockers worsen SUI in women**, and α-agonists, anticholinergics and opioids impair emptying. Diuretics, SGLT2 inhibitors and lithium cause polyuria. **ACE inhibitors cause cough, which worsens SUI**.

## 3 · Physical examination

- Atrophic (oestrogen-deficient) tissue is pale, flat, dry and without rugae.

### SUI testing

- AUA/SUFU 2023 says: **objectively demonstrate SUI with a comfortably full bladder before surgery**.

*See the paper answer **Female SUI: continence mechanism, classification, evaluation (UDS) and management, including male SUI surgery** (Section 6, Ch 108).*

**Also, from the condensed book (not in the paper answer):**

- The Q-tip test is now mostly for teaching and trials; **visual urethral mobility** (meatal rotation < or **>30°**) is routine.

### POP examination

- Examine in lithotomy **and standing**, while straining, to see the worst prolapse.
- Use a **split Graves speculum** to isolate each compartment in turn. If the blade is expelled with posterior foreshortening, there is **level I loss, meaning vault prolapse**; if it stays, there is an isolated rectocele or enterocele.

### POP-Q system

*See the paper answer **Female pelvic floor anatomy and pelvic organ prolapse: evaluation and surgical repair** (Section 6, Ch 101).*

**Also, from the condensed book (not in the paper answer):**

- Total vaginal length (tvl) is measured with the prolapse reduced, and the stage is set by the single most distal point.

| Stage | POP-Q criteria | Baden–Walker grade (by eye) |
|---|---|---|
| 0 | Aa, Ba, Ap, Bp −3; C or D ≤ −(tvl − 2) | 0: normal |
| I | Leading edge < −1 cm | 1: upper vagina to midplane |
| II | Leading edge −1 to +1 cm | 2: midplane to hymen |
| III | > +1 cm but < +(tvl − 2) | 3: through/beyond hymen |
| IV | ≥ +(tvl − 2): complete eversion | 4: complete eversion |

> **KEY POINT**
>
> Stage II is a 2-cm window around the hymen, and the III/IV boundary depends on tvl. Example: tvl 8, Bp +5 lies between +1 and +6, so **stage III**.

### Other elements

- The **bulbocavernosus reflex** (S2–S4) is present in only about **70%** of normal women, so its absence alone is not pathological.
- For **prior mesh**, IUGA/ICS terms distinguish extrusion (through vaginal epithelium) from perforation (into bladder, urethra or another viscus).

## 4 · Supplemental evaluation

*The uncomplicated patient needs few tests.*

**Urinalysis (UA) and post-void residual (PVR) are universal.** For the AUA/SUFU 2023 index patient having SUI surgery, **UA, PVR and objective SUI demonstration** are required;

cystoscopy, urodynamics (UDS) and imaging are not routine. In non-neurogenic OAB only UA is required (PVR optional).

*See the paper answer* ***Female SUI: continence mechanism, classification, evaluation (UDS) and management, including male SUI surgery*** *(Section 6, Ch 108).*

**Also, from the condensed book (not in the paper answer):**

- Unexplained haematuria or pyuria also makes a patient non-index.

## Symptom quantification

- A 3-day **bladder diary** is both diagnostic and therapeutic (see Ch 114), and its largest void equals functional capacity.
- Use the ICIQ-UI SF for UI and the **PFDI / PFIQ** for POP.
- **Pad tests** are mainly for research. In the ICS 1-hour test, 500 mL is drunk in <15 min, standard activities follow, and the pad is weighed.

| | **1-hour test** | **24-hour test** |
|---|---|---|
| Normal | <1.4 g | <4.4 g |
| Mild | 1–10 g | 4–20 g |
| Moderate | 11–50 g | 21–74 g |
| Severe | >50 g | ≥75 g |

- In the **double-dye tampon test** (oral phenazopyridine and intravesical methylene blue), orange staining means a ureterovaginal and blue a vesicovaginal fistula.
- Measure **PVR** by bladder scan and repeat it. There is no agreed cut-off, but asymptomatic women are almost always <100 mL; a raised PVR changes counselling before SUI surgery or antimuscarinics.
- **UDS** are not needed in the index SUI patient (see Ch 107).
- **Cystoscopy** is not routine; use it for haematuria, urgency or irritative symptoms, recurrent UTI, prior sling or mesh, and pelvic radiation.

## Imaging

| **Modality** | **Main indication** |
|---|---|
| Voiding cystourethrogram | Reflux, diverticula, cystocele |
| Renal ultrasound | Hydronephrosis: chronic retention, **severe POP (ureteric kinking)**, fistula, neurogenic bladder |
| Translabial ultrasound | **Modality of choice for mesh** (CT and MRI poor); levator defects |
| MRI / dynamic MRI | **Urethral diverticulum (standard of care)**; complex or recurrent POP; symptoms out of proportion to findings |

- A diverticulum gives the **3 Ds**: dysuria, postvoid dribbling and dyspareunia (see Ch 122).

## 5 · Management pathway

**ALGORITHM**

1. History, examination, UA and PVR.
2. Exclude DIAPPERS causes, UTI, obstruction and tumour.
3. Classify the UI and grade the bother.
4. If there are red flags (pain, haematuria, recurrent UTI, pelvic mass, prior radiation or pelvic surgery, prolapse beyond the introitus, fistula), refer for specialist work-up.
5. If not, start initial management and reassess.
6. Add tests only if they change management; then surgery.

| Diagnosis | Initial management |
|---|---|
| SUI | Lifestyle and behavioural therapy including pelvic floor muscle training (PFMT) (see Ch 114); † pessary; surgery if these fail |
| UUI / OAB | Lifestyle, behavioural therapy; antimuscarinic trial (see Ch 113) |
| Significant PVR | Treat constipation, review drugs; catheter drainage if PVR 200–500 mL, then reassess |
| Symptomatic POP | † Observation, PFMT, pessary or surgery, chosen by bother and goals |

## 6 · Landmark trials

| Trial | Design | Result | Take-home |
|---|---|---|---|
| † VALUE | Randomised controlled trial, uncomplicated demonstrable SUI: office evaluation with or without UDS | Office evaluation alone non-inferior | Preoperative UDS not needed in index SUI |

# 114 Conservative Management of Urinary Incontinence: Behavioural and Pelvic Floor Therapy, Urethral and Pelvic Devices

## 1 · Definition and principles

- **Conservative treatment** is any therapy that is neither surgery nor drugs.
- **Behavioural treatment** changes behaviour or environment, or teaches skills. It is effective and **essentially risk free**, so it is first-line in all guidelines for stress (SUI), urgency (UUI) and mixed (MUI) urinary incontinence, overactive bladder (OAB) and nocturia.
- Programmes are multicomponent and individualised, combining education, pelvic floor muscle training (PFMT), bladder training (BT), biofeedback (BF), electrical stimulation (E-stim) and lifestyle advice.
- Outcome is unpredictable, so offer it to anyone willing; dementia patients benefit from caregiver-led programmes.

| | **Outlet-directed** | **Bladder-directed** |
|---|---|---|
| Target | Urethral closure | Bladder function via voiding habits |
| Examples | PFMT; behavioural training with urge suppression | BT; delayed voiding |
| | **Patient-dependent** | **Caregiver-dependent** |
| Examples | BT, PFMT, behavioural training | Timed voiding, habit training, prompted voiding |
| Needs | Intact function, learning, motivation | Knowledgeable, motivated caregiver |
| Suits | Cognitively and functionally intact | Impaired, nursing home |

> **KEY POINT**
>
> Why first-line if surgery cures SUI more often? Conservative therapy is risk free and at least as effective as drugs. This asymmetry of harm drives every recommendation.

## 2 · Assessment

- A **bladder diary for 3–7 days** records intake, voids and volumes, urgency, leaks and pads. It estimates **functional capacity**, and its **self-monitoring effect** makes it assessment, outcome measure and treatment at once.
- **Education** comes first: anatomy and the mechanism of the patient's own leakage. Improvement is gradual and needs daily practice.

## 3 · Lifestyle modifications

*Cheap adjuncts; evidence is weak except for weight loss.*

### Fluid management

- **Inadequate intake is more common** than excess; self-restriction causes dehydration, urinary tract infection (UTI), constipation and concentrated irritant urine.
- Excess intake worsens frequency, so reduce it if 24-h output is >~**2100 mL**.

- **Target** a **normal range, not restriction**: **30 mL/kg/day**, minimum **1500 mL**, at set times.
- Change the **timing**, not the total: stop fluids **3–4 h before bed** for nocturia. For oedema-driven nocturia, use stockings or leg elevation in the late afternoon, or a mid-to-late afternoon loop diuretic.

### Caffeine and irritants

- Caffeine is a **diuretic and irritant** that raises filling detrusor pressure and is a risk factor for detrusor overactivity (DO).
- About **250 mg/day** (≈2 cups) is associated with moderate–severe UI in men.
- Only a subset is caffeine-sensitive. Try a 3–5-day reduction, tapering with decaf to avoid withdrawal headache.

### Bowel function

- Constipation and straining are linked to lower urinary tract symptoms (LUTS) through shared innervation, cross-sensitisation and straining-induced **pudendal denervation**. **Faecal impaction** can cause OAB or overflow.
- Advise fibre **with adequate fluid**, exercise, and regular evacuation after meals.

### Obesity, smoking, diet

- Obesity roughly doubles UI risk. **Loss of 5–10% body weight** sustained for 12 months substantially reduces UI, mainly SUI; bariatric surgery resolves UI in morbid obesity.
- Behavioural weight loss in overweight women is the **only grade A lifestyle recommendation**.
- For smoking, no UI recommendation is possible (grade D); advise cessation for general health. A fruit, vegetable and whole-grain diet helps prevent UI after menopause (B).

## 4 · Pelvic floor muscle training (PFMT)

*Most failures come from a wrong contraction.*

- PFMT, introduced by **Arnold Kegel in 1948**, is the cornerstone for SUI, UUI, MUI, urgency and frequency.
- **Mechanism**: contraction lifts the urethra and compresses it against the symphysis. Training causes striated muscle **hypertrophy**, which raises urethral pressure and improves bladder-neck support. There is also a **skill** route (the knack).

### Assessment of PFM function

- A correct contraction gives **perineal elevation**; perineal descent means the patient is straining.
- On **digital vaginal exam**, levator ani lies just above the hymen at **4 and 8 o'clock**. Assess tone, symmetry, hold and quick flicks, and note **overactivity** (hypertonic, cannot relax).
- Scales: **Modified Oxford** is the most used; **Brink** scores pressure, duration and displacement.

### Teaching

- Say: "Tighten as if stopping gas, pull up and in", without using buttocks or abdomen. For men: "move the penis up and down".

- The commonest errors are **co-contraction** (rectus, gluteals) and **Valsalva substitution**; counting aloud prevents breath-holding.
- **Verify correct contraction before prescribing.** Failure to isolate the muscle is probably the commonest cause of poor outcome; a pamphlet does not ensure it.

### Regimen

No regimen is ideal; good results come with **45–60** contraction–relaxation pairs per day.

**ALGORITHM**

1. Empty the bladder.
2. **Lying**: 5 quick (1–2 s) and 5 sustained (5, building to 10 s) contractions.
3. Repeat **sitting**, then **standing** (total 30).
4. Rest as long as each contraction (work:rest **1:1 to 1:2**).
5. Do this twice daily = **60/day**.

- **The knack**: contract **just before and during** a cough, sneeze or lift. It reduces leakage immediately.

### Evidence

- PFMT beats no treatment for SUI, UUI and MUI, short and long term; pure SUI does best.
- **Supervised training beats unsupervised** (grade A), and **age is no deterrent**.
- Antenatal PFMT in continent women prevents late-pregnancy and postpartum UI; give supervised postpartum PFMT if UI persists ~3 months.
- **After radical prostatectomy (RP)**, PFMT hastens continence in the first 3 months, and behavioural training helped men leaking years after RP. Adding E-stim or BF does not help.
- In **neurological OAB** (multiple sclerosis, incomplete spinal cord injury, Parkinson disease, stroke), PFMT is safe and first-line (B). PFMT adds to tolterodine for UUI.

### Cones and devices

- **Weighted vaginal cones** (2 × 20 min/day) work by prompting a reflex contraction, but they **cannot be retained with POP or a wide introitus** and are no better than PFMT. Motion-sensing, app-guided devices can distinguish lift from strain and modestly improve results.

## 5 · Urge suppression

- A well-timed PFM contraction **reflexly inhibits the detrusor**, whereas rushing raises bladder pressure.

**ALGORITHM**

1. When the urge comes, do not rush.
2. Stand still or sit, and relax.
3. Do several quick PFM squeezes, then distract yourself.
4. The urge subsides like a wave; then walk normally to void.
5. Once mastered, wait 5 min after the urge, then extend.

- Used alone for UUI, it gives a mean **60–80%** reduction in incontinence, and also reduces urgency, frequency and nocturia.

- For **nocturia**, suppress the urge on waking; void if it has not passed in 1–2 min.
- It is at least as good as antimuscarinics and adds to drugs. In men on an α-blocker, adding behavioural therapy reduced nocturia more than adding an antimuscarinic.

## 6 · Biofeedback (BF)

- BF is a **teaching technique**, not a treatment. It shows PFM activity in real time by manometry or surface electromyography (EMG); EMG separates PFM from abdominal activity, whereas pressure BF cannot separate lift from strain.
- BF **adds no clear benefit to supervised PFMT**. Use it for very weak PFMs, poor proprioception, an uncertain contraction, or motivation.

## 7 · Electrical stimulation (E-stim)

- Non-implanted stimulation (vaginal or anal probe, or surface electrodes) produces **PFM contraction** and **detrusor inhibition**.

| | **Long-term (chronic)** | **Maximal** |
|---|---|---|
| Intensity | Below sensory threshold | Just below pain threshold |
| Aim | Detrusor inhibition via pudendal afferents | Urethral closure (direct and reflex striated contraction) |
| Dose | 6–12 h/day for months | 15–30 min, several times a week to twice daily |

| **Indication** | **Frequency** |
|---|---|
| UUI | **5–20 Hz** (low) |
| SUI (including post-prostatectomy) | **20–50 Hz** (higher) |
| MUI | ~20 Hz or alternating |
| All | Pulse 200–1000 µs, rectangular, biphasic |

- E-stim beats sham but **adds nothing to PFMT**, so PFMT is first-line (A). Its **niche** is women who **cannot contract voluntarily**.
- **Tibial nerve stimulation** for UUI or OAB: transcutaneous equals percutaneous in efficacy, with fewer adverse effects (see Ch 115).

> **KEY POINT**
>
> E-stim is dosed by goal. Sub-sensory, low-frequency, prolonged stimulation inhibits the detrusor; near-painful, higher-frequency, brief stimulation contracts the sphincter. Volitional PFMT does both, so E-stim earns its place only where volitional contraction is absent.

## 8 · Bladder training and scheduled voiding

**P1** *Describe pathophysiology of OAB. Briefly outline management & describe nonmedical non-pharmacological management — BT breaks urgency, frequent voiding, reduced capacity cycle; void by clock not urge (dissociates voiding from urgency)*

- BT also works for SUI and MUI; it is grade A first-line in women, but evidence in men is insufficient.
- Adding behavioural training to drugs gives benefit that persists after the drug stops (UITN).

**ALGORITHM**

1. Start with a diary.
2. Set an initial interval of ~1 h while awake (30 min if needed).
3. Void on waking, at each interval and at bedtime, but not between.
4. If leaks and urgency are controlled, increase by **15–30 min/week**.
5. Target an interval of **3–4 h**; review weekly with reinforcement.

- BT beats no treatment for UUI, SUI and MUI, and the Agency for Healthcare Research and Quality (AHRQ) rates it equal to PFMT.
- **Delayed voiding** has no schedule: at each urge, suppress it, **wait 5 min**, and extend gradually. BT is triggered by the clock; delayed voiding by the urge plus a wait.

### Caregiver-dependent toileting: elderly, nursing homes

| Programme | Patient role | Key features |
|---|---|---|
| Timed voiding | Passive | Fixed 2–4 h by day; dryness goal. Also for impaired sensation and infrequent voiders. ~20% of frail elders become dry, 30–50% improve; a 2–3-day trial predicts success |
| Habit training | Passive | Schedule individualised from the diary, shorter than the usual interval, to **pre-empt** expected leaks; suits diuretic or time-patterned UI |
| Prompted voiding (PV) | Responsive | **Monitoring**, then **prompting**, then **praise**; every 2–2.5 h. 25–40% of nursing-home residents respond (daytime leaks ≤1); a 3-day trial predicts |
| Functional incidental training | Responsive | PV plus mobility and strength exercise |
| Bladder training | Active | Lengthens interval; needs ability and motivation |

- **Only BT lengthens the interval**; caregiver programmes aim at dryness, and PV success depends more on the caregiver than the patient.
- First exclude reversible causes (DIAPPERS) and check post-void residual (PVR) (see Ch 105).

## 9 · High-tone PFM dysfunction

*A tight pelvic floor needs relaxing, not strengthening.*

- Raised tone causes pelvic pain, hesitancy and incomplete emptying. Treat by **down-training**: contract, then relax, with work:rest **1:4 or longer**; BF is useful.

## 10 · Mechanical devices

Devices are **intravaginal** (support) or **intraurethral/meatal** (occlude); no device is proven superior.

### Continence pessaries

- **Rationale**: the pessary **stabilises the bladder neck, lengthens the urethra and compresses it against the pubis**, like a sling. The knob of the ring sits at the urethrovesical junction.
- **Types** for SUI: incontinence **ring** (with knob), **dish** and **Hodge**. For pelvic organ prolapse (POP), support pessaries (ring, Gehrung) suit milder prolapse and space-filling ones

(Gellhorn, cube, donut) suit advanced POP.

- **Fitting**: choose the largest comfortable size, one finger-breadth from the wall. The patient walks, coughs, strains and voids before leaving, with no expulsion, pain or obstruction.
- **Contraindications**: active pelvic infection, severe ulceration, silicone or latex allergy, **likely non-compliance with follow-up**.
- **Complications**: discharge and odour. Neglected use causes **erosion, impaction, vesicovaginal or rectovaginal fistula**, and bleeding.
- **Care**: follow-up every **24 weeks** is acceptable (versus 12; grade C), inspecting for erosion each time.
- About half improve. The International Consultation on Incontinence (ICI) says pessary or PFMT may be offered; pessary with PFMT beats PFMT alone for POP symptoms. See **ATLAS** under trials.

### Other devices

- **Uresta** (reusable bell-shaped pessary) and **Impressa** (disposable tampon-like device) give suburethral support. Urethral inserts are balloon plugs for pure SUI, with risks of UTI, haematuria and migration.
- Use pads when treatment fails or is declined, and prevent **incontinence-associated dermatitis**.
- **An indwelling catheter is the last resort**: unmanageable retention, urine-contaminated sacral or perineal wounds, terminal care, or preference. Clean intermittent self-catheterisation (CISC) is preferred for retention.

## 11 · Vaginal oestrogen

- For SUI with atrophy, PFMT with vaginal oestrogen beats oestrogen alone (C). Local oestrogen improves urgency, frequency, recurrent UTI and pessary tolerance (see Ch 113).

## 12 · Adherence, prevention and delivery

- **Adherence caps efficacy** and declines over time; realistic expectations, follow-up, reminders and linking exercise to daily cues help.
- **Prevention in childbearing**: supervised antenatal PFMT (12 weeks from 16–24 weeks' gestation) prevents late-pregnancy and early postpartum UI (A). Offer a postnatal programme after forceps or a baby ≥**4000 g**. Antenatal PFMT does **not** prevent postnatal prolapse.
- **Prevention at prostatectomy**: PFMT before surgery, with or without early postoperative PFMT, shortens time to continence.

## 13 · Evidence grading: key ICI recommendations

| Grade (Oxford) | Meaning |
|---|---|
| A | Consistent level 1 evidence; effectively mandatory |
| B | Consistent level 2/3 studies or majority of randomised controlled trial (RCT) evidence |
| C | Level 4, or Delphi-processed expert opinion |
| D | **No recommendation possible** (not "weak") |

| ICI 2023 recommendation | Grade |
|---|---|
| Supervised PFMT first-line for UI in women of all ages | A |
| PFMT over E-stim as first-line | A |
| BT first-line for women with UI | A |
| Behavioural weight loss for overweight women with UI | A |
| Antenatal supervised PFMT in continent women to prevent UI | A |
| PFMT reduces POP symptoms (stage reduction uncertain) | A; B |
| PFMT over drugs first-line (fewer adverse effects, less dropout) | B |
| PFMT before surgery (surgery more effective but riskier) | B |
| Adding BF, cones or E-stim to PFMT: not routine | B/C |
| Fluid adequacy and caffeine reduction | B |
| Smoking cessation for UI | D |

- AHRQ (2018): behavioural therapy is **more likely than anticholinergics to cure UUI** (high strength).

## 14 · Landmark trials

| Trial | Design | Result | Take-home |
|---|---|---|---|
| First behavioural vs drug RCT | Older women, UUI: behavioural vs titrated drug | Behavioural fewer leaks, more satisfaction | Behavioural at least equal to drugs for UUI |
| MOTIVE | Men with OAB, no obstruction: behavioural vs oxybutynin extended release (ER) | Equivalent reduction in voids | Behavioural therapy works in men |
| UITN (BE-DRI) | Urgency-predominant UI: tolterodine ER with or without behavioural | Combination better satisfaction, bother; persisted after drug stopped | Add behavioural training to drugs |
| ATLAS | Stress-predominant UI: pessary vs behavioural vs both | 3 mo: behavioural better satisfaction; combination not better; no difference at 12 mo | Pessary reasonable alternative; behavioural first |
| Weight-loss RCT (PRIDE) | Overweight women: 6-month intensive weight loss vs education | Greater UI reduction | Recommend 5–10% weight loss (A) |
| GBT | Women with UI: one 2-h bladder-health class vs none | Modestly better ICIQ-SF score to 12 months | Group classes cheap, scalable |

# 116 Retropubic Suspension Surgery for Incontinence in Females

## 1 · Mechanism

- Stress urinary incontinence (SUI) arises from **urethral hypermobility** and **intrinsic sphincter deficiency (ISD)**, alone or combined.
- Suspension restores **pressure transmission** and a firm backstop for compression.

> **KEY POINT**
>
> Suspension corrects **hypermobility, not the sphincter**: it lifts and stabilises bladder neck and proximal urethra, so presupposes a mobile urethra. A fixed ISD urethra needs compression (sling, artificial urinary sphincter (AUS)).

## 2 · Patient selection

- Do a stress test supine, then standing if negative; a Q-tip angle >30° means hypermobility.
- AUA/SUFU minimum: **post-void residual (PVR) and urinalysis**; urodynamics only if leak is not shown. Valsalva leak point pressure <60 or maximal urethral closure pressure <20 cm $H_2O$ suggests ISD.
- **Indications** (Burch remains an AUA/SUFU option with midurethral sling (MUS), autologous sling and bulking): SUI with hypermobility, or the stress part of mixed incontinence (MUI); **concomitant abdominal surgery** (sacrocolpopexy); **difficult vaginal access**; **mesh aversion**.

| Contraindications | Raise failure risk |
|---|---|
| Immobile vagina (cannot reach Cooper's) | Baseline urgency, preoperative anticholinergic |
| Immobile urethra (negative Q-tip ≈ **5-fold** failure): use sling or AUS | Obesity, menopause, age >60 |
| ISD alone; "lead-pipe" (fixed, open) urethra | Prior hysterectomy or SUI surgery; advanced POP; MUI |

## 3 · Anatomy

- **Retzius** is an avascular preperitoneal space between symphysis and bladder. **Cooper's ligament** lies on the superior pubic ramus.
- The **arcus tendineus fascia pelvis (ATFP)** runs from symphysis to ischial spine.

## 4 · Burch colposuspension

Paravaginal tissue is sutured to Cooper's; sutures elevate, and **fibrosis maintains** the repair.

**ALGORITHM**

1. Low lithotomy, prep including vagina, Foley.
2. Pfannenstiel or low midline incision.
3. Enter Retzius; sweep fat laterally.
4. Find bladder neck by Foley balloon and vaginal finger.
5. Dissect bladder off vagina **laterally**; expose Cooper's.
6. Place sutures, then cystoscopy.

---

## SAQ Burch colposuspension

Paper answer · P1

**Type:** SAQ (10 marks)
**Source:** 2022-Summer Paper 1, Q3b (related: 2017-Winter Paper 3, Q2)

### Definition

- Retropubic suspension (**Burch, 1961**) of the paraurethral and para-bladder-neck vaginal wall to **Cooper's (iliopectineal) ligament**; Tanagho (1976) modified it.

### Principle

- SUI (stress urinary incontinence) from **urethral hypermobility**: the urethra drops out of the retropubic position. Elevation restores **pressure transmission** and a backstop; scarring holds it.
- It treats hypermobility, not ISD (intrinsic sphincter deficiency).

### Indications

- SUI or stress-predominant MUI (mixed urinary incontinence) with hypermobility, especially with **concomitant abdominal surgery** (sacrocolpopexy), **mesh refusal**, or poor vaginal access.
- **Contraindications:** immobile urethra or vagina, ISD alone, pure urgency incontinence.

### Work-up

- Cough stress test, PVR (post-void residual), urinalysis; urodynamics only if SUI is not demonstrated.

### Steps (open)

1. Low lithotomy, Foley; Pfannenstiel incision; enter the **space of Retzius**.
2. Vaginal finger elevates the vagina; sweep bladder medially. **Avoid midline dissection** over urethra.
3. Two to four 0 or 1-0 delayed-absorbable or non-absorbable sutures each side through vaginal wall (not lumen): distal at mid-urethra, proximal at bladder neck, placed laterally.
4. Pass through Cooper's ligament; tie with vaginal elevation, without tension; leave about **two fingerbreadths** between bladder neck and pubis.
5. **Cystoscopy:** ureteric efflux, no suture in bladder. - Laparoscopic and robotic versions match open short-term results.

### Outcomes

- Cure 85–90% at 1 year, about 70% at 5 years (Cochrane).

- **SISTEr (2007):** autologous fascial sling more effective but more voiding dysfunction and UTI.
- **Ward and Hilton:** TVT (tension-free vaginal tape) equivalent to colposuspension; meta-analyses favour midurethral slings.
- **CARE (2006):** prophylactic Burch at sacrocolpopexy reduced postoperative SUI.

### Complications

- Voiding dysfunction 2–27%, de novo storage symptoms 8–27%, **de novo prolapse** 2.5–27%; bladder or ureteric injury; bleeding.
- **Also know:** MMK (Marshall–Marchetti–Krantz; osteitis pubis 0.7–2.5%), paravaginal repair, VOS (vagino-obturator shelf); failure predictors: BMI (body mass index) >25, age >60, prior surgery.

**Diagram to draw:** sagittal pelvis with sutures from paravaginal fascia to Cooper's ligament.

> **Key points to remember:**
> - Burch 1961: vaginal wall to Cooper's ligament; for hypermobility.
> - Cure 85–90% at 1 year, ~70% at 5 years.
> - Niche: concomitant abdominal surgery, mesh avoidance.

**Also, from the condensed book (not in the paper answer):**

- Tie **distal first** and **loosely** (Tanagho), leaving **≈ 2 fingerbreadths** between bladder neck and pubis; over-tightening obstructs.

> **KEY POINT**
>
> **Danger points**: never dissect midline over urethra or bladder neck (Santorini bleeding, injury, obstruction if sutures medial). Passing through Cooper's may bleed suddenly: use pressure, clips, sutures.

| Approach | Bottom line |
|---|---|
| Open (Pfannenstiel) | Cure **85–90% at 1 y, ≈70% at 5 y** |
| Laparoscopic / robotic | Subjective cure equivalent to open to 18 months |
| Needle suspension (Pereyra, Stamey, Raz) | Abandoned: sutures tore through |

## 5 · Other retropubic procedures

*A higher anchor means more obstruction.*

| Technique | Anchor | Holds by | Obstruction |
|---|---|---|---|
| Burch | Cooper's ligament | Sutures; high V-shaped angle | Greater |
| Marshall–Marchetti–Krantz (MMK) | Symphyseal periosteum | Sutures; high | Greater |
| Paravaginal (Richardson) | ATFP | Scarring; flatter | Less |
| Vagino-obturator shelf (VOS) | Obturator fascia (or iliopectineal) | Scarring; flatter | Less |

**KEY POINT**

AUS in women is second-line for ISD (recurrent or neurogenic). Place it at the bladder neck, preferably **abdominally**; transvaginal placement raises infection and erosion.

## 6 · Perioperative care

- **Cystoscopy is mandatory**: no suture in bladder or urethra, and bilateral efflux. If efflux is absent, release sutures.
- For incomplete emptying, use clean intermittent catheterisation (CIC), preferred to suprapubic, until PVR is consistently **<150 mL**.

## 7 · Outcomes and complications

- Burch roughly equals retropubic MUS but is slower and causes more voiding dysfunction.
- Pubovaginal sling (PVS) is more effective but more obstructive (SISTEr).
- Add prophylactic Burch at sacrocolpopexy in stress-continent women (CARE); with existing SUI, MUS beats Burch.
- Voiding dysfunction (2–27%) is treated by CIC, then urethrolysis (Ch 118). De novo overactive bladder (OAB, 8–27%) needs obstruction excluded.
- **Posterior prolapse** (2.5–27%) follows because anterior elevation opens the cul-de-sac.
- **Osteitis pubis** is unique to MMK (0.7–2.5%) and causes suprapubic pain at 2–12 weeks; treat with rest and analgesia.

## 8 · Landmark trials

| Trial | Design | Result | Take-home |
|---|---|---|---|
| SISTEr | Randomised trial (RCT), 655: Burch vs autologous fascial PVS | PVS higher cure at 2 and 5 y; more urinary infection, voiding dysfunction, de novo UUI | Sling more effective, more morbid |
| Ward–Hilton (TVT vs colposuspension) | First RCT, 348 | No difference to 5 y; TVT more perforation; Burch more voiding dysfunction, later POP | MUS equals Burch, less invasive |
| CARE | RCT, 322 stress-continent, sacrocolpopexy with or without Burch | Less postoperative SUI with Burch, sustained 2 y | Add prophylactic Burch at sacrocolpopexy |

# 118 Slings: Autologous, Biologic, Synthetic and Mid-Urethral

## 1 · How slings work — theories of continence

*The theory decides where the sling sits and how tight.*

*See the paper answer **Female SUI: continence mechanism, classification, evaluation (UDS) and management, including male SUI surgery** (Section 6, Ch 108).*

**Also, from the condensed book (not in the paper answer):**

- Supporting integral theory, incontinent women have ≈**40% less collagen** in skin and round ligament MCQ .
- **Urethral hypermobility is a SIGN** of damaged support, **not the cause** of stress urinary incontinence (SUI).
- A pubovaginal sling (PVS) sits under **minimal tension at the bladder neck** and rebuilds the hammock. A mid-urethral sling (MUS) lies **loose under the mid-urethra** and arrests the posterior wall. Neither should obstruct at rest.

### Historical milestones MCQ

- McGuire reintroduced the fascial PVS (1978). The tension-free vaginal tape (**TVT**, 1995) was the first retropubic MUS and defined the two complications **unique to MUS**: urinary tract mesh penetration and vaginal exposure. The transobturator route followed in 2001 and the single-incision mini-sling (SIMS) in 2006.

### Guideline status (AUA/SUFU)

- MUS for the index patient is a **strong recommendation**, grade A. Either route is conditional, grade A; SIMS is conditional, grade B.
- Mesh-specific risks **and** non-mesh alternatives must be discussed before a synthetic MUS (Clinical Principle).

## 2 · Preoperative assessment and patient selection

*See the paper answer **Female SUI: continence mechanism, classification, evaluation (UDS) and management, including male SUI surgery** (Section 6, Ch 108).*

**Also, from the condensed book (not in the paper answer):**

- Q-tip movement **>30°** on straining defines hypermobility.
- Further evaluation, including urodynamic studies (UDS), is needed for unclear diagnosis, neurogenic bladder, urgency-predominant mixed urinary incontinence (MUI), high post-void residual (PVR), POP-Q stage ≥3 without SUI on reduction, or prior surgery. Cystoscopy is not part of the index evaluation.

### Urodynamic studies — the controversy

- UDS may be **omitted** in the index patient with demonstrable SUI (conditional, grade B), based on **ValUE**.
- A maximal urethral closure pressure (MUCP) **<20 cm $H_2O$** defines type 3 SUI, that is intrinsic sphincter deficiency (**ISD**). A Valsalva leak point pressure (VLPP) **<60 cm $H_2O$** also

means ISD; a value **>90** means little or no ISD.

- ISD is a **matter of degree** and now largely a clinical diagnosis; it matters most with a **hypomobile urethra**.
- **Detrusor overactivity (DO)** is the most consequential bladder finding and may improve or worsen, so consider treating it first. Detrusor underactivity predicts voiding difficulty but should **not exclude** surgery.
- Offer non-surgical options to all: weight loss, **pelvic floor muscle training (PFMT)** before surgery for most, and pessaries.

## 3 · Choosing the sling

| Situation | Preferred option | Reasoning |
|---|---|---|
| Index SUI, mobile urethra | MUS (retropubic or transobturator); SIMS acceptable | Equal efficacy, least morbidity |
| ISD with mobile urethra | Retropubic preferred over transobturator | Meta-analyses favour retropubic |
| Fixed, immobile urethra | Retropubic MUS, autologous PVS, bulking, adjustable sling | AUA Expert Opinion; mobility predicts MUS success |
| Recurrent SUI after failed MUS | Retropubic MUS or autologous PVS | Retropubic better than transobturator for recurrence |
| Mesh complication, urethral injury, fistula, diverticulum, reconstruction | **Autologous PVS** | No mesh near a damaged urethra; adds tissue |
| Declines mesh | Autologous PVS or Burch see Ch 116 | — |
| Neurogenic SUI on clean intermittent catheterisation (CIC) | Autologous PVS (may be tight) | Only once UDS confirm safe capacity and compliance |
| Concomitant pelvic organ prolapse (POP) surgery | MUS, PVS or Burch | Conditional, grade C |

## 4 · Pubovaginal sling

- **Indications:** primary or recurrent SUI from **hypermobility and ISD**, stress-predominant MUI, anteroapical POP, neurogenic SUI, **urethral reconstruction** (trauma, mesh perforation, interposition at fistula or diverticulum repair), and failed Burch or MUS.
- **Autologous fascial PVS is the gold standard for all forms of SUI.** It gives **dynamic compression without obstruction**; too much tension causes retention and erosion.
- Width **<2 cm** compresses without a constricting band.

| Material | Behaviour | Status |
|---|---|---|
| **Autologous** — rectus fascia, fascia lata | Best ingrowth and neovascularisation, minimal reaction; negligible erosion | **Gold standard**; cost is harvest pain and wound complications |
| Allograft | Processing weakens it; variable, less durable | Mostly withdrawn |
| Xenograft (porcine, small intestinal submucosa (SIS), bovine) | Weaker, encapsulated; cure falls with time; porcine inferior in trials | Mostly withdrawn |

| Material | Behaviour | Status |
|---|---|---|
| Synthetic at bladder neck | Urethral perforation **15×** and vaginal exposure **14×** other materials (especially woven polyester) | **No longer used at bladder neck** |
| Vaginal wall sling | Weak; inclusion cysts, vaginal shortening | Historical |

**KEY POINT**

**Fibrosis paradox**: more fibrosis should mean better support, but the inflammation that drives it also degrades the sling and causes erosion. Autologous fascia remodels with minimal inflammation, the ideal balance.

- **Rectus or fascia lata:** strength and cure are equal. Fascia lata avoids abdominal wound problems but causes thigh pain; use it after **ventral hernia repair**.

## 5 · Autologous PVS — operative procedure

- **Counsel** about voiding dysfunction and de novo urgency, and teach CIC in selected women. Give one dose of cefazolin and mechanical thromboprophylaxis.

**ALGORITHM**

1. Through a 6–7 cm Pfannenstiel incision, mark and excise a **2 × 8 cm** rectus graft.
2. Close the fascia with No. 1 polydioxanone (PDS).
3. Clean the graft, stitch No. 1 PDS to each end and keep it in saline.

- Keep the transverse incision **≥2 cm above the symphysis** for tension-free closure. A fascia lata graft is taken over the iliotibial band, and the fascia lata is not closed.
- **Vaginal part:** make an **inverted-U incision**. With the **bladder empty**, perforate the **endopelvic fascia** and finger-dissect the retropubic space.

**ALGORITHM**

1. With the bladder empty, pass Stamey needles **from above**, on the back of the pubis, into the vaginal incision.
2. Perform **cystoscopy (70° lens)**. If there is an injury, repass **more laterally** and re-scope after each pass.
3. Confirm ureteric efflux, centre the graft and tag its ends above.

- Do any POP repair **before tensioning** and remove the weighted speculum.
- **Tension:** tie the sutures over the rectus with **two fingerbreadths** of slack, deliberately **loose**. Tie tighter only if lifelong CIC is planned.
- Postoperatively, do a voiding trial on day 1; if it fails, use Foley or CIC and retry within 5 days. No heavy lifting and no intercourse for **6 weeks**.

## 6 · PVS outcomes

*PVS rarely fails for stress leakage; "failure" is usually urgency.*

- Results are durable even in complex patients (radiation, spinal cord injury, trauma). Cure stays high even with MUCP <4 cm $H_2O$, so **severe ISD is not a contraindication**.
- **Recurrent SUI:** a repeat PVS, or a PVS after failed MUS, succeeds in most.

- The commonest "failure" is **storage symptoms or urgency urinary incontinence (UUI)**; de novo UUI occurs in 2–22%, predicted by prior POP surgery.
- **Cochrane, PVS versus MUS:** PVS is probably no better, maybe less effective; MUS is shorter, with fewer complications (except bladder perforation) and less voiding and storage dysfunction.
- A damaged urethra does worst, yet PVS still dries most. In **MUI**, PVS cure equals that of pure SUI but the urgency outcome is unpredictable.

## 7 · PVS complications and management

### Voiding dysfunction and bladder outlet obstruction (BOO)

- Incidence is 2.5–35%, **more than Burch** (SISTEr: all outlet procedures were in the PVS arm). Retention >4 weeks is ≈8%; **permanent retention is ≤5%**.
- The usual cause is **excess tension**; poor contractility and **straining to void** add to it.
- It presents as weak stream, retention, **storage symptoms only**, recurrent urinary tract infection (UTI) or pain, even with a low PVR. Exclude new POP kinking the urethra.
- Prediction is unreliable, so a low detrusor pressure means **counselling about CIC**, not refusing surgery. Evaluate by examination, PVR, cystoscopy and selective video-UDS; a high-pressure, low-flow pattern is often **absent in women**.

> **KEY POINT**
>
> **The key diagnostic criterion is the temporal relationship** between surgery and symptoms, not a urodynamic number. Preoperative UDS does not predict obstruction (SISTEr).

**ALGORITHM**

1. Most transient retention resolves within **10 days**: use catheter or CIC, timed voiding, PFMT, anticholinergic.
2. Autologous obstruction often improves, so wait up to **3 months**.
3. If retention is complete, act earlier: at **<6 weeks**, loosen the sling under anaesthesia by downward urethral traction.
4. If **>6 weeks** or conservative care fails, then do sling incision or urethrolysis.

- Downward loosening is **not for synthetic slings**. Dilation risks fibrosis, and **transurethral bladder-neck incision fails** because the sling is extraluminal.

| Procedure | Features |
|---|---|
| **Sling incision** (transvaginal) | 84–100% success; least morbid, so **first choice**; ≈20% recurrent SUI, not prevented by interposition |
| Transvaginal urethrolysis | Least effective for PVS: cannot relieve suburethral compression |
| Suprameatal urethrolysis | Better access to the lateral wings |
| Retropubic urethrolysis | Best results; after failed transvaginal attempt |

- Urethrolysis succeeds in 65–93%, with recurrent SUI in 0–19%. Failure is usually **inadequate lysis**, and a repeat relieves ≈90%.
- **Storage symptoms persist** in over half: use an antimuscarinic or β3-agonist, **sacral neuromodulation (SNM)**, or onabotulinumtoxinA 100 U.

- Recurrent SUI after lysis is treated with bulking, repeat PVS or MUS.

### Perforation, exposure and other complications

- Perforation is rare with fascia. Urethral perforation needs **sling excision and repair** (with or without Martius flap); bladder erosion is removed endoscopically.
- An exposed synthetic PVS is removed, and recurrent SUI is treated with a second, **autologous** PVS.
- Mortality is ≈**3 per 10,000** anti-incontinence procedures; harvest-site **incisional hernia** occurs in ≈1%.

## 8 · Mid-urethral sling — mechanism and materials

- The MUS **secures the posterior urethral wall**, with dynamic **kinking**. Mesh is fixed along its **whole length**, so 50–93% stay dry after midline incision.
- Preoperative **immobility predicts failure**; the ideal is a loose sling with a mobile urethra. **Exception, SIMS:** success needs restricted mobility, so it is set tighter.

> **KEY POINT**
>
> **Why does a loose tape work?** At rest it does not touch the urethra, so voiding is unobstructed. On coughing the mobile urethra descends onto the tape; the posterior wall is arrested, the urethra kinks and closes. A fixed urethra never meets the tape, and tightening the tape to compensate obstructs.

- The material is **type 1 (Amid) macroporous monofilament polypropylene** with pores **>75 µm**, which admits macrophages and fibrovascular ingrowth. Microporous tapes (<50 µm) had high exposure and were withdrawn.

## 9 · MUS operative procedures

*Each route has its own danger structures.*

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### SAQ Mid-urethral slings: TOT and TVT

Paper answer · P1 · P2 · merged from 2

*Asked as:* P2 SAQ "Mid-urethral slings for SUI: TOT, TVT and TOT versus TVT" · P1 SAQ "TOT in SUI"

#### Definition and history

- Both are polypropylene **MUS (mid-urethral slings)** for female SUI (stress urinary incontinence), based on the **integral theory** (Petros and Ulmsten, 1990): a loose mid-urethral tape replaces the pubourethral ligaments as a backboard on straining.
- **TVT (tension-free vaginal tape, retropubic):** retropubic polypropylene MUS; Ulmsten, 1995–1998.
- **TOT (transobturator tape):** a synthetic MUS of macroporous monofilament polypropylene, passed through the obturator foramina instead of the retropubic space. Delorme (2001) designed it to avoid bladder, bowel and vessel injury.

#### Mechanism

- **Integral theory:** pubourethral ligaments, the suburethral vaginal hammock and pubococcygeus maintain continence at the mid-urethra. Hypermobility is a sign of their damage, not the cause.

- The loose tape at the mid-urethra supports the **posterior urethral wall**, which is compressed against it on straining (dynamic kinking); it is not tight at rest.
- TOT anchors in the obturator muscles (**obturator internus and externus**).

### Material

- **Type 1 (Amid) macroporous monofilament polypropylene**, pores >75 µm: allows macrophage entry and fibrous ingrowth, so it resists infection.

### Indications and evaluation

- SUI or stress-predominant mixed incontinence after failed PFMT (pelvic floor muscle training); index SUI with urethral hypermobility.
- TVT preferred over TOT in **ISD (intrinsic sphincter deficiency)** or recurrent SUI.
- History, examination, **objective stress test**, urinalysis, PVR (post-void residual); urodynamics only if complicated.
- Counsel about permanent mesh, mesh risks and alternatives (colposuspension, autologous sling, bulking) (AUA/SUFU [American Urological Association/Society of Urodynamics and Female Urology]).

### Technique: differences

| | TVT (retropubic) | TOT (transobturator) |
|---|---|---|
| Path | Vagina to retropubic space to suprapubic skin, hugging the pubis; cystoscopy essential | Labiocrural fold to obturator membrane to vagina (outside-in), or reverse (inside-out) |
| Structures at risk | Bladder, bowel, iliac and epigastric vessels | Obturator nerves and vessels, vaginal fornix |

### TVT technique

1. Local, spinal or general anaesthesia; lithotomy.
2. Two suprapubic stab incisions 2 cm lateral to midline; 1.5 cm midline vaginal incision 1.5 cm from the meatus; minimal dissection, endopelvic fascia not opened.
3. **Bottom-up** 5 mm trocar hugging the back of the pubis, through the retropubic space to the skin (catheter guide deflects the bladder).
4. **Cystoscopy with a 70° lens, full bladder**, after each pass; if perforated, re-pass more laterally.
5. Tension **loosely** with a clamp between tape and urethra; remove sheath.

### TOT technique (outside-in)

1. Lithotomy with hips hyperflexed; 1.5 cm midline vaginal incision 1.5 cm below the meatus; dissect laterally to the ischiopubic ramus.
2. Groin stab incision in the labiocrural fold, at the notch between inferior pubic ramus and **adductor longus tendon**.
3. Trocar perforates the obturator membrane (felt as a give), turns medially onto the finger and exits the vaginal incision.
4. Cystourethroscopy; check the vaginal fornices (sulci).
5. Tension-free: a clamp or scissors pass easily between tape and urethra. - **Inside-out** (de Leval; vagina to thigh): fewer vaginal perforations, more voiding dysfunction.

### Outcomes

- TVT: Ulmsten (1998): 91% cured at 1 year. **Ward–Hilton:** equal to colposuspension at 5 years. Long-term cure varies 48–97% by definition.
- **TOMUS (Richter, 2010):** 597 women; 12-month objective cure **80.8% retropubic vs 77.7% transobturator** (not significantly different, equivalent); voiding dysfunction needing surgery 2.7% vs 0%.
- TOMUS at 5 years: 51.3% vs 43.4%; equivalence not shown; satisfaction similar.
- **Cochrane review (Ford, 2017):** cure similar up to and beyond 5 years; repeat surgery possibly commoner after TOT.
- TOT is poorer with **ISD** or a **fixed urethra**: retropubic sling preferred.

### Complications: comparison (Cochrane, 2017)

| Complication | TVT (retropubic) | TOT |
|---|---|---|
| Bladder perforation | **4.5%** | **0.6%** |
| Groin pain | 1.3% | **6.4%** |
| Voiding dysfunction, bleeding, vascular injury | More | Less |
| Mesh exposure | About 2% | About 2% |
| Long-term repeat surgery | Less | Possibly more |

### TVT complications

| Complication | Notes |
|---|---|
| Bladder perforation | 4.5% (TOT 0.6%); re-pass and continue |
| Urethral injury | Abandon mesh; staged repair |
| Bleeding/haematoma | 2–3%; rarely bowel or vascular injury |
| Voiding dysfunction | Surgery needed 2.7%; incise tape in midline within 3 months |
| Mesh exposure | About 2% |
| Others | De novo urgency, suprapubic pain (groin pain less than TOT) |

### TOT complications

- Groin or thigh pain (usually temporary; NSAIDs [non-steroidal anti-inflammatory drugs]), vaginal mesh exposure, voiding dysfunction (incise sling within 4 weeks), de novo urgency, UTI (urinary tract infection), rarely thigh abscess.
- Urethral injury during either sling: **do not place mesh** (AUA guideline).

### Choosing between TVT and TOT

- **TVT preferred: ISD** (ALPP [abdominal leak point pressure] <60 $cmH_2O$ or MUCP [maximum urethral closure pressure] <20 $cmH_2O$), fixed urethra, recurrent SUI. Failure is 12 times likelier after TOT with low leak point pressure (O'Connor, 2006).
- **TOT preferred:** previous retropubic surgery, obesity, avoiding bladder injury.
- AUA: either may be offered to the index patient.
- **Also know:** SIMS (single-incision mini-slings; non-inferior to TOT); mesh warnings (FDA [Food and Drug Administration] 2010; UK pause 2018): counsel about alternatives (Burch colposuspension, autologous fascial sling, bulking agents).

### Diagram to draw

- Coronal pelvis: U-shaped retropubic TVT at mid-urethra versus flat TOT, the tape under the mid-urethra through both obturator foramina.

**Key points to remember:**

- Integral theory; loose tape at mid-urethra.
- Type 1 macroporous monofilament polypropylene.
- Delorme 2001; TOT anchors in obturator muscles; fewer bladder injuries than retropubic.
- TOMUS: cure equivalent at 1 year (80.8% retropubic vs 77.7% transobturator).
- TOT: less bladder injury (0.6% vs 4.5%), more groin pain (6.4% vs 1.3%); worse in ISD.
- Cystoscopy after each TVT trocar pass; bladder perforation 4.5%.
- Prefer retropubic TVT in ISD or fixed urethra.

**Also, from the condensed book (not in the paper answer):**

- Document counselling about the **permanent implant**, mesh-specific complications and non-mesh alternatives.
- Bottom-up beats top-down, with better subjective cure and fewer perforations and exposures.

**ALGORITHM**

1. Infiltrate local; make two suprapubic stabs ≈2 cm from midline.
2. Make a 1.5 cm midline vaginal incision 1.5 cm from the meatus.
3. Make minimal lateral tunnels to the endopelvic fascia, **not perforating** it (unlike PVS).
4. The catheter guide deflects the bladder to the opposite side.
5. Pass the trocar through Retzius and rectus, **hugging the back of the pubis**; repeat on the other side.
6. Perform cystoscopy.
7. Tension loosely (clamp between tape and urethra), remove sheath, trim.

- Keep the trocar in **metal-to-bone contact** with the back of the pubis, so it stays extravesical and extraperitoneal.
- **Transobturator, outside-in (TOT):** hips **hyperflexed (120°)**; puncture the labiocrural fold at the adductor longus notch, pierce the obturator membrane and turn around the ramus onto the finger. Check the vaginal sulcus for a button-hole.
- **Inside-out (TVT-O)** has the same cure, with more voiding dysfunction and fewer vaginal perforations.
- **SIMS** uses only a vaginal incision and anchors in **obturator internus**; set it **tighter**, close to the urethra.

| Feature | Retropubic (TVT) | Transobturator | SIMS |
|---|---|---|---|
| Path / anchor | Retzius, rectus fascia; U-shape | Obturator foramen; flatter hammock | Obturator internus only |
| Cure (index SUI) | Equal to transobturator up to 5 y; best long-term data (>17 y) | Equal; slightly more repeat surgery | Non-inferior to transobturator (TVT-Secur excluded) |

| Feature | Retropubic (TVT) | Transobturator | SIMS |
|---|---|---|---|
| ISD / recurrent SUI | **Preferred** | Worse with low VLPP | Limited data |
| Bladder perforation | **4.5%** | **0.6%** | Rare |
| Vascular/visceral injury, bleeding, voiding dysfunction | More | Less | Less |
| Pain | Suprapubic 2.9% | **Groin 6.4%** vs 1.3% retropubic | Least |
| Vaginal exposure | About 2% | About 2% | Similar (excluding TVT-Secur) |
| Cystoscopy | Mandatory (70°) | Recommended; check the urethra | Essential (30° suffices) |

## 10 · Intraoperative cystoscopy

### SAQ Management of perforated mid urethral sling

Paper answer · P1

**Type:** SAQ (10 marks)
**Source:** 2023-Winter Paper 1, Q3a

#### Terminology (IUGA/ICS, International Urogynecological Association / International Continence Society)

- **Perforation:** mesh in a hollow organ (bladder, urethra). **Exposure:** mesh seen through unhealed vaginal epithelium. **Extrusion:** passage out of a tissue. "Erosion" is no longer used.
- Two settings: **intraoperative trocar injury** and **late mesh perforation**.

#### A. Intraoperative trocar injury

- Bladder injury: 4.5% retropubic versus **0.6% transobturator**; registries 2.7–3.8%.
- **Detection:** cystoscopy with a **70° lens and full bladder distension** after every trocar pass.

| Injury | Management |
|---|---|
| **Bladder** | Withdraw trocar, **repass more laterally**, confirm on cystoscopy, complete the sling; catheter overnight to 72 h. Benign; no later mesh perforation. |
| **Urethra** | **Do not place mesh** (AUA/SUFU, American Urological Association / Society of Urodynamics, Female Pelvic Medicine and Urogenital Reconstruction). Repair, catheter drainage; **staged** anti-incontinence surgery after healing. |

#### B. Late perforation of bladder or urethra

- **Presentation:** recurrent UTI (urinary tract infection), haematuria, urgency, pain, stones on mesh, voiding dysfunction, fistula. Often months later.
- **Evaluation:** examination, **cystourethroscopy**, translabial ultrasound (best imaging for mesh); exclude fistula.

#### Management

- **Bladder:** endoscopic excision (scissors, resection or holmium laser) flush with the wall; or transvesical laparoscopic or robotic excision; open cystotomy if large.
- **Urethra:** transvaginal excision of mesh in the field and layered urethral repair; **Martius flap** if complex or fistula.
- **Continence after excision:** SUI recurs in many (44–100% after synthetic sling revision). Options: concomitant **autologous pubovaginal sling** (Shah 2013: good results) or staged. A damaged urethra predicts poor continence.
- Avoid urethral dilation after mesh slings; it can cause perforation.

**Prevention**

- Empty bladder; trocar hugging the pubis; bottom-up retropubic route; transobturator route when suitable.

**Also know:** Amid type I macroporous polypropylene integrates best; chronic pain or infection may need excision from both sides of the bone.

**Diagram to draw:** retropubic and transobturator trocar paths with bladder at risk.

> **Key points to remember:**
> - Bladder trocar injury: repass laterally, continue.
> - Urethral injury: abandon mesh; stage the repair.
> - Late perforation: excise mesh, repair, consider autologous sling.

**Also, from the condensed book (not in the paper answer):**

- Inspect the urethra too after transobturator and SIMS passes. A bladder injury is benign: leave a catheter overnight to 72 h.

## 11 · MUS outcomes

*These are the figures quoted when counselling.*

- Retropubic slings are the most studied (17–20 years); objective cure stays high and subjective cure is lower. Cochrane (81 trials): retropubic and transobturator cure are equal from short to long term.
- TVT equals colposuspension at 5 years, with less de novo urgency and less later POP.
- In TOMUS at 5 years, satisfaction stayed high (79% retropubic, 85% transobturator) despite falling objective success.
- Long term, **de novo urgency rises** (≈⅓ at 10–20 years); retention is 0–7%.
- In Cochrane, SIMS did worse than retropubic, driven by **TVT-Secur (withdrawn)**; others about equal transobturator, with less pain. The key long-term endpoint is **reoperation**.

| Group | Key points |
|---|---|
| **MUI** | Urgency resolves in ≈60–75% (best when stress-predominant). De novo urgency risk rises with age, obesity, nocturia, DO and low capacity. Persistent DO: transobturator < retropubic < PVS. New storage lower urinary tract symptoms (LUTS) may mean BOO |
| **ISD** | About 50% cure with urodynamic ISD. An immobile urethra predicts failure more strongly than a low VLPP, and low VLPP is not a contraindication to retropubic MUS. Transobturator: 12× odds of persistent SUI with low VLPP |

| Group | Key points |
|---|---|
| **Recurrent SUI** | No level 1 evidence; repeat MUS is commonest, retropubic better than transobturator. Cure ≈81–90% but below primary surgery (≈62% vs 86%); more urgency, and more bladder perforation after prior retropubic surgery |
| **Concomitant POP** | A concomitant sling lowers postoperative SUI and reoperation but raises voiding dysfunction. About 30% of SUI resolves after POP repair alone. A prophylactic sling only after counselling (OPUS) |
| **Elderly** | Safe; similar or lower cure, more urgency and voiding dysfunction. Q-tip >30°: 90% continent; <30°: 42%; <10°: 80% stay incontinent. Cure ≈64% at ≥75 y vs 89% at 55–64 y. **Frailty, not age**, predicts readmission. Do not exclude by age (AUA Expert Opinion) |
| **Obese** | Success >80% but lower (≈87% vs 66% at 5 y), worse with BMI >35; more wound infection, fewer bladder punctures |

## 12 · MUS complications and management

> **KEY POINT**
>
> **Concurrent or staged sling at POP repair**. Concurrent: one operation and less SUI (≈39% falls to 8–19%), but more obstruction, storage symptoms and mesh events. Staged: avoids an unnecessary sling but commits a third to a second operation. Decide after testing for occult SUI with the prolapse reduced.

- Bladder trocar injury is ≈2–4%, almost all retropubic; risk rises with prior retropubic surgery and thin habitus. Vascular (≈0.07%) and bowel (≈0.04%) injuries are rare, but deaths have been reported.
- †Unexplained hypotension, falling haemoglobin, pain, ileus or sepsis after a retropubic sling needs urgent imaging or exploration.
- Bleeding occurs in 2–3%, more with retropubic slings. A retropubic haematoma is usually **asymptomatic, so observe**.
- **UTI is the commonest complication** (3–4% at 30 days). It is less after transobturator slings and marks a worse course.
- Vaginal wound infection is 0.7–2%; rarely there is abscess or **adductor myositis**, which may present years later.
- **Groin pain** after transobturator slings affects 16–26% early and ≈4% at 12 months. Treat with NSAIDs, physiotherapy and trigger-point injection; if refractory, excise, and **complete removal** relieves more than partial.

### Voiding dysfunction and retention

- The tape is **too tight** or **too proximal**, or POP kinks the urethra. Obstruction is commoner after retropubic slings (TOMUS: 2.7% vs 0%). Evaluate by **temporal relation**, examination, PVR, cystoscopy and selective UDS.

**ALGORITHM**

1. Retention or high PVR: CIC or catheter; most resolve within 2 weeks.
2. If obstruction persists, perform a **transvaginal midline sling incision at 2–4 weeks**.
3. Beyond 3 months the tape is fixed, so excise its lateral portion.
4. Formal urethrolysis is rarely needed.

- **Urethral dilation is harmful; do not perform it**. After incision 70–100% stay continent. **Early** release gives better voiding and **less** recurrent SUI.

**KEY POINT**

**Why is the timing different from PVS?** Fascial obstruction often settles, so one waits up to 3 months. Mesh is fixed by ingrowth and rarely loosens; late release needs wider dissection and causes more recurrent SUI, so release within ≈4 weeks.

### Storage symptoms, exposure and sexual function

- De novo overactive bladder (OAB) is ≈6–12%, equal for all MUS types. A clear temporal link means **partial BOO first**, and **cystoscopy is mandatory**. Treat with behavioural measures and drugs, then sling revision, SNM, percutaneous tibial nerve stimulation (PTNS) or botulinum toxin.
- Vaginal exposure is ≈2%; †it causes discharge, bleeding, dyspareunia or partner discomfort. †Small exposures get topical oestrogen with or without excision; larger ones are excised. Urinary tract mesh needs excision and repair, with no immediate new mesh see Ch 119.
- De novo dyspareunia is ≈3–14%, and sling removal may relieve it. Many women improve as **coital incontinence** stops.

## 13 · Adjustable slings

- **Remeex** (mesh on sutures to a suprapubic varitensor, adjusted after surgery) is a niche option for **complex SUI**. Cure is ≈85–90%, but complications reach 32–51% and there is no level 1 evidence.

## 14 · Assessing the failed or troublesome sling

**ALGORITHM**

1. Define the symptom: SUI, storage, voiding, or pain/discharge.
2. Establish the **timeline** in relation to surgery.
3. Examine for exposure, POP, urethral mobility and tenderness over the tape.
4. Check PVR and urine culture, perform **cystoscopy**, and add UDS selectively.
5. Tailor treatment to the cause.

- If SUI recurs with a mobile urethra, repeat a retropubic MUS or do a PVS; if the urethra is fixed or there is ISD, use PVS, bulking or an adjustable sling.
- If storage symptoms have a temporal link, treat obstruction first. Pain or exposure needs excision (complete for pain) see Ch 119.

## 15 · Mesh controversy and regulatory status

- The **2010 FDA** safety communication on mesh for POP and SUI was based on **MAUDE** database reports. Professional bodies still endorse MUS with documented counselling.
- †The 2019 FDA order stopping transvaginal mesh concerned **POP**, not MUS. The UK (2018) paused MUS use pending safeguards see Ch 119.

## 16 · Landmark trials

| Trial | Design | Result | Take-home |
|---|---|---|---|
| **ValUE** | RCT, non-inferiority, 630, uncomplicated SUI: UDS vs office evaluation | Office evaluation non-inferior at 1 y | UDS may be omitted in the index patient |
| **SISTEr** | RCT, 655: rectus PVS vs Burch | PVS higher cure (47% vs 38% at 2 y); more voiding dysfunction, UTI and UUI; all BOO surgery in PVS arm; persisted at 5 y | PVS more effective but more obstructive |
| **Ward–Hilton** | RCT: TVT vs open colposuspension | Equal to 5 y; TVT more bladder injury; Burch more morbidity and POP | MUS replaced Burch |
| **TOMUS** | RCT, 597: retropubic vs transobturator | 80.8% vs 77.7% (1 y, equivalent); surgery for voiding dysfunction 2.7% vs 0%; †more groin and neurological symptoms with transobturator; 5 y 51% vs 43% (equivalence not met) | Either; retropubic slightly more durable, more voiding dysfunction |
| **OPUS** | RCT (327): POP repair with or without prophylactic retropubic MUS in stress-continent women | Incontinence at 12 months 27% vs 43%; NNT 6.3; more perforation, UTI and bleeding | Reduces SUI but adds risk, so counsel |

# 117 Vaginal and Abdominal Reconstructive Surgery for Pelvic Organ Prolapse

## 1 · Goals and principles

*Prolapse is a quality-of-life problem, so operate for bother.*

*See the paper answer **Female pelvic floor anatomy and pelvic organ prolapse: evaluation and surgical repair** (Section 6, Ch 101).*

**Also, from the condensed book (not in the paper answer):**

- **Success means absence of symptoms** (no bulge), which drives satisfaction more than anatomy does.
- Obliterative surgery is contraindicated if intercourse is desired. Open and laparoscopic routes have equal complications, but laparoscopic recovery is faster.
- **The first operation is the best chance**, because planes are then lost and recurrence rises with each re-attempt. Preoperative education improves satisfaction, and declining surgery is a legitimate choice.

## 2 · Choice of procedure by compartment

*See the paper answer **Female pelvic floor anatomy and pelvic organ prolapse: evaluation and surgical repair** (Section 6, Ch 101).*

**Also, from the condensed book (not in the paper answer):**

- Enterocele (point D) is closed by culdoplasty: **McCall** vaginally, **Halban** or **Moschcowitz** abdominally.

| Compartment | POP-Q point | Organ |
|---|---|---|
| Anterior | Aa / Ba | Urethrocele / cystocele |
| Middle | C / D | Cervix or cuff / enterocele |
| Posterior | Ap / Bp | Rectocele / enterocele; perineal body |

| Site | Vaginal | Abdominal |
|---|---|---|
| Aa — urethra | Anterior repair; bladder neck suspension; sling | Retropubic urethropexy |
| Ba — bladder | Anterior repair; paravaginal; colpocleisis; wedge colpectomy | Paravaginal; sacrocolpopexy |
| C — cervix/cuff | Uterosacral ligament suspension (USLS); iliococcygeus; sacrospinous ligament fixation (SSLF); Manchester; hysteropexy; vaginal hysterectomy; colpocleisis | Hysterectomy; USLS; sacrocolpopexy; uterine suspension |
| D — cul-de-sac | **McCall** culdoplasty | **Halban**, **Moschcowitz** culdoplasty |
| Ap — posterior | Posterior repair; site-specific repair | Colpoperineopexy |

**KEY POINT**

Only paravaginal repair and uterosacral suspension are done by both routes. McCall is vaginal; Halban (linear sutures) and Moschcowitz (purse-string, which may angulate the ureter) are abdominal.

## 3 · Anterior compartment

*Anterior repair has the highest recurrence.*

- A defect may be **central** (midline thinning of the pubocervical fascia) or **lateral/paravaginal** (detached from the arcus tendineus fasciae pelvis, ATFP); most are combined. **Anterior repair does not suspend the apex.**

**ALGORITHM**

1. In lithotomy, catheterise, hydrodissect and incise in the midline from apex to bladder neck.
2. Dissect the epithelium off the pubocervical fascia laterally to the ATFP and reduce the bladder.
3. Plicate with interrupted 2-0/3-0 delayed-absorbable sutures from bladder neck to apex.
4. Trim judiciously and close; give intravenous (IV) dye, do **cystoscopy** and pack.

- Deep sutures may enter or kink the bladder or ureter, so completion cystoscopy is mandatory.
- Anterior repair is abandoned for stress urinary incontinence (SUI): AUA meta-analysis failure approached **40%**.
- Recurrence depends on the **definition**: 30–46% for "POP-Q ≥2" versus 9–23% for "beyond hymen".
- Polypropylene lowers recurrence versus native tissue but causes far more graft complications without better QoL. Hence the **withdrawal of transvaginal synthetic POP mesh** (US 2019) see Ch 119.
- A concomitant sling **halves anterior recurrence**, more than any graft.

| Complication | Point / management |
|---|---|
| De novo SUI | Rises with stage: stage 2 **41%** , early 3 **53%** , advanced 3–4 **66%** ; point Ba predicts it |
| Detrusor overactivity | De novo in 5–7%; pre-existing overactivity resolves in up to 63% |
| Retention | Usually from the sling: clean intermittent catheterisation (CIC); if persistent, sling incision or urethrolysis |
| Bleeding (>350 mL) | Bleeding means the wrong plane; worse at redo |
| Bladder injury | Drain the bladder before perforating endopelvic fascia; two-layer closure; Martius if irradiated |
| Others | Ureteric obstruction, vesicovaginal fistula (VVF), vaginal shortening, dyspareunia |

**KEY POINT**

**Confirming ureteric efflux**: indigo carmine (blue, IV), fluorescein 10% (green, IV) or phenazopyridine (orange, oral). If there is no efflux, pass a ureteric catheter or remove plication sutures. Never close without proving patency.

### Paravaginal repair

- Reattach the pubocervical fascia to the ATFP with non-absorbable sutures 1 cm apart (**5–7** vaginal, **4–6** abdominal), then do cystoscopy.
- Lateral recurrence is 2–8%, but central recurrence is ~**22%**, so these are independent failures. The abdominal route is more durable.
- **Neither colporrhaphy nor paravaginal repair treats SUI.** The vaginal route has more serious complications (transfusion 12%).

## 4 · Occult SUI and the concomitant sling

- High-stage anterior POP kinks the urethra and masks SUI; SUI on prolapse reduction occurs in 8–66% of continent women.
- Test **with prolapse reduced** (pack, speculum or pessary); sensitivity is low and specificity high.
- **Staged** argument: sling obstruction needing intervention (8.5%) equals intervention for SUI without a sling (8.3%).
- The **OPUS calculator** uses 7 predictors (accuracy 0.72 vs 0.54 for the stress test). In **CUPIDO**, urodynamics added nothing in asymptomatic women.

**KEY POINT**

**A sling also helps the prolapse**: distal anterior support makes the cystocele repair last. **Why not everyone?** In OPUS it caused more bladder perforation, UTI, bleeding and incomplete emptying. Weigh per patient.

**ALGORITHM**

1. In advanced anterior POP, do a reduction stress test, with or without urodynamic studies (UDS).
2. If SUI is symptomatic, occult or urodynamic, then offer a concomitant procedure.
3. If she is continent and the test is negative, then weigh individual risk and benefit (OPUS calculator).
4. The sling is **selective, not routine**.

## 5 · Apical compartment

*The apex is the cornerstone of prolapse repair.*

### Uterosacral ligament suspension (USLS)

- **The ureter lies 4 cm from the ligament at the sacrum but only 0.9 cm at the cervix**. Injury is usually **kinking**, not ligation. Sutures near the sacrum can entrap S1–S4 roots, causing buttock and posterior thigh pain.
- The **safe zone** is the intermediate portion, ~1 cm posterior, at the level of the ischial spine; pass needles **lateral to medial**.

**ALGORITHM**

1. After vaginal hysterectomy, place cuff stays at 3 and 9 o'clock, excise the enterocele sac, pack bowel and palpate the spines.
2. Place Allis clamps at 5 and 7 o'clock and pass **2–4** non-absorbable sutures per side.
3. Do any anterior repair or sling now.
4. Bring the sutures through the apical pubocervical and rectovaginal fascia and tie.
5. Perform **cystoscopy after tying, before trimming**. If there is no efflux, remove the most lateral ipsilateral suture first.

- Compared with SSLF, USLS keeps the midline axis, closes the cul-de-sac and risks the pudendal vessels less.
- Apical success is ~98% and anterior ~81%; ureteric kinking or injury occurs in **1–11%**.
- **OPTIMAL**: USLS equals SSLF (~65% at 2 years on a strict composite), and perioperative pelvic floor muscle training (PFMT) adds nothing.

## Sacrospinous ligament fixation (SSLF)

- SSLF is extraperitoneal and preserves length. The ligament lies under coccygeus; the pudendal bundle runs round the spine and the inferior gluteal vessels lie behind.
- The **safe zone** is **1.5–3.0 cm medial to the ischial spine**, at limited depth. Too lateral risks the pudendal bundle; too medial hits weak ligament and sacral nerves.

**ALGORITHM**

1. Mark the apex, incise in the midline, enter the perirectal space (rectum medial) and palpate spine and ligament.
2. Place retractors and pass 2 sutures (Capio or Miya hook).
3. Complete other repairs, then attach the sutures to the apex (pulley stitch if permanent).
4. Close the vagina halfway and tie the apex flush, with **no suture bridge**.
5. Do cystoscopy and close.

- SSLF is usually right-sided, with 2 sutures 2 cm apart. The **pulley stitch** keeps the knot out of the lumen.
- Cure is ~85% and apical recurrence ~5%; stage 4 predicts recurrence.
- **Drawback:** the axis is deflected posteriorly and to the right, causing **anterior recurrence** (~18%).
- Gluteal pain occurs in **15%** and resolves in 2–3 months. Pudendal injury, rectal injury and haemorrhage can also occur.

> **KEY POINT**
>
> Both ligaments are safest in their **middle third** and dangerous at their bony ends: the intermediate portion for USLS, and 1.5–3 cm medial to the spine for SSLF.

**Iliococcygeus suspension** is for deficient uterosacrals, with sutures **1 cm distal to the ischial spine**; anterior recurrence is commonest.

## Sacrocolpopexy (SCP)

- **Indications:** other defects (including SUI) repairable at one sitting, failed vaginal repair, young sexually active women, isolated vault prolapse, or wish for the most durable repair. It

maximises vaginal length without distortion.

- Use **type I macroporous monofilament polypropylene**; fascia lata recurs in ~1/3 versus <1/10 with polypropylene.

**ALGORITHM**

1. Low lithotomy with a vaginal stent or EEA (end-to-end anastomosis) sizer; sigmoid to the left.
2. Incise the peritoneum over the promontory and along the **right** of the rectum (watch the right ureter) down to the anterior longitudinal ligament.
3. If there is a large enterocele, then add a Halban culdoplasty.
4. Dissect the bladder off the anterior vagina (6–8 cm) and the rectum off the posterior wall.
5. Attach a Y-mesh to both walls **without entering the lumen**, tension it, and fix it with 2–3 permanent sutures at **S1**.
6. **Retroperitonealise** the mesh, add an anti-incontinence procedure if indicated, and perform cystoscopy.

- Fixation too low (S3–4) risks the presacral venous plexus; too high risks the middle sacral vessels and the **left common iliac vein**.
- Avoid hypersuspension, which causes pain and de novo SUI. Sutures or tacks into the promontory can cause spondylodiscitis.
- Open, laparoscopic and robotic SCP succeed in 87–100%; the robot saves little blood but costs more.
- Failures are mainly **posterior or distal**, with almost no apical retreatment; ~20% later need vaginal repair, so counsel.

| SCP complication | Rate / management |
|---|---|
| Haemorrhage (presacral) | ~4%; †pressure, sutures, bone wax, sterile thumbtacks |
| Cystotomy / enterotomy / ureter | 3% / 1.6% / 1%; bladder is the commonest robotic visceral injury |
| Ileus / small bowel obstruction | 3.6% / 1.1% |
| Mesh exposure | **~10% by 7 years** (E-CARE); more with microporous mesh; excision see Ch 119 |

- **Hysterectomy at SCP:** total hysterectomy raises exposure, so go **supracervical** or close the cuff meticulously.
- Cochrane, vaginal apical repair vs SCP: the vaginal route gives more prolapse awareness (relative risk 2.1), repeat surgery (2.3), recurrence (1.9), SUI (1.9) and dyspareunia (2.5), yet it suits many older women.

**ALGORITHM**

1. If she is elderly or comorbid and not sexually active, then offer colpocleisis.
2. If she is fit, has primary prolapse and prefers the vaginal route, then offer USLS or SSLF, with or without vaginal hysterectomy or hysteropexy.
3. If young or sexually active, or with recurrence, short vagina or high stage, then offer SCP (minimally invasive).

## 6 · Hysterectomy vs uterine preservation

- †Hysterectomy alone does not cure prolapse, so always resuspend the apex.
- Sacrospinous hysteropexy is non-inferior to vaginal hysterectomy with USLS.
- †Preservation needs a normal uterus: exclude abnormal bleeding, endometrial or cervical disease and Lynch syndrome, and keep cervical screening up to date.

## 7 · Obliterative surgery — colpocleisis

*For frail older women it is quick, low-risk and effective.*

- **Total colpocleisis** (after hysterectomy) denudes and inverts the whole vagina. **Le Fort** (uterus in place) excises anterior and posterior rectangles and leaves **lateral channels** for drainage.
- Both need **high perineorrhaphy and levator myorrhaphy** and cystoscopy. If the uterus stays, evaluate the endometrium; hysterectomy removes cancer and **pyometra** risk but raises complications.
- Spare ≥**1.5 cm** of distal anterior wall, because urethral traction causes SUI. Add a sling for SUI or occult SUI.
- Success is ~90–98% and satisfaction >90%, but regret reaches **13%**.

## 8 · Posterior compartment

*Bowel symptoms often persist after rectocele repair.*

- Symptoms are bulge, splinting or digitation, trapping (**not faecal incontinence**) and dyspareunia. First exclude slow transit and **anismus**.
- **Benefit is predicted** by bulge, splinting and a large rectocele, but not by constipation. **Constipation often persists**.
- **The transvaginal route beats transanal** (recurrence 7% versus 66%).
- **Midline fascial plication**, with or without perineal body repair, is the International Consultation on Incontinence (ICI) procedure of choice. In **site-specific** repair the commonest defect is **transverse**.
- Perineorrhaphy should leave a smooth contour, because ridges cause dyspareunia.

**ALGORITHM**

1. Repair anterior and apical compartments first; hydrodissect and incise in the midline.
2. Dissect the rectovaginal fascia off the epithelium; if the hernia is large, place a purse-string at its base.
3. Plicate in the midline (2-0 absorbable), incorporate the perineal body, trim and close.

- **Levator plication has been abandoned** because dyspareunia reaches 50%. **Grafts** are not supported (complications RR 1.8).

## 9 · Complications across POP surgery — summary

| Problem | Prevention / management |
|---|---|
| Ureteric injury (USLS highest, 1–11%) | Bites lateral to medial; cystoscopy after tying; remove the most lateral suture; stent; reimplant |
| Bladder/rectal injury | Correct plane, drained bladder, rectal finger; layered closure |

| Problem | Prevention / management |
|---|---|
| Mesh exposure | Type I mesh, full-thickness flaps, supracervical hysterectomy; oestrogen or excision |
| Recurrence | Anterior after USLS/SSLF, posterior after SCP; most are asymptomatic |
| Buttock pain | Middle-third placement; observe up to 3 months; nerve injection |
| De novo SUI / retention | Reduction testing; selective sling; CIC; sling incision |

## 10 · Landmark trials

| Trial | Design | Result | Take-home |
|---|---|---|---|
| CARE | RCT: SCP with or without Burch, continent women, stage 2–4 | SUI 24% vs 44% at 3 months | Prophylactic Burch reduces de novo SUI |
| E-CARE | 7-year extension | SUI 62% vs 77%; mesh exposure ~10%; composite failure 34–48% | The benefit is a delay; exposure rises with time |
| OPUS | RCT: vaginal POP repair with MUS vs sham | 27% vs 43% at 12 months; more perforation, UTI, bleeding, retention | Selective sling |
| OPTIMAL | RCT: USLS vs SSLF, with or without PFMT | Equal (~65%) at 2 y; PFMT no effect | Native apical repairs are equivalent |
| Bilateral SSLF vs abdominal SCP | RCT | Earlier failure and 3× cystocele reoperation after SSLF | Abdominal more durable |
| Transvaginal vs transanal rectocele | RCT | Recurrence 7% vs 66% | Repair vaginally |

# 119 Genitourinary Complications of Mesh and Their Repair

## 1 · Mesh, terminology and regulation

- Unlike the abdominal wall, the pelvic floor is mobile, contaminated and close to viscera. Mesh harms are pain, exposure, extrusion and injury to nearby structures.

| Amid type | Pore | Example | Behaviour |
|---|---|---|---|
| **I** | Macroporous >75 µm | Knitted monofilament polypropylene | **Only recommended mesh** |
| II | Microporous ~10 µm | Polytetrafluoroethylene (PTFE, Gore-Tex) | Encapsulated; >30% full-length slings excised |
| III | Macropores in microporous base | Mersilene, Mycromesh | Extrusion ~20% |
| IV | Submicronic | Silicone-coated | Extrusion/sinus up to 40% |

**KEY POINT**

**Why pore size matters**: bacteria (~1 µm) enter pores that macrophages (~50 µm) cannot, so infection is sealed from immunity. Large pores (~1500 µm) allow fibrocollagenous integration. Woven polypropylene (Marlex) is stiff and extrudes. Mesh adds no cancer or autoimmunity.

*See the paper answer **Management of perforated mid urethral sling** (Section 6, Ch 118).*

**Also, from the condensed book (not in the paper answer):**

- IUGA/ICS **CTS (Category–Time–Site) coding**: Category 1–3 vaginal (1 no separation; 2 exposure ≤1 cm; 3 >1 cm or extrusion), 4 urinary tract, 5 bowel, 6 skin/musculoskeletal, 7 systemic; suffix A asymptomatic, B symptomatic, C infection, D abscess. Time T1 ≤48 h, T2 48 h–2 months, T3 2–12 months, T4 >12 months. Site S1 vaginal suture line, S2 vagina away from it, S3 trocar path, S4 other skin/musculoskeletal, S5 intra-abdominal.
- The FDA classed transvaginal POP mesh as **class III (2016)** and it stopped selling in **2019**; slings and sacrocolpopexy (**transabdominal POP mesh, class II**) are exempt and proven safe, whereas mini-slings are not.

## 2 · Risk factors and prevention

- **Contraindications to vaginal mesh** are simultaneous urethrovaginal fistula repair, urethral injury or mesh perforation, and diverticulectomy. With **intraoperative urethral injury, abandon mesh** and place a sling later.
- Relative risks are prior fistula or diverticulum surgery, radiation, oestrogen-deficient tissue, smoking, obesity and age extremes; in men, radiotherapy and urethrotomy.
- POP mesh risk rises with high mesh volume, **shallow vaginal dissection**, hysterectomy, diabetes and sexual activity.

## 3 · Evaluation

**ALGORITHM**

1. History and **prior operative notes**.
2. Speculum and pelvic exam: exposure, bands, point tenderness, levator hypertonicity.
3. If haematuria, recurrent UTI, storage LUTS or voiding pain, then cystourethroscopy.
4. If voiding dysfunction, then urodynamics (UDS).
5. Translabial ultrasound for a sling; **MRI** for sacrocolpopexy mesh (invisible on CT); CT for haematoma.

Time never excludes a complication; slings can expose decades later.

## 4 · Midurethral sling complications

### Vaginal exposure

- It follows infection, haematoma, excess tension or superficial placement, and presents with bleeding, discharge, dyspareunia, partner pain or UTI.
- A small early exposure gets **topical oestrogen** (1 g cream nightly for 2 weeks, then 2–3 times weekly; or estradiol 10 µg tablet).
- **Excision**: raise a flap, expose the full width of mesh, pass an instrument behind, divide, excise at least **1 cm** around the defect and close. Avoid blind lateral dissection, which causes vulval haematoma after transobturator (TO) slings.

### Pain

- Causes are shrinkage, tension, obturator neuralgia and pelvic floor hypertonia.
- **Immediate severe pain**: exclude haematoma and consider **early removal** before ingrowth.
- Delayed focal pain gets pelvic floor physical therapy, trigger point injection and oestrogen, then segment excision (60–90% improve). Diffuse late pain reflects central sensitisation, and excision may fail.
- Remove **as much vaginal mesh as possible** (81% pain-free). For TO **groin pain**, add a thigh incision to obturator externus.

### Bladder perforation

- Cystoscopy with a **full bladder** detects trocar puncture. A missed one causes storage LUTS, haematuria, UTI, stones or vesicovaginal fistula (VVF).
- **Endoscopic** removal has low morbidity, but mesh stays in the wall and re-extrudes. **Open or robotic transvesical** removal excises 2 cm beyond the defect with two-layer closure and a catheter for 2–3 weeks; re-extrusion is minimal.

### Urethral perforation

- Causes are trocar injury, over-tensioning and **urethral dilation** (avoid it; divide an obstructing sling). It presents with voiding pain, retention or haematuria.
- Repair: inverted-U flap, peel the mesh off, minimal urethrotomy, layered closure with or without a **Martius** flap, and non-overlapping suture lines. A stricture needs buccal urethroplasty.
- After excision, SUI occurs in **42%**, fewer with a simultaneous fascial sling.

### Voiding dysfunction

- Midline **sling division** resolves retention in 95% and symptoms in 75%. Urgency incontinence persists in up to 76% and is de novo in up to 45%. Recurrent SUI (20–74%) is treated by bulking, repeat MUS or fascial sling.
- Persistent symptoms need UDS, then wider excision and urethrolysis.

> **KEY POINT**
>
> **How much to remove**: retention needs midline division; exposure at least 1 cm each side; pain maximal vaginal removal, rarely arms. **Never remove asymptomatic mesh**; more removal means more bleeding and more POP and SUI recurrence.

## 5 · Transvaginal POP mesh

- In the largest randomised trial (n = 1358), native and mesh repair were equal at 6 years (reoperation 12% vs 10%); 8% of the mesh arm needed excision.
- Exposure is 10–12% and removal 5–8%. **Contraction** (~12%) is unique to POP mesh and causes tightness and dyspareunia. Bladder perforation is 0.7–3.5%, rectal 0.15% and fistula 0.2%.
- Treat **conservatively** (observe, oestrogen) if <1 cm, mildly symptomatic and uninfected; only **size** predicts success. After office trimming, 73% later need theatre.
- **Excision**: raise thick flaps, then clamp behind the mesh and divide, which is the **most dangerous step**. Cut arms under tension rather than chasing them, then repair natively.
- Afterwards, cystocele recurs in 19%, 20% need repeat excision and 3% develop VVF.

## 6 · Sacrocolpopexy mesh

- Exposure is ~**10.5% at 7 years**, higher with total hysterectomy (23% vs 5%) and stage 3–4.
- Features are discharge, granulation, dyspareunia, cuff abscess, bladder or bowel perforation and **discitis**; image cross-sectionally.
- Cut an **exposed suture** below the knot in the office. Oestrogen for small exposures and vaginal excision often fail; partial colpocleisis succeeds in ~50% but is unsuitable if sexually active.
- **Abdominal removal** suits failed vaginal repair, pain, infection, discitis or ureteric complication. **Remove the entire infected mesh**, including sacral sutures, because residual mesh causes sinus, abscess and fistula; bowel injury occurs in up to 29%.

## 7 · Male slings

- **Advance XP** causes retention in 14–21%, usually for days; incise one arm if it persists. **Virtue** causes perineal or scrotal pain in 14–17%, chronic in 0–7%.
- **Urethral extrusion** usually follows a perforation missed without intraoperative cystoscopy. Excise lateral to the urethra and repair primarily or with buccal graft, avoiding transection.

# 121 Urinary Tract Fistulae

## 1 · Definition and universal principles

- A **urinary tract fistula** is an abnormal epithelial- or mesothelial-lined connection from the urinary tract to another cavity (bowel, vessel, genital tract) or skin.
- Causes are iatrogenic (commonest in the developed world), malignancy, inflammation, infection, ischaemia, trauma and childbirth.
- It persists because **urine keeps flowing through a tract that cannot heal**, so it epithelialises. Every principle diverts flow or restores healing, and the **first repair has the best chance**.

| Management (whole course) — Table 121.1 | Operation (any route) — Table 121.2 |
|---|---|
| Optimise nutrition | Evaluate concomitant injury; rule out malignancy |
| Control infection | Adequate exposure; remove all foreign material |
| Continuous drainage (catheter/stent) | Tension-free closure; vascularised edges |
| Eliminate obstruction | Multiple, watertight, **non-overlapping** layers |
| Consider malignancy — **always biopsy** if any history | Consider interposition; drain after |

> **KEY POINT**
>
> Trap: a fistula may be the only sign of recurrent cancer, so biopsy the tract after prior malignancy or radiation.

## 2 · Vesicovaginal fistula (VVF) — aetiology

VVF is the commonest acquired urinary fistula (**>75%** of genitourinary fistulae).

| | Developing world — obstetric (>90%) | Developed world — iatrogenic |
|---|---|---|
| Cause | Obstructed labour; young, small pelvis, poor nutrition | Unrecognised surgical injury; **>75%** after hysterectomy |
| Mechanism | Fetal head impacted for days causes pressure necrosis and a large defect | Dissection, cautery, clamp crush, bladder caught in cuff closure |
| Extent | Trigone, urethra, ureter often involved; "obstructed labour injury complex" (renal, rectal, orthopaedic, neurological, psychosocial) | Solitary, **supratrigonal at vaginal cuff** |

- **Hysterectomy** causes VVF in ≈ 0.8/1000, lowest after vaginal and highest after laparoscopic surgery. Distorted anatomy and concomitant prolapse, incontinence or mesh surgery raise risk; immediate recognition and repair prevent fistula.
- **Universal cystoscopy at hysterectomy** raises intraoperative detection but has an inconsistent effect on delayed injury. AUGS advises it with any prolapse or anti-incontinence procedure, AAGL after laparoscopic or robotic hysterectomy, and ACOG makes no recommendation after benign hysterectomy.
- **Radiation** causes obliterative endarteritis, fibrosis and necrosis. VVF follows 0.6–2% of cervical cancer radiotherapy, typically **1.5–2 years** later (up to 30); risk is dose-dependent and worse in smokers.

- Other causes are mesh, **bevacizumab** (anti-VEGF) and cancer, and rarely pessary, intrauterine device, infection or endometriosis.

## 3 · Classification of VVF

Of >25 systems, the WHO uses simple vs complex.

| Axis | Goh | Waaldijk |
|---|---|---|
| Basis | Distance of distal edge from external urethral meatus | Closing (continence) mechanism and urethral involvement |
| Types | 1: >3.5 cm · 2: 2.5–3.5 · 3: 1.5–2.5 · 4: <1.5 cm | I: closing mechanism spared · IIA: no (sub)total urethral involvement · IIB: (sub)total urethral involvement (A/B = without/with circumferential defect) · III: miscellaneous (ureteric) |
| Size | a <1.5 · b 1.5–3 · c >3 cm | Small <2, medium 2–3, large 4–5, extensive ≥6 cm |
| Tissue | i none/mild fibrosis · ii moderate/severe · iii special (radiation, ureteric, circumferential, previous repair) | — |

**KEY POINT**

Trap: Goh type 1 is farthest from the meatus (best) and type 4 closest (worst); closer means worse closure and continence. Goh predicts closure better. Always describe size, location (trigonal vs supratrigonal), tissue loss, and ureteric and urethral involvement.

## 4 · Diagnosis of VVF

- The classic symptom is **continuous leakage day and night** in any position; a small tract may leak only positionally.
- **Timing points to mechanism**: a missed cystotomy leaks in the first days, thermal or crush injury at **7–10 days**, and radiation after years. Fever, flank pain or ileus suggest urinoma or ureteric injury.
- On speculum the defect is usually at the vaginal apex after hysterectomy, and vaginal fluid creatinine approximates urine creatinine. Biopsy friable tissue after cancer or radiation, and optimise oestrogen, smoking, glucose and nutrition.
- **Single dye test**: backfill ~300 mL methylene blue, insert a tampon and walk 15–60 min. Proximal stain means VVF, distal stain means urethrovaginal fistula or SUI, and dye from the cervical os means vesicouterine fistula.
- **Double dye test**: oral phenazopyridine 100–200 mg plus blue in the bladder, then a tampon. **Orange means ureterovaginal fistula (UVF)**, blue means VVF, and both means both.
- **Cystoscopy is integral** (sensitivity ~92%) and shows size, number, orifices, bladder neck, mesh and capacity, and allows biopsy. Vaginoscopy shows the vaginal side. A cystogram (lateral view) or T2 MRI helps difficult cases.
- **Upper tract imaging is mandatory**, because ureteric injury coexists in up to **12%**. Use CT urography with a delayed phase, or MR urography if contrast allergy or renal failure. **Partial ureteric obstruction with VVF means UVF until proven otherwise.** If unclear, do retrograde pyelogram and stent, or percutaneous nephrostomy (PCN).

- Urodynamics helps planning: 10–15% have SUI needing an autologous sling, and radiation may reduce capacity and compliance.

> **KEY POINT**
>
> Work-up order: is it urine (creatinine), where is it from (dye tests, endoscopy), and what else is damaged (hence compulsory upper tract imaging). A preoperative PCN also allows antegrade ureteroscopy to find the ureter in scar.

## 5 · VVF — timing, conservative care, repair

### LAQ Vesicovaginal fistula: management, timing of repair, eponymous techniques and recurrent VVF

Paper answer · P1 · P2 · P3 · merged from 3

*Asked as:* P3 LAQ "Vesicovaginal fistula (VVF): management options and early vs delayed repair" · P2 SAQ "VVF repair: early versus delayed timing, and surgery for recurrent VVF" · P1 SAQ "Eponymous techniques of VVF repair"

#### Definition and aetiology

- VVF (vesicovaginal fistula): an epithelialised bladder–vagina tract (communication) causing continuous leakage; the commonest acquired urinary fistula (>75% of genitourinary fistulae).
- **Developing world (low-resource settings):** obstetric (obstructed labour) in >90%; the fetal head impacted for days causes pressure necrosis.
- **Developed world (high-resource settings):** iatrogenic in >75%, mainly after hysterectomy, usually supratrigonal at the vaginal cuff.
- Other causes: radiation, malignancy, mesh, foreign body.
- Common mechanism: **ischaemic necrosis**; continuous urine flow keeps the tract open until it epithelialises with time.
- Time to presentation: cystotomy, days; thermal injury, 7–10 days; radiation, months to years.

#### Classification

| Goh (distance from meatus) | Size | Tissue |
|---|---|---|
| 1: >3.5 cm; 2: 2.5–3.5; 3: 1.5–2.5; 4: <1.5 cm | A <1.5 cm; B 1.5–3 cm; C >3 cm | i mild fibrosis; ii moderate/severe; iii radiation, ureteric, circumferential, previous repair |
| - **Also know:** Waaldijk (closing mechanism involvement); WHO (World Health Organization) simple vs complex. | | |

#### Evaluation (guides choice)

- Speculum; fluid creatinine.
- **Dye tests:** single (blue backfill, tampon); **double** (oral phenazopyridine plus blue): orange = UVF (ureterovaginal fistula), blue = VVF.
- **Cystoscopy and vaginoscopy:** number, size, relation to ureteric orifices and bladder neck, mesh; **biopsy the tract if any history of cancer** (the fistula may signal recurrence).

- **Upper tract imaging (CTU, computed tomography urography) is mandatory:** to exclude a coexisting ureteric (ureterovaginal) fistula (up to 12–25%).
- Urodynamics: capacity, compliance, SUI (stress urinary incontinence) in 10–15%.
- Prepare: treat UTI (urinary tract infection), optimise nutrition, vaginal oestrogen for atrophy; counsel about residual incontinence, dyspareunia and possible reoperation.

### Principles of management

- Optimise nutrition, treat infection, drain continuously, relieve obstruction, exclude malignancy.
- Surgery: excise scar, wide mobilisation, tension-free, multilayer, **non-overlapping**, watertight closure; remove foreign material; consider interposition; drain afterwards. **The first repair has the best chance.**
- At repair: treat infection; ureteric stents if within 1 cm of a ureter; vaginal or abdominal route (open, laparoscopic, robotic); **interposition flap** (Martius, omentum, peritoneum) if irradiated or recurrent; catheter drainage for about 2 weeks (≥10–14 days).

### Option 1: Conservative and endoscopic (conservative trial while waiting)

- **Catheter drainage** cures about 15%. Best if small (<1 cm, ideally <3 mm), fresh, started (catheterised) within **3 weeks**, and non-irradiated. Once epithelialised (≥6 weeks), closure is unlikely. EAU (European Association of Urology): if tried, at least 12 weeks.
- **Fulguration** plus 2 weeks' drainage for tiny tracts <3.5 mm.
- **Sealants** (fibrin glue, cyanoacrylate, collagen): tiny series; not recommended by EAU.

### Timing: early versus delayed repair (the debate)

| | **Early repair** | **Delayed repair** |
|---|---|---|
| Definition (varies) | Days, <6 weeks or <3 months | Traditionally ≥3 months |
| Rationale | Clean iatrogenic tissue heals well; ends wetness and distress early | Inflammation settles, necrotic tissue sloughs, margins become clear |
| Evidence | **Comparable closure** in most series; Waaldijk: immediate obstetric repair gave very high first-attempt closure | One small series: more recurrence when repaired <6 weeks |
| Drawback | Friable tissue if inflamed | Prolonged distress, no better outcome |

- Some authors avoid the **4–12 week** window (active healing phase).
- **EAU consensus:** no minimum delay if there is **no active infection and no recent irradiation**.
- **Timing by cause:**
- **Clean iatrogenic / post-hysterectomy:** early repair.
- **Obstetric with necrosis (necrotic edges):** catheter trial, then repair at about 3 months.
- **Radiation:** wait **≥12 months** after radiotherapy; biopsy to exclude recurrence; interposition flap; consider diversion if the bladder is poor.
- **Recurrent (redo) fistula:** wait until inflammation settles (commonly about 3 months); radiation ≥12 months after radiotherapy; failed radiation repairs were redone after **≥6 months**.

### Option 2: Surgical route

| Vaginal | Abdominal (open, laparoscopic, robotic) |
|---|---|
| Low, distal or trigonal, accessible apex | Supratrigonal, narrow vagina, no descent |
| Ureters distant | Ureter involved, needs reimplantation |
| Less pain, blood loss, shorter stay | Augmentation, **Tanagho bladder flap** or diversion possible |
| Flap-splitting, offset flaps | Large (≥4 cm primary, **≥2 cm recurrent**), radiation |
| Failed **Latzko** can be cured by a second Latzko | **O'Conor** bivalving finds the scarred plane |
| Risk: vaginal shortening | |
| - First-attempt closure about **90% by either route**; choose the route the surgeon knows best. | |

### Eponymous and named techniques: transvaginal

- **Sims:** first successful systematic vaginal repair (silver wire).
- **Flap-splitting (Raz):**
- Ureteric stents if within 1 cm; Foley through the tract for traction.
- Inverted J or U incision with a 1–3 mm cuff; raise anterior and posterior vaginal flaps.
- Layer 1: bladder (3-0 or 4-0), including the tract edge. Layer 2: perpendicular perivesical layer. Fill test 200–300 mL.
- Optional Martius or peritoneal flap; advance the opposite vaginal flap ≥2 cm beyond. Closure ≥90%.
- **Latzko (1942), high partial colpocleisis:**
- Denude vaginal epithelium 1–2 cm around the tract; the bladder is **not opened or dissected**.
- Close the vaginal walls over the tract in layers.
- ≥90% success in non-irradiated post-hysterectomy apical VVF; favoured for radiation VVF and tracts near the ureters. Drawback: **vaginal shortening and dyspareunia**.
- **Also know:** vaginal cuff excision (all apical scar excised as an inverted cone).

### Eponymous and named techniques: abdominal

- **O'Conor and Sokol (1951), suprapubic extraperitoneal:**
- Midline cystotomy at the dome, **bivalved down** to the fistula; ureteric catheters.
- Excise the tract; separate bladder from vagina 2–3 cm beyond.
- Close the vagina; **omental or peritoneal flap** between; close the bladder in layers; suprapubic catheter.
- Modifications: **transvesical** (limited cystotomy) and **extravesical** (no cystotomy; fewer LUTS [lower urinary tract symptoms]).
- **Laparoscopic (1994) and robotic (2005):** equal closure (85–100%) with less blood loss, shorter stay and less morbidity; vaginal stent, backfill, ICG (indocyanine green) perfusion check. Robotic repair of recurrences gives high closure.

### Interposition flaps (radiation, recurrent, ischaemic or large VVF)

| Flap | Blood supply | Route | Use |
|---|---|---|---|
| **Martius (1928)** (labial fibrofatty pad) | Internal pudendal (inferior), external pudendal (superior) | Vaginal | Mid–distal VVF, bladder neck; does not reach the apex |
| Peritoneal (vaginal) | Local; pouch of Douglas fold | Vaginal or abdominal | High apical post-hysterectomy VVF |
| **Omentum (J-flap)** | Gastroepiploic arteries | Abdominal | Radiation, recurrent, large |
| Gracilis, Singapore, rectus | Myocutaneous | Vaginal/perineal | Refractory, large tissue loss |
| - RCT (randomised controlled trial): Martius added nothing in simple VVF, with donor-site pain in nearly a third. | | | |
| - **Interposition is widely used in recurrent fistula** (accepted, though not proven superior). | | | |
| - **Also know:** rectus abdominis flap (last resort). | | | |

### Recurrent VVF

- Persists after a failed repair; redo tissue is scarred and poorly vascularised, so the first repair has the best chance.
- Failure causes: tension, ischaemia, overlapping suture lines, infection, poor drainage, **malignancy, radiation**, missed second tract.
- Before redo: biopsy, cystoscopy and vaginoscopy, CT urography and dye test to exclude a **ureterovaginal fistula**, urodynamics; treat infection, optimise nutrition.
- Route: abdominal for recurrent ≥2 cm; a failed Latzko can be redone vaginally; robotic repair gives high closure.
- Interposition flap; catheter ≥10–14 days, then cystogram.
- More than one repair predicts residual incontinence.

### Postoperative care

- Catheter ≥10–14 days (urethral and/or suprapubic); antimuscarinic or β3-agonist for spasm; cystogram before removal; stents out at 4–6 weeks.

### Option 3: Urinary diversion (when repair is unwise)

- Multiple failures, severe radiation damage, massive tissue loss: conduit or continent reservoir/diversion.
- Unfit: ureteric occlusion with nephrostomies.
- Resource-poor: ureterosigmoidostomy as a last resort.

### Outcomes

- Residual urgency or incontinence despite closure (Goh 3–4 predicts; more than one repair predicts); retention (vertical bladder closure); dyspareunia (narrow vagina, fistula ≥3 cm). ICS (International Continence Society): pelvic floor physiotherapy.

### Diagrams to draw

- Flap-splitting repair: perpendicular bladder layers, Martius flap, advanced vaginal flap.
- Latzko denuded area versus flap-splitting layers.
- Timing timeline: early window, 4–12 week zone, 3 months (obstetric), 12 months (radiation).
- Martius flap between bladder and vaginal closures.

**Key points to remember:**

- Image upper tracts; exclude ureteric fistula and cancer (biopsy before redo).
- Catheter trial only if small, fresh (<3 weeks), non-irradiated; works only if started within 3 weeks.
- Vaginal and abdominal routes both close about 90%.
- Early repair for clean iatrogenic VVF; closure comparable to delayed.
- Delay ~3 months for obstetric, ≥12 months for radiation fistula.
- First repair has the highest success.
- Interposition flap (Martius, peritoneum, omentum) for complex and recurrent VVF.
- Diversion after multiple failures or severe radiation.
- Latzko: colpocleisis, bladder not opened, vagina shortened.
- O'Conor: bivalve bladder to the fistula, interpose omentum.
- Martius for distal, omentum or peritoneum for apical VVF.

**Also, from the condensed book (not in the paper answer):**

- Size >3 cm predicts failure, and radiation strongly predicts failure with ureteric stricture, poor compliance and sphincter deficiency, so consider diversion.
- Apical VVF (8–10 cm from introitus) lies beyond Martius reach, so use peritoneum or omentum.

| Setting | Timing of repair | Rationale |
|---|---|---|
| Obstetric (traditional) | ≥3 months, catheter meanwhile | Inflammation settles, necrosis sloughs |
| Iatrogenic, clean tissue | Early; European Association of Urology (EAU): no minimum delay if no infection or recent radiation | Comparable closure; less distress |
| Radiation-induced | ≥**12 months** after radiation ends | Late injury still evolving |

- **Catheter** drainage is favourable for tracts <1 cm catheterised within **3 weeks**, and unfavourable at ≥6 weeks or with radiation or malignancy. The EAU trial lasts ≥**12 weeks**. Sealants are possible for <1 cm in poor surgical candidates.
- **Location**: within **1 cm** of a ureter, stent it; if the ureter is involved, reimplant abdominally. Bladder neck involvement needs reconstruction, plus a pubovaginal sling if needed.
- **Route**: there are no RCTs, and first-attempt closure is ≈ **90%** either way. Transvaginal repair has less blood loss, pain and stay.
- Abdominal repair suits large fistulae (≥4 cm primary, ≥2 cm recurrent), urethral, bladder-neck or ureteric involvement, a non-capacious vagina, radiation, or no lithotomy. Vaginal repair risks shortening and dyspareunia, whereas abdominal repair spares sexual function.

**ALGORITHM**

1. **Transvaginal flap-splitting (Table 121.5)**
2. Cystoscopy: stent close ureters; wire through the tract.
3. Suprapubic catheter (SPC) and urethral catheter; episiotomy if narrow.
4. Foley ≥8 Fr through the fistula for traction; hydrodissect.
5. **J- or U-incision** with a 1–3 mm cuff; raise wide anterior and posterior flaps.
6. Layer 1: 3-0/4-0 delayed absorbable. Layer 2: **perpendicular** to layer 1.
7. Leak test 200–300 mL; Martius or peritoneal flap if needed.
8. Advance the opposite flap ≥2 cm beyond (layer 3); cystoscopy for efflux.

- **Postoperative drainage is the most important part**: urethral catheter, often with SPC, for ≥10–14 days. Give an anticholinergic or β3-agonist for spasms, remove stents at 4–6 weeks, and advise pelvic rest for ~6 weeks.
- **Latzko** leaves the bladder unopened and the tract unexcised: denude 1–2 cm and close in two or more layers. It succeeds in ≥90% of non-irradiated cases and suits close orifices, but shortens the vagina.

**ALGORITHM**

1. **O'Conor (Table 121.7)**
2. Ureteric catheters; low midline extraperitoneal approach.
3. Sagittal dome cystotomy; **bivalve** down around the fistula; excise the tract.
4. Separate bladder from vagina ≥2–3 cm beyond; close the vagina.
5. Place **omentum or peritoneum** beyond the vaginal suture line.
6. Close bladder in 1–2 layers; SPC with or without urethral catheter; fill test.
7. Drain 10–14 days, then cystogram.

- O'Conor causes more de novo urgency and LUTS from long suture lines. Transvesical repair suits low-lying fistulae, and extravesical repair suits supratrigonal fistulae. Robotic O'Conor uses omentum or peritoneum, with indocyanine green to check edge vascularity.

## 6 · Interposition flaps

- Flaps reinforce, fill dead space and import blood supply. Use them for **complex VVF** (radiation, recurrent, ischaemic obstetric, poor tissue); an RCT showed **no benefit of Martius** in simple VVF.
- **Martius** is supplied by the internal pudendal (inferior), external pudendal (superior) and obturator (lateral) vessels. Keep the **superior** pedicle for VVF (inferior for rectovaginal). It reaches mid-distal VVF, bladder neck and urethra, but **not the apex**.
- A vaginal peritoneal flap suits the apex, and **omentum** suits radiation, recurrent or large fistulae. Rectus, gracilis and Singapore flaps are a last resort.
- After repair, residual incontinence (predicted by Goh 3–4 and more than one repair) drives quality of life, and ICS advises physiotherapy for all obstetric fistula. Divert if the fistula is unrepairable.

# 7 · Ureterovaginal fistula (UVF)

## LAQ Urological complications of obstetric-gynaecological surgery: uretero- and urethrovaginal fistula

Paper answer · P3 · merged from 2

*Asked as:* P3 LAQ "Urological complications in OBS-GYN and uretero-vaginal fistula" · P3 LAQ "Urological complications of OBGY surgery and urethrovaginal fistula"

*Assumed meaning: OBS-GYN / OBGY = obstetric and gynaecological practice and surgery.*

### Part A: Urological complications in OBS-GYN

- The bladder, distal ureter and urethra lie against the cervix, uterine vessels and anterior vaginal wall. Risk rises with distorted anatomy: fibroids, endometriosis, cancer, previous caesarean, radiation, heavy bleeding, prolapse or mesh surgery.

#### By setting

| Setting | Complications |
|---|---|
| **Obstetric** | **Obstructed labour** (fetal head impacted for days): pressure necrosis, so large VVF (vesicovaginal fistula), urethral loss, ureteric involvement; over 90% of VVF in the developing world. **Caesarean**: bladder injury, ureteric injury, **vesicouterine fistula** (Youssef syndrome: cyclical haematuria, amenorrhoea, continence preserved). Post-partum or post-epidural **retention** (overdistension) |
| **Gynaecological surgery** | **Bladder injury** (commonest organ). **Ureteric injury** (over 90% distal third; ligation commonest). **Fistulae**: VVF (over 75% follow hysterectomy in the developed world), UVF (ureterovaginal fistula), urethrovaginal fistula |
| **Radical hysterectomy** | Pelvic plexus injury: retention, poor compliance; manage with CIC (clean intermittent catheterisation) |
| **Prolapse and continence surgery** | Ureteric kinking at apical suspension; mesh or sling exposure, perforation, outlet obstruction, pain |
| **Radiotherapy** | Radiation cystitis, VVF, ureteric stricture |

#### By complication: key facts

| Complication | Key facts |
|---|---|
| **Bladder injury** (commonest organ injured) | Caesarean 0.27–0.3%; hysterectomy about 0.3–0.75% (laparoscopic highest); retropubic tape 3–9%. Repair: debride, 2-layer closure, fill test, catheter 7–14 days; check ureters if trigonal |
| **Ureteric injury** | Gynaecology causes over half of iatrogenic injuries; over 90% distal third. Hysterectomy overall 0.2–2.4%: vaginal 0.02–0.5%, abdominal 0.03–2%, minimally invasive 0.13–6%; prolapse surgery 1.7–4.3%. **Ligation** is the commonest gynaecological mechanism |
| **Fistulae** | VVF about 0.8 per 1000 hysterectomies; UVF; vesicouterine fistula after caesarean (Youssef syndrome); urethrovaginal fistula |
| **Voiding dysfunction** | Radical hysterectomy (pelvic plexus injury) 16–80% historically: retention, poor compliance; CIC and wait. Post-partum or post-epidural overdistension |
| **Mesh and sling** | Exposure, erosion into bladder or urethra, outlet obstruction, pain, dyspareunia |

| Complication | Key facts |
|---|---|
| **Radiation** | Radiation cystitis; VVF in 0.6–2% after cervical cancer radiotherapy; ureteric stricture |

- **Ureteric danger sites (four):** pelvic brim (near the infundibulopelvic ligament); **under the uterine artery** in the cardinal ligament; lateral to (beside) the uterosacral ligament; anterolateral vaginal fornix near the UVJ (ureterovesical junction). Most injuries are in the **distal 6 cm**.
- **Prevention and recognition:** haemostasis and anatomy; know the danger sites; prophylactic stents only selectively, if high risk (they aid recognition but do not prevent injury); cystoscopy with dye (IV indigo carmine or oral phenazopyridine) after hysterectomy where indicated, minimally invasive hysterectomy, and prolapse or continence surgery.
- Only about one-third of ureteric injuries are recognised at open surgery. A missed injury presents with fever, ileus, flank pain, urinoma or leakage; do CTU (computed tomography urography) with delayed films, then retrograde stenting. Distal repair: ureteroneocystostomy ± psoas hitch or Boari flap (see Part B).

## Part B: Ureterovaginal fistula (UVF)

### Definition

- Abnormal communication between the ureter and the vagina, causing continuous leakage.

### Aetiology and pathogenesis

- Usually an **unrecognised injury to the distal third of the ureter** during pelvic surgery. About **three-quarters follow an often uneventful abdominal hysterectomy**.
- **Mechanisms:** ligation, crush, laceration, thermal injury, **devascularisation** or ischaemia. The injured segment necroses and leaks into the vaginal cuff, which is the weakest point.
- Diagnosis is delayed in 18–88%; **delayed recognition raises fistula risk about fourfold**.
- Other causes: obstetric, cancer, radiation, endometriosis, foreign body.

### Clinical features

- **Continuous leakage with normal voiding** (the other kidney fills the bladder), usually appearing **within 4 weeks**, often with fever, flank pain or ileus, as thermal or ischaemic injury necroses.

| | VVF | UVF |
|---|---|---|
| Voiding | Reduced | Normal |
| Upper tract | Normal | Hydronephrosis common |
| First line | Catheter, repair | Stent |

### Investigations (confirm before treatment)

| Test | Method and reading |
|---|---|
| **Double dye test** | Oral phenazopyridine 100–200 mg (orange urine); after 1 hour, blue dye instilled into the bladder; tampon; walk about. **Orange = UVF**; blue = VVF; both = both |
| **Three-swab test** | Blue dye in the bladder; three swabs stacked in the vagina. Blue upper swab = VVF; **wet, unstained swab = UVF** |

| Test | Method and reading |
|---|---|
| **CT urography, delayed phase** | Hydronephrosis, extravasation; MR (magnetic resonance) urography if contrast is contraindicated |
| **Cystoscopy + RGP (retrograde pyelogram)** | Abrupt cut-off or contrast leak; stent at the same sitting |
| - **Exclude a coexisting VVF** (up to 25% of VVF have a UVF) and bilateral injury. | |

**Management**

- **Goals:** dryness and preservation of renal function. Management has moved from nephrectomy, through reimplantation, to endourology.
- **1. Endourological (first line): stent first**
- **Retrograde double-J stent as soon as possible** if a wire passes. Stenting **within 2 weeks** of detection works best, succeeding far more often than at 2–6 weeks, because the ureter heals over the stent before fibrosis.
- Wire will not pass: **PCN (percutaneous nephrostomy)** with antegrade stent, or rendezvous.
- Stent for **4–6 weeks**, then remove and re-image. Most stentable fistulae close; a stricture may follow.
- **2. Surgical repair (stent fails or impossible)**
- **Timing:** no consensus. Early repair equals delayed: early repair took less time and blood loss with equal success; a systematic review found equal success before and after 6 weeks. Delay only if tissue is still inflamed.
- **Ureteroneocystostomy** (distal third, so usually feasible):
  1. Transperitoneal or extraperitoneal approach; identify the ureter above the fistula.
  2. Divide above the diseased segment; dissect as distally as possible for length; **preserve periureteric adventitia**, especially laterally where vessels enter.
  3. Mobilise the bladder; spatulate; tension-free mucosa-to-mucosa anastomosis over a double-J stent; refluxing or tunnelled.
  4. Drain; catheter 7–14 days.
- **Adding length:**

| Technique | Extra reach |
|---|---|
| **Psoas hitch** (sutures parallel to psoas fibres, spare genitofemoral nerve) | Up to about 5 cm |
| **Boari flap** (base ≥4 cm, length:base ≤3:1) | 10–15 cm |
| Downward nephropexy; ileal ureter | Longer defects |
| - Open, **laparoscopic and robotic repair give equal results**; vaginal reimplantation reported. **Success ≥90%.** | |
| - The vaginal opening is left to heal; no separate vaginal repair is needed. | |

**Algorithm** `Leak + normal voiding -> exclude VVF (dye test, cystoscopy) -> CT urography / RGP -> wire/stent passes? Yes -> double-J 4-6 wk -> re-image ->`

`healed / persistent No -> PCN + antegrade stent -> fails -> ureteroneocystostomy + psoas hitch / Boari flap`

**Complications and follow-up**

- Hydronephrosis, infection, ischaemic stricture.
- Ultrasound 1–2 weeks after stent removal; some add a renal scan at 6 weeks. Stricture usually appears within a year but may appear **up to 30 months** later, so follow beyond this.

### Part C: Urethrovaginal fistula

**Aetiology**

- Developing world: **obstructed labour**, often with a VVF; also trauma, radiation, catheter injury.
- Developed world (iatrogenic): **urethral diverticulectomy** (fistula in 1–8%; causes 20–25% of urethrovaginal fistulae), slings, bulking agents, anterior repair or mesh, indwelling catheters (pressure necrosis), pessaries, radiation, trauma; also gender-affirming and paediatric genital surgery.

**Clinical features**

- Depend on site relative to the sphincter. **Distal** fistula: often asymptomatic, split stream, vaginal voiding, post-void dribble. **Proximal** (bladder neck): continuous leakage or SUI (stress urinary incontinence).
- Dyspareunia, recurrent UTI (urinary tract infection), perineal irritation; partner discomfort with exposed mesh.

**Evaluation**

- **Speculum examination** usually suffices; **methylene blue test** if rugae hide it (distal stain on tampon). Assess prolapse, SUI, atrophy.
- **VCUG (voiding cystourethrogram), lateral view;** VUDS (videourodynamics) for associated SUI or detrusor overactivity.
- **Cystourethroscopy** (flexible is better: outflow at the lens distends the urethra): size, site, bladder neck, exposed mesh, other fistulae. **About 20% have a separate VVF.** Image the upper tract.
- Goh location (distance from external meatus): type 1 >3.5 cm, 2: 2.5–3.5, 3: 1.5–2.5, 4: <1.5 cm. Closer means worse closure and continence.
- Optimise: treat UTI, vaginal oestrogen, nutrition; biopsy if previous cancer.

**Management**

- Asymptomatic distal fistula: may need no repair. Timing: early if tissue is healthy; otherwise after inflammation settles.
- **Standard: transvaginal rotational-flap repair:**

| Step | Details |
|---|---|
| 1. Exposure | Lithotomy (prone gives better anterior exposure); cystourethroscopy, cannulate tract; suprapubic catheter (Lowsley tractor), urethral catheter |
| 2. Incision | Hydrodissect; circumscribe tract with 1–2 mm cuff, minimal excision; inverted **J- or U-shaped** vaginal flap incision |
| 3. Flaps | Raise vaginal flap; mobilise periurethral tissue widely to bladder neck; open retropubic space if sling planned |

| Step | Details |
|---|---|
| 4. Closure | **Close urethra** over a urethral sound: **4-0 absorbable** sutures, transverse or longitudinal; **second layer perpendicular**; buccal graft or flap onlay for large defects; leak test |
| 5. Interposition | **Martius flap** (labial fat pad, external pudendal pedicle) most often; omentum, gracilis, rectus fascia occasionally |
| 6. Cover | Advance vaginal flap beyond closure (2-0) |

- **Autologous pubovaginal sling** for SUI or bladder-neck involvement; **never synthetic** (erodes the ischaemic urethra). Remove any mesh in the field.
- **Near-total urethral loss:** tubularised anterior bladder flap neourethra. Irreparable cases: diversion.
- **After repair:** urethral catheter out at 2–3 days; suprapubic catheter ≥14 days; lateral VCUG before removal; nothing per vagina for 6 weeks.

### Outcomes and complications

- Closure **90–95%**, but continence only **50–80%**; postoperative SUI in up to half. Sling then or staged later.
- Urethral stricture, recurrence, dyspareunia.

### Diagrams to draw

- Ureter passing under the uterine artery; double dye test; ureteroneocystostomy with psoas hitch.
- Urethrovaginal repair: inverted-U vaginal flap, two perpendicular urethral layers, Martius flap.

**Key points to remember:**

- Bladder is the commonest organ injured; obstructed labour causes most fistulae worldwide.
- Ureteric injury is mostly distal and missed.
- UVF: distal ureteric injury at hysterectomy; continuous leak within 4 weeks with normal voiding.
- Orange tampon = UVF on double dye test; exclude coexisting VVF.
- Stent early (within 2 weeks); remove at 4–6 weeks.
- Reimplant with psoas hitch or Boari flap if stenting fails; success ≥90%; follow 30 months.
- Urethrovaginal fistula: diverticulectomy is the commonest Western cause; obstetric in developing countries.
- Look for a coexisting VVF (about 20%).
- Layered closure plus Martius flap; autologous sling only.
- Closure 90–95% (closure is easy); continence only 50–80% (continence is hard).

**Also, from the condensed book (not in the paper answer):**

- Up to **25%** of VVF have a coexisting UVF. Prophylactic stents are selective (e.g. caesarean hysterectomy for placenta accreta).

**ALGORITHM**

1. Suspect UVF; exclude VVF; confirm with CTU or retrograde pyelogram.
2. Stent; remove at 4–6 weeks and reimage.
3. If resolved, stop. If persistent or stenting fails, then do ureteroneocystostomy.

**KEY POINT**

UVF: wet but voids normally. The ureter heals over a stent if reached early, so the detection-to-stent interval largely decides whether surgery is needed.

## 8 · Vesicouterine fistula (VUF)

- It follows **low transverse caesarean** and joins the uterine body above the internal os to the posterior bladder; a competent cervix **preserves continence**.
- **Youssef syndrome** is menouria, amenorrhoea, first-trimester miscarriage and continence. Bladder endometriosis is the differential of menouria.
- Dye appears from the cervical os, and cystoscopy shows a posterior supratrigonal defect; add cystogram, hysterosalpingography or MRI.
- Spontaneous closure is ~5%. After 3–4 months, separate transabdominally and close in layers with omentum or peritoneum, or do hysterectomy if the family is complete. Deliver later pregnancies by caesarean.

## 9 · Urethrovaginal fistula

*See the paper answer **Urological complications of obstetric-gynaecological surgery: uretero- and urethrovaginal fistula** (Section 6, Ch 121).*

## 10 · Rectourethral fistula (RUF)

### LAQ Recto-urethral (recto-prostatic) fistula: etiopathogenesis, diagnosis and management

Paper answer · P3

**Asked as:**

- Etio-pathogenesis, diagnosis and management of Recto-urethral fistula? — *Summer 2020 Paper III Q1 (LAQ)*
- Etiopathogenesis, diagnosis and management of Recto-Prostatic fistula — *Winter 2025 Paper III Q1 (LAQ)* (also answered in 2020-Summer Paper 3, Q1, as recto-urethral fistula)

**Type:** LAQ (25 marks)

*Assumed meaning (recto-prostatic): rectourethral fistula at the level of the prostatic urethra or the vesicourethral anastomosis.*

**Definition**

- **RUF (rectourethral fistula):** a communication between the posterior urethra / prostatic urethra (or the vesicourethral anastomosis after prostatectomy) and the distal rectum.
- Rare but very morbid. Almost never seen in females, because the vagina lies between urethra and rectum.

## Aetiology

| Group | Causes |
|---|---|
| Congenital | Anorectal malformation in boys (rectobulbar or rectoprostatic fistula) |
| Surgical (commonest acquired) | **RP (radical prostatectomy)**; historically TURP (transurethral resection of the prostate) and open (simple) prostatectomy |
| Radiation and energy | **EBRT (external beam radiotherapy) plus brachytherapy** (commonest), then brachytherapy alone, then pelvic radiotherapy; cryotherapy; HIFU (high-intensity focused ultrasound): low after primary, up to about 9% after salvage HIFU |
| Others | Trauma (penetrating injury, pelvic fracture); Crohn disease; advanced rectal or prostate cancer |

## Pathogenesis

- **Surgical:** the rectum is torn during apical or posterior dissection behind Denonvilliers fascia, especially with small glands and salvage surgery. An unrecognised tear (near the apex) leaks urine (urine leaks into it), an abscess forms and the tract epithelialises.
- Rectal injury at RP: laparoscopic about 1%, open and robotic under 0.5%, salvage RP about 7%. (Another source: rectal injury is commonest with retropubic and laparoscopic RP, lowest with robotic RP, and was over 15% in historical open salvage series.)
- **Radiation or ablation:** obliterative endarteritis makes the tissue hypoxic and fibrotic. Necrosis follows months to years later. These fistulae are large and heal poorly.
- **Congenital:** abnormal cloacal development leaves the rectum ending in the posterior urethra.

## Classification: Martini (after prostate cancer treatment)

| Axis | Category | Definition |
|---|---|---|
| Stage (size on CT (computed tomography) urography or cystography) | I / II / III | ≤1.5 cm / >1.5 cm / any size with sphincter damage |
| Grade (tissue) | 0 | Surgical injury, no prior non-surgical treatment |
| | 1 | Prior radiation, brachytherapy, HIFU or other energy |
| | 2 | Salvage RP or salvage ablation |
| - **Tissue quality (grade), not size, chiefly decides the repair.** | | |

## Clinical features

- **Faecaluria and pneumaturia** (most typical); urine per rectum on voiding.
- Recurrent UTI (urinary tract infection), dysuria, urosepsis; coexisting SUI (stress urinary incontinence).

## Diagnosis

- **DRE (digital rectal examination):** may feel the defect on the anterior rectal wall.
- **Cystoscopy and proctosigmoidoscopy:** show the tract and tissue vitality; **biopsy** to exclude recurrent cancer.

- **VCUG (voiding cystourethrogram) or RUG (retrograde urethrogram):** usually conclusive; gives size and site for planning.
- Upper-tract imaging to exclude ureteric injury; pelvic MRI (magnetic resonance imaging) for complex cases.
- **UDS (urodynamics)** with rectal packing: fistula repair alone will not cure severe SUI; an AUS (artificial urinary sphincter) may be needed later; a later AUS erodes more often.

### Prevention

- At RP: a recognised tear without gross spillage → two-layer (multilayer) closure, air test, **omental or peritoneal interposition**, drain, antibiotics; colostomy if (previously) irradiated.
- Radiotherapy: a **hydrogel rectal spacer** lowers rectal dose and late rectal toxicity (randomised trial: late rectal toxicity 2.0% vs 7.0%).

### Management

- **Principles:** control sepsis; divert urine ± faeces; let inflammation settle (about 12 weeks); exclude cancer; then excise the tract and close urethra and rectum separately in **tension-free, non-overlapping layers with vascularised (interposition) tissue interposed**.
- **Conservative:** no sepsis and no faecaluria → catheter (urethral or suprapubic), low-residue diet ± colostomy. A simple fistula may close in about 12 weeks; irradiated or complex ones rarely do.
- **Surgery:** no single accepted operation; choice depends on site, radiation and prior repair. **A diverting colostomy usually comes first.**

| Approach | Essentials | Strengths | Weaknesses |
|---|---|---|---|
| **Perineal** (Young; Goodwin) | Separate and close urethra and rectum separately; **gracilis flap** between (or dartos; buccal graft) | High closure; most stomas reversed | Urinary incontinence common |
| **Transsphincteric (York-Mason)** | Prone jackknife; sphincter divided in tagged layers; through anterior rectal wall; three-layer closure | Avoids scarred plane; spares sexual function | Poor membranous exposure; no flap; faecal incontinence up to 20% |
| **Abdominal** (transvesical, extravesical, O'Conor-like) / robotic | Omental flap | Large or irradiated fistulae, extensive resection | Morbid; poor deep-pelvic view; small robotic series |
| **Laparoscopic / robotic** | Open pouch of Douglas; close rectum in two layers; omental or peritoneal flap; can combine with perineal gracilis | Deep-pelvic vision | Small series only |
| **Salvage** | Cystoprostatectomy, ileal conduit ± permanent colostomy | Unreconstructable irradiated field | Major surgery |

`RUF → sepsis control, catheter ± colostomy, biopsy small, non-irradiated, no faecaluria → conservative ~12 weeks persistent, grade 0 → perineal (gracilis) or York-Mason irradiated or large → abdominal/robotic (omentum) or perineal (gracilis) unreconstructable → urinary + faecal diversion Congenital → newborn colostomy → posterior sagittal anorectoplasty`

- **PSARP (posterior sagittal anorectoplasty)** separates the rectum from the urethra and places it within (in) the sphincter complex.

### Complications

- **Recurrence** (the greatest challenge), urinary or faecal incontinence, urethral stricture, ED (erectile dysfunction), wound infection, permanent stoma.

**Diagram to draw:** sagittal pelvis showing (with) the fistula and a gracilis flap between urethra and rectum.

> **Key points to remember:**
> - Most acquired RUFs follow prostate cancer treatment.
> - Martini: stage = size; grade = tissue (surgery, radiation, salvage).
> - VCUG or RUG plus endoscopic biopsy to exclude cancer.
> - Divert first; perineal (gracilis) or York-Mason for surgical fistulae; irradiated ones need imported tissue or diversion.

| Martini | Category |
|---|---|
| Stage (size) | I ≤1.5 cm · II >1.5 cm · III any size with sphincter damage |
| Grade (tissue) | 0 surgical injury only · 1 prior radiation/energy · 2 salvage radical prostatectomy (RP) or salvage ablation |

## 11 · Vesicoenteric fistula (VEF)

- Colovesical is commonest. M : F is **3 : 1** because the uterus shields the bladder.
- Causes in order are **diverticulitis**, **Crohn disease** (ileovesical) and colorectal cancer (10–15% of colovesical fistulae), then bladder or prostate cancer, radiation and appendicitis.
- It causes **pneumaturia** (50–70%), faecaluria and polymicrobial UTI. **Gouverneur syndrome** is suprapubic pain, frequency, dysuria and tenesmus.

| Test | Role |
|---|---|
| Poppy seed test | Highest detection of presence; cheap; no location |
| **CT** (preferred) | Air in bladder, wall thickening, adherent bowel; shows cause |
| Cystoscopy | Nonspecific erythema/oedema |
| Colonoscopy | Poor for tract; **excludes cancer** |
| MRI | Crohn, anatomy |

- Treat conservatively if symptoms are minimal and disease benign (especially Crohn); spontaneous closure is <2%.
- **Resect the diseased bowel with the fistula** with single-stage anastomosis; use Hartmann for abscess, contamination, advanced cancer or radiation. Leave or oversew a small bladder defect, do **partial cystectomy** if malignant, and interpose omentum.

> **KEY POINT**
>
> VEF is a bowel disease presenting in the bladder: remove the diseased bowel, and the bladder usually heals itself.

**Renocolic fistula**: spontaneous ones are mostly renal (xanthogranulomatous pyelonephritis [XGP]) and need nephrectomy; small iatrogenic ones after PCNL settle with antibiotics, drainage and nil by mouth.

## 12 · Urocutaneous fistulae

- **Vesicocutaneous** fistula follows long-term SPC, open prostatectomy or radiation, and persists with infection, foreign body, tumour or bladder outlet obstruction (**BOO**). Drain and relieve BOO; if unhealed, do fistulectomy or partial cystectomy.
- **Urethrocutaneous** fistula is the commonest hypospadias complication (up to 20%). Wait ≥**6 months** and exclude distal stricture. Close <1 cm locally; ≥1 cm or redo needs vascularised flaps (**tunica vaginalis** better than dartos).
- **Nephrocutaneous** fistula follows stones, XGP, prior surgery or tuberculosis. The kidney is non-functioning in ~90%, so do **nephrectomy**.

## 13 · Pubovesical fistula (PVF)

- A tract from bladder, prostate or urethra to the pubic symphysis causes **osteomyelitis**. It follows prostate radiotherapy plus repeated outlet procedures, median ~10 years after radiation, and also artificial urinary sphincter surgery.
- It presents with **pubic pain on walking**, abscess, urosepsis or sinus. **MRI is most sensitive and specific**; bone biopsy guides antibiotics because urine culture is unreliable.
- Conservative care usually fails. Debride bone, excise the fistula, fill with a rectus or omental flap, and give 4–6 weeks of antibiotics; if bladder or sphincter function is poor, do **cystoprostatectomy and diversion**.

## 14 · Urovascular fistulae

- **Arteriocalyceal** fistula follows PCNL, transplant or trauma; treat by **angioembolisation**.
- **Ureteroarterial fistula (UAF)** is an emergency. Most (85%) are secondary to long-term stents, pelvic surgery, radiation, cancer or vascular grafts. It presents with **intermittent haematuria**, and ~20% are unstable.
- Diagnose with **angiography**, provoking by stent removal only with endovascular treatment ready; a negative CT angiogram does not exclude it.
- Treat with an **endovascular stent graft** with or without internal iliac embolisation; use open surgery for infection, abscess or enteric involvement.

## 15 · Renal urine leak and anastomotic leak

- After **trauma**, a leak occurs in 1–7%. Use CT with a 10-min delayed phase; a non-opacified distal ureter suggests ureteropelvic junction avulsion. Most resolve, and persistent leaks need stent or nephrostomy.
- After **partial nephrectomy** (0.5–20%), a central tumour predicts leak. Drain, adding stent or nephrostomy if needed.
- **Vesicourethral anastomotic leak after RP** (Patil): I extraperitoneal <6 cm, II extraperitoneal ≥6 cm, III intraperitoneal. Prolong the catheter, then withdraw the drain, then place stents or nephrostomies; reoperate last.

## 16 · Fistula after gender-affirming surgery

- After phalloplasty or metoidioplasty, urethrocutaneous fistula and stricture are common; **primary colpectomy** roughly halves fistula. Small ones close with a catheter. After transfeminine surgery, neovaginal fistula is rare and is repaired with interposition after diversion.

## 17 · Landmark trials

| Trial | Design | Result | Take-home |
|---|---|---|---|
| Martius flap in simple obstetric VVF | RCT, repair ± Martius | No difference in closure; ~⅓ donor-site pain/numbness | Reserve Martius for complex fistula |
| Prophylactic ureteric stents in gynaecological surgery | RCT (+ meta-analysis) | No/small reduction in injury (number needed to treat 224) | Selective, not routine, stenting |

# 122 Bladder and Female Urethral Diverticula

## 1 · Bladder diverticula — definition and pathology

*See the paper answer **Bladder diverticulum.** (Section 12, Ch 124).*

**Also, from the condensed book (not in the paper answer):**

- Diverticular cancer: radical cystectomy gives the same stage-for-stage survival; upstaging is common.
- A diverticulum is urothelium herniating through the muscularis propria; an outer pseudocapsule gives the surgical plane.
- With **no functional muscle, the sac cannot contract**, so post-void residual (PVR) is high and stasis leads to infection, stones and malignancy.
- About 90% occur in adults; M : F ≈ **9 : 1**.

| Feature | Congenital | Acquired |
|---|---|---|
| Age / number | <10 years; solitary | Men >60; often multiple |
| Site | Posterolateral to ureteric orifice (~90% near ureterovesical junction [UVJ]); vesicoureteral reflux (VUR) common | Ureterovesical hiatus commonest; anywhere |
| Cause | Detrusor weakness at UVJ | Bladder outlet obstruction (**BOO**) or neurogenic — almost always |
| Wall / presentation | Smooth; acute urinary tract infection (UTI) | Trabeculated; incidental (lower urinary tract symptoms [LUTS], haematuria, UTI) |
| Malignancy | Not increased | Increased |

- Up to 60% of congenital cases have a syndrome, neuropathic bladder or BOO, e.g. **Menkes**, **Williams**, **Ehlers-Danlos**, prune-belly or posterior urethral valves.
- Acquired cases are due to benign prostatic hyperplasia in ~**70%**; in women, causes include dysfunctional voiding, prolapse, bladder neck obstruction and post-sling obstruction.
- A **Hutch diverticulum** is paraureteric, in a neuropathic bladder with VUR.

> **KEY POINT**
>
> Congenital: weak wall, normal pressure. Acquired: normal wall, high pressure, so an acquired diverticulum sends the search back to the outlet.

## 2 · Bladder diverticula — evaluation

- Symptoms are nonspecific (incomplete emptying, double voiding).
- Do rectal examination, PSA, urinalysis, culture, **cytology**, endoscopy and upper and lower tract imaging, and seek BOO or occult neuropathy.
- Voiding cystourethrogram (**VCUG**) shows size, VUR and emptying. **Paradoxic enlargement** means the sac fills on voiding and refills the bladder afterwards.
- CT or MRI shows number, masses and relation to ureters and rectum.
- Hydroureteronephrosis (HUN) occurs in ~7% of adults, from compression, underlying dysfunction or VUR; the pelvic ureter deviates **medially**.

- Video-urodynamics (**VUDS**) shows a **pressure-sink effect**: contractility looks falsely low, and a raised PVR may be diverticular urine.
- **Cystoscopy** inspects the whole sac with cytology from it. Biopsy with extreme care, as perforation seeds tumour.
- With VUR and a congenital diverticulum, reimplant **selectively**. Treating the cause may make excision unnecessary.

> **KEY POINT**
>
> Two jobs: find the cause (outlet or neurogenic) and exclude cancer in a sac each test sees imperfectly, so tests are combined.

## 3 · Malignancy in bladder diverticula

- Prevalence is 0.8–10%, driven by stasis and inflammation. Urothelial carcinoma is **70–80%** and squamous cell carcinoma (SCC) 20–25%.
- With **no muscularis propria**, spread is early and extravesical, transurethral resection (TUR) risks perforation, and staging is inaccurate.
- **Radical cystectomy** gives the same stage-for-stage survival. Partial cystectomy or diverticulectomy suits selected low-grade, low-stage tumours.

> **KEY POINT**
>
> Treat a diverticular tumour as potentially invasive from the outset; missing muscle explains early spread, perforation risk and inaccurate staging.

## 4 · Bladder diverticula — management

*Treat the outlet first; operate for complications, not size.*

**ALGORITHM**

1. Assess with history, cystoscopy, cytology and VCUG.
2. If a lesion is suspicious, then biopsy and give cancer therapy.
3. If not, image the upper tract. If HUN or symptoms, then do urodynamics and treat BOO, then reassess.
4. If a problem from the diverticulum persists: fit → diverticulectomy; unfit → clean intermittent catheterisation (CIC).
5. If asymptomatic and uncomplicated, then survey.

- **Diverticulectomy indications**: refractory LUTS, relapsing UTI, **stones**, **carcinoma or premalignancy**, upper tract deterioration (obstruction, VUR). **Size alone is not an indication.**
- **Treat the outlet first or simultaneously**; a small sac may then empty, and concomitant prostatectomy is common.
- If observing, counsel about cancer and survey with symptoms, cytology and cystoscopy. The unfit with poor emptying need CIC or a catheter.
- **Endoscopic** treatment for the unfit resects the neck to muscle, but large sacs do poorly.
- Diverticulectomy is open, laparoscopic or robotic: use an ipsilateral ureteric catheter, dissect in the pseudocapsule, close the neck in layers, and reimplant a ureter entering the sac.

## 5 · Female urethral diverticulum (UD) — anatomy and aetiology

### SAQ Female urethral diverticula

Paper answer · P1

**Type:** SAQ (10 marks)
**Source:** Practice Set 13 Paper 1, Q3e

#### Definition

- **UD (urethral diverticulum):** a urine-filled periurethral sac within the periurethral fascia, connected to the urethra by an **ostium**.
- Third to seventh decades; prevalence up to 1–6%.

#### Aetiology

- Mostly **acquired**. **Routh theory** (most accepted): infection of **periurethral glands** leads to duct obstruction, suburethral abscess, rupture into the lumen and an epithelialised sac.
- Hence the ostium is **posterolateral at mid or distal urethra** in ≥90%.
- **Also know:** childbirth trauma theory (but 20–30% are nulliparous).

#### Clinical features

- "Three Ds" (dysuria, dyspareunia, post-void dribbling) in only about 5%.
- Commonest: frequency, urgency, pain, **recurrent UTI (urinary tract infection)**.
- Tender anterior vaginal wall mass; about one-third incontinent.
- **Complications:** stones (4–10%); 10% show metaplasia, dysplasia or carcinoma (**adenocarcinoma** commonest).

#### Evaluation

- Urine culture; **cystourethroscopy** (ostium seen in under half).
- **Imaging sensitivity: MRI (magnetic resonance imaging) 97%**, double-balloon urethrography 95%, ultrasound 82%, VCUG (voiding cystourethrogram) 74%.
- MRI shows saddlebag or circumferential extension.
- **Videourodynamics:** true SUI (stress urinary incontinence) versus dribbling from the sac.
- **Differential diagnosis:** Skene cyst (distal, distorts meatus, no communication), Gartner cyst, leiomyoma, caruncle, bulking agent.

#### Classification

- **L/N/S/C3 (Leach):** location, number, size, configuration, communication, continence.
- **Also know:** Leng–McGuire (absent fascia = pseudodiverticulum).

#### Management

- **Asymptomatic or unfit:** suppressive antibiotics, post-void stripping; counsel about cancer.
- **Symptomatic: transvaginal diverticulectomy.**
- **Principles:** 1. Sterile urine; topical oestrogen if atrophic. 2. **Inverted-U** anterior vaginal wall flap. 3. **Preserve periurethral fascia** as a separate layer. 4. Excise the **entire sac** and close the ostium. 5. **Watertight** urethral closure over a catheter (as small as 12 Fr). 6. **Multilayered, non-overlapping** closure; obliterate dead space. 7. **Martius flap** if poor tissue.
- **Bothersome SUI:** concomitant **autologous pubovaginal sling**; **never synthetic mesh**.

- **Also know:** Spence–Duckett marsupialisation for distal lesions only.
- Pericatheter VCUG at 14–21 days.

### Complications

- Recurrence 10–22%; de novo SUI 7–16%; urethrovaginal fistula 1–8%; stricture; recurrent UTI.

**Diagram to draw:** sagittal section of a mid-urethral sac and its ostium.

> **Key points to remember:**
> - Periurethral gland infection (Routh); posterolateral ostium.
> - MRI most sensitive (97%).
> - Preserve fascia; non-overlapping layers; Martius flap.
> - No synthetic mesh at diverticulectomy.

**Also, from the condensed book (not in the paper answer):**

- Configuration is simple in ~73%, saddlebag in ~17% and circumferential in ~10%; ~50% are mid-urethral and 6–19% multiple.
- Carcinoma in UD is **adenocarcinoma** > urothelial > SCC. Local excision alone recurs, so use anterior exenteration or cystourethrectomy with or without radiotherapy.

- **Periurethral glands** lie posterolaterally, with ducts draining into the **distal third**; UD lie between the leaves of the urethropelvic ligament. The striated sphincter is mid-urethral, so plan repair for continence.
- In the gland theory, a gland abscess herniates into the fascia, usually **ventrally**, then ruptures into the lumen; about a third are proximal, however.

## 6 · UD — presentation and evaluation

- All three Ds occur in only ~**5%**. Features are irritative LUTS, pain, recurrent UTI, mass (~⅓), incontinence (~⅓), haematuria and discharge; 10–20% are asymptomatic. **Size does not track symptoms**, and UD is often misdiagnosed for years.
- In pregnancy, give antibiotics or aspirate, and repair after delivery.
- A mass lies 1–3 cm inside the introitus, and a **hard mass suggests stone or cancer**. Check atrophy, stress test and prolapse.
- **Cystourethroscopy** shows the ostium at **4 and 8 o'clock** mid-urethra in 15–89%; not seeing it must not stop work-up.
- **VUDS** shows SUI in up to 50% and separates true SUI from dribbling from the sac.
- **MRI is best** (T2 high, maps extent) but may miss small cancers. Ultrasound shows an anechoic sac, and VCUG underestimates size.

> **KEY POINT**
>
> UD symptoms mimic common LUTS and do not track size, so suspicion and examination make the diagnosis. Imaging confirms and **maps** it, especially extension towards the bladder neck (bladder or ureteric injury, postoperative incontinence).

## 7 · Differential diagnosis of periurethral masses

| Lesion | Clue | Treatment |
|---|---|---|
| UD | Mid/proximal; communicates | Excision, reconstruction |
| Leiomyoma | Firm, mobile, solid; oestrogen-dependent | Transvaginal enucleation |
| Skene cyst/abscess | Lateral to and distorts meatus; non-communicating | Aspiration, marsupialisation, excision |
| Gartner duct cyst | Anterolateral; may receive ectopic ureter — image upper tract | Observe, sclerotherapy, excision |
| Vaginal wall cyst | Müllerian commonest | Excise if symptomatic |
| Urethral prolapse | Red "doughnut"; prepubertal, postmenopausal | Oestrogen; excision |
| Caruncle | Red, posterior meatal lip; postmenopausal | Oestrogen; excise if atypical |
| Bulking agent | Injection history | Excise if symptomatic |

Skene gland adenocarcinoma may raise PSA.

> **KEY POINT**
>
> Two questions sort most periurethral masses: does it communicate (only UD), and where is it relative to the meatus (Skene, prolapse and caruncle lie at the meatus)? A firm, mobile, solid mass is a leiomyoma.

## 8 · Classification and SUI

| System | Basis | Use |
|---|---|---|
| **L/N/S/C3** (Leach) | Location, Number, Size, Configuration, Communication, Continence | Standard; not validated |
| Ginsburg–Genadry | Location | Distal: marsupialisation; proximal: excision |
| Leng–McGuire | Periurethral fascia present? | Absent = **pseudodiverticulum**, needing flap/graft |

- Significant de novo stress urinary incontinence (SUI) occurs in **7–16%**; risks are size ≥30 mm, a proximal or circumferential sac and a short urethra. Most settles within 12 months.
- About 50% of preoperative SUI improves after excision, so operate only for **bothersome** SUI, using an **autologous pubovaginal sling, never synthetic**.

## 9 · Surgical repair of UD

- Operate for symptoms, stones or suspected cancer; treat the asymptomatic or unfit with suppressive antibiotics and postvoid stripping.
- **Spence-Duckett marsupialisation** suits only distal-third UD in women who are not sexually active.
- Principles (Box 122.2): vascularised flap, separate fascia, excise ostium and entire sac, watertight multilayer non-overlapping closure, close dead space, and **preserve or create continence**.

**ALGORITHM**

1. Lithotomy and Foley; suprapubic catheter (**SPC) at the start**, never the end (it would cross the repair).
2. Make an **inverted-U incision** and raise the vaginal flap off the fascia.
3. Incise the fascia **transversely** and raise fascial flaps.
4. Excise the entire sac to the ostium, including the inflamed urethral edge.
5. Close the urethra watertight over a catheter (as small as 12 Fr).
6. Close fascia **perpendicular** to it, obliterating dead space.
7. If tissue is poor or fascia attenuated, then add a **Martius** flap.
8. Advance the vaginal flap (3 layers; 4 with Martius); antibiotic pack.

- A pseudodiverticulum needs a Martius flap or fascial graft. Do a **pericatheter VCUG at 14–21 days**, repeated weekly if leaking.

## 10 · Complications

| Complication | Incidence | Notes |
|---|---|---|
| Recurrent UD | **10–22%** (pooled ~10%) | Incomplete excision, dead space; circumferential, complex, proximal, multiple |
| Incontinence | 1.7–16% | Conservative 12 months, then autologous sling |
| Urethrovaginal fistula | 0.9–8.3% | Distal: may need no repair; proximal: repair ± Martius |
| Stricture | 0–5% | — |
| Recurrent UTI | up to 31% | Investigate for recurrence |
| Others | — | Distal urethral necrosis, bladder/ureteric injury, dyspareunia |

Redo excision gives less relief and more fistula, so image persistent symptoms for recurrence or stricture.

> **KEY POINT**
>
> Complete sac removal, watertight closure, off-axis fascia and obliterated dead space prevent the three failures: recurrence, fistula, incontinence.

| Feature | Bladder diverticulum | Female UD |
|---|---|---|
| Origin | Herniation through detrusor | Infected periurethral gland ruptured into lumen |
| Cause to treat | BOO / neurogenic | Infection; coexisting SUI |
| Commonest cancer | Urothelial (then SCC) | Adenocarcinoma |
| Key study | Cystoscopy + cytology; VCUG | MRI |
| Operate for | Complications, not size | Symptoms or complications |
| Operation | Diverticulectomy + outlet treatment | Transvaginal excision, layered reconstruction |

# 77 Sexual Function and Dysfunction in the Female

## SAQ Female sexual dysfunction

Paper answer · P3

**Type:** SAQ (10 marks)
**Source:** Practice Set 1 Paper 3, Q3f

### Definition and prevalence

- FSD (female sexual dysfunction): persistent desire, arousal, orgasm or pain problems causing **personal distress**.
- About **40%** of women in their lifetime; **HSDD (hypoactive sexual desire disorder) commonest**. Almost half of women with UI (urinary incontinence) have FSD.

### Physiology and response models

- Arousal: pudendal afferents, pelvic parasympathetic efferents → **NO (nitric oxide)–cGMP (cyclic guanosine monophosphate)** → engorgement and lubrication. Oestrogen maintains tissues; testosterone drives desire.
- **Masters and Johnson** (linear): excitement, plateau, orgasm, resolution; Kaplan added desire. **Basson** (circular): desire is often responsive.

### Classification

| DSM-5 (Diagnostic and Statistical Manual, 2013) | ICSM (International Consultation on Sexual Medicine)/ISSWSH (International Society for the Study of Women's Sexual Health) |
|---|---|
| Female sexual interest/arousal disorder (≥3 of 6 criteria) | HSDD and arousal disorder kept separate (cognitive vs genital arousal) |
| Female orgasmic disorder | Orgasm disorders |
| Genito-pelvic pain/penetration disorder | Genital-pelvic pain dysfunction |
| – | PGAD (persistent genital arousal disorder) |
| - ≥6 months with distress; lifelong or acquired; generalised or situational. | |

### Causes

- Hormonal (menopause, GSM [genitourinary syndrome of menopause]), drugs (SSRIs [selective serotonin reuptake inhibitors], antihypertensives), diabetes, depression, relationship factors.
- Urological: incontinence, prolapse, sling dyspareunia, cystectomy nerve or clitoral injury.

### Evaluation

- Clinician must ask. **PLISSIT** (permission, limited information, specific suggestions, intensive therapy); BSSC-W (Brief Sexual Symptom Checklist for Women) screen; **FSFI (Female Sexual Function Index, 19 items, six domains)**.
- Examination: atrophy, lichen sclerosus, cotton-swab test, pelvic floor tone. Laboratory tests rarely help.

### Management

- Treat causes; education; sex therapy (sensate focus, mindfulness).
- **HSDD (premenopausal, FDA [US Food and Drug Administration] approved): flibanserin** 100 mg at bedtime (5-HT1A agonist/5-HT2A antagonist; alcohol caution); **bremelanotide** 1.75 mg subcutaneously on demand (melanocortin agonist). Postmenopausal: transdermal testosterone (off-label).
- **Arousal:** testosterone; PDE5 (phosphodiesterase type 5) inhibitors only with neurovascular disease.
- **Orgasm:** education, directed masturbation.
- **Pain/vestibulodynia:** pelvic floor physiotherapy, dilators, biofeedback, topical lidocaine, vaginal oestrogen; vestibulectomy if refractory.
- **Also know:** ospemifene, intravaginal prasterone (menopausal dyspareunia).

**Diagram to draw:** Basson circular sexual response model.

> **Key points to remember:**
> - About 40% prevalence; HSDD commonest; distress required.
> - DSM-5 merges desire and arousal, dyspareunia and vaginismus.
> - Flibanserin and bremelanotide: only FDA-approved HSDD drugs (premenopausal).

**Also, from the condensed book (not in the paper answer):**

- Flibanserin: about half respond by 8 weeks; REMS. Bremelanotide: ≤8 doses/month; hyperpigmentation. Testosterone: stop at 6 months if no benefit.
- Genital-sparing cystectomy allows sex at about 6 vs 14 weeks; combined anterior and posterior repair increases dyspareunia; TOT spares the dorsal clitoral nerve.

## 1 · Functional anatomy

The **distal vagina, clitoris and urethra** share blood supply and nerves and respond as one unit, the **clitoral complex**, so urethral or vaginal surgery can change sexual function.

- The **clitoris** is the penile homologue, **separate from the urethra**; its two corpora continue as crura on the ischiopubic rami.
- The glans is very sensitive but **not erectile**. The dorsal nerves lie **superficially on the dorsum**, outside the tunica, so hood or clitoral surgery can injure them.
- **Erectile** tissue is the clitoral body, crura and vestibular bulbs; labia minora, glans, periurethral tissue and vaginal walls are vascular but do not engorge.
- **Skene** glands are the prostate homologue (PSA positive) and **Bartholin** glands the bulbourethral homologue. MCQ The G-spot has no agreed anatomical basis.
- Nerve supply of the labium majus and the glans draining **directly to deep inguinal nodes** are minor exam points.

## 2 · Physiology of the sexual response

**ALGORITHM**

1. Touch on the clitoral complex travels in pudendal (somatic) afferents to **S2–S4**.
2. Parasympathetic efferents (nervi erigentes, then cavernous nerves) dilate arterioles, causing engorgement and lubrication.

The reflex has **somatic afferents and visceral efferents**. **Orgasm** is cerebral, with **sympathetic** output (T12–L1) contracting the uterus and pudendal reflexes contracting pelvic striated muscle.

**KEY POINT**

Pudendal injury removes the **trigger**; pelvic plexus or cavernous nerve injury removes the **response**.

**ALGORITHM**

1. Neuronal nitric oxide synthase (nNOS) makes nitric oxide (NO), which activates guanylate cyclase to make cyclic GMP (cGMP).
2. cGMP lowers $Ca^{2+}$, so smooth muscle relaxes and tissue engorges.
3. Phosphodiesterase type 5 (PDE5) degrades cGMP; vasoactive intestinal polypeptide (VIP) works with NO.

**Desire** balances **excitation** (dopamine, noradrenaline, melanocortins, oxytocin, testosterone, oestrogen) against **inhibition** (serotonin, opioids, prolactin, progesterone); licensed HSDD drugs act on this balance.

| | **Oestrogen** | **Testosterone** |
|---|---|---|
| Role | Maintains the **substrate**: epithelium, collagen, lubrication, acidic pH, NOS/VIP | Drives **desire**; inactive when bound to sex hormone–binding globulin (SHBG) |
| Deficiency | Thin, dry, fibrotic vagina causing dyspareunia; risk rises at oestradiol **≤50 pg/mL** | Lower libido, arousal, sensation and orgasm |
| Treatment risk | — | Virilisation: early reversible (acne, hirsutism); late irreversible (voice, alopecia, clitoromegaly) |

| **Model** | **Stages** | **Key idea** |
|---|---|---|
| Masters & Johnson (1966), linear | Excitement, plateau, orgasm, resolution | **No desire phase** |
| Kaplan (1977), linear | Desire, arousal, orgasm | Adds desire |
| Whipple & Brash-McGreer, circular | Seduction, sensations, surrender, reflection | Reflection feeds next seduction |
| **Basson (2002), circular** | Neutrality, receptivity, arousal, **responsive desire**, satisfaction | Spontaneous desire **not needed**; intimacy is the motive |

**KEY POINT**

Under Basson, lack of spontaneous desire (common after menopause) is not a dysfunction, which is why DSM-5 merged desire and arousal.

## 3 · Classification

DSM-5 requires **≥6 months** and **clinically significant distress**; low desire without distress is not a disorder.

| Domain | DSM-IV-TR | DSM-5 |
|---|---|---|
| Desire and arousal | HSDD; female sexual arousal disorder (FSAD); sexual aversion disorder | **Female sexual interest/arousal disorder (FSIAD)**, **≥3 of 6**: less interest; fewer fantasies; less initiation/receptivity; less pleasure, response to cues or sensation (in 75–100% of encounters). Aversion dropped |
| Orgasm | Female orgasmic disorder (FOD) after normal excitement | FOD: delay, infrequency, absence **or lower intensity** in **75–100%**; normal-excitement requirement removed |
| Pain | Dyspareunia; vaginismus | **Genito-pelvic pain/penetration disorder (GPPPD)**, ≥1 of: penetration difficulty; pain; fear of pain; pelvic floor muscle (PFM) tensing |

- **ICSM 2016** keeps HSDD and FSAD **separate** and adds **persistent genital arousal disorder (PGAD)** and orgasmic illness syndrome; **ISSWSH** adds a urological **genital arousal disorder**.
- About **40%** have a lifetime sexual problem but only about **8–12%** a distressing disorder. **HSDD is commonest**, FOD second.

## 4 · Causes (biopsychosocial)

- **Biological**: diabetes, hypertension, coronary disease, thyroid disease, hyperprolactinaemia, renal failure, spinal cord injury (SCI), multiple sclerosis (MS), vulval dermatoses, incontinence, pregnancy, breastfeeding and menopause.
- **Psychological**: depression (bidirectional, so screen every HSDD), anxiety, body image, abuse, conflict and the **partner's erectile dysfunction or premature ejaculation**.
- **Drugs**: SSRIs, SNRIs, antipsychotics, antihypertensives, antiandrogens, GnRH agonists, opiates and alcohol. MCQ Combined hormonal contraceptives and oral oestrogens raise SHBG.
- **Genitourinary syndrome of menopause (GSM)** causes dryness, burning, dyspareunia (often **vestibular**), urgency, dysuria and recurrent UTI. Dryness causes pain, which lowers desire, so **treat the tissue first**.
- **Cancer treatment** harms by castration, anti-oestrogens (aromatase inhibitors cause severe GSM), pelvic radiotherapy (stenosis) and nerve injury.

## 5 · Evaluation

Women rarely volunteer sexual problems, so the **clinician must ask**, gender-neutrally.

**ALGORITHM**

1. **P**ermission.
2. **L**imited **I**nformation: cycle, ageing, drugs.
3. **S**pecific **S**uggestions: lubricants, **vaginal oestrogen**, positions, vibrators.
4. **I**ntensive **T**herapy means **referral** (PLISSIT).

- **DSDS** (Decreased Sexual Desire Screener) is a **validated** 5-item screen; yes to items 1–4 with no contributing cause means **generalised acquired HSDD**. The ACOG BSSC-W is not validated.
- **FSFI** (19 items, 6 domains) is most used; FSDS-R measures **distress**; **PISQ-12/IR** is the **only validated** tool for prolapse and incontinence.
- History: lifelong or acquired, generalised or situational, the **order problems appeared**, drugs, surgery, LUTS, prolapse and the **partner's** function.
- Examination is essential with pain. **pH ≥4.5** without infection or semen means atrophy. Do a **Q-tip test** of the vestibule and **retract the hood** for adhesions.
- Lichen sclerosus leads to introital stenosis; lichen planus (**Wickham striae**) to vaginal stenosis.
- **Overactive** PFM causes dyspareunia and **underactive** PFM causes incontinence and prolapse; fornix tenderness suggests endometriosis.
- Labs are rarely needed. Do **not** measure a **random testosterone**: there is no female range and no level predicts HSDD.

## 6 · Hypoactive sexual desire disorder

- ISSWSH defines HSDD as ≥6 months of lost desire **with distress**, not due to drug, illness, pain, abuse or partner. **Exclude relationship conflict first.**
- HSDD peaks in **midlife** and is defined by distress, not hormone level.

**ALGORITHM**

1. Screen (DSDS), then history and examination.
2. **Correct modifiable factors**: drugs, depression, GSM, pain, partner's ED.
3. Education; sex or couples therapy.
4. If **generalised acquired** HSDD persists, then drug; review at 8 weeks.
5. **Refer** if situational HSDD fails counselling, or psychiatric illness or sexual trauma.

| | **Flibanserin** | **Bremelanotide** |
|---|---|---|
| Mechanism | Central **5-HT1A agonist / 5-HT2A antagonist**: lowers serotonin, raises dopamine/noradrenaline (removes the brake) | Central **melanocortin agonist** releasing dopamine (presses the accelerator) |
| Licence | FDA: only **premenopausal** acquired generalised HSDD | |
| Dose | **100 mg at bedtime** daily; about half respond by 8 weeks | **1.75 mg subcutaneous** ≥45 min before sex; ≤1/24 h, **≤8/month** |
| Side effects | Dizziness, somnolence, nausea | Nausea, flushing, headache, **hyperpigmentation** |
| Cautions | **Alcohol** (hypotension, syncope); strong CYP3A4 inhibitors; REMS restricted prescribing | Uncontrolled hypertension or cardiovascular disease |

- **Transdermal testosterone** (off-label) improves desire, arousal, orgasm and satisfaction in **postmenopausal HSDD**, with no cardiovascular signal but unknown long-term safety.
- The 2019 Global Consensus targets **premenopausal physiological levels**, checks levels at baseline and 3–6 weeks, and says **stop at 6 months** if there is no benefit. Bupropion (off-label) is an alternative.

## 7 · Arousal disorder and PGAD

- **Subjective and genital arousal correlate poorly**, so vasodilators raise flow without felt arousal. Use mindfulness-based sex-focused CBT.
- **Transdermal testosterone** has the best evidence. PDE5 inhibitors help only with neurovascular impairment (type 1 diabetes, SCI, MS, SSRI-induced). **Bupropion 150 mg twice daily** treats SSRI-induced dysfunction.
- **PGAD** is distressing genital arousal for hours to days, **not relieved by orgasm** and without desire. Causes include Tarlov cysts, pelvic varices and **stopping an SSRI**; treat with CBT, duloxetine or pregabalin.

## 8 · Female orgasmic disorder

- FOD is a **diagnosis of exclusion requiring distress**: exclude drugs, disease, stress and **inadequate stimulation** (including the partner's ED/PE).
- **No orgasm at coitus without clitoral stimulation is normal variation.**
- **Education is the cornerstone**, with directed masturbation, sensate focus and mindfulness. If arousal is intact, increase stimulation (vibrators; MCQ EROS clitoral vacuum is the only FDA-cleared FSD device).

**ALGORITHM**

1. **SSRI-induced dysfunction**: was it present before treatment?
2. If not, wait for tolerance (~4 months).
3. If it persists at 3–6 months, then lower the dose, or switch to or add **bupropion** or mirtazapine; consider sildenafil.

## 9 · Genito-pelvic pain / penetration disorder

- **Dyspareunia** is superficial or deep (endometriosis, prolapse, pelvic floor). **Vaginismus** is involuntary contraction preventing penetration.
- **Vulvodynia** is vulval burning pain for **≥3 months** without a cause; it is localised or generalised, provoked or unprovoked.
- **Provoked vestibulodynia (PVD)** is commonest (about **10–12%** of women). Primary PVD starts at first penetration; secondary PVD (about two-thirds) follows pain-free sex.
- The mechanism is vestibular nerve proliferation with a **hypertonic pelvic floor** and fear-avoidance.
- **Clitorodynia**: hood adhesions trap keratin and form painful pseudocysts, so **examine the glans routinely**.

**ALGORITHM**

1. Education and vulval care.
2. Lidocaine 5% or EMLA **10 min before intercourse**.
3. **Multimodal pelvic floor physical therapy** (myofascial release, down-training, biofeedback, graded **dilators**) with psychological/couples therapy.
4. If hypo-oestrogenic, then vestibular oestrogen.
5. If localised PVD is refractory, then **vestibulectomy**.

- **Multimodal physical therapy is standard of care** for PVD; EMG biofeedback is best if the pelvic floor is overactive.
- Oestrogen deficiency **increases** vulval nerve density, so **topical beats systemic and vestibular beats vaginal**.
- Second-line options are amitriptyline, gabapentin and intralevator botulinum toxin. **Vestibulectomy** beats non-surgical therapy in localised neuroproliferative disease.

### Treatment of GSM

**ALGORITHM**

1. Non-hormonal lubricants/moisturisers and regular activity (NAMS first line).
2. Then **low-dose vaginal oestrogen** (minimal absorption, no progestogen) or vaginal DHEA (prasterone 6.5 mg nightly).
3. If vaginal products are impossible, then ospemifene 60 mg orally (SERM, vaginal agonist; VTE risk).
4. If vasomotor symptoms, then systemic hormone therapy.

**KEY POINT**

After **breast cancer**, use non-hormonal measures first; vaginal oestrogen only after oncology discussion (aromatase inhibitors).

## 10 · Pelvic surgery and sexual function

Surgery harms by three routes: **nerves** (lateral vaginal wall, pelvic plexus), **clitoral blood supply** (distal urethra) and **vaginal shape**. Counsel **before and after**.

- **Radical cystectomy**: loss of desire is commonest, and about half can have intercourse. Preserve function by saving the bundles on the **lateral vaginal walls**, dividing the urethra at the **bladder neck**, and keeping vaginal dimensions.
- **Genital-sparing** cystectomy (if oncologically safe) lets sex resume at **~6 vs 14 weeks**; diversion type is irrelevant. **Never trade oncological control.** See Ch 101
- **Hysterectomy** for benign disease leaves function unchanged or better; after **rectal surgery** dysfunction is common, and nearly universal after abdominoperineal resection.
- **Prolapse repair**: bother and body image, not stage, predict function. Dyspareunia is commoner after vaginal than abdominal repair, and after anterior mesh than **sacrocolpopexy**.
- **Combined anterior and posterior repair increases dyspareunia**, so stage it in sexually active women. **Keep length, avoid excess skin excision, preserve the introitus.** See Ch 119
- **SUI surgery**: about a third improve and most are unchanged; **curing coital incontinence** predicts improvement. Retropubic tape (**TVT**) may denervate the vaginal wall, while transobturator tape (**TOT**) spares the dorsal clitoral nerve. Dyspareunia is a strong reason for sling removal.
- **Coital incontinence** (about 1 in 10 incontinent women): leakage at penetration is stress; at orgasm, detrusor overactivity.
- **Bladder pain syndrome causes the worst FSD**, through pain, PFM overactivity and vulvodynia. **Sacral neuromodulation** improves sexual function whatever the cause.

- **Genital cosmetic surgery** has poor evidence; screen for **body dysmorphic disorder**.

## 11 · Special populations

Ask about orientation and gender identity; offer fertility preservation before transition. Oestrogen risks thromboembolism; testosterone risks erythrocytosis (haematocrit **>50%**), so check haematocrit and testosterone 3-monthly in year 1, then 1–2 times a year. SCI impairs sensation, lubrication and orgasm. See Ch 126

## 12 · Landmark trials

| Trial | Design | Result | Take-home |
|---|---|---|---|
| Flibanserin phase III | RCTs vs placebo, premenopausal | More satisfying events and desire, less distress | First FDA-approved drug for HSDD |
| RECONNECT (bremelanotide) | 2 phase III RCTs | More desire, less distress | On-demand option for premenopausal HSDD |
| Bupropion for SSRI-induced dysfunction (Cochrane) | Meta-analysis | Benefit at 150 mg twice daily | Add or switch to bupropion |

# ★ Putting it together: female pelvic decisions

## A · Incontinent woman

**ALGORITHM**

1. **Classify** (history, diary): stress (SUI), urgency (UUI), mixed, overflow, functional. Check stress test, prolapse (POP), residual, urinalysis (Ch 105).
2. **Urodynamics** are not needed for uncomplicated demonstrable SUI (VALUE). Use them if mixed, voiding dysfunction, prior surgery, neurological or high-grade POP.
3. **Conservative first:** weight, fluids, supervised pelvic floor and bladder training, pessary, vaginal oestrogen (Ch 114).
4. If this fails, then **SUI surgery** (Ch 116, 118); for UUI, drugs, botulinum toxin, neuromodulation.

## B · Choosing an SUI operation

| Operation | Best for | Drawbacks |
|---|---|---|
| Retropubic mid-urethral sling (MUS) | Standard; intrinsic sphincter deficiency | Perforation (cystoscopy mandatory), voiding dysfunction |
| Transobturator MUS | Avoiding the retropubic space | Groin/thigh pain; weaker in severe SUI |
| Autologous pubovaginal sling | Recurrent SUI, urethral repair, mesh declined | More voiding dysfunction |
| Burch | Mesh declined; concurrent abdominal surgery | Posterior prolapse, de novo urgency |
| Bulking agent | Frail, minimal surgery | Lower, short-lived cure |

## C · Prolapse: which repair?

**ALGORITHM**

1. **POP-Q** stage II is a leading edge within 1 cm of the hymen (Ch 105).
2. Observe if asymptomatic; offer pessary or surgery if symptomatic.
3. **Support the apex**: uterosacral/sacrospinous suspension or sacrocolpopexy (most durable) (Ch 117).
4. If **frail, not sexually active**, then colpocleisis.
5. If **occult SUI**, then discuss a concomitant sling (OPUS: fewer leaks, more complications).

## D · Vesicovaginal fistula

**KEY POINT**

The first repair has the best chance. Do a double-dye test, upper tract imaging and, if irradiated, biopsy. Post-hysterectomy fistulae are supratrigonal and mostly repaired vaginally; go abdominal for ureteric involvement, large or high fistulae, or failed vaginal repair. Use tension-free, non-overlapping layers, a vascularised flap (Martius or omentum) for poor tissue, then a cystogram (Ch 121).

## E · Mesh complications

**ALGORITHM**

1. If suspected, examine and cystoscope.
2. If exposure is small, then oestrogen with or without trimming; otherwise excise and close.
3. If mesh is in bladder or urethra, then remove and repair without new mesh.
4. Remove only enough to solve the problem, never asymptomatic mesh (Ch 119).

# Other paper questions in this section

*Asked in the papers but not tied to a specific place in the book above.*

## SAQ Endoscopic procedure of SUI

Paper answer · P1

**Type:** SAQ (10 marks)
**Source:** Practice Set 13 Paper 1, Q3f

*Read as endoscopic injection of urethral bulking agents.*

### Introduction

- **SUI (stress urinary incontinence)** results from urethral hypermobility and **ISD (intrinsic sphincter deficiency)**.
- **Bulking agents**, injected submucosally under cystoscopic vision, are the least invasive surgical option.

### Mechanism

- Bulk at the proximal urethra improves **mucosal coaptation** and resistance.
- It does not correct hypermobility, but most SUI includes some ISD.

### Indications

- ISD with a **fixed, immobile urethra** (low VLPP [Valsalva leak point pressure] <60 $cmH_2O$, historically).
- Women wanting a **low-risk** procedure; elderly or unfit; future pregnancy.
- Persistent leakage after a sling; neurogenic SUI (safer for bladder compliance than a sling).
- **Avoid in:** active UTI (urinary tract infection), untreated detrusor overactivity.

### Agents

| Agent | Note |
|---|---|
| Polyacrylamide hydrogel (Bulkamid) | Non-particulate |
| Polydimethylsiloxane (Macroplastique) | Silicone particles |
| Calcium hydroxylapatite; carbon-coated zirconium beads | Particulate |
| Bovine collagen (skin test); PTFE (polytetrafluoroethylene); autologous fat | Withdrawn or abandoned: allergy, migration, granuloma, embolism |

### Technique

1. Sterile urine; local anaesthesia; cystoscope (0–30° lens) in the urethra.
2. **Transurethral:** needle into the submucosa about 1–1.5 cm distal to the bladder neck; or **periurethral:** needle alongside the urethra, watched cystoscopically.
3. Inject at 2–4 sites until the mucosal cushions **coapt in the midline**.
4. Avoid an indwelling catheter (it moulds the bulk); use CISC (clean intermittent self-catheterisation) for retention.

### Outcomes

- Early improvement is common, but cure is lower than with slings and **declines with time**; repeat injections are often needed.

- **Cochrane review (Kirchin 2017):** limited evidence of benefit.
- **AUA/SUFU (American Urological Association/Society of Urodynamics, Female Pelvic Medicine and Urogenital Reconstruction):** an option, with counselling on durability.
- **Men:** EAU (European Association of Urology): **do not offer** for post-prostatectomy incontinence.

### Complications

- Transient retention, UTI, dysuria, haematuria, urgency.
- Sterile abscess, migration, erosion; may mimic a urethral diverticulum on imaging.

**Diagram to draw:** cystoscopic view of coapted mucosal cushions after injection.

> **Key points to remember:**
> - Bulking improves coaptation; does not correct hypermobility.
> - Best for fixed urethra with ISD, or unfit patients.
> - Less durable than slings; repeat injections common.
> - Not recommended for post-prostatectomy incontinence.

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## SAQ Endometriosis in urology

Paper answer · P1

**Type:** SAQ (10 marks)
**Source:** Practice Set 22 SAQ Sheet, Q3d

### Definition and epidemiology

- Endometriosis is endometrial glands and stroma outside the uterus. Urinary tract involvement occurs in about 1–5% of women with endometriosis, mostly with deep disease.
- Sites: **bladder** (commonest, about 80%) > ureter (about 15%) > kidney, urethra (rare).

### Pathogenesis

- Retrograde menstruation (Sampson), coelomic metaplasia, lymphovascular spread, and **iatrogenic implantation** at pelvic surgery (e.g. caesarean section).
- Bladder lesions start on the **serosa** (vesicouterine pouch) and invade the detrusor.
- Ureteric disease: **extrinsic** (periureteric fibrosis, commoner) or intrinsic (muscularis or mucosa); usually distal, left more often.

### Clinical features

- **Bladder:** cyclical haematuria or frequency, urgency and dysuria with menses; suprapubic pain. Differential: BPS (bladder pain syndrome), vesicouterine fistula (menouria).
- **Ureter:** often **silent**; hydronephrosis and loss of renal function may go unnoticed.

### Evaluation

- Transvaginal and transabdominal ultrasound gauge size and depth; **renal ultrasound in every woman with deep endometriosis**.
- **MRI (magnetic resonance imaging):** extent; high T1 signal from blood products.
- **Cystoscopy:** limited value (disease is extravesical); may show a bluish submucosal lesion. Biopsy excludes carcinoma.

- CTU (computed tomography urography) and MAG3 (mercaptoacetyltriglycine) renogram for obstruction and split function.
- Staging: rASRM (revised American Society for Reproductive Medicine) I–IV; #Enzian records bladder and ureter.

### Management (with gynaecologist)

- **Medical:** progestins (dienogest), combined pill, GnRH (gonadotrophin-releasing hormone) analogues. Symptomatic relief; not for obstructive ureteric disease.
- **Bladder: partial cystectomy** (full-thickness excision with margin), laparoscopic or robotic; stent ureters if near. TUR (transurethral resection) is incomplete, with recurrence and perforation risk.
- **Ureter:**
- Extrinsic, mild: ureterolysis.
- Intrinsic or stricture: segmental resection with ureteroureterostomy, or **reimplantation with psoas hitch or Boari flap**.
- Non-functioning kidney: nephrectomy.
- Treat all pelvic disease at the same operation to lower recurrence.
- **Also know:** risk factor for VVF (vesicovaginal fistula) and ureteric injury at hysterectomy; rare malignant change (clear-cell, endometrioid).

**Diagram to draw:** endometriotic nodule in the vesicouterine pouch invading the detrusor.

> **Key points to remember:**
> - Bladder commonest; cyclical haematuria.
> - Ureteric disease silent: screen kidneys.
> - Partial cystectomy, not TUR.

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## SAQ Urethral caruncle.

Paper answer · P2

**Type:** SAQ (10 marks)
**Source:** 2016-Summer Paper 2, Q3f

### Definition

- A benign, red, fleshy, **inflammatory polypoid lesion of the posterior lip of the distal female urethra**. Not a true tumour, but a benign mimic.
- Common; seen mainly in **postmenopausal women**.

### Aetiology and pathogenesis

- **Oestrogen deficiency (GSM, genitourinary syndrome of menopause)**: atrophy and laxity of distal urethral mucosa.
- Related to **mucosal prolapse**: the everted posterior mucosa is chronically irritated, causing haemorrhage, necrosis and inflammatory growth.

### Pathology

- Classical types: **papillomatous, angiomatous and granulomatous**; inflamed vascular stroma.

### Clinical features

- Often asymptomatic, found on examination.
- **Spotting on underwear**, bleeding, pain, dysuria; may thrombose into a dark, tender mass.
- Examination: small, red, soft mass at the **6 o'clock** meatal margin; the rest of the meatus looks normal.

### Differential diagnosis

| Lesion | Distinguishing feature |
|---|---|
| **Urethral mucosal prolapse** | **Circumferential "doughnut"** around whole meatus; prepubertal girls and postmenopausal women |
| **Urethral carcinoma** | Firm, ulcerated, indurated, fixed; bleeding mass; biopsy |
| **Urethral diverticulum** | Mid or proximal anterior vaginal wall mass; dysuria, dyspareunia, dribbling; MRI (magnetic resonance imaging) |
| Skene gland cyst or abscess | Lateral to meatus; distorts it; does not communicate |
| Others | Melanoma, lymphoma, TB (tuberculosis), condyloma, intestinal metaplasia; postmenopausal bleeding from the uterus |
| - Caruncle is listed as a possible risk factor for female urethral cancer, so atypical lesions must be biopsied. | |

### Evaluation

- Clinical diagnosis; exclude a gynaecological bleeding source.
- **Excision biopsy** if atypical, indurated, ulcerated, enlarging or refractory; cystourethroscopy if in doubt.

### Treatment

- **Conservative (first line)**: **topical vaginal oestrogen** (shrinks most), anti-inflammatory creams, sitz baths. Ask about breast or endometrial cancer before oestrogen.
- **Surgical excision** for large, symptomatic, thrombosed, refractory or suspicious lesions: 1. Grasp the tip and apply traction to expose the base. 2. Excise at the base. 3. Reapproximate mucosa with absorbable sutures if the defect is large. 4. Foley catheter for a few days.
- Complications: bleeding, recurrence, meatal stenosis.

**Diagram to draw:** meatus with a 6 o'clock caruncle compared with circumferential prolapse.

> **Key points to remember:**
> - Postmenopausal posterior-lip lesion; oestrogen deficiency.
> - Prolapse is circumferential; caruncle is focal.
> - Topical oestrogen first; excise if atypical or refractory.

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## SAQ Acute urinary retention in females.

Paper answer · P4

**Type:** SAQ (10 marks)
**Source:** 2017-Summer Paper 4, Q3f

## Introduction

- AUR (acute urinary retention) is sudden inability to void. It is uncommon in women and is often non-obstructive, so search for the cause.

## Causes

| Group | Examples |
|---|---|
| Obstructive | Large cystocele or prolapse; tight sling after incontinence surgery; urethral diverticulum, stricture; pelvic mass, **retroverted gravid uterus**, haematocolpos; cancer |
| Functional | **Fowler syndrome**; dysfunctional voiding; primary bladder neck obstruction (rare) |
| Neurogenic | MS (multiple sclerosis; may present first with retention), cauda equina, transverse myelitis, diabetes |
| Inflammatory | Cystitis, vulvovaginitis, **genital herpes**, Bartholin abscess |
| Drugs | Anticholinergics, opioids, tricyclics, calcium-channel blockers |
| Postoperative/postpartum; psychogenic | Epidural, pain; diagnosis of exclusion |

## Fowler syndrome

- Young women (20–35) with painless retention and bladder capacity over 1 litre.
- **Urethral sphincter EMG (electromyography):** complex repetitive discharges, so the sphincter fails to relax. Urethral closure pressure >100 cm $H_2O$.

## Evaluation

- **History:** drugs, surgery, delivery, neurological and bowel symptoms, genital lesions.
- **Examination:** pelvic (prolapse, mass); neurological (perineal sensation, anal tone).
- **Tests:** drained volume; urine culture; creatinine; pregnancy test; ultrasound.
- MRI (magnetic resonance imaging) spine if neurological signs; cystoscopy.
- **Later:** video-urodynamics (female obstruction, e.g. Qmax ≤12 mL/s with Pdet ≥20 cm $H_2O$); sphincter EMG.

## Management

- **Immediate:** urethral catheter; suprapubic if urethral access fails.
- **Treat the cause:** stop drugs; treat infection or herpes; pessary or prolapse repair; reposition the uterus; sling division; diverticulectomy.
- **TWOC (trial without catheter)** after correction.
- **Persistent retention:**
- **CISC (clean intermittent self-catheterisation)** rather than an indwelling catheter.
- **SNM (sacral neuromodulation):** about 70% success in Fowler syndrome.
- Bethanechol not recommended. Bladder neck incision only with caution (incontinence).

**Diagram to draw:** flowchart: AUR → catheterise → find cause → treat → TWOC → CISC or SNM.

**Key points to remember:**

- Look for prolapse, sling obstruction, pelvic mass, neurology and drugs.
- Fowler: abnormal sphincter EMG; SNM about 70%.
- CISC for persistent retention.

# Rapid Revision Sheet · Section 6

### Female pelvic floor anatomy and pelvic organ prolapse: evaluation and surgical repair (LAQ)

- Levator ani = puborectalis, pubococcygeus, iliococcygeus; S3–S5.
- DeLancey: I suspension (uterosacral–cardinal; failure causes uterine prolapse), II attachment (ATFP), III fusion (urethral hypermobility); level I is the keystone.
- Always restore apical support.
- Enhorning 1961, DeLancey 1994, Petros & Ulmsten 1990, McGuire ISD 1978.
- Integral theory: mid-urethral sling; hammock: backing, not repositioning.
- Burch: Cooper's ligament; TOT: obturator foramen; PVS: bladder neck.
- Bulge is most specific; splinting means high grade.
- Central cystocele rugae lost, paravaginal preserved; POP-Q stage II = −1 to +1 cm (±1 cm of hymen).
- Sacrocolpopexy most durable; colpocleisis for frail elderly.
- Cystoscopy after repair; screen for occult SUI.
- Pessary first; sacrospinous hysteropexy 1.5–3 cm medial to the ischial spine.
- Manchester amputates the cervix: poor choice for future pregnancy.

### Female SUI: continence mechanism, classification, evaluation (UDS) and management, including male SUI surgery (LAQ)

- Levator ani = puborectalis, pubococcygeus, iliococcygeus; S3–5.
- DeLancey: I suspension, II attachment (ATFP), III fusion.
- Continence zone is the mid-urethra; submucosal plexus gives up to 30% of closure.
- Blaivas types 0, I, IIA, IIB, III; type III = ISD.
- Q-tip >30° = hypermobility; VLPP / ALPP <60 $cmH_2O$ = ISD; MUCP ≤20 $cmH_2O$.
- Burch corrects position; slings and AUS correct function.
- VALUE (ValUE) trial: UDS not needed in uncomplicated (index) SUI.
- Use UDS when it changes treatment: mixed, failed surgery, voiding dysfunction.
- PFMT first; MUS first-line surgery.
- SISTEr (fascial sling > Burch); TOMUS (retropubic = transobturator).
- PVS for ISD; urethral injury → no mesh.
- Men: AUS gold standard; sling for mild, non-irradiated.
- AUS PRB 61–70 $cmH_2O$; deactivate 4–6 weeks.
- Radiation or severe male SUI: AUS first; MASTER: sling non-inferior.

### Burch colposuspension (SAQ)

- Burch 1961: vaginal wall to Cooper's ligament; for hypermobility.
- Cure 85–90% at 1 year, ~70% at 5 years.
- Niche: concomitant abdominal surgery, mesh avoidance.

### Mid-urethral slings: TOT and TVT (SAQ)

- Integral theory; loose tape at mid-urethra.
- Type 1 macroporous monofilament polypropylene.
- Delorme 2001; TOT anchors in obturator muscles; fewer bladder injuries than retropubic.

- TOMUS: cure equivalent at 1 year (80.8% retropubic vs 77.7% transobturator).
- TOT: less bladder injury (0.6% vs 4.5%), more groin pain (6.4% vs 1.3%); worse in ISD.
- Cystoscopy after each TVT trocar pass; bladder perforation 4.5%.
- Prefer retropubic TVT in ISD or fixed urethra.

**Management of perforated mid urethral sling (SAQ)**

- Bladder trocar injury: repass laterally, continue.
- Urethral injury: abandon mesh; stage the repair.
- Late perforation: excise mesh, repair, consider autologous sling.

**Vesicovaginal fistula: management, timing of repair, eponymous techniques and recurrent VVF (LAQ)**

- Image upper tracts; exclude ureteric fistula and cancer (biopsy before redo).
- Catheter trial only if small, fresh (<3 weeks), non-irradiated; works only if started within 3 weeks.
- Vaginal and abdominal routes both close about 90%.
- Early repair for clean iatrogenic VVF; closure comparable to delayed.
- Delay ~3 months for obstetric, ≥12 months for radiation fistula.
- First repair has the highest success.
- Interposition flap (Martius, peritoneum, omentum) for complex and recurrent VVF.
- Diversion after multiple failures or severe radiation.
- Latzko: colpocleisis, bladder not opened, vagina shortened.
- O'Conor: bivalve bladder to the fistula, interpose omentum.
- Martius for distal, omentum or peritoneum for apical VVF.

**Urological complications of obstetric-gynaecological surgery: uretero- and urethrovaginal fistula (LAQ)**

- Bladder is the commonest organ injured; obstructed labour causes most fistulae worldwide.
- Ureteric injury is mostly distal and missed.
- UVF: distal ureteric injury at hysterectomy; continuous leak within 4 weeks with normal voiding.
- Orange tampon = UVF on double dye test; exclude coexisting VVF.
- Stent early (within 2 weeks); remove at 4–6 weeks.
- Reimplant with psoas hitch or Boari flap if stenting fails; success ≥90%; follow 30 months.
- Urethrovaginal fistula: diverticulectomy is the commonest Western cause; obstetric in developing countries.
- Look for a coexisting VVF (about 20%).
- Layered closure plus Martius flap; autologous sling only.
- Closure 90–95% (closure is easy); continence only 50–80% (continence is hard).

**Recto-urethral (recto-prostatic) fistula: etiopathogenesis, diagnosis and management (LAQ)**

- Most acquired RUFs follow prostate cancer treatment.
- Martini: stage = size; grade = tissue (surgery, radiation, salvage).
- VCUG or RUG plus endoscopic biopsy to exclude cancer.

- Divert first; perineal (gracilis) or York-Mason for surgical fistulae; irradiated ones need imported tissue or diversion.

**Female urethral diverticula (SAQ)**

- Periurethral gland infection (Routh); posterolateral ostium.
- MRI most sensitive (97%).
- Preserve fascia; non-overlapping layers; Martius flap.
- No synthetic mesh at diverticulectomy.

**Female sexual dysfunction (SAQ)**

- About 40% prevalence; HSDD commonest; distress required.
- DSM-5 merges desire and arousal, dyspareunia and vaginismus.
- Flibanserin and bremelanotide: only FDA-approved HSDD drugs (premenopausal).

**Endoscopic procedure of SUI (SAQ)**

- Bulking improves coaptation; does not correct hypermobility.
- Best for fixed urethra with ISD, or unfit patients.
- Less durable than slings; repeat injections common.
- Not recommended for post-prostatectomy incontinence.

**Endometriosis in urology (SAQ)**

- Bladder commonest; cyclical haematuria.
- Ureteric disease silent: screen kidneys.
- Partial cystectomy, not TUR. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

**Urethral caruncle. (SAQ)**

- Postmenopausal posterior-lip lesion; oestrogen deficiency.
- Prolapse is circumferential; caruncle is focal.
- Topical oestrogen first; excise if atypical or refractory.

**Acute urinary retention in females. (SAQ)**

- Look for prolapse, sling obstruction, pelvic mass, neurology and drugs.
- Fowler: abnormal sphincter EMG; SNM about 70%.
- CISC for persistent retention. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

SECTION 7

# Voiding Dysfunction

## 102 Surgical, Radiographic and Endoscopic Anatomy of the Male Pelvis

### 1 · Bony pelvis

- The sacrum and two innominate bones (**ilium, ischium, pubis**, fused at the acetabulum) meet at the sacroiliac (SI) joints behind and the symphysis in front.
- The **sacrospinous** ligament divides the sciatic foramen into **greater and lesser**.
- The pelvic brim is **50–60°** to horizontal. The **arcuate line** separates false from true pelvis. The iliac crest is at **L4**.

### 2 · Lower abdominal wall

#### Fascia and extravasation

- The umbilicus is the **T10** dermatome, at L3. **Scarpa's** fascia fuses with fascia lata about 1 cm below the inguinal ligament and continues as **Colles fascia**, then dartos.
- Colles is fused to the ischiopubic rami and perineal membrane, so **anterior urethral injury** fills penis and scrotum with blood and urine as a **"butterfly" perineal haematoma**. It spares the thigh but can spread **upward** deep to Scarpa's to the flank.
- †With Buck's fascia intact the haematoma stays on the shaft; a butterfly means Buck's is breached.

#### Rectus sheath

| | Above arcuate line | Below arcuate line |
|---|---|---|
| Anterior | External oblique and part of internal oblique | All three aponeuroses |
| Posterior | Rest of internal oblique and transversus | **Absent**: transversalis fascia and peritoneum only |

- Tendinous intersections tether rectus to the anterior sheath, so it can be **divided transversely without retracting**.
- Segmental nerves enter **laterally**, so a paramedian incision lateral to rectus causes atrophy and ventral hernia.

#### Inguinal canal

- It is **4 cm** long and carries the **spermatic cord and ilioinguinal nerve**. The deep ring is midway between the anterior superior iliac spine and pubic tubercle, **lateral to the inferior epigastric vessels**.
- The internal oblique and transversus roof closes onto the floor as a **shutter mechanism**.
- Direct hernias are **medial** and indirect hernias **lateral** to the inferior epigastrics. At **radical orchidectomy** the cord is ligated at the internal ring, so the stump retracts retroperitoneally.

### Peritoneal folds (laparoscopic map)

| Fold (medial to lateral) | Contains | Significance |
|---|---|---|
| Median | Urachus | Bladder apex to umbilicus |
| Medial (paired) | Obliterated umbilical arteries | Incise **lateral** to them to reach obturator fossa and retropubic space |
| Lateral (paired) | Inferior epigastric vessels | Mark deep ring and hernia type |

## 3 · Pelvic floor and fascia

- Obturator internus fascia thickens into the **tendinous arch of levator ani**, the origin of the pelvic diaphragm (levator ani and coccygeus).
- Levator ani comprises **puborectalis** (a sling behind the anorectal junction for **faecal continence**), pubococcygeus and iliococcygeus. It is tonically contracted, relaxes to void, and **Kegel exercises** increase its tone.
- **Parietal fascia** lines the walls and **visceral fascia** sheaths the organs; their lateral attachments, the **endopelvic fascia**, carry pelvic vessels and are the gateway to the prostate.
- The **arcus tendineus fascia pelvis** marks the parietal–visceral junction and anchors urethral and bladder supports.

## 4 · Perineum

- The **perineal membrane** spans the pubic arch, is pierced by the urethra and supports the sphincter.
- The **perineal body** is the midline keystone, blending with Denonvilliers; injury at perineal prostatectomy causes urinary and faecal incontinence.
- Perineal prostatectomy MCQ : **Young** divides the central tendon, leaving the external sphincter on the rectum; **Belt** passes **under** it.
- The **internal pudendal artery** runs in **Alcock's canal** and becomes the dorsal penile artery.
- The **pudendal nerve (S2–S4)** supplies both external sphincters, levator ani, ischio- and bulbospongiosus, ending as the dorsal penile nerve.

## 5 · Pelvic vessels and lymphatics

- The **external iliac** gives the inferior epigastric and deep circumflex iliac. **Accessory obturator vessels** from the inferior epigastric cross the obturator fossa and are a PLND hazard.
- **Internal iliac**: the posterior trunk is parietal (superior gluteal, ascending lumbar, lateral sacral); the anterior trunk is visceral (umbilical, obturator, middle rectal, inferior vesical, internal pudendal, inferior gluteal).
- The patent umbilical gives the **superior vesical** and artery of the vas. The **inferior vesical** (not internal pudendal) supplies the neurovascular bundle (NVB).
- The iliac veins join **behind the internal iliac artery**, a site of major PLND bleeding.

- The **dorsal venous complex (DVC)** enters under the symphysis in front of the striated sphincter; ligating it at prostatectomy can damage the sphincter.
- Lymph often **crosses sides**. External iliac nodes drain bladder and urachus; internal iliac nodes drain prostate and bladder. The **node of Cloquet** is the **distal limit of PLND**.

## 6 · Pelvic nerves

| Nerve | Roots | Key point |
|---|---|---|
| Ilioinguinal | L1 | Through inguinal canal to anterior scrotum |
| Genitofemoral | L1–L2 | On psoas; **lateral limit of PLND**; genital branch supplies cremaster |
| Femoral | L2–L4 | Injured by psoas hitch, retractor blades, inguinal dissection |
| Obturator | L2–L4 | Iliac bifurcation to obturator canal; PLND injury gives **weak adduction** |
| Pudendal | S2–S4 | Perineal sensation; levator, urinary and anal sphincters |
| Nervi erigentes | S2–S4 | Parasympathetic to bladder, rectum |

> **KEY POINT**
>
> Anterior scrotum is L1 (ilioinguinal, genitofemoral); posterior scrotum is sacral (pudendal, posterior femoral cutaneous).

*See the paper answer **Prostate anatomy, neurovascular bundles and surgical concepts in localised CA prostate** (Section 16, Ch 148).*

**Also, from the condensed book (not in the paper answer):**

- The NVB lies at **5 and 7 o'clock** at the apex; injury at prostatectomy or rectal resection causes erectile dysfunction.

## 7 · Pelvic viscera

### Rectum

- The rectum begins at **S3**. Opening the **rectovesical pouch** exposes the seminal vesicles in the posterior robotic approach. **Denonvilliers fascia** separates rectum from prostate.
- The distal rectum near the apex is the **commonest rectal injury site** at prostatectomy.

### Bladder

- The bladder is extraperitoneal except the dome, which leads to the **urachus** (adenocarcinoma, fistula, cyst).
- Circular bladder-neck fibres form the **internal sphincter**; these **α1-adrenergic** fibres maintain continence and **antegrade ejaculation**, and their damage adds to post-prostatectomy incontinence.
- Lymph goes mainly to the **external iliac** nodes, and also to internal iliac and obturator nodes, bilaterally.

### Ureterovesical junction

- The intramural ureter runs **1.5–2 cm** obliquely in **Waldeyer's sheath**; being narrow, it traps stones.
- It is a **flap valve** in which detrusor backing compresses the ureter. Weak backing causes **vesicoureteral reflux (VUR)**, and a weak hiatus allows a **Hutch diverticulum**.

- The pelvic ureter narrows at the iliac crossing and passes **behind the vas**. Its blood supply enters **laterally**, so stripping the lateral attachments causes stricture.

## 8 · Endoscopic anatomy

**ALGORITHM**

1. Under local anaesthetic, inspect bladder, orifices and trigone with a flexible cystoscope.
2. Retroflex at the neck: intravesical prostatic protrusion, hidden tumours.
3. Find the verumontanum and ejaculatory ducts; the external sphincter coapts just distal, then the bulbar urethra.

**KEY POINT**

The verumontanum is the sentinel at transurethral resection of the prostate (TURP): resection distal to it injures the external sphincter and causes incontinence.

## 9 · Radiographic anatomy

- A **phlebolith** mimics a distal ureteric or bladder stone.
- On a **cystogram**, absent lateral distension suggests an extraperitoneal pelvic haematoma or post-prostatectomy lymphocele.
- Voiding cystourethrography (**VCUG**) best shows the posterior urethra, as the sphincters relax. On retrograde urethrography (**RUG**), steady pressure overcomes the external sphincter.
- The membranous urethra is the **shortest, narrowest** segment and is not a stricture.
- Computed tomography (**CT**) needs contrast except for **stones**. In **extraperitoneal rupture**, urine lies in the retropubic and perivesical spaces.
- Malignant nodes are rounded and heterogeneous; size is most specific at ≥**1 cm** short axis.
- **CT urogram** excretory images are taken at **7–10 min**.
- Magnetic resonance imaging (**MRI**) has the best soft-tissue contrast. A distended bladder wall over **3 mm** is thickened.

# 103 Physiology and Pharmacology of the Bladder and Urethra

## 1 · Overview

- The lower urinary tract (LUT) has two functions, **storage** (continence) and **emptying** (micturition), integrating the central nervous system (CNS), peripheral nerves, detrusor smooth muscle (DSM), urothelium, urethra, pelvic floor and external urethral sphincter (EUS).

> **KEY POINT**
>
> **Why single-organ therapy fails:** bladder and urethra act reciprocally, so boosting contractility in detrusor underactivity fails unless the urethra also relaxes. Afferent modulation acts on both (reduce it for overactivity, augment it for underactivity), making it the most promising therapeutic lever.

## 2 · Functional anatomy

### Bladder

- The **male bladder neck** is a complete smooth muscle ring with dense sympathetic supply; it closes in ejaculation, ensuring **antegrade ejaculation**. Women lack this collar.
- The **lamina propria** is the "functional centre", containing the afferent plexus, interstitial myofibroblasts and capillaries.

### Stroma

- The wall is about **50% collagen, 2% elastin**, mainly types I, III and IV.
- **Poorly compliant bladders** have more connective tissue relative to muscle and a higher **type III : type I collagen** ratio, so capacity falls.

### Urethra and sphincters

- The urethra has smooth muscle (lissosphincter) and an **intramural striated rhabdosphincter (EUS)**, separate from the periurethral pelvic floor.
- The male EUS is a crescent in front of the prostate, becoming a horseshoe distally. The female EUS is a horseshoe over the ventral urethra, and a subepithelial vascular plexus forms a **"urethral seal"**.

| Feature | Slow-twitch | Fast-twitch |
|---|---|---|
| Role | Sustained resting tone | Rapid extra squeeze on coughing |
| Fatigue / metabolism | Slow; oxidative | Rapid; anaerobic |
| Where | Intramural rhabdosphincter, giving **~50% of static urethral resistance** | Periurethral pelvic floor (mixed) |
| Rhabdosphincter mix | Male **65% slow / 35% fast**; female **87% slow / 13% fast** | |

## 3 · Urothelium: barrier and sensor

*The urothelium senses stretch; its damage drives urgency and pain.*

### Barrier

- **Umbrella cells** carry **uroplakin** plaques, which form the barrier and are the binding site of **type 1-fimbriated E. coli**. **Tight junctions** block paracellular flow.
- Loss of umbrella cells (infective or radiation cystitis) lets solutes leak in, so symptoms worsen with distension.

### Sensor–transducer function

Urothelial cells behave like sensory neurons, carrying purinergic, adrenergic, muscarinic and TRP receptors. On stretch they release mediators onto afferents, directly or via myofibroblasts.

| Mediator | Released by / when | Effect |
|---|---|---|
| **ATP** | Stretch; more with injury, inflammation, ageing | Excites suburothelial **P2X3** afferents, giving fullness and pain; P2X3-null mice are hyporeflexic |
| **Nitric oxide (NO)** | Especially inflammation | Blocks afferent $Ca^{2+}$ channels: **afferent inhibition** |
| **Acetylcholine (ACh)**, non-neuronal | Basal; more with stretch, ageing | Modulates afferents and DSM, so antimuscarinics may act partly here |
| Prostaglandins (PGs) | Urothelium is the main source | Detrusor regulation, cytoprotection |
| Inhibitory factor (unknown) | Basal | Reduces muscarinic contraction; not NO or PG |

- The afferent threshold is set by the **ATP (excitatory) : NO (inhibitory) balance**. ATP is raised in detrusor overactivity (DO) and in bladder pain syndrome/interstitial cystitis (BPS/IC).
- **Suburothelial myofibroblasts** are linked by connexin 43 (**Cx43**) gap junctions and respond to ATP and ACh; their M2/M3 expression correlates with urgency in idiopathic DO.

> **KEY POINT**
>
> Damaged urothelium causes symptoms twice over: by **solute leak** into the lamina propria and by **altered signalling** (more ATP, less NO) to afferents.

## 4 · Detrusor smooth muscle physiology

*Each contraction step is a drug target.*

- Not every detrusor cell is innervated, so **gap junctions** spread excitation. $Ca^{2+}$–calmodulin drives **myosin phosphorylation**, and activity is mostly tonic.

**ALGORITHM**

1. ACh binds **M3** receptors (Gq).
2. Intracellular $Ca^{2+}$ rises via L-type entry and phospholipase C (PLC), whose IP3 releases sarcoplasmic reticulum (SR) $Ca^{2+}$.
3. $Ca^{2+}$–calmodulin activates myosin light chain kinase (MLCK), which phosphorylates MLC20.
4. Cross-bridges cycle and the muscle contracts.
5. As $Ca^{2+}$ falls, myosin light chain phosphatase (MLCP) dephosphorylates MLC20 and the muscle relaxes.

- Tone depends on $Ca^{2+}$ and the **MLCK : MLCP balance**. In **calcium sensitisation**, RhoA-activated **Rho kinase** and protein kinase C (PKC) inhibit MLCP, so force rises without more $Ca^{2+}$.
- **L-type voltage-dependent $Ca^{2+}$ channels (VDCC)** are the main trigger for contraction (blocked by nifedipine, diltiazem, verapamil).
- $K^+$ channels repolarise the cell, and **BK is the key brake**: $Ca^{2+}$ sparks open BK, hyperpolarising the cell. BK knockout gives overactive bladder (OAB), and BK is reduced in neurogenic DO (NDO).

| G protein | Receptors | Effect |
|---|---|---|
| Gq | M1, M3, M5 ("odd"); α1; AT II; H1; 5-HT2 | IP3 and DAG release $Ca^{2+}$ and activate PKC: contraction |
| Gi/o | M2, M4 ("even") | Lower cAMP |
| Gs | β3 | Raise cAMP, activating PKA: relaxation |

- **Purinergic (P2X1) contraction** is minimal normally but appears in obstructed, OAB and ageing bladders. It suppresses muscarinic coupling, explaining detrusor hyperactivity with impaired contractility (**DHIC**): urgency with incomplete emptying.
- The detrusor is deliberately **poorly coupled**, protecting filling. More Cx43 in bladder outlet obstruction (BOO) and NDO, induced by cytokines and nerve growth factor (NGF), allows whole-bladder myogenic contraction, which is DO.

**KEY POINT**

**Motor–sensory coupling:** spontaneous micromotions generate afferent bursts perceived as urgency. Antimuscarinics and mirabegron reduce non-voiding contractions and so reduce sensory input; efferent and afferent functions are inseparable.

## 5 · Receptors and drug targets

| Receptor | Location | Function | Drug target |
|---|---|---|---|
| **M3** (Gq) | Detrusor; urothelium; salivary glands | **Mediates contraction** | Antimuscarinics |
| **M2** (Gi) | Detrusor (**numerically predominant**) | Amplifies M3 by opposing β-cAMP relaxation; the balance shifts from M3 to M2 in neurogenic and obstructed bladders | Non-selective antimuscarinics |

| Receptor | Location | Function | Drug target |
|---|---|---|---|
| **β3** (Gs) | Detrusor (**most expressed adrenoceptor**); afferents | Relaxes during storage; increases compliance | Mirabegron, †vibegron |
| **α1A** | Urethra, prostate | Urethral closure; female resting tone | α-blockers (benign prostatic hyperplasia, BPH); α-agonists increase closure |
| **α1D** | Bladder (predominant α); spinal cord | Storage symptoms of obstruction | Tamsulosin improves storage lower urinary tract symptoms (LUTS) |
| α2 | Prejunctional urethra; ganglia | Reduces noradrenaline release | — |
| **P2X3** | Suburothelial afferents | Senses ATP; increased in BPS/IC | AF-219 (proof-of-concept) |
| P2X1 | Detrusor | Purinergic contraction; increased in BOO, ageing | Potential for DHIC |
| **TRPV1** | C fibres; urothelium | Senses heat >43 °C, protons, capsaicin; increased in NDO | Capsaicin and resiniferatoxin (RTX) desensitise |
| TRPM8 | Afferents | Senses cold (bladder cooling reflex) | Experimental |
| NK1 | Spinal cord | Afferent transmission | Aprepitant reduced urgency (trial) |
| Nicotinic | EUS | Sphincter contraction | — |

## LAQ Overactive bladder: pathophysiology, pharmacotherapy and non-pharmacological management

Paper answer · P1 · P2 · P3 · merged from 4

*Asked as:* P1 LAQ "Describe pathophysiology of OAB. Briefly outline management &describe nonmedical non-pharmacological management" · P2 SAQ "Overactive bladder and its drug therapy" · P3 SAQ "OAB Pharmacotherapy: Darifenacin vs Solifenacin vs Tolterodine, and Mirabegron" · P1 SAQ "Compare and contrast various detrusor relaxants in management of OAB"

### Definition

- **OAB (overactive bladder), ICS (International Continence Society): urgency**, usually with frequency and nocturia, with or without UUI (urgency urinary incontinence), in the absence of UTI (urinary tract infection) or other obvious pathology.
- **OAB dry** versus **OAB wet** (with UUI).
- OAB is a **symptom** diagnosis. DO (detrusor overactivity) is a **urodynamic** finding; only 44–69% of patients with urgency show DO.

### Pathophysiology

All mechanisms converge on **more afferent signal reaching awareness**: the bladder turns the signal up, or the CNS (central nervous system) fails to turn it down.

| Mechanism | Explanation |
|---|---|
| **Afferent (sensory)** | Aδ fibres normally sense filling. Unmyelinated **C fibres** are "silent", but become spontaneously active and hypersensitive in pathological states |
| **Urothelial** | Urothelium acts as a transducer: mechanical, inflammatory or chemical stimuli release ATP (adenosine triphosphate), acetylcholine and other mediators onto suburothelial afferents |
| **Myogenic** | Patchy denervation causes receptor upregulation and spontaneous contraction; exaggerated **micromotions** cause urgency, and coordinated spread across the wall gives DO |
| **Neurogenic / central** | Forebrain and PAG–PMC (periaqueductal grey–pontine micturition centre) normally inhibit voiding; loss of this inhibition lets "afferent noise" be felt as urgency |
| **Other** | Bladder–bowel crosstalk (constipation), ischaemia–reperfusion, chronic inflammation, altered urinary microbiome |

- **Risk factors and associations:** ageing, female sex, BOO (bladder outlet obstruction: an animal model, but only an indirect cause in humans; DO persists in up to 60% after TURP [transurethral resection of the prostate]), anxiety and depression, metabolic syndrome, poor toileting habits.

### Evaluation

- **Minimum:** history, examination, **urinalysis**.
- Bladder diary (3–7 days; P2: 3 days), questionnaires, PVR (post-void residual) when indicated.
- Urodynamics and cystoscopy only for complicated or refractory cases, or if urodynamics will change an invasive decision.

### Outline of management (AUA/SUFU [American Urological Association/Society of Urodynamics, Female Pelvic Medicine and Urogenital Reconstruction]; not a compulsory sequence)

`Education + behavioural therapy (first line) ↓ (4–12 weeks) Antimuscarinic or β3-agonist (mirabegron, vibegron); combinations ↓ PTNS / onabotulinumtoxinA / SNM (third line) ↓ Augmentation cystoplasty or urinary diversion`

| Line | Treatment |
|---|---|
| Behavioural | Education; reduce fluids by about 25%; cut caffeine; weight loss; **bladder training** (timed voids, interval increased by 15–30 minutes); **PFMT (pelvic floor muscle training)** with urge suppression: 50–80% fewer leaks |
| Drugs | **Antimuscarinics** (oxybutynin, solifenacin, extended-release trospium, darifenacin, fesoterodine, tolterodine); **β3-agonists** (mirabegron, vibegron); combinations |
| Third line | **PTNS (percutaneous tibial nerve stimulation)** weekly × 12; **onabotulinumtoxinA 100 U** intradetrusor; **SNM (sacral neuromodulation)** at S3 (success about 70–80%; revision >30% at 5 years) |
| Last resort | Augmentation cystoplasty or urinary diversion |

### Non-pharmacological management

#### 1. Education and lifestyle

- Explain normal bladder function; set realistic goals.

- Reduce fluid intake by about 25% if excessive; cut caffeine; treat constipation.
- **Weight loss:** about 8 kg loss in obese women reduced UUI episodes.

**2. Bladder training (BT)** (breaks the cycle of urgency, frequent voiding and smaller capacity)

1. Keep a bladder diary.
2. Start a fixed interval (often 1 hour while awake).
3. Void by the clock, not by urge.
4. Increase by 15–30 minutes a week as control improves.
5. Target interval 3–4 hours. - Works by **uncoupling the void from the urge**.

**3. PFMT and urge suppression**

- Kegel exercises: quick 1–2 s flicks and 5–10 s holds, twice daily.
- A pelvic floor contraction **reflexly inhibits the detrusor**. Urge suppression: stop, stay still, squeeze quickly several times, relax, let the urge pass, then walk calmly to the toilet.
- PFMT and BT reduce incontinence episodes by **50–80%**.
- **Also know:** biofeedback helps patients who cannot find the muscle; electrical stimulation at 5–20 Hz inhibits the detrusor; timed and prompted voiding for cognitively impaired patients.

**4. PTNS**

- 34-gauge needle 3–4 cm above the medial malleolus; 20 Hz, 200 µs; confirm great-toe flexion or sole tingling. **30 min weekly for 12 weeks**, then maintenance.
- The posterior tibial nerve (L4–S3) shares sacral roots with the bladder. **SUmiT trial:** 55% vs 21% (sham) responders.

**5. SNM**

- Mechanism: stimulates **afferents** that modulate brainstem and forebrain control.
- **Test phase:** PNE (percutaneous nerve evaluation, 5–7 days) or **staged tined lead** at **S3** (anal "bellows" and great-toe flexion). ≥50% improvement on diary leads to implanting the pulse generator in the buttock.
- Success about 70–80%. **InSite trial:** 61% vs 42% (standard medical therapy). Revision >30% at 5 years; MRI (magnetic resonance imaging)-conditional rechargeable devices now available.

**6. Intradetrusor onabotulinumtoxinA**

- Cleaves SNAP-25 (synaptosomal-associated protein 25), blocking acetylcholine release; also reduces afferent signalling.
- **100 U in 10 mL at 20 sites** cystoscopically; effect lasts about 6–9 months.
- About 25% become completely dry. Risks: UTI about 20%, retention needing CISC (clean intermittent self-catheterisation). **ROSETTA:** greater UUI reduction than SNM at 6 months, but more UTI.

**7. Last resort:** augmentation cystoplasty or urinary diversion.

### Drug therapy: place and mechanisms

- Drugs are second line, after behavioural therapy, as first-line pharmacotherapy (antimuscarinics and β3 agonists alongside each other; AUA/SUFU). Two main classes: **antimuscarinics** and **β3-adrenoceptor agonists**.

- **Antimuscarinics:** acetylcholine acts on **M3** receptors (raising intracellular calcium) to contract the detrusor; the commoner M2 enhances it. Blockade reduces involuntary contractions and afferent urgency signalling.
- Muscarinic side effects by subtype: **M1** (brain) cognition; **M2** heart rate; **M3** (glands, gut) dry mouth, constipation.
- Selectivity is by **receptor subtype** (M3) or **tissue** (bladder over salivary gland); tissue selectivity matters more.
- **β3 agonists:** β3 is the commonest (most highly expressed) adrenoceptor in human detrusor. Activation raises **cAMP (cyclic adenosine monophosphate)** and relaxes the detrusor during storage, sparing the voiding contraction.

### Agents

| Drug | Distinguishing feature | Usual dose |
|---|---|---|
| Oxybutynin | Non-selective; direct relaxant; metabolite causes most dry mouth; patch, gel | 5 mg two to three times daily; ER (extended-release) |
| Tolterodine | Bladder (over salivary) selective; QT prolongation at high dose | 2 mg twice daily (IR, immediate-release); ER 4 mg daily |
| Fesoterodine | Prodrug of tolterodine's active metabolite | 4–8 mg daily |
| Solifenacin | 10-fold M3 > M2; long half-life | 5–10 mg daily |
| Darifenacin | About 30-fold M3-selective | 7.5–15 mg daily |
| Trospium | Quaternary; does not cross the blood–brain barrier (little brain entry) | 20 mg twice daily |
| Mirabegron (β3) | Inhibits CYP2D6 (a liver enzyme): metoprolol interaction; small blood-pressure rise | 25–50 mg daily |
| Vibegron (β3) | No CYP2D6 interaction | 75 mg daily |

### Darifenacin vs solifenacin vs tolterodine

| Feature | Darifenacin | Solifenacin | Tolterodine |
|---|---|---|---|
| Selectivity | **Most M3-selective (about 30-fold)**; not tissue-selective | M3 about 10-fold over M2; bladder-selective | Non-subtype-selective; **bladder-selective** |
| Dose | 7.5–15 mg daily (ER) | 5–10 mg daily | 2 mg twice daily (IR); 4 mg daily (ER) |
| Half-life | Moderate (ER once daily) | **Long** (about 2–3 days): once daily | Short; active metabolite 5-HMT (5-hydroxymethyl tolterodine) |
| Metabolism | CYP3A4 (a liver enzyme) and CYP2D6 | CYP3A4 | CYP2D6 (and 3A4) |
| Dry mouth | Dose-related | Dose-related | Lower with ER |
| Constipation | **Highest** (gut M3) | Moderate–high | Lower |
| CNS | Spares M1: **preferred in elderly** | Better persistence | High dose: QT prolongation |
| Dose limit | Lower dose with CYP3A4 inhibitors or liver disease | 5 mg in severe renal or moderate liver impairment | Halve with renal or liver impairment |

- **Efficacy is similar**; response is individual, so switch agent or dose.

- **STAR trial:** flexible solifenacin beat tolterodine ER for urgency and incontinence.

### Mirabegron (β3 agonist)

- First **β3-adrenoceptor agonist** for OAB; FDA (Food and Drug Administration) approval 2012.
- **Mechanism:** activation of adenylyl cyclase raises **cAMP**, relaxing the detrusor during **storage**. Reduces non-voiding contractions and afferent firing. The voiding contraction (muscarinic M3) is spared, so PVR barely rises.
- **Dose:** oral, extended release, once daily: start **25 mg**, increase to **50 mg** after 4–8 weeks. **25 mg maximum** in severe renal impairment (eGFR [estimated glomerular filtration rate] 15–29) or moderate hepatic impairment; avoid in end-stage renal disease and severe hepatic impairment.
- **Moderate CYP2D6 inhibitor:** raises levels of metoprolol, desipramine, flecainide; monitor digoxin.
- **Efficacy:** phase III trials (SCORPIO, ARIES, CAPRICORN): fewer incontinence episodes and micturitions than placebo. **TAURUS** (12 months): efficacy similar to tolterodine, with less dry mouth.
- **Adverse effects:** hypertension, tachycardia, headache, nasopharyngitis, UTI. **Dry mouth and constipation near placebo rates**, far below antimuscarinics.
- **Contraindicated:** severe uncontrolled hypertension (≥180/110 mmHg). Caution with arrhythmia, and with BOO when combined with antimuscarinics (retention).
- Other uses: add-on to α-blocker in men with BPH (benign prostatic hyperplasia); neurogenic detrusor overactivity.
- **Also know:** vibegron 75 mg (no CYP2D6 interaction).

### Antimuscarinics versus β3 agonists

| Feature | Antimuscarinics | β3 agonists |
|---|---|---|
| Efficacy | Similar | Similar (TAURUS: mirabegron ≈ tolterodine) |
| Dry mouth, constipation, blurred vision | Common; main reason for stopping | Near placebo |
| Cognition | Central M1 blockade; **dementia risk 10–65% higher with use >3 months** | No anticholinergic burden |
| Retention | Up to 3% in men | Low |
| Contraindications | Urinary or gastric retention, uncontrolled narrow-angle glaucoma | Uncontrolled (severe) hypertension (≥180/110 mmHg) |

### Choice of drug and special groups

- **Elderly and cognitive impairment:** antimuscarinic use >3 months is linked to 10–65% more new-onset dementia (SUFU white paper). Prefer a **β3 agonist** (no anticholinergic burden), or ER trospium, darifenacin or fesoterodine. Beers criteria: avoid antimuscarinics in dementia.
- ER forms cause less dry mouth. Poor persistence is the main problem; switch dose or drug before third-line therapy.

- **Combination:** solifenacin plus mirabegron beats each alone (**SYNERGY, SYNERGY II**), without extra cardiovascular effects; in non-responders it equals solifenacin 10 mg with less dry mouth (**BESIDE**).
- **Men:** α-blocker first; add an antimuscarinic (if PVR is low) or β3 agonist; retention up to 3%, so check PVR.
- **Also know:** vaginal oestrogen in atrophy; desmopressin for nocturnal polyuria; imipramine (children, mixed incontinence); intravesical oxybutynin; refractory OAB: onabotulinumtoxinA 100 U and neuromodulation.

### Diagrams to draw

- Bladder showing urothelial afferents, C fibres and micromotions, with brainstem and forebrain (cortical) inhibitory pathways.
- Detrusor cell: acetylcholine → M3 → contraction; noradrenaline → β3 → cAMP → relaxation.
- Muscarinic subtypes M1 (brain), M2 (heart), M3 (bladder, gland, gut) with each drug's selectivity.

**Key points to remember:**

- OAB is defined by urgency; DO is a urodynamic finding.
- Final pathway: increased afferent signal (urothelial, myogenic) or failed central inhibition.
- Behavioural therapy first: BT and PFMT with urge suppression give 50–80% fewer leaks; antimuscarinic or β3-agonist next.
- M3 mediates contraction; β3 raises cAMP for storage-phase relaxation; voiding spared.
- Equal efficacy; β3 avoids dry mouth and dementia risk (10–65% with antimuscarinics >3 months).
- Darifenacin most M3-selective; tolterodine bladder-selective; solifenacin longest half-life.
- STAR: solifenacin beat tolterodine ER.
- Mirabegron 25 mg → 50 mg; avoid if blood pressure ≥180/110; CYP2D6 interaction (metoprolol).
- Elderly: mirabegron (β3), trospium or darifenacin preferred over other antimuscarinics.
- Solifenacin + mirabegron combination works (SYNERGY).
- Refractory OAB: PTNS (SUmiT 55% vs 21%), SNM (InSite 61% vs 42%), Botox 100 U.

**Also, from the condensed book (not in the paper answer):**

- **Oxybutynin**: its metabolite **desethyloxybutynin** causes most of the dry mouth, so the controlled-release form is better tolerated.
- CNS-penetrant agents block M1, causing delirium and cognitive decline that is reversible within weeks of stopping.
- Outlet drugs are **α-agonists** (ephedrine, midodrine); systemic oestrogen may worsen urinary incontinence (UI).

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## SAQ Duloxetin

Paper answer · P3

**Type:** SAQ (10 marks)

**Source:** Practice Set 4 Paper 3, Q3e

*Read as: duloxetine, mainly its role in stress urinary incontinence.*

### Introduction

- **Duloxetine** is an SNRI (serotonin–noradrenaline reuptake inhibitor), also an antidepressant.
- Licensed in Europe (not USA) for **moderate to severe SUI (stress urinary incontinence) in women**; off-label in men.

### Mechanism

- During storage, bladder afferents drive glutamate excitation of pudendal motor neurons in **Onuf's nucleus (S2–S4)**, which keep the striated urethral sphincter closed.
- Serotonin (5-HT2) and noradrenaline (α1) amplify this glutamate drive. Duloxetine blocks their reuptake, so **sphincter tone and the sneeze-induced continence reflex increase**.
- In voiding, glutamate drive switches off, so retention does not occur.
- Central effects also reduce bladder activity.

### Dose

- **40 mg twice daily**; starting at 20 mg twice daily for 2 weeks reduces nausea. Taper on stopping.

### Evidence

- Phase III RCTs (randomised controlled trials) in women: about 50% median fall in incontinence episodes versus about 30–40% with placebo; better quality of life; **cure rare**.
- Benefit is temporary and lasts only while taken.
- Male PPI (post-prostatectomy incontinence): small, short-term benefit.

### Guideline position

- **EAU (European Association of Urology):** may be offered to women for **temporary** improvement, not cure; warn about side effects. Men: offer for SUI (weak), and **inform of off-label use and adverse effects** (strong).
- **NICE (National Institute for Health and Care Excellence):** not first line; not routinely second line; only if surgery is unwanted or unsuitable.
- Adding PFMT (pelvic floor muscle training): evidence conflicting.

### Adverse effects

- **Nausea** (commonest, early), dry mouth, dizziness, insomnia, constipation; **high discontinuation**.
- Raised blood pressure, hyponatraemia (elderly), suicidal ideation, discontinuation syndrome.

### Contraindications and interactions

- MAOIs (monoamine oxidase inhibitors), including **methylene blue** (serotonin toxicity); CYP1A2 inhibitors (fluvoxamine, ciprofloxacin); hepatic impairment; severe renal impairment; uncontrolled hypertension or narrow-angle glaucoma.

**Diagram to draw:** Onuf's nucleus with 5-HT and noradrenaline terminals driving the pudendal nerve.

**Key points to remember:**

- SNRI acting on Onuf's nucleus to increase sphincter tone.
- 40 mg twice daily; titrate from 20 mg to reduce nausea.
- Temporary benefit only; approved in Europe, not USA.

## 6 · Peripheral innervation

*See the paper answer **Voiding dysfunction in children: neurophysiology and differential diagnosis of frequency/urgency** (Section 5, Ch 47).*

**Also, from the condensed book (not in the paper answer):**

- Parasympathetic outflow is **S2–S4**; sympathetic is †**T10–L2**.
- Some parasympathetic postganglionic neurons are **intramural**, so cauda equina or pelvic plexus injury **decentralises rather than denervates** the bladder.
- C fibres are mucosal, mostly **silent** and capsaicin/RTX-sensitive, and they drive a new **spinal reflex** after spinal cord injury (SCI), inflammation, BOO or ageing.
- Afferent sensitisation by ATP, NGF and inflammatory mediators leads to central sensitisation, and **cross-organ sensitisation** explains BPS/IC overlapping irritable bowel syndrome. μ-opioids raise sphincter tone, causing postoperative retention.

## 7 · Central control

*Lesion level predicts the bladder pattern.*

- The pontine micturition centre (**PMC**) is **Barrington's nucleus** ("M region"); lesions (stroke, multiple sclerosis, tumour) cause **retention**. It does **not** project to Onuf's nucleus but relaxes the sphincter **indirectly** via GABA/glycine interneurons.
- The **L-region** (ventrolateral pons) is the storage centre and projects to Onuf's nucleus.
- The periaqueductal grey (**PAG**) relays afferents to the PMC, forming a binary "on–off" switch; without higher control the bladder empties at a critical volume, as in infants.
- In the **forebrain**, the **insula** registers desire and the **prefrontal cortex** decides, keeping the reflex under **tonic suppression**. Frontal white-matter lesions cause **incontinence**.

**KEY POINT**

**Lesion localisation:** a forebrain lesion removes tonic inhibition, giving DO with preserved coordination. A pontine lesion gives retention. A suprasacral cord lesion gives DO with detrusor–sphincter dyssynergia (**DSD**). A sacral or cauda lesion gives a decentralised, acontractile bladder (see Ch 104).

## 8 · Storage and emptying phases

- Storage pressure stays low through intrinsic wall properties, quiet parasympathetic outflow and a **sympathetic storage reflex**, which is weak in humans.
- **Guarding reflex**: filling recruits pudendal motoneurons, raising EUS activity, and on coughing urethral pressure rises **before** pressure transmission. Its suppression in voiding is supraspinal, so SCI causes **DSD**.
- **Sphincter-to-bladder reflex**: pudendal afferents inhibit bladder preganglionic neurons, the basis of "holding on" and neuromodulation.

**ALGORITHM**

1. Tension-receptor (Aδ) firing exceeds threshold.
2. If the cortex permits, the PAG/PMC switch flips.
3. Sympathetic and pudendal outflow switch off; the **urethral sphincter relaxes first**, and urethral smooth muscle relaxes (NO).
4. The detrusor contracts (parasympathetic) and flow begins.
5. Urethral afferents reinforce the contraction until the bladder is empty.

- **Urethra-to-bladder reflexes**: urethral flow facilitates contraction, linking stress urinary incontinence (SUI) with detrusor instability.
- After SCI, weeks of spinal shock (retention) precede an NGF-driven **C-fibre spinal reflex**, giving NDO with DSD.
- **Pdet = Pves − Pabd**. Detrusor power (Pdet × flow, Q) is roughly constant, so voiding **Pdet reflects outlet resistance, not contractility**.
- Women with SUI void at Pdet at Qmax ≈ **19 $cmH_2O$**, healthy men at ≈ **60–70 $cmH_2O$**. Low pressure in women does not mean impaired contractility, and male nomograms are invalid in women.

## 9 · Urethral closure mechanisms

*Each SUI treatment targets one closure force.*

- α-blockade lowers urethral pressure by about **one third**, so smooth muscle (α1A) tone matters.
- **Mucosal seal**: lamina propria vessels fill to seal the lumen, and oestrogen increases this flow.
- **Passive pressure transmission** reaches the proximal urethra; in the **hammock hypothesis (DeLancey)** it is compressed against a firm posterior fascial layer.
- **Active** guarding adds a fast-twitch reflex squeeze on coughing.

> **KEY POINT**
>
> Continence = **active muscle tone + passive anatomical coaptation**. The integral theory led to the mid-urethral sling (MUS), the SUI standard, yet DeLancey's data suggest intrinsic sphincter function matters more than support (see Ch 109).

## 10 · Compliance

### SAQ Urinary bladder compliance

Paper answer · P3

**Type:** SAQ (10 marks)
**Source:** 2017-Summer Paper 3, Q3e

#### Definition

- **Compliance** is the change in volume for a change in detrusor pressure: **C = ΔV / ΔPdet**, in **mL/$cmH_2O$**.
- It reflects **low-pressure storage** during filling.

### Physiological basis

- **Viscoelastic wall**: smooth muscle, collagen (about 50% of the wall) and elastin (about 2%). Coiled collagen straightens as the bladder fills.
- More collagen or less elastin makes the wall stiff.

### Measurement (filling cystometry)

- ICS (International Continence Society) points: Pdet at **start of filling** and at **cystometric capacity** (or before a leak-causing contraction), **excluding detrusor contractions**.
- **No agreed normal value**; compliance depends on capacity. Below **20 mL/cm$H_2O$** (some use 10–20) suggests poor accommodation. In children, normal is often taken as **>15 mL/cm$H_2O$**.

### Pitfalls

| Pitfall | Effect | Fix |
|---|---|---|
| Fast filling | False low compliance | Slow or stop filling; pressure falls |
| Sustained low-amplitude detrusor contraction | Mimics low compliance | Stop filling: pressure returns to baseline |
| **Pop-off**: VUR (vesicoureteric reflux), diverticulum, leaking outlet | Falsely good compliance | Video-urodynamics; occlude outlet |
| - A small bladder can give a low number without danger (**compliance paradox**). **Absolute storage pressure matters more than the ratio.** | | |

### Causes of low compliance

- **Neurogenic**: spina bifida, spinal cord injury, lower motor neuron and decentralisation injury (radical pelvic surgery).
- **Outlet obstruction**: PUV (posterior urethral valves), long-standing BPO (benign prostatic obstruction).
- **Structural**: radiation cystitis, tuberculosis, schistosomiasis, multiple bladder operations.

### Consequences and DLPP

- **DLPP (detrusor leak point pressure)**: lowest Pdet at which leakage occurs without contraction or strain.
- **McGuire (1981)**: DLPP **>40 cm$H_2O$** in myelodysplastic children led to hydronephrosis or VUR in 85%. Not a validated safe limit (20 cm$H_2O$ is more sensitive); aim as low as possible.
- A strong outlet keeps the patient dry but transmits high pressure to the kidneys.

### Management

- **Clean intermittent catheterisation + antimuscarinics** (first line).
- **Intradetrusor botulinum toxin**.
- **Augmentation cystoplasty** if refractory; vesicostomy in infants.
- Repeat urodynamics to confirm safety.

**Diagram to draw:** cystometrogram: normal flat versus steep rising filling curve.

**Key points to remember:**

- C = ΔV/ΔPdet; <20 mL/cm$H_2O$ suggests poor compliance.
- McGuire: DLPP >40 cm$H_2O$, 85% upper-tract damage.
- Pop-offs (VUR, diverticulum) hide true pressure; use video-urodynamics.

**Also, from the condensed book (not in the paper answer):**

- Normal filling keeps pressure **<10 cm$H_2O$**. Determinants are viscoelasticity, active tone and geometry.
- †A detrusor leak point pressure (DLPP) ≥ **40 cm$H_2O$** in a low-compliance neurogenic bladder predicts upper tract damage (see Ch 104).

## 11 · Pathophysiology links

- **Idiopathic OAB**: DO is found in a minority (11% in one cohort), so symptoms reflect dysregulated sensation: more ATP/TRPV1 and Cx43, less BK, and NGF plasticity.
- **BOO** causes hypertrophy and partial denervation, which lead to **muscarinic supersensitivity** and more P2X1 and Cx43. NGF stays raised, which explains **persistent storage symptoms after prostate surgery**.
- **BPS/IC** shows more NGF, ATP/P2X3, TRPV1 and mast cells; tanezumab and intravesical RTX failed (see Ch 113).
- **Underactive bladder (UAB)** is a symptom complex, whereas detrusor underactivity is the urodynamic finding.
- A post-void residual (PVR) ≥ **300 mL** warrants evaluation, as do bothersome LUTS, hydronephrosis, renal insufficiency or culture-proven urinary tract infection (UTI); PVR <300 mL does not prove normal contractility.

## 12 · Ageing

- Ageing causes **hypofunction**: contractility, sensation and closure pressure fall. Capacity does not fall per se, so small capacity means coexisting DO.
- **α-contraction increases (α1D) and β-relaxation decreases**, so noradrenaline contracts the aged bladder.
- The aged CNS is prone to overactivity, giving urgency with poor emptying (DHIC). †Prefer β3 agonists or low-CNS-penetrant agents in the elderly.

## 13 · Neuromodulation and botulinum toxin — mechanisms

- Sacral neuromodulation (**SNM**), pudendal stimulation and posterior tibial nerve stimulation (**PTNS**) treat refractory OAB **and** non-obstructive retention by acting on **afferent axons**.
- In OAB, somatic input inhibits preganglionics and transfer to the PMC; in retention, it switches off an overactive **guarding reflex**.
- **OnabotulinumtoxinA (OBTX)**: the light chain cleaves **SNAP-25**, reducing ACh release. It also lowers sensory receptors, neuropeptides and urothelial ATP.
- RTX desensitises TRPV1 **without prior excitation**, unlike capsaicin.

## 14 · Landmark trials

| Trial | Design | Result | Take-home |
|---|---|---|---|
| ABC | RCT: intradetrusor OBTX vs oral antimuscarinic, idiopathic OAB with urgency urinary incontinence (UUI) | Equal fall in daily UUI; OBTX gave complete dryness more often | Antimuscarinic: dry mouth; OBTX: catheterisation, UTI |
| ROSETTA | RCT: OBTX 200 U vs SNM, women with refractory non-neurogenic OAB, 2 years | Comparable efficacy; OBTX more often dry | OBTX more UTIs (note 200 U dose) |

# 104 Pathophysiology and Classification of Lower Urinary Tract Dysfunction

## 1 · Normal function: two phases

| Storage requires | Voiding requires |
|---|---|
| Low detrusor pressure (normal compliance), appropriate sensation | Detrusor contraction of adequate magnitude and duration |
| Outlet closed at rest and on straining | Fall in smooth and striated sphincter resistance |
| No detrusor overactivity (DO) | No anatomic obstruction |

- **Storage is mostly passive**. More collagen (inflammation, bladder outlet obstruction [BOO], decentralisation) gives **drug-unresponsive** low compliance.
- Storage relies on α1 bladder-neck tone, β3 detrusor relaxation and the spinal **guarding reflex**; voiding uses pelvic parasympathetic outflow coordinated by the **pons**.
- **Stress continence** needs equal pressure transmission to the proximal urethra, active reflex closure and a **mucosal seal**.

## 2 · Pathophysiology

Every failure is of the **bladder, outlet, or both**: the basis of the **Wein functional classification**.

### Storage failure

- **Bladder overactivity** means involuntary contractions, low compliance, or both.
- **Overactive bladder (OAB)** (International Continence Society, ICS) is urgency with or without urgency urinary incontinence (UUI), usually with frequency and nocturia. With UUI assume DO, as urodynamic studies often miss it.
- **Hypersensitivity** without overactivity occurs in interstitial cystitis/bladder pain syndrome (IC/BPS). **Outlet underactivity** follows denervation, surgery, trauma, ageing or loss of female support.

| | Hypermobility (genuine stress urinary incontinence, SUI) | Intrinsic sphincter deficiency (ISD) |
|---|---|---|
| At rest | Outlet competent | Bladder neck poorly functional or open |
| Cause | Poor pelvic support | Prior surgery, scarring, neurological |
| Treatment | Correct hypermobility | Coaptation or compression |

> **KEY POINT**
>
> Urinary incontinence can occur from ISD alone, but **never from hypermobility alone**. In **DeLancey's hammock**, a firm suburethral layer lets abdominal pressure compress the urethra.

### Emptying failure

- **Bladder underactivity** is neurogenic, myogenic (overdistension, drugs) or psychogenic.

- **BOO** (ICS) is low flow and/or raised post-void residual (PVR) **with high detrusor pressure**. Male causes are prostate, bladder neck contracture and stricture; female causes are prolapse and slings.
- **Functional** obstruction includes striated sphincter dyssynergia (commonest), smooth sphincter dyssynergia (**autonomic dysreflexia**), primary bladder neck obstruction and dysfunctional voiding (**non-neurogenic**).

## 3 · Functional classification (Wein) — 2 × 2 table

| | **Because of the bladder** | **Because of the outlet** |
|---|---|---|
| **Failure to store** | DO; low compliance; hypersensitivity; underactivity with overflow | Hypermobility; ISD. (Plus **fistula**) |
| Treatment | Bladder training; antimuscarinics, β3-agonists, onabotulinumtoxinA; neuromodulation; augmentation | Pelvic floor training; α-agonists, duloxetine; bulking; slings; artificial urinary sphincter (AUS) |
| **Failure to empty** | Underactivity: neurogenic, myogenic, psychogenic | Anatomic obstruction; sphincter dyssynergia; bladder neck dysfunction |
| Treatment | Valsalva/Credé; parasympathomimetics; electrical stimulation | α-blockers, 5α-reductase inhibitors (5ARIs); prostatectomy, incision; sphincter onabotulinumtoxinA, sphincterotomy |
| **Circumvent** | Pads or collecting devices, antidiuretics, catheterisation, diversion | |

> **KEY POINT**
>
> Bladder underactivity appears under **both** storage (overflow) and emptying. MCQ An α-agonist tightens the outlet (storage); an α-blocker relaxes it (emptying). OnabotulinumtoxinA in the detrusor treats storage; in the sphincter it treats emptying.

**Combined deficits** (T10 paraplegia: DO and dyssynergia) are managed by external collection, or by **abolishing DO** plus clean intermittent catheterisation (CIC).

## 4 · Urodynamics by phase and site

| **Phase** | **Bladder** | **Outlet** |
|---|---|---|
| Storage | Filling cystometry; detrusor leak point pressure (DLPP) | Urethral pressure profile; Valsalva leak point pressure (VLPP) |
| Voiding | Voiding cystometry (Pdet) | Fluoroscopy; sphincter electromyography (EMG) |

Flowmetry and PVR assess both phases but **cannot localise** the fault.

## 5 · Other classifications

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### LAQ Pathophysiology, classification and management of LUTS.

Paper answer · P2

**Type:** LAQ (25 marks)
**Source:** 2024-Winter Paper 2, Q1 (medical and surgical management of BPH also answered in 2015-Summer Paper 2, Q1)

### Definition

- **LUTS (lower urinary tract symptoms):** symptoms of the storage, voiding and post-micturition phases; not disease-specific; up to half of older men.
- **BPH (benign prostatic hyperplasia)** is histology; BPE is enlargement; BPO is prostatic obstruction; **BOO (bladder outlet obstruction)** has any cause. Not every man with LUTS has BPO.

### Classification

#### 1. By phase (ICS, International Continence Society)

| Storage | Voiding | Post-micturition |
|---|---|---|
| Frequency, urgency, **nocturia** (waking ≥1 time to void), urgency incontinence | Weak or split stream, hesitancy, intermittency, straining, terminal dribble | Feeling of incomplete emptying, post-micturition dribble |

- Storage symptoms form **OAB (overactive bladder)**: urgency ± urgency incontinence, usually with frequency and nocturia; most bothersome.

#### 2. Functional (Wein)

| | Because of the bladder | Because of the outlet |
|---|---|---|
| Failure to store | DO (detrusor overactivity), low compliance, hypersensitivity | Sphincter weakness (SUI, stress urinary incontinence) |
| Failure to empty | DUA (detrusor underactivity) | BPO, stricture, bladder neck contracture, sphincter dyssynergia |

**3. Severity:** IPSS (International Prostate Symptom Score), 7 questions scored 0–5 plus a quality-of-life question: mild 0–7, moderate 8–19, severe 20–35.

**4. By cause:** prostate, bladder, urethra, neurological, polyuria, infection, stone, tumour.

### Pathophysiology

- **Prostate:** BPH is an increase in cell number of stroma and glands in the **transition zone** and periurethral region. **Type 2 5α-reductase** in stromal fibroblasts makes **DHT (dihydrotestosterone)**; androgens are permissive, not mitogenic. Ageing, inflammation and metabolic syndrome contribute.
- **Static component:** gland bulk; the **capsule** transmits expansion to the urethra (TUIP, transurethral incision of the prostate, relieves obstruction without removing tissue).
- **Dynamic component: α1A-adrenergic** smooth-muscle tone; the basis of α-blockers.
- Size does not correlate with symptoms or obstruction.
- **Bladder response:** hypertrophy → compensation → decompensation. Collagen causes trabeculation and low compliance; DO causes storage symptoms; late detrusor failure causes retention. One-third keep storage symptoms after surgery.
- **Also know:** nocturnal polyuria (night output >33% of 24 h over age 65); BPO complications: AUR (acute urinary retention), UTI (urinary tract infection), stones, haematuria, renal failure.

### Evaluation (EAU, European Association of Urology, men ≥40)

- **Basic:** history (comorbidity, drugs, sexual function), IPSS, DRE (digital rectal examination) and neurological check, **urinalysis (mandatory)**, PVR (post-void residual), PSA (prostate-specific antigen) if life expectancy >10 years.

- **Frequency–volume chart, 3 days:** for storage symptoms or nocturia.
- **Before treatment:** uroflowmetry (Qmax, maximum flow rate, <10 mL/s suggests BOO); ultrasound for volume and median lobe.
- **Selected:** creatinine and renal ultrasound if large PVR; cystoscopy for haematuria, stricture or before PUL (prostatic urethral lift).
- **Pressure–flow study** (separates BOO from DUA) before surgery if voided volume <150 mL, PVR >300 mL, age <50 or >80, or failed surgery.

### Management: EAU algorithm

```
Bothersome LUTS? No -> watchful waiting, lifestyle advice Absolute surgical indication? Yes -> surgery Otherwise: Nocturnal polyuria -> lifestyle +/- desmopressin Storage predominant -> antimuscarinic or beta3-agonist Prostate >40 mL -> 5ARI (5-alpha-reductase inhibitor) +/- alpha1-blocker Others -> alpha1-blocker or PDE5 inhibitor Residual storage -> add antimuscarinic or beta3-agonist Failure or complication -> surgery
```

- **Watchful waiting:** education, fluid 1.5–2 L/day (none 2 h before bed), less caffeine and alcohol, double voiding, bladder training; yearly review.

| Drug | Example | Mechanism and notes |
|---|---|---|
| α1-blocker | Tamsulosin 0.4 mg, silodosin 8 mg, alfuzosin 10 mg | Relax prostatic smooth muscle; acts in days; dizziness, ejaculatory dysfunction, IFIS (intraoperative floppy iris syndrome) |
| 5ARI | Finasteride 5 mg, dutasteride 0.5 mg | Lower DHT; volume falls about 20% over 6–12 months; **PSA halves**; reduces AUR and surgery |
| Antimuscarinic | Solifenacin, tolterodine | Block detrusor M3 receptors; for storage symptoms if PVR <200 mL |
| β3-agonist | Mirabegron 50 mg, vibegron | Relax detrusor during storage |
| PDE5I (phosphodiesterase-5 inhibitor) | Tadalafil 5 mg daily | Smooth-muscle relaxation; useful with ED (erectile dysfunction) |

- **MTOPS:** progression cut by doxazosin 39%, finasteride 34%, combination 66%; **CombAT** confirmed. Combine when volume >30–40 mL and PSA >1.5 ng/mL.

### Surgery

- **Indications:** refractory retention, recurrent UTI, bladder stones, recurrent haematuria, BPO-related renal insufficiency; failed medical therapy.

| Prostate volume | Options |
|---|---|
| <30 mL | TUIP, TURP (transurethral resection of the prostate) |
| 30–80 mL | **TURP (reference standard)**, bipolar TURP, PVP (photoselective vaporisation of the prostate), enucleation, PUL, Rezum (water vapour), Aquablation |
| >80 mL | **HoLEP (holmium laser enucleation)** or thulium enucleation, open or robotic simple prostatectomy |

- Ejaculation preserved: PUL, Rezum. Anticoagulated: PVP, HoLEP. PAE (prostatic artery embolisation) in selected men.

- **TURP complications:** bleeding, **TUR syndrome** (hyponatraemia with glycine; avoided by bipolar saline), retrograde ejaculation, stricture, incontinence.
- **Also know:** iTind; female LUTS follow OAB and SUI guidelines.

**Diagram to draw:** BPH → BPE → BPO → BOO → bladder changes → LUTS.

> **Key points to remember:**
> - ICS: storage, voiding, post-micturition; IPSS 0–7, 8–19, 20–35.
> - Static (bulk, capsule) plus dynamic (α1A) obstruction; size ≠ symptoms.
> - 5ARI for >40 mL; MTOPS combination cut progression 66%.
> - TURP 30–80 mL; HoLEP or simple prostatectomy for >80 mL.

**Also, from the condensed book (not in the paper answer):**
- Chronic obstruction causes hypertrophy, collagen deposition, diverticula and DO, so intervene promptly.

### ICS urodynamic classification (most precise)

| Domain | Storage | Voiding |
|---|---|---|
| Detrusor | Normal; DO (neurogenic or idiopathic) | Normal; underactive; acontractile |
| Sensation, capacity, compliance | Normal, high/increased, low/reduced, absent | — |
| Urethra | Normal or incompetent closure | Normal; BOO, dysfunctional voiding, detrusor–sphincter dyssynergia (DSD) |

- **Maximum cystometric capacity is when one can no longer delay** voiding.
- DSD is detrusor contraction with involuntary striated sphincter contraction. † **Lapides** and **Bors–Comarr** are older neurological classifications. MCQ

## 6 · Nocturia and neurogenic dysfunction

- **Polyuria** is >**40 mL/kg/24 h**. **Nocturnal polyuria** is a night-to-24-h volume ratio >**20%** in the **young, >33% elderly**; causes include loss of the nocturnal vasopressin rise and sleep apnoea.
- In **IC/BPS**, normally silent **C fibres** become hyperactive through central sensitisation.

| Lesion | Detrusor | Sphincter |
|---|---|---|
| Suprapontine | DO | Synergic |
| Suprasacral spinal | DO | DSD, raised PVR |
| Sacral | Acontractile | Impaired |

> **KEY POINT**
>
> DSD needs an intact sacral arc **disconnected from the pons**, so it is suprasacral, never suprapontine.

# 106 Evaluation and Management of Males With Urinary Incontinence

## 1 · Definition and determinants

- **Urinary incontinence (UI)** (International Continence Society, ICS) is any involuntary loss of urine. In men the **external urethral sphincter (EUS) is most important**.
- Risk factors include **age (major, independent)**, lower urinary tract symptoms (LUTS), infection, cognitive or neurological impairment, obesity, diabetes, prostatectomy and drugs.
- Reversible causes, **DIAPPERS**: Delirium, Infection, Atrophic urethritis, Pharmacological, Psychiatric, Endocrine, Reduced mobility, Stool impaction.
- Male:female is about 1:2. **Urgency UI (UUI) predominates**, then stress UI (SUI), then mixed UI (MUI).

## 2 · Types

| Type (ICS) | Mechanism | Male context |
|---|---|---|
| SUI: on effort, cough | Sphincter deficiency | Almost only **after prostatectomy** |
| UUI: with urgency | Detrusor overactivity (DO) overcomes a normal outlet | ~**90%** have DO on urodynamics; it is "wet" overactive bladder (OAB) |
| MUI | DO and sphincter deficiency | Uncommon; may follow prostatectomy |
| Nocturnal enuresis | Wetting in sleep | New in an adult man: suspect **high-pressure chronic retention** and renal failure |
| Continuous | Fistula, exstrophy/epispadias, ectopic ureter | Rare |
| With chronic retention | Residual ≥**300 mL** | "Overflow" is no longer an ICS term; spinal cord injury (SCI) at or above T6 risks autonomic dysreflexia |

## 3 · Evaluation

- The history covers timing, triggers, prior prostate or pelvic surgery, neurology, bowel and sexual function, drugs and the **most bothersome symptom**.
- Examine the abdomen, genitalia, prostate (digital rectal examination, DRE), anal tone, neurology and cognition, and do a **cough test supine then standing**.
- Record **body mass index (BMI)**: weight loss may help SUI, and high abdominal pressure can overcome an artificial urinary sphincter (AUS) cuff.
- **Dipstick** all men; haematuria needs cytology, cystoscopy and upper-tract imaging.
- In a **bladder diary**, **3 days equal 7 days**; include a weekend.
- The International Prostate Symptom Score (IPSS) does **not measure incontinence**, so use the **ICIQ-UI Short Form** (0–21). A **pad test** is not routine and cannot separate UUI from SUI.

| Severity | Pads/day | Pad weight/day |
|---|---|---|
| Mild | 1 | <200 g |

| Severity | Pads/day | Pad weight/day |
|---|---|---|
| Moderate | 2–4 | 200–400 g |
| Severe | >4 | >400 g |

- **Flow and post-void residual (PVR)**: flow cannot separate bladder outflow obstruction (BOO) from underactivity.
- Check prostate-specific antigen (PSA) only if life expectancy is >10 years and it would change management.
- Do urodynamic studies (**UDS) only if they alter management**, i.e. before invasive treatment (ICI, EAU and AUA guidance) (see Ch 107).
- **Urodynamic stress incontinence (USI)** is leakage on effort without detrusor contraction. **Cough-associated DO incontinence** mimics SUI.

> **KEY POINT**
>
> Every man: history, examination, BMI, dipstick, diary, questionnaire, flow, PVR. Optional: PSA, bloods, imaging, endoscopy, invasive UDS, usually just before surgery.

## 4 · Urgency urinary incontinence

> **ALGORITHM**
>
> 1. Lifestyle, bladder training and pelvic floor muscle training (PFMT) ≥6 weeks (ideally 3 months).
> 2. Antimuscarinic or β3-agonist, or both.
> 3. Urodynamics optional; then onabotulinumtoxinA, sacral neuromodulation (SNM) or posterior tibial nerve stimulation (PTNS).
> 4. Last resort: augmentation or ileal conduit.

- Cut fluid by **25%** if intake is >1 L/day, and avoid caffeine.
- **Antimuscarinics** have similar efficacy, and extended-release forms are preferred. **Anticholinergic burden** for ≥3 months raises dementia risk by about 50%.
- **β3-agonists** are co-first-line with less dry mouth; with mirabegron, **monitor blood pressure**. In **cognitive impairment, use a β3-agonist or early surgery**.
- Mirabegron plus solifenacin beats escalating solifenacin. Tadalafil is the only phosphodiesterase-5 inhibitor for LUTS with erectile dysfunction.

| | OnabotulinumtoxinA | SNM | PTNS |
|---|---|---|---|
| Delivery | Cystoscopic, **100 U** (200 U neurogenic DO) | S3 lead, implanted generator | Needle behind medial malleolus, 30 min weekly × 12 |
| Strength | Repeat every 6–9 months | Most effective, durable >10 years | Least invasive; better for dry OAB |
| Drawback | Retention in ~**10%**, so the man must accept self-catheterisation | Most invasive, costliest | Frequent visits |

- If the SNM test gives ≥**50%** improvement, a permanent generator is implanted (see Ch 115).

## 5 · Stress urinary incontinence

**ALGORITHM**

1. Supervised PFMT ≥3 months, with or without biofeedback and electrical stimulation (better started pre-op); consider duloxetine.
2. Exclude anastomotic or urethral stricture and bladder dysfunction (UDS).
3. Choose surgery by severity.

*See the paper answer **Post-prostatectomy incontinence** (Section 8, Ch 123).*

**Also, from the condensed book (not in the paper answer):**

- After radical prostatectomy wait **6–12 months** before surgery.
- The **AUS is the gold standard**, with about 80% continence (Scott 1973 MCQ), and is activated at about 6 weeks; †deactivate it before catheterisation or endoscopy to prevent erosion.
- Radiotherapy, severe leakage and prior transurethral resection worsen fixed-sling results, so choose an AUS. **ProACT** is **contraindicated after radiotherapy**.

| EAU 2023 recommendation (post-prostatectomy incontinence, PPI) | Grade |
|---|---|
| Offer AUS for moderate–severe PPI | Strong |
| Warn of AUS/ProACT complications, mechanical failure, explantation | Strong |
| Inform of duloxetine side effects, off-label use | Strong |
| Fixed transobturator sling for mild–moderate PPI | Weak |
| Do not offer bulking agents; no ProACT after radiotherapy | Weak |

## 6 · Mixed and other types

- In **MUI**, treat the most bothersome component first; DO may mask SUI, so treat DO and repeat UDS.
- In high-pressure chronic retention with enuresis, **catheterise first**, then relieve BOO.
- Enuresis with a low residual is usually OAB and is treated with an antimuscarinic or β3-agonist, with or without desmopressin.
- Postmicturition dribble is treated with **urethral milking**.
- **Red flags** needing a specialist are haematuria, pain, recurrent infection, pelvic radiotherapy, abnormal DRE and high PVR.

## 7 · Landmark trials

| Trial | Design | Result | Take-home |
|---|---|---|---|
| MASTER | Non-inferiority randomised controlled trial (RCT), male sling vs AUS for urodynamic SUI after prostatectomy | Equivalent at 12 months; secondary outcomes favoured AUS; few completely dry | Sling is an acceptable alternative; counsel realistically |

# 107 Urodynamic and Videourodynamic Evaluation of the Lower Urinary Tract

## 1 · Definition and role

- **Urodynamics (UDS)** is the dynamic study of urine transport, storage and evacuation. Use it when simple tests fail, to **diagnose the cause of symptoms** and to assess **diseases that silently damage the upper tracts** such as spinal cord injury (SCI), multiple sclerosis (MS) and radiation cystitis.

### Wein functional classification — the frame for every study

| Site | Overactive | Underactive |
|---|---|---|
| Bladder | Failure to store | Failure to empty |
| Outlet | Failure to empty | Failure to store |

- Set the questions first and aim to **reproduce the symptom**; an unrecorded abnormality is not excluded.

> **KEY POINT**
>
> **Box 107.1 — Urodynamic risk factors** (treat even if asymptomatic): impaired compliance; detrusor–external sphincter dyssynergia (DESD); detrusor–internal sphincter dyssynergia (DISD); high-pressure detrusor overactivity (DO) throughout filling; detrusor leak point pressure (DLPP) **> 40 cm$H_2O$**; poor emptying with high storage pressure.

## 2 · Preparation

- **Decide medication status in advance**: test on regular dosing to assess drug effect, or stop the drug first to uncover aetiology.
- **ICS pre-invasive work-up**: symptom and bother score, **3-day frequency–volume chart**, free flow and postvoid residual (PVR).
- **Urinalysis before every invasive study**; postpone for symptomatic urinary tract infection (UTI).

### Antibiotic prophylaxis (AUA / SUFU Best Practice Policy)

The "index" patient needs **no routine prophylaxis** (level 1). Non-index patients get a **single dose within 1 h**: oral trimethoprim–sulfamethoxazole first; alternatives include a cephalosporin, amoxicillin–clavulanate, an aminoglycoside or a fluoroquinolone.

| Prophylaxis YES | Prophylaxis NO |
|---|---|
| Neurogenic LUT dysfunction (NLUTD); bladder outlet obstruction (BOO) or raised PVR; age > 70; **asymptomatic bacteriuria**; immunosuppression (especially renal transplant); chronic catheters including clean intermittent catheterisation (CIC); high-risk orthopaedic implant | Diabetes; **recurrent UTI history**; recent hospitalisation; obesity; menopause; gender; prior urological surgery; cardiac valve disease (not routinely) |

Trap: asymptomatic bacteriuria is treated here, unlike in most of urology.

## 3 · Non-invasive components

### SAQ Uroflowmetry

Paper answer · P2 · P3 · merged from 2

*Asked as:* P2 SAQ "Uroflowmetry." · P3 SAQ "Uroflowmetry"

#### Definition

- A non-invasive urodynamic test that records the **volume of urine voided per unit time** (mL/s). It reflects detrusor contraction and outlet resistance together.

#### Method

- Patient comes with a **comfortably full bladder** (normal desire) and voids in private, in the usual (preferred) position.
- **Flowmeters:** weight transducer (load cell), rotating disc, capacitance dipstick.
- **Voided volume ≥150 mL** (Qmax is unreliable below 125–150 mL). Repeat 2–3 times; ask whether the void was typical and compare with the frequency–volume chart.
- Measure **PVR** (post-void residual) by ultrasound afterwards.

#### Parameters

| Parameter | Meaning |
|---|---|
| **Qmax** (maximum flow rate) | Most useful (key) value; normal men >15 mL/s, women >20 mL/s; read from the smoothed curve, ignoring spikes |
| Qave (average flow rate) | Volume ÷ flow time; less useful and less accurate for obstruction |
| Voided volume (VV) | Must be ≥150 mL |
| Flow time vs voiding time | Voiding time includes interruptions |
| Time to Qmax | Normally early, in the first third of the void |
| Voiding efficiency | VV ÷ (VV + PVR) × 100 |

#### Curve patterns (ICCS, International Children's Continence Society)

| Pattern | Suggests |
|---|---|
| **Bell-shaped** | Normal |
| Tower (high, short) | Overactive bladder |
| **Plateau (flattened, box, prolonged)** | Fixed obstruction: **urethral stricture**, PUV (posterior urethral valves) |
| Prolonged, low, skewed | BPO (benign prostatic obstruction) or detrusor underactivity |
| Staccato (never reaches zero) | Sphincter overactivity, dysfunctional voiding |
| **Intermittent / interrupted (reaches zero), straining** | Detrusor underactivity with abdominal straining, obstruction, or DSD (detrusor–sphincter dyssynergia) |

#### Interpretation

- **Qmax <10 mL/s:** obstruction likely (high specificity). **Qmax <15 mL/s:** more sensitive, less specific.

- **Nomograms / volume correction: Siroky** (men), **Liverpool** (men and women, centiles). Corrected Qmax = Qmax ÷ √VV.
- **Artefacts:** "wag" (moving stream), knocking the flowmeter, straining spikes. Read Qmax from the smoothed curve, not single spikes.
- Children: valid if voided volume ≥50% of expected capacity; accept Qmax only if the peak lasts ≥2 s.

### Limitations

- **Cannot separate BOO (bladder outlet obstruction) from DUA (detrusor underactivity)**; only a pressure–flow study can.
- Varies with volume, position, anxiety and between days.

### Uses

- BPH (benign prostatic hyperplasia): evaluation before active treatment and follow-up after medical or surgical treatment.
- Follow-up after **urethroplasty** or optical urethrotomy (stricture recurrence).
- Children: dysfunctional voiding, BBD (bladder and bowel dysfunction), after posterior urethral valve ablation; with patch EMG (electromyography).
- Women: voiding dysfunction, including before stress incontinence surgery.
- Pressure–flow study before BPH surgery if Qmax >10 mL/s (P2: EAU [European Association of Urology]; P3: AUA [American Urological Association], "may be done").

### Diagram to draw

- Flow curves: normal bell, tower, plateau (stricture), prolonged low (BPO), interrupted/intermittent.

> **Key points to remember:**
> - Voided volume ≥150 mL; Qmax is the key value.
> - Plateau = fixed obstruction (stricture); staccato = sphincter; interrupted = weak detrusor or straining.
> - Qmax <10 mL/s suggests BOO but cannot exclude underactivity; Siroky and Liverpool nomograms.
> - Cannot distinguish BOO from DUA.

**Also, from the condensed book (not in the paper answer):**

- Minimum voided volume is **VV 150 mL**, because maximum flow (Qmax) is linear above it and hyperbolic below.
- Normal Qmax is > 15 mL/s in men (< 10 suggests obstruction) and > 20 mL/s in women.

## 4 · Equipment, transducers and quality control

- The minimum is vesical (Pves), abdominal (Pabd) and subtracted detrusor pressure (Pdet) with flow. Infuse saline or contrast, **not $CO_2$**.

| Feature | External water-filled | Air-charged | Microtip |
|---|---|---|---|
| Reference level | **Upper border of symphysis pubis**; re-level on position change | None (no hydrostatic effect) | — |

| Feature | External water-filled | Air-charged | Microtip |
|---|---|---|---|
| Problems | Air bubbles; motion and hydrostatic artefact; underdamped | Overdamped (signals > 3 Hz attenuated), so coughs are blunted | Expensive, reusable |

- Systems are **not interchangeable** (Pdet may differ by up to 10 $cmH_2O$); water-filled systems underpin most nomograms.

> **KEY POINT**
>
> **Quality control** (ICS Good Urodynamic Practice): zero to atmosphere, flush air, and keep resting Pdet near 0. Do a **cough test** before filling, at intervals and before voiding; a biphasic Pdet spike means re-flush or reposition. Rectal contractions cause an unexplained fall in Pdet.

## 5 · Filling cystometry (CMG)

*CMG shows whether storage is at safe, low pressure.*

- Filling to "permission to void" gives the **maximum cystometric capacity (MCC)**. **Pdet = Pves − Pabd**; rectal Pabd is preferred.
- Standard adult filling is **30–100 mL/min**; the ICS physiological rate is < body weight (kg)/4 mL/min. Neurogenic bladders need a physiological rate and body-warm saline (EAU). DO is commoner upright.
- A **normal ("stable")** bladder keeps Pdet near zero with no involuntary contractions. **Sensation** (first sensation, first desire, strong desire, urgency) is the only subjective parameter.

### Detrusor overactivity (DO)

- DO means involuntary detrusor contractions (IDC) during filling, either **idiopathic** or **neurogenic**. The label comes from the history, as tracings look identical.
- Absence does not exclude it: **up to 50%** of women with urgency urinary incontinence (UUI) show no DO, and some asymptomatic volunteers do show it.
- In BOO with overactive bladder (OAB), a **single terminal IDC** predicts storage-symptom resolution after transurethral resection of the prostate (TURP). In neurogenic DO, higher Pdetmax and longer IDCs predict upper-tract deterioration.
- **Cough-associated DO** mimics stress urinary incontinence (SUI) but is detrusor in origin.

### Compliance

*See the paper answer **Urinary bladder compliance** (Section 7, Ch 103).*

**Also, from the condensed book (not in the paper answer):**

- Storage pressure **> 40 $cmH_2O$** itself leads to upper-tract damage.
- Outlet pop-off hides poor compliance, which may appear only after a sling or artificial urinary sphincter (AUS), so occlude the outlet.

## 6 · Leak point pressures

*See the paper answer **Leak point pressures and ALPP** (Section 5, Ch 40).*

| Abdominal leak point pressure, ALPP ($cmH_2O$) | Inference (McGuire) |
|---|---|
| **< 60** | ISD |

| Abdominal leak point pressure, ALPP ($cmH_2O$) | Inference (McGuire) |
|---|---|
| 60–90 | Equivocal |
| ≥ 90 | Little or no ISD (hypermobility) |

- Intrinsic sphincter deficiency (ISD) and hypermobility coexist; ALPP poorly predicts surgical success.
- **Occult SUI** is leakage only on prolapse reduction in a continent woman with pelvic organ prolapse (POP).
- **DLPP > 40 $cmH_2O$** is not a validated safe cut-off (20 is more sensitive in children), so aim **as low as achievable** by lowering bladder pressure or the outlet (e.g. sphincterotomy for DESD), then repeat the UDS.

## 7 · Urethral pressure profilometry (UPP)

- **Maximum urethral closure pressure (MUCP)** is maximum urethral minus intravesical pressure (normal 40–60 $cmH_2O$ in women).
- **MUCP ≤ 20 $cmH_2O$** means ISD ("type III" SUI) and predicts sling or suspension failure.
- The EAU advises against using UPP or leak point pressure to grade severity.

## 8 · Pressure–flow studies (PFS)

*Only PFS tells obstruction from a weak bladder.*

| Pdet | Flow | Interpretation |
|---|---|---|
| Low/normal | High/normal | Unobstructed |
| High | Low | **BOO** |
| Low | Low | Detrusor underactivity (obstruction may be hidden) |

- ICS **detrusor underactivity** is reduced contraction strength or duration, so emptying is prolonged or incomplete. **Acontractile detrusor** replaces "areflexia".

### Men — nomograms and indices

| Index | Formula | Cut-offs |
|---|---|---|
| **Bladder outlet obstruction index (BOOI)** | PdetQmax − 2Qmax | **> 40 obstructed · 20–40 equivocal · < 20 unobstructed** |
| **Bladder contractility index (BCI)** | PdetQmax + 5Qmax | **> 150 strong · 100–150 normal · < 100 weak** |

- BOOI ≥ 40 needs Pdet ≥ 40 even at zero flow, so a decompensated detrusor masks obstruction.
- **Stop tests** estimate isovolumetric pressure (Piso); in men Piso < 50 $cmH_2O$ suggests underactivity.
- About 2/3 of men with benign prostatic obstruction (BPO) have DO; half to two-thirds resolve after relief.
- MCQ In the bethanechol supersensitivity test (Lapides), 2.5 mg is given subcutaneously; a rise **≥ 15 $cmH_2O$** at 100 mL is positive (neurogenic), but the test is unreliable.

### Women

- Women void at low pressure, so the **BOOI and ICS nomogram cannot be used**. Suspect BOO after anti-incontinence surgery or with prolapse.

| Method | Criterion | Comment |
|---|---|---|
| Cut-points | e.g. Qmax ≤ 12 mL/s and PdetQmax ≥ 25 $cmH_2O$ (Defreitas); Chassagne ≤ 15 / ≥ 20 | Anatomic obstruction only; overlap |
| **Videourodynamics, VUDS (Nitti)** | Radiographic obstruction between bladder neck and meatus with a sustained contraction of any magnitude | Localises; works despite weak detrusor |
| **Blaivas–Groutz** | Free Qmax vs Pdetmax; none/mild/moderate/severe | Two separate voids; **overdiagnoses** |
| Solomon–Greenwell | PdetQmax − 2.2Qmax: < 0 unlikely; > 18 almost certain | Includes functional obstruction |

## 9 · EMG and sphincter coordination

- Surface perineal electrodes are standard. Normally EMG quietens and urethral pressure falls **before** Pdet rises.
- The **guarding reflex** (EMG rise during an IDC) is normal, not dyssynergia.

| | DESD | Dysfunctional voiding |
|---|---|---|
| Cause | **Suprasacral cord lesion** (the higher, the likelier) | Learned; no neurological lesion |
| Behaviour | Involuntary sphincter contraction during DO or voiding | Intermittent sphincter contraction during voiding |
| Significance | High pressure causes poor compliance and upper-tract risk | LUT symptoms; fluoroscopy improves accuracy |

- **DISD** needs a suprasacral lesion above the T10–L1 sympathetic outflow and is seen on **VUDS only**. **Primary bladder neck obstruction (PBNO)** is also invisible to EMG and needs fluoroscopy.

## 10 · Videourodynamics

### SAQ Urodynamic studies, including video urodynamics

Paper answer · P2 · P4 · merged from 3

*Asked as:* P2 SAQ "Uro Dynamic Studies." · P2 SAQ "Vesicourodynamics" · P4 SAQ "Video Urodynamics"

*"Vesicourodynamics" is taken as videourodynamics (VUDS): urodynamics with simultaneous fluoroscopic imaging of the bladder and urethra.*

#### Definition and principles

- UDS (urodynamic studies) test bladder and urethral function in storage and voiding.
- ICS (International Continence Society): **decide the question first** and **reproduce the symptom**; a normal study does not exclude disease.
- Before invasive UDS: symptom score, 3-day **frequency–volume chart**, free uroflow and PVR (post-void residual); treat UTI (urinary tract infection) first.

### Components of conventional UDS

| Test | Method | Main values |
|---|---|---|
| **Uroflowmetry** | Voided volume ≥150 mL | Bell curve normal; Qmax <10 mL/s (men) suggests obstruction or weak detrusor; plateau = stricture |
| **PVR** | Bladder scan or catheter | Emptying |
| **Filling cystometry** | Bladder and rectal lines, fill with saline 30–100 mL/min (adults); **Pdet = Pves − Pabd**; zero at atmosphere, transducer at symphysis | Sensation, capacity, **DO (detrusor overactivity)**, **compliance** = ΔV/ΔPdet (<20 mL/cm$H_2O$ impaired) |
| **Leak point pressures** | ALPP (abdominal LPP): cough or Valsalva at 150 mL | Low ALPP = weak sphincter (ISD, intrinsic sphincter deficiency) |
| | DLPP (detrusor LPP) | **>40 cm$H_2O$** risks upper tracts (McGuire) |
| **Pressure–flow study** | Void with lines in | High pressure–low flow = obstruction; low–low = underactivity |
| **EMG (electromyography)** | Perineal surface electrodes | DESD (detrusor–external sphincter dyssynergia) |
| **UPP (urethral pressure profile)** | Catheter withdrawn | Limited value |
| **Videourodynamics** | Above with contrast and fluoroscopy | VUR (vesicoureteric reflux), bladder neck, site of obstruction |

### Indices (men)

- **BOOI (bladder outlet obstruction index)** = PdetQmax − 2Qmax: **>40 obstructed**, 20–40 equivocal, <20 unobstructed.
- **BCI (bladder contractility index)** = PdetQmax + 5Qmax: **>150 strong**, 100–150 normal, **<100 weak**.
- Women: no validated nomogram (Nitti video criteria).

### Indications for UDS

- Male LUTS (lower urinary tract symptoms) before surgery when unclear (young men, equivocal flow, failed TURP [transurethral resection of the prostate]); complex, mixed or recurrent incontinence.
- **Neurogenic bladder** (spinal injury, MS [multiple sclerosis], spina bifida), radiation cystitis, before augmentation.
- Children: neurogenic bladder, PUV (posterior urethral valves); fill at **5–10% of expected capacity per minute**; expected capacity = (age + 2) × 30 mL.

### Video urodynamics (VUDS): definition and set-up

- **VUDS:** multichannel urodynamics (pressure–flow, with or without EMG) recorded at the same time as fluoroscopy, using **contrast as the filling fluid**.
- Vesical and rectal catheters (Pdet [detrusor pressure] = Pves − Pabd), flowmeter, EMG; C-arm (most flexible) or a tilting table.

- Position for the question: **SUI (stress urinary incontinence) views standing**; voiding in the usual position (women sitting, men standing).
- **Screen only at key moments** (provocation, pressure rise, voiding) to limit radiation.

### What fluoroscopy adds

| Situation | VUDS finding |
|---|---|
| NLUTD (neurogenic lower urinary tract dysfunction) | **Gold standard** (EAU); DESD and **DISD** (detrusor–internal sphincter dyssynergia, seen only on VUDS); trabeculation, diverticula |
| **VUR** | Pressure, volume and grade at onset; reflux "pops off" pressure, so compliance looks falsely safe |
| **Bladder neck** | Only way to see and assess it; **PBNO** (primary bladder neck obstruction) in young men and women |
| Female obstruction | **Nitti criteria:** radiographic narrowing between bladder neck and meatus during voiding, with a sustained detrusor contraction |
| Dysfunctional voiding | "Spinning-top" urethra at the external sphincter |
| SUI | Leakage with ALPP; open bladder neck at rest suggests ISD |
| Male obstruction | Localises prostatic obstruction, bladder neck contracture or stricture |

- **PBNO types (young men):** I high pressure, low flow; II normal pressure, low flow with bladder-neck narrowing; III delayed bladder-neck opening.

### Indications for VUDS

- Known or suspected NLUTD; unexplained retention in women; young men with LUTS; complex incontinence; prior radical pelvic surgery or radiation; diversion; renal transplant.
- **Children:** PUV, spinal dysraphism; contrast-enhanced ultrasound VUDS avoids radiation.

### Urodynamic risk factors (upper-tract danger)

- Impaired compliance; DESD; DISD; high-pressure DO throughout filling; DLPP >40 $cmH_2O$; poor emptying with high storage pressure.

### Limitations of VUDS

- Radiation, cost, space, availability, contrast allergy.
- EAU (European Association of Urology): optimum test in NLUTD. AUA (American Urological Association) guideline: lowest radiation dose that answers the question.

### Also know

- Ambulatory UDS; bethanechol test; antibiotic prophylaxis for risk groups.

### Diagrams to draw

- Normal trace: Pves, Pabd, Pdet, flow, EMG.
- VUDS trace beside a voiding cystourethrogram image.

**Key points to remember:**

- Pdet = Pves − Pabd.
- Ask a question first; reproduce the symptom.
- BOOI >40 obstructed; BCI <100 weak.
- DLPP >40 $cmH_2O$: upper-tract risk.
- VUDS = urodynamics + fluoroscopy with contrast.
- VUDS is the only test for bladder neck dysfunction, PBNO and DISD.
- Reflux pop-off makes compliance look falsely good.

**Also, from the condensed book (not in the paper answer):**

- If VUDS is unavailable in NLUTD, do CMG with PFS (EAU).
- Fluoroscopy is critical for PBNO types II and III.

## 11 · Ambulatory urodynamics

- Natural filling shows more DO and better compliance than pump filling; use it for symptoms unexplained by conventional UDS.

## 12 · Indications by condition (AUA/SUFU, EAU)

*Order UDS when the result will change treatment.*

### Female SUI

- In uncomplicated demonstrated SUI (the "index" patient) UDS may be omitted (AUA/SUFU 2023; EAU: not routinely).
- Do UDS in non-index women: prior anti-incontinence or POP surgery, symptom–examination mismatch, voiding dysfunction, significant UUI, raised PVR, unconfirmed SUI or NLUTD.

### Male LUTS

- No UDS is needed to start an α-blocker for lower urinary tract symptoms (LUTS).
- AUA/SUFU: do **PFS when obstruction matters, before invasive, morbid or irreversible treatment**, because without obstruction surgery does worse.
- EAU PFS triggers before invasive treatment are prior failed treatment, inability to void > 150 mL, **Qmax > 10 mL/s**, PVR > 300 mL, and age > 80 or < 50.
- **Impaired compliance is the only absolute urodynamic indication to treat BOO.**

### Women with voiding symptoms; NLUTD

- PFS separates obstruction from impaired contractility; functional obstruction (PBNO, dysfunctional voiding) needs VUDS.
- The NLUTD goal is **low-pressure filling**, even at the cost of catheterisation; only UDS separates DO from poor compliance.
- UDS is essential before treatment and in follow-up in SCI and myelomeningocele.

## 13 · Landmark trials

| Trial | Design | Result | Take-home |
|---|---|---|---|
| ValUE | RCT, 630 women, uncomplicated SUI: office evaluation with or without UDS | Success ~77% both arms | Office evaluation alone non-inferior before SUI surgery |
| Dutch embedded RCT | Discordant UDS randomised: immediate mid-urethral sling vs UDS-tailored | Equal cure | UDS discordance need not change surgery |
| SISTEr | Burch vs pubovaginal sling | ALPP and DO did not predict outcome | UDS adds little in stress-predominant SUI |

# 109 Neuromuscular Dysfunction of the Lower Urinary Tract

## 1 · Principles and classification

*The lesion level predicts the bladder and sphincter pattern.*

- The International Continence Society (ICS) uses **five parameters**: sensation; detrusor (normal, overactive, areflexic, impaired); compliance; smooth sphincter (synergic or dyssynergic); striated sphincter (synergic, dyssynergic, bradykinetic, impaired voluntary control, fixed tone).

| Level | Typical disorders | Detrusor | Smooth sphincter | Striated sphincter |
|---|---|---|---|---|
| Suprapontine | Stroke, tumour, Parkinson disease (PD), dementia, cerebral palsy | Overactive | Synergic | Synergic (with or without impaired voluntary control; PD bradykinesia) |
| Multiple system atrophy (MSA) | — | Overactive, later impaired | **Open** | Denervated |
| Suprasacral cord | Spinal cord injury (SCI), multiple sclerosis (MS), transverse myelitis | Overactive | Synergic (dyssynergic if above T7) | **Dyssynergic** |
| Sacral cord / cauda equina | Sacral SCI, disc, myelodysplasia | Areflexic | Competent non-relaxing; may open | Fixed tone |
| Peripheral | Radical pelvic surgery, diabetes, tabes | Impaired / areflexic | Open (pelvic surgery) | Fixed tone / synergic |

> **KEY POINT**
>
> Dyssynergia appears only when the lesion lies **between the pons and the sacral cord**. An open bladder neck marks autonomic (MSA, radical pelvic surgery, myelodysplasia) or sacral damage. PD and MSA share the detrusor pattern but **not the outlet**.

## 2 · Lesions at or above the brainstem

### Stroke

- **Acutely** there is retention ("cerebral shock"). **Chronically** there is detrusor overactivity (DO) with **synergic sphincters**.
- **True detrusor–sphincter dyssynergia (DSD) does not occur. Pseudodyssynergia** is an electromyographic (EMG) flare from voluntary attempts to inhibit a contraction.
- In men with lower urinary tract symptoms (LUTS), do a full urodynamic study (UDS) before outlet surgery.

### Dementia, brain injury, tumour, others

- **Dementia**: anticholinergics worsen cognition, so a **β3 agonist is strongly preferred** (SUFU).

- **Brain injury** above the pontine micturition centre (PMC) gives DO with synergic sphincters; brainstem injury below the PMC can give DSD. Frontal **tumours** can cause retention.
- **Normal-pressure hydrocephalus** gives dementia, gait ataxia and incontinence, improved by a shunt.
- **Cerebral palsy** gives DO with synergic sphincters, but about half of adults with LUTS have poor compliance or a detrusor leak point pressure (DLPP) ≥40 cm $H_2O$, so evaluate fully.

### Parkinson disease vs multiple system atrophy

- **PD** LUTS occur in 35–70% and are mainly storage symptoms. Sphincter **bradykinesia** mimics DSD.
- Treat with overactive bladder (OAB) drugs with low central effect, such as mirabegron or trospium; onabotulinumtoxinA, tibial nerve stimulation (PTNS) and sacral neuromodulation (SNM) also work.
- **Transurethral resection of the prostate (TURP) in PD is not contraindicated.** The old "post-TURP incontinence in PD" cases were MSA, so confirm PD, exclude MSA and do videourodynamics first.
- **MSA** affects the intermediolateral columns and **Onuf nucleus**, causing early severe LUTS and erectile dysfunction (ED).
- In MSA, DO comes early and poor emptying late. Treating DO worsens emptying, so add clean intermittent catheterisation (CIC); give desmopressin for nocturia. **Prostatectomy is hazardous.**

| Feature | MSA | PD |
|---|---|---|
| Urinary symptoms / ED | Precede or accompany parkinsonism; ED near universal | Years after diagnosis; ED ~⅓ |
| Dominant problem; PVR | UI; high postvoid residual (PVR) | Frequency/urgency; PVR normal |
| Bladder neck | **Open** | Synergic |
| Striated sphincter EMG | **Denervation** | Normal (bradykinesia) |
| Detrusor | DO, then areflexia | DO; areflexia uncommon |
| Outlet surgery | Worsens | Acceptable after work-up |

## 3 · Spinal cord disease

*Cord lesions add dyssynergia and silent upper-tract damage.*

### Multiple sclerosis

- LUTS occur in 50–90% and are the presenting complaint in up to 15%. On UDS, DO is found in ~⅔, DSD in 30–65% of those with DO, and areflexia in ~¼.
- Symptoms do not track the disease, so **baseline UDS is essential**.
- Keep management **reversible**, because the pattern changes: antimuscarinic with CIC, onabotulinumtoxinA, or SNM in selected patients. Credé and Valsalva voiding are contraindicated, and an indwelling catheter is the last resort.

## LAQ Describe neuromuscular dysfunction and management of bladder dysfunction following cervical spine surgery in young adult male

Paper answer · P2

**Type:** LAQ (25 marks)
**Source:** Practice Set 1 Paper 2, Q2

*Assumed meaning: cervical spinal cord injury (traumatic, operated, or from the surgery itself), giving a suprasacral lesion above T6.*

### Introduction

- A cervical cord lesion lies **between the PMC (pontine micturition centre) and the sacral cord**. It causes tetraplegia and the classic suprasacral pattern.
- Complete transection is rare; incomplete lesions are commonest. Manage by **urodynamics, not neurology**.

### Pathophysiology

- The PMC normally coordinates detrusor contraction with sphincter relaxation. Cutting the descending pathway leaves a **spinal reflex**.
- **Why NDO (neurogenic detrusor overactivity) appears:** unmyelinated C-fibre afferents become sensitised (NGF (nerve growth factor)), sprout and drive a new sacral reflex. Bulbospinal inhibition is lost.

| Phase | Detrusor | Outlet | Clinical |
|---|---|---|---|
| **Spinal shock** (usually 6–12 weeks; up to 1–2 years) | Areflexic, smooth bladder | Bladder neck closed; reduced closure pressure | **Retention**; overflow only |
| **Recovery (suprasacral pattern)** | **NDO**, no sensation below lesion | **Striated DSD (detrusor–sphincter dyssynergia)**; lesion above T7–T8 adds **smooth sphincter dyssynergia** | Reflex leakage, incomplete emptying, high pressure |

- **Blaivas DSD types:** type 1, sphincter relaxes at peak pressure (incomplete lesions); type 2, sporadic contractions; type 3, crescendo–decrescendo obstruction (types 2–3 with complete lesions).
- **Consequences:** high storage and voiding pressure, VUR (vesicoureteric reflux) in 17–25%, hydronephrosis, stones (about 7% in 10 years), UTI (urinary tract infection). Untreated, about half of men with DSD deteriorate (worse if complete, continuous DSD).
- Upper tract risk also rises with low compliance (<12.5 mL/cm $H_2O$) and NDO for over a third of filling.
- **VUR:** treat the bladder first (lower the pressure); reflux surgery in a thick bladder is difficult.
- **Also know:** cervical myelopathy may instead give an underactive detrusor.

### Autonomic dysreflexia

- Occurs with lesions **above T6** (cervical 60%), after spinal shock. Bladder or bowel distension triggers unopposed spinal sympathetic outflow, as medullary inhibition is lost.
- **Features:** pounding headache, severe hypertension (systolic rise ≥20 mmHg defines it at urodynamics), bradycardia, sweating and flushing above the lesion. It can cause stroke or seizures.

- **Treatment:** sit up, **drain the bladder** (check catheter), then **nitroglycerin 2% paste** above the lesion; intravenous nitroprusside or labetalol if persistent. Prophylaxis with terazosin or prazosin. Monitor blood pressure during cystoscopy and urodynamics; use spinal or general anaesthesia.

### Evaluation

- History, examination, **hand function** (decides self-catheterisation), transfers, carers.
- Urinalysis, creatinine, **renal ultrasound**. Men with good hand function can self-catheterise; others need carers or a suprapubic catheter.
- **VUDS (videourodynamics) after spinal shock**, with blood pressure monitoring: compliance, NDO, DSD, VUR, **DLPP (detrusor leak point pressure) ≥40 cm $H_2O$** = upper tract risk.
- AUA/SUFU (American Urological Association/Society of Urodynamics, Female Pelvic Medicine and Urogenital Reconstruction) 2021: suprasacral lesions are "unknown risk" until tested; DSD or poor compliance = high risk.

### Management

**Goals:** protect kidneys, low-pressure storage, low-pressure emptying, no infection, continence, social acceptability. Start with the simplest reversible option.

- **Spinal shock: CIC (clean intermittent catheterisation)** preferred; indwelling catheter early is acceptable. Defer irreversible procedures.

| Aim | Options |
|---|---|
| **Emptying** | **CIC: gold standard**, volumes <500 mL; carer CIC in tetraplegics. **Suprapubic catheter** better than long-term urethral catheter. Credé/Valsalva not recommended |
| **Storage (lower pressure)** | **Antimuscarinics** ± mirabegron (β3 agonist); **intradetrusor onabotulinumtoxinA 200 U** if refractory (repeat when the effect wears off) |
| **Refractory small bladder** | **Augmentation ileocystoplasty** ± continent catheterisable channel (Mitrofanoff) |
| **Men unable to do CIC** | **External sphincterotomy** or intrasphincteric botulinum toxin → reflex voiding into condom catheter. High failure; DLPP ≥40 cm $H_2O$ afterwards = failure. Urethral stents abandoned |
| **Neuromodulation** | **SARS (sacral anterior root stimulator, Brindley) + posterior sacral rhizotomy**: abolishes NDO and dysreflexia, allows voiding on demand |
| **Last resort** | Ileovesicostomy; bladder neck closure + suprapubic catheter; ileal conduit |

- **Also know:** sacral neuromodulation is used for neurogenic bladder without SCI (spinal cord injury); hydrophilic catheters delay first UTI.

### Complications and follow-up

- **UTI:** treat only symptomatic bacteriuria (5–14 days); no routine prophylaxis.
- Stones, VUR (treat the bladder first), bladder cancer (no screening cystoscopy, AUA/SUFU 2021).
- Lifelong: yearly review and renal function, upper tract imaging every 1–2 years, repeat urodynamics if anything changes.

`Cervical SCI → spinal shock (retention): CIC → VUDS at 3 months: NDO + DSD → antimuscarinic + CIC → Botox → augmentation → cannot CIC: sphincterotomy + condom / SARS / ileovesicostomy`

**Diagram to draw:** neural axis: PMC, lesion at cervical cord, sympathetic T10–L2, sacral S2–S4, with the resulting NDO + DSD.

> **Key points to remember:**
> - Suprasacral lesion = NDO + striated DSD; above T7–T8 also smooth sphincter dyssynergia.
> - Autonomic dysreflexia above T6: drain bladder first, nitrates next.
> - CIC + bladder relaxation (antimuscarinic, Botox 200 U) is the mainstay.
> - DLPP ≥40 cm $H_2O$ threatens the kidneys.

**Also, from the condensed book (not in the paper answer):**

- Spinal shock usually lasts **6–12 weeks**; bladder reflexes return with the lower-limb deep tendon reflexes.
- Smooth sphincter dyssynergia occurs with cord lesions at **T7–T8** (column T6) or above.
- The sacral cord begins at vertebral column T12–L1, so state column and cord level separately.

| | **Suprasacral (complete)** | **Sacral (complete)** |
|---|---|---|
| Detrusor | Overactive | Areflexic |
| Compliance | Normal initially | Normal or high; may fall |
| Smooth sphincter | Synergic (dyssynergic above T7–T8) | Competent non-relaxing; may later open |
| Striated sphincter | Dyssynergic | Fixed tone, no voluntary control |
| Somatic signs | Spasticity, hyperreflexia | Flaccidity, reduced reflexes |
| Management | Lower pressure (drugs or toxin) and CIC | CIC and keep compliance safe |

- **Neurology does not equal urodynamics**, and thoracolumbar (T10–L2 column) lesions are least predictable, so **manage by UDS, not by level**.
- Upper-tract risks are detrusor fibrosis, leak point pressure ≥75 cm $H_2O$ and DO for >⅓ of filling; DLPP >40 cm $H_2O$ is the classic threshold. Damage can occur **without vesicoureteral reflux (VUR)**.

## Autonomic dysreflexia (AD)

- AD occurs in SCI **above T6** and may appear years later. Triggers include bladder or rectal distension, a blocked catheter, instrumentation, stones, sores and fracture.
- Risk during UDS is highest with complete cervical lesions, DSD and poor compliance.

**ALGORITHM**

1. Sit the patient upright and loosen clothing.
2. **Remove the stimulus**: drain the bladder, unblock the catheter, empty the rectum.
3. If blood pressure stays high, apply nitroglycerin 2% paste above the lesion (first line).
4. If it still persists, give labetalol or intravenous nitroprusside.
5. For prophylaxis, give terazosin or prazosin (α1-blockers).
6. If intractable, consider sacral rhizotomy, sympathectomy or cordectomy.

- Nifedipine (10–20 mg sublingual) is now banned in many centres because of uncontrolled hypotension.

### VUR and women with SCI

- A **chronic urethral catheter is worst** for the upper tracts. VUR artefactually enlarges capacity and lowers pressures on UDS.
- Women paraplegics do CIC, while tetraplegics often need indwelling catheters.

### Other cord disorders

- **Transverse myelitis** causes retention, then DO with DSD, and is managed as SCI. **Cervical myelopathy** gives DO with or without DSD, or underactivity, so UDS guides treatment.
- In **myelomeningocele**, level correlates poorly with UDS. The typical adult has an **areflexic bladder and open bladder neck** with a fixed sphincter, so leaks at a fixed pressure; DSD occurs in 10–15%.
- Poor compliance, high pressure and DO predict renal decline. For stress UI, bulking is safer than a sling or artificial urinary sphincter (AUS) because it does not reduce compliance.
- **Tethered cord** presents after growth spurts, usually with DO. A sensory neurogenic bladder (tabes, B12) has a large capacity.

## 4 · Lesions distal to the cord

*Root and nerve lesions give a weak bladder behind a fixed outlet.*

- **Disc disease** (L4–L5, L5–S1) most often gives a **normally compliant areflexic bladder**. Do **pre-laminectomy UDS** for clinical and medicolegal reasons.
- **Cauda equina syndrome** causes saddle anaesthesia and an acontractile detrusor. **Decompression within 48 h** gives better urinary and rectal recovery.
- **Radical pelvic surgery** (abdominoperineal resection, radical hysterectomy) gives impaired contractility, fixed striated tone, an **open non-functional bladder neck** and lower compliance, so the patient leaks or retains.
- After pelvic surgery, **use CIC and wait 6–12 months**; do **no prostatectomy unless BOO is proven**.
- Herpes zoster and herpes simplex (**Elsberg syndrome**) cause transient retention.

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## SAQ Bladder dysfunction non genitourinary systemic disease

Paper answer · P2

**Type:** SAQ (10 marks)
**Source:** Practice Set 2 Paper 2, Q3a

*Taken as lower urinary tract dysfunction caused by systemic (non-genitourinary) diseases.*

### Principle

- The **site of the lesion**, not the disease name, predicts the bladder:
- Brain above the pons: **DO (detrusor overactivity), no DSD (detrusor–sphincter dyssynergia)**.
- Spinal cord: DO **with DSD**.
- Peripheral nerves or roots: **areflexia, reduced sensation, high PVR (post-void residual)**.
- **Urodynamics** before treatment, because symptoms do not predict findings.

### Diabetes mellitus (most important)

- **Diabetic cystopathy:** sensory then motor autonomic neuropathy. Impaired sensation leads to large capacity, poor contractility, low flow and later high PVR and UTI (urinary tract infection).
- **DO is in fact the commonest urodynamic finding** (early disease or coexisting cerebrovascular disease); hypocontractility comes late.
- Mechanisms: glycation end products, sorbitol pathway, oxidative stress, polyuria.
- Differentiate from BOO (bladder outlet obstruction) by pressure–flow study. Treatment: **timed voiding**, CIC (clean intermittent catheterisation), cautious antimuscarinics.

### Other systemic diseases

| Group | Disease | Usual finding |
|---|---|---|
| Neurological | Stroke | Acute retention, then DO; no true DSD |
| | Parkinson disease | DO; pseudodyssynergia; TURP (transurethral resection of prostate) possible after excluding MSA (multiple system atrophy) |
| | Multiple sclerosis | DO about 38%, DO with DSD 29%, areflexia 26%; changes over time |
| Infective | HIV/AIDS | DO, DSD; areflexia with CMV (cytomegalovirus) |
| | Herpes zoster/simplex, Guillain–Barré | Retention, areflexia; reversible |
| | Spinal tuberculosis, Lyme, HTLV-1 | DO with DSD or areflexia |
| Autoimmune and connective tissue | SLE (systemic lupus erythematosus) | DO; lupus cystitis |
| | Systemic sclerosis | Areflexia, low compliance |
| Infiltrative | Amyloidosis (transthyretin) | Low sensation, underactivity, SUI (stress urinary incontinence) |
| Nutritional | B12 deficiency, alcohol (thiamine) | Mixed; reversible with replacement |
| Endocrine | Hyperthyroidism | Poor emptying; resolves with treatment |

**Also know:** myasthenia gravis; schizophrenia; gastroparesis.

### Management principles

- Treat the disease; reversible therapy.

- Storage: antimuscarinics (trospium, or mirabegron in elderly brains), botulinum toxin, neuromodulation.
- Emptying: CIC; avoid Credé with DSD.

**Diagram to draw:** neuraxis with lesion levels and the bladder pattern at each.

> **Key points to remember:**
> - Lesion level predicts pattern: brain DO; cord DO + DSD; peripheral areflexia.
> - Diabetic cystopathy: low sensation, large capacity, high PVR; DO common early.
> - Urodynamics before treating; reversible treatment first.

**Also, from the condensed book (not in the paper answer):**

- **Guillain–Barré** causes retention; use CIC until recovery.

> **KEY POINT**
>
> Distal lesions hit the peripheral reflex arc, giving areflexia with a non-relaxing or fixed outlet. Compliance falls only when preganglionic parasympathetic fibres are damaged. Hence urgent decompression, waiting and CIC outrank outlet surgery.

## 5 · Specific functional and miscellaneous conditions

### Detrusor–sphincter dyssynergia

#### SAQ Detrusor–external sphincter dyssynergia (DESD / DSD)

Paper answer · P1

**Asked as:**

- External Sphincter Dyssynergia — *Summer 2016 Paper I Q3a (SAQ)*
- Detrusor sphincter dyssynergia — *Practice Set 10 Paper I Q3f (SAQ)* (also answered in 2016-Summer Paper 1, Q3a)

**Type:** SAQ (10 marks)

**Definition**

- **DESD / DSD (detrusor–external sphincter dyssynergia):** the striated sphincter contracts involuntarily (involuntary contraction) during detrusor contraction, causing functional obstruction, high pressure and poor emptying.
- **It requires a lesion between the pons and the sacral cord:** suprasacral spinal cord injury (SCI), MS (multiple sclerosis), transverse myelitis, myelomeningocele. Brain lesions (stroke, Parkinson's) do not cause it (**never with brain lesions**).
- Lesions (at or) above T6 may add bladder-neck (**smooth sphincter**) dyssynergia and autonomic dysreflexia.

**Blaivas classification**

| Type | EMG (electromyography) pattern | Lesion |
|---|---|---|
| 1 | Activity rises with detrusor pressure, then relaxes suddenly at peak | Incomplete |

| Type | EMG (electromyography) pattern | Lesion |
|---|---|---|
| 2 | Sporadic contractions throughout voiding | Complete |
| 3 | Crescendo–decrescendo contraction; obstruction throughout | Complete |
| - Alternative: intermittent versus continuous (continuous means a complete lesion). Type has no relation to lesion level (type is unrelated to level). | | |

### Diagnosis

- **Pressure–flow study with EMG:** EMG rises during the detrusor contraction, with high pressure and interrupted flow. Needle EMG is more reliable than surface.
- **VUDS (videourodynamics):** narrowed (narrow) membranous urethra with dilated posterior urethra; shows VUR (vesicoureteral reflux).
- **Exclude:** guarding reflex (normal); **pseudodyssynergia** (straining or suppressing a contraction); dysfunctional voiding (learned, no neurological lesion; Hinman syndrome in children).

### Significance

- Worse when complete, continuous and **in men**: about half of untreated men get VUR, hydronephrosis, stones or urosepsis. Other risk factors: high filling pressure, indwelling catheter.
- **Management logic:** DSD is both a storage and an emptying failure. Lower bladder pressure, then empty it.
- DLPP (detrusor leak point pressure) ≥40 $cmH_2O$ endangers the kidneys.

### Management

- **First line:** CIC (clean intermittent catheterisation) plus antimuscarinic or β3-agonist; then intradetrusor botulinum toxin.
- **CIC impossible (men):** sphincterotomy or sphincteric botulinum toxin, with condom drainage. The goal is DLPP <40; DLPP ≥40 afterwards means failure; repeat procedures are common.
- Others: suprapubic catheter, Brindley stimulator with deafferentation, diversion. Urethral stents are abandoned; oral drugs disappoint. Credé and Valsalva voiding contraindicated.
- **Also know:** α-blockers for bladder-neck dyssynergia; lifelong upper-tract surveillance.

**Diagram to draw:** urodynamic trace with rising detrusor pressure (Pdet), EMG bursts and interrupted flow.

> **Key points to remember:**
> - Lesion between pons and sacral cord; never brain lesions.
> - Blaivas 1 incomplete; 2 and 3 complete.
> - Worst in men; DLPP ≥40 $cmH_2O$ hostile.
> - CIC plus detrusor relaxation first; sphincterotomy if CIC impossible.

**Also, from the condensed book (not in the paper answer):**

- Blaivas type 1 (incomplete) can be observed unless there is VUR, hydronephrosis or AD; types 2–3 need treatment.
- Without a neurological lesion, doubt DSD (except Hinman syndrome in children).

### Functional outlet disorders

- **Dysfunctional voiding** (Hinman) is striated sphincter contraction during voiding without neurological disease, proven on pressure–flow–EMG.
- **Non-neurogenic chronic retention** is PVR ≥**300 mL** persisting ≥6 months on ≥2 occasions. High-risk patients need drainage, preferably suprapubic.
- **Primary bladder neck obstruction** affects young and middle-aged men with a normal prostate, and videourodynamics localises it. α-blockers help some, but **bladder neck incision is definitive**.
- **Female BOO** is rare. Groutz criteria are free maximum flow (Qmax) ≤12 mL/s and detrusor pressure at Qmax ≥20 cm $H_2O$.
- **Fowler syndrome** is retention in young women with complex repetitive discharges on needle EMG and urethral closure pressure >100 cm $H_2O$. **SNM is the treatment** (~70% succeed).

### Other conditions

- Rarer causes include myasthenia (more incontinence after prostatectomy), Isaacs syndrome, infective myelopathies and lupus cystitis (treated with steroids). After **radiation**, early storage symptoms recover by 6 months.

## 6 · Evaluation and risk stratification

- Under AUA/SUFU 2021 and EAU 2022 neurogenic LUT dysfunction (NLUTD) guidelines, **the primary goal is to prevent renal damage.**
- Everyone gets history, examination (cognition, **hand dexterity**, spasticity), urinalysis, PVR and diaries.
- **Defer full assessment during spinal shock or acute brain injury**, and give no irreversible therapy then.

| Risk (AUA/SUFU) | Typical | Defining findings |
|---|---|---|
| Low (all) | Suprapontine or uncomplicated peripheral | Spontaneous voiding, low PVR, stable; normal imaging and renal function |
| Unknown | Infrapontine–suprasacral; sacral/infrasacral | Storage and voiding symptoms, raised PVR, recurrent UTI; not yet assessed, so do UDS, upper-tract imaging and renal function |
| Moderate | — | Neurogenic retention; DO with incomplete emptying; normal tracts |
| High (any one) | — | Poor compliance, raised storage pressure, DO with DSD, VUR; hydroureteronephrosis, scarring, large stones; abnormal renal function |

**KEY POINT**

Low risk needs no routine imaging, renal function or UDS. DO alone fits low risk (suprapontine, synergic); high risk means DO with infrapontine pressure consequences. Risk is judged by **pressure, compliance and upper tracts**, not by diagnosis.

## 7 · Management

*Start with the simplest, most reversible step.*

**ALGORITHM**

1. Aim for low-pressure storage with regular emptying: CIC with an antimuscarinic or β3 agonist.
2. If refractory, give intradetrusor onabotulinumtoxinA.
3. In selected patients, try neuromodulation.
4. If these fail, do augmentation or continent diversion.
5. As a last resort, do bladder neck closure with a suprapubic catheter, ileovesicostomy or incontinent diversion.

- **CIC is the gold standard**; keep volumes <**500 mL**. Credé and Valsalva are not recommended.
- CIC beats an indwelling urethral catheter, and for a chronic catheter **suprapubic is better than urethral**.
- For refractory neurogenic DO, use **intradetrusor onabotulinumtoxinA** (~60% continue long term) and warn about CIC. Intrasphincteric toxin is not routine.
- The Brindley stimulator is used in complete SCI (see Ch 115). PTNS suits patients who void and have OAB; SNM suits NLUTD without SCI or spina bifida.
- **UTI**: treat bacteriuria only if symptomatic, correct stones and residuals, and give **no prophylaxis** for CIC or indwelling catheters.
- **Bladder cancer** incidence is similar to the general population, but it is often muscle-invasive at diagnosis. **AUA/SUFU 2021 advise no routine surveillance cystoscopy**; scope for haematuria.

## 8 · Follow-up

*Follow-up is lifelong, because upper-tract damage is silent.*

| Body | Schedule |
|---|---|
| AUA/SUFU 2021 (risk-based) | Low risk: none routine. Moderate/high: yearly history, examination and renal function; upper-tract imaging every 1–2 years; repeat UDS for change; after bowel reconstruction, yearly electrolytes, acid–base and imaging |
| EAU | Specialist review every 1–2 years; ultrasound and residual 6-monthly; dipstick 2-monthly; yearly examination and bloods |
| American Paraplegia Society (SCI) | Annual for 5–10 years, then alternate years; renogram: a >20% fall in renal plasma flow means investigate; annual cystoscopy with indwelling catheter (**not endorsed** by AUA/SUFU) |

# 110 Overactive Bladder

## 1 · Definitions and terminology

*OAB is a symptom diagnosis built on urgency.*

*See the paper answer **Overactive bladder: pathophysiology, pharmacotherapy and non-pharmacological management** (Section 7, Ch 103).*

**Also, from the condensed book (not in the paper answer):**

- In overactive bladder (OAB), **urgency is the defining symptom**, hence "urgency" rather than "urge" incontinence.
- OAB is a symptom syndrome, whereas detrusor overactivity (DO) is a urodynamic observation that may be asymptomatic or artefact.

| Term | ICS definition (all patient-reported) | |
|---|---|---|
| Increased daytime frequency | Patient considers they void too often by day; upper normal | **7** voids |
| Nocturia | Waking ≥1 time to void; each void preceded and followed by sleep | |
| Urgency | Sudden, compelling desire to void, difficult to defer | |
| UUI | Involuntary leakage associated with urgency | |

- **Urgency urinary incontinence (UUI) is the most bothersome lower urinary tract symptom (LUTS) in both sexes**. Frequency bothers at ≥**8** daytime voids and nocturia at ≥**2**.
- **Isolated nocturia** usually means nocturnal polyuria (night urine >33% of 24-h output in older adults) or poor sleep.
- **Mixed urinary incontinence (MUI)** is stress UI (SUI) with UUI; SUI with "OAB dry" is **not** MUI.

### OAB versus bladder pain syndrome (BPS)

BPS requires bladder pain, pressure or discomfort with ≥1 urinary symptom; pain is excluded from OAB, though overlap is large.

| Feature | OAB | BPS |
|---|---|---|
| Primary fear | **Leakage** | **Pain** |
| Filling sensation | Phasic urgency waves | Steady, accelerating pain |
| Voided volumes | Variable | More consistent |
| Deferral | Limited, because of leak risk | Possible, at cost of pain |
| Site | Perineum/penile base, urethra/vagina | Suprapubic, with or without perineal |
| UUI | Yes | No |

## 2 · Epidemiology and burden

- Adult OAB affects ≈ **12–29%**, depending on the definition (EpiLUTS highest, EPIC lowest).
- Women have UUI 2–3× as often as men, so the sex gap is in **wet** OAB. Prevalence rises with age, to ~40% of men over 75.

- Fewer than half tell a clinician. About **92%** stop the first antimuscarinic within 2 years, mainly because of **unmet expectations**, then side effects.
- OAB dry progresses to UUI at ~1% a year. Among third-line options, onabotulinumtoxinA 100 U is the most cost-effective and sacral nerve stimulation (SNS) the costliest.

## 3 · Pathophysiology

- **DO is no longer the sine qua non**: only about half with DO have urgency, and **44–69%** with urgency show DO.
- The periaqueductal grey and pontine micturition centre are the main regulatory level. Afferent, urothelial, myogenic (exaggerated micromotions) and central mechanisms all contribute (see pointer).

> **KEY POINT**
>
> Final common pathway: more afferent signal reaches awareness. The bladder or a neighbour turns it up, or the CNS fails to turn it down. Hence DO and obstruction correlate only loosely with OAB.

## 4 · Aetiology and risk factors

- Risks are **ageing** (clearest), female sex, metabolic syndrome, bowel and hormonal factors, depression and anxiety.
- **Bladder outlet obstruction (BOO)** is only an **indirect** cause in humans, since OAB is as common in women, in whom BOO is rare.

## 5 · Clinical assessment

- The **minimum** work-up (AUA/SUFU and EAU) is **history, examination and urinalysis**, followed by empirical treatment; the EAU adds questionnaires and a diary.
- **History** covers bother, SUI, voiding symptoms, haematuria, fluids and caffeine, occult neurology, pelvic surgery or radiotherapy, bowel function (faecal incontinence in ~**50%** of women with OAB), drugs and causes of polyuria.

| Drug class | Relative contraindications |
|---|---|
| Antimuscarinics | Uncontrolled closed-angle glaucoma; cognitive impairment; retention; impaired gastric emptying |
| β3-adrenergic agonists | Uncontrolled hypertension; arrhythmia |

- **Examination**: frailty, cognition and oedema, but the **pelvic and genital examination is most important** (prolapse, atrophy, stress test; digital rectal examination in men; perineal sensation).
- **Urinalysis is mandatory**; microhaematuria needs full evaluation.
- **Postvoid residual (PVR)** is not needed before conservative or oral therapy if uncomplicated; measure it with retention risk, voiding symptoms, prior outlet surgery or neurology.
- Use a **diary** for 3–7 days; **maximum voided volume is the functional capacity.** Questionnaires include ICIQ-OAB and OAB-q; MCQ OAB-V8 is a screening tool, not a severity score.

## 6 · Urodynamics (UDS)

- UDS is **not** in the initial work-up. It is **indicated** when the result decides invasive or irreversible treatment, after multiple failures, or with suspected BOO.
- DO is found in OAB wet in **>90% of men** but is absent in 30–40% of women. It correlates **best with UUI and worst with nocturia**.
- Stop antimuscarinics to show DO, and continue them to explain refractory symptoms.
- **Assess emptying**: each **10 mL** of PVR raises the risk of catheterisation after onabotulinumtoxinA by **27%**. DO persists in up to **60%** after transurethral resection of the prostate (TURP).

## 7 · Management

*Treatment follows the patient's goals.*

- OAB is benign, so **no treatment is acceptable**. Set **realistic expectations** of improvement, not cure. AUA/SUFU lines are **not a compulsory sequence**; allow **4–12 weeks** per therapy.

**ALGORITHM**

1. Confirm OAB and exclude mimics.
2. Educate, then start conservative measures.
3. If not enough, add drugs.
4. If drugs fail, offer tibial nerve stimulation, sacral neuromodulation or onabotulinumtoxinA.
5. Rarely, do augmentation or diversion.
6. Reassess at every step.

- Reducing fluid by ~**25%** helps, and cutting caffeine improves urgency and frequency.
- Pelvic floor muscle training with bladder training reduces leaks by **50–80%**. **Weight loss** reduces SUI and UUI (PRIDE trial).
- Drugs (see Ch 113): response is idiosyncratic, so switch or combine before invasive work-up.
- In **dementia**, antimuscarinic use >**3 months** raises risk by 10–65%, so prefer a β3 agonist.
- In benign prostatic hyperplasia with OAB, an α1-blocker plus antimuscarinic works; retention occurs in ~3%, so take care if PVR is raised.

### Third-line (see Ch 115)

Frailty predicts functional decline and 1-year morbidity of up to 50%, even after minor procedures.

| | **Tibial nerve stimulation** | **OnabotulinumtoxinA** | **SNS** |
|---|---|---|---|
| Efficacy | 60–80% respond | UUI reduced 59%; ~¼ dry | 5-year success 70–80% |
| Drawback | 12 weekly visits; response by weeks 8–10 | Urinary tract infection (UTI) ~20%; catheterisation 0–12% in trials, real-world ~20% women, ≤35% men | Revision >30% at 5 years; battery |
| Risk | Lowest | Moderate | Highest |

- **ICI pathways**: in men, treat the outlet with the bladder; in women, treat the most bothersome MUI component first; in the frail elderly, put comorbidity and carer goals first.

> **KEY POINT**
>
> Escalate by invasiveness and reversibility, not severity alone. OAB is benign, so patient goals, frailty and PVR decide how far to climb.

# 111 Underactive Detrusor

## SAQ Detrusor underactivity and its management

Paper answer · P2 · P4 · merged from 2

*Asked as:* P2 SAQ "Management of detrusor underactivity" · P4 SAQ "Detrussor underactivity."

### Definition

- **DUA (detrusor underactivity), ICS (International Continence Society) 2002:** "a contraction of reduced strength and/or duration, resulting in prolonged bladder emptying and/or failure to achieve complete bladder emptying within a normal time span."
- **Acontractile detrusor:** no contraction.
- **UAB (underactive bladder):** the symptom complex of slow stream, hesitancy and straining, with or without incomplete emptying.
- DUA is common in elderly men and women having urodynamics: found in 40–48% of men over 70.

### Pathophysiology and causes

- Failure may be **myogenic** (myocyte loss, fibrosis), **efferent** (poor nerve activation) or **afferent** (reduced sensation cannot sustain the voiding reflex).

| Type | Examples |
|---|---|
| Neurogenic | Cauda equina, disc prolapse, spinal injury, diabetic neuropathy, multiple system atrophy, early stroke, radical pelvic surgery |
| Myogenic | Long-standing BOO (bladder outlet obstruction), diabetes, ischaemia, ageing |
| Drugs | Anticholinergics, opioids |
| Functional | Fowler syndrome, dysfunctional voiding |

### Evaluation

- Bladder diary, flow rate, **PVR (post-void residual)**, sacral neurological examination, spinal MRI if indicated (deficit), drug review.
- **Pressure-flow study is the only accepted (diagnostic) test:** low detrusor pressure with low Qmax.
- **BCI (bladder contractility index) = Pdet@Qmax (detrusor pressure at maximum flow) + 5 × Qmax:** <100 weak, 100–150 normal, >150 strong. Also PIP (projected isovolumetric pressure), Watts factor, stop test.
- Ambulatory urodynamics if no contraction is seen on the standard study (the "bashful bladder").

### Management

- **Goals:** protect kidneys, prevent UTI (urinary tract infection) and stones, improve symptoms. No drug reliably improves contraction.

| Aim / option | Details |
|---|---|
| Conservative drainage | Timed and double voiding; treat constipation; stop causative drugs |
| **CISC (clean intermittent self-catheterisation)** | **Mainstay** for high PVR; fewer infections than an indwelling catheter. Suprapubic catheter if CISC impossible |

| Aim / option | Details |
|---|---|
| Lower outlet resistance | α-blockers (modest benefit); pelvic-floor biofeedback |
| Outlet surgery (TURP, transurethral resection of the prostate) | In selected obstructed men; little help in low-PVR men; may free catheter-dependent fit men |
| **SNM (sacral neuromodulation)** | For non-obstructive retention, especially Fowler syndrome; needs some detrusor function; intact sensation predicts response |
| Sacral anterior root stimulator (Brindley) | Complete spinal cord injury, with posterior rhizotomy |
| Parasympathomimetics (bethanechol, distigmine) | **Not recommended / not routine**: weak evidence; systemic side effects (bronchospasm, rarely cardiac arrest) |
| Intrasphincteric botulinum toxin | Experimental |
| Diversion | Mitrofanoff when urethral CISC fails; rarely ileal conduit |

- Avoid Credé or Valsalva voiding (reflux risk).
- SNM and parasympathomimetics need some intact detrusor and sensory function; a fibrosed detrusor responds to nothing.

### Investigational

- Intravesical electrotherapy, intravesical prostaglandins, TAC-302, latissimus dorsi detrusor myoplasty, muscle-derived stem cells.

### Diagram to draw

- Pressure-flow plot on the Schäfer nomogram with BCI lines at 100 and 150.

> **Key points to remember:**
> - BCI = Pdet@Qmax + 5Qmax; <100 weak.
> - CISC is the mainstay; no effective drug; parasympathomimetics not recommended.
> - SNM helps non-obstructive retention (Fowler syndrome); intact sensation predicts response.

**Also, from the condensed book (not in the paper answer):**

- Poor surgical outcome: voiding pressure under 45 cm $H_2O$, age over 80, PVR over 1500 mL.
- Significant PVR proposal (over 40% of capacity); multisystem atrophy 52–95%.
- Sphincter toxin 50–100 U; no bladder neck incision in women.

## 1 · Definitions and symptoms

*Drugs and surgery lower outlet resistance but cannot strengthen the detrusor.*

- **Detrusor underactivity (DUA)** (International Continence Society [ICS] 2002, urodynamic) is a contraction of **reduced strength and/or duration**, causing prolonged or incomplete emptying. **No thresholds** are given. It is managed mainly by intermittent self-catheterisation (ISC).
- A **"bashful bladder"** fails to void only during the urodynamic study (UDS), unlike a truly **acontractile detrusor**.

- **Underactive bladder (UAB)** (ICS 2018) is the symptom set: slow stream, hesitancy and straining, with or without incomplete emptying. It has **no pathognomonic symptom** and overlaps with overactive bladder (OAB) and bladder outlet obstruction (BOO).
- A significant postvoid residual (PVR) is not agreed; one proposal is over **40%** of functional capacity (voided plus PVR).

| Phase | Symptoms |
|---|---|
| Voiding | Weak stream, intermittency, hesitancy, straining |
| Post-micturition | Incomplete emptying |
| Storage | Frequency and nocturia, **or** reduced desire with infrequent voiding |
| Very large PVR | Incontinence, especially in sleep |

> **KEY POINT**
>
> **Sensation decides the picture.** Absent sensation makes an infrequent voider, with overflow worse at night. Preserved sensation makes a frequent voider with incomplete emptying and nocturia. DUA may be asymptomatic.

## 2 · Epidemiology

- BOO is rare in women, so low flow with high PVR in a woman suggests DUA.
- UDS series find DUA in men under 50 in **9–28%**, men over 70 in ~**48%** and women in 12–45%.
- It peaks in institutionalised elderly people, often as **detrusor hyperactivity with impaired contractility (DHIC)**: an involuntary contraction emptying under half the bladder.
- Elderly patients often also have detrusor overactivity (DO), BOO or stress incontinence, so do **UDS before lower urinary tract symptom (LUTS) surgery in the elderly**.
- Non-neurogenic male DUA shows **minimal progression** over 10 years.

## 3 · Aetiopathogenesis

Confirmed causes are neurological disease and diabetes; probable ones are BOO, ageing and atherosclerosis.

| Mechanism | Site | Result |
|---|---|---|
| Myogenic | Myocyte, extracellular matrix (ECM) | Weak contraction despite normal drive |
| Efferent | Sacral cord, roots, pelvic nerves | Impaired activation |
| Afferent | Peripheral afferents, spinal columns | **Early termination** of the voiding reflex |
| Brain | Pontine micturition centre (PMC), periaqueductal grey (PAG), limbic, prefrontal | Failed integration |

- Ageing contributes through ECM fibrosis and falling afferent sensation; in the **"stiff balloon" hypothesis**, afferent drive fades after a small void, so the contraction stops early and leaves a PVR.
- **Drug causes are reversible**: anticholinergics (including antihistamines, antipsychotics, tricyclics, antiparkinsonian drugs) and **opioids**.

- †**Fowler syndrome** (young women, non-relaxing sphincter) responds to sacral neuromodulation.
- **BOO** causes **decompensation** in animals, but most men in acute retention have preserved detrusor function.
- **Diabetes**: early diuresis causes hypertrophy and storage symptoms; late damage impairs sensation and emptying.

| Condition | DUA | Key point |
|---|---|---|
| Stroke | ~Half retain acutely | "Cerebral shock"; chronic sequel is DO |
| Parkinson disease | <20% | DO commoner; DUA drug-related |
| **Multisystem atrophy** | **52–95%** | Efferent parasympathetic atrophy; mistaken for Parkinson |
| Multiple sclerosis | ~20% | Lumbosacral plaques |
| Radical pelvic surgery | Hysterectomy: ~72% LUT dysfunction | Nerve-sparing reduces it; many recover by 1 year |
| Guillain–Barré, zoster | — | Reversible |

## 4 · Diagnosis

*Only a pressure-flow study proves a weak detrusor.*

- **Invasive UDS is the only accepted method**. There are no consensus criteria, and nearly all measure strength only.
- Typically maximum flow (Qmax) and detrusor pressure at Qmax (Pdet@Qmax) are both low; isovolumetric estimates correct for their inverse relation.

| Measure | Formula | Thresholds / note |
|---|---|---|
| **Bladder contractility index (BCI)** (Abrams) | Pdet@Qmax + 5Qmax | >150 strong; 100–150 normal; **<100 weak** ; male-derived |
| Projected isovolumetric pressure (PIP) (Schäfer) | Pdet@Qmax + K·Qmax (K 5 in men, 1 in older women) | <50 very weak; 50–100 weak |
| Detrusor contraction coefficient (DECO) | BCI/100 | <1 weak |

- †The **BOO index** is Pdet@Qmax − 2Qmax: over 40 is obstructed and under 20 unobstructed.
- If the patient fails to void on UDS, do **ambulatory UDS**: over four-fifths of apparently acontractile patients contract.

## 5 · Management

*Safe drainage protects the kidneys; no drug reliably helps.*

**ALGORITHM**

1. Evaluate: diary, digital rectal examination, urinalysis, flow, PVR.
2. If there is a neurological deficit, then do spinal MRI.
3. Stop offending drugs; treat constipation.
4. Timed or double voiding.
5. Lower outlet resistance where relevant.
6. If PVR is high, then ISC; if ISC is impossible, then a suprapubic catheter.
7. Selected patients: sacral neuromodulation (SNM), outlet surgery, diversion.

| Aim | Routine | Experimental | Not recommended |
|---|---|---|---|
| Drainage | Timed/double voiding; catheterisation; diversion | — | Credé, Valsalva, reflex voiding |
| Lower outlet resistance | Pelvic floor biofeedback; α-blockers; outlet surgery | Sphincter botulinum toxin | — |
| Improve detrusor function | SNM; anterior root stimulator | Intravesical prostanoids; intravesical electrotherapy; myoplasty | **Muscarinic agonists; anticholinesterases** |

- **Credé or Valsalva** is used only with an incompetent sphincter, as otherwise it causes reflux.
- **ISC** causes urethral bleeding in ~⅓; keep volumes below ~500 mL. Suprapubic is the best long-term indwelling route.
- **Bethanechol and distigmine** cause bronchospasm, diarrhoea, salivation and **rare fatal cardiac arrest**, and are useless with fibrosis.
- **SNM** works best with a sphincter or pelvic floor disorder and may help DHIC, where treating DO worsens DUA and vice versa (see Ch 115).
- The **Brindley** stimulator, combined with rhizotomy, is for complete spinal cord injury. Intravesical electrotherapy had a negative sham-controlled trial.

**KEY POINT**

Every therapy needs a surviving link: Brindley needs efferents and muscle; neuromodulation and electrotherapy need afferents and muscle. **Preserved sensation predicts response** to bethanechol, toxin and electrotherapy.

- **Sphincter toxin** is given as 50–100 U cystoscopically (men) or periurethrally (women). About half of idiopathic cases improve; it is short-lived and unlicensed.

| | DUA, low PVR, symptomatic | DUA, catheter-dependent |
|---|---|---|
| Reasoning | Emptying at low Pdet, so BOO is unlikely to matter | Lowering resistance may allow voiding |
| Evidence | Transurethral resection of the prostate (TURP) at ~11 years: no benefit over conservative care | Flow rarely normalises, but many void |
| Verdict | **Surgery unlikely to help** | Consider in young, fit patients |

- Outcome in retention is poor with voiding pressure under **45** cm $H_2O$, age over **80** or PVR over **1500 mL**.
- In women, avoid bladder neck incision because of incontinence and stenosis.
- **Diversion**: a continent catheterisable stoma suits young women with idiopathic retention refractory to SNM; an ileal conduit treats renal deterioration from intramural ureteric obstruction. Reduction cystoplasty is abandoned.

# 113 Pharmacologic Management of Lower Urinary Tract Storage and Emptying Failure

## 1 · Framework

Drugs follow the Wein scheme (see Ch 104). Most lower urinary tract (LUT) drugs fail on **selectivity**, because the same receptor in heart and vessels caps the dose.

## 2 · Antimuscarinics and β3-agonists

*First-line overactive bladder (OAB) drugs; side effects and age drive choice.*

*See the paper answer **Overactive bladder: pathophysiology, pharmacotherapy and non-pharmacological management** (Section 7, Ch 103).*

**Also, from the condensed book (not in the paper answer):**

- Beers avoidance of oxybutynin; myasthenia and severe colitis as contraindications.
- MILAI II add-on to mirabegron; combination doubles constipation.
- †Antimuscarinics act mainly in **storage**, so retention is uncommon.
- **Oxybutynin** causes the most dry mouth and has the most central nervous system (CNS) penetration, so avoid it in the elderly (Beers).
- Tolterodine is 2 mg bd or extended-release (ER) 4 mg od. **Darifenacin** is most M3-selective.
- **Trospium** is a renally excreted quaternary amine. Other doses: propiverine 20 mg, imidafenacin 0.1–0.2 mg bd.
- †**Side effects by receptor**: dry mouth, constipation and blurred vision (M3); tachycardia (M2); cognition (**M1**).
- †**Contraindications**: **uncontrolled narrow-angle glaucoma**, urinary or gastric retention, myasthenia gravis, and severe ulcerative colitis or toxic megacolon.
- †In the elderly, cumulative **anticholinergic burden** favours a β3-agonist or non-CNS-penetrant agent. Men with bladder outlet obstruction are safe if postvoid residual (PVR) is low.
- Mirabegron goes from 25 to 50 mg od; avoid it in severe uncontrolled hypertension (≥**180/110** mm Hg). Vibegron needs no titration and had no blood pressure (BP) effect.

**ALGORITHM**

1. †OAB: behavioural therapy, then an antimuscarinic or a β3-agonist; if elderly or at cognitive risk, then choose the β3-agonist first.
2. If the response is poor, then switch or combine.
3. If refractory, then onabotA 100 U, sacral neuromodulation (SNM) or percutaneous tibial nerve stimulation (PTNS).
4. Last resort: augmentation cystoplasty.

## 3 · Combination therapy

| Direction | Evidence | Agents |
|---|---|---|
| Mirabegron added to failing antimuscarinic | SYNERGY, SYNERGY II, BESIDE, MILAI | **Solifenacin only** |
| Antimuscarinic added to failing mirabegron | MILAI II (52 weeks) | Solifenacin 5, propiverine 20, imidafenacin 0.2, tolterodine ER 4 mg |
| Vibegron plus antimuscarinic | Phase IIb (vibegron 100 plus tolterodine 4 mg) | Extrapolated; no trial after monotherapy failure |

> **KEY POINT**
>
> In non-responders, combination **equals** solifenacin 10 mg with less dry mouth: it gains **tolerability, not efficacy**. Heart rate, BP, QT and PVR are unchanged; constipation roughly doubles.

- An **α-blocker plus desmopressin** suits refractory nocturia in benign prostatic hyperplasia (BPH), on uncontrolled evidence.
- Adding **vaginal oestrogen** is **not established**; †systemic oestrogen worsens urinary incontinence (UI).

## 4 · Intradetrusor onabotulinumtoxinA (onabotA)

*The main third-line option.*

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### SAQ Botulinum toxin in urology

Paper answer · P1 · P2 · P3 · P4 · merged from 4

*Asked as:* P1 SAQ "Botulinum toxin in urology: mechanism, indications and doses" · P2 SAQ "Botulinum neurotoxin and its role in urology" · P3 SAQ "Botulinum toxin in urology" · P4 SAQ "Botulinum toxin in urology"

*"Botulimum" / "Botulinim" in the questions is read as botulinum toxin.*

#### Introduction

- Neurotoxin of *Clostridium botulinum*, serotypes A–G. Urology uses **onabotulinumtoxinA (BoNT-A)**, the only licensed formulation; type B is short-acting.
- 150 kDa: a heavy chain (100 kDa; binding and entry) and a zinc-dependent light chain (50 kDa).

#### Mechanism of action

1. The heavy chain binds the cholinergic nerve terminal and the toxin is internalised.
2. The light chain (a zinc protease) enters the cytosol.
3. It **cleaves SNAP-25** (synaptosomal-associated protein 25), a SNARE (vesicle-docking) protein. Vesicles cannot fuse, so **acetylcholine release stops**: chemical denervation. - **Afferent action:** less ATP (adenosine triphosphate), substance P and CGRP (calcitonin gene-related peptide) release; fewer TRPV1 (capsaicin) and P2X3 (purinergic) sensory receptors on C fibres, so urgency and pain fall. - The effect lasts about 6–9 months and wanes as nerve terminals sprout, so injections are repeated; repeat injections stay effective and response is maintained.

### Technique

- Flexible or rigid cystoscopy under local anaesthesia; diluted toxin injected into the detrusor at **about 20 sites (100 U, in 10 mL) or 30 sites (200 U, in 30 mL)**, usually sparing the trigone (reflux concern).
- Sterile urine, antibiotic cover; check PVR (post-void residual) at 2 weeks.

### Urological uses: indications and doses

| Indication | Dose and evidence |
|---|---|
| **Idiopathic OAB (overactive bladder)**, third line | **100 U** (EMBARK): incontinence down about 59%, about 25% dry. **ABC** trial: equal to antimuscarinic, more complete dryness. **ROSETTA**: comparable to SNM (sacral neuromodulation), more UTI (urinary tract infection) (P2, P3: 200 U used, equal to or ≥ SNM; another source: slightly better than SNM) |
| **NDO (neurogenic detrusor overactivity):** SCI (spinal cord injury), MS (multiple sclerosis) | **200 U** (DIGNITY trials): less incontinence, larger capacity, lower detrusor pressure; works with or without antimuscarinics |
| **Paediatric NDO** (spina bifida) | About **10 U/kg (maximum 360 U)**, about 30 sites, in published (Campbell) series; FDA (Food and Drug Administration) label (2021, age ≥5) 6 U/kg, maximum 200 U; continence 65–87%, most avoid augmentation |
| **DSD (detrusor–sphincter dyssynergia)**, underactive detrusor | Intrasphincteric injection 50–100 U, mainly in men unable to do CISC (clean intermittent self-catheterisation) |
| **IC/BPS (interstitial cystitis/bladder pain syndrome)** | 100 U intradetrusor, often trigonal; AUA (American Urological Association) option, off-label |
| **BPH (benign prostatic hyperplasia)** | 100–300 U intraprostatic: an early small trial was positive, but large placebo-controlled RCTs (randomised controlled trials) showed **no benefit over placebo (saline/sham)**; not recommended |

- **Also know:** pelvic floor spasm; paediatric dysfunctional voiding (sphincter); chronic pelvic pain; chronic orchialgia (RCT negative); ED (erectile dysfunction) trials. Units of different toxin products are not interchangeable.

### Complications

- **Retention needing CISC** (the main risk): 0–12% in OAB trials, but about 20% of women and up to 35% of men in practice; commoner in NDO. Each 10 mL rise in baseline PVR raises the risk about 27%. **The patient must accept CISC beforehand.**
- **UTI** about 20%; haematuria; rarely generalised weakness.

### Contraindications

- Active UTI, myasthenia gravis, aminoglycosides, pregnancy, unwillingness to self-catheterise.

### Recent advances

- Paediatric NDO: FDA approved onabotulinumtoxinA for children aged ≥5 years (2021); used from about 2 years of age.

- Other formulations: abobotulinumtoxinA and incobotulinumtoxinA under study for detrusor overactivity; doses are not interchangeable between products.
- Injection-free delivery under research: intravesical liposome-encapsulated toxin and other instillation vehicles.

### Diagram to draw

- Cholinergic synapse (nerve terminal) showing SNAP-25 cleavage by the light chain.

> **Key points to remember:**
> - Cleaves SNAP-25; acts on efferent and afferent nerves.
> - 100 U idiopathic OAB; 200 U NDO; spare trigone.
> - Children: 10 U/kg (max 360 U) in series; FDA label 6 U/kg (max 200 U), approved for age ≥5 years (2021); used from about 2 years of age.
> - Main risks: retention needing CISC, UTI.
> - Other formulations (abobotulinumtoxinA, incobotulinumtoxinA) under study; doses are not interchangeable between products.
> - Injection-free delivery under research: intravesical liposome-encapsulated toxin and other instillation vehicles.

**Also, from the condensed book (not in the paper answer):**

- Repeat ≥12 weeks apart; aminoglycoside caution.
- The dose is **100 U** for idiopathic OAB and **200 U** for neurogenic detrusor overactivity (NDO).
- †Patients must accept clean intermittent catheterisation (CIC) first, especially in non-catheterising NDO.
- The commonest adverse event (AE) is urinary tract infection (UTI). The effect lasts ≈6–9 months.

## 5 · Nocturia: desmopressin †

*See the paper answer **Desmopressin (DDAVP)** (Section 8, Ch 112).*

**Also, from the condensed book (not in the paper answer):**

- Low-dose sublingual desmopressin is 25 µg in women and 50 µg in men; hyponatraemia risk rises at age ≥**65**.

## 6 · Increasing outlet resistance (SUI)

*No drug is standard for stress urinary incontinence (SUI).*

†Pelvic floor muscle training and surgery are first-line.

| Drug | Mechanism | Status |
|---|---|---|
| Duloxetine † | Serotonin–noradrenaline reuptake inhibitor at **Onuf's nucleus**; raises pudendal tone | 40 mg bd; nausea causes discontinuation; Europe, not USA |
| Phenylpropanolamine, midodrine, norepinephrine | Systemic α-agonists | Weak evidence; BP rises first; phenylpropanolamine banned (stroke) |

| Drug | Mechanism | Status |
|---|---|---|
| Imipramine † | Anticholinergic plus α-like outlet effect | Mixed UI, enuresis; orthostatic hypotension, arrhythmia; caution in elderly |
| Clenbuterol, vaginal phenylephrine gel | β-agonist; local α-agonist | Abandoned |

## 7 · Facilitating emptying

- †**Bethanechol** has **no convincing controlled evidence** in detrusor underactivity and causes cholinergic AEs; avoid in asthma, peptic ulcer, bradycardia and obstruction.
- †**α-blockers** help in BPH, primary bladder neck obstruction and neurogenic LUT dysfunction, and aid catheter trials.
- †Sphincteric onabotA is an option for the striated sphincter.
- †**Drugs that cause retention** include anticholinergics, pseudoephedrine, opioids, benzodiazepines, anaesthesia and epidurals; stopping them helps more than adding any drug.

## 8 · Failed drugs

| Agent | Indication | Reason for failure |
|---|---|---|
| Vitamin D3, elocalcitol; calcium antagonists | Urgency UI, OAB | No urodynamic effect |
| $K^+$ channel openers; URO-902 gene therapy | OAB | Dose capped by heart; phase 1 only |
| RimabotulinumtoxinB | Toxin A-refractory DO | Short duration (≤3 months) |
| Intraprostatic onabotA, PRX302, NX-1207 | BPH lower urinary tract symptoms | No better than sham injection |

## 9 · Landmark trials

| Trial | Design | Result | Take-home |
|---|---|---|---|
| SYNERGY / II | Phase III, 12 weeks / 12 months | Solifenacin 5 plus mirabegron 50 better than monotherapy for UI | Effective, safe long-term |
| BESIDE | Solifenacin 5 non-responders | Add-on better than solifenacin 5, equal to solifenacin 10 | Alternative to dose escalation |
| MILAI II | 52 weeks, mirabegron non-responders | Four antimuscarinic add-ons safe, effective | Add antimuscarinic to mirabegron |
| DIGNITY | Pooled RCTs, NDO | OnabotA 200 U equal in anticholinergic users/non-users | Efficacy independent of anticholinergic |
| EMBARK | RCTs, refractory OAB | OnabotA 100 U effective | Standard idiopathic OAB dose |

# 115 Electrical Stimulation and Neuromodulation in Storage and Emptying Failure

## LAQ Neuromodulation for voiding dysfunction, including sacral neuromodulation

Paper answer · P2 · P3 · P4 · merged from 3

*Asked as:* P4 LAQ "Neuromodulation for voiding dysfunction, including sacral neuromodulation" · P2 SAQ "Neuromodulation" · P3 SAQ "Sacral neuromodulation"

*Taken as neuromodulation for lower urinary tract dysfunction.*

### Definition and history

- **Neuromodulation:** electrical stimulation of a peripheral nerve that alters the activity of central (reflex) pathways controlling the bladder. It acts on the nervous system (**afferents**), not directly on muscle.
- Direct muscle stimulation failed (painful, local). Stimulating **afferents** needs less current and affects the whole organ.
- **SNM (sacral neuromodulation)** (Tanagho and Schmidt, about 1985): continuous low-amplitude **S3** stimulation through a tined lead and IPG (implantable pulse generator). PTNS (percutaneous tibial nerve stimulation): FDA (Food and Drug Administration) approval 2000.
- **Rule:** exclude anatomical causes first (obstruction, stones, cancer, fistula). Stimulation cannot correct anatomy or restore muscle.

### Mechanism

- Sacral roots are mixed nerves. Current excites **afferents to the spinal cord and forebrain** (areas of awareness and attention), altering PMC (pontine micturition centre) output. It acts on **afferent** nerves, not motor nerves.
- **OAB (overactive bladder):** heightened central control (afferent input) damps PMC firing, so there are fewer involuntary contractions.
- **Non-obstructive retention:** restores awareness of filling (sensation) and pelvic floor tone, so the PMC switches on and the **EUS (external urethral sphincter) relaxes**. Hence one therapy treats **both** opposite disorders.
- **Tibial:** the nerve carries **S2–S3** fibres; inhibition of OAB-type (C-fibre) reflexes is **opioid-mediated** (naloxone blocks it); long-term cortical plasticity.

### Modalities

| Modality | Target and method | Main use |
|---|---|---|
| **SNM** | Tined lead in **S3** foramen + implanted generator | Refractory OAB, NOUR (non-obstructive urinary retention), faecal incontinence |
| **PTNS** | 34-gauge needle above medial malleolus | Refractory OAB |
| TTNS (transcutaneous tibial) | Surface patch, home use | OAB (less effective) |
| Implantable tibial | Small subcutaneous device over tibial nerve | OAB, avoids clinic visits |

| Modality | Target and method | Main use |
|---|---|---|
| Pudendal | Tined lead in Alcock's canal | IDO (idiopathic detrusor overactivity), NDO (neurogenic DO), SNM failure (off-label) |
| DGNS (dorsal genital nerve stimulation) | Dorsal penile/clitoral nerve | Experimental; NDO |
| IVES (intravesical electrical stimulation) | Intravesical cathode | Underactive bladder in children |
| **SARS (sacral anterior root stimulator)** | Anterior S2–S4 roots + dorsal rhizotomy | Complete SCI (spinal cord injury) |

### A. Sacral neuromodulation: indications

- **Third line** for OAB after behavioural and drug therapy (AUA [American Urological Association] and EAU [European Association of Urology] agree).

| FDA approved | Not approved |
|---|---|
| Refractory UUI (urgency urinary incontinence) 1997 | Bladder pain syndrome |
| Urgency–frequency 1999 | Neurogenic bladder (small studies positive; low-level evidence) |
| **Non-obstructive urinary retention** (e.g. Fowler's syndrome) 1999 | Pelvic pain |
| Faecal incontinence 2011 | |

- **Fowler's syndrome**: young women in retention with abnormal sphincter EMG (electromyography) activity; responds well to SNM.
- Retention needs some detrusor function; success falls with age.

### SNM: technique

**Test phase** (with a bladder diary)

| Test phase | PNE (percutaneous nerve evaluation) | Staged tined lead |
|---|---|---|
| Lead | Temporary, office, local anaesthesia (dislodges) | Permanent tined lead, theatre, fluoroscopy |
| Duration | 5–7 days | Up to 4 weeks |
| Use | OAB | More reliable; higher conversion; preferred for retention (NOUR) |

- **Landmark:** prone; needle into S3 about **9 cm above the coccyx tip, 1–2 cm lateral**.

| Root | Pelvic response | Foot response |
|---|---|---|
| S2 | — | Calf and foot (external) rotation |
| **S3 (target)** | **Bellows**, perineal sensation | **Great-toe flexion** |
| S4 | Bellows/perineal only | None |

- **Implant if ≥50% improvement** in a diary parameter (retention: ≥50% fewer catheterisations); IPG in the buttock.
- **Also know:** 10–16 Hz at sensory threshold; bilateral stimulation no better.

### SNM: outcomes

- **InSite:** success 61% vs 42% with standard medical therapy at 6 months; about 70–80% success at 5 years (another source: about half keep benefit at ≥5 years).
- **ROSETTA:** onabotulinumtoxinA 200 U reduced UUI slightly more (−3.9 vs −3.3 episodes/day) with higher satisfaction, but caused more UTI (urinary tract infection) (24% vs 10%) and CIC (clean intermittent catheterisation, self-catheterisation); SNM removal 9% at 2 years.
- NOUR: most stop catheterising.

### SNM: complications and contraindications

- Pain at the IPG site, lead migration, loss of effect, infection; **revision over 30% at 5 years**; costly (costliest option); battery replacement.
- Contraindications: failed test, inability to use the programmer, sacral anomaly.

### B. Tibial nerve stimulation

- **PTNS technique:** 34-gauge needle **3–4 cm (three fingerbreadths) above the medial malleolus**, behind the tibia; ground pad on the same leg; 20 Hz, 0–10 mA. Confirm by **toe flexion/fanning** or sole tingling. **30 min weekly × 12 weeks**, then maintenance.
- **SUmiT** (n = 220, validated sham): response **55% vs 21%**. Extension: 77% relapse-free at 3 years with tapering maintenance.
- **OrBIT:** PTNS comparable to extended-release tolterodine, with fewer side effects.
- **TTNS:** home self-treatment; skin impedance limits amplitude, so less effective.
- **Implantable devices** remove clinic visits: **BlueWind** (miniature, external control; benefit sustained at 3 years); **eCoin** (coin-sized, battery-powered, office insertion under local anaesthesia).

### C. Other targets

- **Pudendal nerve stimulation:** tined lead via a posterior approach to Alcock's canal. Better than SNM in a blinded crossover; **salvage after failed SNM**. Off-label.
- **DGNS:** purely sensory; acute DO suppression in SCI; **conditional (on-demand)** stimulation possible. Electrode fixation is difficult.
- **IVES (Katona):** intravesical cathode in saline activates Aδ afferents, producing detrusor contraction. Needs some afferent and muscle function; daily sessions; used in children; the sham-controlled trial was negative.
- **Brindley SARS:** stimulates anterior S2–S4 roots for voiding, defecation and erection. **Dorsal rhizotomy** abolishes reflex incontinence and DSD (detrusor–sphincter dyssynergia). The sphincter relaxes faster than the detrusor between bursts, giving **post-stimulus voiding**. Needs intact efferents.
- **Also know:** intravaginal stimulation (SUI, stress urinary incontinence); sacral dermatome stimulation; closed-loop SNM.

### Comparison of third-line OAB options

| | PTNS | OnabotulinumtoxinA 100 U | SNM |
|---|---|---|---|
| Invasiveness | Lowest | Moderate | Implant |
| Drawback | 12 weekly visits | UTI, retention/CIC | Revision, cost |
| Retention | No role | Worsens | **Treats** |

### Recent advances

- Rechargeable stimulators (FDA 2019; ≥15 years); non-rechargeable devices lasting ≥10 years (2021).
- **Full-body (whole-body) MRI (magnetic resonance imaging)-conditional** systems at 1.5 T and 3 T.
- **Implantable tibial nerve stimulators** (BlueWind, eCoin) and transcutaneous tibial stimulation as less invasive alternatives; PTNS and pudendal stimulation as alternatives to SNM.
- Standardised tined-lead placement improves outcomes and lowers revision.
- Bilateral stimulation no better.

### Diagram to draw

- Lead in the S3 foramen with IPG in the buttock, and PTNS needle above the medial malleolus.

> **Key points to remember:**
> - Afferent modulation explains SNM benefit in both OAB and retention.
> - S3: bellows plus great-toe flexion; implant if the test improves ≥50%.
> - SNM revision >30% at 5 years.
> - SUmiT: PTNS 55% vs sham 21%; 12 weekly sessions.
> - ROSETTA: Botox 200 U slightly better for UUI, more UTI and CIC.
> - Rechargeable and long-life non-rechargeable stimulators, lasting at least 10 years.
> - Full-body MRI-conditional devices at 1.5 T and 3 T.
> - Implantable tibial nerve stimulators (BlueWind, eCoin) and transcutaneous tibial stimulation as less invasive alternatives.
> - Standardised tined-lead placement improves outcomes and lowers revision.

**Also, from the condensed book (not in the paper answer):**

- Urodynamic DO does not predict success; PNE gives false negatives.

## 1 · Evolution and mechanism

*One device treats both overactivity and retention.*

- Direct detrusor stimulation failed, so stimulation now uses an **indirect route via afferents**.
- Milestones MCQ : sacral neuromodulation (SNM) ~1985 (Tanagho, Schmidt); percutaneous tibial nerve stimulation (PTNS) derives from acupuncture point **SP6**.
- **Current theory**: sacral afferents reach forebrain awareness areas, which reduces pontine micturition centre (PMC) firing in overactive bladder (OAB); in underactive bladder, raised filling awareness lets the PMC switch on.
- **Paradox**: in retention it works mainly by **striated sphincter relaxation**, not detrusor contraction.

## 2 · Sacral neuromodulation

- **Third-line** for idiopathic OAB (American Urological Association/Society of Urodynamics [AUA/SUFU] and European Association of Urology [EAU] agree).
- First get a frequency–volume diary and exclude anatomical causes, urinary tract infection (UTI) and a significant postvoid residual (PVR). Urodynamic detrusor overactivity (DO) **does not predict** success.
- In **non-obstructive urinary retention (NOUR)**, PVR is the only objective outcome. Fowler syndrome predicts response; age lowers the odds.

| Food and Drug Administration (FDA) approved | Not approved |
|---|---|
| Urgency urinary incontinence (UUI) | Bladder pain syndrome |
| Urgency–frequency | Pelvic pain syndrome |
| Chronic NOUR | Neurogenic bladder |

- †Faecal incontinence is also approved.
- Test success is ≥**50%** improvement in ≥1 diary parameter against a 3–5-day baseline; in retention, catheterisation falls ≥50%, ideally to none.
- The **S3 target** is ≈9 cm above the coccyx tip, 1–2 cm lateral. S3 gives anal "bellows" and **great-toe flexion**; S2 gives calf or foot rotation; S4 is perineal only.
- Percutaneous nerve evaluation (PNE) is an office test with a temporary lead for 5–7 days; lead migration lowers conversion, and it is not used in NOUR.
- The staged approach places a permanent tined quadripolar lead under fluoroscopy, tested for up to 4 weeks, with the generator in the buttock at stage 2.
- Programming is 210 µs, 10–16 Hz, at sensory threshold. Cycling equals continuous, and **bilateral is not superior** (randomised controlled trial, RCT).
- Battery life is ≥**10 years** (non-rechargeable) or ≥15 years (rechargeable). Devices are whole-body **MRI**-conditional at **1.5 and 3 T**.
- Complications: revision ~3% and explantation ~9% at 2 years (ROSETTA). †Site or leg pain is reprogrammed, then revised; infection needs explant.

## 3 · Tibial nerve stimulation

**ALGORITHM**

1. Supine; insert a 34-gauge needle ≈3–4 cm above the medial malleolus, behind the tibia, at 60–90°; ground pad on the same leg.
2. Stimulate at 20 Hz, 200 µs, 0–10 mA.
3. Confirm **great-toe flexion/toe fanning** or sole tingling.
4. Give **12 weekly sessions × 30 min**, then taper to maintenance.

- Indications are drug-refractory or intolerant OAB; it is safe in neurogenic OAB and in children.
- **Transcutaneous tibial nerve stimulation (TTNS)** uses home surface electrodes; it is cheaper but less effective than PTNS, and equals tolterodine with fewer side effects.
- Success is a ≥50% fall in urgency or UUI, and **relapse is likely without maintenance**. Results are better without DO.

## 4 · Other targets

- **Pudendal** stimulation (posterior approach) beat SNM in a blinded crossover and is salvage after failed SNM; it is not FDA approved.
- Dorsal genital nerve, sacral dermatome and intravaginal stimulation have limited roles and low-quality evidence.
- †**Intravesical stimulation** suits neurogenic underactive bladder with incomplete lesions, since afferents must be intact.

## 5 · Sacral anterior root stimulation (Brindley)

*It restores voiding in complete spinal cord injury.*

- Anterior root stimulation gives voiding, defecation and erection. It is combined with **posterior sacral rhizotomy**, which abolishes reflex incontinence and autonomic dysreflexia, raises capacity and compliance, and protects the kidneys.
- †Voiding is "post-stimulus" because the sphincter relaxes faster than the detrusor. †Rhizotomy costs reflex erection, reflex ejaculation and residual sacral sensation.

## 6 · Landmark trials

| Trial | Design | Result | Take-home |
|---|---|---|---|
| SUmiT | RCT, PTNS vs validated sham, 12 weeks, n = 220 | Responders 55% vs 21% | First sham-proven PTNS efficacy; maintenance keeps 77% relapse-free at 3 years |
| InSite | RCT, SNM vs standard medical therapy, mild–moderate OAB | Success 61% vs 42% at 6 months | SNM beats continued drugs |
| ROSETTA | RCT, onabotulinumtoxinA 200 U vs SNM, refractory UUI women | Toxin slightly fewer leaks at 6 months; equal at 2 years; more UTI, some clean intermittent catheterisation (CIC) | Both valid third-line; registered OAB toxin dose is 100 U |

# ★ Putting it together: voiding dysfunction decisions

## A · Functional classification as treatment map

| | Bladder | Outlet |
|---|---|---|
| **Failure to store** | Overactivity, low compliance: drugs, onabotulinumtoxinA, neuromodulation, augmentation | Sphincter weakness: pelvic floor training, duloxetine, sling, artificial sphincter, bulking |
| **Failure to empty** | Underactive, acontractile: clean intermittent catheterisation (CIC); sacral neuromodulation (SNM) for non-obstructive retention | Obstruction or dyssynergia: α-blocker, outlet surgery, sphincterotomy, CIC |

If neither is fixable, use a catheter or diversion (see Ch 104).

## B · Lesion level and urodynamics

| Lesion | Detrusor | Striated sphincter | Key risk |
|---|---|---|---|
| Above pons (stroke, Parkinson disease, dementia) | Overactive | Synergic | Urgency incontinence; low upper-tract risk |
| Pons–sacral cord (spinal cord injury [SCI], multiple sclerosis) | Overactive | **Dyssynergic** | High pressure damages upper tracts; autonomic dysreflexia (AD) if above T6 |
| Sacral, cauda equina, peripheral (diabetes, pelvic surgery) | Areflexic, reduced sensation | Fixed tone | Retention, overflow |

**KEY POINT**

A leak point or storage pressure ≥40 $cmH_2O$, or poor compliance, threatens the kidneys and must be treated (see Ch 107, 109).

## C · OAB ladder

**ALGORITHM**

1. Behavioural measures, bladder training and pelvic floor training.
2. Antimuscarinic or β3-agonist (β3 in the elderly). If partial response, then combine (see Ch 113).
3. If refractory, then onabotulinumtoxinA (100 U idiopathic, 200 U neurogenic; warn about CIC), tibial nerve stimulation or SNM (see Ch 115).
4. Last resort: augmentation or diversion.

## D · SCI bladder

**ALGORITHM**

1. **Spinal shock**: catheter until reflexes return.
2. **Then** urodynamics; CIC plus antimuscarinic or toxin for high-pressure overactivity.
3. If CIC is impossible, then sphincterotomy or diversion; survey upper tracts lifelong.
4. **AD** (above T6): hypertension, bradycardia, headache, flushing. Sit up, remove the stimulus and give a short-acting antihypertensive (see Ch 109).

# Other paper questions in this section

*Asked in the papers but not tied to a specific place in the book above.*

## SAQ EMG

Paper answer · P1

**Type:** SAQ (10 marks)

**Source:** Practice Set 9 Paper 1, Q3b

*EMG is taken as urodynamic electromyography of the striated urethral/anal sphincter and pelvic floor.*

### Definition and principle

- EMG (electromyography) records the electrical potentials of depolarising striated muscle membranes, here the **external urethral sphincter and pelvic floor**, during filling and voiding.
- It shows **coordination** between detrusor and sphincter, not sphincter strength. The bladder neck (smooth muscle) is invisible to EMG.
- External anal sphincter activity mirrors the urethral sphincter, except in some neurological diseases.

### Electrodes

| Surface (patch) | Concentric needle |
|---|---|
| Perianal patches at 3 and 9 o'clock (ICS [International Continence Society] standard: two skin electrodes plus reference) | Needle into the sphincter (girls beside the meatus; boys transperineal) |
| Comfortable; poorer signal; movement artefact | Best signal; painful; dislodges |
| Coordination only | Also motor unit analysis: acute versus chronic denervation |

### Normal pattern

- Filling: activity **rises gradually** (recruitment, guarding reflex).
- Voiding: EMG **falls silent before** detrusor pressure rises and stays silent until flow ends.
- **Lag time** (uroflow with patch EMG in children): relaxation to flow 2–6 s is normal; <2 s suggests DO (detrusor overactivity); >6 s suggests primary bladder neck dysfunction.

### Abnormal patterns

| Finding | Meaning |
|---|---|
| **DSD (detrusor–sphincter dyssynergia)** | Sphincter contracts during detrusor contraction; needs a **suprasacral** cord lesion (SCI [spinal cord injury], MS [multiple sclerosis]). Blaivas types 1–3 |
| Dysfunctional voiding | Same tracing without neurological lesion (learned) |
| Pseudodyssynergia | EMG rise from straining or trying to suppress a contraction |
| Denervation potentials | Sacral cord or cauda equina lesion; MSA (multiple system atrophy) |
| Complex repetitive discharges, decelerating bursts | **Fowler syndrome** (retention in young women); predicts SNM (sacral neuromodulation) response |

### Clinical uses

- Neurogenic bladder (spina bifida, SCI); voiding dysfunction in children; female retention; before sphincter surgery.
- **Limitation:** surface EMG misdiagnoses some dysfunctional voiders; **VUDS (videourodynamics)** is more accurate.

**Diagram to draw:** tracing of Pdet, flow and EMG: EMG silent during normal voiding versus active in DSD.

> **Key points to remember:**
> - EMG judges coordination, not strength.
> - True DSD needs a suprasacral lesion.
> - Lag time 2–6 s normal.
> - Fowler syndrome: complex repetitive discharges on needle EMG.

# Rapid Revision Sheet · Section 7

### Overactive bladder: pathophysiology, pharmacotherapy and non-pharmacological management (LAQ)

- OAB is defined by urgency; DO is a urodynamic finding.
- Final pathway: increased afferent signal (urothelial, myogenic) or failed central inhibition.
- Behavioural therapy first: BT and PFMT with urge suppression give 50–80% fewer leaks; antimuscarinic or β3-agonist next.
- M3 mediates contraction; β3 raises cAMP for storage-phase relaxation; voiding spared.
- Equal efficacy; β3 avoids dry mouth and dementia risk (10–65% with antimuscarinics >3 months).
- Darifenacin most M3-selective; tolterodine bladder-selective; solifenacin longest half-life.
- STAR: solifenacin beat tolterodine ER.
- Mirabegron 25 mg → 50 mg; avoid if blood pressure ≥180/110; CYP2D6 interaction (metoprolol).
- Elderly: mirabegron (β3), trospium or darifenacin preferred over other antimuscarinics.
- Solifenacin + mirabegron combination works (SYNERGY).
- Refractory OAB: PTNS (SUmiT 55% vs 21%), SNM (InSite 61% vs 42%), Botox 100 U.

### Duloxetin (SAQ)

- SNRI acting on Onuf's nucleus to increase sphincter tone.
- 40 mg twice daily; titrate from 20 mg to reduce nausea.
- Temporary benefit only; approved in Europe, not USA. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

### Urinary bladder compliance (SAQ)

- C = ΔV/ΔPdet; <20 mL/cm$H_2O$ suggests poor compliance.
- McGuire: DLPP >40 cm$H_2O$, 85% upper-tract damage.
- Pop-offs (VUR, diverticulum) hide true pressure; use video-urodynamics.

### Pathophysiology, classification and management of LUTS. (LAQ)

- ICS: storage, voiding, post-micturition; IPSS 0–7, 8–19, 20–35.
- Static (bulk, capsule) plus dynamic (α1A) obstruction; size ≠ symptoms.
- 5ARI for >40 mL; MTOPS combination cut progression 66%.
- TURP 30–80 mL; HoLEP or simple prostatectomy for >80 mL.

### Uroflowmetry (SAQ)

- Voided volume ≥150 mL; Qmax is the key value.
- Plateau = fixed obstruction (stricture); staccato = sphincter; interrupted = weak detrusor or straining.
- Qmax <10 mL/s suggests BOO but cannot exclude underactivity; Siroky and Liverpool nomograms.
- Cannot distinguish BOO from DUA.

### Urodynamic studies, including video urodynamics (SAQ)

- Pdet = Pves − Pabd.

- Ask a question first; reproduce the symptom.
- BOOI >40 obstructed; BCI <100 weak.
- DLPP >40 cm$H_2O$: upper-tract risk.
- VUDS = urodynamics + fluoroscopy with contrast.
- VUDS is the only test for bladder neck dysfunction, PBNO and DISD.
- Reflux pop-off makes compliance look falsely good.

**Describe neuromuscular dysfunction and management of bladder dysfunction following cervical spine surgery in young adult male (LAQ)**

- Suprasacral lesion = NDO + striated DSD; above T7–T8 also smooth sphincter dyssynergia.
- Autonomic dysreflexia above T6: drain bladder first, nitrates next.
- CIC + bladder relaxation (antimuscarinic, Botox 200 U) is the mainstay.
- DLPP ≥40 cm $H_2O$ threatens the kidneys.

**Bladder dysfunction non genitourinary systemic disease (SAQ)**

- Lesion level predicts pattern: brain DO; cord DO + DSD; peripheral areflexia.
- Diabetic cystopathy: low sensation, large capacity, high PVR; DO common early.
- Urodynamics before treating; reversible treatment first.

**Detrusor–external sphincter dyssynergia (DESD / DSD) (SAQ)**

- Lesion between pons and sacral cord; never brain lesions.
- Blaivas 1 incomplete; 2 and 3 complete.
- Worst in men; DLPP ≥40 cm$H_2O$ hostile.
- CIC plus detrusor relaxation first; sphincterotomy if CIC impossible.

**Detrusor underactivity and its management (SAQ)**

- BCI = Pdet@Qmax + 5Qmax; <100 weak.
- CISC is the mainstay; no effective drug; parasympathomimetics not recommended.
- SNM helps non-obstructive retention (Fowler syndrome); intact sensation predicts response.

**Botulinum toxin in urology (SAQ)**

- Cleaves SNAP-25; acts on efferent and afferent nerves.
- 100 U idiopathic OAB; 200 U NDO; spare trigone.
- Children: 10 U/kg (max 360 U) in series; FDA label 6 U/kg (max 200 U), approved for age ≥5 years (2021); used from about 2 years of age.
- Main risks: retention needing CISC, UTI.
- Other formulations (abobotulinumtoxinA, incobotulinumtoxinA) under study; doses are not interchangeable between products.
- Injection-free delivery under research: intravesical liposome-encapsulated toxin and other instillation vehicles.

**Neuromodulation for voiding dysfunction, including sacral neuromodulation (LAQ)**

- Afferent modulation explains SNM benefit in both OAB and retention.
- S3: bellows plus great-toe flexion; implant if the test improves ≥50%.
- SNM revision >30% at 5 years.
- SUmiT: PTNS 55% vs sham 21%; 12 weekly sessions.
- ROSETTA: Botox 200 U slightly better for UUI, more UTI and CIC.

- Rechargeable and long-life non-rechargeable stimulators, lasting at least 10 years.
- Full-body MRI-conditional devices at 1.5 T and 3 T.
- Implantable tibial nerve stimulators (BlueWind, eCoin) and transcutaneous tibial stimulation as less invasive alternatives.
- Standardised tined-lead placement improves outcomes and lowers revision.

**EMG (SAQ)**

- EMG judges coordination, not strength.
- True DSD needs a suprasacral lesion.
- Lag time 2–6 s normal.
- Fowler syndrome: complex repetitive discharges on needle EMG.

SECTION 8

# Nocturia & Male Incontinence

## 112 Nocturia

### 1 · Definition and terminology

- **Nocturia** (International Continence Society, ICS, 2019) is the number of voids during the **main sleep period**, from intending to sleep until intending to rise. The period follows the **individual's own schedule**, so shift workers are included.
- The definition does **not exclude 1 void per night** and does **not incorporate bother**.
- ICS 2019 dropped "wake **to void**"; non-urological causes are **leverage for interdisciplinary care, not grounds to exclude**.
- Guidance is from the European Association of Urology (EAU) and ICS; the AUA overactive bladder (OAB) guideline calls **isolated nocturia a separate entity**.

#### Key diary terms

| Term | Definition / formula | Meaning |
|---|---|---|
| Actual nightly voids (ANV) | Voids in the main sleep period | Severity count |
| First uninterrupted sleep period (FUSP) | Hours from sleep onset to the first night void | Key sleep measure |
| Nocturnal urine volume (NUV) | Night voids **plus the first morning void** | Volume made during sleep |
| Nocturnal polyuria index (NPi) | NUV / 24-h volume | Proportion made at night |
| Nocturnal urine production (NUP) | NUV / hours of sleep | Hourly rate |
| Maximum voided volume (MVV); nocturnal MVV (NMVV) | Largest void in 24 h; largest in the sleep period | Proxies for functional capacity |
| Nocturia index (Ni) | NUV / MVV | >1: night output exceeds capacity |
| Modified Ni (MNi) | NUV / NMVV | Uses night data only |
| Predicted nightly voids (PNV) | Ni − 1 | Voids expected if the bladder filled fully each time |
| Nocturnal bladder capacity index (NBCi) | ANV − PNV | >0: patient voids below capacity |

**Trap**: the first morning void is **split**. Its volume goes into NUV, but it counts as daytime frequency, not nocturia.

### 2 · Why nocturia matters

- **It is common**: up to **3 in 5** people >70 years void ≥2 times nightly; age is the most robust correlate.
- **Bother** is reported by most people with ≥**2** voids, but **any nocturia that brings a patient to a urologist is clinically significant**.

- **Sleep**: nocturia is the **commonest cause of adult sleep disruption**. Most first voids occur within **2–3 h** of sleep onset, breaking early slow-wave (N3) sleep and causing daytime fatigue.
- **Falls and hip fracture** are linked to nocturia more than to other lower urinary tract symptoms (LUTS).
- **Systemic disease**: nocturia may herald cardiometabolic (strongest), renal, endocrine or neurological disease; ≥2 voids is linked to higher mortality.

> **KEY POINT**
>
> The first void breaks restorative slow-wave sleep; the symptom may be the visible edge of systemic disease.

## 3 · Pathophysiology — production, capacity and sleep

Remember **"Sleep CALM"**: **S**leep disorders, **C**omorbidities, **A**ctions (behaviour, lifestyle), **L**ower urinary tract dysfunction (LUTD) and **M**edications.

- Every factor upsets **night production**, **functional capacity** or **sleep**; the sleep link is **bidirectional** ("convenience" voids).

### Nocturnal polyuria (NP) and global polyuria

| Criterion | Threshold | Comment |
|---|---|---|
| NPi (ICS) | > **20%** in young adults to > **33%** in older adults | Most used, but discriminates poorly |
| NUP (ICS alternative) | > **90 mL/h** | More specific for production-driven nocturia |
| Global polyuria | > **40 mL/kg/24 h** (>2.8 L at 70 kg) | Uncommon but serious; **always investigate** |

- **Krimpen study**: NPi >33% was found in 92% of older men with ≥2 voids, but also in 70% of men **without** nocturia.
- Excess night production contributes to **most adult nocturia**.

### Functional capacity and the mismatch

- Normal capacity is usually >300–400 mL; **150–300 mL** suggests diminished capacity, which may be **24-hour** (all voids small) or **night-specific** (partly convenience voids).
- **Ni is the strongest diary determinant of severity and of treatment response**.

> **ALGORITHM**
>
> 1. Worked example:
> 2. NUV is 800 mL and MVV is 200 mL, so Ni = 4.
> 3. PNV = Ni − 1 = 3.
> 4. The diary shows 6 night voids, so NBCi = 6 − 3 = **3**: the patient voids well below capacity.

Investigate diminished night capacity when NBCi is >**1.3**; a value >**2** is strongly associated with severe nocturia.

## 4 · The bladder diary

EAU (strong): use a diary for female LUTS, and for male LUTS with storage symptoms or nocturia.

**ALGORITHM**

1. Give the patient a measuring container.
2. Record time and volume of every void and every drink.
3. Mark bedtime (intending to sleep) and rising time.
4. Grade urgency and note leakage for each void.

- International Consultation on Incontinence (ICI) urgency codes: **0** no need (social void), **1** normal desire, **2** urgency that passed, **3** urgency but dry, **4** urgency with leak.
- **Diary ANV is the gold standard for severity**; questionnaires overestimate it. **FUSP is the most important sleep measure**.
- Duration: EAU advises **3 days** (Male guideline: at least 3); because of "diary fatigue", fewer than half complete it.
- A **nocturnal-only diary** lets MNi substitute for Ni but **cannot diagnose global polyuria**; EAU does not recommend it for women.

**KEY POINT**

Every cause works through production against capacity, gated by sleep; their ratio (Ni) predicts severity better than either alone.

## 5 · Evaluation — the Sleep CALM work-up

Do a basic work-up in all, then tailored tests and referral. The **TANGO** questionnaire screens all domains.

| Factor | History / examination | Tests (basic, then ancillary) |
|---|---|---|
| Sleep | Habits, apnoea, restless legs, pain, mood | STOP-BANG and sleep diary; then polysomnography, sleep specialist |
| Comorbidities | Blood pressure, volume overload, neurological signs | Metabolic panel; then HbA1c, ECG, echo, natriuretic peptide, urine albumin, calcium, thyroid, water deprivation |
| Actions | Fluid, alcohol, caffeine, sodium; body mass index (BMI) | Fluid intake diary |
| LUTD | LUTS, prior surgery; abdominal, genital, rectal examination | Urinalysis, post-void residual (PVR); then prostate-specific antigen (PSA), uroflowmetry, cystoscopy |
| Medications | Reconcile drugs and **dose timing** | — |

- **Trap**: urology is one row of five; first-line is to **optimise every Sleep CALM abnormality**.

## 6 · Sleep disorders

**Obstructive sleep apnoea (OSA)** is the best-described link; frequent nocturia predicts severe OSA.

| Arm | OSA mechanism | Effect of continuous positive airway pressure (CPAP) |
|---|---|---|
| Production (main) | Hypoxia causes pulmonary vasoconstriction, raising right atrial transmural pressure. This releases **atrial natriuretic peptide (ANP)**, causing natriuresis. | Lowers night production |
| Capacity | Intermittent hypoxia causes oxidative stress and detrusor instability | Less detrusor overactivity |
| Sleep | More light sleep, less slow-wave sleep | Increases slow-wave sleep |

- CPAP reduces nocturia by up to ≈**2 voids** in moderate–severe OSA, among the **largest effects of any intervention**. EAU Female (strong): refer suspected OSA to a sleep clinic.
- **STOP-BANG** scores Snoring, Tired, Observed apnoea, Pressure, BMI >**35**, Age >**50**, Neck >**40 cm** and Gender male. A score ≥**3** means high risk; the tool is highly sensitive.
- Ask the **primary motive for getting up** (urge, or already awake). Melatonin and insomnia therapy may help.

> **KEY POINT**
>
> OSA raises production, lowers capacity and fragments sleep — hence one of the largest nocturia responses to treatment.

## 7 · Comorbidities

- Comorbidities act mainly through **excess night production**. Nocturia may be the **first sign** of hypertension, metabolic syndrome, coronary, diabetic or early kidney disease.
- EAU: consider systemic causes in all; even **non-bothersome nocturia merits a frequency–volume chart**.
- **Volume overload** (e.g. heart failure): leg fluid is mobilised on lying down and excreted at night.
- **Hypertension**: blood pressure normally dips **10–20%** in sleep. **Non-dipping**, from renin–angiotensin–aldosterone dysregulation, causes **nocturnal natriuresis**.

### Disorders of water homeostasis

**Arginine vasopressin (AVP)** acts on collecting-duct **V2 receptors**; defects give **24-hour polyuria**.

| Condition | Defect | Causes |
|---|---|---|
| Central diabetes insipidus (DI) | Reduced AVP secretion | Brain tumour, haemorrhage, trauma, autoimmune |
| Nephrogenic DI | Kidney insensitive to AVP | Drugs, electrolytes, renal failure |
| Primary polydipsia | AVP suppressed by high intake | Idiopathic; psychiatric |

Arginine-stimulated **copeptin** appears superior to water deprivation. Always evaluate global polyuria.

> **KEY POINT**
>
> Comorbidity acts through the kidney (third-spaced fluid, nocturnal natriuresis, failed antidiuresis). Global polyuria and non-bothersome nocturia both warrant looking beyond urology.

## 8 · Actions — lifestyle and behaviour

- **Timing matters more than amount**. Water causes diuresis within ≈1 h; caffeine and alcohol within 2–3 h, and they disturb sleep.
- Restrict fluid, caffeine and alcohol from **3–4 h** before sleep; older adults should drink more by day.
- Also: **sodium** restriction, weight loss, exercise, **compression stockings**, bladder training, pelvic floor muscle training (PFMT), treat constipation.

## 9 · Lower urinary tract dysfunction

- Nocturia accompanies bladder outlet obstruction (BOO), OAB (up to **85%** of trial participants), underactivity, neurogenic bladder and menopause syndrome See Voiding Dysfunction book, Ch 110.
- **Isolated nocturia is not a LUTS**, so it is not expected to respond to bladder or prostate drugs.
- Do urinalysis in all, and **PVR in both sexes**. Check PSA only if it changes management. †Urodynamics are not routine See Voiding Dysfunction book, Ch 107.

### Benign prostatic obstruction

- In benign prostatic hyperplasia (BPH) trials, **α-blockers** carry most of the nocturia effect; 5α-reductase inhibitors (5ARIs) alone are weakest. Gains are fractions of a void with a large placebo response: **statistical, not clinical, significance** See Voiding Dysfunction book, Ch 113.
- **Surgery**: TURP beat microwave therapy, α-blockers and watchful waiting; **counsel** that nocturia may respond sub-optimally.

### Overactive bladder

- Tolterodine was negative for night voids but reduced urge-related voids; **fesoterodine** (nocturnal urgency endpoint) was positive.
- Patients with a **raised NBCi** respond best to an evening dose. β3-agonists give a marginal benefit (≈0.1–0.2 voids over placebo).
- **Behavioural therapy** comes first and matches or beats drugs. Antimuscarinics are on the **Beers Criteria** for people ≥65 years, the core nocturia population.

| Therapy | EAU Male LUTS | EAU Female LUTS |
|---|---|---|
| α1-blocker | Offer for nocturia with LUTS (weak) | — |
| 5ARI | Offer with LUTS and prostate > **40 mL** (weak) | — |
| Phosphodiesterase-5 (PDE5) inhibitor | **Do not offer** | — |
| Antimuscarinic | Offer with OAB (weak) | Offer after counselling (**strong**) |

| Therapy | EAU Male LUTS | EAU Female LUTS |
|---|---|---|
| PFMT; vaginal oestrogen | — | Offer PFMT (**strong**); offer oestrogen (weak) |

**KEY POINT**

LUT drugs act on capacity and outflow, but most nocturia is production-driven — effects are statistically significant, clinically small. Treat the mechanism, not the organ.

## 10 · Medications causing nocturia

| Mechanism | Drugs |
|---|---|
| Increased urine output | Diuretics; SSRIs (block antidiuretic hormone); **calcium channel blockers** (oedema, raised ANP); tetracycline; lithium (reduces aquaporin-2); **SGLT-2 inhibitors** (glycosuria) |
| Insomnia / central nervous system effects | Stimulants; α- and β-blockers; salbutamol, theophylline; corticosteroids, thyroxine; SSRIs; phenytoin |
| Bladder toxicity | Ketamine; tiaprofenic acid; cyclophosphamide |

Adjust dose timing first, or substitute another drug.

## 11 · Management algorithm and refractory nocturia

**ALGORITHM**

1. Take a history, examine, and obtain a diary and basic tests.
2. Identify the Sleep CALM factors.
3. Optimise each factor, with multidisciplinary help.
4. Add conservative measures.
5. If nocturia stays refractory, then consider an antidiuretic or a timed diuretic in selected patients.
6. Repeat the diary at every step.

- Treat interacting causes together (EAU **"Shared Care Pathway"**). **Target night volume even when capacity is small**.

### Desmopressin

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### SAQ Desmopressin (DDAVP)

Paper answer · P3

**Asked as:**

- Desmopressin. — *Winter 2023 Paper III Q3b (SAQ)*
- DDAVP — *Practice Set 3 Paper III Q3c (SAQ)*

**Type:** SAQ (10 marks)
(also answered in 2023-Winter Paper 3, Q3b)

#### Pharmacology

- DDAVP (**1-deamino-8-D-arginine vasopressin**, desmopressin), a synthetic analogue of AVP (arginine vasopressin/antidiuretic hormone).

- Deamination and D-arginine substitution prolong action (give longer action) and make it a **selective V2 agonist (selective V2 agonism) without pressor (V1) effect**.
- **Mechanism:** V2 receptor on collecting-duct cells raises cAMP (cyclic adenosine monophosphate), inserting **aquaporin-2**; water is reabsorbed, so urine is less and concentrated (V2 receptor → cAMP → aquaporin-2 insertion → water reabsorption → less, concentrated urine).
- Endothelial V2 action releases von Willebrand factor and factor VIII (haemostatic use).

### Formulations

| Form | Dose |
|---|---|
| Tablet | 0.2–0.4 mg at bedtime, about 1 h before bed (enuresis) |
| Oral lyophilisate (melt) | 120–240 µg, about 1 h before bed |
| Low-dose nasal spray (nocturia) | 0.83 or 1.66 µg, 30 min before bed |
| Sublingual ODT (orally disintegrating tablet) | 27.7 µg women, 55.3 µg men |
| IV / subcutaneous | 0.3 µg/kg (bleeding) |

### Uses (in urology)

1. **Monosymptomatic nocturnal enuresis:** first line with the alarm. **Rule of thirds**: one-third full, one-third partial, one-third no response. Acts (works) at once but relapses on stopping. Restrict fluids in the evening (avoid evening fluid); add oxybutynin if partial.
2. **Nocturia from nocturnal polyuria** (NPi [nocturnal polyuria index] >20–33%, age-dependent). EAU (European Association of Urology): offer below (under) 65; low dose above (over) 65 with ≥2 voids. Add to α-blocker for refractory nocturia in BPH (benign prostatic hyperplasia).
3. **Neurogenic nocturia** (multiple sclerosis, MSA [multiple system atrophy]).
4. **Bleeding:** uraemic platelet dysfunction, von Willebrand disease (including acquired, in Wilms tumour). Tachyphylaxis after about 3 (repeated) doses.
5. **Diagnosis:** central versus nephrogenic DI (diabetes insipidus).

### Adverse effects

- **Hyponatraemia** (most important): headache, nausea, confusion, seizures.
- Risk factors: **age ≥65**, baseline sodium <135 mmol/L, CrCl (creatinine clearance) <60 mL/min, higher dose; women are more sensitive.
- Most events occur in the **first 2–3 weeks**.

### Monitoring and contraindications

- Sodium at baseline, then within 1 week and at 1 month (EAU adds day 3), and after dose change.
- **Avoid:** hyponatraemia, polydipsia, heart failure, SIADH (syndrome of inappropriate antidiuretic hormone), significant renal impairment, **loop diuretics**.

**Diagram to draw:** V2 receptor–cAMP–aquaporin-2 pathway in the collecting duct.

**Key points to remember:**

- Selective V2 agonist; no pressor effect.
- Enuresis: 0.2–0.4 mg tablet or 120–240 µg melt; rule of thirds.
- Hyponatraemia: age ≥65, low sodium; check sodium early.

**Also, from the condensed book (not in the paper answer):**

- Desmopressin is the **only US-approved nocturia drug**, for **≥2 voids per night due to NP** (level 1b); some NP patients have a **blunted nocturnal AVP rise**.
- †Also contraindicated with systemic or inhaled glucocorticoids and estimated glomerular filtration rate (eGFR) <50 mL/min.

| | **Nasal spray** | **Orally disintegrating tablet (ODT)** |
|---|---|---|
| Dose | 1.66 mcg if <65 with no risk factors; **0.83 mcg** if ≥65 or at risk | **Sex-specific**: women **25 mcg**; men **50 mcg** (75 mcg added hyponatraemia, not efficacy) |
| Timing | ≈30 min before bed | Sublingual, ≈60 min before bed |

- **Women are more sensitive**, because the V2 receptor gene is on the **X chromosome**.
- †Advise minimal fluid from 1 h before until 8 h after the dose.
- Hyponatraemia is defined as sodium <**130 mmol/L**; most cases occur within **3 weeks**.
- **Sodium monitoring** (EAU): baseline, **day 3**, day 7, 1 month, then periodically (US label: baseline, within 7 days and ≈30 days). EAU Female: avoid if baseline sodium is low.
- A **low dose** still works and avoids antidiuresis outlasting sleep. **Age 65 is the pivot.**

### Timed diuretics

- Furosemide acts for 6–8 h. Evening dosing **causes** NP, whereas **morning or early-afternoon** dosing treats it (level 1b).
- Small trials (bumetanide 4–6 h, or furosemide 40 mg 6 h before bed) cut voids; monitor sodium.
- **Trap**: afternoon furosemide with bedtime desmopressin worked in a trial, but loop diuretics are a **contraindication** to desmopressin in US labelling.

**KEY POINT**

Refractory nocturia: move urine production out of the night. The price is sodium balance — dose, age and monitoring define safe use.

## 12 · Landmark trials

| **Trial** | **Design** | **Result** | **Take-home** |
|---|---|---|---|
| MTOPS | Doxazosin, finasteride, combination, placebo | Doxazosin and combination beat placebo; finasteride did not | α-blocker drives benefit |
| VA Cooperative | Terazosin, finasteride, combination, placebo | Only terazosin separated from placebo | 5ARI alone does not help |
| CombAT | Tamsulosin, dutasteride, combination | Combination −0.5 vs −0.3/−0.4 voids | Significant but small |

| Trial | Design | Result | Take-home |
|---|---|---|---|
| Fesoterodine RCT | Nocturnal urgency primary endpoint | Positive | Helps urgency-driven night voids |
| Nasal spray phase III | 0.83/1.66 mcg vs placebo | Both beat placebo | Basis of approval |
| ODT sex-specific RCTs | Women 25; men 50/75 mcg | Positive; 75 mcg caused more hyponatraemia | Sex-specific dosing |

# 123 Surgical Procedures for Sphincteric Incontinence in the Male: Artificial Urinary Sphincter and Perineal Slings

## 1 · Pathophysiology and device mechanisms

- Male **sphincteric (stress) urinary incontinence (SUI)** needs failure of **both** sphincters. Radical prostatectomy (RP), the commonest cause, removes the internal sphincter, leaving the rhabdosphincter.
- Other causes: radiotherapy (RT), pelvic fracture, neuropathy, vesicourethral anastomotic stenosis (VUAS) treatment. See Voiding Dysfunction book, Ch 106.

| Device | Mechanism | Site | Adjustable | Needs residual sphincter |
|---|---|---|---|---|
| Artificial urinary sphincter (AUS, AMS 800) | Circumferential compression, cycled by the patient | Bulbar urethra or bladder neck | No | No |
| Advance / Advance XP | **Repositions** and lengthens the urethra; no compression | Bulbar, transobturator | No | **Yes** |
| Virtue (quadratic) | Prepubic compression and elevation | Bulbar, four arms | No | Less dependent |
| ATOMS (adjustable transobturator male system) | Cushion compression | Bulbar, transobturator | Yes, via scrotal port | Less dependent |
| ProACT | Two para-urethral balloons | Bladder neck | Yes, via scrotal ports | — |

- The **AUS** has a cuff, a scrotal pump and a pressure-regulating balloon (PRB) that sets cuff pressure. The cuff refills passively over minutes.
- A transobturator sling raises leak point pressure **without obstruction** by elevating the bulb and rhabdosphincter.

> **KEY POINT**
>
> Every device compresses the urethra or repositions a functioning sphincter. Trap: ATOMS and Virtue are "slings" but compress — only the Advance depends absolutely on a coapting residual sphincter.

## 2 · Indications and device selection

- Indication: **irreversible sphincter incompetence** with bothersome incontinence (UI) after conservative care. Do not delay surgery beyond **6 months** for severe SUI.
- 24-hour pad weight grades severity: mild **<150 g**, moderate **<400 g**. AUS is preferred if >250 g or after RT (MASTER trial See Ch 106).

| Stratum (AUA/SUFU) | Recommendation |
|---|---|
| Severe UI or **any RT** | AUS first-line; sling only with counselling |
| Moderate, no RT | AUS or male sling; adjustable balloons in specialised centres |

| Stratum (AUA/SUFU) | Recommendation |
|---|---|
| Mild, no RT | Continue pelvic floor muscle training (PFMT), sling, possibly AUS |

- Counsel: ~1 thin pad/day, AUS lifespan **7–10 years**; failed slings mostly go to AUS.
- **A sling needs** mild–moderate SUI, a contractile bladder (if hypocontractile, choose AUS), a non-irradiated urethra and a coapting sphincter.
- **An AUS needs** manual dexterity and cognition, a healthy urethra and adequate bladder compliance.

| Absolute — need an alternative | Relative |
|---|---|
| Low compliance threatening the kidneys; low capacity; urethra too poor (alternatives: clamp, catheter, bladder neck closure, diversion) | Frequent transurethral work (bladder cancer, refractory VUAS, radiation cystitis), because it risks erosion and infection |

- Radiation cystitis needs hyperbaric oxygen and **6 months** free of bleeding before AUS.

> **KEY POINT**
>
> Each device's contraindication mirrors the other's requirement.

## 3 · The high-risk ("fragile") urethra

- Strong erosion risk factors are **prior RT, prior erosion or infection, and urethroplasty** (worse than RT, as transection divides the bulbourethral arteries).
- After a prior erosion, 5-year repeat erosion approaches **50%**.
- Countermeasures: optimise medically, use a **transcorporal (TC) cuff**, a PRB of **51–60 cm H2O**, a cuff 0.5 cm larger, and nightly deactivation.

> **KEY POINT**
>
> Erosion = spongiosal ischaemia under constant pressure. Countermeasures add tissue (TC tunica) or lower pressure — never squeeze harder.

## 4 · Operative technique

- Sterile urine; give IV antibiotics within **60 min** of incision: aminoglycoside 5 mg/kg plus a first/second-generation cephalosporin or vancomycin.
- Place the cuff on the **proximal bulbar urethra, proximal to corporal fusion**, where spongiosum is thickest.

**ALGORITHM**

1. Midline perineal incision; divide bulbospongiosus.
2. Incise Buck fascia and make a 2-cm tunnel around the urethra.
3. Measure and place the cuff (usually 4.0–4.5 cm).
4. Place the PRB preperitoneally and fill it with **23 mL**.
5. Place the pump subdartos in the anterior scrotum.
6. Cycle the device; do urethroscopy.
7. Leave it **deactivated for 4–6 weeks**.

- Less tight cuffs continent better; a **3.5-cm cuff** raises erosion after RT. †Deactivate the cuff before any later catheterisation or endoscopy.

| PRB (cm H2O) | Use |
|---|---|
| 51–60 | High-risk bulbar (RT, prior erosion, atrophy); trades continence for safety |
| **61–70** | Standard bulbar cuff |
| 71–80 | Bladder neck cuff |

- **Tandem cuff**: a second cuff ≥1.5–2 cm distal; not first-line, because the distal cuff erodes more.
- **Transcorporal cuff** (Webster 2002): ventral tunica stays on the dorsal urethra and adds bulk. Use it for a urethra <3.5 cm, revisions, prior erosion or urethroplasty; erosion benefit is unproven.
- **Bladder neck AUS** suits an intact prostate (neuropathic bladder) and is **contraindicated after RP**; cuff ≥8 cm, PRB 71–80.

**ALGORITHM**

1. **Advance XP**:
2. Perineal incision; open bulbospongiosus.
3. Divide the perineal body to release the bulb.
4. Pass the helical trocar outside-in below the adductor longus tendon.
5. Fix the mesh to the spongiosum.
6. Tension under cystoscopy until the mucosa coapts (bulb moves 3–4 cm proximally).

- **ATOMS** leaves bulbospongiosus **intact** and is adjusted in clinic via the scrotal port; **ProACT** balloons are placed at the bladder neck under fluoroscopy.

## 5 · AUS complications and management

**ALGORITHM**

1. **Retention**:
2. Confirm the cuff is deactivated.
3. Insert a small catheter for 24–48 h.
4. If the second trial fails, then place a **suprapubic cystostomy**, avoiding the PRB.
5. If retention lasts weeks, then the cuff is undersized: replace it.

- **Infection** occurs in **1–3%** (up to 10% if high-risk), usually staphylococci; **scrotal pain is the key early sign**. If antibiotics fail, explant all; salvage only if not eroded, septic or immunosuppressed.
- **Erosion** occurs in **2–15%** (up to 35% with several risk factors). Remove **all components**. If <50% of the circumference is lost, leave a catheter ~3 weeks; if >50%, do in situ repair or anastomosis.
- Reimplant after **3–6 months** at a **new site**, using a TC cuff with or without a low-pressure PRB.
- **Atrophy** causes a **gradual** recurrent leak. Downsize, relocate (TC cuff distally) or add a tandem cuff; **never simply raise PRB pressure**.
- **Mechanical failure**: within 3 years, replace the failed part; **after 3 years, replace the entire device**.

| Recurrent leak after AUS | Clue, then how to confirm |
|---|---|
| Deactivated | Sudden; reactivate |
| Fluid leak | Sudden; PRB shrinks on imaging |
| Atrophy | Gradual; PRB volume normal (>20 mL) |
| Erosion / detrusor overactivity | Cystoscopy / urodynamics |

> **KEY POINT**
>
> Sudden = device, gradual = urethra. Manage every complication to keep a salvageable urethra for the next cuff.

## 6 · Sling complications and failure

| Device | Commonest | Erosion / explant |
|---|---|---|
| Transobturator | Transient groin pain; retention or urgency 4–10% | 0–0.4% |
| ATOMS | More complications | Erosion/infection ~10%; explant 10–16% |
| ProACT | **Perforation** (~5%); migration; revision 22% | ~4% |

- Cystoscope every worsening sling. After a failed sling the **AUS is the gold standard**, with results as good as primary AUS.

## 7 · Special circumstances

- A stricture at the cuff may herald erosion.
- With a penile prosthesis, synchronous and staged placement do equally well; if staged, place the AUS first. **Climacturia** responds to a sling or AUS, adding a **mini-jupette** graft if there is erectile dysfunction.

## 8 · Outcomes

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### SAQ Post-prostatectomy incontinence

Paper answer · P2

**Asked as:**

- Post prostatectomy incontinence. — *Summer 2021 Paper II Q3a (SAQ)*
- Prostatectomy incontinence — *Practice Set 11 Paper II Q3f (SAQ)*

**Type:** SAQ (10 marks)

(Practice Set 11 Paper II Q3f: also answered in 2021-Summer Paper 2, Q3a)

#### Definition and incidence

- PPI (post-prostatectomy incontinence) is mainly **SUI (stress urinary incontinence)** after RP (radical prostatectomy), the commonest cause of male SUI. Rates vary with definition ("no pad" vs "safety pad") and reporting; expert no-pad continence is in the mid-90s % at 1 year; continence improves for up to 2 years.

### Pathophysiology

- RP removes the internal sphincter (bladder neck, prostatic urethra), so continence depends (entirely) on the external **rhabdosphincter**.
- Causes: apical injury to rhabdosphincter or pudendal branches (sphincter or nerve injury at the apex), short membranous urethra, afferent denervation; also detrusor overactivity, low compliance, **VUAS (vesicourethral anastomotic stenosis)**.
- Nerve sparing and posterior reconstruction (Rocco stitch) give **earlier, not better final** continence.

### Evaluation

- History (type, severity, radiotherapy, cancer status), bladder diary, **24-hour pad weight**: mild <200 g, moderate 200–400 g, severe >400 g (another source: mild <150 g, moderate <400 g, severe more; thresholds vary).
- **Standing cough test**, grade (grades) 0 to 4 (0–4: leak reported but not seen; delayed drops; early drops; delayed stream; early persistent stream).
- Urinalysis, creatinine, PVR (post-void residual), PSA (prostate-specific antigen). **Cystoscopy** excludes VUAS and judges (shows) residual sphincter coaptation; urodynamics when needed (if mixed symptoms).

### Management

1. **Wait 6–12 months** (improves up to 2 years). Supervised **PFMT (pelvic floor muscle training)** with biofeedback, ideally preoperative (started before surgery); pads, condom catheter or (penile) clamp.
2. **Duloxetine** (off-label). **Bulking agents are not advised.**
3. **Surgery** (AUA/SUFU [American Urological Association/Society of Urodynamics, Female Pelvic Medicine and Urogenital Reconstruction] 2019, EAU [European Association of Urology]):

| Situation | Choice |
|---|---|
| Mild–moderate, no radiotherapy | **Male sling**: AdVance / AdVance XP (repositions the bulbar urethra; needs a coapting sphincter) or adjustable systems (ATOMS [adjustable transobturator male system]) |
| Severe, or any radiotherapy | **AUS (artificial urinary sphincter), AMS 800**: gold standard; cuff pressure 61–70 cm $H_2O$ |
| ProACT balloons | Expert centres; avoid after radiotherapy |

- **MASTER trial:** male sling non-inferior to AUS at 12 months.
- AUS: social continence (0–1 pad) 55–77%; revision 10–29% at 3 years; lasts (lifespan) 8–10 years. Erosion risks: radiotherapy (risk higher after radiotherapy), prior erosion, urethroplasty (use a **transcorporal cuff** for fragile urethra).
- **Also know:** climacturia; treat VUAS first.

**Diagram to draw:** AMS 800 with cuff, scrotal pump and pressure-regulating balloon.

**Key points to remember:**

- Wait 6–12 months with PFMT before surgery.
- Sling for mild–moderate; AUS for severe or irradiated.
- MASTER: sling non-inferior to AUS at 12 months.

*See the paper answer **Female SUI: continence mechanism, classification, evaluation (UDS) and management, including male SUI surgery** (Section 6, Ch 108).*

**Also, from the condensed book (not in the paper answer):**

- Cure clusters at 55–65% for all devices (ProACT ~60–67%); Advance XP results are poorer after RT and in severe UI.

# ★ Putting it together

## A · Nocturia: three diary questions, by danger

**ALGORITHM**

1. **Global polyuria?** If >40 mL/kg/24 h, then investigate for diabetes, diabetes insipidus or polydipsia.
2. **Nocturnal polyuria?** If night fraction >20% (young) or >33% (older), then seek heart failure, sleep apnoea, oedema or evening fluids; consider desmopressin with sodium checks.
3. **Reduced night capacity?** If NBCi >1.3, then treat overactive bladder, obstruction or residual (See Ch 112).

## B · Post-prostatectomy incontinence

**KEY POINT**

Pelvic floor training first; operate on severe leakage by ~6 months. Pad weight: mild <150 g, moderate <400 g. A sling suits mild–moderate leakage, a contractile detrusor and no radiotherapy. Leakage >250 g, radiotherapy or a failed sling needs a sphincter. For erosion risk, add tissue (transcorporal cuff) or lower pressure; never tighten. Remove eroded or infected devices (See Ch 123).

# Rapid Revision Sheet · Section 8

### Desmopressin (DDAVP) (SAQ)

- Selective V2 agonist; no pressor effect.
- Enuresis: 0.2–0.4 mg tablet or 120–240 µg melt; rule of thirds.
- Hyponatraemia: age ≥65, low sodium; check sodium early. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

### Post-prostatectomy incontinence (SAQ)

- Wait 6–12 months with PFMT before surgery.
- Sling for mild–moderate; AUS for severe or irradiated.
- MASTER: sling non-inferior to AUS at 12 months. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*